```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     2 ************************************************************************
                                                     3 *
                                                     4 *        PLO-Extension Facility function ARMODE tests
                                                     5 *
                                                     6 *        James Wekel November 2026
                                                     7 *
                                                     8 *        NOTE: All address space, access-register mode, etc. setup is
                                                     9 *        based on test 'mvcos-001.asm'.
                                                    10 ************************************************************************


                                                    12 ************************************************************************
                                                    13 *
                                                    14 *        basic instruction tests
                                                    15 *
                                                    16 ************************************************************
                                                    17 *  This program tests proper functioning of the PLO-Extension
                                                    18 *  facility functions with ACCESS-REGISTER mode. Specification
                                                    19 *  exceptions are not tested.
                                                    20 *
                                                    21 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                    22 *  obvious coding errors.  None of the tests are thorough.  They are
                                                    23 *  NOT designed to test all aspects of any of the instructions.
                                                    24 *
                                                    25 *-----------------------------------------------------------------
                                                    26 * Programmer's Note:
                                                    27 *
                                                    28 * The following tests have a LOT of duplicate (copy/paste) code
                                                    29 * to allow for minor modifications specific to each test.
                                                    30 *-----------------------------------------------------------------
                                                    31 *
                                                    32 ************************************************************************
                                                    33 *
                                                    34 *  Example Hercules Testcase:
                                                    35 *
                                                    36 *    *Testcase "plo-02-extension-armode
                                                    37 *    *
                                                    38 *    *   PLO-Extension Functions Facility tests
                                                    39 *    *
                                                    40 *    *   # ---------------------------------------------------------
                                                    41 *    *   #  This tests only the basic function of the instruction.
                                                    42 *    *   #  Exceptions are NOT tested.
                                                    43 *    *   # ---------------------------------------------------------
                                                    44 *    *
                                                    45 *    mainsize    2
                                                    46 *    numcpu      1
                                                    47 *    sysclear
                                                    48 *    archlvl     z/Arch
                                                    49 *
                                                    50 *    loadcore    "$(testpath)/"plo-02-extension-armode.core.core" 0x0
                                                    51 *
                                                    52 *    diag8cmd    enable    # (needed for messages to Hercules console)
                                                    53 *    runtest     5         #
                                                    54 *    diag8cmd    disable   # (reset back to default)
                                                    55 *
                                                    56 *    *Done
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                                                      57 *
                                                      58 ********************************************************************************
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                       60 ************************************************************************
                                                       61 *
                                                       62 *                    Memory Map - REAL STORAGE
                                                       63 *
                                                       64
                                                       65 *                  General Layout
                                                       66 *
                                                       67 *                  Description
                                                       68 *                  -----------------------------------------------------
                                                       69 * 0    - 48K    - Absolute page 0 and 1, program code
                                                       70 * 48K  - 128K   - segment and page tables
                                                       71 * 128K - 384K   - Pages for Primary virtual space       at vaddr 20000
                                                       72 * 384K - 640K   - Pages for Secondary virtual space     at vaddr 20000
                                                       73 * 640K - 896K   - Pages for Home virtual space          at vaddr 20000
                                                       74 *
                                                       75 *
                                                       76 *                    Memory Map - REAL STORAGE
                                                       77 *                    ========================
                                                       78 *          Hex
                                                       79 * RAddr    Len     Description
                                                       80 * -----    ----    -----------------------------------------------------
                                                       81 *     0    2000  - Absolute page 0 and 1
                                                       82 *  2000    A000  - Program code
                                                       83 *  C000    1000  - Segment table, primary space
                                                       84 *  D000    1000  - Segment table, secondary space
                                                       85 *  E000    1000  - Segment table, home space
                                                       86 * 10000    4000  - Page tables, primary space
                                                       87 * 14000    4000  - Page tables, secondary space
                                                       88 * 18000    4000  - Page tables, home space
                                                       89 * 19000    1000  - Primary ASTE, DUCT, DU-AL, ALE blocks
                                                       90 * 1A000    1000  - Home ASTE block
                                                       91 * 1B000    5000  - unused; available
                                                       92 * 20000   40000  - Pages backing Primary virtual space at vaddr 20000
                                                       93 * 60000   40000  - Pages backing Secondary virtual space at vaddr 20000
                                                       94 * A0000   40000  - Pages backing Home virtual space at vaddr 20000
                                                       95 *
                                                       96 *
                                                       97 *                      Memory Map - VIRTUAL STORAGE
                                                       98 *                      ============================
                                                       99 *          Hex
                                                      100 * VAddr    Len    Raddr  Key   Description
                                                      101 * -----    ----    ----  ---   --------------------------------------
                                                      102 * 0000    C0000  00000  00 - Common V=R storage (all address spaces)
                                                      103 * 20000   40000  20000  00 - Primary space storage
                                                      104 * 20000   40000  60000  00 - Secondary space storage
                                                      105 * 20000   40000  A0000  00 - Home space storage
                                                      106 *
                                                      107 *
                                                      108 ************************************************************************
                                                      109
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   111 ************************************************************************
                                                   112 *        FCHECK Macro - Is a Facility Bit set?
                                                   113 *
                                                   114 *        If the facility bit is NOT set, an message is issued and
                                                   115 *        the test is skipped.
                                                   116 *
                                                   117 *        Fcheck uses R0, R1 and R2
                                                   118 *
                                                   119 * eg.    FCHECK 134,'vector-packed-decimal'
                                                   120 ************************************************************************
                                                   121          MACRO
                                                   122          FCHECK &BITNO,&NOTSETMSG
                                                   123 .*                      &BITNO : facility bit number to check
                                                   124 .*                      &NOTSETMSG : 'facility name'
                                                   125          LCLA  &FBBYTE           Facility bit in Byte
                                                   126          LCLA  &FBBIT            Facility bit within Byte
                                                   127
                                                   128          LCLA  &L(8)
                                                   129 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                   130
                                                   131 &FBBYTE  SETA  &BITNO/8
                                                   132 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                   133 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                   134
                                                   135          B     X&SYSNDX
                                                   136 *                                      Fcheck data area
                                                   137 *                                      skip messgae
                                                   138 SKT&SYSNDX DC  C'          Skipping tests: '
                                                   139          DC    C&NOTSETMSG
                                                   140          DC    C' facility (bit &BITNO) is not installed.'
                                                   141 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                   142 *                                      facility bits
                                                   143          DS    FD                      gap
                                                   144 FB&SYSNDX DS   4FD
                                                   145          DS    FD                      gap
                                                   146 *
                                                   147 X&SYSNDX EQU *
                                                   148          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                   149          STFLE FB&SYSNDX               get facility bits
                                                   150
                                                   151          XGR   R0,R0
                                                   152          IC    R0,FB&SYSNDX+&FBBYTE    get fbit byte
                                                   153          N     R0,=F'&FBBIT'           is bit set?
                                                   154          BNZ   XC&SYSNDX
                                                   155 *
                                                   156 * facility bit not set, issue message and exit
                                                   157 *
                                                   158          LA    R0,SKL&SYSNDX           message length
                                                   159          LA    R1,SKT&SYSNDX           message address
                                                   160          BAL   R2,MSG
                                                   161
                                                   162          B     EOJ
                                                   163 XC&SYSNDX EQU *
                                                   164          MEND
                                                   165
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     167 * ================================================================================
                                                     168 * LOW CORE
                                                     169 * ================================================================================
                                                     170 *
                               00000000  0004DA73    171 PLOETST  START 0
                               00000000  00000001    172 A000LABL EQU   *
00000000                       00000000              173          USING PLOETST,R0          Low core addressability

                               00000140  00000000    175 SVOLDPSW EQU   PLOETST+X'140'      z/Arch Supervisor call old PSW

00000000                       00000000  000001A0    177          ORG   PLOETST+X'1A0'      z/Architecure RESTART PSW
000001A0  00000001 80000000                         178          DC    X'0000000180000000'
000001A8  00000000 00002000                         179          DC    AD(BEGIN)

000001B0                       000001B0  000001D0    181          ORG   PLOETST+X'1D0'      z/Architecure PROGRAM CHECK PSW
000001D0  00020001 80000000                         182          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                         183          DC    AD(X'DEAD')
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    185 ************************************************************************
                                                    186 *               The actual "PLOETST" program itself...
                                                    187 ************************************************************************
                                                    188 *
                                                    189 *  Architecture Mode: z/Arch
                                                    190 *  Register Usage:
                                                    191 *
                                                    192 *   R0-R4      (work)
                                                    193 *   R5       Testing control table - base current entry
                                                    194 *   R6         (test work) - secondary space
                                                    195 *   R7         (test work) - home space
                                                    196 *   R8       First base register
                                                    197 *   R9       Second base register
                                                    198 *   R10      Third base register
                                                    199 *   R11      Return from test
                                                    200 *   R12      Address of current test
                                                    201 *   R12        (test work) - secondary space
                                                    202 *   R13        (test work) - home space
                                                    203 *   R14      Subroutine call return
                                                    204 *   R15      Secondary Subroutine call return
                                                    205 *            - test: used as PLO R3 (alet register)
                                                    206 *
                                                    207 ************************************************************************

000001E0                      000001E0  00002000    209          ORG   PLOETST+X'2000'     Start of actual test program...

00002000                      00002000              211          USING  BEGIN,R8           FIRST Base Register
00002000                      00003000              212          USING  BEGIN+4096,R9      SECOND Base Register
00002000                      00004000              213          USING  BEGIN+8192,R10     THIRD Base Register

00002000  0580                                      215 BEGIN    BALR  R8,0                Initalize FIRST base register
00002002  0680                                      216          BCTR  R8,0                Initalize FIRST base register
00002004  0680                                      217          BCTR  R8,0                Initalize FIRST base register

00002006  4190 8800                   00000800      219          LA    R9,2048(,R8)        Initalize SECOND base register
0000200A  4190 9800                   00000800      220          LA    R9,2048(,R9)        Initalize SECOND base register
                                                    221
0000200E  41A0 9800                   00000800      222          LA    R10,2048(,R9)       Initalize THIRD base register
00002012  41A0 A800                   00000800      223          LA    R10,2048(,R10)      Initalize THIRD base register
                                                    224
                                                    225 ************************************************************************
                                                    226 * Is PLO-extension facility installed  (bit 87)
                                                    227 ************************************************************************
                                                    228
                                                    229          FCHECK 87,'PLO-extension'
00002016  47F0 8098                   00002098      230+         B     X0001
                                                    231+*                                  Fcheck data area
                                                    232+*                                  skip messgae
0000201A  40404040 40404040                         233+SKT0001  DC    C'          Skipping tests: '
00002034  D7D3D660 85A7A385                         234+         DC    C'PLO-extension'
00002041  40868183 899389A3                         235+         DC    C' facility (bit 87) is not installed.'
                              0000004B  00000001    236+SKL0001  EQU   *-SKT0001
                                                    237+*                                  facility bits
00002068  00000000 00000000                         238+         DS    FD                  gap
00002070  00000000 00000000                         239+FB0001   DS    4FD
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00002090  00000000 00000000                          240+          DS    FD                        gap
                                                     241+*
                                00002098  00000001  242+X0001     EQU   *
00002098  4100 0004                         00000004  243+          LA    R0,((X0001-FB0001)/8)-1
0000209C  B2B0 8070                         00002070  244+          STFLE FB0001                    get facility bits
000020A0  B982 0000                                   245+          XGR   R0,R0
000020A4  4300 807A                         0000207A  246+          IC    R0,FB0001+10              get fbit byte
000020A8  5400 8700                         00002700  247+          N     R0,=F'1'                  is bit set?
000020AC  4770 80C0                         000020C0  248+          BNZ   XC0001
                                                     249+*
                                                     250+* facility bit not set, issue message and exit
                                                     251+*
000020B0  4100 004B                         0000004B  252+          LA    R0,SKL0001                message length
000020B4  4110 801A                         0000201A  253+          LA    R1,SKT0001                message address
000020B8  4520 8600                         00002600  254+          BAL   R2,MSG
000020BC  47F0 86C8                         000026C8  255+          B     EOJ
                                00000020C0  00000001  256+XC0001    EQU   *
                                                     257
                                                     258 ************************************************************************
                                                     259 * Build  Page table: primary space
                                                     260 ************************************************************************
                                                     261 * common 0K - 128K
000020C0  B982 0000                                  262           XGR   R0,R0                     count entries
000020C4  C208 0000 0020                             263           AGFI  R0,(128/4)
000020CA  B982 0011                                  264           XGR   R1,R1                     page table entry value - 0k
000020CE  E320 86E8 0004                    000026E8  265           LG    R2,=AD(PAGPRI)            page table entry address
                                                     266
                                00000020D4  00000001  267 PP01      EQU   *
000020D4  E312 0000 0024                    00000000  268           STG   R1,0(R2)
000020DA  C218 0000 1000                              269           AGFI  R1,X'1000'                next page (+4096)
000020E0  C228 0000 0008                              270           AGFI  R2,8                      next entry addres
000020E6  4600 80D4                         000020D4  271           BCT   R0,PP01
                                                     272
                                                     273 * private real: 128K - 384K (256K)
000020EA  B982 0000                                  274           XGR   R0,R0                     count entries
000020EE  C208 0000 0040                             275           AGFI  R0,(256/4)
000020F4  B982 0011                                  276           XGR   R1,R1                     page table entry value
000020F8  C218 0002 0000                             277           AGFI  R1,(128*k)
                                                     278
                                000020FE  00000001  279 PP02      EQU   *
000020FE  E312 0000 0024                    00000000  280           STG   R1,0(R2)
00002104  C218 0000 1000                              281           AGFI  R1,X'1000'                next page (+4096)
0000210A  C228 0000 0008                              282           AGFI  R2,8                      next entry addres
00002110  4600 80FE                         000020FE  283           BCT   R0,PP02
                                                     284
                                                     285 ************************************************************************
                                                     286 * Build  Page table: secondary space
                                                     287 ************************************************************************
                                                     288 * common 0K - 128K
00002114  B982 0000                                  289           XGR   R0,R0                     count entries
00002118  C208 0000 0020                             290           AGFI  R0,(128/4)
0000211E  B982 0011                                  291           XGR   R1,R1                     page table entry value
00002122  E320 86F0 0004                    000026F0  292           LG    R2,=AD(PAGSEC)            page table entry addresss
                                                     293
                                00002128  00000001  294 PS01      EQU   *
00002128  E312 0000 0024                    00000000  295           STG   R1,0(R2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000212E  C218 0000 1000                             296          AGFI  R1,X'1000'           next page (+4096)
00002134  C228 0000 0008                             297          AGFI  R2,8                 next entry addres
0000213A  4600 8128                   00002128       298          BCT   R0,PS01
                                                     299
                                                     300 * private real: 384K - 640K (256K)
0000213E  B982 0000                                  301          XGR   R0,R0                count entries
00002142  C208 0000 0040                             302          AGFI  R0,(256/4)
00002148  B982 0011                                  303          XGR   R1,R1                page table entry value
0000214C  C218 0006 0000                             304          AGFI  R1,(384*K)
                                                     305
                              00002152  00000001     306 PS02     EQU   *
00002152  E312 0000 0024                   00000000  307          STG   R1,0(R2)
00002158  C218 0000 1000                             308          AGFI  R1,X'1000'           next page (+4096)
0000215E  C228 0000 0008                             309          AGFI  R2,8                 next entry addres
00002164  4600 8152                   00002152       310          BCT   R0,PS02
                                                     311
                                                     312 ********************************************************************************
                                                     313 * Build  Page tables: Home space
                                                     314 ********************************************************************************
                                                     315 * common 0K - 128K
00002168  B982 0000                                  316          XGR   R0,R0                count entries
0000216C  C208 0000 0020                             317          AGFI  R0,(128/4)
00002172  B982 0011                                  318          XGR   R1,R1                page table entry value
00002176  E320 86F8 0004                   000026F8  319          LG    R2,=AD(PAGHOM)       page table entry address
                                                     320
                              0000217C  00000001     321 PH01     EQU   *
0000217C  E312 0000 0024                   00000000  322          STG   R1,0(R2)
00002182  C218 0000 1000                             323          AGFI  R1,X'1000'           next page (+4096)
00002188  C228 0000 0008                             324          AGFI  R2,8                 next entry addres
0000218E  4600 817C                   0000217C       325          BCT   R0,PH01
                                                     326
                                                     327 * private real: 640K - 896K (256K)
00002192  B982 0000                                  328          XGR   R0,R0                count entries
00002196  C208 0000 0040                             329          AGFI  R0,(256/4)
0000219C  B982 0011                                  330          XGR   R1,R1                page table entry value
000021A0  C218 000A 0000                             331          AGFI  R1,(640*K)
                                                     332
                              000021A6  00000001     333 PH02     EQU   *
000021A6  E312 0000 0024                   00000000  334          STG   R1,0(R2)
000021AC  C218 0000 1000                             335          AGFI  R1,X'1000'           next page (+4096)
000021B2  C228 0000 0008                             336          AGFI  R2,8                 next entry addres
000021B8  4600 81A6                   000021A6       337          BCT   R0,PH02
                                                     338
                                                     339 ********************************************************************************
                                                     340 * Setup control
                                                     341 ********************************************************************************
                                                     342
000021BC  9A0F AA00                   00004A00       343          LAM   AR0,AR15,AREGS       Clear all ARs
000021C0  EB0F A980 002F              00004980       344          LCTLG CR0,CR15,CREGS       Load all the CRs
000021C6  8000 8718                   00002718       345          SSM   =X'04'               Turn on DAT
                                                     346 *
000021CA  B600 879C                   0000279C       347          STCTL R0,R0,CTLR0          Store CR0 to enable AFP
000021CE  9604 879D                   0000279D       348          OI    CTLR0+1,X'04'        Turn on AFP bit
000021D2  9602 879D                   0000279D       349          OI    CTLR0+1,X'02'        Turn on Vector bit
000021D6  B700 879C                   0000279C       350          LCTL  R0,R0,CTLR0          Reload updated CR0
                                                     351
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    352 ********************************************************************************
                                                    353 *    Do tests in the PLOETST table
                                                    354 ********************************************************************************
                                                    355
000021DA  58C0 8704                      00002704   356          L     R12,=A(PLOTESTS)    get table of test addresses
                                                    357
                               000021DE  00000001   358 NEXTTEST EQU   *
000021DE  5850 C000                      00000000   359          L     R5,0(0,R12)         get test address
000021E2  1255                                      360          LTR   R5,R5                have a test?
000021E4  4780 82A0                      000022A0   361          BZ    ENDTEST                 done?
                                                    362
000021E8                       00000000             363          USING PLOETEST,R5
                                                    364
                                                    365 * save before a test
000021E8  EB0F 8220 0024                 00002220   366          STMG  R0,R15,SVTST        Save regs before test
                                                    367
                                                    368 * start a test
000021EE  E300 5004 0015                 00000004   369          LGH   R0,TNUM             Save current test number
000021F4  5000 A004                      00004004   370          ST    R0,TESTING
                                                    371
000021F8  45F0 84C8                      000024C8   372          BAL   R15,SETAREYE        set AR regs and copy eye catches
                                                    373
000021FC  E3B0 5000 0014                 00000000   374          LGF   R11,TSUB            get address of test routine
00002202  05BB                                      375          BALR  R11,R11             do test
                                                    376
                                                    377 * restore after a test
00002204  EB0F 8220 0004                 00002220   378          LMG   R0,R15,SVTST        restore regs after test
0000220A  B219 0000                      00000000   379          SAC   PRIMODE             ensure were in Primary Space mode
0000220E  9A0F AA00                      00004A00   380          LAM   AR0,AR15,AREGS      Clear all ARs
                                                    381 *
00002212  41C0 C004                      00000004   382          LA    R12,4(0,R12)        next test address
00002216  47F0 81DE                      000021DE   383          B     NEXTTEST
                                                    384
00002220  00000000 00000000                         385 SVTST    DS    16D                 save area


                                                    387 ********************************************************************************
                                                    388 * end of testing; set ending psw
                                                    389 ********************************************************************************
                               000022A0  00000001   390 ENDTEST  EQU   *
000022A0  5810 A000                      00004000   391          L     R1,FAILED           did a test fail?
000022A4  1211                                      392          LTR   R1,R1
000022A6  4780 86C8                      000026C8   393          BZ    EOJ                 No, exit
000022AA  47F0 86E0                      000026E0   394          B     FAILTEST             Yes, exit with BAD PSW
000022AE  47F0 86C8                      000026C8   395          B     EOJ
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    397 ************************************************************************
                                                    398 * CCMSG:       cc was not as expected
                                                    399 *              R5  = address of test control table
                                                    400 *              R15 = return address
                                                    401 ************************************************************************
                                 000022B2  00000001 402 CCMSG    EQU   *
000022B2  50F0 836C                        0000236C 403          ST    R15,FCCSVR15        Save return address
000022B6  5050 8370                        00002370 404          ST    R5,FCCSVR5          Save R5
000022BA  9002 8378                        00002378 405          STM   R0,R2,FCCSAV        save regs used by MSG
                                                    406
                                                    407 ** check for ignore cc as PLO did not update CC
                                                    408 *
000022BE  9563 500A                        0000000A 409          CLI   CC,99               99 -->  cc not updated
000022C2  478F 0000                        00000000 410          BE    0(R15)
                                                    411
000022C6  5800 8700                        00002700 412          L     R0,=F'1'            set GLOBAL failed test indicator
000022CA  5000 A000                        00004000 413          ST    R0,FAILED
                                                    414 *
                                                    415 * extract CC extracted PSW
                                                    416 *
000022CE  5810 5010                        00000010 417          L     R1,CCPSW
000022D2  8810 000C                        0000000C 418          SRL   R1,12
000022D6  5410 8708                        00002708 419          N     R1,=XL4'3'
000022DA  4210 500C                        0000000C 420          STC   R1,CCFOUND     save cc
                                                    421 *
                                                    422 * FILL IN MESSAGE
                                                    423 *
000022DE  4820 5004                        00000004 424          LH    R2,TNUM             get test number and convert
000022E2  4E20 A161                        00004161 425          CVD   R2,DECNUM
000022E6  D211 A14B A135       0000414B    00004135 426          MVC   PRT3,EDIT
000022EC  DE11 A14B A161       0000414B    00004161 427          ED    PRT3,DECNUM
000022F2  D202 A087 A158       00004087    00004158 428          MVC   CCPRTNUM(3),PRT3+13     fill in message with test #
                                                    429
000022F8  B982 0022                                 430          XGR   R2,R2               get FUN as U8
000022FC  4320 5007                        00000007 431          IC    R2,FUN
00002300  4E20 A161                        00004161 432          CVD   R2,DECNUM           and convert
00002304  D211 A14B A135       0000414B    00004135 433          MVC   PRT3,EDIT
0000230A  DE11 A14B A161       0000414B    00004161 434          ED    PRT3,DECNUM
00002310  D201 A0B3 A159       000040B3    00004159 435          MVC   CCPRTFUN(2),PRT3+14     fill in message with CC field
                                                    436
00002316  B982 0022                                 437          XGR   R2,R2               get CC as U8
0000231A  4320 500A                        0000000A 438          IC    R2,CC
0000231E  4E20 A161                        00004161 439          CVD   R2,DECNUM           and convert
00002322  D211 A14B A135       0000414B    00004135 440          MVC   PRT3,EDIT
00002328  DE11 A14B A161       0000414B    00004161 441          ED    PRT3,DECNUM
0000232E  D200 A0C4 A15A       000040C4    0000415A 442          MVC   CCPRTEXP(1),PRT3+15     fill in message with CC field
                                                    443
00002334  B982 0022                                 444          XGR   R2,R2               get CCFOUND as U8
00002338  4320 500C                        0000000C 445          IC    R2,CCFOUND
0000233C  4E20 A161                        00004161 446          CVD   R2,DECNUM           and convert
00002340  D211 A14B A135       0000414B    00004135 447          MVC   PRT3,EDIT
00002346  DE11 A14B A161       0000414B    00004161 448          ED    PRT3,DECNUM
0000234C  D200 A0D4 A15A       000040D4    0000415A 449          MVC   CCPRTGOT(1),PRT3+15    fill in message with ccfound
                                                    450
00002352  4100 005F                        0000005F 451          LA    R0,CCPRTLNG         message length
00002356  4110 A077                        00004077 452          LA    R1,CCPRTLINE        messagfe address
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000235A  45F0 85C6                     000025C6   453           BAL    R15,RPTERROR
                                                   454
0000235E  9802 8378                     00002378   455           LM     R0,R2,FCCSAV        restore regs
00002362  5850 8370                     00002370   456           L      R5,FCCSVR5          Restore R5
00002366  58F0 836C                     0000236C   457           L      R15,FCCSVR15        Restore return address
0000236A  07FF                                     458           BR     R15                 Return to caller
                                                   459
0000236C  00000000                                 460 FCCSVR15  DC     F'0'                R15 save area
00002370  00000000                                 461 FCCSVR5   DC     F'0'                R5 save area

00002378  00000000 00000000                        463 FCCSAV    DC     2D'0'               R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    465 ************************************************************************
                                                    466 * URMSG:       UR was not as expected
                                                    467 *              R5  = address of test control table
                                                    468 *              R15 = return address
                                                    469 *
                                                    470 *              note: UR 99 - UR was not updated
                                                    471 ************************************************************************
                                 00002388  00000001 472 URMSG    EQU   *
00002388  50F0 8434                        00002434 473          ST    R15,URSVR15         Save return address
0000238C  5050 8438                        00002438 474          ST    R5,URSVR5           Save R5
00002390  9002 8440                        00002440 475          STM   R0,R2,URSAV         save regs used by MSG
                                                    476 *
                                                    477 ** check for ignore UR as PLO did not update UR
                                                    478 *
00002394  9563 5008                        00000008 479          CLI   UR,99               99 -->  UR not updated
00002398  478F 0000                        00000000 480          BE    0(R15)
                                                    481
0000239C  5800 8700                        00002700 482          L     R0,=F'1'            set GLOBAL failed test indicator
000023A0  5000 A000                        00004000 483          ST    R0,FAILED
                                                    484 *
                                                    485 * FILL IN MESSAGE
                                                    486 *
000023A4  4820 5004                        00000004 487          LH    R2,TNUM             get test number and convert
000023A8  4E20 A161                        00004161 488          CVD   R2,DECNUM
000023AC  D211 A14B A135       0000414B    00004135 489          MVC   PRT3,EDIT
000023B2  DE11 A14B A161       0000414B    00004161 490          ED    PRT3,DECNUM
000023B8  D202 A0E6 A158       000040E6    00004158 491          MVC   URPRTNUM(3),PRT3+13 fill in message with test #
                                                    492
000023BE  B982 0022                                 493          XGR   R2,R2               get FUN as U8
000023C2  4320 5007                        00000007 494          IC    R2,FUN
000023C6  4E20 A161                        00004161 495          CVD   R2,DECNUM           and convert
000023CA  D211 A14B A135       0000414B    00004135 496          MVC   PRT3,EDIT
000023D0  DE11 A14B A161       0000414B    00004161 497          ED    PRT3,DECNUM
000023D6  D201 A112 A159       00004112    00004159 498          MVC   URPRTFUN(2),PRT3+14 fill in message with Fun field
                                                    499
000023DC  B982 0022                                 500          XGR   R2,R2               get UR as U8
000023E0  4320 5008                        00000008 501          IC    R2,UR
000023E4  4E20 A161                        00004161 502          CVD   R2,DECNUM           and convert
000023E8  D211 A14B A135       0000414B    00004135 503          MVC   PRT3,EDIT
000023EE  DE11 A14B A161       0000414B    00004161 504          ED    PRT3,DECNUM
000023F4  D200 A123 A15A       00004123    0000415A 505          MVC   URPRTEXP(1),PRT3+15 fill in message with UR field
                                                    506
000023FA  B982 0022                                 507          XGR   R2,R2               get URFOUND as U8
000023FE  4320 5009                        00000009 508          IC    R2,URFOUND
00002402  4E20 A161                        00004161 509          CVD   R2,DECNUM           and convert
00002406  D211 A14B A135       0000414B    00004135 510          MVC   PRT3,EDIT
0000240C  DE11 A14B A161       0000414B    00004161 511          ED    PRT3,DECNUM
00002412  D200 A133 A15A       00004133    0000415A 512          MVC   URPRTGOT(1),PRT3+15 fill in message with urfound
                                                    513
00002418  4100 005F                        0000005F 514          LA    R0,URPRTLNG         message length
0000241C  4110 A0D6                        000040D6 515          LA    R1,URPRTLINE        messagfe address
00002420  45F0 85C6                        000025C6 516          BAL   R15,RPTERROR
                                                    517
00002424  9802 8440                        00002440 518          LM    R0,R2,URSAV         restore regs
00002428  5850 8438                        00002438 519          L     R5,URSVR5           Restore R5
0000242C  58F0 8434                        00002434 520          L     R15,URSVR15         Restore return address
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00002430  07FF                                      521          BR    R15                Return to caller
                                                    522
00002434  00000000                                  523 URSVR15  DC    F'0'               R15 save area
00002438  00000000                                  524 URSVR5   DC    F'0'               R5 save area

00002440  00000000 00000000                         526 URSAV    DC     2D'0'             R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     528 ************************************************************************
                                                     529 * Fail Message to console
                                                     530 *              R5  = address of test control table
                                                     531 *              R15 = return address
                                                     532 ************************************************************************
                                 00002450  00000001  533 FAILMSG  EQU   *
00002450  5800 8700                        00002700  534          L     R0,=F'1'            set GLOBAL failed test indicator
00002454  5000 A000                        00004000  535          ST    R0,FAILED
                                                     536
00002458  4820 5004                        00000004  537          LH    R2,TNUM             get test number and convert
0000245C  4E20 A161                        00004161  538          CVD   R2,DECNUM
00002460  D211 A14B A135       0000414B    00004135  539          MVC   PRT3,EDIT
00002466  DE11 A14B A161       0000414B    00004161  540          ED    PRT3,DECNUM
0000246C  D202 A018 A158       00004018    00004158  541          MVC   FAILTNUM(3),PRT3+13 fill in message with test #
                                                     542
00002472  D259 A01D 8719       0000401D    00002719  543          MVC   FAILTXT(L'FAILTXT),=CL128' '  set to blanks
00002478  B982 0022                                  544          XGR   R2,R2
0000247C  5820 5018                        00000018  545          L     R2,FAILTXTA
00002480  D259 A01D 2000       0000401D    00000000  546          MVC   FAILTXT(L'FAILTXT),0(R2)      load message text
                                                     547
                                                     548 *
                                                     549 *        Use Hercules Diagnose for Message to console
                                                     550 *
                                                     551
00002486  5050 85E8                        000025E8  552          ST     R5,RPTSVR5          Save R5
0000248A  50F0 85E4                        000025E4  553          ST     R15,RPTSAVE         Save return address
0000248E  9002 84B8                        000024B8  554          STM   R0,R2,FLDWSAV       save regs used by MSG
00002492  4100 006F                        0000006F  555          LA    R0,FAILLNG          message length
00002496  4110 A008                        00004008  556          LA    R1,FAILLINE         messagfe address
0000249A  4520 8600                        00002600  557          BAL   R2,MSG              call Hercules console MSG display
                                                     558
0000249E  9802 84B8                        000024B8  559          LM    R0,R2,FLDWSAV       restore regs
000024A2  5850 85E8                        000025E8  560          L     R5,RPTSVR5          Restore R5
000024A6  58F0 85E4                        000025E4  561          L     R15,RPTSAVE         Restore return address
000024AA  07FF                                       562          BR    R15                 Return to caller
                                                     563
000024AC  00000000                                   564 FLSVR15  DC    F'0'                R15 save area
000024B0  00000000                                   565 FLSVR5   DC    F'0'                R5 save area

000024B8  00000000 00000000                          567 FLDWSAV  DC    2D'0'               R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    569 ************************************************************************
                                                    570 *        SETAREYE  - initialize for ARMode
                                                    571 *                                    - Set AR Registers
                                                    572 *                                    - Copy test eye catchers to
                                                    573 *                                      Secondary and Home spaces
                                                    574 ************************************************************************

000024C8                                            576 SETAREYE DS    0F
000024C8                          00000000          577          USING PLOETEST,R5
                                                    578
000024C8  4160 0001                       00000001  579          LA    R6,SECALET          Set Access Regiser for R6
000024CC  B24E 0066                                 580          SAR   AR6,R6
000024D0  4170 0002                       00000002  581          LA    R7,HOMEALET         Set Access Regiser for R7
000024D4  B24E 0077                                 582          SAR   AR7,R7
000024D8  41C0 0001                       00000001  583          LA    R12,SECALET         Set Access Regiser for R12
000024DC  B24E 00CC                                 584          SAR   AR12,R12
000024E0  41D0 0002                       00000002  585          LA    R13,HOMEALET        Set Access Regiser for R13
000024E4  B24E 00DD                                 586          SAR   AR13,R13
000024E8  B219 0200                       00000200  587          SAC   ARMODE
                                                    588 *
                                                    589 * Start test eye catcher
                                                    590 *
000024EC  B904 0015                                 591          LGR   R1,R5               R1: Test - Primary
000024F0  C218 FFFF FFE0                            592          AGFI  R1,-32
000024F6  B904 0061                                 593          LGR   R6,R1               R6: TestC - Secondary
000024FA  B904 0071                                 594          LGR   R7,R1               R7: Test - Home
000024FE  D21F 6000 1000  00000000  00000000  595          MVC   0(32,R6),0(R1)      copy eye catcher
00002504  D203 6000 870C  00000000  0000270C  596          MVC   0(4,R6),=CL4'SEC '    and identify
0000250A  D21F 7000 1000  00000000  00000000  597          MVC   0(32,R7),0(R1)      copy eye catcher
00002510  D203 7000 8710  00000000  00002710  598          MVC   0(4,R7),=CL4'HOME'    and identify
                                                    599 *
                                                    600 * OPSRC eye catcher
                                                    601 *
00002516  E310 501C 0014                  0000001C  602          LGF   R1,OPSRCA           R1: OPSRC - Primary
0000251C  C218 FFFF FFE0                            603          AGFI  R1,-32
00002522  B904 0061                                 604          LGR   R6,R1               R6: OPSRC - Secondary
00002526  B904 0071                                 605          LGR   R7,R1               R7: OPSRC - Home
0000252A  D21F 6000 1000  00000000  00000000  606          MVC   0(32,R6),0(R1)      copy eye catcher
00002530  D203 6000 870C  00000000  0000270C  607          MVC   0(4,R6),=CL4'SEC '    and identify
00002536  D21F 7000 1000  00000000  00000000  608          MVC   0(32,R7),0(R1)      copy eye catcher
0000253C  D203 7000 8710  00000000  00002710  609          MVC   0(4,R7),=CL4'HOME'    and identify
                                                    610 *
                                                    611 * OPDST eye catcher
                                                    612 *
00002542  E310 5020 0014                  00000020  613          LGF   R1,OPDSTA           R1: OPDST - Primary
00002548  C218 FFFF FFE0                            614          AGFI  R1,-32
0000254E  B904 0061                                 615          LGR   R6,R1               R6: OPDST - Secondary
00002552  B904 0071                                 616          LGR   R7,R1               R7: OPDST - Home
00002556  D21F 6000 1000  00000000  00000000  617          MVC   0(32,R6),0(R1)      copy eye catcher
0000255C  D203 6000 870C  00000000  0000270C  618          MVC   0(4,R6),=CL4'SEC '    and identify
00002562  D21F 7000 1000  00000000  00000000  619          MVC   0(32,R7),0(R1)      copy eye catcher
00002568  D203 7000 8710  00000000  00002710  620          MVC   0(4,R7),=CL4'HOME'    and identify
                                                    621 *
                                                    622 * PL eye catcher
                                                    623 *
0000256E  E310 5024 0014                  00000024  624          LGF   R1,PLA              R1: PL - Primary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00002574  C218 FFFF FFE0                           625          AGFI  R1,-32
0000257A  B904 0061                               626          LGR   R6,R1               R6: PL - Secondary
0000257E  B904 0071                               627          LGR   R7,R1               R7: PL - Home
00002582  D21F 6000 1000      00000000  00000000  628          MVC   0(32,R6),0(R1)      copy eye catcher
00002588  D203 6000 870C      00000000  0000270C  629          MVC   0(4,R6),=CL4'SEC '    and identify
0000258E  D21F 7000 1000      00000000  00000000  630          MVC   0(32,R7),0(R1)      copy eye catcher
00002594  D203 7000 8710      00000000  00002710  631          MVC   0(4,R7),=CL4'HOME'    and identify
                                                  632 *
                                                  633 * Test end eye catcher
                                                  634 *
0000259A  E310 502C 0014                0000002C  635          LGF   R1,TENDA            R1: Test end - Primary
000025A0  B904 0061                               636          LGR   R6,R1               R6: Test end - Secondary
000025A4  B904 0071                               637          LGR   R7,R1               R7: Test end - Home
000025A8  D21F 6000 1000      00000000  00000000  638          MVC   0(32,R6),0(R1)      copy eye catcher
000025AE  D203 6000 870C      00000000  0000270C  639          MVC   0(4,R6),=CL4'SEC '    and identify
000025B4  D21F 7000 1000      00000000  00000000  640          MVC   0(32,R7),0(R1)      copy eye catcher
000025BA  D203 7000 8710      00000000  00002710  641          MVC   0(4,R7),=CL4'HOME'    and identify
                                                  642
000025C0  B219 0000                     00000000  643          SAC   PRIMODE
000025C4  07FF                                    644          BR    R15
                                                  645
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    647 ********************************************************************************
                                                    648 *        RPTERROR                  Report instruction test in error
                                                    649 ********************************************************************************

000025C6  50F0 85E4                       000025E4  651 RPTERROR ST    R15,RPTSAVE         Save return address
000025CA  5050 85E8                       000025E8  652          ST    R5,RPTSVR5          Save R5
000025CE  9002 85F0                       000025F0  653          STM   R0,R2,RPTDWSAV      save regs used by MSG
                                                    654 *
                                                    655 *        Use Hercules Diagnose for Message to console
                                                    656 *
000025D2  4520 8600                       00002600  657          BAL   R2,MSG              call Hercules console MSG display
                                                    658
000025D6  9802 85F0                       000025F0  659          LM    R0,R2,RPTDWSAV      restore regs
000025DA  5850 85E8                       000025E8  660          L     R5,RPTSVR5          Restore R5
000025DE  58F0 85E4                       000025E4  661          L     R15,RPTSAVE         Restore return address
000025E2  07FF                                      662          BR    R15                 Return to caller

000025E4  00000000                                  664 RPTSAVE  DC    F'0'                R15 save area
000025E8  00000000                                  665 RPTSVR5  DC    F'0'                R5 save area

000025F0  00000000 00000000                         667 RPTDWSAV DC    2D'0'               R0-R2 save area for MSG call
                                                    668 ********************************************************************************
                                                    669 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                    670 *             R2 = return address
                                                    671 ********************************************************************************

00002600  4900 8714                       00002714  673 MSG      CH    R0,=H'0'            Do we even HAVE a message?
00002604  07D2                                      674          BNHR  R2                  No, ignore

00002606  9002 863C                       0000263C  676          STM   R0,R2,MSGSAVE       Save registers

0000260A  4900 8716                       00002716  678          CH    R0,=AL2(L'MSGMSG)   Message length within limits?
0000260E  47D0 8616                       00002616  679          BNH   MSGOK               Yes, continue
00002612  4100 005F                       0000005F  680          LA    R0,L'MSGMSG         No, set to maximum

00002616  1820                                      682 MSGOK    LR    R2,R0               Copy length to work register
00002618  0620                                      683          BCTR  R2,0                Minus-1 for execute
0000261A  4420 8648                       00002648  684          EX    R2,MSGMVC           Copy message to O/P buffer

0000261E  4120 200A                       0000000A  686          LA    R2,1+L'MSGCMD(,R2)  Calculate true command length
00002622  4110 864E                       0000264E  687          LA    R1,MSGCMD           Point to true command

00002626  83120008                                  689          DC    X'83',X'12',X'0008' Issue Hercules Diagnose X'008'
0000262A  4780 8636                       00002636  690          BZ    MSGRET              Return if successful
                                                    691
0000262E  1222                                      692          LTR   R2,R2               Is Diag8 Ry (R2) 0?
00002630  4780 8636                       00002636  693          BZ    MSGRET                an error occurred but coninue
                                                    694
00002634  0000                                      695          DC    H'0'                CRASH for debugging purposes

00002636  9802 863C                       0000263C  697 MSGRET  LM    R0,R2,MSGSAVE       Restore registers
0000263A  07F2                                      698          BR    R2                  Return to caller
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000263C  00000000 00000000                          700 MSGSAVE  DC    3F'0'               Registers save area
00002648  D200 8657 1000      00002657  00000000    701 MSGMVC   MVC   MSGMSG(0),0(R1)     Executed instruction


0000264E  D4E2C7D5 D6C8405C                          703 MSGCMD   DC    C'MSGNOH * '        *** HERCULES MESSAGE COMMAND ***
00002657  40404040 40404040                          704 MSGMSG   DC    CL95' '             The message text to be displayed
                                                     705
000026B8                                             706          DS    0D
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     708 ************************************************************************
                                                     709 *        Normal completion or Abnormal termination PSWs
                                                     710 ************************************************************************


000026B8  00020001 80000000                          712 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

000026C8  B2B2 86B8                  000026B8        714 EOJ      LPSWE EOJPSW               Normal completion


000026D0  00020001 80000000                          716 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

000026E0  B2B2 86D0                  000026D0        718 FAILTEST LPSWE FAILPSW              Abnormal termination
                                                     719
                                                     720 ************************************************************************
                                                     721 *        Working Storage
                                                     722 ************************************************************************

000026E8                                             724          LTORG ,                    Literals pool
000026E8  00000000 00010000                         725                =AD(PAGPRI)
000026F0  00000000 00014000                         726                =AD(PAGSEC)
000026F8  00000000 00018000                         727                =AD(PAGHOM)
00002700  00000001                                  728                =F'1'
00002704  00004578                                  729                =A(PLOTESTS)
00002708  00000003                                  730                =XL4'3'
0000270C  E2C5C340                                  731                =CL4'SEC '
00002710  C8D6D4C5                                  732                =CL4'HOME'
00002714  0000                                      733                =H'0'
00002716  005F                                      734                =AL2(L'MSGMSG)
00002718  04                                        735                =X'04'
00002719  40404040 40404040                         736                =CL128' '
                                                    737
0000279C  00000000                                  738 CTLR0    DS    F
000027A0  00000000                                  739          DS    F
                                                    740
                                                    741 * Constants
                                                    742
                           00000400  00000001       743 K        EQU   1024                 One KB
                           00001000  00000001       744 PAGE     EQU   (4*K)                Size of one page
                           00010000  00000001       745 K64      EQU   (64*K)               64 KB
                           00100000  00000001       746 MB       EQU   (K*K)                1 MB
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     748 *=========================================================================
                                                     749 *
                                                     750 *  NOTE: start data on an address that is easy to display
                                                     751 *        within Hercules
                                                     752 *
                                                     753 *=========================================================================
                                                     754
000027A4                      000027A4  00004000    755          ORG   PLOETST+X'4000'
00004000  00000000                                   756 FAILED   DC    F'0'                 some test failed?
00004004  00000000                                   757 TESTING  DC    F'0'                 current test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     759 ********************************************************************************
                                                     760 *        TEST failed : instruction data
                                                     761 ********************************************************************************
                                                     762 *
                                                     763 *        failed message and associated editting
                                                     764 *
00004008  40404040 40404040                          765 FAILLINE DC    C'         Test # '
00004018  A7A7A7                                     766 FAILTNUM DC    C'xxx'
0000401B  7A40                                       767          DC    C': '
                            00000015  00000001       768 FAILHLNG EQU   *-FAILLINE          fail line header length
                                                     769
0000401D  40404040 40404040                          770 FAILTXT  DC    CL90' '            Test fail message
                            0000006F  00000001       771 FAILLNG  EQU   *-FAILLINE
                                                     772
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    774 ************************************************************************
                                                    775 *        TEST failed : CC message
                                                    776 ************************************************************************
                                                    777 *
                                                    778 *        failed message and associated editting
                                                    779 *
00004077  40404040 40404040                         780 CCPRTLINE DC   C'         Test # '
00004087  A7A7A7                                    781 CCPRTNUM DC    C'xxx'
0000408A  7A40A699 96958740                         782           DC    c': wrong cc for PLO instruction'
000040A8  404D86A4 9583A389                         783           DC    c' (function '
000040B3  A7A7                                      784 CCPRTFUN DC    CL2'xx'
000040B5  5D                                        785           DC    c')'
000040B6  4085A797 8583A385                         786           DC    C' expected: cc='
000040C4  A7                                        787 CCPRTEXP DC    C'x'
000040C5  6B                                        788           DC    C','
000040C6  40998583 8589A585                         789           DC    C' received: cc='
000040D4  A7                                        790 CCPRTGOT DC    C'x'
000040D5  4B                                        791           DC    C'.'
                            0000005F  00000001      792 CCPRTLNG   EQU   *-CCPRTLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     794 ************************************************************************
                                                     795 *TEST failed : unequal results (UR) field message
                                                     796 ************************************************************************
                                                     797 *
                                                     798 *        failed message and associated editting
                                                     799 *
000040D6  40404040 40404040                          800 URPRTLINE DC   C'         Test # '
000040E6  A7A7A7                                     801 URPRTNUM DC    C'xxx'
000040E9  7A40A699 96958740                          802           DC    c': wrong UR for PLO instruction'
00004107  404D86A4 9583A389                          803           DC    c' (function '
00004112  A7A7                                       804 URPRTFUN DC    CL2'xx'
00004114  5D                                         805           DC    c')'
00004115  4085A797 8583A385                          806           DC    C' expected: UR='
00004123  A7                                         807 URPRTEXP DC    C'x'
00004124  6B                                         808           DC    C','
00004125  40998583 8589A585                          809           DC    C' received: UR='
00004133  A7                                         810 URPRTGOT DC    C'x'
00004134  4B                                         811           DC    C'.'
                            0000005F  00000001       812 URPRTLNG   EQU   *-URPRTLINE
                                                     813
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     815 ************************************************************************
                                                     816 *        TEST failed : message working storge
                                                     817 ************************************************************************
00004135  40212020 20202020                          818 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                     819
00004147  7E7E7E6E                                   820          DC    C'===>'
0000414B  40404040 40404040                          821 PRT3     DC    CL18' '
0000415D  4C7E7E7E                                   822          DC    C'<==='
00004161  00000000 00000000                          823 DECNUM   DS    CL16
                                                     824
                                                     825 * ----------------------------------------------------------------------
                                                     826 * Random Bytes for Test Data (1 KB)
                                                     827 * ----------------------------------------------------------------------
00004178                                             828          DS    0D
00004178  7A1FC308 9DE246B1                          829 RANDX    DC    XL16'7A 1F C3 08 9D E2 46 B1 5C F7 20 8B 64 DA 03 91'
00004188  BE550CA7 3ED87219                          830          DC    XL16'BE 55 0C A7 3E D8 72 19 F4 6B 80 2D C9 14 EF 5A'
00004198  31AC77E0 4B960FD3                          831          DC    XL16'31 AC 77 E0 4B 96 0F D3 68 B9 25 FE 52 8D 17 C6'
000041A8  EB40A16E 09D5832A                          832          DC    XL16'EB 40 A1 6E 09 D5 83 2A F8 5F BC 12 74 C1 9A 36'
000041B8  0DE759B2 43FC861B                          833          DC    XL16'0D E7 59 B2 43 FC 86 1B D0 6C 28 A5 93 4E F1 7D'
000041C8  C81165DE 3B90FA27                          834          DC    XL16'C8 11 65 DE 3B 90 FA 27 54 AD 02 79 E4 B7 4C 88'
000041D8  16D970A3 5ECB34F5                          835          DC    XL16'16 D9 70 A3 5E CB 34 F5 81 0A BD 62 E9 47 9F 23'
000041E8  D65801AF 75CA3DE1                          836          DC    XL16'D6 58 01 AF 75 CA 3D E1 92 6F 18 B4 50 FD 87 2C'
000041F8  A844DB0E 69C235F0                          837          DC    XL16'A8 44 DB 0E 69 C2 35 F0 9B 56 ED 21 7F B8 04 D4'
00004208  63AA1DE6 4895CF30                          838          DC    XL16'63 AA 1D E6 48 95 CF 30 F9 82 5B 13 DC 76 A0 2F'
00004218  8E41F3B5 0B67CE24                          839          DC    XL16'8E 41 F3 B5 0B 67 CE 24 99 DA 53 E8 1A 7C C7 32'
00004228  F68549B0 1ED26A94                          840          DC    XL16'F6 85 49 B0 1E D2 6A 94 3F EB 07 5D A9 70 C4 26'
00004238  DD1084F2 38AB61C5                          841          DC    XL16'DD 10 84 F2 38 AB 61 C5 0C 97 4A E3 7B B6 29 D1'
00004248  5F8A15CC 73F800A6                          842          DC    XL16'5F 8A 15 CC 73 F8 00 A6 42 9C E5 37 BD 68 02 D7'
00004258  904DEA22 7EC059F1                          843          DC    XL16'90 4D EA 22 7E C0 59 F1 0F A4 6D DB 35 88 B3 1C'
00004268  E0712BC9 469EF460                          844          DC    XL16'E0 71 2B C9 46 9E F4 60 AD 18 D5 83 3A BF 54 07'
00004278  A73CF108 6DB249DE                          845          DC    XL16'A7 3C F1 08 6D B2 49 DE 15 83 5A C6 2F 91 E4 70'
00004288  D30B58AC E9347F12                          846          DC    XL16'D3 0B 58 AC E9 34 7F 12 C1 66 9D 42 F8 25 B7 5E'
00004298  81DA3704 9AE26BC5                          847          DC    XL16'81 DA 37 04 9A E2 6B C5 18 F3 4D 76 A0 2C BF 53'
000042A8  E7901D68 C83FA572                          848          DC    XL16'E7 90 1D 68 C8 3F A5 72 0E DB 46 B1 59 F6 23 8C'
000042B8  35CE7A11 E0945FB8                          849          DC    XL16'35 CE 7A 11 E0 94 5F B8 27 D4 63 0A F9 41 AD 86'
000042C8  C31A75EC 509F28D7                          850          DC    XL16'C3 1A 75 EC 50 9F 28 D7 62 B4 0D E8 39 87 FA 14'
000042D8  6FC03295 4ABE07D1                          851          DC    XL16'6F C0 32 95 4A BE 07 D1 7C E5 21 89 56 F2 A9 3D'
000042E8  B648DF10 649C2AF5                          852          DC    XL16'B6 48 DF 10 64 9C 2A F5 73 C7 0F 82 ED 31 A4 5B'
000042F8  D926904F B36AE10C                          853          DC    XL16'D9 26 90 4F B3 6A E1 0C 78 C2 55 AF 13 F7 84 3E'
00004308  69D5209B 47EA7106                          854          DC    XL16'69 D5 20 9B 47 EA 71 06 BD 52 F0 38 8E C9 16 A2'
00004318  5DE62B97 40BC037E                          855          DC    XL16'5D E6 2B 97 40 BC 03 7E D0 19 A8 65 F4 33 8A C4'
00004328  1F74EB57 920AD843                          856          DC    XL16'1F 74 EB 57 92 0A D8 43 B5 6C F1 2E 80 CD 39 A6'
00004338  E3519E24 7BC80DB0                          857          DC    XL16'E3 51 9E 24 7B C8 0D B0 67 F5 32 8F DA 14 46 AC'
00004348  79E90593 2DC658B1                          858          DC    XL16'79 E9 05 93 2D C6 58 B1 7F 30 ED 84 12 A5 6E D2'
00004358  4CF8219A 63BE07D6                          859          DC    XL16'4C F8 21 9A 63 BE 07 D6 50 C3 88 1B F0 35 A9 72'
00004368  DB448D17 E269B72C                          860          DC    XL16'DB 44 8D 17 E2 69 B7 2C 95 0F C1 5A F6 38 AE 73'
00004378  4FA8D21C 9B673EF0                          861          DC    XL16'4F A8 D2 1C 9B 67 3E F0 85 14 C9 72 AD 5F E1 38'
00004388  B72D94FE 610AC357                          862          DC    XL16'B7 2D 94 FE 61 0A C3 57 88 D1 4B AE 23 F9 76 10'
00004398  D8459A3C E78116BF                          863          DC    XL16'D8 45 9A 3C E7 81 16 BF 52 CC 09 74 A1 ED 68 35'
000043A8  FA20B35D 8EC7419C                          864          DC    XL16'FA 20 B3 5D 8E C7 41 9C 6B F4 17 DA 80 2E 53 BD'
000043B8  31CE7A05 E8924FB1                          865          DC    XL16'31 CE 7A 05 E8 92 4F B1 6D 23 FC 58 A7 10 D9 84'
000043C8  65B8E31F 904AC67D                          866          DC    XL16'65 B8 E3 1F 90 4A C6 7D 2B F5 18 A2 DE 73 0C 49'
000043D8  87D43FB9 21EC5608                          867          DC    XL16'87 D4 3F B9 21 EC 56 08 C1 7A AD 34 F2 6E 95 10'
000043E8  CB42F81D 73A60ED5                          868          DC    XL16'CB 42 F8 1D 73 A6 0E D5 89 2C B7 50 EF 14 68 93'
000043F8  3AD186FC 275BE09D                          869          DC    XL16'3A D1 86 FC 27 5B E0 9D 41 B8 12 C7 6F A4 35 EE'
00004408  904CDB18 75A2F63E                          870          DC    XL16'90 4C DB 18 75 A2 F6 3E C9 07 B1 5D E8 24 7A CF'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00004418  16E3588B D032FA6D                          871          DC    XL16'16 E3 58 8B D0 32 FA 6D 97 41 BC 0F 74 A5 E1 2C'
00004428  F78913CE 5AD420B6                          872          DC    XL16'F7 89 13 CE 5A D4 20 B6 7F E2 49 9C 35 A8 01 DD'
00004438  68F12B95 C07E14BA                          873          DC    XL16'68 F1 2B 95 C0 7E 14 BA 53 EF 86 39 D7 42 AC 10'
00004448  B46DE821 F9570CC3                          874          DC    XL16'B4 6D E8 21 F9 57 0C C3 7A DE 45 98 12 FB 64 3E'
00004458  8D30C75B E1A419F2                          875          DC    XL16'8D 30 C7 5B E1 A4 19 F2 76 0F D8 43 BE 95 28 6A'
00004468  F35E80CA 176DB924                          876          DC    XL16'F3 5E 80 CA 17 6D B9 24 E7 41 0A DC 58 92 AF 35'
00004478  34C17DE9 52AB06F8                          877          DC    XL16'34 C1 7D E9 52 AB 06 F8 1E 93 6C D4 27 B0 5F EA'
00004488  7914CE63 A82DF145                          878          DC    XL16'79 14 CE 63 A8 2D F1 45 9B 30 E7 5A 8C D6 12 BF'
00004498  E2580FB4 71DA2983                          879          DC    XL16'E2 58 0F B4 71 DA 29 83 C5 1C F7 6E 90 3D A1 4B'
000044A8  0DFE6738 C2951AB8                          880          DC    XL16'0D FE 67 38 C2 95 1A B8 4F E3 72 DC 06 A9 54 21'
000044B8  BC47D18E 23F5609A                          881          DC    XL16'BC 47 D1 8E 23 F5 60 9A 14 C8 7D E0 35 B2 6F 91'
000044C8  A619FB42 D70C853E                          882          DC    XL16'A6 19 FB 42 D7 0C 85 3E E1 74 28 CA 5D 90 B3 67'
000044D8  12E853AC 39F60BD1                          883          DC    XL16'12 E8 53 AC 39 F6 0B D1 7C A5 24 EF 68 93 4D B7'
000044E8  F02A9E65 C318DB47                          884          DC    XL16'F0 2A 9E 65 C3 18 DB 47 81 5C EA 36 72 0F B9 D4'
000044F8  58A13CE7 942FD06B                          885          DC    XL16'58 A1 3C E7 94 2F D0 6B C5 19 FE 43 87 B2 0D 7A'
00004508  CD7016F9 4AB528E3                          886          DC    XL16'CD 70 16 F9 4A B5 28 E3 8D 51 AC 37 F0 64 9B 12'
00004518  26DE815B C4F7309D                          887          DC    XL16'26 DE 81 5B C4 F7 30 9D EA 14 68 A2 3F D1 75 BC'
00004528  910EF467 2BC853AD                          888          DC    XL16'91 0E F4 67 2B C8 53 AD 18 E7 45 9A D0 36 FB 72'
00004538  AF3DC184 59E217B6                          889          DC    XL16'AF 3D C1 84 59 E2 17 B6 F8 20 D7 6A 93 4C 0F E1'
00004548  73B924DA 058EC741                          890          DC    XL16'73 B9 24 DA 05 8E C7 41 F2 6D 10 AC 57 E8 3B 96'
00004558  1F62D5A8 34EB790C                          891          DC    XL16'1F 62 D5 A8 34 EB 79 0C C0 4A 97 F3 2E BD 61 85'
00004568  E4378AD2 5CF019B7                          892          DC    XL16'E4 37 8A D2 5C F0 19 B7 46 CE 23 9D 70 1B A5 F8'
                                                     893
                                                     894 * ------------------------------------------------------------------
                                                     895 * Test Table is Located here .....
                                                     896 * ------------------------------------------------------------------
00004578                                             897 PLOTESTS DS    0D
                                                     898 *
                                                     899 *        PTTABLE will be located here
                                                     900 *
00004578                  00004578  00004980         901          ORG   *+(256*4)+8         1K for pointers for up to 256 tests
                                                     902 *                                        plus 8 bytes for table end
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     904 ********************************************************************************
                                                     905 *        Adress Space WORKING STORAGE
                                                     906 ********************************************************************************
                                                     907 *
                                                     908 *
                                                     909 *        Control registers
                                                     910 *
00004980                                             911 CREGS    DC    0D'0'
00004980  00000000 04000000                          912 CREG0    DC    X'00000000',X'04000000' Secondary space control bit 37
00004988  00000000 0000C000                          913 CREG1    DC    X'00000000',A(SEGPRI)   Primary ASCE
00004990  00000000 00019100                          914 CREG2    DC    X'00000000',A(DUCT)     Dispatchable Unit Ctl Table
00004998  00000000 C0000001                          915 CREG3    DC    X'00000000',X'C0000001' PKM=C000, Secondary ASN=1
000049A0  00000000 00000000                          916 CREG4    DC    X'00000000',X'00000000' Primary ASN=0
000049A8  00000000 00019000                          917 CREG5    DC    X'00000000',A(PASTEO)   Primary ASTE Origin
000049B0  00000000 00000000                          918 CREG6    DC    X'00000000',X'00000000'
000049B8  00000000 0000D000                          919 CREG7    DC    X'00000000',A(SEGSEC)   Secondary ASCE
000049C0  00000000 00000000                          920 CREG8    DC    X'00000000',X'00000000'
000049C8  00000000 00000000                          921 CREG9    DC    X'00000000',X'00000000'
000049D0  00000000 00000000                          922 CREG10   DC    X'00000000',X'00000000'
000049D8  00000000 00000000                          923 CREG11   DC    X'00000000',X'00000000'
000049E0  00000000 00000000                          924 CREG12   DC    X'00000000',X'00000000'
000049E8  00000000 0000E000                          925 CREG13   DC    X'00000000',A(SEGHOM)   Home ASCE
000049F0  00000000 00000000                          926 CREG14   DC    X'00000000',X'00000000'
000049F8  00000000 00000000                          927 CREG15   DC    X'00000000',X'00000000'

00004A00  00000000 00000000                          929 AREGS    DC    16F'0'                  Init for Access Registers
                                                     930

                                                     932 ********************************************************************************
                                                     933 *        Segment Tables
                                                     934 ********************************************************************************
00004A40                            00004A40  0000C000 935          ORG   A000LABL+X'C000'
0000C000  00000000 00010000                          936 SEGPRI   DC    X'00000000',A(PAGPRI)
                                                     937
0000C008                            0000C008  0000D000 938          ORG   A000LABL+X'D000'
0000D000  00000000 00014000                          939 SEGSEC   DC    X'00000000',A(PAGSEC)
                                                     940
0000D008                            0000D008  0000E000 941          ORG   A000LABL+X'E000'
0000E000  00000000 00018000                          942 SEGHOM   DC    X'00000000',A(PAGHOM)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     944 ********************************************************************************
                                                     945 *        Page Tables
                                                     946 ********************************************************************************
0000E008                      0000E008  00010000   947          ORG   A000LABL+X'10000'
                              00010000  00000001   948 PAGPRI   EQU   *                          Primary Space Page Tables
                                                     949 *
                                                     950 * Addresses 0-FFFF common to all addresses spaces, VIRT=REAL
                                                     951 * Begin primary space only storage V-addrs 10000-1FFFF
                                                     952

00010000                      00010000  00014000   954          ORG   A000LABL+X'14000'
                              00014000  00000001   955 PAGSEC   EQU   *                          Secondary Space Page Tables
                                                     956 *
                                                     957 * Addresses 0-FFFF common to all addresses spaces, VIRT=REAL
                                                     958 * Begin secondary space only storage V-addrs 10000-1FFFF
                                                     959

00014000                      00014000  00018000   961          ORG   A000LABL+X'18000'
                              00018000  00000001   962 PAGHOM   EQU   *                          Home Space Page Tables
                                                     963 *
                                                     964 * Addresses 0-FFFF common to all addresses spaces, VIRT=REAL
                                                     965 * Begin home space only storage V-addrs 10000-1FFFF (V=R)
                                                     966
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                     968 ********************************************************************************
                                                     969 *        Primary ASN Second Table Entry
                                                     970 ********************************************************************************
00018000                       00018000  00019000    971           ORG   A000LABL+X'19000'
00019000                                             972 PASTEO    DS    0XL64                    Primary ASN Second Table Entry
00019000  00000000                                   973           DC    A(0)           +0        ATO
00019004  00000000                                   974           DC    A(0)            4        AX,ATL
00019008  00000000 0000C000                          975           DC    A(0),A(SEGPRI)  8        Primary ASCE (same as CREG1)
00019010  00000000                                   976           DC    A(0)           16        ALD
00019014  00000000                                   977           DC    A(0)           20        ASTESN
00019018  00000000                                   978           DC    A(0)           24        LTD
0001901C  00000000                                   979           DC    A(0)           28        Ctl prog use
00019020  00000000                                   980           DC    A(0)           32        Ctl prog use
00019024  00000000                                   981           DC    A(0)           36        Ctl prog use
00019028  00000000                                   982           DC    A(0)           40        unassigned
0001902C  00000000                                   983           DC    A(0)           44        ASTEIN
00019030  00000000                                   984           DC    A(0)           48        unassigned
00019034  00000000                                   985           DC    A(0)           58        unassigned
00019038  00000000                                   986           DC    A(0)           56        unassigned
0001903C  00000000                                   987           DC    A(0)           60        unassigned


                                                     989 ********************************************************************************
                                                     990 *        Dispatchable Unit Control Table (DUCT)
                                                     991 ********************************************************************************
                                                     992 *
                                                     993 *        This DUCT is used by the primary space programming
                                                     994 *        when in Access Register mode in order to use the DU-AL.
                                                     995 *
00019040                       00019040  00019100    996           ORG   A000LABL+X'19100'
00019100                                             997 DUCT      DS    0XL64                    Dispatchable Unit Control Tbl
00019100  00000000                                   998           DC    A(0)      +0             BASTEO
00019104  00000000                                   999           DC    A(0)       4             SSASTEO
00019108  00000000                                  1000           DC    A(0)       8             unassigned
0001910C  00000000                                  1001           DC    A(0)      12             SSASTESN
00019110  00019200                                  1002 DUALD    DC    A(DUAL)   16             DU-AL origin
00019114  00000000                                  1003           DC    A(0)      20             PSW key masks
00019118  00000000                                  1004           DC    A(0)      24             unassigned
0001911C  00000000                                  1005           DC    A(0)      28             unassigned
00019120  00000000                                  1006           DC    A(0)      32             Return addr high
00019124  00000000                                  1007           DC    A(0)      36             Return addr low
00019128  00000000                                  1008           DC    A(0)      40             unassigned
0001912C  00000000                                  1009           DC    A(0)      44             TRCB
00019130  00000000                                  1010           DC    A(0)      48             unassigned
00019134  00000000                                  1011           DC    A(0)      52             unassigned
00019138  00000000                                  1012           DC    A(0)      56             unassigned
0001913C  00000000                                  1013           DC    A(0)      60             unassigned
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    1015 ************************************************************************
                                                    1016 *        Dispatchable Unit - Access List (DU-AL)
                                                    1017 ************************************************************************
                                                    1018 *
                                                    1019 *         8 access list entries, only entry 2 is valid (AR ALET = 2)
                                                    1020 *
00019140                        00019140  00019200  1021          ORG   A000LABL+X'19200'
                                00019200  00000001  1022 DUAL     EQU   *                        DU Access List
00019200  80000000 00000000                         1023 ALE0     DC    X'80',15X'00'            ALE 0 invalid
00019210  80000000 00000000                         1024 ALE1     DC    X'80',15X'00'            ALE 1 invalid
                                                    1025 *
00019220                                            1026 ALE2     DS    0XL16                    ALE 2 -> HOME space
00019220  00000000                                  1027          DC    A(0)                     I,FO,P,ALESN,ALEAX all 0
00019224  00000000                                  1028          DC    A(0)                     unassigned
00019228  0001A000                                  1029          DC    A(HASTEO)                Home space ASTE Origin
0001922C  00000000                                  1030          DC    A(0)                     ASTESN seq # set to 0
                                                    1031 *
00019230  80000000 00000000                         1032 ALE3     DC    X'80',15X'00'            ALE 3 invalid
00019240  80000000 00000000                         1033 ALE4     DC    X'80',15X'00'            ALE 4 invalid
00019250  80000000 00000000                         1034 ALE5     DC    X'80',15X'00'            ALE 5 invalid
00019260  80000000 00000000                         1035 ALE6     DC    X'80',15X'00'            ALE 6 invalid
00019270  80000000 00000000                         1036 ALE7     DC    X'80',15X'00'            ALE 7 invalid


                                                    1038 ************************************************************************
                                                    1039 *        Home ASN Second Table Entry (HASTE)
                                                    1040 ************************************************************************
                                                    1041 *        The HASTE is needed for ALET 2's ALE entry above
                                                    1042 *
00019280                        00019280  0001A000  1043          ORG   A000LABL+X'1A000'
0001A000                                            1044 HASTEO   DS    0XL64                    Home ASN Second Table Entry
0001A000  00000000                                  1045          DC    A(0)           +0        ATO
0001A004  00000000                                  1046          DC    A(0)            4        AX,ATL
0001A008  00000000 0000E000                         1047          DC    A(0),A(SEGHOM)  8        Home ASCE (same as CREG13)
0001A010  00000000                                  1048          DC    A(0)           16        ALD
0001A014  00000000                                  1049          DC    A(0)           20        ASTESN
0001A018  00000000                                  1050          DC    A(0)           24        LTD
0001A01C  00000000                                  1051          DC    A(0)           28        Ctl prog use
0001A020  00000000                                  1052          DC    A(0)           32        Ctl prog use
0001A024  00000000                                  1053          DC    A(0)           36        Ctl prog use
0001A028  00000000                                  1054          DC    A(0)           40        unassigned
0001A02C  00000000                                  1055          DC    A(0)           44        ASTEIN
0001A030  00000000                                  1056          DC    A(0)           48        unassigned
0001A034  00000000                                  1057          DC    A(0)           58        unassigned
0001A038  00000000                                  1058          DC    A(0)           56        unassigned
0001A03C  00000000                                  1059          DC    A(0)           60        unassigned
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          1061 ********************************************************************************
                                          1062 *        SET ADDRESS SPACE CONTROL (SAC) constants
                                          1063 ********************************************************************************
                                          1064
                             00000000  00000001  1065 PRIMODE  EQU   0                   Primary mode
                             00000100  00000001  1066 SECMODE  EQU   256                 Secondary mode
                             00000300  00000001  1067 HOMEMODE EQU   768                 Home space mode
                             00000200  00000001  1068 ARMODE   EQU   512                 Enter AR mode


                                          1070 ********************************************************************************
                                          1071 *        alet token constants
                                          1072 ********************************************************************************
                                          1073
                             00000000  00000001  1074 PRIALET  EQU   0                   Special alet token - primary space
                             00000001  00000001  1075 SECALET  EQU   1                   Special alet token - secondary space
                             00000002  00000001  1076 HOMEALET EQU   2                   Special alet token - home space
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1078 ********************************************************************************
                                                   1079 *        PLOETEST DSECT
                                                   1080 ********************************************************************************


                                                   1082 PLOETEST DSECT ,
00000000  00000000                                 1083 TSUB     DC    A(0)               pointer  to test
00000004  0000                                     1084 TNUM     DC    H'00'              Test Number
00000006  00                                       1085          DC    XL1'00'
00000007  00                                       1086 FUN      DC    HL1'00'            Function
00000008  00                                       1087 UR       DC    HL1'00'            expected unequal results (UR)
00000009  00                                       1088 URFOUND  DS    HL1'00'            extracted UR from PL
                                                   1089
0000000A  00                                       1090 CC       DC    HL1'00'            cc
0000000B  00                                       1091 CCMASK   DC    HL1'00'            not expected CC mask
                                                   1092
0000000C  00                                       1093 CCFOUND  DS    X                  extracted cc
00000010  00000000 00000000                        1094 CCPSW    DS    2F                 extract PSW after test (has CC)
                                                   1095
00000018  00000000                                 1096 FAILTXTA DC    A(0)               address of test fail message
0000001C  00000000                                 1097 OPSRCA   DC    A(0)               address of operand source
                           00000020  00000001      1098 OPREPA   EQU   *                  address of operand replacement
00000020  00000000                                 1099 OPDSTA   DC    A(0)               address of operand destination
00000024  00000000                                 1100 PLA      DC    A(0)               address of parameter list
00000028  00000000                                 1101 RANDXA   DC    A(0)               address of random source data
0000002C  00000000                                 1102 TENDA    DC    A(0)               address of test end eye catcher
                                                   1103
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                    1105 ********************************************************************************
                                                    1106 *     Macros to help build test tables
                                                    1107 ********************************************************************************


                                                    1109 *-------------------------------------------------------------------------
                                                    1110 * TSTLBL : insert a label with a test number suffix
                                                    1111 *          - &name: the label name
                                                    1112 *-------------------------------------------------------------------------
                                                    1113 *                                      eye catch (32 bytes) for octo check
                                                    1114          MACRO
                                                    1115          TSTLBL  &NAME
                                                    1116          GBLA  &TNUM
                                                    1117          LCLC  &X
                                                    1118 &X       SETC  '&NAME&TNUM'
                                                    1119          DC    CL32'PRI   >>>&X<<<'      eye catcher
                                                    1120 &X       DS    0C
                                                    1121          MEND
                                                    1122
                                                    1123 *-------------------------------------------------------------------------
                                                    1124 * TSTEND : end a test
                                                    1125 *-------------------------------------------------------------------------
                                                    1126          MACRO
                                                    1127          TSTEND  &NAME
                                                    1128          GBLA  &TNUM
                                                    1129 TEND&TNUM  DC  CL32'PRI   END of test: &TNUM <<<'
                                                    1130          MEND
                                                    1131
                                                    1132 *-------------------------------------------------------------------------
                                                    1133 * TSTCODE : The code for the test follows the label
                                                    1134 *-------------------------------------------------------------------------
                                                    1135          MACRO
                                                    1136          TSTCODE  &NAME
                                                    1137          GBLA  &TNUM
                                                    1138          LCLC  &X
                                                    1139 &X       SETC  'CODE&TNUM'
                                                    1140 &X       DS    0F
                                                    1141          MEND
                                                    1142
                                                    1143 *-------------------------------------------------------------------------
                                                    1144 * TSTDATA: The data for the test follows.
                                                    1145 *          Data is aligned to the 1.5K boundary after the TSTHDR
                                                    1146 *-------------------------------------------------------------------------
                                                    1147          MACRO
                                                    1148          TSTDATA  &NAME
                                                    1149          GBLA  &THDRNUM
                                                    1150          ORG   PLOETST+(128*K)+&THDRNUM*(3*K)+(K+512)
                                                    1151          MEND
                                                    1152
                                                    1153 *-------------------------------------------------------------------------
                                                    1154 * TSTHDR: The start of a PLOE function test
                                                    1155 *         Test is aligned to 3K boundary
                                                    1156 *              (1.5K data/code; 1.5K instruction test data)
                                                    1157 *
                                                    1158 *        Note: PLOETEST DSECT needs to be in-sync with TSTHDR
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                     1159 *-------------------------------------------------------------------
                                                     1160          MACRO
                                                     1161          TSTHDR  &CC,&UR,&FUN
                                                     1162 .*                      &CC    - expected CC
                                                     1163 .*                      &UR    - expected unequal results
                                                     1164 .*                      &FUN   - function
                                                     1165
                                                     1166          LCLA  &XCC(4)  &CC has mask values for FAILED condition codes
                                                     1167 &XCC(1)  SETA  7                  CC != 0
                                                     1168 &XCC(2)  SETA  11                 CC != 1
                                                     1169 &XCC(3)  SETA  13                 CC != 2
                                                     1170 &XCC(4)  SETA  14                 CC != 3
                                                     1171
                                                     1172          GBLA  &TNUM
                                                     1173 &TNUM    SETA  &TNUM+1
                                                     1174
                                                     1175          GBLA  &THDRNUM
                                                     1176 &THDRNUM SETA  &THDRNUM+1
                                                     1177
                                                     1178          ORG   PLOETST+(128*K)+&THDRNUM*(3*K)
                                                     1179          DC    CL32'PRI   START of test: &TNUM'
                                                     1180
                                                     1181          USING *,R5                base for test data and test routine
                                                     1182          USING PLOETEST,R5
                                                     1183
                                                     1184 T&TNUM   DC    A(CODE&TNUM)        address of test code
                                                     1185          DC    H'&TNUM'            test number
                                                     1186          DC    X'00'
                                                     1187          DC    HL1'&FUN'           Function
                                                     1188          DC    HL1'&UR'            expected unequal results (UR)
                                                     1189          DC    HL1'00'             found UR
                                                     1190
                                                     1191          DC    HL1'&CC'            cc
                                                     1192          AIF   ('&CC' EQ '99').NOCC
                                                     1193          DC    HL1'&XCC(&CC+1)'    not expected CC mask
                                                     1194          AGO   .CCCONT
                                                     1195 .NOCC    ANOP
                                                     1196          DC    HL1'00'             cc not checked
                                                     1197 .CCCONT  ANOP
                                                     1198
                                                     1199          DS    X                   extracted cc
                                                     1200          DS    2F                  extract PSW after test (has CC)
                                                     1201          DC    A(FMSG&TNUM)        address of test fail message
                                                     1202          DC    A(OPSRC&TNUM)       address of operand source
                                                     1203          DC    A(OPDST&TNUM)       address of operand destination
                                                     1204          DC    A(PL&TNUM)          address of parameter list
                                                     1205          DC    A(RANDX+&TNUM)      address of random source data
                                                     1206          DC    A(TEND&TNUM)        address of test end eye catche
                                                     1207 .*
                                                     1208          MEND
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                    1210 *-------------------------------------------------------------------
                                                    1211 * PTTABLE: table of pointers to each test
                                                    1212 *-------------------------------------------------------------------
                                                    1213 *
                                                    1214 *        generate table of pointers to individual tests
                                                    1215 *
                                                    1216          MACRO
                                                    1217          PTTABLE
                                                    1218          GBLA  &TNUM
                                                    1219          LCLA  &CUR
                                                    1220 &CUR     SETA  1
                                                    1221 .*
                                                    1222 TTABLE   DS    0F
                                                    1223 .LOOP    ANOP
                                                    1224 .*
                                                    1225          DC    A(T&CUR)
                                                    1226 .*
                                                    1227 &CUR     SETA  &CUR+1
                                                    1228          AIF   (&CUR LE &TNUM).LOOP
                                                    1229 *
                                                    1230          DC    A(0)                END OF TABLE
                                                    1231          DC    A(0)
                                                    1232 .*
                                                    1233          MEND
                                                    1234
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                                                     1236 ********************************************************************************
                                                     1237 *        PLO-Ext tests
                                                     1238 ********************************************************************************
                                00000000  0004DA73   1239 PLOETST  CSECT
                                                     1240         PRINT DATA
                                                     1241 * ------------------------------------------------------------------------
                                                     1242 * NOTE:  Test PLO-Extension functions with ALET operands
                                                     1243 *        - random source created in Pri, Sec and Home spaces
                                                     1244 *
                                                     1245 *        A test is structured as follows:
                                                     1246 *
                                                     1247 *        TSTHDR
                                                     1248 *        TSTCODE
                                                     1249 *
                                                     1250 *        ... code to test the function
                                                     1251 *
                                                     1252 *        TSTDATA                    test data is located here
                                                     1253 *        TSTLBL OPDST               Operand Destination
                                                     1254 *        DC    XL128'0'
                                                     1255 *
                                                     1256 *        TSTLBL OPSRC               Operand Source
                                                     1257 *        DC    XL128'0'
                                                     1258 *
                                                     1259 *        TSTLBL PL                  Parameter List
                                                     1260 *        DC    XL256'00'
                                                     1261 *
                                                     1262 *        TSTLBL FMSG                fail message
                                                     1263 *        DC    C'failed PLO-Extension '
                                                     1264 *        DC    CLxx'.......'
                                                     1265 *
                                                     1266 *        TSTEND
                                                     1267 *
                                                     1268 * ------------------------------------------------------------------------
                                                     1269 *
                                                     1270 *  Register Usage:
                                                     1271 *
                                                     1272 *   R0-R4      (work)
                                                     1273 *   R5       Testing control table - base current entry
                                                     1274 *   R6         (test work) - secondary space
                                                     1275 *   R7         (test work) - home space
                                                     1276 *   R8       First base register
                                                     1277 *   R9       Second base register
                                                     1278 *   R10      Third base register
                                                     1279 *   R11      Return from test
                                                     1280 *   R12        (test work) - secondary space
                                                     1281 *   R13        (test work) - home space
                                                     1282 *   R14      Subroutine call return
                                                     1283 *   R15      Secondary Subroutine call return
                                                     1284 *            - test: used as PLO R3 (alet register)
                                                     1285 *
                                                     1286 * ------------------------------------------------------------------------
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    1288 * ------------------------------------------------------------------------
                                                    1289 * Test Quadruple Load(QLO: 48) / Store(QSTO: 61) instructions
                                                    1290 * ------------------------------------------------------------------------
                                                    1291          TSTHDR 99,99,99
0001A040                        0001A040  00020C00  1292+         ORG   PLOETST+(128*K)+1*(3*K)
00020C00  D7D9C940 4040E2E3                         1293+         DC    CL32'PRI   START of test: 1'
00020C08  C1D9E340 968640A3
00020C10  85A2A37A 40F14040
00020C18  40404040 40404040
00020C20                        00020C20            1294+         USING *,R5                base for test data and test routine
00020C20                        00000000            1295+         USING PLOETEST,R5
00020C20  00020C50                                  1296+T1       DC    A(CODE1)            address of test code
00020C24  0001                                      1297+         DC    H'1'                test number
00020C26  00                                        1298+         DC    X'00'
00020C27  63                                        1299+         DC    HL1'99'             Function
00020C28  63                                        1300+         DC    HL1'99'             expected unequal results (UR)
00020C29  00                                        1301+         DC    HL1'00'             found UR
00020C2A  63                                        1302+         DC    HL1'99'             cc
00020C2B  00                                        1303+         DC    HL1'00'             cc not checked
00020C2C  00                                        1304+         DS    X                   extracted cc
00020C30  00000000 00000000                         1305+         DS    2F                  extract PSW after test (has CC)
00020C38  00021480                                  1306+         DC    A(FMSG1)            address of test fail message
00020C3C  000212C0                                  1307+         DC    A(OPSRC1)           address of operand source
00020C40  00021220                                  1308+         DC    A(OPDST1)           address of operand destination
00020C44  00021360                                  1309+         DC    A(PL1)              address of parameter list
00020C48  00004179                                  1310+         DC    A(RANDX+1)          address of random source data
00020C4C  000214F5                                  1311+         DC    A(TEND1)            address of test end eye catche
                                                    1312
                                                    1313          TSTCODE
00020C50                                            1314+CODE1    DS    0F
                                                    1315
00020C50  E310 501C 0014                0000001C    1316          LGF   R1,OPSRCA           R1: OPSRC - Primary
00020C56  B904 0061                                 1317          LGR   R6,R1               R6: OPSRC - Secondary
00020C5A  B904 0071                                 1318          LGR   R7,R1               R7: OPSRC - Home
                                                    1319
00020C5E  E320 5028 0014                00000028    1320          LGF   R2,RANDXA           R2: address random data
00020C64  E330 5024 0014                00000024    1321          LGF   R3,PLA              R3: address PL
                                                    1322
00020C6A  B219 0200                     00000200    1323          SAC   ARMODE
00020C6E  D21F 1000 2000        00000000  00000000  1324          MVC   0(32,R1),0(R2)      load Op1 = Primary
00020C74  D21F 6020 2020        00000020  00000020  1325          MVC   32(32,R6),32(R2)    load Op2 = Secondary
00020C7A  D21F 7040 2040        00000040  00000040  1326          MVC   64(32,R7),64(R2)    load Op3 = Home
00020C80  D21F 6060 2060        00000060  00000060  1327          MVC   96(32,R6),96(R2)    load Op4 = Secondary
00020C86  B219 0000                     00000000    1328          SAC   PRIMODE
                                                    1329 *
                                                    1330 * Quadruple Load(QLO: 48)
                                                    1331 *
00020C8A  E313 0000 0024                00000000    1332          STG   R1,0(R3)            load OP1 address in PL
00020C90  E54C 3020 0000                00000020    1333          MVHI  32(R3),PRIALET
                                                    1334
00020C96  4111 0020                     00000020    1335          LA    R1,32(R1)
00020C9A  E313 0008 0024                00000008    1336          STG   R1,8(R3)            load OP2 address in PL
00020CA0  E54C 3024 0001                00000024    1337          MVHI  36(R3),SECALET
                                                    1338
00020CA6  4111 0020                     00000020    1339          LA    R1,32(R1)
00020CAA  E313 0010 0024                00000010    1340          STG   R1,16(R3)           load OP3 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00020CB0  E54C 3028 0002                 00000028  1341          MVHI  40(R3),HOMEALET
                                                   1342
00020CB6  4111 0020                      00000020  1343          LA    R1,32(R1)
00020CBA  E313 0018 0024                 00000018  1344          STG   R1,24(R3)           load OP4 address in PL
00020CC0  E54C 302C 0001                 0000002C  1345          MVHI  44(R3),SECALET
                                                   1346
00020CC6  B982 0000                                1347          XGR   R0,R0
00020CCA  4100 0030                      00000030  1348          LA    R0,48
                                                   1349
00020CCE  B219 0200                      00000200  1350          SAC   ARMODE
00020CD2  EE0F 0000 3000      00000000   00000000  1351          PLO   0,0,R15,0(R3)       Quadruple Load(QLO: 48)
00020CD8  B219 0000                      00000000  1352          SAC   PRIMODE
                                                   1353 *
                                                   1354 * Quadruple Store(QSTO: 61)
                                                   1355 *
00020CDC  E310 5020 0014                 00000020  1356          LGF   R1,OPDSTA           R1:  OPDST - Primary
00020CE2  E330 5024 0014                 00000024  1357          LGF   R3,PLA
                                                   1358
00020CE8  E313 0000 0024                 00000000  1359          STG   R1,0(R3)            load OP1 address in PL
00020CEE  E54C 3020 0002                 00000020  1360          MVHI  32(R3),HOMEALET
                                                   1361
00020CF4  4111 0020                      00000020  1362          LA    R1,32(R1)
00020CF8  E313 0008 0024                 00000008  1363          STG   R1,8(R3)            load OP2 address in PL
00020CFE  E54C 3024 0002                 00000024  1364          MVHI  36(R3),HOMEALET
                                                   1365
00020D04  4111 0020                      00000020  1366          LA    R1,32(R1)
00020D08  E313 0010 0024                 00000010  1367          STG   R1,16(R3)           load OP3 address in PL
00020D0E  E54C 3028 0001                 00000028  1368          MVHI  40(R3),SECALET
                                                   1369
00020D14  4111 0020                      00000020  1370          LA    R1,32(R1)
00020D18  E313 0018 0024                 00000018  1371          STG   R1,24(R3)           load OP4 address in PL
00020D1E  E54C 302C 0000                 0000002C  1372          MVHI  44(R3),PRIALET
                                                   1373
00020D24  B982 0000                                1374          XGR   R0,R0
00020D28  4100 003D                      0000003D  1375          LA    R0,61
                                                   1376
00020D2C  B219 0200                      00000200  1377          SAC   ARMODE
00020D30  EE0F 0000 3000      00000000   00000000  1378          PLO   0,0,R15,0(R3)       Quadruple Store(QSTO: 61)
00020D36  B219 0000                      00000000  1379          SAC   PRIMODE
                                                   1380 *
                                                   1381 * compare results
                                                   1382 *                                  reset registers
00020D3A  E310 501C 0014                 0000001C  1383          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00020D40  B904 0061                               1384          LGR   R6,R1               R6:  OPSRC - Secondary
00020D44  B904 0071                               1385          LGR   R7,R1               R7:  OPSRC - Home
                                                   1386
00020D48  E340 5020 0014                 00000020  1387          LGF   R4,OPDSTA           R4:  OPDST - Primary
00020D4E  B904 00C4                               1388          LGR   R12,R4              R12: OPDST - Secondary
00020D52  B904 00D4                               1389          LGR   R13,R4              R13: OPDST - Home
                                                   1390
00020D56  B219 0200                      00000200  1391          SAC   ARMODE
00020D5A  D51F 1000 D000      00000000   00000000  1392          CLC   0(32,R1),0(R13)
00020D60  4770 5166                      00020D86  1393          BNE   F4861               OP1 Different, error
00020D64  D51F 6020 D020      00000020   00000020  1394          CLC   32(32,R6),32(R13)
00020D6A  4770 5166                      00020D86  1395          BNE   F4861               OP2 Different, error
00020D6E  D51F 7040 C040      00000040   00000040  1396          CLC   64(32,R7),64(R12)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00020D74  4770 5166                      00020D86  1397           BNE   F4861               OP3 Different, error
00020D78  D51F 6060 4060      00000060   00000060  1398           CLC   96(32,R6),96(R4)
00020D7E  B219 0000                      00000000  1399           SAC   PRIMODE
00020D82  478B 0000                      00000000  1400           BE    0(R11)              OP4 same, return
                                                   1401
                              00020D86   00000001  1402 F4861     EQU   *
00020D86  B219 0000                      00000000  1403           SAC   PRIMODE
00020D8A  45F0 8450                      00002450  1404           BAL   R15,FAILMSG
00020D8E  07FB                                     1405           BR    R11
                                                   1406
                                                   1407           TSTDATA
00020D90                      00020D90   00021200  1408+          ORG   PLOETST+(128*K)+1*(3*K)+(K+512)
                                                   1409           TSTLBL OPDST              Operand Destination (Results)
00021200  D7D9C940 40406E6E                        1410+          DC    CL32'PRI   >>>OPDST1<<<'  eye catcher
00021208  6ED6D7C4 E2E3F14C
00021210  4C4C4040 40404040
00021218  40404040 40404040
00021220                                           1411+OPDST1    DS    0C
00021220  00000000 00000000                        1412           DC    XL128'0'
00021228  00000000 00000000
00021230  00000000 00000000
00021238  00000000 00000000
00021240  00000000 00000000
00021248  00000000 00000000
00021250  00000000 00000000
00021258  00000000 00000000
00021260  00000000 00000000
00021268  00000000 00000000
00021270  00000000 00000000
00021278  00000000 00000000
00021280  00000000 00000000
00021288  00000000 00000000
00021290  00000000 00000000
00021298  00000000 00000000
                                                   1413
                                                   1414           TSTLBL OPSRC              Operand Source
000212A0  D7D9C940 40406E6E                        1415+          DC    CL32'PRI   >>>OPSRC1<<<'  eye catcher
000212A8  6ED6D7E2 D9C3F14C
000212B0  4C4C4040 40404040
000212B8  40404040 40404040
000212C0                                           1416+OPSRC1    DS    0C
000212C0  00000000 00000000                        1417           DC    XL128'0'
000212C8  00000000 00000000
000212D0  00000000 00000000
000212D8  00000000 00000000
000212E0  00000000 00000000
000212E8  00000000 00000000
000212F0  00000000 00000000
000212F8  00000000 00000000
00021300  00000000 00000000
00021308  00000000 00000000
00021310  00000000 00000000
00021318  00000000 00000000
00021320  00000000 00000000
00021328  00000000 00000000
00021330  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00021338  00000000 00000000                          1418
                                                    1419          TSTLBL PL                 Parameter List
00021340  D7D9C940 40406E6E                         1420+         DC    CL32'PRI   >>>PL1<<<'     eye catcher
00021348  6ED7D3F1 4C4C4C40
00021350  40404040 40404040
00021358  40404040 40404040
00021360                                            1421+PL1      DS    0C
00021360  00000000 00000000                         1422          DC    XL256'00'
00021368  00000000 00000000
00021370  00000000 00000000
00021378  00000000 00000000
00021380  00000000 00000000
00021388  00000000 00000000
00021390  00000000 00000000
00021398  00000000 00000000
000213A0  00000000 00000000
000213A8  00000000 00000000
000213B0  00000000 00000000
000213B8  00000000 00000000
000213C0  00000000 00000000
000213C8  00000000 00000000
000213D0  00000000 00000000
000213D8  00000000 00000000
000213E0  00000000 00000000
000213E8  00000000 00000000
000213F0  00000000 00000000
000213F8  00000000 00000000
00021400  00000000 00000000
00021408  00000000 00000000
00021410  00000000 00000000
00021418  00000000 00000000
00021420  00000000 00000000
00021428  00000000 00000000
00021430  00000000 00000000
00021438  00000000 00000000
00021440  00000000 00000000
00021448  00000000 00000000
00021450  00000000 00000000
00021458  00000000 00000000
                                                    1423
                                                    1424          TSTLBL FMSG               fail message
00021460  D7D9C940 40406E6E                         1425+         DC    CL32'PRI   >>>FMSG1<<<'   eye catcher
00021468  6EC6D4E2 C7F14C4C
00021470  4C404040 40404040
00021478  40404040 40404040
00021480                                            1426+FMSG1    DS    0C
00021480  86818993 858440D7                         1427          DC    C'failed PLO-Extension '
00021488  D3D660C5 A7A38595
00021490  A2899695 40
00021495  D8A48184 99A49793                         1428          DC    CL96'Quadruple Load(QLO: 48) / Store(QSTO: 61)       '
0002149D  8540D396 81844DD8
000214A5  D3D67A40 F4F85D40
000214AD  6140E2A3 9699854D
000214B5  D8E2E3D6 7A40F6F1
000214BD  5D404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000214C5  40404040 40404040
000214CD  40404040 40404040
000214D5  40404040 40404040
000214DD  40404040 40404040
000214E5  40404040 40404040
000214ED  40404040 40404040
                                        1429
                                        1430          TSTEND
000214F5  D7D9C940 4040C5D5             1431+TEND1    DC    CL32'PRI   END of test: 1 <<<'
000214FD  C4409686 40A385A2
00021505  A37A40F1 404C4C4C
0002150D  40404040 40404040
                                        1432
                                        1433 * ------------------------------------------------------------------
                                        1434 * Test Quadruple Load(QLX: 47) / Store(QSTX: 60) instructions
                                        1435 * ------------------------------------------------------------------
                                        1436          TSTHDR 99,99,99
00021515                  00021515  00021800  1437+         ORG   PLOETST+(128*K)+2*(3*K)
00021800  D7D9C940 4040E2E3             1438+         DC    CL32'PRI   START of test: 2'
00021808  C1D9E340 968640A3
00021810  85A2A37A 40F24040
00021818  40404040 40404040
00021820                  00021820              1439+         USING *,R5                base for test data and test routine
00021820                  00000000              1440+         USING PLOETEST,R5
00021820  00021850                              1441+T2       DC    A(CODE2)            address of test code
00021824  0002                                  1442+         DC    H'2'                test number
00021826  00                                    1443+         DC    X'00'
00021827  63                                    1444+         DC    HL1'99'             Function
00021828  63                                    1445+         DC    HL1'99'             expected unequal results (UR)
00021829  00                                    1446+         DC    HL1'00'             found UR
0002182A  63                                    1447+         DC    HL1'99'             cc
0002182B  00                                    1448+         DC    HL1'00'             cc not checked
0002182C  00                                    1449+         DS    X                   extracted cc
00021830  00000000 00000000                     1450+         DS    2F                  extract PSW after test (has CC)
00021838  00022080                              1451+         DC    A(FMSG2)            address of test fail message
0002183C  00021EC0                              1452+         DC    A(OPSRC2)           address of operand source
00021840  00021E20                              1453+         DC    A(OPDST2)           address of operand destination
00021844  00021F60                              1454+         DC    A(PL2)              address of parameter list
00021848  0000417A                              1455+         DC    A(RANDX+2)          address of random source data
0002184C  000220F5                              1456+         DC    A(TEND2)            address of test end eye catche
                                                1457
                                                1458          TSTCODE
00021850                                        1459+CODE2    DS    0F
                                                1460
00021850  E310 501C 0014                0000001C  1461          LGF   R1,OPSRCA           R1: OPSRC - Primary
00021856  B904 0061                             1462          LGR   R6,R1               R6: OPSRC - Secondary
0002185A  B904 0071                             1463          LGR   R7,R1               R7: OPSRC - Home
                                                1464
0002185E  E320 5028 0014                00000028  1465          LGF   R2,RANDXA           R2: address random data
00021864  E330 5024 0014                00000024  1466          LGF   R3,PLA              R3: address PL
                                                1467
0002186A  B219 0200                     00000200  1468          SAC   ARMODE
0002186E  D20F 1000 2000        00000000 00000000 1469          MVC   0(16,R1),0(R2)      load Op1 = Primary
00021874  D20F 6020 2020        00000020 00000020 1470          MVC   32(16,R6),32(R2)    load Op2 = Secondary
0002187A  D20F 7040 2040        00000040 00000040 1471          MVC   64(16,R7),64(R2)    load Op3 = Home
00021880  D20F 6060 2060        00000060 00000060 1472          MVC   96(16,R6),96(R2)    load Op4 = Secondary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00021886  B219 0000                     00000000  1473          SAC   PRIMODE
                                                   1474 *
                                                   1475 * Quadruple Load(QLX: 47)
                                                   1476 *
0002188A  E313 0000 0024                 00000000  1477          STG   R1,0(R3)            load OP1 address in PL
00021890  E54C 3020 0000                 00000020  1478          MVHI  32(R3),PRIALET
                                                   1479
00021896  4111 0020                      00000020  1480          LA    R1,32(R1)
0002189A  E313 0008 0024                 00000008  1481          STG   R1,8(R3)            load OP2 address in PL
000218A0  E54C 3024 0001                 00000024  1482          MVHI  36(R3),SECALET
                                                   1483
000218A6  4111 0020                      00000020  1484          LA    R1,32(R1)
000218AA  E313 0010 0024                 00000010  1485          STG   R1,16(R3)           load OP3 address in PL
000218B0  E54C 3028 0002                 00000028  1486          MVHI  40(R3),HOMEALET
                                                   1487
000218B6  4111 0020                      00000020  1488          LA    R1,32(R1)
000218BA  E313 0018 0024                 00000018  1489          STG   R1,24(R3)           load OP4 address in PL
000218C0  E54C 302C 0001                 0000002C  1490          MVHI  44(R3),SECALET
                                                   1491
000218C6  B982 0000                               1492          XGR   R0,R0
000218CA  4100 002F                      0000002F  1493          LA    R0,47
                                                   1494
000218CE  B219 0200                      00000200  1495          SAC   ARMODE
000218D2  EE0F 0000 3000       00000000  00000000  1496          PLO   0,0,R15,0(R3)       Quadruple Load(QLX: 47)
000218D8  B219 0000                      00000000  1497          SAC   PRIMODE
                                                   1498 *
                                                   1499 * Quadruple Store(QSTX: 60)
                                                   1500 *
000218DC  E310 5020 0014                 00000020  1501          LGF   R1,OPDSTA           R1:  OPDST - Primary
000218E2  E330 5024 0014                 00000024  1502          LGF   R3,PLA
                                                   1503
000218E8  E313 0000 0024                 00000000  1504          STG   R1,0(R3)            load OP1 address in PL
000218EE  E54C 3020 0002                 00000020  1505          MVHI  32(R3),HOMEALET
                                                   1506
000218F4  4111 0020                      00000020  1507          LA    R1,32(R1)
000218F8  E313 0008 0024                 00000008  1508          STG   R1,8(R3)            load OP2 address in PL
000218FE  E54C 3024 0002                 00000024  1509          MVHI  36(R3),HOMEALET
                                                   1510
00021904  4111 0020                      00000020  1511          LA    R1,32(R1)
00021908  E313 0010 0024                 00000010  1512          STG   R1,16(R3)           load OP3 address in PL
0002190E  E54C 3028 0001                 00000028  1513          MVHI  40(R3),SECALET
                                                   1514
00021914  4111 0020                      00000020  1515          LA    R1,32(R1)
00021918  E313 0018 0024                 00000018  1516          STG   R1,24(R3)           load OP4 address in PL
0002191E  E54C 302C 0000                 0000002C  1517          MVHI  44(R3),PRIALET
                                                   1518
00021924  B982 0000                               1519          XGR   R0,R0
00021928  4100 003C                      0000003C  1520          LA    R0,60
                                                   1521
0002192C  B219 0200                      00000200  1522          SAC   ARMODE
00021930  EE0F 0000 3000       00000000  00000000  1523          PLO   0,0,R15,0(R3)       Quadruple Store(QSTX: 60)
00021936  B219 0000                      00000000  1524          SAC   PRIMODE
                                                   1525 *
                                                   1526 * compare results
                                                   1527 *                                  reset registers
0002193A  E310 501C 0014                 0000001C  1528          LGF   R1,OPSRCA           R1:  OPSRC - Primary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00021940  B904 0061                             1529           LGR   R6,R1               R6:  OPSRC - Secondary
00021944  B904 0071                             1530           LGR   R7,R1               R7:  OPSRC - Home
                                                1531
00021948  E340 5020 0014                 00000020 1532           LGF   R4,OPDSTA           R4:  OPDST - Primary
0002194E  B904 00C4                             1533           LGR   R12,R4              R12: OPDST - Secondary
00021952  B904 00D4                             1534           LGR   R13,R4              R13: OPDST - Home
                                                1535
00021956  B219 0200                             1536           SAC   ARMODE
0002195A  D50F 1000 D000       00000000 00000000 1537           CLC   0(16,R1),0(R13)
00021960  4770 5166                      00021986 1538           BNE   F4760               OP1 Different, error
00021964  D50F 6020 D020       00000020 00000020 1539           CLC   32(16,R6),32(R13)
0002196A  4770 5166                      00021986 1540           BNE   F4760               OP2 Different, error
0002196E  D50F 7040 C040       00000040 00000040 1541           CLC   64(16,R7),64(R12)
00021974  4770 5166                      00021986 1542           BNE   F4760               OP3 Different, error
00021978  D50F 6060 4060       00000060 00000060 1543           CLC   96(16,R6),96(R4)
0002197E  B219 0000                      00000000 1544           SAC   PRIMODE
00021982  478B 0000                      00000000 1545           BE    0(R11)              OP4 same, return
                                                1546
                                         00021986 00000001 1547 F4760    EQU   *
00021986  B219 0000                      00000000 1548           SAC   PRIMODE
0002198A  45F0 8450                      00002450 1549           BAL   R15,FAILMSG
0002198E  07FB                                   1550           BR    R11
                                                1551
                                                1552           TSTDATA
00021990                         00021990 00021E00 1553+          ORG   PLOETST+(128*K)+2*(3*K)+(K+512)
                                                1554           TSTLBL OPDST              Operand Destination (Results)
00021E00  D7D9C940 40406E6E                     1555+          DC    CL32'PRI   >>>OPDST2<<<'  eye catcher
00021E08  6ED6D7C4 E2E3F24C
00021E10  4C4C4040 40404040
00021E18  40404040 40404040
00021E20                                        1556+OPDST2    DS    0C
00021E20  00000000 00000000                     1557           DC    XL128'0'
00021E28  00000000 00000000
00021E30  00000000 00000000
00021E38  00000000 00000000
00021E40  00000000 00000000
00021E48  00000000 00000000
00021E50  00000000 00000000
00021E58  00000000 00000000
00021E60  00000000 00000000
00021E68  00000000 00000000
00021E70  00000000 00000000
00021E78  00000000 00000000
00021E80  00000000 00000000
00021E88  00000000 00000000
00021E90  00000000 00000000
00021E98  00000000 00000000
                                                1558
                                                1559           TSTLBL OPSRC              Operand Source
00021EA0  D7D9C940 40406E6E                     1560+          DC    CL32'PRI   >>>OPSRC2<<<'  eye catcher
00021EA8  6ED6D7E2 D9C3F24C
00021EB0  4C4C4040 40404040
00021EB8  40404040 40404040
00021EC0                                        1561+OPSRC2    DS    0C
00021EC0  00000000 00000000                     1562           DC    XL128'0'
00021EC8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00021ED0  00000000 00000000
00021ED8  00000000 00000000
00021EE0  00000000 00000000
00021EE8  00000000 00000000
00021EF0  00000000 00000000
00021EF8  00000000 00000000
00021F00  00000000 00000000
00021F08  00000000 00000000
00021F10  00000000 00000000
00021F18  00000000 00000000
00021F20  00000000 00000000
00021F28  00000000 00000000
00021F30  00000000 00000000
00021F38  00000000 00000000
                                        1563
                                        1564          TSTLBL PL                 Parameter List
00021F40  D7D9C940 40406E6E             1565+         DC    CL32'PRI   >>>PL2<<<'     eye catcher
00021F48  6ED7D3F2 4C4C4C40
00021F50  40404040 40404040
00021F58  40404040 40404040
00021F60                                1566+PL2      DS    0C
00021F60  00000000 00000000             1567          DC    XL256'00'
00021F68  00000000 00000000
00021F70  00000000 00000000
00021F78  00000000 00000000
00021F80  00000000 00000000
00021F88  00000000 00000000
00021F90  00000000 00000000
00021F98  00000000 00000000
00021FA0  00000000 00000000
00021FA8  00000000 00000000
00021FB0  00000000 00000000
00021FB8  00000000 00000000
00021FC0  00000000 00000000
00021FC8  00000000 00000000
00021FD0  00000000 00000000
00021FD8  00000000 00000000
00021FE0  00000000 00000000
00021FE8  00000000 00000000
00021FF0  00000000 00000000
00021FF8  00000000 00000000
00022000  00000000 00000000
00022008  00000000 00000000
00022010  00000000 00000000
00022018  00000000 00000000
00022020  00000000 00000000
00022028  00000000 00000000
00022030  00000000 00000000
00022038  00000000 00000000
00022040  00000000 00000000
00022048  00000000 00000000
00022050  00000000 00000000
00022058  00000000 00000000
                                        1568
                                        1569          TSTLBL FMSG               fail message
00022060  D7D9C940 40406E6E             1570+         DC    CL32'PRI   >>>FMSG2<<<'   eye catcher
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022068  6EC6D4E2 C7F24C4C
00022070  4C404040 40404040
00022078  40404040 40404040
00022080                                          1571+FMSG2    DS    0C
00022080  86818993 858440D7                       1572          DC    C'failed PLO-Extension '
00022088  D3D660C5 A7A38595
00022090  A2899695 40
00022095  D8A48184 99A49793                       1573          DC    CL96'Quadruple Load(QLX: 47) / Store(QSTX: 60)'
0002209D  8540D396 81844DD8
000220A5  D3E77A40 F4F75D40
000220AD  6140E2A3 9699854D
000220B5  D8E2E3E7 7A40F6F0
000220BD  5D404040 40404040
000220C5  40404040 40404040
000220CD  40404040 40404040
000220D5  40404040 40404040
000220DD  40404040 40404040
000220E5  40404040 40404040
000220ED  40404040 40404040
                                                  1574
                                                  1575          TSTEND
000220F5  D7D9C940 4040C5D5                       1576+TEND2    DC    CL32'PRI   END of test: 2 <<<'
000220FD  C4409686 40A385A2
00022105  A37A40F2 404C4C4C
0002210D  40404040 40404040
                                                  1577
                                                  1578 * ------------------------------------------------------------------
                                                  1579 * Test Quadruple Load(QLG: 46) / Store(QSTG: 59) instructions
                                                  1580 * ------------------------------------------------------------------
                                                  1581          TSTHDR 99,99,99
00022115                        00022115  00022400  1582+         ORG   PLOETST+(128*K)+3*(3*K)
00022400  D7D9C940 4040E2E3                       1583+         DC    CL32'PRI   START of test: 3'
00022408  C1D9E340 968640A3
00022410  85A2A37A 40F34040
00022418  40404040 40404040
00022420                        00022420          1584+         USING *,R5                base for test data and test routine
00022420                        00000000          1585+         USING PLOETEST,R5
00022420  00022450                                1586+T3       DC    A(CODE3)            address of test code
00022424  0003                                    1587+         DC    H'3'                test number
00022426  00                                      1588+         DC    X'00'
00022427  63                                      1589+         DC    HL1'99'             Function
00022428  63                                      1590+         DC    HL1'99'             expected unequal results (UR)
00022429  00                                      1591+         DC    HL1'00'             found UR
0002242A  63                                      1592+         DC    HL1'99'             cc
0002242B  00                                      1593+         DC    HL1'00'             cc not checked
0002242C  00                                      1594+         DS    X                   extracted cc
00022430  00000000 00000000                       1595+         DS    2F                  extract PSW after test (has CC)
00022438  00022C80                                1596+         DC    A(FMSG3)            address of test fail message
0002243C  00022AC0                                1597+         DC    A(OPSRC3)           address of operand source
00022440  00022A20                                1598+         DC    A(OPDST3)           address of operand destination
00022444  00022B60                                1599+         DC    A(PL3)              address of parameter list
00022448  0000417B                                1600+         DC    A(RANDX+3)          address of random source data
0002244C  00022CF5                                1601+         DC    A(TEND3)            address of test end eye catche
                                                  1602
                                                  1603          TSTCODE
00022450                                          1604+CODE3    DS    0F
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1605
00022450  E310 501C 0014              0000001C  1606          LGF   R1,OPSRCA           R1: OPSRC - Primary
00022456  B904 0061                             1607          LGR   R6,R1               R6: OPSRC - Secondary
0002245A  B904 0071                             1608          LGR   R7,R1               R7: OPSRC - Home
                                                1609
0002245E  E320 5028 0014              00000028  1610          LGF   R2,RANDXA           R2: address random data
00022464  E330 5024 0014              00000024  1611          LGF   R3,PLA              R3: address PL
                                                1612
0002246A  B219 0200                   00000200  1613          SAC   ARMODE
0002246E  D207 1000 2000      00000000 00000000 1614          MVC   0(8,R1),0(R2)       load Op1 = Primary
00022474  D207 6020 2020      00000020 00000020 1615          MVC   32(8,R6),32(R2)     load Op2 = Secondary
0002247A  D207 7040 2040      00000040 00000040 1616          MVC   64(8,R7),64(R2)     load Op3 = Home
00022480  D207 6060 2060      00000060 00000060 1617          MVC   96(8,R6),96(R2)     load Op4 = Secondary
00022486  B219 0000                   00000000  1618          SAC   PRIMODE
                                                1619 *
                                                1620 * Quadruple Load(QLX: 46)
                                                1621 *
0002248A  E313 0000 0024              00000000  1622          STG   R1,0(R3)            load OP1 address in PL
00022490  E54C 3020 0000              00000020  1623          MVHI  32(R3),PRIALET
                                                1624
00022496  4111 0020                   00000020  1625          LA    R1,32(R1)
0002249A  E313 0008 0024              00000008  1626          STG   R1,8(R3)            load OP2 address in PL
000224A0  E54C 3024 0001              00000024  1627          MVHI  36(R3),SECALET
                                                1628
000224A6  4111 0020                   00000020  1629          LA    R1,32(R1)
000224AA  E313 0010 0024              00000010  1630          STG   R1,16(R3)           load OP3 address in PL
000224B0  E54C 3028 0002              00000028  1631          MVHI  40(R3),HOMEALET
                                                1632
000224B6  4111 0020                   00000020  1633          LA    R1,32(R1)
000224BA  E313 0018 0024              00000018  1634          STG   R1,24(R3)           load OP4 address in PL
000224C0  E54C 302C 0001              0000002C  1635          MVHI  44(R3),SECALET
                                                1636
000224C6  B982 0000                             1637          XGR   R0,R0
000224CA  4100 002E                   0000002E  1638          LA    R0,46
                                                1639
000224CE  B219 0200                   00000200  1640          SAC   ARMODE
000224D2  EE0F 0000 3000      00000000 00000000 1641          PLO   0,0,R15,0(R3)       Quadruple Load(QLG: 46)
000224D8  B219 0000                   00000000  1642          SAC   PRIMODE
                                                1643 *
                                                1644 * Quadruple Store(QSTX: 59)
                                                1645 *
000224DC  E310 5020 0014              00000020  1646          LGF   R1,OPDSTA           R1:  OPDST - Primary
000224E2  E330 5024 0014              00000024  1647          LGF   R3,PLA
                                                1648
000224E8  E313 0000 0024              00000000  1649          STG   R1,0(R3)            load OP1 address in PL
000224EE  E54C 3020 0002              00000020  1650          MVHI  32(R3),HOMEALET
                                                1651
000224F4  4111 0020                   00000020  1652          LA    R1,32(R1)
000224F8  E313 0008 0024              00000008  1653          STG   R1,8(R3)            load OP2 address in PL
000224FE  E54C 3024 0002              00000024  1654          MVHI  36(R3),HOMEALET
                                                1655
00022504  4111 0020                   00000020  1656          LA    R1,32(R1)
00022508  E313 0010 0024              00000010  1657          STG   R1,16(R3)           load OP3 address in PL
0002250E  E54C 3028 0001              00000028  1658          MVHI  40(R3),SECALET
                                                1659
00022514  4111 0020                   00000020  1660          LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022518  E313 0018 0024                 00000018  1661          STG   R1,24(R3)           load OP4 address in PL
0002251E  E54C 302C 0000                 0000002C  1662          MVHI  44(R3),PRIALET
                                                   1663
00022524  B982 0000                              1664          XGR   R0,R0
00022528  4100 003B                      0000003B  1665          LA    R0,59
                                                   1666
0002252C  B219 0200                      00000200  1667          SAC   ARMODE
00022530  EE0F 0000 3000      00000000   00000000  1668          PLO   0,0,R15,0(R3)       Quadruple Store(QSTG: 59)
00022536  B219 0000                      00000000  1669          SAC   PRIMODE
                                                   1670 *
                                                   1671 * compare results
                                                   1672 *                                 reset registers
0002253A  E310 501C 0014                 0000001C  1673          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00022540  B904 0061                              1674          LGR   R6,R1               R6:  OPSRC - Secondary
00022544  B904 0071                              1675          LGR   R7,R1               R7:  OPSRC - Home
                                                   1676
00022548  E340 5020 0014                 00000020  1677          LGF   R4,OPDSTA           R4:  OPDST - Primary
0002254E  B904 00C4                              1678          LGR   R12,R4              R12: OPDST - Secondary
00022552  B904 00D4                              1679          LGR   R13,R4              R13: OPDST - Home
                                                   1680
00022556  B219 0200                      00000200  1681          SAC   ARMODE
0002255A  D507 1000 D000      00000000   00000000  1682          CLC   0(8,R1),0(R13)
00022560  4770 5166                      00022586  1683          BNE   F4659               OP1 Different, error
00022564  D507 6020 D020      00000020   00000020  1684          CLC   32(8,R6),32(R13)
0002256A  4770 5166                      00022586  1685          BNE   F4659               OP2 Different, error
0002256E  D507 7040 C040      00000040   00000040  1686          CLC   64(8,R7),64(R12)
00022574  4770 5166                      00022586  1687          BNE   F4659               OP3 Different, error
00022578  D507 6060 4060      00000060   00000060  1688          CLC   96(8,R6),96(R4)
0002257E  B219 0000                      00000000  1689          SAC   PRIMODE
00022582  478B 0000                      00000000  1690          BE    0(R11)              OP4 same, return
                                                   1691
                                                   1692
                              00022586   00000001  1693 F4659    EQU   *
00022586  B219 0000                      00000000  1694          SAC   PRIMODE
0002258A  45F0 8450                      00002450  1695          BAL   R15,FAILMSG
0002258E  07FB                                     1696          BR    R11
                                                   1697
                                                   1698          TSTDATA
00022590                      00022590   00022A00  1699+         ORG   PLOETST+(128*K)+3*(3*K)+(K+512)
                                                   1700          TSTLBL OPDST              Operand Destination (Results)
00022A00  D7D9C940 40406E6E                        1701+         DC    CL32'PRI   >>>OPDST3<<<'  eye catcher
00022A08  6ED6D7C4 E2E3F34C
00022A10  4C4C4040 40404040
00022A18  40404040 40404040
00022A20                                           1702+OPDST3   DS    0C
00022A20  00000000 00000000                        1703          DC    XL128'0'
00022A28  00000000 00000000
00022A30  00000000 00000000
00022A38  00000000 00000000
00022A40  00000000 00000000
00022A48  00000000 00000000
00022A50  00000000 00000000
00022A58  00000000 00000000
00022A60  00000000 00000000
00022A68  00000000 00000000
00022A70  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00022A78  00000000 00000000
00022A80  00000000 00000000
00022A88  00000000 00000000
00022A90  00000000 00000000
00022A98  00000000 00000000
                                          1704
                                          1705          TSTLBL OPSRC              Operand Source
00022AA0  D7D9C940 40406E6E               1706+         DC    CL32'PRI   >>>OPSRC3<<<'   eye catcher
00022AA8  6ED6D7E2 D9C3F34C
00022AB0  4C4C4040 40404040
00022AB8  40404040 40404040
00022AC0                                  1707+OPSRC3   DS    0C
00022AC0  00000000 00000000               1708          DC    XL128'0'
00022AC8  00000000 00000000
00022AD0  00000000 00000000
00022AD8  00000000 00000000
00022AE0  00000000 00000000
00022AE8  00000000 00000000
00022AF0  00000000 00000000
00022AF8  00000000 00000000
00022B00  00000000 00000000
00022B08  00000000 00000000
00022B10  00000000 00000000
00022B18  00000000 00000000
00022B20  00000000 00000000
00022B28  00000000 00000000
00022B30  00000000 00000000
00022B38  00000000 00000000
                                          1709
                                          1710          TSTLBL PL                 Parameter List
00022B40  D7D9C940 40406E6E               1711+         DC    CL32'PRI   >>>PL3<<<'      eye catcher
00022B48  6ED7D3F3 4C4C4C40
00022B50  40404040 40404040
00022B58  40404040 40404040
00022B60                                  1712+PL3      DS    0C
00022B60  00000000 00000000               1713          DC    XL256'00'
00022B68  00000000 00000000
00022B70  00000000 00000000
00022B78  00000000 00000000
00022B80  00000000 00000000
00022B88  00000000 00000000
00022B90  00000000 00000000
00022B98  00000000 00000000
00022BA0  00000000 00000000
00022BA8  00000000 00000000
00022BB0  00000000 00000000
00022BB8  00000000 00000000
00022BC0  00000000 00000000
00022BC8  00000000 00000000
00022BD0  00000000 00000000
00022BD8  00000000 00000000
00022BE0  00000000 00000000
00022BE8  00000000 00000000
00022BF0  00000000 00000000
00022BF8  00000000 00000000
00022C00  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00022C08  00000000 00000000
00022C10  00000000 00000000
00022C18  00000000 00000000
00022C20  00000000 00000000
00022C28  00000000 00000000
00022C30  00000000 00000000
00022C38  00000000 00000000
00022C40  00000000 00000000
00022C48  00000000 00000000
00022C50  00000000 00000000
00022C58  00000000 00000000
                                        1714
                                        1715          TSTLBL FMSG               fail message
00022C60  D7D9C940 40406E6E             1716+         DC    CL32'PRI   >>>FMSG3<<<'   eye catcher
00022C68  6EC6D4E2 C7F34C4C
00022C70  4C404040 40404040
00022C78  40404040 40404040
00022C80                                1717+FMSG3    DS    0C
00022C80  86818993 858440D7             1718          DC    C'failed PLO-Extension '
00022C88  D3D660C5 A7A38595
00022C90  A2899695 40
00022C95  D8A48184 99A49793             1719          DC    CL96'Quadruple Load(QLG: 46) / Store(QSTG: 59)'
00022C9D  8540D396 81844DD8
00022CA5  D3C77A40 F4F65D40
00022CAD  6140E2A3 9699854D
00022CB5  D8E2E3C7 7A40F5F9
00022CBD  5D404040 40404040
00022CC5  40404040 40404040
00022CCD  40404040 40404040
00022CD5  40404040 40404040
00022CDD  40404040 40404040
00022CE5  40404040 40404040
00022CED  40404040 40404040
                                        1720
                                        1721          TSTEND
00022CF5  D7D9C940 4040C5D5             1722+TEND3    DC    CL32'PRI   END of test: 3 <<<'
00022CFD  C4409686 40A385A2
00022D05  A37A40F3 404C4C4C
00022D0D  40404040 40404040
                                        1723
                                        1724 * ------------------------------------------------------------------
                                        1725 * Test Quadruple Load(QL: 45) / Store(QST: 58) instructions
                                        1726 * ------------------------------------------------------------------
                                        1727          TSTHDR 99,99,99
00022D15                  00022D15  00023000  1728+         ORG   PLOETST+(128*K)+4*(3*K)
00023000  D7D9C940 4040E2E3             1729+         DC    CL32'PRI   START of test: 4'
00023008  C1D9E340 968640A3
00023010  85A2A37A 40F44040
00023018  40404040 40404040
00023020                  00023020      1730+         USING *,R5                base for test data and test routine
00023020                  00000000      1731+         USING PLOETEST,R5
00023020  00023050                      1732+T4       DC    A(CODE4)            address of test code
00023024  0004                          1733+         DC    H'4'                test number
00023026  00                            1734+         DC    X'00'
00023027  63                            1735+         DC    HL1'99'             Function
00023028  63                            1736+         DC    HL1'99'             expected unequal results (UR)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00023029  00                                       1737+          DC    HL1'00'             found UR
0002302A  63                                       1738+          DC    HL1'99'             cc
0002302B  00                                       1739+          DC    HL1'00'             cc not checked
0002302C  00                                       1740+          DS    X                   extracted cc
00023030  00000000 00000000                        1741+          DS    2F                  extract PSW after test (has CC)
00023038  00023880                                 1742+          DC    A(FMSG4)            address of test fail message
0002303C  000236C0                                 1743+          DC    A(OPSRC4)           address of operand source
00023040  00023620                                 1744+          DC    A(OPDST4)           address of operand destination
00023044  00023760                                 1745+          DC    A(PL4)              address of parameter list
00023048  0000417C                                 1746+          DC    A(RANDX+4)          address of random source data
0002304C  000238F5                                 1747+          DC    A(TEND4)            address of test end eye catche
                                                   1748
                                                   1749           TSTCODE
00023050                                           1750+CODE4     DS    0F
                                                   1751
00023050  E310 501C 0014                 0000001C  1752           LGF   R1,OPSRCA           R1: OPSRC - Primary
00023056  B904 0061                                1753           LGR   R6,R1               R6: OPSRC - Secondary
0002305A  B904 0071                                1754           LGR   R7,R1               R7: OPSRC - Home
                                                   1755
0002305E  E320 5028 0014                 00000028  1756           LGF   R2,RANDXA           R2: address random data
00023064  E330 5024 0014                 00000024  1757           LGF   R3,PLA              R3: address PL
                                                   1758
0002306A  B219 0200                      00000200  1759           SAC   ARMODE
0002306E  D203 1000 2000       00000000  00000000  1760           MVC   0(4,R1),0(R2)       load Op1 = Primary
00023074  D203 6020 2020       00000020  00000020  1761           MVC   32(4,R6),32(R2)     load Op2 = Secondary
0002307A  D203 7040 2040       00000040  00000040  1762           MVC   64(4,R7),64(R2)     load Op3 = Home
00023080  D203 6060 2060       00000060  00000060  1763           MVC   96(4,R6),96(R2)     load Op4 = Secondary
00023086  B219 0000                      00000000  1764           SAC   PRIMODE
                                                   1765 *
                                                   1766 * Quadruple Load(QLX: 45`
                                                   1767 *
0002308A  E313 0000 0024                 00000000  1768           STG   R1,0(R3)            load OP1 address in PL
00023090  E54C 3020 0000                 00000020  1769           MVHI  32(R3),PRIALET
                                                   1770
00023096  4111 0020                      00000020  1771           LA    R1,32(R1)
0002309A  E313 0008 0024                 00000008  1772           STG   R1,8(R3)            load OP2 address in PL
000230A0  E54C 3024 0001                 00000024  1773           MVHI  36(R3),SECALET
                                                   1774
000230A6  4111 0020                      00000020  1775           LA    R1,32(R1)
000230AA  E313 0010 0024                 00000010  1776           STG   R1,16(R3)           load OP3 address in PL
000230B0  E54C 3028 0002                 00000028  1777           MVHI  40(R3),HOMEALET
                                                   1778
000230B6  4111 0020                      00000020  1779           LA    R1,32(R1)
000230BA  E313 0018 0024                 00000018  1780           STG   R1,24(R3)           load OP4 address in PL
000230C0  E54C 302C 0001                 0000002C  1781           MVHI  44(R3),SECALET
                                                   1782
000230C6  B982 0000                                1783           XGR   R0,R0
000230CA  4100 002D                      0000002D  1784           LA    R0,45
                                                   1785
000230CE  B219 0200                      00000200  1786           SAC   ARMODE
000230D2  EE0F 0000 3000       00000000  00000000  1787           PLO   0,0,R15,0(R3)       Quadruple Load(QL: 45)
000230D8  B219 0000                      00000000  1788           SAC   PRIMODE
                                                   1789 *
                                                   1790 * Quadruple Store(QST: 58)
                                                   1791 *
000230DC  E310 5020 0014                 00000020  1792           LGF   R1,OPDSTA           R1:  OPDST - Primary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000230E2  E330 5024 0014                 00000024  1793           LGF   R3,PLA
                                                   1794
000230E8  E313 0000 0024                 00000000  1795           STG   R1,0(R3)            load OP1 address in PL
000230EE  E54C 3020 0002                 00000020  1796           MVHI  32(R3),HOMEALET
                                                   1797
000230F4  4111 0020                      00000020  1798           LA    R1,32(R1)
000230F8  E313 0008 0024                 00000008  1799           STG   R1,8(R3)            load OP2 address in PL
000230FE  E54C 3024 0002                 00000024  1800           MVHI  36(R3),HOMEALET
                                                   1801
00023104  4111 0020                      00000020  1802           LA    R1,32(R1)
00023108  E313 0010 0024                 00000010  1803           STG   R1,16(R3)           load OP3 address in PL
0002310E  E54C 3028 0001                 00000028  1804           MVHI  40(R3),SECALET
                                                   1805
00023114  4111 0020                      00000020  1806           LA    R1,32(R1)
00023118  E313 0018 0024                 00000018  1807           STG   R1,24(R3)           load OP4 address in PL
0002311E  E54C 302C 0000                 0000002C  1808           MVHI  44(R3),PRIALET
                                                   1809
00023124  B982 0000                                1810           XGR   R0,R0
00023128  4100 003A                      0000003A  1811           LA    R0,58
                                                   1812
0002312C  B219 0200                      00000200  1813           SAC   ARMODE
00023130  EE0F 0000 3000      00000000   00000000  1814           PLO   0,0,R15,0(R3)       Quadruple Store(QST: 58)
00023136  B219 0000                      00000000  1815           SAC   PRIMODE
                                                   1816 *
                                                   1817 * compare results
                                                   1818 *                                  reset registers
0002313A  E310 501C 0014                 0000001C  1819           LGF   R1,OPSRCA           R1:  OPSRC - Primary
00023140  B904 0061                                1820           LGR   R6,R1               R6:  OPSRC - Secondary
00023144  B904 0071                                1821           LGR   R7,R1               R7:  OPSRC - Home
                                                   1822
00023148  E340 5020 0014                 00000020  1823           LGF   R4,OPDSTA           R4:  OPDST - Primary
0002314E  B904 00C4                                1824           LGR   R12,R4              R12: OPDST - Secondary
00023152  B904 00D4                                1825           LGR   R13,R4              R13: OPDST - Home
                                                   1826
00023156  B219 0200                      00000200  1827           SAC   ARMODE
0002315A  D503 1000 D000      00000000   00000000  1828           CLC   0(4,R1),0(R13)
00023160  4770 5166                      00023186  1829           BNE   F4558               OP1 Different, error
00023164  D503 6020 D020      00000020   00000020  1830           CLC   32(4,R6),32(R13)
0002316A  4770 5166                      00023186  1831           BNE   F4558               OP2 Different, error
0002316E  D503 7040 C040      00000040   00000040  1832           CLC   64(4,R7),64(R12)
00023174  4770 5166                      00023186  1833           BNE   F4558               OP3 Different, error
00023178  D503 6060 4060      00000060   00000060  1834           CLC   96(4,R6),96(R4)
0002317E  B219 0000                      00000000  1835           SAC   PRIMODE
00023182  478B 0000                      00000000  1836           BE    0(R11)              OP4 same, return
                                                   1837
                              00023186   00000001  1838 F4558     EQU   *
00023186  B219 0000                      00000000  1839           SAC   PRIMODE
0002318A  45F0 8450                      00002450  1840           BAL   R15,FAILMSG
0002318E  07FB                                     1841           BR    R11
                                                   1842
                                                   1843           TSTDATA
00023190                      00023190   00023600  1844+          ORG   PLOETST+(128*K)+4*(3*K)+(K+512)
                                                   1845           TSTLBL OPDST              Operand Destination (Results)
00023600  D7D9C940 40406E6E                        1846+          DC    CL32'PRI   >>>OPDST4<<<'  eye catcher
00023608  6ED6D7C4 E2E3F44C
00023610  4C4C4040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00023618  40404040 40404040
00023620                                             1847+OPDST4   DS    0C
00023620  00000000 00000000                          1848          DC    XL128'0'
00023628  00000000 00000000
00023630  00000000 00000000
00023638  00000000 00000000
00023640  00000000 00000000
00023648  00000000 00000000
00023650  00000000 00000000
00023658  00000000 00000000
00023660  00000000 00000000
00023668  00000000 00000000
00023670  00000000 00000000
00023678  00000000 00000000
00023680  00000000 00000000
00023688  00000000 00000000
00023690  00000000 00000000
00023698  00000000 00000000
                                                     1849
                                                     1850          TSTLBL OPSRC              Operand Source
000236A0  D7D9C940 40406E6E                          1851+         DC    CL32'PRI   >>>OPSRC4<<<'  eye catcher
000236A8  6ED6D7E2 D9C3F44C
000236B0  4C4C4040 40404040
000236B8  40404040 40404040
000236C0                                             1852+OPSRC4   DS    0C
000236C0  00000000 00000000                          1853          DC    XL128'0'
000236C8  00000000 00000000
000236D0  00000000 00000000
000236D8  00000000 00000000
000236E0  00000000 00000000
000236E8  00000000 00000000
000236F0  00000000 00000000
000236F8  00000000 00000000
00023700  00000000 00000000
00023708  00000000 00000000
00023710  00000000 00000000
00023718  00000000 00000000
00023720  00000000 00000000
00023728  00000000 00000000
00023730  00000000 00000000
00023738  00000000 00000000
                                                     1854
                                                     1855          TSTLBL PL                 Parameter List
00023740  D7D9C940 40406E6E                          1856+         DC    CL32'PRI   >>>PL4<<<'     eye catcher
00023748  6ED7D3F4 4C4C4C40
00023750  40404040 40404040
00023758  40404040 40404040
00023760                                             1857+PL4      DS    0C
00023760  00000000 00000000                          1858          DC    XL256'00'
00023768  00000000 00000000
00023770  00000000 00000000
00023778  00000000 00000000
00023780  00000000 00000000
00023788  00000000 00000000
00023790  00000000 00000000
00023798  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000237A0  00000000 00000000
000237A8  00000000 00000000
000237B0  00000000 00000000
000237B8  00000000 00000000
000237C0  00000000 00000000
000237C8  00000000 00000000
000237D0  00000000 00000000
000237D8  00000000 00000000
000237E0  00000000 00000000
000237E8  00000000 00000000
000237F0  00000000 00000000
000237F8  00000000 00000000
00023800  00000000 00000000
00023808  00000000 00000000
00023810  00000000 00000000
00023818  00000000 00000000
00023820  00000000 00000000
00023828  00000000 00000000
00023830  00000000 00000000
00023838  00000000 00000000
00023840  00000000 00000000
00023848  00000000 00000000
00023850  00000000 00000000
00023858  00000000 00000000
                                          1859
                                          1860          TSTLBL FMSG                fail message
00023860  D7D9C940 40406E6E               1861+         DC     CL32'PRI   >>>FMSG4<<<'   eye catcher
00023868  6EC6D4E2 C7F44C4C
00023870  4C404040 40404040
00023878  40404040 40404040
00023880                                  1862+FMSG4    DS     0C
00023880  86818993 858440D7               1863          DC     C'failed PLO-Extension '
00023888  D3D660C5 A7A38595
00023890  A2899695 40
00023895  D8A48184 99A49793               1864          DC     CL96'Quadruple Load(QL: 45) / Store(QST: 58)'
0002389D  8540D396 81844DD8
000238A5  D37A40F4 F55D4061
000238AD  40E2A396 99854DD8
000238B5  E2E37A40 F5F85D40
000238BD  40404040 40404040
000238C5  40404040 40404040
000238CD  40404040 40404040
000238D5  40404040 40404040
000238DD  40404040 40404040
000238E5  40404040 40404040
000238ED  40404040 40404040
                                          1865
                                          1866          TSTEND
000238F5  D7D9C940 4040C5D5               1867+TEND4    DC     CL32'PRI   END of test: 4 <<<'
000238FD  C4409686 40A385A2
00023905  A37A40F4 404C4C4C
0002390D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1869 *=============================================================================
                                                   1870
                                                   1871 * ---------------------------------------------------------------------
                                                   1872 * Test Triple Load(TLO: 44) / Store(TSTO: 57) instructions
                                                   1873 * ---------------------------------------------------------------------
                                                   1874          TSTHDR 99,99,99
00023915                         00023915  00023C00 1875+         ORG   PLOETST+(128*K)+5*(3*K)
00023C00  D7D9C940 4040E2E3                        1876+         DC    CL32'PRI   START of test: 5'
00023C08  C1D9E340 968640A3
00023C10  85A2A37A 40F54040
00023C18  40404040 40404040
00023C20                         00023C20          1877+         USING *,R5                base for test data and test routine
00023C20                         00000000          1878+         USING PLOETEST,R5
00023C20  00023C50                                 1879+T5       DC    A(CODE5)            address of test code
00023C24  0005                                     1880+         DC    H'5'                test number
00023C26  00                                       1881+         DC    X'00'
00023C27  63                                       1882+         DC    HL1'99'             Function
00023C28  63                                       1883+         DC    HL1'99'             expected unequal results (UR)
00023C29  00                                       1884+         DC    HL1'00'             found UR
00023C2A  63                                       1885+         DC    HL1'99'             cc
00023C2B  00                                       1886+         DC    HL1'00'             cc not checked
00023C2C  00                                       1887+         DS    X                   extracted cc
00023C30  00000000 00000000                        1888+         DS    2F                  extract PSW after test (has CC)
00023C38  00024480                                 1889+         DC    A(FMSG5)            address of test fail message
00023C3C  000242C0                                 1890+         DC    A(OPSRC5)           address of operand source
00023C40  00024220                                 1891+         DC    A(OPDST5)           address of operand destination
00023C44  00024360                                 1892+         DC    A(PL5)              address of parameter list
00023C48  0000417D                                 1893+         DC    A(RANDX+5)          address of random source data
00023C4C  000244F5                                 1894+         DC    A(TEND5)            address of test end eye catche
                                                   1895
                                                   1896          TSTCODE
00023C50                                           1897+CODE5    DS    0F
                                                   1898
00023C50  E310 501C 0014                 0000001C  1899          LGF   R1,OPSRCA           R1: OPSRC - Primary
00023C56  B904 0061                                1900          LGR   R6,R1               R6: OPSRC - Secondary
00023C5A  B904 0071                                1901          LGR   R7,R1               R7: OPSRC - Home
                                                   1902
00023C5E  E320 5028 0014                 00000028  1903          LGF   R2,RANDXA           R2: address random data
00023C64  E330 5024 0014                 00000024  1904          LGF   R3,PLA              R3: address PL
                                                   1905
00023C6A  B219 0200                      00000200  1906          SAC   ARMODE
00023C6E  D21F 1000 2000       00000000  00000000  1907          MVC   0(32,R1),0(R2)      load Op1 = Primary
00023C74  D21F 6020 2020       00000020  00000020  1908          MVC   32(32,R6),32(R2)    load Op2 = Secondary
00023C7A  D21F 7040 2040       00000040  00000040  1909          MVC   64(32,R7),64(R2)    load Op3 = Home
00023C80  B219 0000                      00000000  1910          SAC   PRIMODE
                                                   1911 *
                                                   1912 * Triple Load(TLO: 44)
                                                   1913 *
00023C84  E313 0000 0024                 00000000  1914          STG   R1,0(R3)            load OP1 address in PL
00023C8A  E54C 3020 0000                 00000020  1915          MVHI  32(R3),PRIALET
                                                   1916
00023C90  4111 0020                      00000020  1917          LA    R1,32(R1)
00023C94  E313 0008 0024                 00000008  1918          STG   R1,8(R3)            load OP2 address in PL
00023C9A  E54C 3024 0001                 00000024  1919          MVHI  36(R3),SECALET
                                                   1920
00023CA0  4111 0020                      00000020  1921          LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00023CA4  E313 0010 0024                 00000010  1922          STG   R1,16(R3)           load OP3 address in PL
00023CAA  E54C 3028 0002                 00000028  1923          MVHI  40(R3),HOMEALET
                                                   1924
00023CB0  B982 0000                              1925          XGR   R0,R0
00023CB4  4100 002C                      0000002C  1926          LA    R0,44
                                                   1927
00023CB8  B219 0200                      00000200  1928          SAC   ARMODE
00023CBC  EE0F 0000 3000      00000000   00000000  1929          PLO   0,0,R15,0(R3)       Triple Load(TLO: 44)
00023CC2  B219 0000                      00000000  1930          SAC   PRIMODE
                                                   1931 *
                                                   1932 * Triple Store(TSTO: 57)
                                                   1933 *
00023CC6  E310 5020 0014                 00000020  1934          LGF   R1,OPDSTA           R1:  OPDST - Primary
00023CCC  E330 5024 0014                 00000024  1935          LGF   R3,PLA
                                                   1936
00023CD2  E313 0000 0024                 00000000  1937          STG   R1,0(R3)            load OP1 address in PL
00023CD8  E54C 3020 0002                 00000020  1938          MVHI  32(R3),HOMEALET
                                                   1939
00023CDE  4111 0020                      00000020  1940          LA    R1,32(R1)
00023CE2  E313 0008 0024                 00000008  1941          STG   R1,8(R3)            load OP2 address in PL
00023CE8  E54C 3024 0002                 00000024  1942          MVHI  36(R3),HOMEALET
                                                   1943
00023CEE  4111 0020                      00000020  1944          LA    R1,32(R1)
00023CF2  E313 0010 0024                 00000010  1945          STG   R1,16(R3)           load OP3 address in PL
00023CF8  E54C 3028 0001                 00000028  1946          MVHI  40(R3),SECALET
00023CFE  B982 0000                              1947          XGR   R0,R0
00023D02  4100 0039                      00000039  1948          LA    R0,57
                                                   1949
00023D06  B219 0200                      00000200  1950          SAC   ARMODE
00023D0A  EE0F 0000 3000      00000000   00000000  1951          PLO   0,0,R15,0(R3)       Triple Store(TSTO: 57)
00023D10  B219 0000                      00000000  1952          SAC   PRIMODE
                                                   1953 *
                                                   1954 * compare results
                                                   1955 *                                  reset registers
00023D14  E310 501C 0014                 0000001C  1956          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00023D1A  B904 0061                              1957          LGR   R6,R1               R6:  OPSRC - Secondary
00023D1E  B904 0071                              1958          LGR   R7,R1               R7:  OPSRC - Home
                                                   1959
00023D22  E340 5020 0014                 00000020  1960          LGF   R4,OPDSTA           R4:  OPDST - Primary
00023D28  B904 00C4                              1961          LGR   R12,R4              R12: OPDST - Secondary
00023D2C  B904 00D4                              1962          LGR   R13,R4              R13: OPDST - Home
                                                   1963
00023D30  B219 0200                      00000200  1964          SAC   ARMODE
00023D34  D51F 1000 D000      00000000   00000000  1965          CLC   0(32,R1),0(R13)
00023D3A  4770 5136                      00023D56  1966          BNE   F4457               OP1 Different, error
00023D3E  D51F 6020 D020      00000020   00000020  1967          CLC   32(32,R6),32(R13)
00023D44  4770 5136                      00023D56  1968          BNE   F4457               OP2 Different, error
00023D48  D51F 7040 C040      00000040   00000040  1969          CLC   64(32,R7),64(R12)
00023D4E  B219 0000                      00000000  1970          SAC   PRIMODE
00023D52  478B 0000                      00000000  1971          BE    0(R11)              OP3 same, return
                                                   1972
                                00023D56  00000001  1973 F4457     EQU   *
00023D56  B219 0000                      00000000  1974          SAC   PRIMODE
00023D5A  45F0 8450                      00002450  1975          BAL   R15,FAILMSG
00023D5E  07FB                                   1976          BR    R11
                                                   1977
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   1978           TSTDATA
00023D60                      00023D60  00024200  1979+          ORG   PLOETST+(128*K)+5*(3*K)+(K+512)
                                                   1980           TSTLBL OPDST              Operand Destination (Results)
00024200  D7D9C940 40406E6E                        1981+          DC    CL32'PRI   >>>OPDST5<<<'  eye catcher
00024208  6ED6D7C4 E2E3F54C
00024210  4C4C4040 40404040
00024218  40404040 40404040
00024220                                           1982+OPDST5    DS    0C
00024220  00000000 00000000                        1983           DC    XL128'0'
00024228  00000000 00000000
00024230  00000000 00000000
00024238  00000000 00000000
00024240  00000000 00000000
00024248  00000000 00000000
00024250  00000000 00000000
00024258  00000000 00000000
00024260  00000000 00000000
00024268  00000000 00000000
00024270  00000000 00000000
00024278  00000000 00000000
00024280  00000000 00000000
00024288  00000000 00000000
00024290  00000000 00000000
00024298  00000000 00000000
                                                   1984
                                                   1985           TSTLBL OPSRC              Operand Source
000242A0  D7D9C940 40406E6E                        1986+          DC    CL32'PRI   >>>OPSRC5<<<'  eye catcher
000242A8  6ED6D7E2 D9C3F54C
000242B0  4C4C4040 40404040
000242B8  40404040 40404040
000242C0                                           1987+OPSRC5    DS    0C
000242C0  00000000 00000000                        1988           DC    XL128'0'
000242C8  00000000 00000000
000242D0  00000000 00000000
000242D8  00000000 00000000
000242E0  00000000 00000000
000242E8  00000000 00000000
000242F0  00000000 00000000
000242F8  00000000 00000000
00024300  00000000 00000000
00024308  00000000 00000000
00024310  00000000 00000000
00024318  00000000 00000000
00024320  00000000 00000000
00024328  00000000 00000000
00024330  00000000 00000000
00024338  00000000 00000000
                                                   1989
                                                   1990           TSTLBL PL                 Parameter List
00024340  D7D9C940 40406E6E                        1991+          DC    CL32'PRI   >>>PL5<<<'     eye catcher
00024348  6ED7D3F5 4C4C4C40
00024350  40404040 40404040
00024358  40404040 40404040
00024360                                           1992+PL5       DS    0C
00024360  00000000 00000000                        1993           DC    XL256'00'
00024368  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00024370  00000000 00000000
00024378  00000000 00000000
00024380  00000000 00000000
00024388  00000000 00000000
00024390  00000000 00000000
00024398  00000000 00000000
000243A0  00000000 00000000
000243A8  00000000 00000000
000243B0  00000000 00000000
000243B8  00000000 00000000
000243C0  00000000 00000000
000243C8  00000000 00000000
000243D0  00000000 00000000
000243D8  00000000 00000000
000243E0  00000000 00000000
000243E8  00000000 00000000
000243F0  00000000 00000000
000243F8  00000000 00000000
00024400  00000000 00000000
00024408  00000000 00000000
00024410  00000000 00000000
00024418  00000000 00000000
00024420  00000000 00000000
00024428  00000000 00000000
00024430  00000000 00000000
00024438  00000000 00000000
00024440  00000000 00000000
00024448  00000000 00000000
00024450  00000000 00000000
00024458  00000000 00000000
                                      1994
                                      1995          TSTLBL FMSG               fail message
00024460  D7D9C940 40406E6E           1996+         DC    CL32'PRI   >>>FMSG5<<<'   eye catcher
00024468  6EC6D4E2 C7F54C4C
00024470  4C404040 40404040
00024478  40404040 40404040
00024480                              1997+FMSG5    DS    0C
00024480  86818993 858440D7           1998          DC    C'failed PLO-Extension '
00024488  D3D660C5 A7A38595
00024490  A2899695 40
00024495  E3998997 938540D3           1999          DC    CL96'Triple Load(TLO: 44) / Store(TSTO: 57)'
0002449D  9681844D E3D3D67A
000244A5  40F4F45D 406140E2
000244AD  A3969985 4DE3E2E3
000244B5  D67A40F5 F75D4040
000244BD  40404040 40404040
000244C5  40404040 40404040
000244CD  40404040 40404040
000244D5  40404040 40404040
000244DD  40404040 40404040
000244E5  40404040 40404040
000244ED  40404040 40404040
                                      2000
                                      2001          TSTEND
000244F5  D7D9C940 4040C5D5           2002+TEND5    DC    CL32'PRI   END of test: 5 <<<'
000244FD  C4409686 40A385A2
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00024505  A37A40F5 404C4C4C
0002450D  40404040 40404040
                                          2003
                                          2004 * ------------------------------------------------------------------
                                          2005 * Test Triple Load(TLX: 43) / Store(TSTX: 56) instructions
                                          2006 * ------------------------------------------------------------------
                                          2007          TSTHDR 99,99,99
00024515                    00024515  00024800  2008+         ORG   PLOETST+(128*K)+6*(3*K)
00024800  D7D9C940 4040E2E3               2009+         DC    CL32'PRI   START of test: 6'
00024808  C1D9E340 968640A3
00024810  85A2A37A 40F64040
00024818  40404040 40404040
00024820                    00024820      2010+         USING *,R5                base for test data and test routine
00024820                    00000000      2011+         USING PLOETEST,R5
00024820  00024850                        2012+T6       DC    A(CODE6)            address of test code
00024824  0006                            2013+         DC    H'6'                test number
00024826  00                              2014+         DC    X'00'
00024827  63                              2015+         DC    HL1'99'             Function
00024828  63                              2016+         DC    HL1'99'             expected unequal results (UR)
00024829  00                              2017+         DC    HL1'00'             found UR
0002482A  63                              2018+         DC    HL1'99'             cc
0002482B  00                              2019+         DC    HL1'00'             cc not checked
0002482C  00                              2020+         DS    X                   extracted cc
00024830  00000000 00000000               2021+         DS    2F                  extract PSW after test (has CC)
00024838  00025080                        2022+         DC    A(FMSG6)            address of test fail message
0002483C  00024EC0                        2023+         DC    A(OPSRC6)           address of operand source
00024840  00024E20                        2024+         DC    A(OPDST6)           address of operand destination
00024844  00024F60                        2025+         DC    A(PL6)              address of parameter list
00024848  0000417E                        2026+         DC    A(RANDX+6)          address of random source data
0002484C  000250F5                        2027+         DC    A(TEND6)            address of test end eye catche
                                          2028
                                          2029          TSTCODE
00024850                                  2030+CODE6    DS    0F
                                          2031
00024850  E310 501C 0014        0000001C  2032          LGF   R1,OPSRCA           R1: OPSRC - Primary
00024856  B904 0061                       2033          LGR   R6,R1               R6: OPSRC - Secondary
0002485A  B904 0071                       2034          LGR   R7,R1               R7: OPSRC - Home
                                          2035
0002485E  E320 5028 0014        00000028  2036          LGF   R2,RANDXA           R2: address random data
00024864  E330 5024 0014        00000024  2037          LGF   R3,PLA              R3: address PL
                                          2038
0002486A  B219 0200             00000200  2039          SAC   ARMODE
0002486E  D20F 1000 2000  00000000 00000000 2040          MVC   0(16,R1),0(R2)      load Op1 = Primary
00024874  D20F 6020 2020  00000020 00000020 2041          MVC   32(16,R6),32(R2)    load Op2 = Secondary
0002487A  D20F 7040 2040  00000040 00000040 2042          MVC   64(16,R7),64(R2)    load Op3 = Home
00024880  B219 0000             00000000  2043          SAC   PRIMODE
                                          2044 *
                                          2045 * Triple Load(TLX: 43)
                                          2046 *
00024884  E313 0000 0024        00000000  2047          STG   R1,0(R3)            load OP1 address in PL
0002488A  E54C 3020 0000        00000020  2048          MVHI  32(R3),PRIALET
                                          2049
00024890  4111 0020             00000020  2050          LA    R1,32(R1)
00024894  E313 0008 0024        00000008  2051          STG   R1,8(R3)            load OP2 address in PL
0002489A  E54C 3024 0001        00000024  2052          MVHI  36(R3),SECALET
                                          2053
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000248A0  4111 0020                    00000020  2054          LA    R1,32(R1)
000248A4  E313 0010 0024               00000010  2055          STG   R1,16(R3)          load OP3 address in PL
000248AA  E54C 3028 0002               00000028  2056          MVHI  40(R3),HOMEALET
000248B0  B982 0000                              2057          XGR   R0,R0
000248B4  4100 002B                    0000002B  2058          LA    R0,43
                                                 2059
000248B8  B219 0200                    00000200  2060          SAC   ARMODE
000248BC  EE0F 0000 3000      00000000 00000000  2061          PLO   0,0,R15,0(R3)       Triple Load(TLX: 43)
000248C2  B219 0000                    00000000  2062          SAC   PRIMODE
                                                 2063 *
                                                 2064 * Triple Store(TSTX: 56)
                                                 2065 *
000248C6  E310 5020 0014               00000020  2066          LGF   R1,OPDSTA           R1:  OPDST - Primary
000248CC  E330 5024 0014               00000024  2067          LGF   R3,PLA
                                                 2068
000248D2  E313 0000 0024               00000000  2069          STG   R1,0(R3)            load OP1 address in PL
000248D8  E54C 3020 0002               00000020  2070          MVHI  32(R3),HOMEALET
                                                 2071
000248DE  4111 0020                    00000020  2072          LA    R1,32(R1)
000248E2  E313 0008 0024               00000008  2073          STG   R1,8(R3)            load OP2 address in PL
000248E8  E54C 3024 0002               00000024  2074          MVHI  36(R3),HOMEALET
                                                 2075
000248EE  4111 0020                    00000020  2076          LA    R1,32(R1)
000248F2  E313 0010 0024               00000010  2077          STG   R1,16(R3)           load OP3 address in PL
000248F8  E54C 3028 0001               00000028  2078          MVHI  40(R3),SECALET
                                                 2079
000248FE  B982 0000                              2080          XGR   R0,R0
00024902  4100 0038                    00000038  2081          LA    R0,56
                                                 2082
00024906  B219 0200                    00000200  2083          SAC   ARMODE
0002490A  EE0F 0000 3000      00000000 00000000  2084          PLO   0,0,R15,0(R3)       Triple Store(TSTX: 56)
00024910  B219 0000                    00000000  2085          SAC   PRIMODE
                                                 2086 *
                                                 2087 * compare results
                                                 2088 *
                                                 2089 *                                  reset registers
00024914  E310 501C 0014               0000001C  2090          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002491A  B904 0061                              2091          LGR   R6,R1               R6:  OPSRC - Secondary
0002491E  B904 0071                              2092          LGR   R7,R1               R7:  OPSRC - Home
                                                 2093
00024922  E340 5020 0014               00000020  2094          LGF   R4,OPDSTA           R4:  OPDST - Primary
00024928  B904 00C4                              2095          LGR   R12,R4              R12: OPDST - Secondary
0002492C  B904 00D4                              2096          LGR   R13,R4              R13: OPDST - Home
                                                 2097
00024930  B219 0200                    00000200  2098          SAC   ARMODE
00024934  D50F 1000 D000      00000000 00000000  2099          CLC   0(16,R1),0(R13)
0002493A  4770 5136                    00024956  2100          BNE   F4356               OP1 Different, error
0002493E  D50F 6020 D020      00000020 00000020  2101          CLC   32(16,R6),32(R13)
00024944  4770 5136                    00024956  2102          BNE   F4356               OP2 Different, error
00024948  D50F 7040 C040      00000040 00000040  2103          CLC   64(16,R7),64(R12)
0002494E  B219 0000                    00000000  2104          SAC   PRIMODE
00024952  478B 0000                    00000000  2105          BE    0(R11)              OP3 same, return
                                                 2106
                                                 2107
                              00024956 00000001  2108 F4356    EQU   *
00024956  B219 0000                    00000000  2109          SAC   PRIMODE
```

```
  LOC         OBJECT CODE        ADDR1     ADDR2     STMT
0002495A  45F0 8450                        00002450  2110           BAL   R15,FAILMSG
0002495E  07FB                             00000000  2111           BR    R11
                                           00000000  2112
                                           00000000  2113           TSTDATA
00024960                       00024960    00024E00  2114+          ORG   PLOETST+(128*K)+6*(3*K)+(K+512)
                                           00000000  2115           TSTLBL OPDST              Operand Destination (Results)
00024E00  D7D9C940 40406E6E                00000000  2116+          DC    CL32'PRI   >>>OPDST6<<<'  eye catcher
00024E08  6ED6D7C4 E2E3F64C                00000000
00024E10  4C4C4040 40404040                00000000
00024E18  40404040 40404040                00000000
00024E20                                   00000000  2117+OPDST6    DS    0C
00024E20  00000000 00000000                00000000  2118           DC    XL128'0'
00024E28  00000000 00000000                00000000
00024E30  00000000 00000000                00000000
00024E38  00000000 00000000                00000000
00024E40  00000000 00000000                00000000
00024E48  00000000 00000000                00000000
00024E50  00000000 00000000                00000000
00024E58  00000000 00000000                00000000
00024E60  00000000 00000000                00000000
00024E68  00000000 00000000                00000000
00024E70  00000000 00000000                00000000
00024E78  00000000 00000000                00000000
00024E80  00000000 00000000                00000000
00024E88  00000000 00000000                00000000
00024E90  00000000 00000000                00000000
00024E98  00000000 00000000                00000000
                                           00000000  2119
                                           00000000  2120           TSTLBL OPSRC              Operand Source
00024EA0  D7D9C940 40406E6E                00000000  2121+          DC    CL32'PRI   >>>OPSRC6<<<'  eye catcher
00024EA8  6ED6D7E2 D9C3F64C                00000000
00024EB0  4C4C4040 40404040                00000000
00024EB8  40404040 40404040                00000000
00024EC0                                   00000000  2122+OPSRC6    DS    0C
00024EC0  00000000 00000000                00000000  2123           DC    XL128'0'
00024EC8  00000000 00000000                00000000
00024ED0  00000000 00000000                00000000
00024ED8  00000000 00000000                00000000
00024EE0  00000000 00000000                00000000
00024EE8  00000000 00000000                00000000
00024EF0  00000000 00000000                00000000
00024EF8  00000000 00000000                00000000
00024F00  00000000 00000000                00000000
00024F08  00000000 00000000                00000000
00024F10  00000000 00000000                00000000
00024F18  00000000 00000000                00000000
00024F20  00000000 00000000                00000000
00024F28  00000000 00000000                00000000
00024F30  00000000 00000000                00000000
00024F38  00000000 00000000                00000000
                                           00000000  2124
                                           00000000  2125           TSTLBL PL                 Parameter List
00024F40  D7D9C940 40406E6E                00000000  2126+          DC    CL32'PRI   >>>PL6<<<'     eye catcher
00024F48  6ED7D3F6 4C4C4C40                00000000
00024F50  40404040 40404040                00000000
00024F58  40404040 40404040                00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00024F60                                             2127+PL6      DS    0C
00024F60  00000000 00000000                          2128          DC    XL256'00'
00024F68  00000000 00000000
00024F70  00000000 00000000
00024F78  00000000 00000000
00024F80  00000000 00000000
00024F88  00000000 00000000
00024F90  00000000 00000000
00024F98  00000000 00000000
00024FA0  00000000 00000000
00024FA8  00000000 00000000
00024FB0  00000000 00000000
00024FB8  00000000 00000000
00024FC0  00000000 00000000
00024FC8  00000000 00000000
00024FD0  00000000 00000000
00024FD8  00000000 00000000
00024FE0  00000000 00000000
00024FE8  00000000 00000000
00024FF0  00000000 00000000
00024FF8  00000000 00000000
00025000  00000000 00000000
00025008  00000000 00000000
00025010  00000000 00000000
00025018  00000000 00000000
00025020  00000000 00000000
00025028  00000000 00000000
00025030  00000000 00000000
00025038  00000000 00000000
00025040  00000000 00000000
00025048  00000000 00000000
00025050  00000000 00000000
00025058  00000000 00000000
                                                     2129
                                                     2130          TSTLBL FMSG               fail message
00025060  D7D9C940 40406E6E                          2131+         DC    CL32'PRI   >>>FMSG6<<<'   eye catcher
00025068  6EC6D4E2 C7F64C4C
00025070  4C404040 40404040
00025078  40404040 40404040
00025080                                             2132+FMSG6    DS    0C
00025080  86818993 858440D7                          2133          DC    C'failed PLO-Extension '
00025088  D3D660C5 A7A38595
00025090  A2899695 40
00025095  E3998997 938540D3                          2134          DC    CL96'Triple Load(TLX: 43) / Store(TSTX: 56)'
0002509D  9681844D E3D3E77A
000250A5  40F4F35D 406140E2
000250AD  A3969985 4DE3E2E3
000250B5  E77A40F5 F65D4040
000250BD  40404040 40404040
000250C5  40404040 40404040
000250CD  40404040 40404040
000250D5  40404040 40404040
000250DD  40404040 40404040
000250E5  40404040 40404040
000250ED  40404040 40404040
                                                     2135
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2136          TSTEND
000250F5  D7D9C940 4040C5D5                         2137+TEND6    DC    CL32'PRI   END of test: 6 <<<'
000250FD  C4409686 40A385A2
00025105  A37A40F6 404C4C4C
0002510D  40404040 40404040
                                                    2138
                                                    2139 * ------------------------------------------------------------------------
                                                    2140 * Test Triple Load(TLG: 42) / Store(TSTG: 55) instructions
                                                    2141 * ------------------------------------------------------------------------
                                                    2142          TSTHDR 99,99,99
00025115                          00025115  00025400  2143+          ORG   PLOETST+(128*K)+7*(3*K)
00025400  D7D9C940 4040E2E3                         2144+          DC    CL32'PRI   START of test: 7'
00025408  C1D9E340 968640A3
00025410  85A2A37A 40F74040
00025418  40404040 40404040
00025420                          00025420          2145+          USING *,R5                base for test data and test routine
00025420                          00000000          2146+          USING PLOETEST,R5
00025420  00025450                                  2147+T7        DC    A(CODE7)            address of test code
00025424  0007                                      2148+          DC    H'7'                test number
00025426  00                                        2149+          DC    X'00'
00025427  63                                        2150+          DC    HL1'99'             Function
00025428  63                                        2151+          DC    HL1'99'             expected unequal results (UR)
00025429  00                                        2152+          DC    HL1'00'             found UR
0002542A  63                                        2153+          DC    HL1'99'             cc
0002542B  00                                        2154+          DC    HL1'00'             cc not checked
0002542C  00                                        2155+          DS    X                   extracted cc
00025430  00000000 00000000                         2156+          DS    2F                  extract PSW after test (has CC)
00025438  00025C80                                  2157+          DC    A(FMSG7)            address of test fail message
0002543C  00025AC0                                  2158+          DC    A(OPSRC7)           address of operand source
00025440  00025A20                                  2159+          DC    A(OPDST7)           address of operand destination
00025444  00025B60                                  2160+          DC    A(PL7)              address of parameter list
00025448  0000417F                                  2161+          DC    A(RANDX+7)          address of random source data
0002544C  00025CF5                                  2162+          DC    A(TEND7)            address of test end eye catche
                                                    2163
                                                    2164          TSTCODE
00025450                                            2165+CODE7    DS    0F
                                                    2166
00025450  E310 501C 0014                  0000001C  2167          LGF   R1,OPSRCA           R1: OPSRC - Primary
00025456  B904 0061                               2168          LGR   R6,R1               R6: OPSRC - Secondary
0002545A  B904 0071                               2169          LGR   R7,R1               R7: OPSRC - Home
                                                    2170
0002545E  E320 5028 0014                  00000028  2171          LGF   R2,RANDXA           R2: address random data
00025464  E330 5024 0014                  00000024  2172          LGF   R3,PLA              R3: address PL
                                                    2173
0002546A  B219 0200                       00000200  2174          SAC   ARMODE
0002546E  D207 1000 2000        00000000  00000000  2175          MVC   0(8,R1),0(R2)       load Op1 = Primary
00025474  D207 6020 2020        00000020  00000020  2176          MVC   32(8,R6),32(R2)     load Op2 = Secondary
0002547A  D207 7040 2040        00000040  00000040  2177          MVC   64(8,R7),64(R2)     load Op3 = Home
00025480  B219 0000                       00000000  2178          SAC   PRIMODE
                                                    2179 *
                                                    2180 * Triple Load(TLG: 42)
                                                    2181 *
00025484  E313 0000 0024                  00000000  2182          STG   R1,0(R3)            load OP1 address in PL
0002548A  E54C 3020 0000                  00000020  2183          MVHI  32(R3),PRIALET
                                                    2184
00025490  4111 0020                       00000020  2185          LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00025494  E313 0008 0024                 00000008  2186          STG   R1,8(R3)            load OP2 address in PL
0002549A  E54C 3024 0001                 00000024  2187          MVHI  36(R3),SECALET
                                                   2188
000254A0  4111 0020                      00000020  2189          LA    R1,32(R1)
000254A4  E313 0010 0024                 00000010  2190          STG   R1,16(R3)           load OP3 address in PL
000254AA  E54C 3028 0002                 00000028  2191          MVHI  40(R3),HOMEALET
                                                   2192
000254B0  B982 0000                              0  2193          XGR   R0,R0
000254B4  4100 002A                      0000002A  2194          LA    R0,42
                                                   2195
000254B8  B219 0200                      00000200  2196          SAC   ARMODE
000254BC  EE0F 0000 3000      00000000   00000000  2197          PLO   0,0,R15,0(R3)       Triple Load(TLG: 42)
000254C2  B219 0000                      00000000  2198          SAC   PRIMODE
                                                   2199 *
                                                   2200 * Triple Store(TSTG: 55)
                                                   2201 *
000254C6  E310 5020 0014                 00000020  2202          LGF   R1,OPDSTA           R1:  OPDST - Primary
000254CC  E330 5024 0014                 00000024  2203          LGF   R3,PLA
                                                   2204
000254D2  E313 0000 0024                 00000000  2205          STG   R1,0(R3)            load OP1 address in PL
000254D8  E54C 3020 0002                 00000020  2206          MVHI  32(R3),HOMEALET
                                                   2207
000254DE  4111 0020                      00000020  2208          LA    R1,32(R1)
000254E2  E313 0008 0024                 00000008  2209          STG   R1,8(R3)            load OP2 address in PL
000254E8  E54C 3024 0002                 00000024  2210          MVHI  36(R3),HOMEALET
                                                   2211
000254EE  4111 0020                      00000020  2212          LA    R1,32(R1)
000254F2  E313 0010 0024                 00000010  2213          STG   R1,16(R3)           load OP3 address in PL
000254F8  E54C 3028 0001                 00000028  2214          MVHI  40(R3),SECALET
                                                   2215
000254FE  B982 0000                              0  2216          XGR   R0,R0
00025502  4100 0037                      00000037  2217          LA    R0,55
                                                   2218
00025506  B219 0200                      00000200  2219          SAC   ARMODE
0002550A  EE0F 0000 3000      00000000   00000000  2220          PLO   0,0,R15,0(R3)       Triple Store(TSTG: 55)
00025510  B219 0000                      00000000  2221          SAC   PRIMODE
                                                   2222 *
                                                   2223 * compare results
                                                   2224 *                                  reset registers
00025514  E310 501C 0014                 0000001C  2225          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002551A  B904 0061                              1  2226          LGR   R6,R1               R6:  OPSRC - Secondary
0002551E  B904 0071                              1  2227          LGR   R7,R1               R7:  OPSRC - Home
                                                   2228
00025522  E340 5020 0014                 00000020  2229          LGF   R4,OPDSTA           R4:  OPDST - Primary
00025528  B904 00C4                              4  2230          LGR   R12,R4              R12: OPDST - Secondary
0002552C  B904 00D4                              4  2231          LGR   R13,R4              R13: OPDST - Home
                                                   2232
00025530  B219 0200                      00000200  2233          SAC   ARMODE
00025534  D507 1000 D000      00000000   00000000  2234          CLC   0(8,R1),0(R13)
0002553A  4770 5136                      00025556  2235          BNE   F4255               OP1 Different, error
0002553E  D507 6020 D020      00000020   00000020  2236          CLC   32(8,R6),32(R13)
00025544  4770 5136                      00025556  2237          BNE   F4255               OP2 Different, error
00025548  D507 7040 C040      00000040   00000040  2238          CLC   64(8,R7),64(R12)
0002554E  B219 0000                      00000000  2239          SAC   PRIMODE
00025552  478B 0000                      00000000  2240          BE    0(R11)              OP3 same, return
                                                   2241
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               00025556  00000001  2242 F4255    EQU   *
00025556  B219 0000                      00000000  2243          SAC   PRIMODE
0002555A  45F0 8450                      00002450  2244          BAL   R15,FAILMSG
0002555E  07FB                                     2245          BR    R11
                                                   2246
                                                   2247          TSTDATA
00025560                       00025560  00025A00  2248+         ORG   PLOETST+(128*K)+7*(3*K)+(K+512)
                                                   2249          TSTLBL OPDST              Operand Destination (Results)
00025A00  D7D9C940 40406E6E                        2250+         DC    CL32'PRI   >>>OPDST7<<<'  eye catcher
00025A08  6ED6D7C4 E2E3F74C
00025A10  4C4C4040 40404040
00025A18  40404040 40404040
00025A20                                           2251+OPDST7   DS    0C
00025A20  00000000 00000000                        2252          DC    XL128'0'
00025A28  00000000 00000000
00025A30  00000000 00000000
00025A38  00000000 00000000
00025A40  00000000 00000000
00025A48  00000000 00000000
00025A50  00000000 00000000
00025A58  00000000 00000000
00025A60  00000000 00000000
00025A68  00000000 00000000
00025A70  00000000 00000000
00025A78  00000000 00000000
00025A80  00000000 00000000
00025A88  00000000 00000000
00025A90  00000000 00000000
00025A98  00000000 00000000
                                                   2253
                                                   2254          TSTLBL OPSRC              Operand Source
00025AA0  D7D9C940 40406E6E                        2255+         DC    CL32'PRI   >>>OPSRC7<<<'  eye catcher
00025AA8  6ED6D7E2 D9C3F74C
00025AB0  4C4C4040 40404040
00025AB8  40404040 40404040
00025AC0                                           2256+OPSRC7   DS    0C
00025AC0  00000000 00000000                        2257          DC    XL128'0'
00025AC8  00000000 00000000
00025AD0  00000000 00000000
00025AD8  00000000 00000000
00025AE0  00000000 00000000
00025AE8  00000000 00000000
00025AF0  00000000 00000000
00025AF8  00000000 00000000
00025B00  00000000 00000000
00025B08  00000000 00000000
00025B10  00000000 00000000
00025B18  00000000 00000000
00025B20  00000000 00000000
00025B28  00000000 00000000
00025B30  00000000 00000000
00025B38  00000000 00000000
                                                   2258
                                                   2259          TSTLBL PL                 Parameter List
00025B40  D7D9C940 40406E6E                        2260+         DC    CL32'PRI   >>>PL7<<<'     eye catcher
00025B48  6ED7D3F7 4C4C4C40
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00025B50  40404040 40404040
00025B58  40404040 40404040
00025B60                                         2261+PL7      DS    0C
00025B60  00000000 00000000                      2262          DC    XL256'00'
00025B68  00000000 00000000
00025B70  00000000 00000000
00025B78  00000000 00000000
00025B80  00000000 00000000
00025B88  00000000 00000000
00025B90  00000000 00000000
00025B98  00000000 00000000
00025BA0  00000000 00000000
00025BA8  00000000 00000000
00025BB0  00000000 00000000
00025BB8  00000000 00000000
00025BC0  00000000 00000000
00025BC8  00000000 00000000
00025BD0  00000000 00000000
00025BD8  00000000 00000000
00025BE0  00000000 00000000
00025BE8  00000000 00000000
00025BF0  00000000 00000000
00025BF8  00000000 00000000
00025C00  00000000 00000000
00025C08  00000000 00000000
00025C10  00000000 00000000
00025C18  00000000 00000000
00025C20  00000000 00000000
00025C28  00000000 00000000
00025C30  00000000 00000000
00025C38  00000000 00000000
00025C40  00000000 00000000
00025C48  00000000 00000000
00025C50  00000000 00000000
00025C58  00000000 00000000
                                         2263
                                         2264          TSTLBL FMSG               fail message
00025C60  D7D9C940 40406E6E              2265+         DC    CL32'PRI   >>>FMSG7<<<'   eye catcher
00025C68  6EC6D4E2 C7F74C4C
00025C70  4C404040 40404040
00025C78  40404040 40404040
00025C80                                         2266+FMSG7    DS    0C
00025C80  86818993 858440D7                      2267          DC    C'failed PLO-Extension '
00025C88  D3D660C5 A7A38595
00025C90  A2899695 40
00025C95  E3998997 938540D3              2268          DC    CL96'Triple Load(TLG: 42) / Store(TSTG: 55)'
00025C9D  9681844D E3D3C77A
00025CA5  40F4F25D 406140E2
00025CAD  A3969985 4DE3E2E3
00025CB5  C77A40F5 F55D4040
00025CBD  40404040 40404040
00025CC5  40404040 40404040
00025CCD  40404040 40404040
00025CD5  40404040 40404040
00025CDD  40404040 40404040
00025CE5  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00025CED  40404040 40404040                        2269
                                                   2270          TSTEND
00025CF5  D7D9C940 4040C5D5                        2271+TEND7    DC    CL32'PRI   END of test: 7 <<<'
00025CFD  C4409686 40A385A2
00025D05  A37A40F7 404C4C4C
00025D0D  40404040 40404040
                                                   2272
                                                   2273 * ------------------------------------------------------------------
                                                   2274 * Test Triple Load(TL: 41) / Store(TST: 54) instructions
                                                   2275 * ------------------------------------------------------------------
                                                   2276          TSTHDR 99,99,99
00025D15                          00025D15  00026000 2277+         ORG   PLOETST+(128*K)+8*(3*K)
00026000  D7D9C940 4040E2E3                        2278+         DC    CL32'PRI   START of test: 8'
00026008  C1D9E340 968640A3
00026010  85A2A37A 40F84040
00026018  40404040 40404040
00026020                          00026020         2279+         USING *,R5                base for test data and test routine
00026020                          00000000         2280+         USING PLOETEST,R5
00026020  00026050                                 2281+T8       DC    A(CODE8)            address of test code
00026024  0008                                     2282+         DC    H'8'                test number
00026026  00                                       2283+         DC    X'00'
00026027  63                                       2284+         DC    HL1'99'             Function
00026028  63                                       2285+         DC    HL1'99'             expected unequal results (UR)
00026029  00                                       2286+         DC    HL1'00'             found UR
0002602A  63                                       2287+         DC    HL1'99'             cc
0002602B  00                                       2288+         DC    HL1'00'             cc not checked
0002602C  00                                       2289+         DS    X                   extracted cc
00026030  00000000 00000000                        2290+         DS    2F                  extract PSW after test (has CC)
00026038  00026880                                 2291+         DC    A(FMSG8)            address of test fail message
0002603C  000266C0                                 2292+         DC    A(OPSRC8)           address of operand source
00026040  00026620                                 2293+         DC    A(OPDST8)           address of operand destination
00026044  00026760                                 2294+         DC    A(PL8)              address of parameter list
00026048  00004180                                 2295+         DC    A(RANDX+8)          address of random source data
0002604C  000268F5                                 2296+         DC    A(TEND8)            address of test end eye catche
                                                   2297
                                                   2298          TSTCODE
00026050                                           2299+CODE8    DS    0F
                                                   2300
00026050  E310 501C 0014                0000001C   2301          LGF   R1,OPSRCA           R1: OPSRC - Primary
00026056  B904 0061                                2302          LGR   R6,R1               R6: OPSRC - Secondary
0002605A  B904 0071                                2303          LGR   R7,R1               R7: OPSRC - Home
                                                   2304
0002605E  E320 5028 0014                00000028   2305          LGF   R2,RANDXA           R2: address random data
00026064  E330 5024 0014                00000024   2306          LGF   R3,PLA              R3: address PL
                                                   2307
0002606A  B219 0200                     00000200   2308          SAC   ARMODE
0002606E  D203 1000 2000      00000000  00000000   2309          MVC   0(4,R1),0(R2)       load Op1 = Primary
00026074  D203 6020 2020      00000020  00000020   2310          MVC   32(4,R6),32(R2)     load Op2 = Secondary
0002607A  D203 7040 2040      00000040  00000040   2311          MVC   64(4,R7),64(R2)     load Op3 = Home
00026080  B219 0000                     00000000   2312          SAC   PRIMODE
                                                   2313 *
                                                   2314 * Triple Load(TL: 41)
                                                   2315 *
00026084  E313 0000 0024                00000000   2316          STG   R1,0(R3)            load OP1 address in PL
0002608A  E54C 3020 0000                00000020   2317          MVHI  32(R3),PRIALET
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   2318
00026090  4111 0020                    00000020  2319          LA    R1,32(R1)
00026094  E313 0008 0024                00000008  2320          STG   R1,8(R3)           load OP2 address in PL
0002609A  E54C 3024 0001                00000024  2321          MVHI  36(R3),SECALET
                                                   2322
000260A0  4111 0020                    00000020  2323          LA    R1,32(R1)
000260A4  E313 0010 0024                00000010  2324          STG   R1,16(R3)          load OP3 address in PL
000260AA  E54C 3028 0002                00000028  2325          MVHI  40(R3),HOMEALET
                                                   2326
000260B0  B982 0000                              2327          XGR   R0,R0
000260B4  4100 0029                    00000029  2328          LA    R0,41
                                                   2329
000260B8  B219 0200                    00000200  2330          SAC   ARMODE
000260BC  EE0F 0000 3000      00000000  00000000  2331          PLO   0,0,R15,0(R3)      Triple Load(TL: 41)
000260C2  B219 0000                    00000000  2332          SAC   PRIMODE
                                                   2333 *
                                                   2334 * Triple Store(TST: 54)
                                                   2335 *
000260C6  E310 5020 0014                00000020  2336          LGF   R1,OPDSTA          R1:  OPDST - Primary
000260CC  E330 5024 0014                00000024  2337          LGF   R3,PLA
                                                   2338
000260D2  E313 0000 0024                00000000  2339          STG   R1,0(R3)           load OP1 address in PL
000260D8  E54C 3020 0002                00000020  2340          MVHI  32(R3),HOMEALET
                                                   2341
000260DE  4111 0020                    00000020  2342          LA    R1,32(R1)
000260E2  E313 0008 0024                00000008  2343          STG   R1,8(R3)           load OP2 address in PL
000260E8  E54C 3024 0002                00000024  2344          MVHI  36(R3),HOMEALET
                                                   2345
000260EE  4111 0020                    00000020  2346          LA    R1,32(R1)
000260F2  E313 0010 0024                00000010  2347          STG   R1,16(R3)          load OP3 address in PL
000260F8  E54C 3028 0001                00000028  2348          MVHI  40(R3),SECALET
                                                   2349
000260FE  B982 0000                              2350          XGR   R0,R0
00026102  4100 0036                    00000036  2351          LA    R0,54
                                                   2352
00026106  B219 0200                    00000200  2353          SAC   ARMODE
0002610A  EE0F 0000 3000      00000000  00000000  2354          PLO   0,0,R15,0(R3)      Triple Store(TST: 54)
00026110  B219 0000                    00000000  2355          SAC   PRIMODE
                                                   2356 *
                                                   2357 * compare results
                                                   2358 *
                                                   2359 *                               reset registers
00026114  E310 501C 0014                0000001C  2360          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0002611A  B904 0061                              2361          LGR   R6,R1              R6:  OPSRC - Secondary
0002611E  B904 0071                              2362          LGR   R7,R1              R7:  OPSRC - Home
                                                   2363
00026122  E340 5020 0014                00000020  2364          LGF   R4,OPDSTA          R4:  OPDST - Primary
00026128  B904 00C4                              2365          LGR   R12,R4             R12: OPDST - Secondary
0002612C  B904 00D4                              2366          LGR   R13,R4             R13: OPDST - Home
                                                   2367
00026130  B219 0200                    00000200  2368          SAC   ARMODE
00026134  D503 1000 D000      00000000  00000000  2369          CLC   0(4,R1),0(R13)
0002613A  4770 5136                    00026156  2370          BNE   F4154              OP1 Different, error
0002613E  D503 6020 D020      00000020  00000020  2371          CLC   32(4,R6),32(R13)
00026144  4770 5136                    00026156  2372          BNE   F4154              OP2 Different, error
00026148  D503 7040 C040      00000040  00000040  2373          CLC   64(4,R7),64(R12)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002614E  B219 0000                     00000000  2374           SAC   PRIMODE
00026152  478B 0000                     00000000  2375           BE    0(R11)              OP3 same, return
                                                  2376
                              00026156  00000001  2377 F4154     EQU   *
00026156  B219 0000                     00000000  2378           SAC   PRIMODE
0002615A  45F0 8450                     00002450  2379           BAL   R15,FAILMSG
0002615E  07FB                                    2380           BR    R11
                                                  2381
                                                  2382           TSTDATA
00026160                      00026160  00026600  2383+          ORG   PLOETST+(128*K)+8*(3*K)+(K+512)
                                                  2384           TSTLBL OPDST              Operand Destination (Results)
00026600  D7D9C940 40406E6E                       2385+          DC    CL32'PRI   >>>OPDST8<<<'  eye catcher
00026608  6ED6D7C4 E2E3F84C
00026610  4C4C4040 40404040
00026618  40404040 40404040
00026620                                          2386+OPDST8    DS    0C
00026620  00000000 00000000                       2387           DC    XL128'0'
00026628  00000000 00000000
00026630  00000000 00000000
00026638  00000000 00000000
00026640  00000000 00000000
00026648  00000000 00000000
00026650  00000000 00000000
00026658  00000000 00000000
00026660  00000000 00000000
00026668  00000000 00000000
00026670  00000000 00000000
00026678  00000000 00000000
00026680  00000000 00000000
00026688  00000000 00000000
00026690  00000000 00000000
00026698  00000000 00000000
                                                  2388
                                                  2389           TSTLBL OPSRC              Operand Source
000266A0  D7D9C940 40406E6E                       2390+          DC    CL32'PRI   >>>OPSRC8<<<'  eye catcher
000266A8  6ED6D7E2 D9C3F84C
000266B0  4C4C4040 40404040
000266B8  40404040 40404040
000266C0                                          2391+OPSRC8    DS    0C
000266C0  00000000 00000000                       2392           DC    XL128'0'
000266C8  00000000 00000000
000266D0  00000000 00000000
000266D8  00000000 00000000
000266E0  00000000 00000000
000266E8  00000000 00000000
000266F0  00000000 00000000
000266F8  00000000 00000000
00026700  00000000 00000000
00026708  00000000 00000000
00026710  00000000 00000000
00026718  00000000 00000000
00026720  00000000 00000000
00026728  00000000 00000000
00026730  00000000 00000000
00026738  00000000 00000000
                                                  2393
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   2394           TSTLBL PL                  Parameter List
00026740  D7D9C940 40406E6E                        2395+          DC     CL32'PRI   >>>PL8<<<'     eye catcher
00026748  6ED7D3F8 4C4C4C40
00026750  40404040 40404040
00026758  40404040 40404040
00026760                                           2396+PL8       DS     0C
00026760  00000000 00000000                        2397           DC     XL256'00'
00026768  00000000 00000000
00026770  00000000 00000000
00026778  00000000 00000000
00026780  00000000 00000000
00026788  00000000 00000000
00026790  00000000 00000000
00026798  00000000 00000000
000267A0  00000000 00000000
000267A8  00000000 00000000
000267B0  00000000 00000000
000267B8  00000000 00000000
000267C0  00000000 00000000
000267C8  00000000 00000000
000267D0  00000000 00000000
000267D8  00000000 00000000
000267E0  00000000 00000000
000267E8  00000000 00000000
000267F0  00000000 00000000
000267F8  00000000 00000000
00026800  00000000 00000000
00026808  00000000 00000000
00026810  00000000 00000000
00026818  00000000 00000000
00026820  00000000 00000000
00026828  00000000 00000000
00026830  00000000 00000000
00026838  00000000 00000000
00026840  00000000 00000000
00026848  00000000 00000000
00026850  00000000 00000000
00026858  00000000 00000000
                                                   2398
                                                   2399           TSTLBL FMSG                fail message
00026860  D7D9C940 40406E6E                        2400+          DC     CL32'PRI   >>>FMSG8<<<'   eye catcher
00026868  6EC6D4E2 C7F84C4C
00026870  4C404040 40404040
00026878  40404040 40404040
00026880                                           2401+FMSG8     DS     0C
00026880  86818993 858440D7                        2402           DC     C'failed PLO-Extension '
00026888  D3D660C5 A7A38595
00026890  A2899695 40
00026895  E3998997 938540D3                        2403           DC     CL96'Triple Load(TL: 41) / Store(TST: 54)'
0002689D  9681844D E3D37A40
000268A5  F4F15D40 6140E2A3
000268AD  9699854D E3E2E37A
000268B5  40F5F45D 40404040
000268BD  40404040 40404040
000268C5  40404040 40404040
000268CD  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000268D5  40404040 40404040
000268DD  40404040 40404040
000268E5  40404040 40404040
000268ED  40404040 40404040
                                          2404
                                          2405           TSTEND
000268F5  D7D9C940 4040C5D5               2406+TEND8     DC    CL32'PRI   END of test: 8 <<<'
000268FD  C4409686 40A385A2
00026905  A37A40F8 404C4C4C
0002690D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2408 *=============================================================================
                                                    2409
                                                    2410 * -----------------------------------------------------------------------
                                                    2411 * Test Double Load(DLO: 40) / Store(DSTO: 53) instructions
                                                    2412 * -----------------------------------------------------------------------
                                                    2413          TSTHDR 99,99,99
00026915                        00026915  00026C00  2414+         ORG   PLOETST+(128*K)+9*(3*K)
00026C00  D7D9C940 4040E2E3                         2415+         DC    CL32'PRI   START of test: 9'
00026C08  C1D9E340 968640A3
00026C10  85A2A37A 40F94040
00026C18  40404040 40404040
00026C20                        00026C20            2416+         USING *,R5                base for test data and test routine
00026C20                        00000000            2417+         USING PLOETEST,R5
00026C20  00026C50                                  2418+T9       DC    A(CODE9)            address of test code
00026C24  0009                                      2419+         DC    H'9'                test number
00026C26  00                                        2420+         DC    X'00'
00026C27  63                                        2421+         DC    HL1'99'             Function
00026C28  63                                        2422+         DC    HL1'99'             expected unequal results (UR)
00026C29  00                                        2423+         DC    HL1'00'             found UR
00026C2A  63                                        2424+         DC    HL1'99'             cc
00026C2B  00                                        2425+         DC    HL1'00'             cc not checked
00026C2C  00                                        2426+         DS    X                   extracted cc
00026C30  00000000 00000000                         2427+         DS    2F                  extract PSW after test (has CC)
00026C38  00027480                                  2428+         DC    A(FMSG9)            address of test fail message
00026C3C  000272C0                                  2429+         DC    A(OPSRC9)           address of operand source
00026C40  00027220                                  2430+         DC    A(OPDST9)           address of operand destination
00026C44  00027360                                  2431+         DC    A(PL9)              address of parameter list
00026C48  00004181                                  2432+         DC    A(RANDX+9)          address of random source data
00026C4C  000274F5                                  2433+         DC    A(TEND9)            address of test end eye catche
                                                    2434
                                                    2435          TSTCODE
00026C50                                            2436+CODE9    DS    0F
                                                    2437
00026C50  E310 501C 0014                 0000001C   2438          LGF   R1,OPSRCA           R1: OPSRC - Primary
00026C56  B904 0061                                 2439          LGR   R6,R1               R6: OPSRC - Secondary
00026C5A  B904 0071                                 2440          LGR   R7,R1               R7: OPSRC - Home
                                                    2441
00026C5E  E320 5028 0014                 00000028   2442          LGF   R2,RANDXA           R2: address random data
00026C64  E330 5024 0014                 00000024   2443          LGF   R3,PLA              R3: address PL
                                                    2444
00026C6A  B219 0200                      00000200   2445          SAC   ARMODE
00026C6E  D21F 1000 2000       00000000  00000000   2446          MVC   0(32,R1),0(R2)      load Op1 = Primary
00026C74  D21F 6020 2020       00000020  00000020   2447          MVC   32(32,R6),32(R2)    load Op2 = Secondary
00026C7A  B219 0000                      00000000   2448          SAC   PRIMODE
                                                    2449 *
                                                    2450 * Double Load(DLO: 40)
                                                    2451 *
00026C7E  E313 0000 0024                 00000000   2452          STG   R1,0(R3)            load OP1 address in PL
00026C84  E54C 3020 0000                 00000020   2453          MVHI  32(R3),PRIALET
                                                    2454
00026C8A  4111 0020                      00000020   2455          LA    R1,32(R1)
00026C8E  E313 0008 0024                 00000008   2456          STG   R1,8(R3)            load OP2 address in PL
00026C94  E54C 3024 0001                 00000024   2457          MVHI  36(R3),SECALET
                                                    2458
00026C9A  B982 0000                                 2459          XGR   R0,R0
00026C9E  4100 0028                      00000028   2460          LA    R0,40
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2461
00026CA2  B219 0200                     00000200  2462          SAC   ARMODE
00026CA6  EE0F 0000 3000      00000000  00000000  2463          PLO   0,0,R15,0(R3)       Double Load(DLO: 40)
00026CAC  B219 0000                     00000000  2464          SAC   PRIMODE
                                                    2465 *
                                                    2466 * Double Store(DSTO: 53)
                                                    2467 *
00026CB0  E310 5020 0014                00000020  2468          LGF   R1,OPDSTA           R1:  OPDST - Primary
00026CB6  E330 5024 0014                00000024  2469          LGF   R3,PLA
                                                    2470
00026CBC  E313 0000 0024                00000000  2471          STG   R1,0(R3)            load OP1 address in PL
00026CC2  E54C 3020 0002                00000020  2472          MVHI  32(R3),HOMEALET
                                                    2473
00026CC8  4111 0020                     00000020  2474          LA    R1,32(R1)
00026CCC  E313 0008 0024                00000008  2475          STG   R1,8(R3)            load OP2 address in PL
00026CD2  E54C 3024 0002                00000024  2476          MVHI  36(R3),HOMEALET
                                                    2477
00026CD8  B982 0000                               2478          XGR   R0,R0
00026CDC  4100 0035                     00000035  2479          LA    R0,53
                                                    2480
00026CE0  B219 0200                     00000200  2481          SAC   ARMODE
00026CE4  EE0F 0000 3000      00000000  00000000  2482          PLO   0,0,R15,0(R3)       Double Store(DSTO: 53)
00026CEA  B219 0000                     00000000  2483          SAC   PRIMODE
                                                    2484 *
                                                    2485 * compare results
                                                    2486 *                                reset registers
00026CEE  E310 501C 0014                0000001C  2487          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00026CF4  B904 0061                               2488          LGR   R6,R1               R6:  OPSRC - Secondary
00026CF8  B904 0071                               2489          LGR   R7,R1               R7:  OPSRC - Home
                                                    2490
00026CFC  E340 5020 0014                00000020  2491          LGF   R4,OPDSTA           R4:  OPDST - Primary
00026D02  B904 00C4                               2492          LGR   R12,R4              R12: OPDST - Secondary
00026D06  B904 00D4                               2493          LGR   R13,R4              R13: OPDST - Home
                                                    2494
00026D0A  B219 0200                     00000200  2495          SAC   ARMODE
00026D0E  D51F 1000 D000      00000000  00000000  2496          CLC   0(32,R1),0(R13)
00026D14  4770 5106                     00026D26  2497          BNE   F4053               OP1 Different, error
00026D18  D51F 6020 D020      00000020  00000020  2498          CLC   32(32,R6),32(R13)
00026D1E  B219 0000                     00000000  2499          SAC   PRIMODE
00026D22  478B 0000                     00000000  2500          BE    0(R11)              OP2 same, return
                                                    2501
                                00026D26  00000001  2502 F4053   EQU   *
00026D26  B219 0000                     00000000  2503          SAC   PRIMODE
00026D2A  45F0 8450                     00002450  2504          BAL   R15,FAILMSG
00026D2E  07FB                                    2505          BR    R11
                                                    2506
                                                    2507          TSTDATA
00026D30                        00026D30  00027200  2508+         ORG   PLOETST+(128*K)+9*(3*K)+(K+512)
                                                    2509          TSTLBL OPDST              Operand Destination (Results)
00027200  D7D9C940 40406E6E                         2510+         DC    CL32'PRI   >>>OPDST9<<<'  eye catcher
00027208  6ED6D7C4 E2E3F94C
00027210  4C4C4040 40404040
00027218  40404040 40404040
00027220                                            2511+OPDST9   DS    0C
00027220  00000000 00000000                         2512          DC    XL128'0'
00027228  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00027230  00000000 00000000
00027238  00000000 00000000
00027240  00000000 00000000
00027248  00000000 00000000
00027250  00000000 00000000
00027258  00000000 00000000
00027260  00000000 00000000
00027268  00000000 00000000
00027270  00000000 00000000
00027278  00000000 00000000
00027280  00000000 00000000
00027288  00000000 00000000
00027290  00000000 00000000
00027298  00000000 00000000
                                          2513
                                          2514           TSTLBL OPSRC              Operand Source
000272A0  D7D9C940 40406E6E               2515+          DC     CL32'PRI   >>>OPSRC9<<<'   eye catcher
000272A8  6ED6D7E2 D9C3F94C
000272B0  4C4C4040 40404040
000272B8  40404040 40404040
000272C0                                  2516+OPSRC9    DS     0C
000272C0  00000000 00000000               2517           DC     XL128'0'
000272C8  00000000 00000000
000272D0  00000000 00000000
000272D8  00000000 00000000
000272E0  00000000 00000000
000272E8  00000000 00000000
000272F0  00000000 00000000
000272F8  00000000 00000000
00027300  00000000 00000000
00027308  00000000 00000000
00027310  00000000 00000000
00027318  00000000 00000000
00027320  00000000 00000000
00027328  00000000 00000000
00027330  00000000 00000000
00027338  00000000 00000000
                                       2518
                                       2519           TSTLBL PL                 Parameter List
00027340  D7D9C940 40406E6E            2520+          DC     CL32'PRI   >>>PL9<<<'     eye catcher
00027348  6ED7D3F9 4C4C4C40
00027350  40404040 40404040
00027358  40404040 40404040
00027360                               2521+PL9       DS     0C
00027360  00000000 00000000            2522           DC     XL256'00'
00027368  00000000 00000000
00027370  00000000 00000000
00027378  00000000 00000000
00027380  00000000 00000000
00027388  00000000 00000000
00027390  00000000 00000000
00027398  00000000 00000000
000273A0  00000000 00000000
000273A8  00000000 00000000
000273B0  00000000 00000000
000273B8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000273C0  00000000 00000000
000273C8  00000000 00000000
000273D0  00000000 00000000
000273D8  00000000 00000000
000273E0  00000000 00000000
000273E8  00000000 00000000
000273F0  00000000 00000000
000273F8  00000000 00000000
00027400  00000000 00000000
00027408  00000000 00000000
00027410  00000000 00000000
00027418  00000000 00000000
00027420  00000000 00000000
00027428  00000000 00000000
00027430  00000000 00000000
00027438  00000000 00000000
00027440  00000000 00000000
00027448  00000000 00000000
00027450  00000000 00000000
00027458  00000000 00000000
                                        2523
                                        2524          TSTLBL FMSG                fail message
00027460  D7D9C940 40406E6E             2525+         DC    CL32'PRI   >>>FMSG9<<<'   eye catcher
00027468  6EC6D4E2 C7F94C4C
00027470  4C404040 40404040
00027478  40404040 40404040
00027480                                2526+FMSG9    DS    0C
00027480  86818993 858440D7             2527          DC    C'failed PLO-Extension '
00027488  D3D660C5 A7A38595
00027490  A2899695 40
00027495  C496A482 938540D3             2528          DC    CL96'Double Load(DLO: 40) / Store(DSTO: 53)'
0002749D  9681844D C4D3D67A
000274A5  40F4F05D 406140E2
000274AD  A3969985 4DC4E2E3
000274B5  D67A40F5 F35D4040
000274BD  40404040 40404040
000274C5  40404040 40404040
000274CD  40404040 40404040
000274D5  40404040 40404040
000274DD  40404040 40404040
000274E5  40404040 40404040
000274ED  40404040 40404040
                                        2529
                                        2530          TSTEND
000274F5  D7D9C940 4040C5D5             2531+TEND9    DC    CL32'PRI   END of test: 9 <<<'
000274FD  C4409686 40A385A2
00027505  A37A40F9 404C4C4C
0002750D  40404040 40404040
                                        2532
                                        2533 * ------------------------------------------------------------------
                                        2534 * Test Double Load(DLX: 39) / Store(DSTX: 52) instructions
                                        2535 * ------------------------------------------------------------------
                                        2536          TSTHDR 99,99,99
00027515                        00027515  00027800      2537+         ORG   PLOETST+(128*K)+10*(3*K)
00027800  D7D9C940 4040E2E3             2538+         DC    CL32'PRI   START of test: 10'
00027808  C1D9E340 968640A3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00027810  85A2A37A 40F1F040
00027818  40404040 40404040
00027820                     00027820            2539+          USING *,R5                base for test data and test routine
00027820                     00000000            2540+          USING PLOETEST,R5
00027820  00027850                               2541+T10       DC    A(CODE10)           address of test code
00027824  000A                                   2542+          DC    H'10'               test number
00027826  00                                     2543+          DC    X'00'
00027827  63                                     2544+          DC    HL1'99'             Function
00027828  63                                     2545+          DC    HL1'99'             expected unequal results (UR)
00027829  00                                     2546+          DC    HL1'00'             found UR
0002782A  63                                     2547+          DC    HL1'99'             cc
0002782B  00                                     2548+          DC    HL1'00'             cc not checked
0002782C  00                                     2549+          DS    X                   extracted cc
00027830  00000000 00000000                      2550+          DS    2F                  extract PSW after test (has CC)
00027838  00028080                               2551+          DC    A(FMSG10)           address of test fail message
0002783C  00027EC0                               2552+          DC    A(OPSRC10)          address of operand source
00027840  00027E20                               2553+          DC    A(OPDST10)          address of operand destination
00027844  00027F60                               2554+          DC    A(PL10)             address of parameter list
00027848  00004182                               2555+          DC    A(RANDX+10)         address of random source data
0002784C  000280F5                               2556+          DC    A(TEND10)           address of test end eye catche
                                                 2557
                                                 2558           TSTCODE
00027850                                         2559+CODE10    DS    0F
                                                 2560 *                                   load random source to Secondary space
                                                 2561
00027850  E310 501C 0014           0000001C      2562           LGF   R1,OPSRCA              R1: OPSRC - Primary
00027856  B904 0061                              2563           LGR   R6,R1               R6: OPSRC - Secondary
0002785A  B904 0071                              2564           LGR   R7,R1               R7: OPSRC - Home
                                                 2565
0002785E  E320 5028 0014           00000028      2566           LGF   R2,RANDXA           R2: address random data
00027864  E330 5024 0014           00000024      2567           LGF   R3,PLA              R3: address PL
                                                 2568
0002786A  B219 0200                00000200      2569           SAC   ARMODE
0002786E  D20F 1000 2000  00000000  00000000      2570           MVC   0(16,R1),0(R2)      load Op1 = Primary
00027874  D20F 6020 2020  00000020  00000020      2571           MVC   32(16,R6),32(R2)    load Op2 = Secondary
0002787A  B219 0000                00000000      2572           SAC   PRIMODE
                                                 2573 *
                                                 2574 * Double Load(DLX: 39)
                                                 2575 *
0002787E  E313 0000 0024           00000000      2576           STG   R1,0(R3)            load OP1 address in PL
00027884  E54C 3020 0000           00000020      2577           MVHI  32(R3),PRIALET
                                                 2578
0002788A  4111 0020                00000020      2579           LA    R1,32(R1)
0002788E  E313 0008 0024           00000008      2580           STG   R1,8(R3)            load OP2 address in PL
00027894  E54C 3024 0001           00000024      2581           MVHI  36(R3),SECALET
                                                 2582
0002789A  B982 0000                              2583           XGR   R0,R0
0002789E  4100 002B                0000002B      2584           LA    R0,43
                                                 2585
000278A2  B219 0200                00000200      2586           SAC   ARMODE
000278A6  EE0F 0000 3000  00000000  00000000      2587           PLO   0,0,R15,0(R3)       Double Load(DLX: 39)
000278AC  B219 0000                00000000      2588           SAC   PRIMODE
                                                 2589 *
                                                 2590 * Double Store(DSTX: 52)
                                                 2591 *
000278B0  E310 5020 0014           00000020      2592           LGF   R1,OPDSTA           R1:  OPDST - Primary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000278B6  E330 5024 0014                 00000024  2593          LGF   R3,PLA
                                                   2594
000278BC  E313 0000 0024                 00000000  2595          STG   R1,0(R3)            load OP1 address in PL
000278C2  E54C 3020 0002                 00000020  2596          MVHI  32(R3),HOMEALET
                                                   2597
000278C8  4111 0020                      00000020  2598          LA    R1,32(R1)
000278CC  E313 0008 0024                 00000008  2599          STG   R1,8(R3)            load OP2 address in PL
000278D2  E54C 3024 0002                 00000024  2600          MVHI  36(R3),HOMEALET
                                                   2601
000278D8  B982 0000                                2602          XGR   R0,R0
000278DC  4100 0034                      00000034  2603          LA    R0,52
                                                   2604
000278E0  B219 0200                      00000200  2605          SAC   ARMODE
000278E4  EE0F 0000 3000       00000000  00000000  2606          PLO   0,0,R15,0(R3)       Double Store(DSTX: 52)
000278EA  B219 0000                      00000000  2607          SAC   PRIMODE
                                                   2608 *
                                                   2609 * compare results
                                                   2610 *
000278EE  E310 501C 0014                 0000001C  2611          LGF   R1,OPSRCA           R1:  OPSRC - Primary
000278F4  B904 0061                                2612          LGR   R6,R1               R6:  OPSRC - Secondary
000278F8  B904 0071                                2613          LGR   R7,R1               R7:  OPSRC - Home
                                                   2614
000278FC  E340 5020 0014                 00000020  2615          LGF   R4,OPDSTA           R4:  OPDST - Primary
00027902  B904 00C4                                2616          LGR   R12,R4              R12: OPDST - Secondary
00027906  B904 00D4                                2617          LGR   R13,R4              R13: OPDST - Home
                                                   2618
0002790A  B219 0200                      00000200  2619          SAC   ARMODE
0002790E  D50F 1000 D000       00000000  00000000  2620          CLC   0(16,R1),0(R13)
00027914  4770 5106                      00027926  2621          BNE   F4352               OP1 Different, error
00027918  D50F 6020 D020       00000020  00000020  2622          CLC   32(16,R6),32(R13)
0002791E  B219 0000                      00000000  2623          SAC   PRIMODE
00027922  478B 0000                      00000000  2624          BE    0(R11)              OP2 same, return
                                                   2625
                                00027926  00000001  2626 F4352   EQU   *
00027926  B219 0000                      00000000  2627          SAC   PRIMODE
0002792A  45F0 8450                      00002450  2628          BAL   R15,FAILMSG
0002792E  07FB                                     2629          BR    R11
                                                   2630
                                                   2631          TSTDATA
00027930                        00027930  00027E00  2632+         ORG   PLOETST+(128*K)+10*(3*K)+(K+512)
                                                   2633          TSTLBL OPDST              Operand Destination (Results)
00027E00  D7D9C940 40406E6E                        2634+         DC    CL32'PRI   >>>OPDST10<<<' eye catcher
00027E08  6ED6D7C4 E2E3F1F0
00027E10  4C4C4C40 40404040
00027E18  40404040 40404040
00027E20                                           2635+OPDST10  DS    0C
00027E20  00000000 00000000                        2636          DC    XL128'0'
00027E28  00000000 00000000
00027E30  00000000 00000000
00027E38  00000000 00000000
00027E40  00000000 00000000
00027E48  00000000 00000000
00027E50  00000000 00000000
00027E58  00000000 00000000
00027E60  00000000 00000000
00027E68  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00027E70  00000000 00000000
00027E78  00000000 00000000
00027E80  00000000 00000000
00027E88  00000000 00000000
00027E90  00000000 00000000
00027E98  00000000 00000000
                                          2637
                                          2638           TSTLBL OPSRC              Operand Source
00027EA0  D7D9C940 40406E6E               2639+          DC    CL32'PRI   >>>OPSRC10<<<' eye catcher
00027EA8  6ED6D7E2 D9C3F1F0
00027EB0  4C4C4C40 40404040
00027EB8  40404040 40404040
00027EC0                                  2640+OPSRC10   DS    0C
00027EC0  00000000 00000000               2641           DC    XL128'0'
00027EC8  00000000 00000000
00027ED0  00000000 00000000
00027ED8  00000000 00000000
00027EE0  00000000 00000000
00027EE8  00000000 00000000
00027EF0  00000000 00000000
00027EF8  00000000 00000000
00027F00  00000000 00000000
00027F08  00000000 00000000
00027F10  00000000 00000000
00027F18  00000000 00000000
00027F20  00000000 00000000
00027F28  00000000 00000000
00027F30  00000000 00000000
00027F38  00000000 00000000
                                          2642
                                          2643           TSTLBL PL                 Parameter List
00027F40  D7D9C940 40406E6E               2644+          DC    CL32'PRI   >>>PL10<<<'    eye catcher
00027F48  6ED7D3F1 F04C4C4C
00027F50  40404040 40404040
00027F58  40404040 40404040
00027F60                                  2645+PL10      DS    0C
00027F60  00000000 00000000               2646           DC    XL256'00'
00027F68  00000000 00000000
00027F70  00000000 00000000
00027F78  00000000 00000000
00027F80  00000000 00000000
00027F88  00000000 00000000
00027F90  00000000 00000000
00027F98  00000000 00000000
00027FA0  00000000 00000000
00027FA8  00000000 00000000
00027FB0  00000000 00000000
00027FB8  00000000 00000000
00027FC0  00000000 00000000
00027FC8  00000000 00000000
00027FD0  00000000 00000000
00027FD8  00000000 00000000
00027FE0  00000000 00000000
00027FE8  00000000 00000000
00027FF0  00000000 00000000
00027FF8  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00028000  00000000 00000000
00028008  00000000 00000000
00028010  00000000 00000000
00028018  00000000 00000000
00028020  00000000 00000000
00028028  00000000 00000000
00028030  00000000 00000000
00028038  00000000 00000000
00028040  00000000 00000000
00028048  00000000 00000000
00028050  00000000 00000000
00028058  00000000 00000000
                                        2647
                                        2648          TSTLBL FMSG               fail message
00028060  D7D9C940 40406E6E             2649+         DC    CL32'PRI   >>>FMSG10<<<'  eye catcher
00028068  6EC6D4E2 C7F1F04C
00028070  4C4C4040 40404040
00028078  40404040 40404040
00028080                                2650+FMSG10   DS    0C
00028080  86818993 858440D7             2651          DC    C'failed PLO-Extension '
00028088  D3D660C5 A7A38595
00028090  A2899695 40
00028095  C496A482 938540D3             2652          DC    CL96'Double Load(DLX: 39) / Store(DSTX: 52)'
0002809D  9681844D C4D3E77A
000280A5  40F3F95D 406140E2
000280AD  A3969985 4DC4E2E3
000280B5  E77A40F5 F25D4040
000280BD  40404040 40404040
000280C5  40404040 40404040
000280CD  40404040 40404040
000280D5  40404040 40404040
000280DD  40404040 40404040
000280E5  40404040 40404040
000280ED  40404040 40404040
                                        2653
                                        2654          TSTEND
000280F5  D7D9C940 4040C5D5             2655+TEND10   DC    CL32'PRI   END of test: 10 <<<'
000280FD  C4409686 40A385A2
00028105  A37A40F1 F0404C4C
0002810D  4C404040 40404040
                                        2656
                                        2657 * ------------------------------------------------------------------
                                        2658 * Test Double Store(DSTG: 51) instruction
                                        2659 * ------------------------------------------------------------------
                                        2660          TSTHDR 99,99,51
00028115                  00028115  00028400  2661+         ORG   PLOETST+(128*K)+11*(3*K)
00028400  D7D9C940 4040E2E3             2662+         DC    CL32'PRI   START of test: 11'
00028408  C1D9E340 968640A3
00028410  85A2A37A 40F1F140
00028418  40404040 40404040
00028420                  00028420      2663+         USING *,R5                base for test data and test routine
00028420                  00000000      2664+         USING PLOETEST,R5
00028420  00028450                      2665+T11      DC    A(CODE11)           address of test code
00028424  000B                          2666+         DC    H'11'               test number
00028426  00                            2667+         DC    X'00'
00028427  33                            2668+         DC    HL1'51'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00028428  63                                      2669+          DC    HL1'99'             expected unequal results (UR)
00028429  00                                      2670+          DC    HL1'00'             found UR
0002842A  63                                      2671+          DC    HL1'99'             cc
0002842B  00                                      2672+          DC    HL1'00'             cc not checked
0002842C  00                                      2673+          DS    X                   extracted cc
00028430  00000000 00000000                       2674+          DS    2F                  extract PSW after test (has CC)
00028438  00028C80                                2675+          DC    A(FMSG11)           address of test fail message
0002843C  00028AC0                                2676+          DC    A(OPSRC11)          address of operand source
00028440  00028A20                                2677+          DC    A(OPDST11)          address of operand destination
00028444  00028B60                                2678+          DC    A(PL11)             address of parameter list
00028448  00004183                                2679+          DC    A(RANDX+11)         address of random source data
0002844C  00028CF5                                2680+          DC    A(TEND11)           address of test end eye catche
                                                  2681
                                                  2682           TSTCODE
00028450                                          2683+CODE11    DS    0F
                                                  2684 *                                   load random source
00028450  E310 501C 0014              0000001C    2685           LGF   R1,OPSRCA           R1: OPSRC - Primary
00028456  B904 0061                               2686           LGR   R6,R1               R6: OPSRC - Secondary
0002845A  B904 0071                               2687           LGR   R7,R1               R7: OPSRC - Home
                                                  2688
0002845E  E320 5028 0014              00000028    2689           LGF   R2,RANDXA           R2: address random data
00028464  E330 5024 0014              00000024    2690           LGF   R3,PLA              R3: address PL
                                                  2691
0002846A  B219 0200                   00000200    2692           SAC   ARMODE
0002846E  D207 1000 2000      00000000 00000000   2693           MVC   0(8,R1),0(R2)       load Op1 = Primary
00028474  D207 6020 2020      00000020 00000020   2694           MVC   32(8,R6),32(R2)     load Op2 = Secondary
                                                  2695 *
                                                  2696 * load of op1-data, op2-data
                                                  2697 *
0002847A  E321 0000 0004              00000000    2698                  LG    R2,0(R1)            Get OP1 from src-Primary
00028480  E323 0040 0024              00000040    2699                  STG   R2,64(R3)           load OP1-Data in PL
                                                  2700
00028486  E320 6020 0004              00000020    2701                  LG    R2,32(0,R6)         Get OP2 from src-Secondary
0002848C  E320 3048 0024              00000048    2702                  STG   R2,72(0,R3)         load OP2-Data in PL
00028492  B219 0000                   00000000    2703                  SAC   PRIMODE
                                                  2704 *
                                                  2705 * Double Store(DSTG: 51)
                                                  2706 *
00028496  E310 5020 0014              00000020    2707                  LGF   R1,OPDSTA           R1:  OPDST - Primary
0002849C  E330 5024 0014              00000024    2708                  LGF   R3,PLA
                                                  2709
000284A2  E313 0000 0024              00000000    2710                  STG   R1,0(R3)            load OP1 address in PL
000284A8  E54C 3020 0002              00000020    2711                  MVHI  32(R3),HOMEALET
                                                  2712
000284AE  4111 0020                   00000020    2713                  LA    R1,32(R1)
000284B2  E313 0008 0024              00000008    2714                  STG   R1,8(R3)            load OP2 address in PL
000284B8  E54C 3024 0002              00000024    2715                  MVHI  36(R3),HOMEALET
                                                  2716
000284BE  B982 0000                               2717                  XGR   R0,R0
000284C2  4100 0033                   00000033    2718                  LA    R0,51
                                                  2719
000284C6  B219 0200                   00000200    2720                  SAC   ARMODE
000284CA  EE0F 0000 3000      00000000 00000000   2721                  PLO   0,0,R15,0(R3)       Double Store(DSTG: 51)
000284D0  B219 0000                   00000000    2722                  SAC   PRIMODE
                                                  2723 *
                                                  2724 * compare results
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2725 *
                                                    2726 *                                reset registers
000284D4  E310 501C 0014                 0000001C  2727          LGF   R1,OPSRCA         R1:  OPSRC - Primary
000284DA  B904 0061                               2728          LGR   R6,R1             R6:  OPSRC - Secondary
000284DE  B904 0071                               2729          LGR   R7,R1             R7:  OPSRC - Home
                                                  2730
000284E2  E340 5020 0014                 00000020  2731          LGF   R4,OPDSTA         R4:  OPDST - Primary
000284E8  B904 00C4                               2732          LGR   R12,R4            R12: OPDST - Secondary
000284EC  B904 00D4                               2733          LGR   R13,R4            R13: OPDST - Home
                                                  2734
000284F0  B219 0200                      00000200  2735          SAC   ARMODE
000284F4  D507 1000 D000       00000000  00000000  2736          CLC   0(8,R1),0(R13)
000284FA  4770 50EC                      0002850C  2737          BNE   FXX51             OP1 Different, error
000284FE  D507 6020 D020       00000020  00000020  2738          CLC   32(8,R6),32(R13)
00028504  B219 0000                      00000000  2739          SAC   PRIMODE
00028508  478B 0000                      00000000  2740          BE    0(R11)            OP2 same, return
                                                  2741
                               0002850C  00000001  2742 FXX51    EQU   *
0002850C  B219 0000                      00000000  2743          SAC   PRIMODE
00028510  45F0 8450                      00002450  2744          BAL   R15,FAILMSG
00028514  07FB                                    2745          BR    R11
                                                  2746
                                                  2747          TSTDATA
00028516                       00028516  00028A00  2748+         ORG   PLOETST+(128*K)+11*(3*K)+(K+512)
                                                  2749          TSTLBL OPDST            Operand Destination (Results)
00028A00  D7D9C940 40406E6E                       2750+         DC    CL32'PRI   >>>OPDST11<<<' eye catcher
00028A08  6ED6D7C4 E2E3F1F1
00028A10  4C4C4C40 40404040
00028A18  40404040 40404040
00028A20                                          2751+OPDST11  DS    0C
00028A20  00000000 00000000                       2752          DC    XL128'0'
00028A28  00000000 00000000
00028A30  00000000 00000000
00028A38  00000000 00000000
00028A40  00000000 00000000
00028A48  00000000 00000000
00028A50  00000000 00000000
00028A58  00000000 00000000
00028A60  00000000 00000000
00028A68  00000000 00000000
00028A70  00000000 00000000
00028A78  00000000 00000000
00028A80  00000000 00000000
00028A88  00000000 00000000
00028A90  00000000 00000000
00028A98  00000000 00000000
                                                  2753
                                                  2754          TSTLBL OPSRC              Operand Source
00028AA0  D7D9C940 40406E6E                       2755+         DC    CL32'PRI   >>>OPSRC11<<<' eye catcher
00028AA8  6ED6D7E2 D9C3F1F1
00028AB0  4C4C4C40 40404040
00028AB8  40404040 40404040
00028AC0                                          2756+OPSRC11  DS    0C
00028AC0  00000000 00000000                       2757          DC    XL128'0'
00028AC8  00000000 00000000
00028AD0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00028AD8  00000000 00000000
00028AE0  00000000 00000000
00028AE8  00000000 00000000
00028AF0  00000000 00000000
00028AF8  00000000 00000000
00028B00  00000000 00000000
00028B08  00000000 00000000
00028B10  00000000 00000000
00028B18  00000000 00000000
00028B20  00000000 00000000
00028B28  00000000 00000000
00028B30  00000000 00000000
00028B38  00000000 00000000
                                          2758
                                          2759          TSTLBL PL                 Parameter List
00028B40  D7D9C940 40406E6E               2760+         DC    CL32'PRI   >>>PL11<<<'    eye catcher
00028B48  6ED7D3F1 F14C4C4C
00028B50  40404040 40404040
00028B58  40404040 40404040
00028B60                                  2761+PL11     DS    0C
00028B60  00000000 00000000               2762          DC    XL256'00'
00028B68  00000000 00000000
00028B70  00000000 00000000
00028B78  00000000 00000000
00028B80  00000000 00000000
00028B88  00000000 00000000
00028B90  00000000 00000000
00028B98  00000000 00000000
00028BA0  00000000 00000000
00028BA8  00000000 00000000
00028BB0  00000000 00000000
00028BB8  00000000 00000000
00028BC0  00000000 00000000
00028BC8  00000000 00000000
00028BD0  00000000 00000000
00028BD8  00000000 00000000
00028BE0  00000000 00000000
00028BE8  00000000 00000000
00028BF0  00000000 00000000
00028BF8  00000000 00000000
00028C00  00000000 00000000
00028C08  00000000 00000000
00028C10  00000000 00000000
00028C18  00000000 00000000
00028C20  00000000 00000000
00028C28  00000000 00000000
00028C30  00000000 00000000
00028C38  00000000 00000000
00028C40  00000000 00000000
00028C48  00000000 00000000
00028C50  00000000 00000000
00028C58  00000000 00000000
                                          2763
                                          2764          TSTLBL FMSG               fail message
00028C60  D7D9C940 40406E6E               2765+         DC    CL32'PRI   >>>FMSG11<<<'  eye catcher
00028C68  6EC6D4E2 C7F1F14C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00028C70  4C4C4040 40404040
00028C78  40404040 40404040
00028C80                                          2766+FMSG11   DS    0C
00028C80  86818993 858440D7                       2767          DC    C'failed PLO-Extension '
00028C88  D3D660C5 A7A38595
00028C90  A2899695 40
00028C95  C496A482 938540E2                       2768          DC    CL96'Double Store(DSTX: 51)'
00028C9D  A3969985 4DC4E2E3
00028CA5  E77A40F5 F15D4040
00028CAD  40404040 40404040
00028CB5  40404040 40404040
00028CBD  40404040 40404040
00028CC5  40404040 40404040
00028CCD  40404040 40404040
00028CD5  40404040 40404040
00028CDD  40404040 40404040
00028CE5  40404040 40404040
00028CED  40404040 40404040
                                                  2769
                                                  2770          TSTEND
00028CF5  D7D9C940 4040C5D5                       2771+TEND11   DC    CL32'PRI   END of test: 11 <<<'
00028CFD  C4409686 40A385A2
00028D05  A37A40F1 F1404C4C
00028D0D  4C404040 40404040
                                                  2772
                                                  2773 * ------------------------------------------------------------------
                                                  2774 * Test Double Store(DST: 50) instruction
                                                  2775 * ------------------------------------------------------------------
                                                  2776          TSTHDR 99,99,50
00028D15                          00028D15  00029000  2777+         ORG   PLOETST+(128*K)+12*(3*K)
00029000  D7D9C940 4040E2E3                       2778+         DC    CL32'PRI   START of test: 12'
00029008  C1D9E340 968640A3
00029010  85A2A37A 40F1F240
00029018  40404040 40404040
00029020                          00029020        2779+         USING *,R5                base for test data and test routine
00029020                          00000000        2780+         USING PLOETEST,R5
00029020  00029050                                2781+T12      DC    A(CODE12)           address of test code
00029024  000C                                    2782+         DC    H'12'               test number
00029026  00                                      2783+         DC    X'00'
00029027  32                                      2784+         DC    HL1'50'             Function
00029028  63                                      2785+         DC    HL1'99'             expected unequal results (UR)
00029029  00                                      2786+         DC    HL1'00'             found UR
0002902A  63                                      2787+         DC    HL1'99'             cc
0002902B  00                                      2788+         DC    HL1'00'             cc not checked
0002902C  00                                      2789+         DS    X                   extracted cc
00029030  00000000 00000000                       2790+         DS    2F                  extract PSW after test (has CC)
00029038  00029880                                2791+         DC    A(FMSG12)           address of test fail message
0002903C  000296C0                                2792+         DC    A(OPSRC12)          address of operand source
00029040  00029620                                2793+         DC    A(OPDST12)          address of operand destination
00029044  00029760                                2794+         DC    A(PL12)             address of parameter list
00029048  00004184                                2795+         DC    A(RANDX+12)         address of random source data
0002904C  000298F5                                2796+         DC    A(TEND12)           address of test end eye catche
                                                  2797
                                                  2798          TSTCODE
00029050                                          2799+CODE12   DS    0F
                                                  2800 *                                    load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00029050  E310 501C 0014                 0000001C  2801          LGF   R1,OPSRCA           R1: OPSRC - Primary
00029056  B904 0061                              2802          LGR   R6,R1               R6: OPSRC - Secondary
0002905A  B904 0071                              2803          LGR   R7,R1               R7: OPSRC - Home
                                                 2804
0002905E  E320 5028 0014                 00000028  2805          LGF   R2,RANDXA           R2: address random data
00029064  E330 5024 0014                 00000024  2806          LGF   R3,PLA              R3: address PL
                                                 2807
0002906A  B219 0200                      00000200  2808          SAC   ARMODE
0002906E  D203 1000 2000       00000000  00000000  2809          MVC   0(4,R1),0(R2)       load Op1 = Primary
00029074  D203 6020 2020       00000020  00000020  2810          MVC   32(4,R6),32(R2)     load Op2 = Secondary
                                                 2811 *
                                                 2812 * load of op1-data, op2-data
                                                 2813 *
0002907A  5821 0000                      00000000  2814          L     R2,0(R1)            Get OP1 from src-Primary
0002907E  5023 0040                      00000040  2815          ST    R2,64(R3)           load OP1-Data in PL
                                                 2816
00029082  5820 6020                      00000020  2817          L     R2,32(0,R6)         Get OP2 from src-Secondary
00029086  5020 3048                      00000048  2818          ST    R2,72(0,R3)         load OP2-Data in PL
0002908A  B219 0000                      00000000  2819          SAC   PRIMODE
                                                 2820 *
                                                 2821 * Double Store(DSTG: 50
                                                 2822 *
0002908E  E310 5020 0014                 00000020  2823          LGF   R1,OPDSTA           R1:  OPDST - Primary
00029094  E330 5024 0014                 00000024  2824          LGF   R3,PLA
                                                 2825
0002909A  E313 0000 0024                 00000000  2826          STG   R1,0(R3)            load OP1 address in PL
000290A0  E54C 3020 0002                 00000020  2827          MVHI  32(R3),HOMEALET
                                                 2828
000290A6  4111 0020                      00000020  2829          LA    R1,32(R1)
000290AA  E313 0008 0024                 00000008  2830          STG   R1,8(R3)            load OP2 address in PL
000290B0  E54C 3024 0002                 00000024  2831          MVHI  36(R3),HOMEALET
                                                 2832
000290B6  B982 0000                              2833          XGR   R0,R0
000290BA  4100 0032                      00000032  2834          LA    R0,50
                                                 2835
000290BE  B219 0200                      00000200  2836          SAC   ARMODE
000290C2  EE0F 0000 3000       00000000  00000000  2837          PLO   0,0,R15,0(R3)       Double Store(DSTG: 50)
000290C8  B219 0000                      00000000  2838          SAC   PRIMODE
                                                 2839 *
                                                 2840 * compare results
                                                 2841 *
                                                 2842 *                                  reset registers
000290CC  E310 501C 0014                 0000001C  2843          LGF   R1,OPSRCA           R1:  OPSRC - Primary
000290D2  B904 0061                              2844          LGR   R6,R1               R6:  OPSRC - Secondary
000290D6  B904 0071                              2845          LGR   R7,R1               R7:  OPSRC - Home
                                                 2846
000290DA  E340 5020 0014                 00000020  2847          LGF   R4,OPDSTA           R4:  OPDST - Primary
000290E0  B904 00C4                              2848          LGR   R12,R4              R12: OPDST - Secondary
000290E4  B904 00D4                              2849          LGR   R13,R4              R13: OPDST - Home
                                                 2850
000290E8  B219 0200                      00000200  2851          SAC   ARMODE
000290EC  D503 1000 D000       00000000  00000000  2852          CLC   0(4,R1),0(R13)
000290F2  4770 50E4                      00029104  2853          BNE   FXX50               OP1 Different, error
000290F6  D503 6020 D020       00000020  00000020  2854          CLC   32(4,R6),32(R13)
000290FC  B219 0000                      00000000  2855          SAC   PRIMODE
00029100  478B 0000                      00000000  2856          BE    0(R11)              OP2 same, return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    2857
                               00029104  00000001  2858 FXX50    EQU   *
00029104  B219 0000                      00000000  2859          SAC   PRIMODE
00029108  45F0 8450                      00002450  2860          BAL   R15,FAILMSG
0002910C  07FB                                     2861          BR    R11
                                                   2862
                                                   2863          TSTDATA
0002910E                       0002910E  00029600  2864+          ORG   PLOETST+(128*K)+12*(3*K)+(K+512)
                                                   2865          TSTLBL OPDST              Operand Destination (Results)
00029600  D7D9C940 40406E6E                        2866+         DC    CL32'PRI   >>>OPDST12<<<' eye catcher
00029608  6ED6D7C4 E2E3F1F2
00029610  4C4C4C40 40404040
00029618  40404040 40404040
00029620                                           2867+OPDST12  DS    0C
00029620  00000000 00000000                        2868          DC    XL128'0'
00029628  00000000 00000000
00029630  00000000 00000000
00029638  00000000 00000000
00029640  00000000 00000000
00029648  00000000 00000000
00029650  00000000 00000000
00029658  00000000 00000000
00029660  00000000 00000000
00029668  00000000 00000000
00029670  00000000 00000000
00029678  00000000 00000000
00029680  00000000 00000000
00029688  00000000 00000000
00029690  00000000 00000000
00029698  00000000 00000000
                                                   2869
                                                   2870          TSTLBL OPSRC              Operand Source
000296A0  D7D9C940 40406E6E                        2871+         DC    CL32'PRI   >>>OPSRC12<<<' eye catcher
000296A8  6ED6D7E2 D9C3F1F2
000296B0  4C4C4C40 40404040
000296B8  40404040 40404040
000296C0                                           2872+OPSRC12  DS    0C
000296C0  00000000 00000000                        2873          DC    XL128'0'
000296C8  00000000 00000000
000296D0  00000000 00000000
000296D8  00000000 00000000
000296E0  00000000 00000000
000296E8  00000000 00000000
000296F0  00000000 00000000
000296F8  00000000 00000000
00029700  00000000 00000000
00029708  00000000 00000000
00029710  00000000 00000000
00029718  00000000 00000000
00029720  00000000 00000000
00029728  00000000 00000000
00029730  00000000 00000000
00029738  00000000 00000000
                                                   2874
                                                   2875          TSTLBL PL                 Parameter List
00029740  D7D9C940 40406E6E                        2876+         DC    CL32'PRI   >>>PL12<<<'    eye catcher
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00029748  6ED7D3F1 F24C4C4C
00029750  40404040 40404040
00029758  40404040 40404040
00029760                                           2877+PL12     DS    0C
00029760  00000000 00000000                        2878          DC    XL256'00'
00029768  00000000 00000000
00029770  00000000 00000000
00029778  00000000 00000000
00029780  00000000 00000000
00029788  00000000 00000000
00029790  00000000 00000000
00029798  00000000 00000000
000297A0  00000000 00000000
000297A8  00000000 00000000
000297B0  00000000 00000000
000297B8  00000000 00000000
000297C0  00000000 00000000
000297C8  00000000 00000000
000297D0  00000000 00000000
000297D8  00000000 00000000
000297E0  00000000 00000000
000297E8  00000000 00000000
000297F0  00000000 00000000
000297F8  00000000 00000000
00029800  00000000 00000000
00029808  00000000 00000000
00029810  00000000 00000000
00029818  00000000 00000000
00029820  00000000 00000000
00029828  00000000 00000000
00029830  00000000 00000000
00029838  00000000 00000000
00029840  00000000 00000000
00029848  00000000 00000000
00029850  00000000 00000000
00029858  00000000 00000000
                                           2879
                                           2880          TSTLBL FMSG               fail message
00029860  D7D9C940 40406E6E                2881+         DC    CL32'PRI   >>>FMSG12<<<'  eye catcher
00029868  6EC6D4E2 C7F1F24C
00029870  4C4C4040 40404040
00029878  40404040 40404040
00029880                                           2882+FMSG12   DS    0C
00029880  86818993 858440D7                        2883          DC    C'failed PLO-Extension '
00029888  D3D660C5 A7A38595
00029890  A2899695 40
00029895  C496A482 938540E2                        2884          DC    CL96'Double Store(DSTG: 50)'
0002989D  A3969985 4DC4E2E3
000298A5  C77A40F5 F05D4040
000298AD  40404040 40404040
000298B5  40404040 40404040
000298BD  40404040 40404040
000298C5  40404040 40404040
000298CD  40404040 40404040
000298D5  40404040 40404040
000298DD  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000298E5  40404040 40404040
000298ED  40404040 40404040
                                                   2885
                                                   2886          TSTEND
000298F5  D7D9C940 4040C5D5                        2887+TEND12   DC    CL32'PRI   END of test: 12 <<<'
000298FD  C4409686 40A385A2
00029905  A37A40F1 F2404C4C
0002990D  4C404040 40404040
                                                   2888
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   2890 *=============================================================================
                                                   2891
                                                   2892 * -----------------------------------------------------------------------
                                                   2893 * Test Load(LO: 38) / Store(STO: 49) instructions
                                                   2894 * -----------------------------------------------------------------------
                                                   2895          TSTHDR 99,99,99
00029915                         00029915  00029C00 2896+         ORG   PLOETST+(128*K)+13*(3*K)
00029C00  D7D9C940 4040E2E3                        2897+         DC    CL32'PRI   START of test: 13'
00029C08  C1D9E340 968640A3
00029C10  85A2A37A 40F1F340
00029C18  40404040 40404040
00029C20                         00029C20          2898+         USING *,R5                base for test data and test routine
00029C20                         00000000          2899+         USING PLOETEST,R5
00029C20  00029C50                                 2900+T13      DC    A(CODE13)           address of test code
00029C24  000D                                     2901+         DC    H'13'               test number
00029C26  00                                       2902+         DC    X'00'
00029C27  63                                       2903+         DC    HL1'99'             Function
00029C28  63                                       2904+         DC    HL1'99'             expected unequal results (UR)
00029C29  00                                       2905+         DC    HL1'00'             found UR
00029C2A  63                                       2906+         DC    HL1'99'             cc
00029C2B  00                                       2907+         DC    HL1'00'             cc not checked
00029C2C  00                                       2908+         DS    X                   extracted cc
00029C30  00000000 00000000                        2909+         DS    2F                  extract PSW after test (has CC)
00029C38  0002A480                                 2910+         DC    A(FMSG13)           address of test fail message
00029C3C  0002A2C0                                 2911+         DC    A(OPSRC13)          address of operand source
00029C40  0002A220                                 2912+         DC    A(OPDST13)          address of operand destination
00029C44  0002A360                                 2913+         DC    A(PL13)             address of parameter list
00029C48  00004185                                 2914+         DC    A(RANDX+13)         address of random source data
00029C4C  0002A4F5                                 2915+         DC    A(TEND13)           address of test end eye catche
                                                   2916
                                                   2917          TSTCODE
00029C50                                           2918+CODE13   DS    0F
                                                   2919 *                                  load random source to Secondary space
                                                   2920
00029C50  E310 501C 0014                 0000001C  2921          LGF   R1,OPSRCA             R1: OPSRC - Primary
00029C56  B904 0061                                2922          LGR   R6,R1                 R6: OPSRC - Secondary
00029C5A  B904 0071                                2923          LGR   R7,R1                 R7: OPSRC - Home
                                                   2924
00029C5E  E320 5028 0014                 00000028  2925          LGF   R2,RANDXA             R2: address random data
00029C64  E330 5024 0014                 00000024  2926          LGF   R3,PLA                R3: address PL
                                                   2927
00029C6A  B219 0200                      00000200  2928          SAC   ARMODE
00029C6E  D21F 6000 2000       00000000  00000000  2929          MVC   0(32,R6),0(R2)        load Op1 = Secondary
00029C74  B219 0000                      00000000  2930          SAC   PRIMODE
                                                   2931 *
                                                   2932 * Load(LO: 38)
                                                   2933 *
00029C78  E313 0000 0024                 00000000  2934          STG   R1,0(R3)              load OP1 address in PL
00029C7E  E54C 3020 0001                 00000020  2935          MVHI  32(R3),SECALET
                                                   2936
00029C84  B982 0000                                2937          XGR   R0,R0
00029C88  4100 0026                      00000026  2938          LA    R0,38
                                                   2939
00029C8C  B219 0200                      00000200  2940          SAC   ARMODE
00029C90  EE0F 0000 3000       00000000  00000000  2941          PLO   0,0,R15,0(R3)         Load(LO: 38)
00029C96  B219 0000                      00000000  2942          SAC   PRIMODE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          2943 *
                                          2944 * Store(STO: 49)
                                          2945 *
00029C9A  E310 5020 0014                  00000020  2946          LGF   R1,OPDSTA           R1:  OPDST - Primary
00029CA0  E330 5024 0014                  00000024  2947          LGF   R3,PLA
                                          2948
00029CA6  E313 0000 0024                  00000000  2949          STG   R1,0(R3)            load OP1 address in PL
00029CAC  E54C 3020 0002                  00000020  2950          MVHI  32(R3),HOMEALET
                                          2951
00029CB2  B982 0000                                 2952          XGR   R0,R0
00029CB6  4100 0031                       00000031  2953          LA    R0,49
                                          2954
00029CBA  B219 0200                       00000200  2955          SAC   ARMODE
00029CBE  EE0F 0000 3000      00000000    00000000  2956          PLO   0,0,R15,0(R3)       Store(STO: 49)
00029CC4  B219 0000                       00000000  2957          SAC   PRIMODE
                                          2958 *
                                          2959 * compare results
                                          2960 *
00029CC8  E310 501C 0014                  0000001C  2961          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00029CCE  B904 0061                                 2962          LGR   R6,R1               R6:  OPSRC - Secondary
00029CD2  B904 0071                                 2963          LGR   R7,R1               R7:  OPSRC - Home
                                          2964
00029CD6  E340 5020 0014                  00000020  2965          LGF   R4,OPDSTA           R4:  OPDST - Primary
00029CDC  B904 00C4                                 2966          LGR   R12,R4              R12: OPDST - Secondary
00029CE0  B904 00D4                                 2967          LGR   R13,R4              R13: OPDST - Home
                                          2968
00029CE4  B219 0200                       00000200  2969          SAC   ARMODE
00029CE8  D51F 6000 D000      00000000    00000000  2970          CLC   0(32,R6),0(R13)
00029CEE  B219 0000                       00000000  2971          SAC   PRIMODE
00029CF2  478B 0000                       00000000  2972          BE    0(R11)              OP1 same, return
                                          2973
00029CF6  45F0 8450                       00002450  2974          BAL   R15,FAILMSG
00029CFA  07FB                                      2975          BR    R11
                                          2976
                                          2977          TSTDATA
00029CFC                      00029CFC    0002A200  2978+         ORG   PLOETST+(128*K)+13*(3*K)+(K+512)
                                          2979          TSTLBL OPDST              Operand Destination (Results)
0002A200  D7D9C940 40406E6E               2980+         DC    CL32'PRI   >>>OPDST13<<<' eye catcher
0002A208  6ED6D7C4 E2E3F1F3
0002A210  4C4C4C40 40404040
0002A218  40404040 40404040
0002A220                                  2981+OPDST13  DS    0C
0002A220  00000000 00000000               2982          DC    XL128'0'
0002A228  00000000 00000000
0002A230  00000000 00000000
0002A238  00000000 00000000
0002A240  00000000 00000000
0002A248  00000000 00000000
0002A250  00000000 00000000
0002A258  00000000 00000000
0002A260  00000000 00000000
0002A268  00000000 00000000
0002A270  00000000 00000000
0002A278  00000000 00000000
0002A280  00000000 00000000
0002A288  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002A290  00000000 00000000
0002A298  00000000 00000000
                                          2983
                                          2984          TSTLBL OPSRC              Operand Source
0002A2A0  D7D9C940 40406E6E               2985+         DC    CL32'PRI   >>>OPSRC13<<<' eye catcher
0002A2A8  6ED6D7E2 D9C3F1F3
0002A2B0  4C4C4C40 40404040
0002A2B8  40404040 40404040
0002A2C0                                  2986+OPSRC13  DS    0C
0002A2C0  00000000 00000000               2987          DC    XL128'0'
0002A2C8  00000000 00000000
0002A2D0  00000000 00000000
0002A2D8  00000000 00000000
0002A2E0  00000000 00000000
0002A2E8  00000000 00000000
0002A2F0  00000000 00000000
0002A2F8  00000000 00000000
0002A300  00000000 00000000
0002A308  00000000 00000000
0002A310  00000000 00000000
0002A318  00000000 00000000
0002A320  00000000 00000000
0002A328  00000000 00000000
0002A330  00000000 00000000
0002A338  00000000 00000000
                                          2988
                                          2989          TSTLBL PL                 Parameter List
0002A340  D7D9C940 40406E6E               2990+         DC    CL32'PRI   >>>PL13<<<'    eye catcher
0002A348  6ED7D3F1 F34C4C4C
0002A350  40404040 40404040
0002A358  40404040 40404040
0002A360                                  2991+PL13     DS    0C
0002A360  00000000 00000000               2992          DC    XL256'00'
0002A368  00000000 00000000
0002A370  00000000 00000000
0002A378  00000000 00000000
0002A380  00000000 00000000
0002A388  00000000 00000000
0002A390  00000000 00000000
0002A398  00000000 00000000
0002A3A0  00000000 00000000
0002A3A8  00000000 00000000
0002A3B0  00000000 00000000
0002A3B8  00000000 00000000
0002A3C0  00000000 00000000
0002A3C8  00000000 00000000
0002A3D0  00000000 00000000
0002A3D8  00000000 00000000
0002A3E0  00000000 00000000
0002A3E8  00000000 00000000
0002A3F0  00000000 00000000
0002A3F8  00000000 00000000
0002A400  00000000 00000000
0002A408  00000000 00000000
0002A410  00000000 00000000
0002A418  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0002A420  00000000 00000000
0002A428  00000000 00000000
0002A430  00000000 00000000
0002A438  00000000 00000000
0002A440  00000000 00000000
0002A448  00000000 00000000
0002A450  00000000 00000000
0002A458  00000000 00000000
                                          2993
                                          2994          TSTLBL FMSG               fail message
0002A460  D7D9C940 40406E6E               2995+         DC    CL32'PRI   >>>FMSG13<<<'  eye catcher
0002A468  6EC6D4E2 C7F1F34C
0002A470  4C4C4040 40404040
0002A478  40404040 40404040
0002A480                                  2996+FMSG13   DS    0C
0002A480  86818993 858440D7               2997          DC    C'failed PLO-Extension '
0002A488  D3D660C5 A7A38595
0002A490  A2899695 40
0002A495  D3968184 4DD3D67A               2998          DC    CL96'Load(LO: 38) / Store(STO: 49)'
0002A49D  40F3F85D 406140E2
0002A4A5  A3969985 4DE2E3D6
0002A4AD  7A40F4F9 5D404040
0002A4B5  40404040 40404040
0002A4BD  40404040 40404040
0002A4C5  40404040 40404040
0002A4CD  40404040 40404040
0002A4D5  40404040 40404040
0002A4DD  40404040 40404040
0002A4E5  40404040 40404040
0002A4ED  40404040 40404040
                                          2999
                                          3000          TSTEND
0002A4F5  D7D9C940 4040C5D5               3001+TEND13   DC    CL32'PRI   END of test: 13 <<<'
0002A4FD  C4409686 40A385A2
0002A505  A37A40F1 F3404C4C
0002A50D  4C404040 40404040
                                          3002
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3004 *=============================================================================
                                                   3005
                                                   3006 * -----------------------------------------------------------------------
                                                   3007 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 1st NE
                                                   3008 * -----------------------------------------------------------------------
                                                   3009          TSTHDR 2,1,37             1st Operand NOT Equal (cc=2, UR=1)
0002A515                          0002A515  0002A800 3010+         ORG   PLOETST+(128*K)+14*(3*K)
0002A800  D7D9C940 4040E2E3                        3011+         DC    CL32'PRI   START of test: 14'
0002A808  C1D9E340 968640A3
0002A810  85A2A37A 40F1F440
0002A818  40404040 40404040
0002A820                          0002A820         3012+         USING *,R5                base for test data and test routine
0002A820                          00000000         3013+         USING PLOETEST,R5
0002A820  0002A850                                 3014+T14      DC    A(CODE14)           address of test code
0002A824  000E                                     3015+         DC    H'14'               test number
0002A826  00                                       3016+         DC    X'00'
0002A827  25                                       3017+         DC    HL1'37'             Function
0002A828  01                                       3018+         DC    HL1'1'              expected unequal results (UR)
0002A829  00                                       3019+         DC    HL1'00'             found UR
0002A82A  02                                       3020+         DC    HL1'2'              cc
0002A82B  0D                                       3021+         DC    HL1'13'             not expected CC mask
0002A82C  00                                       3022+         DS    X                   extracted cc
0002A830  00000000 00000000                        3023+         DS    2F                  extract PSW after test (has CC)
0002A838  0002B1E0                                 3024+         DC    A(FMSG14)           address of test fail message
0002A83C  0002AF40                                 3025+         DC    A(OPSRC14)          address of operand source
0002A840  0002AE20                                 3026+         DC    A(OPDST14)          address of operand destination
0002A844  0002B060                                 3027+         DC    A(PL14)             address of parameter list
0002A848  00004186                                 3028+         DC    A(RANDX+14)         address of random source data
0002A84C  0002B255                                 3029+         DC    A(TEND14)           address of test end eye catche
                                                   3030
                                                   3031          TSTCODE
0002A850                                           3032+CODE14   DS    0F
                                                   3033 *                                  load random source
0002A850  E310 501C 0014                 0000001C  3034          LGF   R1,OPSRCA           R1: OPSRC - Primary
0002A856  B904 0061                                3035          LGR   R6,R1               R6: OPSRC - Secondary
0002A85A  B904 0071                                3036          LGR   R7,R1               R7: OPSRC - Home
                                                   3037
0002A85E  E320 5028 0014                 00000028  3038          LGF   R2,RANDXA           R2: address random data
0002A864  E330 5024 0014                 00000024  3039          LGF   R3,PLA              R3: address PL
                                                   3040
0002A86A  B219 0200                      00000200  3041          SAC   ARMODE
0002A86E  D21F 1000 2000       00000000  00000000  3042          MVC   0(32,R1),0(R2)      load Op1 = Primary
0002A874  D21F 6020 2020       00000020  00000020  3043          MVC   32(32,R6),32(R2)    load Op2 = Secondary
0002A87A  D21F 7040 2040       00000040  00000040  3044          MVC   64(32,R7),64(R2)    load Op3 = Home
0002A880  D21F 6060 2060       00000060  00000060  3045          MVC   96(32,R6),96(R2)    load Op4 = Secondary
                                                   3046 *
                                                   3047 * Quadruple Compare and Swap (QCSO: 37)
                                                   3048 * Build Parambter List
                                                   3049 *
0002A886  D27F 3040 2000       00000040  00000000  3050          MVC   64(128,R3),0(R2)    Load PL OP1-Op4 Compare
0002A88C  D27F 30C0 2080       000000C0  00000080  3051          MVC   192(128,R3),128(R2) Load PL Op1-Op4 Replacement
                                                   3052 ****
                                                   3053 * force a failure by changing OP1
                                                   3054 ****
0002A892  D207 1000 2100       00000000  00000100  3055          MVC   0(8,R1),256(R2)     some random data
                                                   3056 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3057
0002A898  E313 0000 0024                00000000  3058          STG   R1,0(R3)            load OP1 address in PL (pri space)
0002A89E  E54C 3020 0000                00000020  3059          MVHI  32(R3),PRIALET
                                                    3060
0002A8A4  4111 0020                     00000020  3061          LA    R1,32(R1)
0002A8A8  E313 0008 0024                00000008  3062          STG   R1,8(R3)            load OP2 address in PL (sec space)
0002A8AE  E54C 3024 0001                00000024  3063          MVHI  36(R3),SECALET
                                                    3064
0002A8B4  4111 0020                     00000020  3065          LA    R1,32(R1)
0002A8B8  E313 0010 0024                00000010  3066          STG   R1,16(R3)           load OP3 address in PL (home space)
0002A8BE  E54C 3028 0002                00000028  3067          MVHI  40(R3),HOMEALET
                                                    3068
0002A8C4  4111 0020                     00000020  3069          LA    R1,32(R1)
0002A8C8  E313 0018 0024                00000018  3070          STG   R1,24(R3)           load OP4 address in PL (sec space)
0002A8CE  E54C 302C 0001                0000002C  3071          MVHI  44(R3),SECALET
                                                    3072
0002A8D4  B982 0000                               3073          XGR   R0,R0
0002A8D8  4100 0025                     00000025  3074          LA    R0,37
                                                    3075
0002A8DC  B219 0200                     00000200  3076          SAC   ARMODE
0002A8E0  EE0F 0000 3000  00000000      00000000  3077          PLO   0,0,R15,0(R3)       Quadruple C/S (QCSO: 37)
0002A8E6  B219 0000                     00000000  3078          SAC   PRIMODE
                                                    3079
0002A8EA  B98D 0020                               3080          EPSW  R2,R0               extract psw
0002A8EE  5020 5010                     00000010  3081          ST    R2,CCPSW              to save CC
0002A8F2  E310 500B 0076                0000000B  3082          LB    R1,CCMASK           (failure CC mask)
0002A8F8  8910 0004                     00000004  3083          SLL   R1,4                (shift to BC instr CC position)
0002A8FC  4410 513C                     0002A95C  3084          EX    R1,F371TSTCC          fail if...
                                                    3085
                          0002A900      00000001  3086 F371CKUR EQU   *
0002A900  D200 5009 3030  00000009      00000030  3087          MVC   URFOUND,48(R3)      Extract UR
0002A906  D500 5008 3030  00000008      00000030  3088          CLC   UR,48(R3)
0002A90C  4780 50F4                     0002A914  3089          BE    F371CKRES
0002A910  45F0 8388                     00002388  3090          BAL   R15,URMSG
                                                    3091 *
                                                    3092 * compare results
                                                    3093 *
                          0002A914      00000001  3094 F371CKRES EQU   *
                                                    3095 *                              reset registers
0002A914  E310 501C 0014                0000001C  3096          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002A91A  B904 0061                               3097          LGR   R6,R1               R6:  OPSRC - Secondary
0002A91E  B904 0071                               3098          LGR   R7,R1               R7:  OPSRC - Home
                                                    3099
0002A922  B219 0200                     00000200  3100          SAC   ARMODE
0002A926  D51F 1000 3040  00000000      00000040  3101          CLC   0(32,R1),64(R3)
0002A92C  4770 5132                     0002A952  3102          BNE   FXX137              OP1C Different, error
0002A930  D51F 6020 3060  00000020      00000060  3103          CLC   32(32,R6),96(R3)
0002A936  4770 5132                     0002A952  3104          BNE   FXX137              OP2C Different, error
0002A93A  D51F 7040 3080  00000040      00000080  3105          CLC   64(32,R7),128(R3)
0002A940  4770 5132                     0002A952  3106          BNE   FXX137              OP3C Different, error
0002A944  D51F 6060 30A0  00000060      000000A0  3107          CLC   96(32,R6),160(R3)
0002A94A  B219 0000                     00000000  3108          SAC   PRIMODE
0002A94E  478B 0000                     00000000  3109          BE    0(R11)              OP4C same, return
                                                    3110
                          0002A952      00000001  3111 FXX137   EQU   *
0002A952  B219 0000                     00000000  3112          SAC   PRIMODE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002A956  45F0 8450                     00002450  3113           BAL   R15,FAILMSG
0002A95A  07FB                                    3114           BR    R11
                                                  3115
0002A95C  4700 5140                     0002A960  3116 F371TSTCC BC    0,F371CCMSG        fail if unexpected condition code
                                0002A960  00000001  3117 F371CCMSG EQU   *
0002A960  45F0 82B2                     000022B2  3118           BAL   R15,CCMSG
0002A964  47F0 50E0                     0002A900  3119           B     F371CKUR
                                                  3120
                                                  3121           TSTDATA
0002A968                        0002A968  0002AE00  3122+          ORG   PLOETST+(128*K)+14*(3*K)+(K+512)
                                                  3123           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002AE00  D7D9C940 40406E6E                       3124+          DC    CL32'PRI   >>>OPDST14<<<' eye catcher
0002AE08  6ED6D7C4 E2E3F1F4
0002AE10  4C4C4C40 40404040
0002AE18  40404040 40404040
0002AE20                                          3125+OPDST14   DS    0C
0002AE20  00000000 00000000                       3126           DC    XL256'0'
0002AE28  00000000 00000000
0002AE30  00000000 00000000
0002AE38  00000000 00000000
0002AE40  00000000 00000000
0002AE48  00000000 00000000
0002AE50  00000000 00000000
0002AE58  00000000 00000000
0002AE60  00000000 00000000
0002AE68  00000000 00000000
0002AE70  00000000 00000000
0002AE78  00000000 00000000
0002AE80  00000000 00000000
0002AE88  00000000 00000000
0002AE90  00000000 00000000
0002AE98  00000000 00000000
0002AEA0  00000000 00000000
0002AEA8  00000000 00000000
0002AEB0  00000000 00000000
0002AEB8  00000000 00000000
0002AEC0  00000000 00000000
0002AEC8  00000000 00000000
0002AED0  00000000 00000000
0002AED8  00000000 00000000
0002AEE0  00000000 00000000
0002AEE8  00000000 00000000
0002AEF0  00000000 00000000
0002AEF8  00000000 00000000
0002AF00  00000000 00000000
0002AF08  00000000 00000000
0002AF10  00000000 00000000
0002AF18  00000000 00000000
                                                  3127
                                                  3128           TSTLBL OPSRC              Source: OP1-OP4
0002AF20  D7D9C940 40406E6E                       3129+          DC    CL32'PRI   >>>OPSRC14<<<' eye catcher
0002AF28  6ED6D7E2 D9C3F1F4
0002AF30  4C4C4C40 40404040
0002AF38  40404040 40404040
0002AF40                                          3130+OPSRC14   DS    0C
0002AF40  00000000 00000000                       3131           DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002AF48  00000000 00000000
0002AF50  00000000 00000000
0002AF58  00000000 00000000
0002AF60  00000000 00000000
0002AF68  00000000 00000000
0002AF70  00000000 00000000
0002AF78  00000000 00000000
0002AF80  00000000 00000000
0002AF88  00000000 00000000
0002AF90  00000000 00000000
0002AF98  00000000 00000000
0002AFA0  00000000 00000000
0002AFA8  00000000 00000000
0002AFB0  00000000 00000000
0002AFB8  00000000 00000000
0002AFC0  00000000 00000000
0002AFC8  00000000 00000000
0002AFD0  00000000 00000000
0002AFD8  00000000 00000000
0002AFE0  00000000 00000000
0002AFE8  00000000 00000000
0002AFF0  00000000 00000000
0002AFF8  00000000 00000000
0002B000  00000000 00000000
0002B008  00000000 00000000
0002B010  00000000 00000000
0002B018  00000000 00000000
0002B020  00000000 00000000
0002B028  00000000 00000000
0002B030  00000000 00000000
0002B038  00000000 00000000
                                        3132
                                        3133          TSTLBL PL                 Parameter List
0002B040  D7D9C940 40406E6E             3134+         DC    CL32'PRI   >>>PL14<<<'    eye catcher
0002B048  6ED7D3F1 F44C4C4C
0002B050  40404040 40404040
0002B058  40404040 40404040
0002B060                                3135+PL14     DS    0C
0002B060  00000000 00000000             3136          DC    XL352'00'
0002B068  00000000 00000000
0002B070  00000000 00000000
0002B078  00000000 00000000
0002B080  00000000 00000000
0002B088  00000000 00000000
0002B090  00000000 00000000
0002B098  00000000 00000000
0002B0A0  00000000 00000000
0002B0A8  00000000 00000000
0002B0B0  00000000 00000000
0002B0B8  00000000 00000000
0002B0C0  00000000 00000000
0002B0C8  00000000 00000000
0002B0D0  00000000 00000000
0002B0D8  00000000 00000000
0002B0E0  00000000 00000000
0002B0E8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002B0F0  00000000 00000000
0002B0F8  00000000 00000000
0002B100  00000000 00000000
0002B108  00000000 00000000
0002B110  00000000 00000000
0002B118  00000000 00000000
0002B120  00000000 00000000
0002B128  00000000 00000000
0002B130  00000000 00000000
0002B138  00000000 00000000
0002B140  00000000 00000000
0002B148  00000000 00000000
0002B150  00000000 00000000
0002B158  00000000 00000000
0002B160  00000000 00000000
0002B168  00000000 00000000
0002B170  00000000 00000000
0002B178  00000000 00000000
0002B180  00000000 00000000
0002B188  00000000 00000000
0002B190  00000000 00000000
0002B198  00000000 00000000
0002B1A0  00000000 00000000
0002B1A8  00000000 00000000
0002B1B0  00000000 00000000
0002B1B8  00000000 00000000
                                      3137
                                      3138          TSTLBL FMSG               fail message
0002B1C0  D7D9C940 40406E6E           3139+         DC    CL32'PRI   >>>FMSG14<<<'  eye catcher
0002B1C8  6EC6D4E2 C7F1F44C
0002B1D0  4C4C4040 40404040
0002B1D8  40404040 40404040
0002B1E0                              3140+FMSG14   DS    0C
0002B1E0  86818993 858440D7           3141          DC    C'failed PLO-Extension '
0002B1E8  D3D660C5 A7A38595
0002B1F0  A2899695 40
0002B1F5  D8A48184 99A49793           3142          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 1st NE'
0002B1FD  8540C396 94978199
0002B205  85408195 8440E2A6
0002B20D  8197404D D8C3E2D6
0002B215  7A40F3F7 5D7A40F1
0002B21D  A2A340D5 C5404040
0002B225  40404040 40404040
0002B22D  40404040 40404040
0002B235  40404040 40404040
0002B23D  40404040 40404040
0002B245  40404040 40404040
0002B24D  40404040 40404040
                                      3143
                                      3144          TSTEND
0002B255  D7D9C940 4040C5D5           3145+TEND14   DC    CL32'PRI   END of test: 14 <<<'
0002B25D  C4409686 40A385A2
0002B265  A37A40F1 F4404C4C
0002B26D  4C404040 40404040
                                      3146
                                      3147 * ------------------------------------------------------------------
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   3148 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 2nd NE
                                                   3149 * ------------------------------------------------------------------
                                                   3150          TSTHDR 2,2,37             2nd Operand NOT Equal (cc=2, UR=2)
0002B275                       0002B275  0002B400  3151+         ORG   PLOETST+(128*K)+15*(3*K)
0002B400  D7D9C940 4040E2E3                        3152+         DC    CL32'PRI   START of test: 15'
0002B408  C1D9E340 968640A3
0002B410  85A2A37A 40F1F540
0002B418  40404040 40404040
0002B420                       0002B420          3153+         USING *,R5                base for test data and test routine
0002B420                       00000000          3154+         USING PLOETEST,R5
0002B420  0002B450                               3155+T15      DC    A(CODE15)           address of test code
0002B424  000F                                   3156+         DC    H'15'               test number
0002B426  00                                     3157+         DC    X'00'
0002B427  25                                     3158+         DC    HL1'37'             Function
0002B428  02                                     3159+         DC    HL1'2'              expected unequal results (UR)
0002B429  00                                     3160+         DC    HL1'00'             found UR
0002B42A  02                                     3161+         DC    HL1'2'              cc
0002B42B  0D                                     3162+         DC    HL1'13'             not expected CC mask
0002B42C  00                                     3163+         DS    X                   extracted cc
0002B430  00000000 00000000                      3164+         DS    2F                  extract PSW after test (has CC)
0002B438  0002BDE0                               3165+         DC    A(FMSG15)           address of test fail message
0002B43C  0002BB40                               3166+         DC    A(OPSRC15)          address of operand source
0002B440  0002BA20                               3167+         DC    A(OPDST15)          address of operand destination
0002B444  0002BC60                               3168+         DC    A(PL15)             address of parameter list
0002B448  00004187                               3169+         DC    A(RANDX+15)         address of random source data
0002B44C  0002BE55                               3170+         DC    A(TEND15)           address of test end eye catche
                                                   3171
                                                   3172          TSTCODE
0002B450                                         3173+CODE15   DS    0F
                                                   3174 *                                  load random source
0002B450  E310 501C 0014             0000001C    3175          LGF   R1,OPSRCA           R1: OPSRC - Primary
0002B456  B904 0061                              3176          LGR   R6,R1               R6: OPSRC - Secondary
0002B45A  B904 0071                              3177          LGR   R7,R1               R7: OPSRC - Home
                                                   3178
0002B45E  E320 5028 0014             00000028    3179          LGF   R2,RANDXA           R2: address random data
0002B464  E330 5024 0014             00000024    3180          LGF   R3,PLA              R3: address PL
                                                   3181
0002B46A  B219 0200                  00000200    3182          SAC   ARMODE
0002B46E  D21F 1000 2000   00000000  00000000    3183          MVC   0(32,R1),0(R2)      load Op1 = Primary
0002B474  D21F 6020 2020   00000020  00000020    3184          MVC   32(32,R6),32(R2)    load Op2 = Secondary
0002B47A  D21F 7040 2040   00000040  00000040    3185          MVC   64(32,R7),64(R2)    load Op3 = Home
0002B480  D21F 6060 2060   00000060  00000060    3186          MVC   96(32,R6),96(R2)    load Op4 = Secondary
                                                   3187 *
                                                   3188 * Quadruple Compare and Swap (QCSO: 37)
                                                   3189 * Build Parambter List
                                                   3190 *
0002B486  D27F 3040 2000   00000040  00000000    3191          MVC   64(128,R3),0(R2)    Load PL OP1-Op4 Compare
0002B48C  D27F 30C0 2080   000000C0  00000080    3192          MVC   192(128,R3),128(R2) Load PL Op1-Op4 Replacement
                                                   3193 ****
                                                   3194 * force a failure by changing the 2nd operand
                                                   3195 ****
0002B492  D207 6020 2100   00000020  00000100    3196          MVC   32(8,R6),256(R2)
                                                   3197 ****
                                                   3198
0002B498  E313 0000 0024             00000000    3199          STG   R1,0(R3)            load OP1 address in PL
0002B49E  E54C 3020 0000             00000020    3200          MVHI  32(R3),PRIALET
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3201
0002B4A4  4111 0020                     00000020  3202          LA    R1,32(R1)
0002B4A8  E313 0008 0024                00000008  3203          STG   R1,8(R3)           load OP2 address in PL
0002B4AE  E54C 3024 0001                00000024  3204          MVHI  36(R3),SECALET
                                                    3205
0002B4B4  4111 0020                     00000020  3206          LA    R1,32(R1)
0002B4B8  E313 0010 0024                00000010  3207          STG   R1,16(R3)          load OP3 address in PL
0002B4BE  E54C 3028 0002                00000028  3208          MVHI  40(R3),HOMEALET
                                                    3209
0002B4C4  4111 0020                     00000020  3210          LA    R1,32(R1)
0002B4C8  E313 0018 0024                00000018  3211          STG   R1,24(R3)          load OP4 address in PL
0002B4CE  E54C 302C 0001                0000002C  3212          MVHI  44(R3),SECALET
                                                    3213
0002B4D4  B982 0000                               3214          XGR   R0,R0
0002B4D8  4100 0025                     00000025  3215          LA    R0,37
                                                    3216
0002B4DC  B219 0200                     00000200  3217          SAC   ARMODE
0002B4E0  EE0F 0000 3000  00000000      00000000  3218          PLO   0,0,R15,0(R3)      Quadruple C/S (QCSO: 37)
0002B4E6  B219 0000                     00000000  3219          SAC   PRIMODE
                                                    3220
0002B4EA  B98D 0020                               3221          EPSW  R2,R0              extract psw
0002B4EE  5020 5010                     00000010  3222          ST    R2,CCPSW             to save CC
0002B4F2  E310 500B 0076                0000000B  3223          LB    R1,CCMASK          (failure CC mask)
0002B4F8  8910 0004                     00000004  3224          SLL   R1,4               (shift to BC instr CC position)
0002B4FC  4410 513C                     0002B55C  3225          EX    R1,F372TSTCC         fail if...
                                                    3226
                          0002B500  00000001  3227 F372CKUR  EQU   *
0002B500  D200 5009 3030  00000009  00000030  3228          MVC   URFOUND,48(R3)     Extract UR
0002B506  D500 5008 3030  00000008  00000030  3229          CLC   UR,48(R3)
0002B50C  4780 50F4                     0002B514  3230          BE    F372CKRES
0002B510  45F0 8388                     00002388  3231          BAL   R15,URMSG
                                                    3232 *
                                                    3233 * compare results
                                                    3234 *
                          0002B514  00000001  3235 F372CKRES EQU   *
                                                    3236 *                         reset addresses
0002B514  E310 501C 0014                0000001C  3237          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0002B51A  B904 0061                               3238          LGR   R6,R1              R6:  OPSRC - Secondary
0002B51E  B904 0071                               3239          LGR   R7,R1              R7:  OPSRC - Home
                                                    3240
0002B522  B219 0200                     00000200  3241          SAC   ARMODE
0002B526  D51F 1000 3040  00000000      00000040  3242          CLC   0(32,R1),64(R3)
0002B52C  4770 5132                     0002B552  3243          BNE   FXX237             OP1C Different, error
0002B530  D51F 6020 3060  00000020      00000060  3244          CLC   32(32,R6),96(R3)
0002B536  4770 5132                     0002B552  3245          BNE   FXX237             OP2C Different, error
0002B53A  D51F 7040 3080  00000040      00000080  3246          CLC   64(32,R7),128(R3)
0002B540  4770 5132                     0002B552  3247          BNE   FXX237             OP3C Different, error
0002B544  D51F 6060 30A0  00000060      000000A0  3248          CLC   96(32,R6),160(R3)
0002B54A  B219 0000                     00000000  3249          SAC   PRIMODE
0002B54E  478B 0000                     00000000  3250          BE    0(R11)             OP4Csame, return
                                                    3251
                          0002B552  00000001  3252 FXX237   EQU   *
0002B552  B219 0000                     00000000  3253          SAC   PRIMODE
0002B556  45F0 8450                     00002450  3254          BAL   R15,FAILMSG
0002B55A  07FB                                    3255          BR    R11
                                                    3256
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0002B55C  4700 5140                       0002B560  3257 F372TSTCC BC    0,F372CCMSG        fail if unexpected condition code
                                0002B560  00000001  3258 F372CCMSG EQU   *
0002B560  45F0 82B2                       000022B2  3259           BAL   R15,CCMSG
0002B564  47F0 50E0                       0002B500  3260           B     F372CKUR
                                                    3261
                                                    3262           TSTDATA
0002B568                        0002B568  0002BA00  3263+          ORG   PLOETST+(128*K)+15*(3*K)+(K+512)
                                                    3264           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002BA00  D7D9C940 40406E6E                         3265+          DC    CL32'PRI   >>>OPDST15<<<' eye catcher
0002BA08  6ED6D7C4 E2E3F1F5
0002BA10  4C4C4C40 40404040
0002BA18  40404040 40404040
0002BA20                                            3266+OPDST15   DS    0C
0002BA20  00000000 00000000                         3267           DC    XL256'0'
0002BA28  00000000 00000000
0002BA30  00000000 00000000
0002BA38  00000000 00000000
0002BA40  00000000 00000000
0002BA48  00000000 00000000
0002BA50  00000000 00000000
0002BA58  00000000 00000000
0002BA60  00000000 00000000
0002BA68  00000000 00000000
0002BA70  00000000 00000000
0002BA78  00000000 00000000
0002BA80  00000000 00000000
0002BA88  00000000 00000000
0002BA90  00000000 00000000
0002BA98  00000000 00000000
0002BAA0  00000000 00000000
0002BAA8  00000000 00000000
0002BAB0  00000000 00000000
0002BAB8  00000000 00000000
0002BAC0  00000000 00000000
0002BAC8  00000000 00000000
0002BAD0  00000000 00000000
0002BAD8  00000000 00000000
0002BAE0  00000000 00000000
0002BAE8  00000000 00000000
0002BAF0  00000000 00000000
0002BAF8  00000000 00000000
0002BB00  00000000 00000000
0002BB08  00000000 00000000
0002BB10  00000000 00000000
0002BB18  00000000 00000000
                                                    3268
                                                    3269           TSTLBL OPSRC              Source: OP1-OP4
0002BB20  D7D9C940 40406E6E                         3270+          DC    CL32'PRI   >>>OPSRC15<<<' eye catcher
0002BB28  6ED6D7E2 D9C3F1F5
0002BB30  4C4C4C40 40404040
0002BB38  40404040 40404040
0002BB40                                            3271+OPSRC15   DS    0C
0002BB40  00000000 00000000                         3272           DC    XL256'0'
0002BB48  00000000 00000000
0002BB50  00000000 00000000
0002BB58  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002BB60  00000000 00000000
0002BB68  00000000 00000000
0002BB70  00000000 00000000
0002BB78  00000000 00000000
0002BB80  00000000 00000000
0002BB88  00000000 00000000
0002BB90  00000000 00000000
0002BB98  00000000 00000000
0002BBA0  00000000 00000000
0002BBA8  00000000 00000000
0002BBB0  00000000 00000000
0002BBB8  00000000 00000000
0002BBC0  00000000 00000000
0002BBC8  00000000 00000000
0002BBD0  00000000 00000000
0002BBD8  00000000 00000000
0002BBE0  00000000 00000000
0002BBE8  00000000 00000000
0002BBF0  00000000 00000000
0002BBF8  00000000 00000000
0002BC00  00000000 00000000
0002BC08  00000000 00000000
0002BC10  00000000 00000000
0002BC18  00000000 00000000
0002BC20  00000000 00000000
0002BC28  00000000 00000000
0002BC30  00000000 00000000
0002BC38  00000000 00000000
                                          3273
                                          3274           TSTLBL PL                  Parameter List
0002BC40  D7D9C940 40406E6E               3275+          DC    CL32'PRI   >>>PL15<<<'    eye catcher
0002BC48  6ED7D3F1 F54C4C4C
0002BC50  40404040 40404040
0002BC58  40404040 40404040
0002BC60                                  3276+PL15      DS    0C
0002BC60  00000000 00000000               3277           DC    XL352'00'
0002BC68  00000000 00000000
0002BC70  00000000 00000000
0002BC78  00000000 00000000
0002BC80  00000000 00000000
0002BC88  00000000 00000000
0002BC90  00000000 00000000
0002BC98  00000000 00000000
0002BCA0  00000000 00000000
0002BCA8  00000000 00000000
0002BCB0  00000000 00000000
0002BCB8  00000000 00000000
0002BCC0  00000000 00000000
0002BCC8  00000000 00000000
0002BCD0  00000000 00000000
0002BCD8  00000000 00000000
0002BCE0  00000000 00000000
0002BCE8  00000000 00000000
0002BCF0  00000000 00000000
0002BCF8  00000000 00000000
0002BD00  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002BD08  00000000 00000000
0002BD10  00000000 00000000
0002BD18  00000000 00000000
0002BD20  00000000 00000000
0002BD28  00000000 00000000
0002BD30  00000000 00000000
0002BD38  00000000 00000000
0002BD40  00000000 00000000
0002BD48  00000000 00000000
0002BD50  00000000 00000000
0002BD58  00000000 00000000
0002BD60  00000000 00000000
0002BD68  00000000 00000000
0002BD70  00000000 00000000
0002BD78  00000000 00000000
0002BD80  00000000 00000000
0002BD88  00000000 00000000
0002BD90  00000000 00000000
0002BD98  00000000 00000000
0002BDA0  00000000 00000000
0002BDA8  00000000 00000000
0002BDB0  00000000 00000000
0002BDB8  00000000 00000000
                                      3278
                                      3279          TSTLBL FMSG              fail message
0002BDC0  D7D9C940 40406E6E           3280+         DC    CL32'PRI   >>>FMSG15<<<'  eye catcher
0002BDC8  6EC6D4E2 C7F1F54C
0002BDD0  4C4C4040 40404040
0002BDD8  40404040 40404040
0002BDE0                              3281+FMSG15   DS    0C
0002BDE0  86818993 858440D7           3282          DC    C'failed PLO-Extension '
0002BDE8  D3D660C5 A7A38595
0002BDF0  A2899695 40
0002BDF5  D8A48184 99A49793           3283          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 2nd NE'
0002BDFD  8540C396 94978199
0002BE05  85408195 8440E2A6
0002BE0D  8197404D D8C3E2D6
0002BE15  7A40F3F7 5D7A40F2
0002BE1D  958440D5 C5404040
0002BE25  40404040 40404040
0002BE2D  40404040 40404040
0002BE35  40404040 40404040
0002BE3D  40404040 40404040
0002BE45  40404040 40404040
0002BE4D  40404040 40404040
                                      3284
                                      3285          TSTEND
0002BE55  D7D9C940 4040C5D5           3286+TEND15   DC    CL32'PRI   END of test: 15 <<<'
0002BE5D  C4409686 40A385A2
0002BE65  A37A40F1 F5404C4C
0002BE6D  4C404040 40404040
                                      3287
                                      3288 * ------------------------------------------------------------------
                                      3289 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 3rd NE
                                      3290 * ------------------------------------------------------------------
                                      3291          TSTHDR 2,3,37             3rd Operand NOT Equal (cc=2, UR=3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002BE75                     0002BE75  0002C000  3292+          ORG   PLOETST+(128*K)+16*(3*K)
0002C000  D7D9C940 4040E2E3                       3293+          DC    CL32'PRI   START of test: 16'
0002C008  C1D9E340 968640A3
0002C010  85A2A37A 40F1F640
0002C018  40404040 40404040
0002C020                     0002C020             3294+          USING *,R5                 base for test data and test routine
0002C020                     00000000             3295+          USING PLOETEST,R5
0002C020  0002C050                                3296+T16       DC    A(CODE16)            address of test code
0002C024  0010                                    3297+          DC    H'16'                test number
0002C026  00                                      3298+          DC    X'00'
0002C027  25                                      3299+          DC    HL1'37'              Function
0002C028  03                                      3300+          DC    HL1'3'               expected unequal results (UR)
0002C029  00                                      3301+          DC    HL1'00'              found UR
0002C02A  02                                      3302+          DC    HL1'2'               cc
0002C02B  0D                                      3303+          DC    HL1'13'              not expected CC mask
0002C02C  00                                      3304+          DS    X                    extracted cc
0002C030  00000000 00000000                       3305+          DS    2F                   extract PSW after test (has CC)
0002C038  0002C9E0                                3306+          DC    A(FMSG16)            address of test fail message
0002C03C  0002C740                                3307+          DC    A(OPSRC16)           address of operand source
0002C040  0002C620                                3308+          DC    A(OPDST16)           address of operand destination
0002C044  0002C860                                3309+          DC    A(PL16)              address of parameter list
0002C048  00004188                                3310+          DC    A(RANDX+16)          address of random source data
0002C04C  0002CA55                                3311+          DC    A(TEND16)            address of test end eye catche
                                                  3312
                                                  3313           TSTCODE
0002C050                                          3314+CODE16    DS    0F
                                                  3315 *                                     load random source
0002C050  E310 501C 0014                0000001C  3316           LGF   R1,OPSRCA            R1: OPSRC - Primary
0002C056  B904 0061                               3317           LGR   R6,R1                R6: OPSRC - Secondary
0002C05A  B904 0071                               3318           LGR   R7,R1                R7: OPSRC - Home
                                                  3319
0002C05E  E320 5028 0014                00000028  3320           LGF   R2,RANDXA            R2: address random data
0002C064  E330 5024 0014                00000024  3321           LGF   R3,PLA               R3: address PL
                                                  3322
0002C06A  B219 0200                     00000200  3323           SAC   ARMODE
0002C06E  D21F 1000 2000       00000000 00000000  3324           MVC   0(32,R1),0(R2)       load Op1 = Primary
0002C074  D21F 6020 2020       00000020 00000020  3325           MVC   32(32,R6),32(R2)     load Op2 = Secondary
0002C07A  D21F 7040 2040       00000040 00000040  3326           MVC   64(32,R7),64(R2)     load Op3 = Home
0002C080  D21F 6060 2060       00000060 00000060  3327           MVC   96(32,R6),96(R2)     load Op4 = Secondary
                                                  3328 *
                                                  3329 * Quadruple Compare and Swap (QCSO: 37)
                                                  3330 * Build Parambter List
                                                  3331 *
                                                  3332
0002C086  D27F 3040 2000       00000040 00000000  3333           MVC   64(128,R3),0(R2)     Load PL OP1-Op4 Compare
0002C08C  D27F 30C0 2080       000000C0 00000080  3334           MVC   192(128,R3),128(R2) Load PL Op1-Op4 Replacement
                                                  3335 ****
                                                  3336 * force a failure by changing the 3nd operand
                                                  3337 ****
0002C092  D207 7040 2100       00000040 00000100  3338           MVC   64(8,R7),256(R2)      some random data
                                                  3339 ****
                                                  3340
0002C098  E313 0000 0024                00000000  3341           STG   R1,0(R3)             load OP1 address in PL
0002C09E  E54C 3020 0000                00000020  3342           MVHI  32(R3),PRIALET
                                                  3343
0002C0A4  4111 0020                     00000020  3344           LA    R1,32(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002C0A8  E313 0008 0024                 00000008  3345           STG   R1,8(R3)            load OP2 address in PL
0002C0AE  E54C 3024 0001                 00000024  3346           MVHI  36(R3),SECALET
                                                   3347
0002C0B4  4111 0020                      00000020  3348           LA    R1,32(R1)
0002C0B8  E313 0010 0024                 00000010  3349           STG   R1,16(R3)           load OP3 address in PL
0002C0BE  E54C 3028 0002                 00000028  3350           MVHI  40(R3),HOMEALET
                                                   3351
0002C0C4  4111 0020                      00000020  3352           LA    R1,32(R1)
0002C0C8  E313 0018 0024                 00000018  3353           STG   R1,24(R3)           load OP4 address in PL
0002C0CE  E54C 302C 0001                 0000002C  3354           MVHI  44(R3),SECALET
                                                   3355
0002C0D4  B982 0000                                3356           XGR   R0,R0
0002C0D8  4100 0025                      00000025  3357           LA    R0,37
                                                   3358
0002C0DC  B219 0200                      00000200  3359           SAC   ARMODE
0002C0E0  EE0F 0000 3000      00000000   00000000  3360           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSO): 37
0002C0E6  B219 0000                      00000000  3361           SAC   PRIMODE
                                                   3362
0002C0EA  B98D 0020                                3363           EPSW  R2,R0               extract psw
0002C0EE  5020 5010                      00000010  3364           ST    R2,CCPSW              to save CC
0002C0F2  E310 500B 0076                 0000000B  3365           LB    R1,CCMASK           (failure CC mask)
0002C0F8  8910 0004                      00000004  3366           SLL   R1,4                (shift to BC instr CC position)
0002C0FC  4410 513C                      0002C15C  3367           EX    R1,F373TSTCC            fail if...
                                                   3368
                                0002C100  00000001  3369 F373CKUR  EQU   *
0002C100  D200 5009 3030      00000009   00000030  3370           MVC   URFOUND,48(R3)      Extract UR
0002C106  D500 5008 3030      00000008   00000030  3371           CLC   UR,48(R3)
0002C10C  4780 50F4                      0002C114  3372           BE    F373CKRES
0002C110  45F0 8388                      00002388  3373           BAL   R15,URMSG
                                                   3374 *
                                                   3375 * compare results
                                                   3376 *
                                0002C114  00000001  3377 F373CKRES EQU   *
                                                   3378 *                                  reset addresses
0002C114  E310 501C 0014                 0000001C  3379           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002C11A  B904 0061                                3380           LGR   R6,R1               R6:  OPSRC - Secondary
0002C11E  B904 0071                                3381           LGR   R7,R1               R7:  OPSRC - Home
                                                   3382
0002C122  B219 0200                      00000200  3383           SAC   ARMODE
0002C126  D51F 1000 3040      00000000   00000040  3384           CLC   0(32,R1),64(R3)
0002C12C  4770 5132                      0002C152  3385           BNE   F373XX              OP1C Different, error
0002C130  D51F 6020 3060      00000020   00000060  3386           CLC   32(32,R6),96(R3)
0002C136  4770 5132                      0002C152  3387           BNE   F373XX              OP2C Different, error
0002C13A  D51F 7040 3080      00000040   00000080  3388           CLC   64(32,R7),128(R3)
0002C140  4770 5132                      0002C152  3389           BNE   F373XX              OP3C Different, error
0002C144  D51F 6060 30A0      00000060   000000A0  3390           CLC   96(32,R6),160(R3)
0002C14A  B219 0000                      00000000  3391           SAC   PRIMODE
0002C14E  478B 0000                      00000000  3392           BE    0(R11)              OP4Csame, return
                                                   3393
                                0002C152  00000001  3394 F373XX   EQU   *
0002C152  B219 0000                      00000000  3395           SAC   PRIMODE
0002C156  45F0 8450                      00002450  3396           BAL   R15,FAILMSG
0002C15A  07FB                                     3397           BR    R11
                                                   3398
0002C15C  4700 5140                      0002C160  3399 F373TSTCC BC    0,F373CCMSG         fail if unexpected condition code
                                0002C160  00000001  3400 F373CCMSG EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002C160  45F0 82B2                     000022B2  3401           BAL   R15,CCMSG
0002C164  47F0 50E0                     0002C100  3402           B     F373CKUR
                                                  3403
                                                  3404           TSTDATA
0002C168                    0002C168  0002C600  3405+           ORG   PLOETST+(128*K)+16*(3*K)+(K+512)
                                                  3406           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002C600  D7D9C940 40406E6E               3407+           DC    CL32'PRI   >>>OPDST16<<<' eye catcher
0002C608  6ED6D7C4 E2E3F1F6
0002C610  4C4C4C40 40404040
0002C618  40404040 40404040
0002C620                                  3408+OPDST16    DS    0C
0002C620  00000000 00000000               3409            DC    XL256'0'
0002C628  00000000 00000000
0002C630  00000000 00000000
0002C638  00000000 00000000
0002C640  00000000 00000000
0002C648  00000000 00000000
0002C650  00000000 00000000
0002C658  00000000 00000000
0002C660  00000000 00000000
0002C668  00000000 00000000
0002C670  00000000 00000000
0002C678  00000000 00000000
0002C680  00000000 00000000
0002C688  00000000 00000000
0002C690  00000000 00000000
0002C698  00000000 00000000
0002C6A0  00000000 00000000
0002C6A8  00000000 00000000
0002C6B0  00000000 00000000
0002C6B8  00000000 00000000
0002C6C0  00000000 00000000
0002C6C8  00000000 00000000
0002C6D0  00000000 00000000
0002C6D8  00000000 00000000
0002C6E0  00000000 00000000
0002C6E8  00000000 00000000
0002C6F0  00000000 00000000
0002C6F8  00000000 00000000
0002C700  00000000 00000000
0002C708  00000000 00000000
0002C710  00000000 00000000
0002C718  00000000 00000000
                                          3410
                                          3411            TSTLBL OPSRC              Source: OP1-OP4
0002C720  D7D9C940 40406E6E               3412+           DC    CL32'PRI   >>>OPSRC16<<<' eye catcher
0002C728  6ED6D7E2 D9C3F1F6
0002C730  4C4C4C40 40404040
0002C738  40404040 40404040
0002C740                                  3413+OPSRC16    DS    0C
0002C740  00000000 00000000               3414            DC    XL256'0'
0002C748  00000000 00000000
0002C750  00000000 00000000
0002C758  00000000 00000000
0002C760  00000000 00000000
0002C768  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002C770  00000000 00000000
0002C778  00000000 00000000
0002C780  00000000 00000000
0002C788  00000000 00000000
0002C790  00000000 00000000
0002C798  00000000 00000000
0002C7A0  00000000 00000000
0002C7A8  00000000 00000000
0002C7B0  00000000 00000000
0002C7B8  00000000 00000000
0002C7C0  00000000 00000000
0002C7C8  00000000 00000000
0002C7D0  00000000 00000000
0002C7D8  00000000 00000000
0002C7E0  00000000 00000000
0002C7E8  00000000 00000000
0002C7F0  00000000 00000000
0002C7F8  00000000 00000000
0002C800  00000000 00000000
0002C808  00000000 00000000
0002C810  00000000 00000000
0002C818  00000000 00000000
0002C820  00000000 00000000
0002C828  00000000 00000000
0002C830  00000000 00000000
0002C838  00000000 00000000
                             3415
                             3416          TSTLBL PL                 Parameter List
0002C840  D7D9C940 40406E6E  3417+         DC    CL32'PRI   >>>PL16<<<'    eye catcher
0002C848  6ED7D3F1 F64C4C4C
0002C850  40404040 40404040
0002C858  40404040 40404040
0002C860                     3418+PL16     DS    0C
0002C860  00000000 00000000  3419          DC    XL352'00'
0002C868  00000000 00000000
0002C870  00000000 00000000
0002C878  00000000 00000000
0002C880  00000000 00000000
0002C888  00000000 00000000
0002C890  00000000 00000000
0002C898  00000000 00000000
0002C8A0  00000000 00000000
0002C8A8  00000000 00000000
0002C8B0  00000000 00000000
0002C8B8  00000000 00000000
0002C8C0  00000000 00000000
0002C8C8  00000000 00000000
0002C8D0  00000000 00000000
0002C8D8  00000000 00000000
0002C8E0  00000000 00000000
0002C8E8  00000000 00000000
0002C8F0  00000000 00000000
0002C8F8  00000000 00000000
0002C900  00000000 00000000
0002C908  00000000 00000000
0002C910  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002C918  00000000 00000000
0002C920  00000000 00000000
0002C928  00000000 00000000
0002C930  00000000 00000000
0002C938  00000000 00000000
0002C940  00000000 00000000
0002C948  00000000 00000000
0002C950  00000000 00000000
0002C958  00000000 00000000
0002C960  00000000 00000000
0002C968  00000000 00000000
0002C970  00000000 00000000
0002C978  00000000 00000000
0002C980  00000000 00000000
0002C988  00000000 00000000
0002C990  00000000 00000000
0002C998  00000000 00000000
0002C9A0  00000000 00000000
0002C9A8  00000000 00000000
0002C9B0  00000000 00000000
0002C9B8  00000000 00000000
                                      3420
                                      3421          TSTLBL FMSG                fail message
0002C9C0  D7D9C940 40406E6E           3422+         DC    CL32'PRI   >>>FMSG16<<<'   eye catcher
0002C9C8  6EC6D4E2 C7F1F64C
0002C9D0  4C4C4040 40404040
0002C9D8  40404040 40404040
0002C9E0                              3423+FMSG16   DS    0C
0002C9E0  86818993 858440D7           3424          DC    C'failed PLO-Extension '
0002C9E8  D3D660C5 A7A38595
0002C9F0  A2899695 40
0002C9F5  D8A48184 99A49793           3425          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 3rd NE'
0002C9FD  8540C396 94978199
0002CA05  85408195 8440E2A6
0002CA0D  8197404D D8C3E2D6
0002CA15  7A40F3F7 5D7A40F3
0002CA1D  998440D5 C5404040
0002CA25  40404040 40404040
0002CA2D  40404040 40404040
0002CA35  40404040 40404040
0002CA3D  40404040 40404040
0002CA45  40404040 40404040
0002CA4D  40404040 40404040
                                      3426
                                      3427          TSTEND
0002CA55  D7D9C940 4040C5D5           3428+TEND16   DC    CL32'PRI   END of test: 16 <<<'
0002CA5D  C4409686 40A385A2
0002CA65  A37A40F1 F6404C4C
0002CA6D  4C404040 40404040
                                      3429
                                      3430 * ------------------------------------------------------------------
                                      3431 * Test Quadruple Compare and Swap (QCSO: 37) instruction : 4th NE
                                      3432 * ------------------------------------------------------------------
                                      3433          TSTHDR 2,4,37              4th Operand NOT Equal (cc=2, UR=4)
0002CA75                 0002CA75  0002CC00  3434+         ORG   PLOETST+(128*K)+17*(3*K)
0002CC00  D7D9C940 4040E2E3           3435+         DC    CL32'PRI   START of test: 17'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002CC08  C1D9E340 968640A3
0002CC10  85A2A37A 40F1F740
0002CC18  40404040 40404040
0002CC20                     0002CC20            3436+          USING *,R5                base for test data and test routine
0002CC20                     00000000            3437+          USING PLOETEST,R5
0002CC20  0002CC50                               3438+T17       DC    A(CODE17)           address of test code
0002CC24  0011                                   3439+          DC    H'17'               test number
0002CC26  00                                     3440+          DC    X'00'
0002CC27  25                                     3441+          DC    HL1'37'             Function
0002CC28  04                                     3442+          DC    HL1'4'              expected unequal results (UR)
0002CC29  00                                     3443+          DC    HL1'00'             found UR
0002CC2A  02                                     3444+          DC    HL1'2'              cc
0002CC2B  0D                                     3445+          DC    HL1'13'             not expected CC mask
0002CC2C  00                                     3446+          DS    X                   extracted cc
0002CC30  00000000 00000000                      3447+          DS    2F                  extract PSW after test (has CC)
0002CC38  0002D5E0                               3448+          DC    A(FMSG17)           address of test fail message
0002CC3C  0002D340                               3449+          DC    A(OPSRC17)          address of operand source
0002CC40  0002D220                               3450+          DC    A(OPDST17)          address of operand destination
0002CC44  0002D460                               3451+          DC    A(PL17)             address of parameter list
0002CC48  00004189                               3452+          DC    A(RANDX+17)         address of random source data
0002CC4C  0002D655                               3453+          DC    A(TEND17)           address of test end eye catche
                                                 3454
                                                 3455           TSTCODE
0002CC50                                         3456+CODE17    DS    0F
                                                 3457 *                                   load random source
0002CC50  E310 501C 0014           0000001C      3458           LGF   R1,OPSRCA           R1: OPSRC - Primary
0002CC56  B904 0061                              3459           LGR   R6,R1               R6: OPSRC - Secondary
0002CC5A  B904 0071                              3460           LGR   R7,R1               R7: OPSRC - Home
                                                 3461
0002CC5E  E320 5028 0014           00000028      3462           LGF   R2,RANDXA           R2: address random data
0002CC64  E330 5024 0014           00000024      3463           LGF   R3,PLA              R3: address PL
                                                 3464
0002CC6A  B219 0200                00000200      3465           SAC   ARMODE
0002CC6E  D21F 1000 2000  00000000 00000000      3466           MVC   0(32,R1),0(R2)      load Op1 = Primary
0002CC74  D21F 6020 2020  00000020 00000020      3467           MVC   32(32,R6),32(R2)    load Op2 = Secondary
0002CC7A  D21F 7040 2040  00000040 00000040      3468           MVC   64(32,R7),64(R2)    load Op3 = Home
0002CC80  D21F 6060 2060  00000060 00000060      3469           MVC   96(32,R6),96(R2)    load Op4 = Secondary
                                                 3470 *
                                                 3471 * Quadruple Compare and Swap (QCSO: 37)
                                                 3472 * Build Parambter List
                                                 3473 *
0002CC86  D27F 3040 2000  00000040 00000000      3474           MVC   64(128,R3),0(R2)    Load PL OP1-Op4 Compare
0002CC8C  D27F 30C0 2080  000000C0 00000080      3475           MVC   192(128,R3),128(R2) Load PL Op1-Op4 Replacement
                                                 3476 ****
                                                 3477 * force a failure by changing the 4th operand
                                                 3478 ****
0002CC92  D207 6060 2100  00000060 00000100      3479           MVC   96(8,R6),256(R2)     some random data
                                                 3480 ****
                                                 3481
0002CC98  E313 0000 0024           00000000      3482           STG   R1,0(R3)            load OP1 address in PL
0002CC9E  E54C 3020 0000           00000020      3483           MVHI  32(R3),PRIALET
                                                 3484
0002CCA4  4111 0020                00000020      3485           LA    R1,32(R1)
0002CCA8  E313 0008 0024           00000008      3486           STG   R1,8(R3)            load OP2 address in PL
0002CCAE  E54C 3024 0001           00000024      3487           MVHI  36(R3),SECALET
                                                 3488
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002CCB4  4111 0020                    00000020  3489           LA    R1,32(R1)
0002CCB8  E313 0010 0024               00000010  3490           STG   R1,16(R3)           load OP3 address in PL
0002CCBE  E54C 3028 0002               00000028  3491           MVHI  40(R3),HOMEALET
                                                 3492
0002CCC4  4111 0020                    00000020  3493           LA    R1,32(R1)
0002CCC8  E313 0018 0024               00000018  3494           STG   R1,24(R3)           load OP4 address in PL
0002CCCE  E54C 302C 0001               0000002C  3495           MVHI  44(R3),SECALET
                                                 3496
0002CCD4  B982 0000                              3497           XGR   R0,R0
0002CCD8  4100 0025                    00000025  3498           LA    R0,37
                                                 3499
0002CCDC  B219 0200                    00000200  3500           SAC   ARMODE
0002CCE0  EE0F 0000 3000  00000000     00000000  3501           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSO: 37)
0002CCE6  B219 0000                    00000000  3502           SAC   PRIMODE
                                                 3503
0002CCEA  B98D 0020                              3504           EPSW  R2,R0               extract psw
0002CCEE  5020 5010                    00000010  3505           ST    R2,CCPSW              to save CC
0002CCF2  E310 500B 0076               0000000B  3506           LB    R1,CCMASK           (failure CC mask)
0002CCF8  8910 0004                    00000004  3507           SLL   R1,4                (shift to BC instr CC position)
0002CCFC  4410 513C                    0002CD5C  3508           EX    R1,F374TSTCC          fail if...
                                                 3509
                          0002CD00     00000001  3510 F374CKUR  EQU   *
0002CD00  D200 5009 3030  00000009     00000030  3511           MVC   URFOUND,48(R3)      Extract UR
0002CD06  D500 5008 3030  00000008     00000030  3512           CLC   UR,48(R3)
0002CD0C  4780 50F4                    0002CD14  3513           BE    F374CKRES
0002CD10  45F0 8388                    00002388  3514           BAL   R15,URMSG
                                                 3515 *
                                                 3516 * compare results
                                                 3517 *
                          0002CD14     00000001  3518 F374CKRES EQU   *
                                                 3519 *                                 reset addresses
0002CD14  E310 501C 0014               0000001C  3520           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002CD1A  B904 0061                              3521           LGR   R6,R1               R6:  OPSRC - Secondary
0002CD1E  B904 0071                              3522           LGR   R7,R1               R7:  OPSRC - Home
                                                 3523
0002CD22  B219 0200                    00000200  3524           SAC   ARMODE
0002CD26  D51F 1000 3040  00000000     00000040  3525           CLC   0(32,R1),64(R3)
0002CD2C  4770 5132                    0002CD52  3526           BNE   F374XX              OP1C Different, error
0002CD30  D51F 6020 3060  00000020     00000060  3527           CLC   32(32,R6),96(R3)
0002CD36  4770 5132                    0002CD52  3528           BNE   F374XX              OP2C Different, error
0002CD3A  D51F 7040 3080  00000040     00000080  3529           CLC   64(32,R7),128(R3)
0002CD40  4770 5132                    0002CD52  3530           BNE   F374XX              OP3C Different, error
0002CD44  D51F 6060 30A0  00000060     000000A0  3531           CLC   96(32,R6),160(R3)
0002CD4A  B219 0000                    00000000  3532           SAC   PRIMODE
0002CD4E  478B 0000                    00000000  3533           BE    0(R11)              OP4Csame, return
                                                 3534
                          0002CD52     00000001  3535 F374XX    EQU   *
0002CD52  B219 0000                    00000000  3536           SAC   PRIMODE
0002CD56  45F0 8450                    00002450  3537           BAL   R15,FAILMSG
0002CD5A  07FB                                   3538           BR    R11
                                                 3539
0002CD5C  4700 5140                    0002CD60  3540 F374TSTCC BC    0,F374CCMSG         fail if unexpected condition code
                          0002CD60     00000001  3541 F374CCMSG EQU   *
0002CD60  45F0 82B2                    000022B2  3542           BAL   R15,CCMSG
0002CD64  47F0 50E0                    0002CD00  3543           B     F374CKUR
                                                 3544
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3545           TSTDATA
0002CD68                      0002CD68  0002D200  3546+          ORG   PLOETST+(128*K)+17*(3*K)+(K+512)
                                                    3547           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002D200  D7D9C940 40406E6E                         3548+          DC    CL32'PRI   >>>OPDST17<<<' eye catcher
0002D208  6ED6D7C4 E2E3F1F7
0002D210  4C4C4C40 40404040
0002D218  40404040 40404040
0002D220                                            3549+OPDST17   DS    0C
0002D220  00000000 00000000                         3550           DC    XL256'0'
0002D228  00000000 00000000
0002D230  00000000 00000000
0002D238  00000000 00000000
0002D240  00000000 00000000
0002D248  00000000 00000000
0002D250  00000000 00000000
0002D258  00000000 00000000
0002D260  00000000 00000000
0002D268  00000000 00000000
0002D270  00000000 00000000
0002D278  00000000 00000000
0002D280  00000000 00000000
0002D288  00000000 00000000
0002D290  00000000 00000000
0002D298  00000000 00000000
0002D2A0  00000000 00000000
0002D2A8  00000000 00000000
0002D2B0  00000000 00000000
0002D2B8  00000000 00000000
0002D2C0  00000000 00000000
0002D2C8  00000000 00000000
0002D2D0  00000000 00000000
0002D2D8  00000000 00000000
0002D2E0  00000000 00000000
0002D2E8  00000000 00000000
0002D2F0  00000000 00000000
0002D2F8  00000000 00000000
0002D300  00000000 00000000
0002D308  00000000 00000000
0002D310  00000000 00000000
0002D318  00000000 00000000
                                                    3551
                                                    3552           TSTLBL OPSRC              Source: OP1-OP4
0002D320  D7D9C940 40406E6E                         3553+          DC    CL32'PRI   >>>OPSRC17<<<' eye catcher
0002D328  6ED6D7E2 D9C3F1F7
0002D330  4C4C4C40 40404040
0002D338  40404040 40404040
0002D340                                            3554+OPSRC17   DS    0C
0002D340  00000000 00000000                         3555           DC    XL256'0'
0002D348  00000000 00000000
0002D350  00000000 00000000
0002D358  00000000 00000000
0002D360  00000000 00000000
0002D368  00000000 00000000
0002D370  00000000 00000000
0002D378  00000000 00000000
0002D380  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002D388  00000000 00000000
0002D390  00000000 00000000
0002D398  00000000 00000000
0002D3A0  00000000 00000000
0002D3A8  00000000 00000000
0002D3B0  00000000 00000000
0002D3B8  00000000 00000000
0002D3C0  00000000 00000000
0002D3C8  00000000 00000000
0002D3D0  00000000 00000000
0002D3D8  00000000 00000000
0002D3E0  00000000 00000000
0002D3E8  00000000 00000000
0002D3F0  00000000 00000000
0002D3F8  00000000 00000000
0002D400  00000000 00000000
0002D408  00000000 00000000
0002D410  00000000 00000000
0002D418  00000000 00000000
0002D420  00000000 00000000
0002D428  00000000 00000000
0002D430  00000000 00000000
0002D438  00000000 00000000
                                        3556
                                        3557          TSTLBL PL                 Parameter List
0002D440  D7D9C940 40406E6E             3558+         DC    CL32'PRI   >>>PL17<<<'    eye catcher
0002D448  6ED7D3F1 F74C4C4C
0002D450  40404040 40404040
0002D458  40404040 40404040
0002D460                                3559+PL17     DS    0C
0002D460  00000000 00000000             3560          DC    XL352'00'
0002D468  00000000 00000000
0002D470  00000000 00000000
0002D478  00000000 00000000
0002D480  00000000 00000000
0002D488  00000000 00000000
0002D490  00000000 00000000
0002D498  00000000 00000000
0002D4A0  00000000 00000000
0002D4A8  00000000 00000000
0002D4B0  00000000 00000000
0002D4B8  00000000 00000000
0002D4C0  00000000 00000000
0002D4C8  00000000 00000000
0002D4D0  00000000 00000000
0002D4D8  00000000 00000000
0002D4E0  00000000 00000000
0002D4E8  00000000 00000000
0002D4F0  00000000 00000000
0002D4F8  00000000 00000000
0002D500  00000000 00000000
0002D508  00000000 00000000
0002D510  00000000 00000000
0002D518  00000000 00000000
0002D520  00000000 00000000
0002D528  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002D530  00000000 00000000
0002D538  00000000 00000000
0002D540  00000000 00000000
0002D548  00000000 00000000
0002D550  00000000 00000000
0002D558  00000000 00000000
0002D560  00000000 00000000
0002D568  00000000 00000000
0002D570  00000000 00000000
0002D578  00000000 00000000
0002D580  00000000 00000000
0002D588  00000000 00000000
0002D590  00000000 00000000
0002D598  00000000 00000000
0002D5A0  00000000 00000000
0002D5A8  00000000 00000000
0002D5B0  00000000 00000000
0002D5B8  00000000 00000000
                                       3561
                                       3562          TSTLBL FMSG               fail message
0002D5C0  D7D9C940 40406E6E            3563+         DC    CL32'PRI   >>>FMSG17<<<'  eye catcher
0002D5C8  6EC6D4E2 C7F1F74C
0002D5D0  4C4C4040 40404040
0002D5D8  40404040 40404040
0002D5E0                               3564+FMSG17   DS    0C
0002D5E0  86818993 858440D7            3565          DC    C'failed PLO-Extension '
0002D5E8  D3D660C5 A7A38595
0002D5F0  A2899695 40
0002D5F5  D8A48184 99A49793            3566          DC    CL96'Quadruple Compare and Swap (QCSO: 37): 4th NE'
0002D5FD  8540C396 94978199
0002D605  85408195 8440E2A6
0002D60D  8197404D D8C3E2D6
0002D615  7A40F3F7 5D7A40F4
0002D61D  A38840D5 C5404040
0002D625  40404040 40404040
0002D62D  40404040 40404040
0002D635  40404040 40404040
0002D63D  40404040 40404040
0002D645  40404040 40404040
0002D64D  40404040 40404040
                                       3567
                                       3568          TSTEND
0002D655  D7D9C940 4040C5D5            3569+TEND17   DC    CL32'PRI   END of test: 17 <<<'
0002D65D  C4409686 40A385A2
0002D665  A37A40F1 F7404C4C
0002D66D  4C404040 40404040
                                       3570
                                       3571
                                       3572 * ------------------------------------------------------------------
                                       3573 * Test Quadruple Compare and Swap (QCSO: 37) instruction : All EQ
                                       3574 * ------------------------------------------------------------------
                                       3575          TSTHDR 0,99,37             All Equal (cc=0, UR= unchanged)
0002D675                  0002D675  0002D800  3576+         ORG   PLOETST+(128*K)+18*(3*K)
0002D800  D7D9C940 4040E2E3            3577+         DC    CL32'PRI   START of test: 18'
0002D808  C1D9E340 968640A3
0002D810  85A2A37A 40F1F840
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002D818  40404040 40404040
0002D820                     0002D820              3578+          USING *,R5                base for test data and test routine
0002D820                     00000000              3579+          USING PLOETEST,R5
0002D820  0002D850                                 3580+T18       DC    A(CODE18)           address of test code
0002D824  0012                                     3581+          DC    H'18'               test number
0002D826  00                                       3582+          DC    X'00'
0002D827  25                                       3583+          DC    HL1'37'             Function
0002D828  63                                       3584+          DC    HL1'99'             expected unequal results (UR)
0002D829  00                                       3585+          DC    HL1'00'             found UR
0002D82A  00                                       3586+          DC    HL1'0'              cc
0002D82B  07                                       3587+          DC    HL1'7'              not expected CC mask
0002D82C  00                                       3588+          DS    X                   extracted cc
0002D830  00000000 00000000                        3589+          DS    2F                  extract PSW after test (has CC)
0002D838  0002E1E0                                 3590+          DC    A(FMSG18)           address of test fail message
0002D83C  0002DF40                                 3591+          DC    A(OPSRC18)          address of operand source
0002D840  0002DE20                                 3592+          DC    A(OPDST18)          address of operand destination
0002D844  0002E060                                 3593+          DC    A(PL18)             address of parameter list
0002D848  0000418A                                 3594+          DC    A(RANDX+18)         address of random source data
0002D84C  0002E255                                 3595+          DC    A(TEND18)           address of test end eye catche
                                                   3596
                                                   3597           TSTCODE
0002D850                                           3598+CODE18    DS    0F
                                                   3599 *                                   load random source
0002D850  E310 501C 0014                0000001C   3600           LGF   R1,OPSRCA           R1: OPSRC - Primary
0002D856  B904 0061                                3601           LGR   R6,R1               R6: OPSRC - Secondary
0002D85A  B904 0071                                3602           LGR   R7,R1               R7: OPSRC - Home
                                                   3603
0002D85E  E320 5028 0014                00000028   3604           LGF   R2,RANDXA           R2: address random data
0002D864  E330 5024 0014                00000024   3605           LGF   R3,PLA              R3: address PL
                                                   3606
0002D86A  B219 0200                     00000200   3607           SAC   ARMODE
0002D86E  D21F 1000 2000      00000000  00000000   3608           MVC   0(32,R1),0(R2)      load Op1 = Primary
0002D874  D21F 6020 2020      00000020  00000020   3609           MVC   32(32,R6),32(R2)    load Op2 = Secondary
0002D87A  D21F 7040 2040      00000040  00000040   3610           MVC   64(32,R7),64(R2)    load Op3 = Home
0002D880  D21F 6060 2060      00000060  00000060   3611           MVC   96(32,R6),96(R2)    load Op4 = Secondary
                                                   3612 *
                                                   3613 * Quadruple Compare and Swap (QCSO: 37)
                                                   3614 * Build Parambter List
                                                   3615 *
0002D886  D27F 3040 2000      00000040  00000000   3616           MVC   64(128,R3),0(R2)    Load PL OP1-Op4 Compare
0002D88C  D27F 30C0 2080      000000C0  00000080   3617           MVC   192(128,R3),128(R2) Load PL Op1-Op4 Replacement
                                                   3618 ****
                                                   3619 * Do NOT force a failure by changing an operand
                                                   3620 ****
                                                   3621 *           MVC   0(8,R1),256(R2)     some random data
                                                   3622 ****
                                                   3623
0002D892  E313 0000 0024                00000000   3624           STG   R1,0(R3)            load OP1 address in PL
0002D898  E54C 3020 0000                00000020   3625           MVHI  32(R3),PRIALET
                                                   3626
0002D89E  4111 0020                     00000020   3627           LA    R1,32(R1)
0002D8A2  E313 0008 0024                00000008   3628           STG   R1,8(R3)            load OP2 address in PL
0002D8A8  E54C 3024 0001                00000024   3629           MVHI  36(R3),SECALET
                                                   3630
0002D8AE  4111 0020                     00000020   3631           LA    R1,32(R1)
0002D8B2  E313 0010 0024                00000010   3632           STG   R1,16(R3)           load OP3 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002D8B8  E54C 3028 0002                 00000028  3633          MVHI  40(R3),HOMEALET          (NOTE: Home space)
                                                   3634
0002D8BE  4111 0020                      00000020  3635          LA    R1,32(R1)
0002D8C2  E313 0018 0024                 00000018  3636          STG   R1,24(R3)            load OP4 address in PL
0002D8C8  E54C 302C 0001                 0000002C  3637          MVHI  44(R3),SECALET
                                                   3638
0002D8CE  B982 0000                                3639          XGR   R0,R0
0002D8D2  4100 0025                      00000025  3640          LA    R0,37
                                                   3641
0002D8D6  B219 0200                      00000200  3642          SAC   ARMODE
0002D8DA  EE0F 0000 3000       00000000  00000000  3643          PLO   0,0,R15,0(R3)        Quadruple C/S (QCSO: 37)
0002D8E0  B219 0000                      00000000  3644          SAC   PRIMODE
                                                   3645
0002D8E4  B98D 0020                                3646          EPSW  R2,R0                extract psw
0002D8E8  5020 5010                      00000010  3647          ST    R2,CCPSW               to save CC
0002D8EC  E310 500B 0076                 0000000B  3648          LB    R1,CCMASK            (failure CC mask)
0002D8F2  8910 0004                      00000004  3649          SLL   R1,4                 (shift to BC instr CC position)
0002D8F6  4410 5136                      0002D956  3650          EX    R1,F375TSTCC           fail if...
                                                   3651
                                0002D8FA  00000001  3652 F375CKUR  EQU   *
0002D8FA  D200 5009 3030       00000009  00000030  3653          MVC   URFOUND,48(R3)       Extract UR
0002D900  D500 5008 3030       00000008  00000030  3654          CLC   UR,48(R3)
0002D906  4780 50EE                      0002D90E  3655          BE    F375CKRES
0002D90A  45F0 8388                      00002388  3656          BAL   R15,URMSG
                                                   3657 *
                                                   3658 * compare results
                                                   3659 *
                                0002D90E  00000001  3660 F375CKRES EQU   *
                                                   3661 *                                 reset addresses
0002D90E  E310 501C 0014                 0000001C  3662          LGF   R1,OPSRCA            R1:  OPSRC - Primary
0002D914  B904 0061                              3663          LGR   R6,R1                R6:  OPSRC - Secondary
0002D918  B904 0071                              3664          LGR   R7,R1                R7:  OPSRC - Home
                                                   3665
                                                   3666 * All equal; replacement values copied to operand source
0002D91C  B219 0200                      00000200  3667          SAC   ARMODE
0002D920  D51F 1000 30C0       00000000  000000C0  3668          CLC   0(32,R1),192(R3)
0002D926  4770 512C                      0002D94C  3669          BNE   F375XX               OP1R Different, error
0002D92A  D51F 6020 30E0       00000020  000000E0  3670          CLC   32(32,R6),224(R3)
0002D930  4770 512C                      0002D94C  3671          BNE   F375XX               OP2R Different, error
0002D934  D51F 7040 3100       00000040  00000100  3672          CLC   64(32,R7),256(R3)
0002D93A  4770 512C                      0002D94C  3673          BNE   F375XX               OP3R Different, error
0002D93E  D51F 6060 3120       00000060  00000120  3674          CLC   96(32,R6),288(R3)
0002D944  B219 0000                      00000000  3675          SAC   PRIMODE
0002D948  478B 0000                      00000000  3676          BE    0(R11)               OP4R same, return
                                                   3677
                                0002D94C  00000001  3678 F375XX   EQU   *
0002D94C  B219 0000                      00000000  3679          SAC   PRIMODE
0002D950  45F0 8450                      00002450  3680          BAL   R15,FAILMSG
0002D954  07FB                                     3681          BR    R11
                                                   3682
0002D956  4700 513A                      0002D95A  3683 F375TSTCC BC    0,F375CCMSG          fail if unexpected condition code
                                0002D95A  00000001  3684 F375CCMSG EQU   *
0002D95A  45F0 82B2                      000022B2  3685          BAL   R15,CCMSG
0002D95E  47F0 50DA                      0002D8FA  3686          B     F375CKUR
                                                   3687
                                                   3688          TSTDATA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002D962                     0002D962  0002DE00  3689+         ORG   PLOETST+(128*K)+18*(3*K)+(K+512)
                                                 3690          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002DE00  D7D9C940 40406E6E                      3691+         DC    CL32'PRI   >>>OPDST18<<<' eye catcher
0002DE08  6ED6D7C4 E2E3F1F8
0002DE10  4C4C4C40 40404040
0002DE18  40404040 40404040
0002DE20                                         3692+OPDST18  DS    0C
0002DE20  00000000 00000000                      3693          DC    XL256'0'
0002DE28  00000000 00000000
0002DE30  00000000 00000000
0002DE38  00000000 00000000
0002DE40  00000000 00000000
0002DE48  00000000 00000000
0002DE50  00000000 00000000
0002DE58  00000000 00000000
0002DE60  00000000 00000000
0002DE68  00000000 00000000
0002DE70  00000000 00000000
0002DE78  00000000 00000000
0002DE80  00000000 00000000
0002DE88  00000000 00000000
0002DE90  00000000 00000000
0002DE98  00000000 00000000
0002DEA0  00000000 00000000
0002DEA8  00000000 00000000
0002DEB0  00000000 00000000
0002DEB8  00000000 00000000
0002DEC0  00000000 00000000
0002DEC8  00000000 00000000
0002DED0  00000000 00000000
0002DED8  00000000 00000000
0002DEE0  00000000 00000000
0002DEE8  00000000 00000000
0002DEF0  00000000 00000000
0002DEF8  00000000 00000000
0002DF00  00000000 00000000
0002DF08  00000000 00000000
0002DF10  00000000 00000000
0002DF18  00000000 00000000
                                                 3694
                                                 3695          TSTLBL OPSRC              Source: OP1-OP4
0002DF20  D7D9C940 40406E6E                      3696+         DC    CL32'PRI   >>>OPSRC18<<<' eye catcher
0002DF28  6ED6D7E2 D9C3F1F8
0002DF30  4C4C4C40 40404040
0002DF38  40404040 40404040
0002DF40                                         3697+OPSRC18  DS    0C
0002DF40  00000000 00000000                      3698          DC    XL256'0'
0002DF48  00000000 00000000
0002DF50  00000000 00000000
0002DF58  00000000 00000000
0002DF60  00000000 00000000
0002DF68  00000000 00000000
0002DF70  00000000 00000000
0002DF78  00000000 00000000
0002DF80  00000000 00000000
0002DF88  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002DF90  00000000 00000000
0002DF98  00000000 00000000
0002DFA0  00000000 00000000
0002DFA8  00000000 00000000
0002DFB0  00000000 00000000
0002DFB8  00000000 00000000
0002DFC0  00000000 00000000
0002DFC8  00000000 00000000
0002DFD0  00000000 00000000
0002DFD8  00000000 00000000
0002DFE0  00000000 00000000
0002DFE8  00000000 00000000
0002DFF0  00000000 00000000
0002DFF8  00000000 00000000
0002E000  00000000 00000000
0002E008  00000000 00000000
0002E010  00000000 00000000
0002E018  00000000 00000000
0002E020  00000000 00000000
0002E028  00000000 00000000
0002E030  00000000 00000000
0002E038  00000000 00000000
                                          3699
                                          3700          TSTLBL PL                 Parameter List
0002E040  D7D9C940 40406E6E               3701+         DC    CL32'PRI   >>>PL18<<<'    eye catcher
0002E048  6ED7D3F1 F84C4C4C
0002E050  40404040 40404040
0002E058  40404040 40404040
0002E060                                  3702+PL18     DS    0C
0002E060  00000000 00000000               3703          DC    XL352'00'
0002E068  00000000 00000000
0002E070  00000000 00000000
0002E078  00000000 00000000
0002E080  00000000 00000000
0002E088  00000000 00000000
0002E090  00000000 00000000
0002E098  00000000 00000000
0002E0A0  00000000 00000000
0002E0A8  00000000 00000000
0002E0B0  00000000 00000000
0002E0B8  00000000 00000000
0002E0C0  00000000 00000000
0002E0C8  00000000 00000000
0002E0D0  00000000 00000000
0002E0D8  00000000 00000000
0002E0E0  00000000 00000000
0002E0E8  00000000 00000000
0002E0F0  00000000 00000000
0002E0F8  00000000 00000000
0002E100  00000000 00000000
0002E108  00000000 00000000
0002E110  00000000 00000000
0002E118  00000000 00000000
0002E120  00000000 00000000
0002E128  00000000 00000000
0002E130  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0002E138  00000000 00000000
0002E140  00000000 00000000
0002E148  00000000 00000000
0002E150  00000000 00000000
0002E158  00000000 00000000
0002E160  00000000 00000000
0002E168  00000000 00000000
0002E170  00000000 00000000
0002E178  00000000 00000000
0002E180  00000000 00000000
0002E188  00000000 00000000
0002E190  00000000 00000000
0002E198  00000000 00000000
0002E1A0  00000000 00000000
0002E1A8  00000000 00000000
0002E1B0  00000000 00000000
0002E1B8  00000000 00000000
                                        3704
                                        3705          TSTLBL FMSG               fail message
0002E1C0  D7D9C940 40406E6E             3706+         DC    CL32'PRI   >>>FMSG18<<<'  eye catcher
0002E1C8  6EC6D4E2 C7F1F84C
0002E1D0  4C4C4040 40404040
0002E1D8  40404040 40404040
0002E1E0                                3707+FMSG18   DS    0C
0002E1E0  86818993 858440D7             3708          DC    C'failed PLO-Extension '
0002E1E8  D3D660C5 A7A38595
0002E1F0  A2899695 40
0002E1F5  D8A48184 99A49793             3709          DC    CL96'Quadruple Compare and Swap (QCSO: 37): All EQ'
0002E1FD  8540C396 94978199
0002E205  85408195 8440E2A6
0002E20D  8197404D D8C3E2D6
0002E215  7A40F3F7 5D7A40C1
0002E21D  939340C5 D8404040
0002E225  40404040 40404040
0002E22D  40404040 40404040
0002E235  40404040 40404040
0002E23D  40404040 40404040
0002E245  40404040 40404040
0002E24D  40404040 40404040
                                        3710
                                        3711          TSTEND
0002E255  D7D9C940 4040C5D5             3712+TEND18   DC    CL32'PRI   END of test: 18 <<<'
0002E25D  C4409686 40A385A2
0002E265  A37A40F1 F8404C4C
0002E26D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   3714 *============================================================================
                                                   3715
                                                   3716 * ----------------------------------------------------------------------
                                                   3717 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 1st NE
                                                   3718 * ----------------------------------------------------------------------
                                                   3719          TSTHDR 2,1,36             1st Operand NOT Equal (cc=2, UR=1)
0002E275                          0002E275  0002E400 3720+         ORG   PLOETST+(128*K)+19*(3*K)
0002E400  D7D9C940 4040E2E3                        3721+         DC    CL32'PRI   START of test: 19'
0002E408  C1D9E340 968640A3
0002E410  85A2A37A 40F1F940
0002E418  40404040 40404040
0002E420                          0002E420         3722+         USING *,R5                base for test data and test routine
0002E420                          00000000         3723+         USING PLOETEST,R5
0002E420  0002E450                                 3724+T19      DC    A(CODE19)           address of test code
0002E424  0013                                     3725+         DC    H'19'               test number
0002E426  00                                       3726+         DC    X'00'
0002E427  24                                       3727+         DC    HL1'36'             Function
0002E428  01                                       3728+         DC    HL1'1'              expected unequal results (UR)
0002E429  00                                       3729+         DC    HL1'00'             found UR
0002E42A  02                                       3730+         DC    HL1'2'              cc
0002E42B  0D                                       3731+         DC    HL1'13'             not expected CC mask
0002E42C  00                                       3732+         DS    X                   extracted cc
0002E430  00000000 00000000                        3733+         DS    2F                  extract PSW after test (has CC)
0002E438  0002EDE0                                 3734+         DC    A(FMSG19)           address of test fail message
0002E43C  0002EB40                                 3735+         DC    A(OPSRC19)          address of operand source
0002E440  0002EA20                                 3736+         DC    A(OPDST19)          address of operand destination
0002E444  0002EC60                                 3737+         DC    A(PL19)             address of parameter list
0002E448  0000418B                                 3738+         DC    A(RANDX+19)         address of random source data
0002E44C  0002EE55                                 3739+         DC    A(TEND19)           address of test end eye catche
                                                   3740
                                                   3741          TSTCODE
0002E450                                           3742+CODE19   DS    0F
                                                   3743 *                                  load random source
0002E450  E310 501C 0014                0000001C   3744          LGF   R1,OPSRCA           R1: OPSRC - Primary
0002E456  B904 0061                                3745          LGR   R6,R1               R6: OPSRC - Secondary
0002E45A  B904 0071                                3746          LGR   R7,R1               R7: OPSRC - Home
                                                   3747
0002E45E  E320 5028 0014                00000028   3748          LGF   R2,RANDXA           R2: address random data
0002E464  E330 5024 0014                00000024   3749          LGF   R3,PLA              R3: address PL
                                                   3750
0002E46A  B219 0200                     00000200   3751          SAC   ARMODE
0002E46E  D20F 1000 2000  00000000      00000000   3752          MVC   0(16,R1),0(R2)      load Op1 = Primary
0002E474  D20F 6010 2010  00000010      00000010   3753          MVC   16(16,R6),16(R2)    load Op2 = Secondary
0002E47A  D20F 7020 2020  00000020      00000020   3754          MVC   32(16,R7),32(R2)    load Op3 = Home
0002E480  D20F 6030 2030  00000030      00000030   3755          MVC   48(16,R6),48(R2)    load Op4 = Secondary
                                                   3756 *
                                                   3757 * Quadruple Compare and Swap (QCSX: 36)
                                                   3758 * Build Parambter List
                                                   3759 *
0002E486  D23F 3040 2000  00000040      00000000   3760          MVC   64(64,R3),0(R2)     Load PL OP1-Op4 Compare
0002E48C  D23F 3080 2080  00000080      00000080   3761          MVC   128(64,R3),128(R2)  Load PL Op1-Op4 Replacement
                                                   3762 ****
                                                   3763 * force a failure by changing the first operand
                                                   3764 ****
0002E492  D207 1000 2100  00000000      00000100   3765          MVC   0(8,R1),256(R2)     some random data
                                                   3766 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    3767
0002E498  E313 0000 0024              00000000  3768          STG   R1,0(R3)            load OP1 address in PL
0002E49E  E54C 3020 0000              00000020  3769          MVHI  32(R3),PRIALET
                                                    3770
0002E4A4  4111 0010                   00000010  3771          LA    R1,16(R1)
0002E4A8  E313 0008 0024              00000008  3772          STG   R1,8(R3)            load OP2 address in PL
0002E4AE  E54C 3024 0001              00000024  3773          MVHI  36(R3),SECALET
                                                    3774
0002E4B4  4111 0010                   00000010  3775          LA    R1,16(R1)
0002E4B8  E313 0010 0024              00000010  3776          STG   R1,16(R3)           load OP3 address in PL
0002E4BE  E54C 3028 0002              00000028  3777          MVHI  40(R3),HOMEALET
                                                    3778
0002E4C4  4111 0010                   00000010  3779          LA    R1,16(R1)
0002E4C8  E313 0018 0024              00000018  3780          STG   R1,24(R3)           load OP4 address in PL
0002E4CE  E54C 302C 0001              0000002C  3781          MVHI  44(R3),SECALET
                                                    3782
0002E4D4  B982 0000                             3783          XGR   R0,R0
0002E4D8  4100 0024                   00000024  3784          LA    R0,36
                                                    3785
0002E4DC  B219 0200                   00000200  3786          SAC   ARMODE
0002E4E0  EE0F 0000 3000  00000000    00000000  3787          PLO   0,0,R15,0(R3)       Quadruple C/S (QCSX: 36)
0002E4E6  B219 0000                   00000000  3788          SAC   PRIMODE
                                                    3789
0002E4EA  B98D 0020                             3790          EPSW  R2,R0               extract psw
0002E4EE  5020 5010                   00000010  3791          ST    R2,CCPSW              to save CC
0002E4F2  E310 500B 0076              0000000B  3792          LB    R1,CCMASK           (failure CC mask)
0002E4F8  8910 0004                   00000004  3793          SLL   R1,4               (shift to BC instr CC position)
0002E4FC  4410 513C                   0002E55C  3794          EX    R1,F361TSTCC          fail if...
                                                    3795
                          0002E500    00000001  3796 F361CKUR EQU   *
0002E500  D200 5009 3030  00000009    00000030  3797          MVC   URFOUND,48(R3)      Extract UR
0002E506  D500 5008 3030  00000008    00000030  3798          CLC   UR,48(R3)
0002E50C  4780 50F4                   0002E514  3799          BE    F361CKRES
0002E510  45F0 8388                   00002388  3800          BAL   R15,URMSG
                                                    3801 *
                                                    3802 * compare results
                                                    3803 *
                          0002E514    00000001  3804 F361CKRES EQU   *
                                                    3805 *                          reset adresse
0002E514  E310 501C 0014              0000001C  3806          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002E51A  B904 0061                             3807          LGR   R6,R1               R6:  OPSRC - Secondary
0002E51E  B904 0071                             3808          LGR   R7,R1               R7:  OPSRC - Home
                                                    3809
0002E522  B219 0200                   00000200  3810          SAC   ARMODE
0002E526  D50F 1000 3040  00000000    00000040  3811          CLC   0(16,R1),64(R3)
0002E52C  4770 5132                   0002E552  3812          BNE   F361XX              OP1C Different, error
0002E530  D50F 6010 3050  00000010    00000050  3813          CLC   16(16,R6),80(R3)
0002E536  4770 5132                   0002E552  3814          BNE   F361XX              OP2C Different, error
0002E53A  D50F 7020 3060  00000020    00000060  3815          CLC   32(16,R7),96(R3)
0002E540  4770 5132                   0002E552  3816          BNE   F361XX              OP3C Different, error
0002E544  D50F 6030 3070  00000030    00000070  3817          CLC   48(16,R6),112(R3)
0002E54A  B219 0000                   00000000  3818          SAC   PRIMODE
0002E54E  478B 0000                   00000000  3819          BE    0(R11)              OP4Csame, return
                                                    3820
                          0002E552    00000001  3821 F361XX   EQU   *
0002E552  B219 0000                   00000000  3822          SAC   PRIMODE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002E556  45F0 8450                     00002450  3823           BAL   R15,FAILMSG
0002E55A  07FB                                    3824           BR    R11
                                                  3825
0002E55C  4700 5140                     0002E560  3826 F361TSTCC BC    0,F361CCMSG        fail if unexpected condition code
                                0002E560  00000001  3827 F361CCMSG EQU   *
0002E560  45F0 82B2                     000022B2  3828           BAL   R15,CCMSG
0002E564  47F0 50E0                     0002E500  3829           B     F361CKUR
                                                  3830
                                                  3831           TSTDATA
0002E568                        0002E568  0002EA00  3832+          ORG   PLOETST+(128*K)+19*(3*K)+(K+512)
                                                  3833           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002EA00  D7D9C940 40406E6E                       3834+          DC    CL32'PRI   >>>OPDST19<<<' eye catcher
0002EA08  6ED6D7C4 E2E3F1F9
0002EA10  4C4C4C40 40404040
0002EA18  40404040 40404040
0002EA20                                          3835+OPDST19   DS    0C
0002EA20  00000000 00000000                       3836           DC    XL256'0'
0002EA28  00000000 00000000
0002EA30  00000000 00000000
0002EA38  00000000 00000000
0002EA40  00000000 00000000
0002EA48  00000000 00000000
0002EA50  00000000 00000000
0002EA58  00000000 00000000
0002EA60  00000000 00000000
0002EA68  00000000 00000000
0002EA70  00000000 00000000
0002EA78  00000000 00000000
0002EA80  00000000 00000000
0002EA88  00000000 00000000
0002EA90  00000000 00000000
0002EA98  00000000 00000000
0002EAA0  00000000 00000000
0002EAA8  00000000 00000000
0002EAB0  00000000 00000000
0002EAB8  00000000 00000000
0002EAC0  00000000 00000000
0002EAC8  00000000 00000000
0002EAD0  00000000 00000000
0002EAD8  00000000 00000000
0002EAE0  00000000 00000000
0002EAE8  00000000 00000000
0002EAF0  00000000 00000000
0002EAF8  00000000 00000000
0002EB00  00000000 00000000
0002EB08  00000000 00000000
0002EB10  00000000 00000000
0002EB18  00000000 00000000
                                                  3837
                                                  3838           TSTLBL OPSRC              Source: OP1-OP4
0002EB20  D7D9C940 40406E6E                       3839+          DC    CL32'PRI   >>>OPSRC19<<<' eye catcher
0002EB28  6ED6D7E2 D9C3F1F9
0002EB30  4C4C4C40 40404040
0002EB38  40404040 40404040
0002EB40                                          3840+OPSRC19   DS    0C
0002EB40  00000000 00000000                       3841           DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002EB48  00000000 00000000
0002EB50  00000000 00000000
0002EB58  00000000 00000000
0002EB60  00000000 00000000
0002EB68  00000000 00000000
0002EB70  00000000 00000000
0002EB78  00000000 00000000
0002EB80  00000000 00000000
0002EB88  00000000 00000000
0002EB90  00000000 00000000
0002EB98  00000000 00000000
0002EBA0  00000000 00000000
0002EBA8  00000000 00000000
0002EBB0  00000000 00000000
0002EBB8  00000000 00000000
0002EBC0  00000000 00000000
0002EBC8  00000000 00000000
0002EBD0  00000000 00000000
0002EBD8  00000000 00000000
0002EBE0  00000000 00000000
0002EBE8  00000000 00000000
0002EBF0  00000000 00000000
0002EBF8  00000000 00000000
0002EC00  00000000 00000000
0002EC08  00000000 00000000
0002EC10  00000000 00000000
0002EC18  00000000 00000000
0002EC20  00000000 00000000
0002EC28  00000000 00000000
0002EC30  00000000 00000000
0002EC38  00000000 00000000
                                        3842
                                        3843          TSTLBL PL                 Parameter List
0002EC40  D7D9C940 40406E6E             3844+         DC    CL32'PRI   >>>PL19<<<'    eye catcher
0002EC48  6ED7D3F1 F94C4C4C
0002EC50  40404040 40404040
0002EC58  40404040 40404040
0002EC60                                3845+PL19     DS    0C
0002EC60  00000000 00000000             3846          DC    XL352'00'
0002EC68  00000000 00000000
0002EC70  00000000 00000000
0002EC78  00000000 00000000
0002EC80  00000000 00000000
0002EC88  00000000 00000000
0002EC90  00000000 00000000
0002EC98  00000000 00000000
0002ECA0  00000000 00000000
0002ECA8  00000000 00000000
0002ECB0  00000000 00000000
0002ECB8  00000000 00000000
0002ECC0  00000000 00000000
0002ECC8  00000000 00000000
0002ECD0  00000000 00000000
0002ECD8  00000000 00000000
0002ECE0  00000000 00000000
0002ECE8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002ECF0  00000000 00000000
0002ECF8  00000000 00000000
0002ED00  00000000 00000000
0002ED08  00000000 00000000
0002ED10  00000000 00000000
0002ED18  00000000 00000000
0002ED20  00000000 00000000
0002ED28  00000000 00000000
0002ED30  00000000 00000000
0002ED38  00000000 00000000
0002ED40  00000000 00000000
0002ED48  00000000 00000000
0002ED50  00000000 00000000
0002ED58  00000000 00000000
0002ED60  00000000 00000000
0002ED68  00000000 00000000
0002ED70  00000000 00000000
0002ED78  00000000 00000000
0002ED80  00000000 00000000
0002ED88  00000000 00000000
0002ED90  00000000 00000000
0002ED98  00000000 00000000
0002EDA0  00000000 00000000
0002EDA8  00000000 00000000
0002EDB0  00000000 00000000
0002EDB8  00000000 00000000
                                      3847
                                      3848          TSTLBL FMSG               fail message
0002EDC0  D7D9C940 40406E6E           3849+         DC    CL32'PRI   >>>FMSG19<<<'  eye catcher
0002EDC8  6EC6D4E2 C7F1F94C
0002EDD0  4C4C4040 40404040
0002EDD8  40404040 40404040
0002EDE0                              3850+FMSG19   DS    0C
0002EDE0  86818993 858440D7           3851          DC    C'failed PLO-Extension '
0002EDE8  D3D660C5 A7A38595
0002EDF0  A2899695 40
0002EDF5  D8A48184 99A49793           3852          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 1st NE'
0002EDFD  8540C396 94978199
0002EE05  85408195 8440E2A6
0002EE0D  8197404D D8C3E2E7
0002EE15  7A40F3F6 5D7A40F1
0002EE1D  A2A340D5 C5404040
0002EE25  40404040 40404040
0002EE2D  40404040 40404040
0002EE35  40404040 40404040
0002EE3D  40404040 40404040
0002EE45  40404040 40404040
0002EE4D  40404040 40404040
                                      3853
                                      3854          TSTEND
0002EE55  D7D9C940 4040C5D5           3855+TEND19   DC    CL32'PRI   END of test: 19 <<<'
0002EE5D  C4409686 40A385A2
0002EE65  A37A40F1 F9404C4C
0002EE6D  4C404040 40404040
                                      3856
                                      3857 * ------------------------------------------------------------------
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   3858 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 2nd NE
                                                   3859 * ------------------------------------------------------------------
                                                   3860          TSTHDR 2,2,36             2nd Operand NOT Equal (cc=2, UR=2)
0002EE75                       0002EE75  0002F000  3861+         ORG   PLOETST+(128*K)+20*(3*K)
0002F000  D7D9C940 4040E2E3                        3862+         DC    CL32'PRI   START of test: 20'
0002F008  C1D9E340 968640A3
0002F010  85A2A37A 40F2F040
0002F018  40404040 40404040
0002F020                       0002F020          3863+         USING *,R5                base for test data and test routine
0002F020                       00000000          3864+         USING PLOETEST,R5
0002F020  0002F050                               3865+T20      DC    A(CODE20)           address of test code
0002F024  0014                                   3866+         DC    H'20'               test number
0002F026  00                                     3867+         DC    X'00'
0002F027  24                                     3868+         DC    HL1'36'             Function
0002F028  02                                     3869+         DC    HL1'2'              expected unequal results (UR)
0002F029  00                                     3870+         DC    HL1'00'             found UR
0002F02A  02                                     3871+         DC    HL1'2'              cc
0002F02B  0D                                     3872+         DC    HL1'13'             not expected CC mask
0002F02C  00                                     3873+         DS    X                   extracted cc
0002F030  00000000 00000000                      3874+         DS    2F                  extract PSW after test (has CC)
0002F038  0002F9E0                               3875+         DC    A(FMSG20)           address of test fail message
0002F03C  0002F740                               3876+         DC    A(OPSRC20)          address of operand source
0002F040  0002F620                               3877+         DC    A(OPDST20)          address of operand destination
0002F044  0002F860                               3878+         DC    A(PL20)             address of parameter list
0002F048  0000418C                               3879+         DC    A(RANDX+20)         address of random source data
0002F04C  0002FA55                               3880+         DC    A(TEND20)           address of test end eye catche
                                                   3881
                                                   3882          TSTCODE
0002F050                                         3883+CODE20   DS    0F
                                                   3884 *                                  load random source
0002F050  E310 501C 0014             0000001C    3885          LGF   R1,OPSRCA           R1: OPSRC - Primary
0002F056  B904 0061                              3886          LGR   R6,R1               R6: OPSRC - Secondary
0002F05A  B904 0071                              3887          LGR   R7,R1               R7: OPSRC - Home
                                                   3888
0002F05E  E320 5028 0014             00000028    3889          LGF   R2,RANDXA           R2: address random data
0002F064  E330 5024 0014             00000024    3890          LGF   R3,PLA              R3: address PL
                                                   3891
0002F06A  B219 0200                  00000200    3892          SAC   ARMODE
0002F06E  D20F 1000 2000   00000000  00000000    3893          MVC   0(16,R1),0(R2)      load Op1 = Primary
0002F074  D20F 6010 2010   00000010  00000010    3894          MVC   16(16,R6),16(R2)    load Op2 = Secondary
0002F07A  D20F 7020 2020   00000020  00000020    3895          MVC   32(16,R7),32(R2)    load Op3 = Home
0002F080  D20F 6030 2030   00000030  00000030    3896          MVC   48(16,R6),48(R2)    load Op4 = Secondary
                                                   3897 *
                                                   3898 * Quadruple Compare and Swap (QCSX: 36)
                                                   3899 * Build Parambter List
                                                   3900 *
0002F086  D23F 3040 2000   00000040  00000000    3901          MVC   64(64,R3),0(R2)     Load PL OP1-Op4 Compare
0002F08C  D23F 3080 2080   00000080  00000080    3902          MVC   128(64,R3),128(R2)  Load PL Op1-Op4 Replacement
                                                   3903 ****
                                                   3904 * force a failure by changing the 2nd operand
                                                   3905 ****
0002F092  D207 6010 2100   00000010  00000100    3906          MVC   16(8,R6),256(R2)     some random data
                                                   3907 ****
0002F098  E313 0000 0024             00000000    3908          STG   R1,0(R3)            load OP1 address in PL
0002F09E  E54C 3020 0000             00000020    3909          MVHI  32(R3),PRIALET
                                                   3910
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002F0A4  4111 0010                     00000010  3911           LA    R1,16(R1)
0002F0A8  E313 0008 0024                00000008  3912           STG   R1,8(R3)            load OP2 address in PL
0002F0AE  E54C 3024 0001                00000024  3913           MVHI  36(R3),SECALET
                                                  3914
0002F0B4  4111 0010                     00000010  3915           LA    R1,16(R1)
0002F0B8  E313 0010 0024                00000010  3916           STG   R1,16(R3)           load OP3 address in PL
0002F0BE  E54C 3028 0002                00000028  3917           MVHI  40(R3),HOMEALET
                                                  3918
0002F0C4  4111 0010                     00000010  3919           LA    R1,16(R1)
0002F0C8  E313 0018 0024                00000018  3920           STG   R1,24(R3)           load OP4 address in PL
0002F0CE  E54C 302C 0001                0000002C  3921           MVHI  44(R3),SECALET
                                                  3922
0002F0D4  B982 0000                               3923           XGR   R0,R0
0002F0D8  4100 0024                     00000024  3924           LA    R0,36
                                                  3925
0002F0DC  B219 0200                     00000200  3926           SAC   ARMODE
0002F0E0  EE0F 0000 3000  00000000      00000000  3927           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSX: 36)
0002F0E6  B219 0000                     00000000  3928           SAC   PRIMODE
                                                  3929
0002F0EA  B98D 0020                               3930           EPSW  R2,R0               extract psw
0002F0EE  5020 5010                     00000010  3931           ST    R2,CCPSW              to save CC
0002F0F2  E310 500B 0076                0000000B  3932           LB    R1,CCMASK           (failure CC mask)
0002F0F8  8910 0004                     00000004  3933           SLL   R1,4                (shift to BC instr CC position)
0002F0FC  4410 513C                     0002F15C  3934           EX    R1,F362TSTCC            fail if...
                                                  3935
                          0002F100      00000001  3936 F362CKUR  EQU   *
0002F100  D200 5009 3030  00000009      00000030  3937           MVC   URFOUND,48(R3)      Extract UR
0002F106  D500 5008 3030  00000008      00000030  3938           CLC   UR,48(R3)
0002F10C  4780 50F4                     0002F114  3939           BE    F362CKRES
0002F110  45F0 8388                     00002388  3940           BAL   R15,URMSG
                                                  3941 *
                                                  3942 * compare results
                                                  3943 *
                          0002F114      00000001  3944 F362CKRES EQU   *
                                                  3945 *                                  reset addresses
0002F114  E310 501C 0014                0000001C  3946           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002F11A  B904 0061                               3947           LGR   R6,R1               R6:  OPSRC - Secondary
0002F11E  B904 0071                               3948           LGR   R7,R1               R7:  OPSRC - Home
                                                  3949
0002F122  B219 0200                     00000200  3950           SAC   ARMODE
0002F126  D50F 1000 3040  00000000      00000040  3951           CLC   0(16,R1),64(R3)
0002F12C  4770 5132                     0002F152  3952           BNE   F362XX              OP1C Different, error
0002F130  D50F 6010 3050  00000010      00000050  3953           CLC   16(16,R6),80(R3)
0002F136  4770 5132                     0002F152  3954           BNE   F362XX              OP2C Different, error
0002F13A  D50F 7020 3060  00000020      00000060  3955           CLC   32(16,R7),96(R3)
0002F140  4770 5132                     0002F152  3956           BNE   F362XX              OP3C Different, error
0002F144  D50F 6030 3070  00000030      00000070  3957           CLC   48(16,R6),112(R3)
0002F14A  B219 0000                     00000000  3958           SAC   PRIMODE
0002F14E  478B 0000                     00000000  3959           BE    0(R11)              OP4Csame, return
                                                  3960
                          0002F152      00000001  3961 F362XX    EQU   *
0002F152  B219 0000                     00000000  3962           SAC   PRIMODE
0002F156  45F0 8450                     00002450  3963           BAL   R15,FAILMSG
0002F15A  07FB                                    3964           BR    R11
                                                  3965
0002F15C  4700 5140                     0002F160  3966 F362TSTCC BC    0,F362CCMSG         fail if unexpected condition code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               0002F160  00000001  3967 F362CCMSG EQU   *
0002F160  45F0 82B2                      000022B2  3968           BAL   R15,CCMSG
0002F164  47F0 50E0                      0002F100  3969           B     F362CKUR
                                                   3970
                                                   3971           TSTDATA
0002F168                     0002F168  0002F600  3972+          ORG   PLOETST+(128*K)+20*(3*K)+(K+512)
                                                   3973           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
0002F600  D7D9C940 40406E6E                        3974+          DC    CL32'PRI   >>>OPDST20<<<' eye catcher
0002F608  6ED6D7C4 E2E3F2F0
0002F610  4C4C4C40 40404040
0002F618  40404040 40404040
0002F620                                           3975+OPDST20   DS    0C
0002F620  00000000 00000000                        3976           DC    XL256'0'
0002F628  00000000 00000000
0002F630  00000000 00000000
0002F638  00000000 00000000
0002F640  00000000 00000000
0002F648  00000000 00000000
0002F650  00000000 00000000
0002F658  00000000 00000000
0002F660  00000000 00000000
0002F668  00000000 00000000
0002F670  00000000 00000000
0002F678  00000000 00000000
0002F680  00000000 00000000
0002F688  00000000 00000000
0002F690  00000000 00000000
0002F698  00000000 00000000
0002F6A0  00000000 00000000
0002F6A8  00000000 00000000
0002F6B0  00000000 00000000
0002F6B8  00000000 00000000
0002F6C0  00000000 00000000
0002F6C8  00000000 00000000
0002F6D0  00000000 00000000
0002F6D8  00000000 00000000
0002F6E0  00000000 00000000
0002F6E8  00000000 00000000
0002F6F0  00000000 00000000
0002F6F8  00000000 00000000
0002F700  00000000 00000000
0002F708  00000000 00000000
0002F710  00000000 00000000
0002F718  00000000 00000000
                                                   3977
                                                   3978           TSTLBL OPSRC              Source: OP1-OP4
0002F720  D7D9C940 40406E6E                        3979+          DC    CL32'PRI   >>>OPSRC20<<<' eye catcher
0002F728  6ED6D7E2 D9C3F2F0
0002F730  4C4C4C40 40404040
0002F738  40404040 40404040
0002F740                                           3980+OPSRC20   DS    0C
0002F740  00000000 00000000                        3981           DC    XL256'0'
0002F748  00000000 00000000
0002F750  00000000 00000000
0002F758  00000000 00000000
0002F760  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0002F768  00000000 00000000
0002F770  00000000 00000000
0002F778  00000000 00000000
0002F780  00000000 00000000
0002F788  00000000 00000000
0002F790  00000000 00000000
0002F798  00000000 00000000
0002F7A0  00000000 00000000
0002F7A8  00000000 00000000
0002F7B0  00000000 00000000
0002F7B8  00000000 00000000
0002F7C0  00000000 00000000
0002F7C8  00000000 00000000
0002F7D0  00000000 00000000
0002F7D8  00000000 00000000
0002F7E0  00000000 00000000
0002F7E8  00000000 00000000
0002F7F0  00000000 00000000
0002F7F8  00000000 00000000
0002F800  00000000 00000000
0002F808  00000000 00000000
0002F810  00000000 00000000
0002F818  00000000 00000000
0002F820  00000000 00000000
0002F828  00000000 00000000
0002F830  00000000 00000000
0002F838  00000000 00000000
                                      3982
                                      3983           TSTLBL PL                  Parameter List
0002F840  D7D9C940 40406E6E           3984+          DC    CL32'PRI   >>>PL20<<<'    eye catcher
0002F848  6ED7D3F2 F04C4C4C
0002F850  40404040 40404040
0002F858  40404040 40404040
0002F860                              3985+PL20      DS    0C
0002F860  00000000 00000000           3986           DC    XL352'00'
0002F868  00000000 00000000
0002F870  00000000 00000000
0002F878  00000000 00000000
0002F880  00000000 00000000
0002F888  00000000 00000000
0002F890  00000000 00000000
0002F898  00000000 00000000
0002F8A0  00000000 00000000
0002F8A8  00000000 00000000
0002F8B0  00000000 00000000
0002F8B8  00000000 00000000
0002F8C0  00000000 00000000
0002F8C8  00000000 00000000
0002F8D0  00000000 00000000
0002F8D8  00000000 00000000
0002F8E0  00000000 00000000
0002F8E8  00000000 00000000
0002F8F0  00000000 00000000
0002F8F8  00000000 00000000
0002F900  00000000 00000000
0002F908  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002F910  00000000 00000000
0002F918  00000000 00000000
0002F920  00000000 00000000
0002F928  00000000 00000000
0002F930  00000000 00000000
0002F938  00000000 00000000
0002F940  00000000 00000000
0002F948  00000000 00000000
0002F950  00000000 00000000
0002F958  00000000 00000000
0002F960  00000000 00000000
0002F968  00000000 00000000
0002F970  00000000 00000000
0002F978  00000000 00000000
0002F980  00000000 00000000
0002F988  00000000 00000000
0002F990  00000000 00000000
0002F998  00000000 00000000
0002F9A0  00000000 00000000
0002F9A8  00000000 00000000
0002F9B0  00000000 00000000
0002F9B8  00000000 00000000
                                        3987
                                        3988          TSTLBL FMSG               fail message
0002F9C0  D7D9C940 40406E6E             3989+         DC    CL32'PRI   >>>FMSG20<<<'  eye catcher
0002F9C8  6EC6D4E2 C7F2F04C
0002F9D0  4C4C4040 40404040
0002F9D8  40404040 40404040
0002F9E0                                3990+FMSG20   DS    0C
0002F9E0  86818993 858440D7             3991          DC    C'failed PLO-Extension '
0002F9E8  D3D660C5 A7A38595
0002F9F0  A2899695 40
0002F9F5  D8A48184 99A49793             3992          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 2nd NE'
0002F9FD  8540C396 94978199
0002FA05  85408195 8440E2A6
0002FA0D  8197404D D8C3E2E7
0002FA15  7A40F3F6 5D7A40F2
0002FA1D  958440D5 C5404040
0002FA25  40404040 40404040
0002FA2D  40404040 40404040
0002FA35  40404040 40404040
0002FA3D  40404040 40404040
0002FA45  40404040 40404040
0002FA4D  40404040 40404040
                                        3993
                                        3994          TSTEND
0002FA55  D7D9C940 4040C5D5             3995+TEND20   DC    CL32'PRI   END of test: 20 <<<'
0002FA5D  C4409686 40A385A2
0002FA65  A37A40F2 F0404C4C
0002FA6D  4C404040 40404040
                                        3996
                                        3997 * ------------------------------------------------------------------
                                        3998 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 3rd NE
                                        3999 * ------------------------------------------------------------------
                                        4000          TSTHDR 2,3,36             3rd Operand NOT Equal (cc=2, UR=3)
0002FA75                  0002FA75  0002FC00  4001+         ORG   PLOETST+(128*K)+21*(3*K)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002FC00  D7D9C940 4040E2E3                        4002+         DC    CL32'PRI   START of test: 21'
0002FC08  C1D9E340 968640A3
0002FC10  85A2A37A 40F2F140
0002FC18  40404040 40404040
0002FC20                     0002FC20              4003+         USING *,R5                base for test data and test routine
0002FC20                     00000000              4004+         USING PLOETEST,R5
0002FC20  0002FC50                                 4005+T21      DC    A(CODE21)           address of test code
0002FC24  0015                                     4006+         DC    H'21'               test number
0002FC26  00                                       4007+         DC    X'00'
0002FC27  24                                       4008+         DC    HL1'36'             Function
0002FC28  03                                       4009+         DC    HL1'3'              expected unequal results (UR)
0002FC29  00                                       4010+         DC    HL1'00'             found UR
0002FC2A  02                                       4011+         DC    HL1'2'              cc
0002FC2B  0D                                       4012+         DC    HL1'13'             not expected CC mask
0002FC2C  00                                       4013+         DS    X                   extracted cc
0002FC30  00000000 00000000                        4014+         DS    2F                  extract PSW after test (has CC)
0002FC38  000305E0                                 4015+         DC    A(FMSG21)           address of test fail message
0002FC3C  00030340                                 4016+         DC    A(OPSRC21)          address of operand source
0002FC40  00030220                                 4017+         DC    A(OPDST21)          address of operand destination
0002FC44  00030460                                 4018+         DC    A(PL21)             address of parameter list
0002FC48  0000418D                                 4019+         DC    A(RANDX+21)         address of random source data
0002FC4C  00030655                                 4020+         DC    A(TEND21)           address of test end eye catche
                                                   4021
                                                   4022          TSTCODE
0002FC50                                           4023+CODE21   DS    0F
                                                   4024 *                                  load random source
0002FC50  E310 501C 0014                0000001C   4025          LGF   R1,OPSRCA           R1: OPSRC - Primary
0002FC56  B904 0061                                4026          LGR   R6,R1               R6: OPSRC - Secondary
0002FC5A  B904 0071                                4027          LGR   R7,R1               R7: OPSRC - Home
                                                   4028
0002FC5E  E320 5028 0014                00000028   4029          LGF   R2,RANDXA           R2: address random data
0002FC64  E330 5024 0014                00000024   4030          LGF   R3,PLA              R3: address PL
                                                   4031
0002FC6A  B219 0200                     00000200   4032          SAC   ARMODE
0002FC6E  D20F 1000 2000      00000000  00000000   4033          MVC   0(16,R1),0(R2)      load Op1 = Primary
0002FC74  D20F 6010 2010      00000010  00000010   4034          MVC   16(16,R6),16(R2)    load Op2 = Secondary
0002FC7A  D20F 7020 2020      00000020  00000020   4035          MVC   32(16,R7),32(R2)    load Op3 = Home
0002FC80  D20F 6030 2030      00000030  00000030   4036          MVC   48(16,R6),48(R2)    load Op4 = Secondary
                                                   4037 *
                                                   4038 * Quadruple Compare and Swap (QCSX: 36)
                                                   4039 * Build Parambter List
                                                   4040 *
0002FC86  D23F 3040 2000      00000040  00000000   4041          MVC   64(64,R3),0(R2)     Load PL OP1-Op4 Compare
0002FC8C  D23F 3080 2080      00000080  00000080   4042          MVC   128(64,R3),128(R2)  Load PL Op1-Op4 Replacement
                                                   4043 ****
                                                   4044 * force a failure by changing the 3nd operand
                                                   4045 ****
0002FC92  D207 7020 2100      00000020  00000100   4046          MVC   32(8,R7),256(R2)
                                                   4047 ****
0002FC98  E313 0000 0024                00000000   4048          STG   R1,0(R3)            load OP1 address in PL
0002FC9E  E54C 3020 0000                00000020   4049          MVHI  32(R3),PRIALET
                                                   4050
0002FCA4  4111 0010                     00000010   4051          LA    R1,16(R1)
0002FCA8  E313 0008 0024                00000008   4052          STG   R1,8(R3)            load OP2 address in PL
0002FCAE  E54C 3024 0001                00000024   4053          MVHI  36(R3),SECALET
                                                   4054
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0002FCB4  4111 0010                    00000010  4055           LA    R1,16(R1)
0002FCB8  E313 0010 0024               00000010  4056           STG   R1,16(R3)           load OP3 address in PL
0002FCBE  E54C 3028 0002               00000028  4057           MVHI  40(R3),HOMEALET
                                                 4058
0002FCC4  4111 0010                    00000010  4059           LA    R1,16(R1)
0002FCC8  E313 0018 0024               00000018  4060           STG   R1,24(R3)           load OP4 address in PL
0002FCCE  E54C 302C 0001               0000002C  4061           MVHI  44(R3),SECALET
                                                 4062
0002FCD4  B982 0000                              4063           XGR   R0,R0
0002FCD8  4100 0024                    00000024  4064           LA    R0,36
                                                 4065
0002FCDC  B219 0200                    00000200  4066           SAC   ARMODE
0002FCE0  EE0F 0000 3000  00000000     00000000  4067           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSX: 36)
0002FCE6  B219 0000                    00000000  4068           SAC   PRIMODE
                                                 4069
0002FCEA  B98D 0020                              4070           EPSW  R2,R0               extract psw
0002FCEE  5020 5010                    00000010  4071           ST    R2,CCPSW              to save CC
0002FCF2  E310 500B 0076               0000000B  4072           LB    R1,CCMASK           (failure CC mask)
0002FCF8  8910 0004                    00000004  4073           SLL   R1,4                (shift to BC instr CC position)
0002FCFC  4410 513C                    0002FD5C  4074           EX    R1,F363TSTCC          fail if...
                                                 4075
                          0002FD00     00000001  4076 F363CKUR  EQU   *
0002FD00  D200 5009 3030  00000009     00000030  4077           MVC   URFOUND,48(R3)      Extract UR
0002FD06  D500 5008 3030  00000008     00000030  4078           CLC   UR,48(R3)
0002FD0C  4780 50F4                    0002FD14  4079           BE    F363CKRES
0002FD10  45F0 8388                    00002388  4080           BAL   R15,URMSG
                                                 4081 *
                                                 4082 * compare results
                                                 4083 *
                          0002FD14     00000001  4084 F363CKRES EQU   *
                                                 4085 *                                 reset addresses
0002FD14  E310 501C 0014               0000001C  4086           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0002FD1A  B904 0061                              4087           LGR   R6,R1               R6:  OPSRC - Secondary
0002FD1E  B904 0071                              4088           LGR   R7,R1               R7:  OPSRC - Home
                                                 4089
0002FD22  B219 0200                    00000200  4090           SAC   ARMODE
0002FD26  D50F 1000 3040  00000000     00000040  4091           CLC   0(16,R1),64(R3)
0002FD2C  4770 5132                    0002FD52  4092           BNE   F363XX              OP1C Different, error
0002FD30  D50F 6010 3050  00000010     00000050  4093           CLC   16(16,R6),80(R3)
0002FD36  4770 5132                    0002FD52  4094           BNE   F363XX              OP2C Different, error
0002FD3A  D50F 7020 3060  00000020     00000060  4095           CLC   32(16,R7),96(R3)
0002FD40  4770 5132                    0002FD52  4096           BNE   F363XX              OP3C Different, error
0002FD44  D50F 6030 3070  00000030     00000070  4097           CLC   48(16,R6),112(R3)
0002FD4A  B219 0000                    00000000  4098           SAC   PRIMODE
0002FD4E  478B 0000                    00000000  4099           BE    0(R11)              OP4Csame, return
                                                 4100
                          0002FD52     00000001  4101 F363XX    EQU   *
0002FD52  B219 0000                    00000000  4102           SAC   PRIMODE
0002FD56  45F0 8450                    00002450  4103           BAL   R15,FAILMSG
0002FD5A  07FB                                   4104           BR    R11
                                                 4105
0002FD5C  4700 5140                    0002FD60  4106 F363TSTCC BC    0,F363CCMSG         fail if unexpected condition code
                          0002FD60     00000001  4107 F363CCMSG EQU   *
0002FD60  45F0 82B2                    000022B2  4108           BAL   R15,CCMSG
0002FD64  47F0 50E0                    0002FD00  4109           B     F363CKUR
                                                 4110
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4111           TSTDATA
0002FD68                      0002FD68  00030200  4112+          ORG   PLOETST+(128*K)+21*(3*K)+(K+512)
                                                    4113           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00030200  D7D9C940 40406E6E                         4114+          DC    CL32'PRI   >>>OPDST21<<<' eye catcher
00030208  6ED6D7C4 E2E3F2F1
00030210  4C4C4C40 40404040
00030218  40404040 40404040
00030220                                            4115+OPDST21   DS    0C
00030220  00000000 00000000                         4116           DC    XL256'0'
00030228  00000000 00000000
00030230  00000000 00000000
00030238  00000000 00000000
00030240  00000000 00000000
00030248  00000000 00000000
00030250  00000000 00000000
00030258  00000000 00000000
00030260  00000000 00000000
00030268  00000000 00000000
00030270  00000000 00000000
00030278  00000000 00000000
00030280  00000000 00000000
00030288  00000000 00000000
00030290  00000000 00000000
00030298  00000000 00000000
000302A0  00000000 00000000
000302A8  00000000 00000000
000302B0  00000000 00000000
000302B8  00000000 00000000
000302C0  00000000 00000000
000302C8  00000000 00000000
000302D0  00000000 00000000
000302D8  00000000 00000000
000302E0  00000000 00000000
000302E8  00000000 00000000
000302F0  00000000 00000000
000302F8  00000000 00000000
00030300  00000000 00000000
00030308  00000000 00000000
00030310  00000000 00000000
00030318  00000000 00000000
                                                    4117
                                                    4118           TSTLBL OPSRC              Source: OP1-OP4
00030320  D7D9C940 40406E6E                         4119+          DC    CL32'PRI   >>>OPSRC21<<<' eye catcher
00030328  6ED6D7E2 D9C3F2F1
00030330  4C4C4C40 40404040
00030338  40404040 40404040
00030340                                            4120+OPSRC21   DS    0C
00030340  00000000 00000000                         4121           DC    XL256'0'
00030348  00000000 00000000
00030350  00000000 00000000
00030358  00000000 00000000
00030360  00000000 00000000
00030368  00000000 00000000
00030370  00000000 00000000
00030378  00000000 00000000
00030380  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00030388  00000000 00000000
00030390  00000000 00000000
00030398  00000000 00000000
000303A0  00000000 00000000
000303A8  00000000 00000000
000303B0  00000000 00000000
000303B8  00000000 00000000
000303C0  00000000 00000000
000303C8  00000000 00000000
000303D0  00000000 00000000
000303D8  00000000 00000000
000303E0  00000000 00000000
000303E8  00000000 00000000
000303F0  00000000 00000000
000303F8  00000000 00000000
00030400  00000000 00000000
00030408  00000000 00000000
00030410  00000000 00000000
00030418  00000000 00000000
00030420  00000000 00000000
00030428  00000000 00000000
00030430  00000000 00000000
00030438  00000000 00000000
                                        4122
                                        4123          TSTLBL PL                 Parameter List
00030440  D7D9C940 40406E6E             4124+         DC    CL32'PRI   >>>PL21<<<'    eye catcher
00030448  6ED7D3F2 F14C4C4C
00030450  40404040 40404040
00030458  40404040 40404040
00030460                                4125+PL21     DS    0C
00030460  00000000 00000000             4126          DC    XL352'00'
00030468  00000000 00000000
00030470  00000000 00000000
00030478  00000000 00000000
00030480  00000000 00000000
00030488  00000000 00000000
00030490  00000000 00000000
00030498  00000000 00000000
000304A0  00000000 00000000
000304A8  00000000 00000000
000304B0  00000000 00000000
000304B8  00000000 00000000
000304C0  00000000 00000000
000304C8  00000000 00000000
000304D0  00000000 00000000
000304D8  00000000 00000000
000304E0  00000000 00000000
000304E8  00000000 00000000
000304F0  00000000 00000000
000304F8  00000000 00000000
00030500  00000000 00000000
00030508  00000000 00000000
00030510  00000000 00000000
00030518  00000000 00000000
00030520  00000000 00000000
00030528  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00030530  00000000 00000000
00030538  00000000 00000000
00030540  00000000 00000000
00030548  00000000 00000000
00030550  00000000 00000000
00030558  00000000 00000000
00030560  00000000 00000000
00030568  00000000 00000000
00030570  00000000 00000000
00030578  00000000 00000000
00030580  00000000 00000000
00030588  00000000 00000000
00030590  00000000 00000000
00030598  00000000 00000000
000305A0  00000000 00000000
000305A8  00000000 00000000
000305B0  00000000 00000000
000305B8  00000000 00000000
                                        4127
                                        4128          TSTLBL FMSG               fail message
000305C0  D7D9C940 40406E6E             4129+         DC    CL32'PRI   >>>FMSG21<<<'  eye catcher
000305C8  6EC6D4E2 C7F2F14C
000305D0  4C4C4040 40404040
000305D8  40404040 40404040
000305E0                                4130+FMSG21   DS    0C
000305E0  86818993 858440D7             4131          DC    C'failed PLO-Extension '
000305E8  D3D660C5 A7A38595
000305F0  A2899695 40
000305F5  D8A48184 99A49793                           4132          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 3rd NE'
000305FD  8540C396 94978199
00030605  85408195 8440E2A6
0003060D  8197404D D8C3E2E7
00030615  7A40F3F6 5D7A40F3
0003061D  998440D5 C5404040
00030625  40404040 40404040
0003062D  40404040 40404040
00030635  40404040 40404040
0003063D  40404040 40404040
00030645  40404040 40404040
0003064D  40404040 40404040
                                        4133
                                        4134          TSTEND
00030655  D7D9C940 4040C5D5             4135+TEND21   DC    CL32'PRI   END of test: 21 <<<'
0003065D  C4409686 40A385A2
00030665  A37A40F2 F1404C4C
0003066D  4C404040 40404040
                                        4136
                                        4137 * ------------------------------------------------------------------------
                                        4138 * Test Quadruple Compare and Swap (QCSX: 36) instruction : 4th NE
                                        4139 * ------------------------------------------------------------------------
                                        4140          TSTHDR 2,4,36             4th Operand NOT Equal (cc=2, UR=4)
00030675                        00030675  00030800    4141+         ORG   PLOETST+(128*K)+22*(3*K)
00030800  D7D9C940 4040E2E3             4142+         DC    CL32'PRI   START of test: 22'
00030808  C1D9E340 968640A3
00030810  85A2A37A 40F2F240
00030818  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00030820                     00030820              4143+          USING *,R5                base for test data and test routine
00030820                     00000000              4144+          USING PLOETEST,R5
00030820  00030850                                 4145+T22       DC    A(CODE22)           address of test code
00030824  0016                                     4146+          DC    H'22'               test number
00030826  00                                       4147+          DC    X'00'
00030827  24                                       4148+          DC    HL1'36'             Function
00030828  04                                       4149+          DC    HL1'4'              expected unequal results (UR)
00030829  00                                       4150+          DC    HL1'00'             found UR
0003082A  02                                       4151+          DC    HL1'2'              cc
0003082B  0D                                       4152+          DC    HL1'13'             not expected CC mask
0003082C  00                                       4153+          DS    X                   extracted cc
00030830  00000000 00000000                        4154+          DS    2F                  extract PSW after test (has CC)
00030838  000311E0                                 4155+          DC    A(FMSG22)           address of test fail message
0003083C  00030F40                                 4156+          DC    A(OPSRC22)          address of operand source
00030840  00030E20                                 4157+          DC    A(OPDST22)          address of operand destination
00030844  00031060                                 4158+          DC    A(PL22)             address of parameter list
00030848  0000418E                                 4159+          DC    A(RANDX+22)         address of random source data
0003084C  00031255                                 4160+          DC    A(TEND22)           address of test end eye catche
                                                   4161
                                                   4162           TSTCODE
00030850                                           4163+CODE22    DS    0F
                                                   4164 *                                   load random source
00030850  E310 501C 0014                 0000001C  4165           LGF   R1,OPSRCA           R1: OPSRC - Primary
00030856  B904 0061                                4166           LGR   R6,R1               R6: OPSRC - Secondary
0003085A  B904 0071                                4167           LGR   R7,R1               R7: OPSRC - Home
                                                   4168
0003085E  E320 5028 0014                 00000028  4169           LGF   R2,RANDXA           R2: address random data
00030864  E330 5024 0014                 00000024  4170           LGF   R3,PLA              R3: address PL
                                                   4171
0003086A  B219 0200                      00000200  4172           SAC   ARMODE
0003086E  D20F 1000 2000       00000000  00000000  4173           MVC   0(16,R1),0(R2)      load Op1 = Primary
00030874  D20F 6010 2010       00000010  00000010  4174           MVC   16(16,R6),16(R2)    load Op2 = Secondary
0003087A  D20F 7020 2020       00000020  00000020  4175           MVC   32(16,R7),32(R2)    load Op3 = Home
00030880  D20F 6030 2030       00000030  00000030  4176           MVC   48(16,R6),48(R2)    load Op4 = Secondary
                                                   4177 *
                                                   4178 * Quadruple Compare and Swap (QCSX: 36)
                                                   4179 * Build Parambter List
                                                   4180 *
00030886  D23F 3040 2000       00000040  00000000  4181           MVC   64(64,R3),0(R2)     Load PL OP1-Op4 Compare
0003088C  D23F 3080 2080       00000080  00000080  4182           MVC   128(64,R3),128(R2)  Load PL Op1-Op4 Replacement
                                                   4183 ****
                                                   4184 * force a failure by changing the 4th operand
                                                   4185 ****
00030892  D207 6030 2100       00000030  00000100  4186           MVC   48(8,R6),256(R2)
                                                   4187 ****
00030898  E313 0000 0024                 00000000  4188           STG   R1,0(R3)            load OP1 address in PL
0003089E  E54C 3020 0000                 00000020  4189           MVHI  32(R3),PRIALET
                                                   4190
000308A4  4111 0010                      00000010  4191           LA    R1,16(R1)
000308A8  E313 0008 0024                 00000008  4192           STG   R1,8(R3)            load OP2 address in PL
000308AE  E54C 3024 0001                 00000024  4193           MVHI  36(R3),SECALET
                                                   4194
000308B4  4111 0010                      00000010  4195           LA    R1,16(R1)
000308B8  E313 0010 0024                 00000010  4196           STG   R1,16(R3)           load OP3 address in PL
000308BE  E54C 3028 0002                 00000028  4197           MVHI  40(R3),HOMEALET
                                                   4198
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000308C4  4111 0010                    00000010  4199           LA    R1,16(R1)
000308C8  E313 0018 0024               00000018  4200           STG   R1,24(R3)           load OP4 address in PL
000308CE  E54C 302C 0001               0000002C  4201           MVHI  44(R3),SECALET
                                                 4202
000308D4  B982 0000                              4203           XGR   R0,R0
000308D8  4100 0024                    00000024  4204           LA    R0,36
                                                 4205
000308DC  B219 0200                    00000200  4206           SAC   ARMODE
000308E0  EE0F 0000 3000       00000000  00000000  4207           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSX: 36)
000308E6  B219 0000                    00000000  4208           SAC   PRIMODE
                                                 4209
000308EA  B98D 0020                              4210           EPSW  R2,R0               extract psw
000308EE  5020 5010                    00000010  4211           ST    R2,CCPSW              to save CC
000308F2  E310 500B 0076               0000000B  4212           LB    R1,CCMASK           (failure CC mask)
000308F8  8910 0004                    00000004  4213           SLL   R1,4                (shift to BC instr CC position)
000308FC  4410 513C                    0003095C  4214           EX    R1,F364TSTCC            fail if...
                                                 4215
                                00030900  00000001  4216 F364CKUR  EQU   *
00030900  D200 5009 3030       00000009  00000030  4217           MVC   URFOUND,48(R3)      Extract UR
00030906  D500 5008 3030       00000008  00000030  4218           CLC   UR,48(R3)
0003090C  4780 50F4                    00030914  4219           BE    F364CKRES
00030910  45F0 8388                    00002388  4220           BAL   R15,URMSG
                                                 4221 *
                                                 4222 * compare results
                                                 4223 *
                                00030914  00000001  4224 F364CKRES EQU   *
                                                 4225 *                                  reset addresses
00030914  E310 501C 0014               0000001C  4226           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003091A  B904 0061                              4227           LGR   R6,R1               R6:  OPSRC - Secondary
0003091E  B904 0071                              4228           LGR   R7,R1               R7:  OPSRC - Home
                                                 4229
00030922  B219 0200                    00000200  4230           SAC   ARMODE
00030926  D50F 1000 3040       00000000  00000040  4231           CLC   0(16,R1),64(R3)
0003092C  4770 5132                    00030952  4232           BNE   FXX437              OP1C Different, error
00030930  D50F 6010 3050       00000010  00000050  4233           CLC   16(16,R6),80(R3)
00030936  4770 5132                    00030952  4234           BNE   FXX437              OP2C Different, error
0003093A  D50F 7020 3060       00000020  00000060  4235           CLC   32(16,R7),96(R3)
00030940  4770 5132                    00030952  4236           BNE   FXX437              OP3C Different, error
00030944  D50F 6030 3070       00000030  00000070  4237           CLC   48(16,R6),112(R3)
0003094A  B219 0000                    00000000  4238           SAC   PRIMODE
0003094E  478B 0000                    00000000  4239           BE    0(R11)              OP4Csame, return
                                                 4240
                                00030952  00000001  4241 FXX437   EQU   *
00030952  B219 0000                    00000000  4242           SAC   PRIMODE
00030956  45F0 8450                    00002450  4243           BAL   R15,FAILMSG
0003095A  07FB                                   4244           BR    R11
                                                 4245
0003095C  4700 5140                    00030960  4246 F364TSTCC BC     0,F364CCMSG        fail if unexpected condition code
                                00030960  00000001  4247 F364CCMSG EQU   *
00030960  45F0 82B2                    000022B2  4248           BAL   R15,CCMSG
00030964  47F0 50E0                    00030900  4249           B     F364CKUR
                                                 4250
                                                 4251           TSTDATA
00030968                        00030968  00030E00  4252+          ORG   PLOETST+(128*K)+22*(3*K)+(K+512)
                                                 4253           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00030E00  D7D9C940 40406E6E                      4254+          DC    CL32'PRI   >>>OPDST22<<<' eye catcher
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00030E08  6ED6D7C4 E2E3F2F2
00030E10  4C4C4C40 40404040
00030E18  40404040 40404040
00030E20                                             4255+OPDST22  DS    0C
00030E20  00000000 00000000                          4256          DC    XL256'0'
00030E28  00000000 00000000
00030E30  00000000 00000000
00030E38  00000000 00000000
00030E40  00000000 00000000
00030E48  00000000 00000000
00030E50  00000000 00000000
00030E58  00000000 00000000
00030E60  00000000 00000000
00030E68  00000000 00000000
00030E70  00000000 00000000
00030E78  00000000 00000000
00030E80  00000000 00000000
00030E88  00000000 00000000
00030E90  00000000 00000000
00030E98  00000000 00000000
00030EA0  00000000 00000000
00030EA8  00000000 00000000
00030EB0  00000000 00000000
00030EB8  00000000 00000000
00030EC0  00000000 00000000
00030EC8  00000000 00000000
00030ED0  00000000 00000000
00030ED8  00000000 00000000
00030EE0  00000000 00000000
00030EE8  00000000 00000000
00030EF0  00000000 00000000
00030EF8  00000000 00000000
00030F00  00000000 00000000
00030F08  00000000 00000000
00030F10  00000000 00000000
00030F18  00000000 00000000
                                                     4257
                                                     4258          TSTLBL OPSRC              Source: OP1-OP4
00030F20  D7D9C940 40406E6E                          4259+         DC    CL32'PRI   >>>OPSRC22<<<' eye catcher
00030F28  6ED6D7E2 D9C3F2F2
00030F30  4C4C4C40 40404040
00030F38  40404040 40404040
00030F40                                             4260+OPSRC22  DS    0C
00030F40  00000000 00000000                          4261          DC    XL256'0'
00030F48  00000000 00000000
00030F50  00000000 00000000
00030F58  00000000 00000000
00030F60  00000000 00000000
00030F68  00000000 00000000
00030F70  00000000 00000000
00030F78  00000000 00000000
00030F80  00000000 00000000
00030F88  00000000 00000000
00030F90  00000000 00000000
00030F98  00000000 00000000
00030FA0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00030FA8  00000000 00000000
00030FB0  00000000 00000000
00030FB8  00000000 00000000
00030FC0  00000000 00000000
00030FC8  00000000 00000000
00030FD0  00000000 00000000
00030FD8  00000000 00000000
00030FE0  00000000 00000000
00030FE8  00000000 00000000
00030FF0  00000000 00000000
00030FF8  00000000 00000000
00031000  00000000 00000000
00031008  00000000 00000000
00031010  00000000 00000000
00031018  00000000 00000000
00031020  00000000 00000000
00031028  00000000 00000000
00031030  00000000 00000000
00031038  00000000 00000000
                                        4262
                                        4263          TSTLBL PL                  Parameter List
00031040  D7D9C940 40406E6E             4264+         DC    CL32'PRI   >>>PL22<<<'    eye catcher
00031048  6ED7D3F2 F24C4C4C
00031050  40404040 40404040
00031058  40404040 40404040
00031060                                4265+PL22     DS    0C
00031060  00000000 00000000             4266          DC    XL352'00'
00031068  00000000 00000000
00031070  00000000 00000000
00031078  00000000 00000000
00031080  00000000 00000000
00031088  00000000 00000000
00031090  00000000 00000000
00031098  00000000 00000000
000310A0  00000000 00000000
000310A8  00000000 00000000
000310B0  00000000 00000000
000310B8  00000000 00000000
000310C0  00000000 00000000
000310C8  00000000 00000000
000310D0  00000000 00000000
000310D8  00000000 00000000
000310E0  00000000 00000000
000310E8  00000000 00000000
000310F0  00000000 00000000
000310F8  00000000 00000000
00031100  00000000 00000000
00031108  00000000 00000000
00031110  00000000 00000000
00031118  00000000 00000000
00031120  00000000 00000000
00031128  00000000 00000000
00031130  00000000 00000000
00031138  00000000 00000000
00031140  00000000 00000000
00031148  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00031150  00000000 00000000
00031158  00000000 00000000
00031160  00000000 00000000
00031168  00000000 00000000
00031170  00000000 00000000
00031178  00000000 00000000
00031180  00000000 00000000
00031188  00000000 00000000
00031190  00000000 00000000
00031198  00000000 00000000
000311A0  00000000 00000000
000311A8  00000000 00000000
000311B0  00000000 00000000
000311B8  00000000 00000000
                                        4267
                                        4268          TSTLBL FMSG               fail message
000311C0  D7D9C940 40406E6E             4269+         DC    CL32'PRI   >>>FMSG22<<<'  eye catcher
000311C8  6EC6D4E2 C7F2F24C
000311D0  4C4C4040 40404040
000311D8  40404040 40404040
000311E0                                4270+FMSG22   DS    0C
000311E0  86818993 858440D7             4271          DC    C'failed PLO-Extension '
000311E8  D3D660C5 A7A38595
000311F0  A2899695 40
000311F5  D8A48184 99A49793             4272          DC    CL96'Quadruple Compare and Swap (QCSX: 36): 4th NE'
000311FD  8540C396 94978199
00031205  85408195 8440E2A6
0003120D  8197404D D8C3E2E7
00031215  7A40F3F6 5D7A40F4
0003121D  A38840D5 C5404040
00031225  40404040 40404040
0003122D  40404040 40404040
00031235  40404040 40404040
0003123D  40404040 40404040
00031245  40404040 40404040
0003124D  40404040 40404040
                                        4273
                                        4274          TSTEND
00031255  D7D9C940 4040C5D5             4275+TEND22   DC    CL32'PRI   END of test: 22 <<<'
0003125D  C4409686 40A385A2
00031265  A37A40F2 F2404C4C
0003126D  4C404040 40404040
                                        4276
                                        4277
                                        4278 * ------------------------------------------------------------------
                                        4279 * Test Quadruple Compare and Swap (QCSX: 36) instruction : All EQ
                                        4280 * ------------------------------------------------------------------
                                        4281          TSTHDR 0,99,36             All Equal (cc=0, UR= unchanged)
00031275                  00031275  00031400  4282+         ORG   PLOETST+(128*K)+23*(3*K)
00031400  D7D9C940 4040E2E3             4283+         DC    CL32'PRI   START of test: 23'
00031408  C1D9E340 968640A3
00031410  85A2A37A 40F2F340
00031418  40404040 40404040
00031420                  00031420      4284+         USING *,R5                base for test data and test routine
00031420                  00000000      4285+         USING PLOETEST,R5
00031420  00031450                      4286+T23      DC    A(CODE23)           address of test code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00031424  0017                                    4287+          DC    H'23'              test number
00031426  00                                      4288+          DC    X'00'
00031427  24                                      4289+          DC    HL1'36'            Function
00031428  63                                      4290+          DC    HL1'99'            expected unequal results (UR)
00031429  00                                      4291+          DC    HL1'00'            found UR
0003142A  00                                      4292+          DC    HL1'0'             cc
0003142B  07                                      4293+          DC    HL1'7'             not expected CC mask
0003142C  00                                      4294+          DS    X                  extracted cc
00031430  00000000 00000000                       4295+          DS    2F                 extract PSW after test (has CC)
00031438  00031DE0                                4296+          DC    A(FMSG23)          address of test fail message
0003143C  00031B40                                4297+          DC    A(OPSRC23)         address of operand source
00031440  00031A20                                4298+          DC    A(OPDST23)         address of operand destination
00031444  00031C60                                4299+          DC    A(PL23)            address of parameter list
00031448  0000418F                                4300+          DC    A(RANDX+23)        address of random source data
0003144C  00031E55                                4301+          DC    A(TEND23)          address of test end eye catche
                                                  4302
                                                  4303     TSTCODE
00031450                                          4304+CODE23    DS    0F
                                                  4305 *                                  load random source
00031450  E310 501C 0014              0000001C    4306           LGF   R1,OPSRCA          R1: OPSRC - Primary
00031456  B904 0061                               4307           LGR   R6,R1              R6: OPSRC - Secondary
0003145A  B904 0071                               4308           LGR   R7,R1              R7: OPSRC - Home
                                                  4309
0003145E  E320 5028 0014              00000028    4310           LGF   R2,RANDXA          R2: address random data
00031464  E330 5024 0014              00000024    4311           LGF   R3,PLA             R3: address PL
                                                  4312
0003146A  B219 0200                   00000200    4313           SAC   ARMODE
0003146E  D20F 1000 2000    00000000  00000000    4314           MVC   0(16,R1),0(R2)     load Op1 = Primary
00031474  D20F 6010 2010    00000010  00000010    4315           MVC   16(16,R6),16(R2)   load Op2 = Secondary
0003147A  D20F 7020 2020    00000020  00000020    4316           MVC   32(16,R7),32(R2)   load Op3 = Home
00031480  D20F 6030 2030    00000030  00000030    4317           MVC   48(16,R6),48(R2)   load Op4 = Secondary
                                                  4318 *
                                                  4319 * Quadruple Compare and Swap (QCSX: 36)
                                                  4320 * Build Parambter List
                                                  4321 *
00031486  D23F 3040 2000    00000040  00000000    4322           MVC   64(64,R3),0(R2)    Load PL OP1-Op4 Compare
0003148C  D23F 3080 2080    00000080  00000080    4323           MVC   128(64,R3),128(R2) Load PL Op1-Op4 Replacement
                                                  4324 ****
                                                  4325 * Do NOT force a failure by changing an operand
                                                  4326 ****
                                                  4327 *           MVC   48(8,R6),256(R2)
                                                  4328 ****
00031492  E313 0000 0024              00000000    4329           STG   R1,0(R3)           load OP1 address in PL
00031498  E54C 3020 0000              00000020    4330           MVHI  32(R3),PRIALET
                                                  4331
0003149E  4111 0010                   00000010    4332           LA    R1,16(R1)
000314A2  E313 0008 0024              00000008    4333           STG   R1,8(R3)           load OP2 address in PL
000314A8  E54C 3024 0001              00000024    4334           MVHI  36(R3),SECALET
                                                  4335
000314AE  4111 0010                   00000010    4336           LA    R1,16(R1)
000314B2  E313 0010 0024              00000010    4337           STG   R1,16(R3)          load OP3 address in PL
000314B8  E54C 3028 0002              00000028    4338           MVHI  40(R3),HOMEALET
                                                  4339
000314BE  4111 0010                   00000010    4340           LA    R1,16(R1)
000314C2  E313 0018 0024              00000018    4341           STG   R1,24(R3)          load OP4 address in PL
000314C8  E54C 302C 0001              0000002C    4342           MVHI  44(R3),SECALET
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    4343
000314CE  B982 0000                                 4344          XGR   R0,R0
000314D2  4100 0024                       00000024  4345          LA    R0,36
                                                    4346
000314D6  B219 0200                       00000200  4347          SAC   ARMODE
000314DA  EE0F 0000 3000      00000000    00000000  4348          PLO   0,0,R15,0(R3)       Quadruple C/S (QCSX: 36)
000314E0  B219 0000                       00000000  4349          SAC   PRIMODE
                                                    4350
000314E4  B98D 0020                                 4351          EPSW  R2,R0               extract psw
000314E8  5020 5010                       00000010  4352          ST    R2,CCPSW              to save CC
000314EC  E310 500B 0076                  0000000B  4353          LB    R1,CCMASK           (failure CC mask)
000314F2  8910 0004                       00000004  4354          SLL   R1,4                (shift to BC instr CC position)
000314F6  4410 5136                       00031556  4355          EX    R1,F365TSTCC            fail if...
                                                    4356
                              000314FA    00000001  4357 F365CKUR  EQU   *
000314FA  D200 5009 3030      00000009    00000030  4358          MVC   URFOUND,48(R3)      Extract UR
00031500  D500 5008 3030      00000008    00000030  4359          CLC   UR,48(R3)
00031506  4780 50EE                       0003150E  4360          BE    F365CKRES
0003150A  45F0 8388                       00002388  4361          BAL   R15,URMSG
                                                    4362 *
                                                    4363 * compare results
                                                    4364 *
                              0003150E    00000001  4365 F365CKRES EQU   *
                                                    4366 *                              reset addresses
0003150E  E310 501C 0014                  0000001C  4367          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00031514  B904 0061                                 4368          LGR   R6,R1               R6:  OPSRC - Secondary
00031518  B904 0071                                 4369          LGR   R7,R1               R7:  OPSRC - Home
                                                    4370
                                                    4371 * All equal; replacement values copied to operand source
                                                    4372
0003151C  B219 0200                       00000200  4373          SAC   ARMODE
00031520  D50F 1000 3080      00000000    00000080  4374          CLC   0(16,R1),128(R3)
00031526  4770 512C                       0003154C  4375          BNE   FXX537              OP1R Different, error
0003152A  D50F 6010 3090      00000010    00000090  4376          CLC   16(16,R6),144(R3)
00031530  4770 512C                       0003154C  4377          BNE   FXX537              OP2R Different, error
00031534  D50F 7020 30A0      00000020    000000A0  4378          CLC   32(16,R7),160(R3)
0003153A  4770 512C                       0003154C  4379          BNE   FXX537              OP3R Different, error
0003153E  D50F 6030 30B0      00000030    000000B0  4380          CLC   48(16,R6),176(R3)
00031544  B219 0000                       00000000  4381          SAC   PRIMODE
00031548  478B 0000                       00000000  4382          BE    0(R11)              OP4R same, return
                                                    4383
                              0003154C    00000001  4384 FXX537   EQU   *
0003154C  B219 0000                       00000000  4385          SAC   PRIMODE
00031550  45F0 8450                       00002450  4386          BAL   R15,FAILMSG
00031554  07FB                                      4387          BR    R11
                                                    4388
00031556  4700 513A                       0003155A  4389 F365TSTCC BC    0,F365CCMSG         fail if unexpected condition code
                              0003155A    00000001  4390 F365CCMSG EQU   *
0003155A  45F0 82B2                       000022B2  4391          BAL   R15,CCMSG
0003155E  47F0 50DA                       000314FA  4392          B     F365CKUR
                                                    4393
                                                    4394          TSTDATA
00031562                      00031562    00031A00  4395+         ORG   PLOETST+(128*K)+23*(3*K)+(K+512)
                                                    4396          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00031A00  D7D9C940 40406E6E                         4397+         DC    CL32'PRI   >>>OPDST23<<<' eye catcher
00031A08  6ED6D7C4 E2E3F2F3
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00031A10  4C4C4C40 40404040
00031A18  40404040 40404040
00031A20                                             4398+OPDST23  DS    0C
00031A20  00000000 00000000                          4399          DC    XL256'0'
00031A28  00000000 00000000
00031A30  00000000 00000000
00031A38  00000000 00000000
00031A40  00000000 00000000
00031A48  00000000 00000000
00031A50  00000000 00000000
00031A58  00000000 00000000
00031A60  00000000 00000000
00031A68  00000000 00000000
00031A70  00000000 00000000
00031A78  00000000 00000000
00031A80  00000000 00000000
00031A88  00000000 00000000
00031A90  00000000 00000000
00031A98  00000000 00000000
00031AA0  00000000 00000000
00031AA8  00000000 00000000
00031AB0  00000000 00000000
00031AB8  00000000 00000000
00031AC0  00000000 00000000
00031AC8  00000000 00000000
00031AD0  00000000 00000000
00031AD8  00000000 00000000
00031AE0  00000000 00000000
00031AE8  00000000 00000000
00031AF0  00000000 00000000
00031AF8  00000000 00000000
00031B00  00000000 00000000
00031B08  00000000 00000000
00031B10  00000000 00000000
00031B18  00000000 00000000
                                                     4400
                                                     4401          TSTLBL OPSRC              Source: OP1-OP4
00031B20  D7D9C940 40406E6E                          4402+         DC    CL32'PRI   >>>OPSRC23<<<' eye catcher
00031B28  6ED6D7E2 D9C3F2F3
00031B30  4C4C4C40 40404040
00031B38  40404040 40404040
00031B40                                             4403+OPSRC23  DS    0C
00031B40  00000000 00000000                          4404          DC    XL256'0'
00031B48  00000000 00000000
00031B50  00000000 00000000
00031B58  00000000 00000000
00031B60  00000000 00000000
00031B68  00000000 00000000
00031B70  00000000 00000000
00031B78  00000000 00000000
00031B80  00000000 00000000
00031B88  00000000 00000000
00031B90  00000000 00000000
00031B98  00000000 00000000
00031BA0  00000000 00000000
00031BA8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00031BB0  00000000 00000000
00031BB8  00000000 00000000
00031BC0  00000000 00000000
00031BC8  00000000 00000000
00031BD0  00000000 00000000
00031BD8  00000000 00000000
00031BE0  00000000 00000000
00031BE8  00000000 00000000
00031BF0  00000000 00000000
00031BF8  00000000 00000000
00031C00  00000000 00000000
00031C08  00000000 00000000
00031C10  00000000 00000000
00031C18  00000000 00000000
00031C20  00000000 00000000
00031C28  00000000 00000000
00031C30  00000000 00000000
00031C38  00000000 00000000
                                          4405
                                          4406          TSTLBL PL                  Parameter List
00031C40  D7D9C940 40406E6E               4407+         DC    CL32'PRI   >>>PL23<<<'    eye catcher
00031C48  6ED7D3F2 F34C4C4C
00031C50  40404040 40404040
00031C58  40404040 40404040
00031C60                                  4408+PL23     DS    0C
00031C60  00000000 00000000               4409          DC    XL352'00'
00031C68  00000000 00000000
00031C70  00000000 00000000
00031C78  00000000 00000000
00031C80  00000000 00000000
00031C88  00000000 00000000
00031C90  00000000 00000000
00031C98  00000000 00000000
00031CA0  00000000 00000000
00031CA8  00000000 00000000
00031CB0  00000000 00000000
00031CB8  00000000 00000000
00031CC0  00000000 00000000
00031CC8  00000000 00000000
00031CD0  00000000 00000000
00031CD8  00000000 00000000
00031CE0  00000000 00000000
00031CE8  00000000 00000000
00031CF0  00000000 00000000
00031CF8  00000000 00000000
00031D00  00000000 00000000
00031D08  00000000 00000000
00031D10  00000000 00000000
00031D18  00000000 00000000
00031D20  00000000 00000000
00031D28  00000000 00000000
00031D30  00000000 00000000
00031D38  00000000 00000000
00031D40  00000000 00000000
00031D48  00000000 00000000
00031D50  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00031D58  00000000 00000000
00031D60  00000000 00000000
00031D68  00000000 00000000
00031D70  00000000 00000000
00031D78  00000000 00000000
00031D80  00000000 00000000
00031D88  00000000 00000000
00031D90  00000000 00000000
00031D98  00000000 00000000
00031DA0  00000000 00000000
00031DA8  00000000 00000000
00031DB0  00000000 00000000
00031DB8  00000000 00000000
                                          4410
                                          4411          TSTLBL FMSG               fail message
00031DC0  D7D9C940 40406E6E               4412+         DC    CL32'PRI   >>>FMSG23<<<'  eye catcher
00031DC8  6EC6D4E2 C7F2F34C
00031DD0  4C4C4040 40404040
00031DD8  40404040 40404040
00031DE0                                  4413+FMSG23   DS    0C
00031DE0  86818993 858440D7               4414          DC    C'failed PLO-Extension '
00031DE8  D3D660C5 A7A38595
00031DF0  A2899695 40
00031DF5  D8A48184 99A49793               4415          DC    CL96'Quadruple Compare and Swap (QCSX: 36): All EQ'
00031DFD  8540C396 94978199
00031E05  85408195 8440E2A6
00031E0D  8197404D D8C3E2E7
00031E15  7A40F3F6 5D7A40C1
00031E1D  939340C5 D8404040
00031E25  40404040 40404040
00031E2D  40404040 40404040
00031E35  40404040 40404040
00031E3D  40404040 40404040
00031E45  40404040 40404040
00031E4D  40404040 40404040
                                          4416
                                          4417          TSTEND
00031E55  D7D9C940 4040C5D5               4418+TEND23   DC    CL32'PRI   END of test: 23 <<<'
00031E5D  C4409686 40A385A2
00031E65  A37A40F2 F3404C4C
00031E6D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4420 *=============================================================================
                                                   4421
                                                   4422 * -----------------------------------------------------------------------
                                                   4423 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 1st NE
                                                   4424 * -----------------------------------------------------------------------
                                                   4425          TSTHDR 2,1,35             1st Operand NOT Equal (cc=2, UR=1)
00031E75                          00031E75  00032000 4426+         ORG   PLOETST+(128*K)+24*(3*K)
00032000  D7D9C940 4040E2E3                        4427+         DC    CL32'PRI   START of test: 24'
00032008  C1D9E340 968640A3
00032010  85A2A37A 40F2F440
00032018  40404040 40404040
00032020                          00032020         4428+         USING *,R5                base for test data and test routine
00032020                          00000000         4429+         USING PLOETEST,R5
00032020  00032050                                 4430+T24      DC    A(CODE24)           address of test code
00032024  0018                                     4431+         DC    H'24'               test number
00032026  00                                       4432+         DC    X'00'
00032027  23                                       4433+         DC    HL1'35'             Function
00032028  01                                       4434+         DC    HL1'1'              expected unequal results (UR)
00032029  00                                       4435+         DC    HL1'00'             found UR
0003202A  02                                       4436+         DC    HL1'2'              cc
0003202B  0D                                       4437+         DC    HL1'13'             not expected CC mask
0003202C  00                                       4438+         DS    X                   extracted cc
00032030  00000000 00000000                        4439+         DS    2F                  extract PSW after test (has CC)
00032038  000329E0                                 4440+         DC    A(FMSG24)           address of test fail message
0003203C  00032740                                 4441+         DC    A(OPSRC24)          address of operand source
00032040  00032620                                 4442+         DC    A(OPDST24)          address of operand destination
00032044  00032860                                 4443+         DC    A(PL24)             address of parameter list
00032048  00004190                                 4444+         DC    A(RANDX+24)         address of random source data
0003204C  00032A55                                 4445+         DC    A(TEND24)           address of test end eye catche
                                                   4446
                                                   4447          TSTCODE
00032050                                           4448+CODE24   DS    0F
                                                   4449 *                                  load random source
00032050  E310 501C 0014                0000001C   4450          LGF   R1,OPSRCA           R1: OPSRC - Primary
00032056  B904 0061                                4451          LGR   R6,R1               R6: OPSRC - Secondary
0003205A  B904 0071                                4452          LGR   R7,R1               R7: OPSRC - Home
                                                   4453
0003205E  E320 5028 0014                00000028   4454          LGF   R2,RANDXA           R2: address random data
00032064  E330 5024 0014                00000024   4455          LGF   R3,PLA              R3: address PL
                                                   4456
0003206A  B219 0200                     00000200   4457          SAC   ARMODE
0003206E  D207 1000 2000        00000000  00000000 4458          MVC   0(8,R1),0(R2)       load Op1 = Primary
00032074  D207 6008 2008        00000008  00000008 4459          MVC   8(8,R6),8(R2)       load Op2 = Secondary
0003207A  D207 7010 2010        00000010  00000010 4460          MVC   16(8,R7),16(R2)     load Op3 = Home
00032080  D207 6018 2018        00000018  00000018 4461          MVC   24(8,R6),24(R2)     load Op4 = Secondary
                                                   4462 *
                                                   4463 * Quadruple Compare and Swap (QCSG: 35)
                                                   4464 * Build Parambter List
                                                   4465 *
00032086  D21F 3040 2000        00000040  00000000 4466          MVC   64(32,R3),0(R2)     Load PL OP1-Op4 Compare
0003208C  D21F 3060 2080        00000060  00000080 4467          MVC   96(32,R3),128(R2)   Load PL Op1-Op4 Replacement
                                                   4468 ****
                                                   4469 * force a failure by changing the first operand
                                                   4470 ****
00032092  D203 1000 2100        00000000  00000100 4471          MVC   0(4,R1),256(R2)     some random data
                                                   4472 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00032098  E313 0000 0024                 00000000  4473          STG   R1,0(R3)            load OP1 address in PL
0003209E  E54C 3020 0000                 00000020  4474          MVHI  32(R3),PRIALET
                                                   4475
000320A4  4111 0008                      00000008  4476          LA    R1,8(R1)
000320A8  E313 0008 0024                 00000008  4477          STG   R1,8(R3)            load OP2 address in PL
000320AE  E54C 3024 0001                 00000024  4478          MVHI  36(R3),SECALET
                                                   4479
000320B4  4111 0008                      00000008  4480          LA    R1,8(R1)
000320B8  E313 0010 0024                 00000010  4481          STG   R1,16(R3)           load OP3 address in PL
000320BE  E54C 3028 0002                 00000028  4482          MVHI  40(R3),HOMEALET
                                                   4483
000320C4  4111 0008                      00000008  4484          LA    R1,8(R1)
000320C8  E313 0018 0024                 00000018  4485          STG   R1,24(R3)           load OP4 address in PL
000320CE  E54C 302C 0001                 0000002C  4486          MVHI  44(R3),SECALET
                                                   4487
000320D4  B982 0000                                4488          XGR   R0,R0
000320D8  4100 0023                      00000023  4489          LA    R0,35
                                                   4490
000320DC  B219 0200                      00000200  4491          SAC   ARMODE
000320E0  EE0F 0000 3000      00000000   00000000  4492          PLO   0,0,R15,0(R3)       Quadruple C/S (QCSG: 35)
000320E6  B219 0000                      00000000  4493          SAC   PRIMODE
                                                   4494
000320EA  B98D 0020                                4495          EPSW  R2,R0               extract psw
000320EE  5020 5010                      00000010  4496          ST    R2,CCPSW              to save CC
000320F2  E310 500B 0076                 0000000B  4497          LB    R1,CCMASK           (failure CC mask)
000320F8  8910 0004                      00000004  4498          SLL   R1,4                (shift to BC instr CC position)
000320FC  4410 513C                      0003215C  4499          EX    R1,F351TSTCC          fail if...
                                                   4500
                                00032100  00000001  4501 F351CKUR EQU   *
00032100  D200 5009 3030      00000009   00000030  4502          MVC   URFOUND,48(R3)      Extract UR
00032106  D500 5008 3030      00000008   00000030  4503          CLC   UR,48(R3)
0003210C  4780 50F4                      00032114  4504          BE    F351CKRES
00032110  45F0 8388                      00002388  4505          BAL   R15,URMSG
                                                   4506 *
                                                   4507 * compare results
                                                   4508 *
                                00032114  00000001  4509 F351CKRES EQU   *
                                                   4510 *                                reset addresses
00032114  E310 501C 0014                 0000001C  4511          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003211A  B904 0061                                4512          LGR   R6,R1               R6:  OPSRC - Secondary
0003211E  B904 0071                                4513          LGR   R7,R1               R7:  OPSRC - Home
                                                   4514
00032122  B219 0200                      00000200  4515          SAC   ARMODE
00032126  D507 1000 3040      00000000   00000040  4516          CLC   0(8,R1),64(R3)
0003212C  4770 5132                      00032152  4517          BNE   F351XX              OP1C Different, error
00032130  D507 6008 3048      00000008   00000048  4518          CLC   8(8,R6),72(R3)
00032136  4770 5132                      00032152  4519          BNE   F351XX              OP2C Different, error
0003213A  D507 7010 3050      00000010   00000050  4520          CLC   16(8,R7),80(R3)
00032140  4770 5132                      00032152  4521          BNE   F351XX              OP3C Different, error
00032144  D507 6018 3058      00000018   00000058  4522          CLC   24(8,R6),88(R3)
0003214A  B219 0000                      00000000  4523          SAC   PRIMODE
0003214E  478B 0000                      00000000  4524          BE    0(R11)              OP4Csame, return
                                                   4525
                                00032152  00000001  4526 F351XX   EQU   *
00032152  B219 0000                      00000000  4527          SAC   PRIMODE
00032156  45F0 8450                      00002450  4528          BAL   R15,FAILMSG
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003215A  07FB                                     4529           BR    R11
                                                   4530
0003215C  4700 5140                      00032160  4531 F351TSTCC BC    0,F351CCMSG        fail if unexpected condition code
                                00032160  00000001  4532 F351CCMSG EQU   *
00032160  45F0 82B2                      000022B2  4533           BAL   R15,CCMSG
00032164  47F0 50E0                      00032100  4534           B     F351CKUR
                                                   4535
                                                   4536           TSTDATA
00032168                        00032168  00032600  4537+          ORG   PLOETST+(128*K)+24*(3*K)+(K+512)
                                                   4538           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00032600  D7D9C940 40406E6E                         4539+          DC    CL32'PRI   >>>OPDST24<<<' eye catcher
00032608  6ED6D7C4 E2E3F2F4
00032610  4C4C4C40 40404040
00032618  40404040 40404040
00032620                                           4540+OPDST24   DS    0C
00032620  00000000 00000000                        4541           DC    XL256'0'
00032628  00000000 00000000
00032630  00000000 00000000
00032638  00000000 00000000
00032640  00000000 00000000
00032648  00000000 00000000
00032650  00000000 00000000
00032658  00000000 00000000
00032660  00000000 00000000
00032668  00000000 00000000
00032670  00000000 00000000
00032678  00000000 00000000
00032680  00000000 00000000
00032688  00000000 00000000
00032690  00000000 00000000
00032698  00000000 00000000
000326A0  00000000 00000000
000326A8  00000000 00000000
000326B0  00000000 00000000
000326B8  00000000 00000000
000326C0  00000000 00000000
000326C8  00000000 00000000
000326D0  00000000 00000000
000326D8  00000000 00000000
000326E0  00000000 00000000
000326E8  00000000 00000000
000326F0  00000000 00000000
000326F8  00000000 00000000
00032700  00000000 00000000
00032708  00000000 00000000
00032710  00000000 00000000
00032718  00000000 00000000
                                                   4542
                                                   4543           TSTLBL OPSRC              Source: OP1-OP4
00032720  D7D9C940 40406E6E                        4544+          DC    CL32'PRI   >>>OPSRC24<<<' eye catcher
00032728  6ED6D7E2 D9C3F2F4
00032730  4C4C4C40 40404040
00032738  40404040 40404040
00032740                                           4545+OPSRC24   DS    0C
00032740  00000000 00000000                        4546           DC    XL256'0'
00032748  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00032750  00000000 00000000
00032758  00000000 00000000
00032760  00000000 00000000
00032768  00000000 00000000
00032770  00000000 00000000
00032778  00000000 00000000
00032780  00000000 00000000
00032788  00000000 00000000
00032790  00000000 00000000
00032798  00000000 00000000
000327A0  00000000 00000000
000327A8  00000000 00000000
000327B0  00000000 00000000
000327B8  00000000 00000000
000327C0  00000000 00000000
000327C8  00000000 00000000
000327D0  00000000 00000000
000327D8  00000000 00000000
000327E0  00000000 00000000
000327E8  00000000 00000000
000327F0  00000000 00000000
000327F8  00000000 00000000
00032800  00000000 00000000
00032808  00000000 00000000
00032810  00000000 00000000
00032818  00000000 00000000
00032820  00000000 00000000
00032828  00000000 00000000
00032830  00000000 00000000
00032838  00000000 00000000
                                      4547
                                      4548           TSTLBL PL                 Parameter List
00032840  D7D9C940 40406E6E           4549+          DC    CL32'PRI   >>>PL24<<<'    eye catcher
00032848  6ED7D3F2 F44C4C4C
00032850  40404040 40404040
00032858  40404040 40404040
00032860                              4550+PL24      DS    0C
00032860  00000000 00000000           4551           DC    XL352'00'
00032868  00000000 00000000
00032870  00000000 00000000
00032878  00000000 00000000
00032880  00000000 00000000
00032888  00000000 00000000
00032890  00000000 00000000
00032898  00000000 00000000
000328A0  00000000 00000000
000328A8  00000000 00000000
000328B0  00000000 00000000
000328B8  00000000 00000000
000328C0  00000000 00000000
000328C8  00000000 00000000
000328D0  00000000 00000000
000328D8  00000000 00000000
000328E0  00000000 00000000
000328E8  00000000 00000000
000328F0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000328F8  00000000 00000000
00032900  00000000 00000000
00032908  00000000 00000000
00032910  00000000 00000000
00032918  00000000 00000000
00032920  00000000 00000000
00032928  00000000 00000000
00032930  00000000 00000000
00032938  00000000 00000000
00032940  00000000 00000000
00032948  00000000 00000000
00032950  00000000 00000000
00032958  00000000 00000000
00032960  00000000 00000000
00032968  00000000 00000000
00032970  00000000 00000000
00032978  00000000 00000000
00032980  00000000 00000000
00032988  00000000 00000000
00032990  00000000 00000000
00032998  00000000 00000000
000329A0  00000000 00000000
000329A8  00000000 00000000
000329B0  00000000 00000000
000329B8  00000000 00000000
                                        4552
                                        4553           TSTLBL FMSG                fail message
000329C0  D7D9C940 40406E6E             4554+          DC    CL32'PRI   >>>FMSG24<<<'  eye catcher
000329C8  6EC6D4E2 C7F2F44C
000329D0  4C4C4040 40404040
000329D8  40404040 40404040
000329E0                                4555+FMSG24    DS    0C
000329E0  86818993 858440D7             4556           DC    C'failed PLO-Extension '
000329E8  D3D660C5 A7A38595
000329F0  A2899695 40
000329F5  D8A48184 99A49793             4557           DC    CL96'Quadruple Compare and Swap (QCSG: 35): 1st NE'
000329FD  8540C396 94978199
00032A05  85408195 8440E2A6
00032A0D  8197404D D8C3E2C7
00032A15  7A40F3F5 5D7A40F1
00032A1D  A2A340D5 C5404040
00032A25  40404040 40404040
00032A2D  40404040 40404040
00032A35  40404040 40404040
00032A3D  40404040 40404040
00032A45  40404040 40404040
00032A4D  40404040 40404040
                                        4558
                                        4559           TSTEND
00032A55  D7D9C940 4040C5D5             4560+TEND24    DC    CL32'PRI   END of test: 24 <<<'
00032A5D  C4409686 40A385A2
00032A65  A37A40F2 F4404C4C
00032A6D  4C404040 40404040
                                        4561
                                        4562 * ------------------------------------------------------------------
                                        4563 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 2nd NE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   4564 * ------------------------------------------------------------------------
                                                   4565          TSTHDR 2,2,35             2nd Operand NOT Equal (cc=2, UR=2)
00032A75                       00032A75  00032C00  4566+         ORG   PLOETST+(128*K)+25*(3*K)
00032C00  D7D9C940 4040E2E3                        4567+         DC    CL32'PRI   START of test: 25'
00032C08  C1D9E340 968640A3
00032C10  85A2A37A 40F2F540
00032C18  40404040 40404040
00032C20                       00032C20            4568+         USING *,R5                base for test data and test routine
00032C20                       00000000            4569+         USING PLOETEST,R5
00032C20  00032C50                                 4570+T25      DC    A(CODE25)           address of test code
00032C24  0019                                     4571+         DC    H'25'               test number
00032C26  00                                       4572+         DC    X'00'
00032C27  23                                       4573+         DC    HL1'35'             Function
00032C28  02                                       4574+         DC    HL1'2'              expected unequal results (UR)
00032C29  00                                       4575+         DC    HL1'00'             found UR
00032C2A  02                                       4576+         DC    HL1'2'              cc
00032C2B  0D                                       4577+         DC    HL1'13'             not expected CC mask
00032C2C  00                                       4578+         DS    X                   extracted cc
00032C30  00000000 00000000                        4579+         DS    2F                  extract PSW after test (has CC)
00032C38  000335E0                                 4580+         DC    A(FMSG25)           address of test fail message
00032C3C  00033340                                 4581+         DC    A(OPSRC25)          address of operand source
00032C40  00033220                                 4582+         DC    A(OPDST25)          address of operand destination
00032C44  00033460                                 4583+         DC    A(PL25)             address of parameter list
00032C48  00004191                                 4584+         DC    A(RANDX+25)         address of random source data
00032C4C  00033655                                 4585+         DC    A(TEND25)           address of test end eye catche
                                                   4586
                                                   4587          TSTCODE
00032C50                                           4588+CODE25   DS    0F
                                                   4589 *                                  load random source
00032C50  E310 501C 0014                0000001C   4590          LGF   R1,OPSRCA           R1: OPSRC - Primary
00032C56  B904 0061                                4591          LGR   R6,R1               R6: OPSRC - Secondary
00032C5A  B904 0071                                4592          LGR   R7,R1               R7: OPSRC - Home
                                                   4593
00032C5E  E320 5028 0014                00000028   4594          LGF   R2,RANDXA           R2: address random data
00032C64  E330 5024 0014                00000024   4595          LGF   R3,PLA              R3: address PL
                                                   4596
00032C6A  B219 0200                     00000200   4597          SAC   ARMODE
00032C6E  D207 1000 2000       00000000 00000000   4598          MVC   0(8,R1),0(R2)       load Op1 = Primary
00032C74  D207 6008 2008       00000008 00000008   4599          MVC   8(8,R6),8(R2)       load Op2 = Secondary
00032C7A  D207 7010 2010       00000010 00000010   4600          MVC   16(8,R7),16(R2)     load Op3 = Home
00032C80  D207 6018 2018       00000018 00000018   4601          MVC   24(8,R6),24(R2)     load Op4 = Secondary
                                                   4602 *
                                                   4603 * Quadruple Compare and Swap (QCSG: 35)
                                                   4604 * Build Parambter List
                                                   4605 *
00032C86  D21F 3040 2000       00000040 00000000   4606          MVC   64(32,R3),0(R2)     Load PL OP1-Op4 Compare
00032C8C  D21F 3060 2080       00000060 00000080   4607          MVC   96(32,R3),128(R2)   Load PL Op1-Op4 Replacement
                                                   4608 ****
                                                   4609 * force a failure by changing the 2nd operand
                                                   4610 ****
00032C92  D203 6008 2100       00000008 00000100   4611          MVC   8(4,R6),256(R2)     some random data
                                                   4612 ****
00032C98  E313 0000 0024                00000000   4613          STG   R1,0(R3)            load OP1 address in PL
00032C9E  E54C 3020 0000                00000020   4614          MVHI  32(R3),PRIALET
                                                   4615
00032CA4  4111 0008                     00000008   4616          LA    R1,8(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00032CA8  E313 0008 0024                 00000008  4617           STG   R1,8(R3)           load OP2 address in PL
00032CAE  E54C 3024 0001                 00000024  4618           MVHI  36(R3),SECALET
                                                   4619
00032CB4  4111 0008                      00000008  4620           LA    R1,8(R1)
00032CB8  E313 0010 0024                 00000010  4621           STG   R1,16(R3)          load OP3 address in PL
00032CBE  E54C 3028 0002                 00000028  4622           MVHI  40(R3),HOMEALET
                                                   4623
00032CC4  4111 0008                      00000008  4624           LA    R1,8(R1)
00032CC8  E313 0018 0024                 00000018  4625           STG   R1,24(R3)          load OP4 address in PL
00032CCE  E54C 302C 0002                 0000002C  4626           MVHI  44(R3),HOMEALET
                                                   4627
00032CD4  B982 0000                                4628           XGR   R0,R0
00032CD8  4100 0023                      00000023  4629           LA    R0,35
                                                   4630
00032CDC  B219 0200                      00000200  4631           SAC   ARMODE
00032CE0  EE0F 0000 3000 00000000        00000000  4632           PLO   0,0,R15,0(R3)      Quadruple C/S (QCSG: 35)
00032CE6  B219 0000                      00000000  4633           SAC   PRIMODE
                                                   4634
00032CEA  B98D 0020                                4635           EPSW  R2,R0              extract psw
00032CEE  5020 5010                      00000010  4636           ST    R2,CCPSW             to save CC
00032CF2  E310 500B 0076                 0000000B  4637           LB    R1,CCMASK          (failure CC mask)
00032CF8  8910 0004                      00000004  4638           SLL   R1,4               (shift to BC instr CC position)
00032CFC  4410 513C                      00032D5C  4639           EX    R1,F352TSTCC         fail if...
                                                   4640
                                 00032D00  00000001  4641 F352CKUR EQU   *
00032D00  D200 5009 3030         00000009  00000030  4642           MVC   URFOUND,48(R3)     Extract UR
00032D06  D500 5008 3030         00000008  00000030  4643           CLC   UR,48(R3)
00032D0C  4780 50F4                        00032D14  4644           BE    F352CKRES
00032D10  45F0 8388                        00002388  4645           BAL   R15,URMSG
                                                   4646 *
                                                   4647 * compare results
                                                   4648 *
                                 00032D14  00000001  4649 F352CKRES EQU   *
                                                   4650 *                             reset addresses
00032D14  E310 501C 0014                   0000001C  4651           LGF   R1,OPSRCA          R1:  OPSRC - Primary
00032D1A  B904 0061                                4652           LGR   R6,R1              R6:  OPSRC - Secondary
00032D1E  B904 0071                                4653           LGR   R7,R1              R7:  OPSRC - Home
                                                   4654
00032D22  B219 0200                      00000200  4655           SAC   ARMODE
00032D26  D507 1000 3040         00000000  00000040  4656           CLC   0(8,R1),64(R3)
00032D2C  4770 5132                        00032D52  4657           BNE   F352XX             OP1C Different, error
00032D30  D507 6008 3048         00000008  00000048  4658           CLC   8(8,R6),72(R3)
00032D36  4770 5132                        00032D52  4659           BNE   F352XX             OP2C Different, error
00032D3A  D507 7010 3050         00000010  00000050  4660           CLC   16(8,R7),80(R3)
00032D40  4770 5132                        00032D52  4661           BNE   F352XX             OP3C Different, error
00032D44  D507 6018 3058         00000018  00000058  4662           CLC   24(8,R6),88(R3)
00032D4A  B219 0000                        00000000  4663           SAC   PRIMODE
00032D4E  478B 0000                        00000000  4664           BE    0(R11)             OP4Csame, return
                                                   4665
                                 00032D52  00000001  4666 F352XX   EQU   *
00032D52  B219 0000                        00000000  4667           SAC   PRIMODE
00032D56  45F0 8450                        00002450  4668           BAL   R15,FAILMSG
00032D5A  07FB                                     4669           BR    R11
                                                   4670
00032D5C  4700 5140                        00032D60  4671 F352TSTCC BC     0,F352CCMSG      fail if unexpected condition code
                                 00032D60  00000001  4672 F352CCMSG EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00032D60  45F0 82B2                     000022B2  4673           BAL   R15,CCMSG
00032D64  47F0 50E0                     00032D00  4674           B     F352CKUR
                                                  4675
                                                  4676           TSTDATA
00032D68                    00032D68    00033200  4677+          ORG   PLOETST+(128*K)+25*(3*K)+(K+512)
                                                  4678           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00033200  D7D9C940 40406E6E               00033200  4679+          DC    CL32'PRI   >>>OPDST25<<<' eye catcher
00033208  6ED6D7C4 E2E3F2F5
00033210  4C4C4C40 40404040
00033218  40404040 40404040
00033220                                  00033220  4680+OPDST25   DS    0C
00033220  00000000 00000000               00033220  4681           DC    XL256'0'
00033228  00000000 00000000               00033228
00033230  00000000 00000000               00033230
00033238  00000000 00000000               00033238
00033240  00000000 00000000               00033240
00033248  00000000 00000000               00033248
00033250  00000000 00000000               00033250
00033258  00000000 00000000               00033258
00033260  00000000 00000000               00033260
00033268  00000000 00000000               00033268
00033270  00000000 00000000               00033270
00033278  00000000 00000000               00033278
00033280  00000000 00000000               00033280
00033288  00000000 00000000               00033288
00033290  00000000 00000000               00033290
00033298  00000000 00000000               00033298
000332A0  00000000 00000000               000332A0
000332A8  00000000 00000000               000332A8
000332B0  00000000 00000000               000332B0
000332B8  00000000 00000000               000332B8
000332C0  00000000 00000000               000332C0
000332C8  00000000 00000000               000332C8
000332D0  00000000 00000000               000332D0
000332D8  00000000 00000000               000332D8
000332E0  00000000 00000000               000332E0
000332E8  00000000 00000000               000332E8
000332F0  00000000 00000000               000332F0
000332F8  00000000 00000000               000332F8
00033300  00000000 00000000               00033300
00033308  00000000 00000000               00033308
00033310  00000000 00000000               00033310
00033318  00000000 00000000               00033318
                                                  4682
                                                  4683           TSTLBL OPSRC              Source: OP1-OP4
00033320  D7D9C940 40406E6E               00033320  4684+          DC    CL32'PRI   >>>OPSRC25<<<' eye catcher
00033328  6ED6D7E2 D9C3F2F5               00033328
00033330  4C4C4C40 40404040               00033330
00033338  40404040 40404040               00033338
00033340                                  00033340  4685+OPSRC25   DS    0C
00033340  00000000 00000000               00033340  4686           DC    XL256'0'
00033348  00000000 00000000               00033348
00033350  00000000 00000000               00033350
00033358  00000000 00000000               00033358
00033360  00000000 00000000               00033360
00033368  00000000 00000000               00033368
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00033370  00000000 00000000
00033378  00000000 00000000
00033380  00000000 00000000
00033388  00000000 00000000
00033390  00000000 00000000
00033398  00000000 00000000
000333A0  00000000 00000000
000333A8  00000000 00000000
000333B0  00000000 00000000
000333B8  00000000 00000000
000333C0  00000000 00000000
000333C8  00000000 00000000
000333D0  00000000 00000000
000333D8  00000000 00000000
000333E0  00000000 00000000
000333E8  00000000 00000000
000333F0  00000000 00000000
000333F8  00000000 00000000
00033400  00000000 00000000
00033408  00000000 00000000
00033410  00000000 00000000
00033418  00000000 00000000
00033420  00000000 00000000
00033428  00000000 00000000
00033430  00000000 00000000
00033438  00000000 00000000
                             4687
                             4688           TSTLBL PL                  Parameter List
00033440  D7D9C940 40406E6E  4689+          DC    CL32'PRI   >>>PL25<<<'    eye catcher
00033448  6ED7D3F2 F54C4C4C
00033450  40404040 40404040
00033458  40404040 40404040
00033460                     4690+PL25      DS    0C
00033460  00000000 00000000  4691           DC    XL352'00'
00033468  00000000 00000000
00033470  00000000 00000000
00033478  00000000 00000000
00033480  00000000 00000000
00033488  00000000 00000000
00033490  00000000 00000000
00033498  00000000 00000000
000334A0  00000000 00000000
000334A8  00000000 00000000
000334B0  00000000 00000000
000334B8  00000000 00000000
000334C0  00000000 00000000
000334C8  00000000 00000000
000334D0  00000000 00000000
000334D8  00000000 00000000
000334E0  00000000 00000000
000334E8  00000000 00000000
000334F0  00000000 00000000
000334F8  00000000 00000000
00033500  00000000 00000000
00033508  00000000 00000000
00033510  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00033518  00000000 00000000
00033520  00000000 00000000
00033528  00000000 00000000
00033530  00000000 00000000
00033538  00000000 00000000
00033540  00000000 00000000
00033548  00000000 00000000
00033550  00000000 00000000
00033558  00000000 00000000
00033560  00000000 00000000
00033568  00000000 00000000
00033570  00000000 00000000
00033578  00000000 00000000
00033580  00000000 00000000
00033588  00000000 00000000
00033590  00000000 00000000
00033598  00000000 00000000
000335A0  00000000 00000000
000335A8  00000000 00000000
000335B0  00000000 00000000
000335B8  00000000 00000000
                                        4692
                                        4693          TSTLBL FMSG                fail message
000335C0  D7D9C940 40406E6E             4694+         DC    CL32'PRI   >>>FMSG25<<<'  eye catcher
000335C8  6EC6D4E2 C7F2F54C
000335D0  4C4C4040 40404040
000335D8  40404040 40404040
000335E0                                4695+FMSG25   DS    0C
000335E0  86818993 858440D7             4696          DC    C'failed PLO-Extension '
000335E8  D3D660C5 A7A38595
000335F0  A2899695 40
000335F5  D8A48184 99A49793             4697          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 2nd NE'
000335FD  8540C396 94978199
00033605  85408195 8440E2A6
0003360D  8197404D D8C3E2C7
00033615  7A40F3F5 5D7A40F2
0003361D  958440D5 C5404040
00033625  40404040 40404040
0003362D  40404040 40404040
00033635  40404040 40404040
0003363D  40404040 40404040
00033645  40404040 40404040
0003364D  40404040 40404040
                                        4698
                                        4699          TSTEND
00033655  D7D9C940 4040C5D5             4700+TEND25   DC    CL32'PRI   END of test: 25 <<<'
0003365D  C4409686 40A385A2
00033665  A37A40F2 F5404C4C
0003366D  4C404040 40404040
                                        4701
                                        4702 * ------------------------------------------------------------------
                                        4703 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 3rd NE
                                        4704 * ------------------------------------------------------------------
                                        4705          TSTHDR 2,3,35              3rd Operand NOT Equal (cc=2, UR=3)
00033675                  00033675  00033800  4706+         ORG   PLOETST+(128*K)+26*(3*K)
00033800  D7D9C940 4040E2E3             4707+         DC    CL32'PRI   START of test: 26'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00033808  C1D9E340 968640A3
00033810  85A2A37A 40F2F640
00033818  40404040 40404040
00033820                    00033820            4708+          USING *,R5                base for test data and test routine
00033820                    00000000            4709+          USING PLOETEST,R5
00033820  00033850                              4710+T26       DC    A(CODE26)           address of test code
00033824  001A                                  4711+          DC    H'26'               test number
00033826  00                                    4712+          DC    X'00'
00033827  23                                    4713+          DC    HL1'35'             Function
00033828  03                                    4714+          DC    HL1'3'              expected unequal results (UR)
00033829  00                                    4715+          DC    HL1'00'             found UR
0003382A  02                                    4716+          DC    HL1'2'              cc
0003382B  0D                                    4717+          DC    HL1'13'             not expected CC mask
0003382C  00                                    4718+          DS    X                   extracted cc
00033830  00000000 00000000                     4719+          DS    2F                  extract PSW after test (has CC)
00033838  000341E0                              4720+          DC    A(FMSG26)           address of test fail message
0003383C  00033F40                              4721+          DC    A(OPSRC26)          address of operand source
00033840  00033E20                              4722+          DC    A(OPDST26)          address of operand destination
00033844  00034060                              4723+          DC    A(PL26)             address of parameter list
00033848  00004192                              4724+          DC    A(RANDX+26)         address of random source data
0003384C  00034255                              4725+          DC    A(TEND26)           address of test end eye catche
                                                4726
                                                4727           TSTCODE
00033850                                        4728+CODE26    DS    0F
                                                4729 *                                   load random source
00033850  E310 501C 0014              0000001C  4730           LGF   R1,OPSRCA           R1: OPSRC - Primary
00033856  B904 0061                             4731           LGR   R6,R1               R6: OPSRC - Secondary
0003385A  B904 0071                             4732           LGR   R7,R1               R7: OPSRC - Home
                                                4733
0003385E  E320 5028 0014              00000028  4734           LGF   R2,RANDXA           R2: address random data
00033864  E330 5024 0014              00000024  4735           LGF   R3,PLA              R3: address PL
                                                4736
0003386A  B219 0200                   00000200  4737           SAC   ARMODE
0003386E  D207 1000 2000  00000000    00000000  4738           MVC   0(8,R1),0(R2)       load Op1 = Primary
00033874  D207 6008 2008  00000008    00000008  4739           MVC   8(8,R6),8(R2)       load Op2 = Secondary
0003387A  D207 7010 2010  00000010    00000010  4740           MVC   16(8,R7),16(R2)     load Op3 = Home
00033880  D207 6018 2018  00000018    00000018  4741           MVC   24(8,R6),24(R2)     load Op4 = Secondary
                                                4742 *
                                                4743 * Quadruple Compare and Swap (QCSG: 35)
                                                4744 * Build Parambter List
                                                4745 *
00033886  D21F 3040 2000  00000040    00000000  4746           MVC   64(32,R3),0(R2)     Load PL OP1-Op4 Compare
0003388C  D21F 3060 2080  00000060    00000080  4747           MVC   96(32,R3),128(R2)   Load PL Op1-Op4 Replacement
                                                4748 ****
                                                4749 * force a failure by changing the 3rd operand
                                                4750 ****
00033892  D203 7010 2100  00000010    00000100  4751           MVC   16(4,R7),256(R2)    some random data
                                                4752 ****
00033898  E313 0000 0024                        4753           STG   R1,0(R3)            load OP1 address in PL
0003389E  E54C 3020 0000                        4754           MVHI  32(R3),PRIALET
                                                4755
000338A4  4111 0008                             4756           LA    R1,8(R1)
000338A8  E313 0008 0024                        4757           STG   R1,8(R3)            load OP2 address in PL
000338AE  E54C 3024 0001                        4758           MVHI  36(R3),SECALET
                                                4759
000338B4  4111 0008                             4760           LA    R1,8(R1)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000338B8  E313 0010 0024                 00000010  4761           STG   R1,16(R3)          load OP3 address in PL
000338BE  E54C 3028 0002                 00000028  4762           MVHI  40(R3),HOMEALET
                                                   4763
000338C4  4111 0008                      00000008  4764           LA    R1,8(R1)
000338C8  E313 0018 0024                 00000018  4765           STG   R1,24(R3)          load OP4 address in PL
000338CE  E54C 302C 0001                 0000002C  4766           MVHI  44(R3),SECALET
                                                   4767
000338D4  B982 0000                                4768           XGR   R0,R0
000338D8  4100 0023                      00000023  4769           LA    R0,35
                                                   4770
000338DC  B219 0200                      00000200  4771           SAC   ARMODE
000338E0  EE0F 0000 3000       00000000  00000000  4772           PLO   0,0,R15,0(R3)      Quadruple C/S (QCSG: 35)
000338E6  B219 0000                      00000000  4773           SAC   PRIMODE
                                                   4774
000338EA  B98D 0020                                4775           EPSW  R2,R0              extract psw
000338EE  5020 5010                      00000010  4776           ST    R2,CCPSW             to save CC
000338F2  E310 500B 0076                 0000000B  4777           LB    R1,CCMASK          (failure CC mask)
000338F8  8910 0004                      00000004  4778           SLL   R1,4               (shift to BC instr CC position)
000338FC  4410 513C                      0003395C  4779           EX    R1,F353TSTCC         fail if...
                                                   4780
                                00033900  00000001  4781 F353CKUR EQU   *
00033900  D200 5009 3030       00000009  00000030  4782           MVC   URFOUND,48(R3)     Extract UR
00033906  D500 5008 3030       00000008  00000030  4783           CLC   UR,48(R3)
0003390C  4780 50F4                      00033914  4784           BE    F353CKRES
00033910  45F0 8388                      00002388  4785           BAL   R15,URMSG
                                                   4786 *
                                                   4787 * compare results
                                                   4788 *
                                00033914  00000001  4789 F353CKRES EQU   *
                                                   4790 *                             reset addresses
00033914  E310 501C 0014                 0000001C  4791           LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003391A  B904 0061                                4792           LGR   R6,R1              R6:  OPSRC - Secondary
0003391E  B904 0071                                4793           LGR   R7,R1              R7:  OPSRC - Home
                                                   4794
00033922  B219 0200                      00000200  4795           SAC   ARMODE
00033926  D507 1000 3040       00000000  00000040  4796           CLC   0(8,R1),64(R3)
0003392C  4770 5132                      00033952  4797           BNE   F353XX             OP1C Different, error
00033930  D507 6008 3048       00000008  00000048  4798           CLC   8(8,R6),72(R3)
00033936  4770 5132                      00033952  4799           BNE   F353XX             OP2C Different, error
0003393A  D507 7010 3050       00000010  00000050  4800           CLC   16(8,R7),80(R3)
00033940  4770 5132                      00033952  4801           BNE   F353XX             OP3C Different, error
00033944  D507 6018 3058       00000018  00000058  4802           CLC   24(8,R6),88(R3)
0003394A  B219 0000                      00000000  4803           SAC   PRIMODE
0003394E  478B 0000                      00000000  4804           BE    0(R11)             OP4Csame, return
                                                   4805
                                00033952  00000001  4806 F353XX   EQU   *
00033952  B219 0000                      00000000  4807           SAC   PRIMODE
00033956  45F0 8450                      00002450  4808           BAL   R15,FAILMSG
0003395A  07FB                                     4809           BR    R11
                                                   4810
0003395C  4700 5140                      00033960  4811 F353TSTCC BC    0,F353CCMSG        fail if unexpected condition code
                                00033960  00000001  4812 F353CCMSG EQU   *
00033960  45F0 82B2                      000022B2  4813           BAL   R15,CCMSG
00033964  47F0 50E0                      00033900  4814           B     F353CKUR
                                                   4815
                                                   4816           TSTDATA
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00033968                     00033968  00033E00  4817+         ORG   PLOETST+(128*K)+26*(3*K)+(K+512)
                                                 4818          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00033E00  D7D9C940 40406E6E                      4819+         DC    CL32'PRI   >>>OPDST26<<<' eye catcher
00033E08  6ED6D7C4 E2E3F2F6
00033E10  4C4C4C40 40404040
00033E18  40404040 40404040
00033E20                                         4820+OPDST26  DS    0C
00033E20  00000000 00000000                      4821          DC    XL256'0'
00033E28  00000000 00000000
00033E30  00000000 00000000
00033E38  00000000 00000000
00033E40  00000000 00000000
00033E48  00000000 00000000
00033E50  00000000 00000000
00033E58  00000000 00000000
00033E60  00000000 00000000
00033E68  00000000 00000000
00033E70  00000000 00000000
00033E78  00000000 00000000
00033E80  00000000 00000000
00033E88  00000000 00000000
00033E90  00000000 00000000
00033E98  00000000 00000000
00033EA0  00000000 00000000
00033EA8  00000000 00000000
00033EB0  00000000 00000000
00033EB8  00000000 00000000
00033EC0  00000000 00000000
00033EC8  00000000 00000000
00033ED0  00000000 00000000
00033ED8  00000000 00000000
00033EE0  00000000 00000000
00033EE8  00000000 00000000
00033EF0  00000000 00000000
00033EF8  00000000 00000000
00033F00  00000000 00000000
00033F08  00000000 00000000
00033F10  00000000 00000000
00033F18  00000000 00000000
                                                 4822
                                                 4823          TSTLBL OPSRC              Source: OP1-OP4
00033F20  D7D9C940 40406E6E                      4824+         DC    CL32'PRI   >>>OPSRC26<<<' eye catcher
00033F28  6ED6D7E2 D9C3F2F6
00033F30  4C4C4C40 40404040
00033F38  40404040 40404040
00033F40                                         4825+OPSRC26  DS    0C
00033F40  00000000 00000000                      4826          DC    XL256'0'
00033F48  00000000 00000000
00033F50  00000000 00000000
00033F58  00000000 00000000
00033F60  00000000 00000000
00033F68  00000000 00000000
00033F70  00000000 00000000
00033F78  00000000 00000000
00033F80  00000000 00000000
00033F88  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00033F90  00000000 00000000
00033F98  00000000 00000000
00033FA0  00000000 00000000
00033FA8  00000000 00000000
00033FB0  00000000 00000000
00033FB8  00000000 00000000
00033FC0  00000000 00000000
00033FC8  00000000 00000000
00033FD0  00000000 00000000
00033FD8  00000000 00000000
00033FE0  00000000 00000000
00033FE8  00000000 00000000
00033FF0  00000000 00000000
00033FF8  00000000 00000000
00034000  00000000 00000000
00034008  00000000 00000000
00034010  00000000 00000000
00034018  00000000 00000000
00034020  00000000 00000000
00034028  00000000 00000000
00034030  00000000 00000000
00034038  00000000 00000000
                                          4827
                                          4828          TSTLBL PL                 Parameter List
00034040  D7D9C940 40406E6E               4829+         DC    CL32'PRI   >>>PL26<<<'    eye catcher
00034048  6ED7D3F2 F64C4C4C
00034050  40404040 40404040
00034058  40404040 40404040
00034060                                  4830+PL26     DS    0C
00034060  00000000 00000000               4831          DC    XL352'00'
00034068  00000000 00000000
00034070  00000000 00000000
00034078  00000000 00000000
00034080  00000000 00000000
00034088  00000000 00000000
00034090  00000000 00000000
00034098  00000000 00000000
000340A0  00000000 00000000
000340A8  00000000 00000000
000340B0  00000000 00000000
000340B8  00000000 00000000
000340C0  00000000 00000000
000340C8  00000000 00000000
000340D0  00000000 00000000
000340D8  00000000 00000000
000340E0  00000000 00000000
000340E8  00000000 00000000
000340F0  00000000 00000000
000340F8  00000000 00000000
00034100  00000000 00000000
00034108  00000000 00000000
00034110  00000000 00000000
00034118  00000000 00000000
00034120  00000000 00000000
00034128  00000000 00000000
00034130  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00034138  00000000 00000000
00034140  00000000 00000000
00034148  00000000 00000000
00034150  00000000 00000000
00034158  00000000 00000000
00034160  00000000 00000000
00034168  00000000 00000000
00034170  00000000 00000000
00034178  00000000 00000000
00034180  00000000 00000000
00034188  00000000 00000000
00034190  00000000 00000000
00034198  00000000 00000000
000341A0  00000000 00000000
000341A8  00000000 00000000
000341B0  00000000 00000000
000341B8  00000000 00000000
                                        4832
                                        4833          TSTLBL FMSG               fail message
000341C0  D7D9C940 40406E6E             4834+         DC    CL32'PRI   >>>FMSG26<<<'  eye catcher
000341C8  6EC6D4E2 C7F2F64C
000341D0  4C4C4040 40404040
000341D8  40404040 40404040
000341E0                                4835+FMSG26   DS    0C
000341E0  86818993 858440D7             4836          DC    C'failed PLO-Extension '
000341E8  D3D660C5 A7A38595
000341F0  A2899695 40
000341F5  D8A48184 99A49793             4837          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 3rd NE'
000341FD  8540C396 94978199
00034205  85408195 8440E2A6
0003420D  8197404D D8C3E2C7
00034215  7A40F3F5 5D7A40F3
0003421D  998440D5 C5404040
00034225  40404040 40404040
0003422D  40404040 40404040
00034235  40404040 40404040
0003423D  40404040 40404040
00034245  40404040 40404040
0003424D  40404040 40404040
                                        4838
                                        4839          TSTEND
00034255  D7D9C940 4040C5D5             4840+TEND26   DC    CL32'PRI   END of test: 26 <<<'
0003425D  C4409686 40A385A2
00034265  A37A40F2 F6404C4C
0003426D  4C404040 40404040
                                        4841
                                        4842 * ------------------------------------------------------------------
                                        4843 * Test Quadruple Compare and Swap (QCSG: 35) instruction : 4th NE
                                        4844 * ------------------------------------------------------------------
                                        4845          TSTHDR 2,4,35             4th Operand NOT Equal (cc=2, UR=4)
00034275                    00034275  00034400  4846+         ORG   PLOETST+(128*K)+27*(3*K)
00034400  D7D9C940 4040E2E3             4847+         DC    CL32'PRI   START of test: 27'
00034408  C1D9E340 968640A3
00034410  85A2A37A 40F2F740
00034418  40404040 40404040
00034420                    00034420              4848+         USING *,R5                base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00034420                     00000000              4849+         USING PLOETEST,R5
00034420  00034450                                 4850+T27      DC    A(CODE27)           address of test code
00034424  001B                                     4851+         DC    H'27'               test number
00034426  00                                       4852+         DC    X'00'
00034427  23                                       4853+         DC    HL1'35'             Function
00034428  04                                       4854+         DC    HL1'4'              expected unequal results (UR)
00034429  00                                       4855+         DC    HL1'00'             found UR
0003442A  02                                       4856+         DC    HL1'2'              cc
0003442B  0D                                       4857+         DC    HL1'13'             not expected CC mask
0003442C  00                                       4858+         DS    X                   extracted cc
00034430  00000000 00000000                        4859+         DS    2F                  extract PSW after test (has CC)
00034438  00034DE0                                 4860+         DC    A(FMSG27)           address of test fail message
0003443C  00034B40                                 4861+         DC    A(OPSRC27)          address of operand source
00034440  00034A20                                 4862+         DC    A(OPDST27)          address of operand destination
00034444  00034C60                                 4863+         DC    A(PL27)             address of parameter list
00034448  00004193                                 4864+         DC    A(RANDX+27)         address of random source data
0003444C  00034E55                                 4865+         DC    A(TEND27)           address of test end eye catche
                                                   4866
                                                   4867          TSTCODE
00034450                                           4868+CODE27   DS    0F
                                                   4869 *                                  load random source
00034450  E310 501C 0014               0000001C    4870          LGF   R1,OPSRCA           R1: OPSRC - Primary
00034456  B904 0061                               4871          LGR   R6,R1               R6: OPSRC - Secondary
0003445A  B904 0071                               4872          LGR   R7,R1               R7: OPSRC - Home
                                                   4873
0003445E  E320 5028 0014               00000028    4874          LGF   R2,RANDXA           R2: address random data
00034464  E330 5024 0014               00000024    4875          LGF   R3,PLA              R3: address PL
                                                   4876
0003446A  B219 0200                    00000200    4877          SAC   ARMODE
0003446E  D207 1000 2000   00000000    00000000    4878          MVC   0(8,R1),0(R2)       load Op1 = Primary
00034474  D207 6008 2008   00000008    00000008    4879          MVC   8(8,R6),8(R2)       load Op2 = Secondary
0003447A  D207 7010 2010   00000010    00000010    4880          MVC   16(8,R7),16(R2)     load Op3 = Home
00034480  D207 6018 2018   00000018    00000018    4881          MVC   24(8,R6),24(R2)     load Op4 = Secondary
                                                   4882 *
                                                   4883 * Quadruple Compare and Swap (QCSG: 35)
                                                   4884 * Build Parambter List
                                                   4885 *
00034486  D21F 3040 2000   00000040    00000000    4886          MVC   64(32,R3),0(R2)     Load PL OP1-Op4 Compare
0003448C  D21F 3060 2080   00000060    00000080    4887          MVC   96(32,R3),128(R2)   Load PL Op1-Op4 Replacement
                                                   4888 ****
                                                   4889 * force a failure by changing the 4th operand
                                                   4890 ****
00034492  D203 6018 2100   00000018    00000100    4891          MVC   24(4,R6),256(R2)    some random data
                                                   4892 ****
                                                   4893
00034498  E313 0000 0024               00000000    4894          STG   R1,0(R3)            load OP1 address in PL
0003449E  E54C 3020 0000               00000020    4895          MVHI  32(R3),PRIALET
                                                   4896
000344A4  4111 0008                    00000008    4897          LA    R1,8(R1)
000344A8  E313 0008 0024               00000008    4898          STG   R1,8(R3)            load OP2 address in PL
000344AE  E54C 3024 0001               00000024    4899          MVHI  36(R3),SECALET
                                                   4900
000344B4  4111 0008                    00000008    4901          LA    R1,8(R1)
000344B8  E313 0010 0024               00000010    4902          STG   R1,16(R3)           load OP3 address in PL
000344BE  E54C 3028 0002               00000028    4903          MVHI  40(R3),HOMEALET
                                                   4904
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000344C4  4111 0008                    00000008  4905          LA    R1,8(R1)
000344C8  E313 0018 0024               00000018  4906          STG   R1,24(R3)           load OP4 address in PL
000344CE  E54C 302C 0001               0000002C  4907          MVHI  44(R3),SECALET
                                                 4908
000344D4  B982 0000                              4909          XGR   R0,R0
000344D8  4100 0023                    00000023  4910          LA    R0,35
                                                 4911
000344DC  B219 0200                    00000200  4912          SAC   ARMODE
000344E0  EE0F 0000 3000      00000000 00000000  4913          PLO   0,0,R15,0(R3)       Quadruple C/S (QCSG: 35)
000344E6  B219 0000                    00000000  4914          SAC   PRIMODE
                                                 4915
000344EA  B98D 0020                              4916          EPSW  R2,R0               extract psw
000344EE  5020 5010                    00000010  4917          ST    R2,CCPSW              to save CC
000344F2  E310 500B 0076               0000000B  4918          LB    R1,CCMASK           (failure CC mask)
000344F8  8910 0004                    00000004  4919          SLL   R1,4                (shift to BC instr CC position)
000344FC  4410 513C                    0003455C  4920          EX    R1,F354TSTCC            fail if...
                                                 4921
                                00034500 00000001  4922 F354CKUR EQU   *
00034500  D200 5009 3030      00000009 00000030  4923          MVC   URFOUND,48(R3)      Extract UR
00034506  D500 5008 3030      00000008 00000030  4924          CLC   UR,48(R3)
0003450C  4780 50F4                    00034514  4925          BE    F354CKRES
00034510  45F0 8388                    00002388  4926          BAL   R15,URMSG
                                                 4927 *
                                                 4928 * compare results
                                                 4929 *
                                00034514 00000001  4930 F354CKRES EQU   *
                                                 4931 *                            reset addresses
00034514  E310 501C 0014               0000001C  4932          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003451A  B904 0061                              4933          LGR   R6,R1               R6:  OPSRC - Secondary
0003451E  B904 0071                              4934          LGR   R7,R1               R7:  OPSRC - Home
                                                 4935
00034522  B219 0200                    00000200  4936          SAC   ARMODE
00034526  D507 1000 3040      00000000 00000040  4937          CLC   0(8,R1),64(R3)
0003452C  4770 5132                    00034552  4938          BNE   F354XX              OP1C Different, error
00034530  D507 6008 3048      00000008 00000048  4939          CLC   8(8,R6),72(R3)
00034536  4770 5132                    00034552  4940          BNE   F354XX              OP2C Different, error
0003453A  D507 7010 3050      00000010 00000050  4941          CLC   16(8,R7),80(R3)
00034540  4770 5132                    00034552  4942          BNE   F354XX              OP3C Different, error
00034544  D507 6018 3058      00000018 00000058  4943          CLC   24(8,R6),88(R3)
0003454A  B219 0000                    00000000  4944          SAC   PRIMODE
0003454E  478B 0000                    00000000  4945          BE    0(R11)              OP4Csame, return
                                                 4946
                                00034552 00000001  4947 F354XX   EQU   *
00034552  B219 0000                    00000000  4948          SAC   PRIMODE
00034556  45F0 8450                    00002450  4949          BAL   R15,FAILMSG
0003455A  07FB                                   4950          BR    R11
                                                 4951
0003455C  4700 5140                    00034560  4952 F354TSTCC BC    0,F354CCMSG         fail if unexpected condition code
                                00034560 00000001  4953 F354CCMSG EQU   *
00034560  45F0 82B2                    000022B2  4954          BAL   R15,CCMSG
00034564  47F0 50E0                    00034500  4955          B     F354CKUR
                                                 4956
                                                 4957          TSTDATA
00034568                        00034568 00034A00  4958+         ORG   PLOETST+(128*K)+27*(3*K)+(K+512)
                                                 4959          TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00034A00  D7D9C940 40406E6E                      4960+         DC    CL32'PRI   >>>OPDST27<<<' eye catcher
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00034A08  6ED6D7C4 E2E3F2F7
00034A10  4C4C4C40 40404040
00034A18  40404040 40404040
00034A20                                         4961+OPDST27  DS    0C
00034A20  00000000 00000000                      4962          DC    XL256'0'
00034A28  00000000 00000000
00034A30  00000000 00000000
00034A38  00000000 00000000
00034A40  00000000 00000000
00034A48  00000000 00000000
00034A50  00000000 00000000
00034A58  00000000 00000000
00034A60  00000000 00000000
00034A68  00000000 00000000
00034A70  00000000 00000000
00034A78  00000000 00000000
00034A80  00000000 00000000
00034A88  00000000 00000000
00034A90  00000000 00000000
00034A98  00000000 00000000
00034AA0  00000000 00000000
00034AA8  00000000 00000000
00034AB0  00000000 00000000
00034AB8  00000000 00000000
00034AC0  00000000 00000000
00034AC8  00000000 00000000
00034AD0  00000000 00000000
00034AD8  00000000 00000000
00034AE0  00000000 00000000
00034AE8  00000000 00000000
00034AF0  00000000 00000000
00034AF8  00000000 00000000
00034B00  00000000 00000000
00034B08  00000000 00000000
00034B10  00000000 00000000
00034B18  00000000 00000000
                                         4963
                                         4964          TSTLBL OPSRC              Source: OP1-OP4
00034B20  D7D9C940 40406E6E              4965+         DC    CL32'PRI   >>>OPSRC27<<<' eye catcher
00034B28  6ED6D7E2 D9C3F2F7
00034B30  4C4C4C40 40404040
00034B38  40404040 40404040
00034B40                                         4966+OPSRC27  DS    0C
00034B40  00000000 00000000                      4967          DC    XL256'0'
00034B48  00000000 00000000
00034B50  00000000 00000000
00034B58  00000000 00000000
00034B60  00000000 00000000
00034B68  00000000 00000000
00034B70  00000000 00000000
00034B78  00000000 00000000
00034B80  00000000 00000000
00034B88  00000000 00000000
00034B90  00000000 00000000
00034B98  00000000 00000000
00034BA0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00034BA8  00000000 00000000
00034BB0  00000000 00000000
00034BB8  00000000 00000000
00034BC0  00000000 00000000
00034BC8  00000000 00000000
00034BD0  00000000 00000000
00034BD8  00000000 00000000
00034BE0  00000000 00000000
00034BE8  00000000 00000000
00034BF0  00000000 00000000
00034BF8  00000000 00000000
00034C00  00000000 00000000
00034C08  00000000 00000000
00034C10  00000000 00000000
00034C18  00000000 00000000
00034C20  00000000 00000000
00034C28  00000000 00000000
00034C30  00000000 00000000
00034C38  00000000 00000000                          4968
                                                     4969          TSTLBL PL                  Parameter List
00034C40  D7D9C940 40406E6E                          4970+         DC    CL32'PRI   >>>PL27<<<'    eye catcher
00034C48  6ED7D3F2 F74C4C4C
00034C50  40404040 40404040
00034C58  40404040 40404040
00034C60                                             4971+PL27     DS    0C
00034C60  00000000 00000000                          4972          DC    XL352'00'
00034C68  00000000 00000000
00034C70  00000000 00000000
00034C78  00000000 00000000
00034C80  00000000 00000000
00034C88  00000000 00000000
00034C90  00000000 00000000
00034C98  00000000 00000000
00034CA0  00000000 00000000
00034CA8  00000000 00000000
00034CB0  00000000 00000000
00034CB8  00000000 00000000
00034CC0  00000000 00000000
00034CC8  00000000 00000000
00034CD0  00000000 00000000
00034CD8  00000000 00000000
00034CE0  00000000 00000000
00034CE8  00000000 00000000
00034CF0  00000000 00000000
00034CF8  00000000 00000000
00034D00  00000000 00000000
00034D08  00000000 00000000
00034D10  00000000 00000000
00034D18  00000000 00000000
00034D20  00000000 00000000
00034D28  00000000 00000000
00034D30  00000000 00000000
00034D38  00000000 00000000
00034D40  00000000 00000000
00034D48  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00034D50  00000000 00000000
00034D58  00000000 00000000
00034D60  00000000 00000000
00034D68  00000000 00000000
00034D70  00000000 00000000
00034D78  00000000 00000000
00034D80  00000000 00000000
00034D88  00000000 00000000
00034D90  00000000 00000000
00034D98  00000000 00000000
00034DA0  00000000 00000000
00034DA8  00000000 00000000
00034DB0  00000000 00000000
00034DB8  00000000 00000000
                                        4973
                                        4974          TSTLBL FMSG                fail message
00034DC0  D7D9C940 40406E6E             4975+         DC    CL32'PRI   >>>FMSG27<<<'  eye catcher
00034DC8  6EC6D4E2 C7F2F74C
00034DD0  4C4C4040 40404040
00034DD8  40404040 40404040
00034DE0                                4976+FMSG27   DS    0C
00034DE0  86818993 858440D7             4977          DC    C'failed PLO-Extension '
00034DE8  D3D660C5 A7A38595
00034DF0  A2899695 40
00034DF5  D8A48184 99A49793             4978          DC    CL96'Quadruple Compare and Swap (QCSG: 35): 4th NE'
00034DFD  8540C396 94978199
00034E05  85408195 8440E2A6
00034E0D  8197404D D8C3E2C7
00034E15  7A40F3F5 5D7A40F4
00034E1D  A38840D5 C5404040
00034E25  40404040 40404040
00034E2D  40404040 40404040
00034E35  40404040 40404040
00034E3D  40404040 40404040
00034E45  40404040 40404040
00034E4D  40404040 40404040
                                        4979
                                        4980          TSTEND
00034E55  D7D9C940 4040C5D5             4981+TEND27   DC    CL32'PRI   END of test: 27 <<<'
00034E5D  C4409686 40A385A2
00034E65  A37A40F2 F7404C4C
00034E6D  4C404040 40404040
                                        4982
                                        4983 * ------------------------------------------------------------------
                                        4984 * Test Quadruple Compare and Swap (QCSG: 35) instruction : All EQ
                                        4985 * ------------------------------------------------------------------
                                        4986          TSTHDR 0,99,35              ALL EQ Equal (cc=0, UR=99
00034E75                    00034E75  00035000  4987+         ORG   PLOETST+(128*K)+28*(3*K)
00035000  D7D9C940 4040E2E3             4988+         DC    CL32'PRI   START of test: 28'
00035008  C1D9E340 968640A3
00035010  85A2A37A 40F2F840
00035018  40404040 40404040
00035020                    00035020              4989+         USING *,R5                 base for test data and test routine
00035020                    00000000              4990+         USING PLOETEST,R5
00035020  00035050                                4991+T28      DC    A(CODE28)            address of test code
00035024  001C                                    4992+         DC    H'28'                test number
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00035026  00                                       4993+          DC    X'00'
00035027  23                                       4994+          DC    HL1'35'             Function
00035028  63                                       4995+          DC    HL1'99'             expected unequal results (UR)
00035029  00                                       4996+          DC    HL1'00'             found UR
0003502A  00                                       4997+          DC    HL1'0'              cc
0003502B  07                                       4998+          DC    HL1'7'              not expected CC mask
0003502C  00                                       4999+          DS    X                   extracted cc
00035030  00000000 00000000                        5000+          DS    2F                  extract PSW after test (has CC)
00035038  000359E0                                 5001+          DC    A(FMSG28)           address of test fail message
0003503C  00035740                                 5002+          DC    A(OPSRC28)          address of operand source
00035040  00035620                                 5003+          DC    A(OPDST28)          address of operand destination
00035044  00035860                                 5004+          DC    A(PL28)             address of parameter list
00035048  00004194                                 5005+          DC    A(RANDX+28)         address of random source data
0003504C  00035A55                                 5006+          DC    A(TEND28)           address of test end eye catche
                                                   5007
                                                   5008           TSTCODE
00035050                                           5009+CODE28    DS    0F
                                                   5010 *                                   load random source
00035050  E310 501C 0014                0000001C   5011           LGF   R1,OPSRCA           R1: OPSRC - Primary
00035056  B904 0061                                5012           LGR   R6,R1               R6: OPSRC - Secondary
0003505A  B904 0071                                5013           LGR   R7,R1               R7: OPSRC - Home
                                                   5014
0003505E  E320 5028 0014                00000028   5015           LGF   R2,RANDXA           R2: address random data
00035064  E330 5024 0014                00000024   5016           LGF   R3,PLA              R3: address PL
                                                   5017
0003506A  B219 0200                     00000200   5018           SAC   ARMODE
0003506E  D207 1000 2000       00000000 00000000   5019           MVC   0(8,R1),0(R2)       load Op1 = Primary
00035074  D207 6008 2008       00000008 00000008   5020           MVC   8(8,R6),8(R2)       load Op2 = Secondary
0003507A  D207 7010 2010       00000010 00000010   5021           MVC   16(8,R7),16(R2)     load Op3 = Home
00035080  D207 6018 2018       00000018 00000018   5022           MVC   24(8,R6),24(R2)     load Op4 = Secondary
                                                   5023 *
                                                   5024 * Quadruple Compare and Swap (QCSG: 35)
                                                   5025 * Build Parambter List
                                                   5026 *
00035086  D21F 3040 2000       00000040 00000000   5027           MVC   64(32,R3),0(R2)     Load PL OP1-Op4 Compare
0003508C  D21F 3060 2080       00000060 00000080   5028           MVC   96(32,R3),128(R2)   Load PL Op1-Op4 Replacement
                                                   5029 ****
                                                   5030 * force a failure by changing the 3rd operand
                                                   5031 ****
                                                   5032 *         MVC   24(4,R6),256(R2)    some random data
                                                   5033 ****
00035092  E313 0000 0024                00000000   5034           STG   R1,0(R3)            load OP1 address in PL
00035098  E54C 3020 0000                00000020   5035           MVHI  32(R3),PRIALET
                                                   5036
0003509E  4111 0008                     00000008   5037           LA    R1,8(R1)
000350A2  E313 0008 0024                00000008   5038           STG   R1,8(R3)            load OP2 address in PL
000350A8  E54C 3024 0001                00000024   5039           MVHI  36(R3),SECALET
                                                   5040
000350AE  4111 0008                     00000008   5041           LA    R1,8(R1)
000350B2  E313 0010 0024                00000010   5042           STG   R1,16(R3)           load OP3 address in PL
000350B8  E54C 3028 0002                00000028   5043           MVHI  40(R3),HOMEALET
                                                   5044
000350BE  4111 0008                     00000008   5045           LA    R1,8(R1)
000350C2  E313 0018 0024                00000018   5046           STG   R1,24(R3)           load OP4 address in PL
000350C8  E54C 302C 0001                0000002C   5047           MVHI  44(R3),SECALET
                                                   5048
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000350CE  B982 0000                            5049           XGR   R0,R0
000350D2  4100 0023                  00000023  5050           LA    R0,35
                                               5051
000350D6  B219 0200                  00000200  5052           SAC   ARMODE
000350DA  EE0F 0000 3000  00000000   00000000  5053           PLO   0,0,R15,0(R3)       Quadruple C/S (QCSG: 35)
000350E0  B219 0000                  00000000  5054           SAC   PRIMODE
                                               5055
000350E4  B98D 0020                            5056           EPSW  R2,R0               extract psw
000350E8  5020 5010                  00000010  5057           ST    R2,CCPSW              to save CC
000350EC  E310 500B 0076             0000000B  5058           LB    R1,CCMASK           (failure CC mask)
000350F2  8910 0004                  00000004  5059           SLL   R1,4                (shift to BC instr CC position)
000350F6  4410 5136                  00035156  5060           EX    R1,F355TSTCC            fail if...
                                               5061
                          000350FA   00000001  5062 F355CKUR  EQU   *
000350FA  D200 5009 3030  00000009   00000030  5063           MVC   URFOUND,48(R3)      Extract UR
00035100  D500 5008 3030  00000008   00000030  5064           CLC   UR,48(R3)
00035106  4780 50EE                  0003510E  5065           BE    F355CKRES
0003510A  45F0 8388                  00002388  5066           BAL   R15,URMSG
                                               5067 *
                                               5068 * compare results
                                               5069 *
                          0003510E   00000001  5070 F355CKRES EQU   *
                                               5071 *                                  reset addresses
0003510E  E310 501C 0014             0000001C  5072           LGF   R1,OPSRCA           R1:  OPSRC - Primary
00035114  B904 0061                            5073           LGR   R6,R1               R6:  OPSRC - Secondary
00035118  B904 0071                            5074           LGR   R7,R1               R7:  OPSRC - Home
                                               5075
                                               5076 * All equal; replacement values copied to operand source
                                               5077
0003511C  B219 0200                  00000200  5078           SAC   ARMODE
00035120  D507 1000 3060  00000000   00000060  5079           CLC   0(8,R1),96(R3)
00035126  4770 512C                  0003514C  5080           BNE   F355XX              OP1R Different, error
0003512A  D507 6008 3068  00000008   00000068  5081           CLC   8(8,R6),104(R3)
00035130  4770 512C                  0003514C  5082           BNE   F355XX              OP2R Different, error
00035134  D507 7010 3070  00000010   00000070  5083           CLC   16(8,R7),112(R3)
0003513A  4770 512C                  0003514C  5084           BNE   F355XX              OP3R Different, error
0003513E  D507 6018 3078  00000018   00000078  5085           CLC   24(8,R6),120(R3)
00035144  B219 0000                  00000000  5086           SAC   PRIMODE
00035148  478B 0000                  00000000  5087           BE    0(R11)              OP4R same, return
                                               5088
                          0003514C   00000001  5089 F355XX    EQU   *
0003514C  B219 0000                  00000000  5090           SAC   PRIMODE
00035150  45F0 8450                  00002450  5091           BAL   R15,FAILMSG
00035154  07FB                                 5092           BR    R11
                                               5093
00035156  4700 513A                  0003515A  5094 F355TSTCC BC    0,F355CCMSG         fail if unexpected condition code
                          0003515A   00000001  5095 F355CCMSG EQU   *
0003515A  45F0 82B2                  000022B2  5096           BAL   R15,CCMSG
0003515E  47F0 50DA                  000350FA  5097           B     F355CKUR
                                               5098
                                               5099           TSTDATA
00035162                  00035162   00035600  5100+          ORG   PLOETST+(128*K)+28*(3*K)+(K+512)
                                               5101           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00035600  D7D9C940 40406E6E                     5102+          DC    CL32'PRI   >>>OPDST28<<<' eye catcher
00035608  6ED6D7C4 E2E3F2F8
00035610  4C4C4C40 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00035618  40404040 40404040
00035620                                             5103+OPDST28  DS    0C
00035620  00000000 00000000                          5104          DC    XL256'0'
00035628  00000000 00000000
00035630  00000000 00000000
00035638  00000000 00000000
00035640  00000000 00000000
00035648  00000000 00000000
00035650  00000000 00000000
00035658  00000000 00000000
00035660  00000000 00000000
00035668  00000000 00000000
00035670  00000000 00000000
00035678  00000000 00000000
00035680  00000000 00000000
00035688  00000000 00000000
00035690  00000000 00000000
00035698  00000000 00000000
000356A0  00000000 00000000
000356A8  00000000 00000000
000356B0  00000000 00000000
000356B8  00000000 00000000
000356C0  00000000 00000000
000356C8  00000000 00000000
000356D0  00000000 00000000
000356D8  00000000 00000000
000356E0  00000000 00000000
000356E8  00000000 00000000
000356F0  00000000 00000000
000356F8  00000000 00000000
00035700  00000000 00000000
00035708  00000000 00000000
00035710  00000000 00000000
00035718  00000000 00000000
                                                     5105
                                                     5106          TSTLBL OPSRC              Source: OP1-OP4
00035720  D7D9C940 40406E6E                          5107+         DC    CL32'PRI   >>>OPSRC28<<<' eye catcher
00035728  6ED6D7E2 D9C3F2F8
00035730  4C4C4C40 40404040
00035738  40404040 40404040
00035740                                             5108+OPSRC28  DS    0C
00035740  00000000 00000000                          5109          DC    XL256'0'
00035748  00000000 00000000
00035750  00000000 00000000
00035758  00000000 00000000
00035760  00000000 00000000
00035768  00000000 00000000
00035770  00000000 00000000
00035778  00000000 00000000
00035780  00000000 00000000
00035788  00000000 00000000
00035790  00000000 00000000
00035798  00000000 00000000
000357A0  00000000 00000000
000357A8  00000000 00000000
000357B0  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000357B8  00000000 00000000
000357C0  00000000 00000000
000357C8  00000000 00000000
000357D0  00000000 00000000
000357D8  00000000 00000000
000357E0  00000000 00000000
000357E8  00000000 00000000
000357F0  00000000 00000000
000357F8  00000000 00000000
00035800  00000000 00000000
00035808  00000000 00000000
00035810  00000000 00000000
00035818  00000000 00000000
00035820  00000000 00000000
00035828  00000000 00000000
00035830  00000000 00000000
00035838  00000000 00000000
                                        5110
                                        5111          TSTLBL PL                 Parameter List
00035840  D7D9C940 40406E6E             5112+         DC    CL32'PRI   >>>PL28<<<'    eye catcher
00035848  6ED7D3F2 F84C4C4C
00035850  40404040 40404040
00035858  40404040 40404040
00035860                                5113+PL28     DS    0C
00035860  00000000 00000000             5114          DC    XL352'00'
00035868  00000000 00000000
00035870  00000000 00000000
00035878  00000000 00000000
00035880  00000000 00000000
00035888  00000000 00000000
00035890  00000000 00000000
00035898  00000000 00000000
000358A0  00000000 00000000
000358A8  00000000 00000000
000358B0  00000000 00000000
000358B8  00000000 00000000
000358C0  00000000 00000000
000358C8  00000000 00000000
000358D0  00000000 00000000
000358D8  00000000 00000000
000358E0  00000000 00000000
000358E8  00000000 00000000
000358F0  00000000 00000000
000358F8  00000000 00000000
00035900  00000000 00000000
00035908  00000000 00000000
00035910  00000000 00000000
00035918  00000000 00000000
00035920  00000000 00000000
00035928  00000000 00000000
00035930  00000000 00000000
00035938  00000000 00000000
00035940  00000000 00000000
00035948  00000000 00000000
00035950  00000000 00000000
00035958  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00035960  00000000 00000000
00035968  00000000 00000000
00035970  00000000 00000000
00035978  00000000 00000000
00035980  00000000 00000000
00035988  00000000 00000000
00035990  00000000 00000000
00035998  00000000 00000000
000359A0  00000000 00000000
000359A8  00000000 00000000
000359B0  00000000 00000000
000359B8  00000000 00000000
                                          5115
                                          5116          TSTLBL FMSG               fail message
000359C0  D7D9C940 40406E6E               5117+         DC    CL32'PRI   >>>FMSG28<<<'  eye catcher
000359C8  6EC6D4E2 C7F2F84C
000359D0  4C4C4040 40404040
000359D8  40404040 40404040
000359E0                                  5118+FMSG28   DS    0C
000359E0  86818993 858440D7               5119          DC    C'failed PLO-Extension '
000359E8  D3D660C5 A7A38595
000359F0  A2899695 40
000359F5  D8A48184 99A49793               5120          DC    CL96'Quadruple Compare and Swap (QCSG: 35): All EQ'
000359FD  8540C396 94978199
00035A05  85408195 8440E2A6
00035A0D  8197404D D8C3E2C7
00035A15  7A40F3F5 5D7A40C1
00035A1D  939340C5 D8404040
00035A25  40404040 40404040
00035A2D  40404040 40404040
00035A35  40404040 40404040
00035A3D  40404040 40404040
00035A45  40404040 40404040
00035A4D  40404040 40404040
                                       5121
                                       5122          TSTEND
00035A55  D7D9C940 4040C5D5            5123+TEND28   DC    CL32'PRI   END of test: 28 <<<'
00035A5D  C4409686 40A385A2
00035A65  A37A40F2 F8404C4C
00035A6D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   5125 *==============================================================================
                                                   5126
                                                   5127 * ------------------------------------------------------------------------
                                                   5128 * Test Quadruple Compare and Swap (QCS: 34) instruction : 1st NE
                                                   5129 * ------------------------------------------------------------------------
                                                   5130          TSTHDR 2,1,34             1st Operand NOT Equal (cc=2, UR=1)
00035A75                          00035A75  00035C00 5131+         ORG   PLOETST+(128*K)+29*(3*K)
00035C00  D7D9C940 4040E2E3                        5132+         DC    CL32'PRI   START of test: 29'
00035C08  C1D9E340 968640A3
00035C10  85A2A37A 40F2F940
00035C18  40404040 40404040
00035C20                          00035C20         5133+         USING *,R5                base for test data and test routine
00035C20                          00000000         5134+         USING PLOETEST,R5
00035C20  00035C50                                 5135+T29      DC    A(CODE29)           address of test code
00035C24  001D                                     5136+         DC    H'29'               test number
00035C26  00                                       5137+         DC    X'00'
00035C27  22                                       5138+         DC    HL1'34'             Function
00035C28  01                                       5139+         DC    HL1'1'              expected unequal results (UR)
00035C29  00                                       5140+         DC    HL1'00'             found UR
00035C2A  02                                       5141+         DC    HL1'2'              cc
00035C2B  0D                                       5142+         DC    HL1'13'             not expected CC mask
00035C2C  00                                       5143+         DS    X                   extracted cc
00035C30  00000000 00000000                        5144+         DS    2F                  extract PSW after test (has CC)
00035C38  000365E0                                 5145+         DC    A(FMSG29)           address of test fail message
00035C3C  00036340                                 5146+         DC    A(OPSRC29)          address of operand source
00035C40  00036220                                 5147+         DC    A(OPDST29)          address of operand destination
00035C44  00036460                                 5148+         DC    A(PL29)             address of parameter list
00035C48  00004195                                 5149+         DC    A(RANDX+29)         address of random source data
00035C4C  00036655                                 5150+         DC    A(TEND29)           address of test end eye catche
                                                   5151
                                                   5152          TSTCODE
00035C50                                           5153+CODE29   DS    0F
                                                   5154 *                                  load random source
00035C50  E310 501C 0014                0000001C   5155          LGF   R1,OPSRCA           R1: OPSRC - Primary
00035C56  B904 0061                                5156          LGR   R6,R1               R6: OPSRC - Secondary
00035C5A  B904 0071                                5157          LGR   R7,R1               R7: OPSRC - Home
                                                   5158
00035C5E  E320 5028 0014                00000028   5159          LGF   R2,RANDXA           R2: address random data
00035C64  E330 5024 0014                00000024   5160          LGF   R3,PLA              R3: address PL
                                                   5161
00035C6A  B219 0200                     00000200   5162          SAC   ARMODE
00035C6E  D203 1000 2000  00000000      00000000   5163          MVC   0(4,R1),0(R2)       OP1 - Primary
00035C74  D203 6004 2004  00000004      00000004   5164          MVC   4(4,R6),4(R2)       OP2 - Secondary
00035C7A  D203 7008 2008  00000008      00000008   5165          MVC   8(4,R7),8(R2)       OP3 - Home
00035C80  D203 600C 200C  0000000C      0000000C   5166          MVC   12(4,R6),12(R2)     OP4 - Secondary
                                                   5167 *
                                                   5168 * Quadruple Compare and Swap (QCS: 34)
                                                   5169 * Build Parambter List
                                                   5170 *
00035C86  D203 3040 2000  00000040      00000000   5171          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
00035C8C  D203 3048 2004  00000048      00000004   5172          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00035C92  D203 3050 2008  00000050      00000008   5173          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
00035C98  D203 3058 200C  00000058      0000000C   5174          MVC   88(4,R3),12(R2)     Load PL OP4-Compare
                                                   5175
00035C9E  D203 3060 2080  00000060      00000080   5176          MVC   96(4,R3),128+0(R2)   Load PL OP1-Replacement
00035CA4  D203 3068 2084  00000068      00000084   5177          MVC   104(4,R3),128+4(R2)  Load PL OP2-Replacement
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00035CAA  D203 3070 2088      00000070  00000088  5178          MVC   112(4,R3),128+8(R2)  Load PL OP3-Replacement
00035CB0  D203 3078 208C      00000078  0000008C  5179          MVC   120(4,R3),128+12(R2) Load PL OP4-Replacement
                                        0000008C  5180 ****
                                        0000008C  5181 * force a failure by changing the 1st operand
                                        0000008C  5182 ****
00035CB6  D201 1000 2100      00000000  00000100  5183          MVC   0(2,R1),256(R2)     some random data
                                        00000100  5184 ****
00035CBC  E313 0000 0024                00000000  5185          STG   R1,0(R3)            load OP1 address in PL
00035CC2  E54C 3020 0000                00000020  5186          MVHI  32(R3),PRIALET
                                        00000026  5187
00035CC8  4111 0004                     00000004  5188          LA    R1,4(R1)
00035CCC  E313 0008 0024                00000008  5189          STG   R1,8(R3)            load OP2 address in PL
00035CD2  E54C 3024 0001                00000024  5190          MVHI  36(R3),SECALET
                                        0000002A  5191
00035CD8  4111 0004                     00000004  5192          LA    R1,4(R1)
00035CDC  E313 0010 0024                00000010  5193          STG   R1,16(R3)           load OP3 address in PL
00035CE2  E54C 3028 0002                00000028  5194          MVHI  40(R3),HOMEALET
                                        0000002E  5195
00035CE8  4111 0004                     00000004  5196          LA    R1,4(R1)
00035CEC  E313 0018 0024                00000018  5197          STG   R1,24(R3)           load OP4 address in PL
00035CF2  E54C 302C 0001                0000002C  5198          MVHI  44(R3),SECALET
                                        00000032  5199
00035CF8  B982 0000                               5200          XGR   R0,R0
00035CFC  4100 0022                     00000022  5201          LA    R0,34
                                                  5202
00035D00  B219 0200                     00000200  5203          SAC   ARMODE
00035D04  EE0F 0000 3000      00000000  00000000  5204          PLO   0,0,R15,0(R3)       Quadruple C/S (QCS: 34)
00035D0A  B219 0000                     00000000  5205          SAC   PRIMODE
                                                  5206
00035D0E  B98D 0020                               5207          EPSW  R2,R0               extract psw
00035D12  5020 5010                     00000010  5208          ST    R2,CCPSW              to save CC
00035D16  E310 500B 0076                0000000B  5209          LB    R1,CCMASK           (failure CC mask)
00035D1C  8910 0004                     00000004  5210          SLL   R1,4                (shift to BC instr CC position)
00035D20  4410 5160                     00035D80  5211          EX    R1,F341TSTCC            fail if...
                                                  5212
                          00035D24      00000001  5213 F341CKUR EQU   *
00035D24  D200 5009 3030  00000009      00000030  5214          MVC   URFOUND,48(R3)      Extract UR
00035D2A  D500 5008 3030  00000008      00000030  5215          CLC   UR,48(R3)
00035D30  4780 5118                     00035D38  5216          BE    F341CKRES
00035D34  45F0 8388                     00002388  5217          BAL   R15,URMSG
                                                  5218 *
                                                  5219 * compare results
                                                  5220 *
                          00035D38      00000001  5221 F341CKRES EQU   *
                                                  5222 *                              reset addresses
00035D38  E310 501C 0014                0000001C  5223          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00035D3E  B904 0061                               5224          LGR   R6,R1               R6:  OPSRC - Secondary
00035D42  B904 0071                               5225          LGR   R7,R1               R7:  OPSRC - Home
                                                  5226
00035D46  B219 0200                     00000200  5227          SAC   ARMODE
00035D4A  D503 1000 3040  00000000      00000040  5228          CLC   0(4,R1),64(R3)
00035D50  4770 5156                     00035D76  5229          BNE   F341XX              OP1C Different, error
00035D54  D503 6004 3048  00000004      00000048  5230          CLC   4(4,R6),72(R3)
00035D5A  4770 5156                     00035D76  5231          BNE   F341XX              OP2C Different, error
00035D5E  D503 7008 3050  00000008      00000050  5232          CLC   8(4,R7),80(R3)
00035D64  4770 5156                     00035D76  5233          BNE   F341XX              OP3C Different, error
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00035D68  D503 600C 3058      0000000C  00000058  5234           CLC   12(4,R6),88(R3)
00035D6E  B219 0000                     00000000  5235           SAC   PRIMODE
00035D72  478B 0000                     00000000  5236           BE    0(R11)              OP4Csame, return
                                                  5237
                                00035D76  00000001  5238 F341XX    EQU   *
00035D76  B219 0000                     00000000  5239           SAC   PRIMODE
00035D7A  45F0 8450                     00002450  5240           BAL   R15,FAILMSG
00035D7E  07FB                                    5241           BR    R11
                                                  5242
00035D80  4700 5164                     00035D84  5243 F341TSTCC BC     0,F341CCMSG         fail if unexpected condition code
                                00035D84  00000001  5244 F341CCMSG EQU   *
00035D84  45F0 82B2                     000022B2  5245           BAL   R15,CCMSG
00035D88  47F0 5104                     00035D24  5246           B     F341CKUR
                                                  5247
                                                  5248           TSTDATA
00035D8C                        00035D8C  00036200  5249+          ORG   PLOETST+(128*K)+29*(3*K)+(K+512)
                                                  5250           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00036200  D7D9C940 40406E6E                       5251+          DC    CL32'PRI   >>>OPDST29<<<' eye catcher
00036208  6ED6D7C4 E2E3F2F9
00036210  4C4C4C40 40404040
00036218  40404040 40404040
00036220                                          5252+OPDST29   DS    0C
00036220  00000000 00000000                       5253           DC    XL256'0'
00036228  00000000 00000000
00036230  00000000 00000000
00036238  00000000 00000000
00036240  00000000 00000000
00036248  00000000 00000000
00036250  00000000 00000000
00036258  00000000 00000000
00036260  00000000 00000000
00036268  00000000 00000000
00036270  00000000 00000000
00036278  00000000 00000000
00036280  00000000 00000000
00036288  00000000 00000000
00036290  00000000 00000000
00036298  00000000 00000000
000362A0  00000000 00000000
000362A8  00000000 00000000
000362B0  00000000 00000000
000362B8  00000000 00000000
000362C0  00000000 00000000
000362C8  00000000 00000000
000362D0  00000000 00000000
000362D8  00000000 00000000
000362E0  00000000 00000000
000362E8  00000000 00000000
000362F0  00000000 00000000
000362F8  00000000 00000000
00036300  00000000 00000000
00036308  00000000 00000000
00036310  00000000 00000000
00036318  00000000 00000000
                                                  5254
                                                  5255           TSTLBL OPSRC              Source: OP1-OP4
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00036320  D7D9C940 40406E6E                         5256+          DC    CL32'PRI   >>>OPSRC29<<<' eye catcher
00036328  6ED6D7E2 D9C3F2F9
00036330  4C4C4C40 40404040
00036338  40404040 40404040
00036340                                            5257+OPSRC29   DS    0C
00036340  00000000 00000000                         5258           DC    XL256'0'
00036348  00000000 00000000
00036350  00000000 00000000
00036358  00000000 00000000
00036360  00000000 00000000
00036368  00000000 00000000
00036370  00000000 00000000
00036378  00000000 00000000
00036380  00000000 00000000
00036388  00000000 00000000
00036390  00000000 00000000
00036398  00000000 00000000
000363A0  00000000 00000000
000363A8  00000000 00000000
000363B0  00000000 00000000
000363B8  00000000 00000000
000363C0  00000000 00000000
000363C8  00000000 00000000
000363D0  00000000 00000000
000363D8  00000000 00000000
000363E0  00000000 00000000
000363E8  00000000 00000000
000363F0  00000000 00000000
000363F8  00000000 00000000
00036400  00000000 00000000
00036408  00000000 00000000
00036410  00000000 00000000
00036418  00000000 00000000
00036420  00000000 00000000
00036428  00000000 00000000
00036430  00000000 00000000
00036438  00000000 00000000
                                                    5259
                                                    5260           TSTLBL PL                  Parameter List
00036440  D7D9C940 40406E6E                         5261+          DC    CL32'PRI   >>>PL29<<<'    eye catcher
00036448  6ED7D3F2 F94C4C4C
00036450  40404040 40404040
00036458  40404040 40404040
00036460                                            5262+PL29      DS    0C
00036460  00000000 00000000                         5263           DC    XL352'00'
00036468  00000000 00000000
00036470  00000000 00000000
00036478  00000000 00000000
00036480  00000000 00000000
00036488  00000000 00000000
00036490  00000000 00000000
00036498  00000000 00000000
000364A0  00000000 00000000
000364A8  00000000 00000000
000364B0  00000000 00000000
000364B8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000364C0  00000000 00000000
000364C8  00000000 00000000
000364D0  00000000 00000000
000364D8  00000000 00000000
000364E0  00000000 00000000
000364E8  00000000 00000000
000364F0  00000000 00000000
000364F8  00000000 00000000
00036500  00000000 00000000
00036508  00000000 00000000
00036510  00000000 00000000
00036518  00000000 00000000
00036520  00000000 00000000
00036528  00000000 00000000
00036530  00000000 00000000
00036538  00000000 00000000
00036540  00000000 00000000
00036548  00000000 00000000
00036550  00000000 00000000
00036558  00000000 00000000
00036560  00000000 00000000
00036568  00000000 00000000
00036570  00000000 00000000
00036578  00000000 00000000
00036580  00000000 00000000
00036588  00000000 00000000
00036590  00000000 00000000
00036598  00000000 00000000
000365A0  00000000 00000000
000365A8  00000000 00000000
000365B0  00000000 00000000
000365B8  00000000 00000000
                             5264
                             5265          TSTLBL FMSG               fail message
000365C0  D7D9C940 40406E6E  5266+         DC    CL32'PRI   >>>FMSG29<<<'  eye catcher
000365C8  6EC6D4E2 C7F2F94C
000365D0  4C4C4040 40404040
000365D8  40404040 40404040
000365E0                     5267+FMSG29   DS    0C
000365E0  86818993 858440D7  5268          DC    C'failed PLO-Extension '
000365E8  D3D660C5 A7A38595
000365F0  A2899695 40
000365F5  D8A48184 99A49793  5269          DC    CL96'Quadruple Compare and Swap (QCS: 34): 1st NE'
000365FD  8540C396 94978199
00036605  85408195 8440E2A6
0003660D  8197404D D8C3E27A
00036615  40F3F45D 7A40F1A2
0003661D  A340D5C5 40404040
00036625  40404040 40404040
0003662D  40404040 40404040
00036635  40404040 40404040
0003663D  40404040 40404040
00036645  40404040 40404040
0003664D  40404040 40404040
                             5270
                             5271          TSTEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00036655  D7D9C940 4040C5D5                        5272+TEND29   DC    CL32'PRI   END of test: 29 <<<'
0003665D  C4409686 40A385A2
00036665  A37A40F2 F9404C4C
0003666D  4C404040 40404040
                                                   5273
                                                   5274 * ------------------------------------------------------------------
                                                   5275 * Test Quadruple Compare and Swap (QCS: 34) instruction : 2nd NE
                                                   5276 * ------------------------------------------------------------------
                                                   5277          TSTHDR 2,2,34             2nd Operand NOT Equal (cc=2, UR=2)
00036675                    00036675  00036800     5278+         ORG   PLOETST+(128*K)+30*(3*K)
00036800  D7D9C940 4040E2E3                        5279+         DC    CL32'PRI   START of test: 30'
00036808  C1D9E340 968640A3
00036810  85A2A37A 40F3F040
00036818  40404040 40404040
00036820                    00036820              5280+         USING *,R5                base for test data and test routine
00036820                    00000000              5281+         USING PLOETEST,R5
00036820  00036850                                5282+T30      DC    A(CODE30)           address of test code
00036824  001E                                    5283+         DC    H'30'               test number
00036826  00                                      5284+         DC    X'00'
00036827  22                                      5285+         DC    HL1'34'             Function
00036828  02                                      5286+         DC    HL1'2'              expected unequal results (UR)
00036829  00                                      5287+         DC    HL1'00'             found UR
0003682A  02                                      5288+         DC    HL1'2'              cc
0003682B  0D                                      5289+         DC    HL1'13'             not expected CC mask
0003682C  00                                      5290+         DS    X                   extracted cc
00036830  00000000 00000000                       5291+         DS    2F                  extract PSW after test (has CC)
00036838  000371E0                                5292+         DC    A(FMSG30)           address of test fail message
0003683C  00036F40                                5293+         DC    A(OPSRC30)          address of operand source
00036840  00036E20                                5294+         DC    A(OPDST30)          address of operand destination
00036844  00037060                                5295+         DC    A(PL30)             address of parameter list
00036848  00004196                                5296+         DC    A(RANDX+30)         address of random source data
0003684C  00037255                                5297+         DC    A(TEND30)           address of test end eye catche
                                                   5298
                                                   5299          TSTCODE
00036850                                          5300+CODE30   DS    0F
                                                   5301 *                                  load random source
00036850  E310 501C 0014              0000001C    5302          LGF   R1,OPSRCA           R1: OPSRC - Primary
00036856  B904 0061                               5303          LGR   R6,R1               R6: OPSRC - Secondary
0003685A  B904 0071                               5304          LGR   R7,R1               R7: OPSRC - Home
                                                   5305
0003685E  E320 5028 0014              00000028    5306          LGF   R2,RANDXA           R2: address random data
00036864  E330 5024 0014              00000024    5307          LGF   R3,PLA              R3: address PL
                                                   5308
0003686A  B219 0200                   00000200    5309          SAC   ARMODE
0003686E  D203 1000 2000    00000000  00000000    5310          MVC   0(4,R1),0(R2)       OP1 - Primary
00036874  D203 6004 2004    00000004  00000004    5311          MVC   4(4,R6),4(R2)       OP2 - Secondary
0003687A  D203 7008 2008    00000008  00000008    5312          MVC   8(4,R7),8(R2)       OP3 - Home
00036880  D203 600C 200C    0000000C  0000000C    5313          MVC   12(4,R6),12(R2)     OP4 - Secondary
                                                   5314 *
                                                   5315 * Quadruple Compare and Swap (QCS: 34)
                                                   5316 * Build Parambter List
                                                   5317 *
00036886  D203 3040 2000    00000040  00000000    5318          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
0003688C  D203 3048 2004    00000048  00000004    5319          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00036892  D203 3050 2008    00000050  00000008    5320          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
00036898  D203 3058 200C    00000058  0000000C    5321          MVC   88(4,R3),12(R2)     Load PL OP4-Compare
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5322
0003689E  D203 3060 2080      00000060  00000080  5323          MVC   96(4,R3),128+0(R2)   Load PL OP1-Replacement
000368A4  D203 3068 2084      00000068  00000084  5324          MVC   104(4,R3),128+4(R2)  Load PL OP2-Replacement
000368AA  D203 3070 2088      00000070  00000088  5325          MVC   112(4,R3),128+8(R2)  Load PL OP3-Replacement
000368B0  D203 3078 208C      00000078  0000008C  5326          MVC   120(4,R3),128+12(R2) Load PL OP4-Replacement
                                                    5327 ****
                                                    5328 * force a failure by changing the 2nd operand
                                                    5329 ****
000368B6  D201 6004 2100      00000004  00000100  5330          MVC   4(2,R6),256(R2)      some random data
                                                    5331 ****
000368BC  E313 0000 0024                00000000  5332          STG   R1,0(R3)             load OP1 address in PL
000368C2  E54C 3020 0000                00000020  5333          MVHI  32(R3),PRIALET
                                                    5334
000368C8  4111 0004                     00000004  5335          LA    R1,4(R1)
000368CC  E313 0008 0024                00000008  5336          STG   R1,8(R3)             load OP2 address in PL
000368D2  E54C 3024 0001                00000024  5337          MVHI  36(R3),SECALET
                                                    5338
000368D8  4111 0004                     00000004  5339          LA    R1,4(R1)
000368DC  E313 0010 0024                00000010  5340          STG   R1,16(R3)            load OP3 address in PL
000368E2  E54C 3028 0002                00000028  5341          MVHI  40(R3),HOMEALET
                                                    5342
000368E8  4111 0004                     00000004  5343          LA    R1,4(R1)
000368EC  E313 0018 0024                00000018  5344          STG   R1,24(R3)            load OP4 address in PL
000368F2  E54C 302C 0001                0000002C  5345          MVHI  44(R3),SECALET
                                                    5346
000368F8  B982 0000                               5347          XGR   R0,R0
000368FC  4100 0022                     00000022  5348          LA    R0,34
                                                    5349
00036900  B219 0200                     00000200  5350          SAC   ARMODE
00036904  EE0F 0000 3000      00000000  00000000  5351          PLO   0,0,R15,0(R3)        Quadruple C/S (QCS: 34)
0003690A  B219 0000                     00000000  5352          SAC   PRIMODE
                                                    5353
0003690E  B98D 0020                               5354          EPSW  R2,R0                extract psw
00036912  5020 5010                     00000010  5355          ST    R2,CCPSW               to save CC
00036916  E310 500B 0076                0000000B  5356          LB    R1,CCMASK            (failure CC mask)
0003691C  8910 0004                     00000004  5357          SLL   R1,4                 (shift to BC instr CC position)
00036920  4410 5160                     00036980  5358          EX    R1,F342TSTCC             fail if...
                                                    5359
                                00036924  00000001  5360 F342CKUR EQU   *
00036924  D200 5009 3030      00000009  00000030  5361          MVC   URFOUND,48(R3)       Extract UR
0003692A  D500 5008 3030      00000008  00000030  5362          CLC   UR,48(R3)
00036930  4780 5118                     00036938  5363          BE    F342CKRES
00036934  45F0 8388                     00002388  5364          BAL   R15,URMSG
                                                    5365 *
                                                    5366 * compare results
                                                    5367 *
                                00036938  00000001  5368 F342CKRES EQU   *
                                                    5369 *                             reset addresses
00036938  E310 501C 0014                0000001C  5370          LGF   R1,OPSRCA            R1:  OPSRC - Primary
0003693E  B904 0061                               5371          LGR   R6,R1                R6:  OPSRC - Secondary
00036942  B904 0071                               5372          LGR   R7,R1                R7:  OPSRC - Home
                                                    5373
00036946  B219 0200                     00000200  5374          SAC   ARMODE
0003694A  D503 1000 3040      00000000  00000040  5375          CLC   0(4,R1),64(R3)
00036950  4770 5156                     00036976  5376          BNE   F342XX               OP1C Different, error
00036954  D503 6004 3048      00000004  00000048  5377          CLC   4(4,R6),72(R3)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003695A  4770 5156                      00036976  5378           BNE   F342XX              OP2C Different, error
0003695E  D503 7008 3050      00000008  00000050  5379           CLC   8(4,R7),80(R3)
00036964  4770 5156                      00036976  5380           BNE   F342XX              OP3C Different, error
00036968  D503 600C 3058      0000000C  00000058  5381           CLC   12(4,R6),88(R3)
0003696E  B219 0000                      00000000  5382           SAC   PRIMODE
00036972  478B 0000                      00000000  5383           BE    0(R11)              OP4Csame, return
                                                   5384
                              00036976  00000001  5385 F342XX    EQU   *
00036976  B219 0000                      00000000  5386           SAC   PRIMODE
0003697A  45F0 8450                      00002450  5387           BAL   R15,FAILMSG
0003697E  07FB                                     5388           BR    R11
                                                   5389
00036980  4700 5164                      00036984  5390 F342TSTCC BC     0,F342CCMSG         fail if unexpected condition code
                              00036984  00000001  5391 F342CCMSG EQU   *
00036984  45F0 82B2                      000022B2  5392           BAL   R15,CCMSG
00036988  47F0 5104                      00036924  5393           B     F342CKUR
                                                   5394
                                                   5395           TSTDATA
0003698C                      0003698C  00036E00  5396+          ORG   PLOETST+(128*K)+30*(3*K)+(K+512)
                                                   5397           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00036E00  D7D9C940 40406E6E                        5398+          DC    CL32'PRI   >>>OPDST30<<<' eye catcher
00036E08  6ED6D7C4 E2E3F3F0
00036E10  4C4C4C40 40404040
00036E18  40404040 40404040
00036E20                                           5399+OPDST30   DS    0C
00036E20  00000000 00000000                        5400           DC    XL256'0'
00036E28  00000000 00000000
00036E30  00000000 00000000
00036E38  00000000 00000000
00036E40  00000000 00000000
00036E48  00000000 00000000
00036E50  00000000 00000000
00036E58  00000000 00000000
00036E60  00000000 00000000
00036E68  00000000 00000000
00036E70  00000000 00000000
00036E78  00000000 00000000
00036E80  00000000 00000000
00036E88  00000000 00000000
00036E90  00000000 00000000
00036E98  00000000 00000000
00036EA0  00000000 00000000
00036EA8  00000000 00000000
00036EB0  00000000 00000000
00036EB8  00000000 00000000
00036EC0  00000000 00000000
00036EC8  00000000 00000000
00036ED0  00000000 00000000
00036ED8  00000000 00000000
00036EE0  00000000 00000000
00036EE8  00000000 00000000
00036EF0  00000000 00000000
00036EF8  00000000 00000000
00036F00  00000000 00000000
00036F08  00000000 00000000
00036F10  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00036F18  00000000 00000000                          5401
                                                     5402           TSTLBL OPSRC              Source: OP1-OP4
00036F20  D7D9C940 40406E6E                          5403+          DC    CL32'PRI   >>>OPSRC30<<<' eye catcher
00036F28  6ED6D7E2 D9C3F3F0
00036F30  4C4C4C40 40404040
00036F38  40404040 40404040
00036F40                                             5404+OPSRC30   DS    0C
00036F40  00000000 00000000                          5405           DC    XL256'0'
00036F48  00000000 00000000
00036F50  00000000 00000000
00036F58  00000000 00000000
00036F60  00000000 00000000
00036F68  00000000 00000000
00036F70  00000000 00000000
00036F78  00000000 00000000
00036F80  00000000 00000000
00036F88  00000000 00000000
00036F90  00000000 00000000
00036F98  00000000 00000000
00036FA0  00000000 00000000
00036FA8  00000000 00000000
00036FB0  00000000 00000000
00036FB8  00000000 00000000
00036FC0  00000000 00000000
00036FC8  00000000 00000000
00036FD0  00000000 00000000
00036FD8  00000000 00000000
00036FE0  00000000 00000000
00036FE8  00000000 00000000
00036FF0  00000000 00000000
00036FF8  00000000 00000000
00037000  00000000 00000000
00037008  00000000 00000000
00037010  00000000 00000000
00037018  00000000 00000000
00037020  00000000 00000000
00037028  00000000 00000000
00037030  00000000 00000000
00037038  00000000 00000000
                                                     5406
                                                     5407           TSTLBL PL                   Parameter List
00037040  D7D9C940 40406E6E                          5408+          DC    CL32'PRI   >>>PL30<<<'    eye catcher
00037048  6ED7D3F3 F04C4C4C
00037050  40404040 40404040
00037058  40404040 40404040
00037060                                             5409+PL30      DS    0C
00037060  00000000 00000000                          5410           DC    XL352'00'
00037068  00000000 00000000
00037070  00000000 00000000
00037078  00000000 00000000
00037080  00000000 00000000
00037088  00000000 00000000
00037090  00000000 00000000
00037098  00000000 00000000
000370A0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000370A8  00000000 00000000
000370B0  00000000 00000000
000370B8  00000000 00000000
000370C0  00000000 00000000
000370C8  00000000 00000000
000370D0  00000000 00000000
000370D8  00000000 00000000
000370E0  00000000 00000000
000370E8  00000000 00000000
000370F0  00000000 00000000
000370F8  00000000 00000000
00037100  00000000 00000000
00037108  00000000 00000000
00037110  00000000 00000000
00037118  00000000 00000000
00037120  00000000 00000000
00037128  00000000 00000000
00037130  00000000 00000000
00037138  00000000 00000000
00037140  00000000 00000000
00037148  00000000 00000000
00037150  00000000 00000000
00037158  00000000 00000000
00037160  00000000 00000000
00037168  00000000 00000000
00037170  00000000 00000000
00037178  00000000 00000000
00037180  00000000 00000000
00037188  00000000 00000000
00037190  00000000 00000000
00037198  00000000 00000000
000371A0  00000000 00000000
000371A8  00000000 00000000
000371B0  00000000 00000000
000371B8  00000000 00000000
                                      5411
                                      5412          TSTLBL FMSG               fail message
000371C0  D7D9C940 40406E6E           5413+         DC    CL32'PRI   >>>FMSG30<<<'  eye catcher
000371C8  6EC6D4E2 C7F3F04C
000371D0  4C4C4040 40404040
000371D8  40404040 40404040
000371E0                              5414+FMSG30   DS    0C
000371E0  86818993 858440D7           5415          DC    C'failed PLO-Extension '
000371E8  D3D660C5 A7A38595
000371F0  A2899695 40
000371F5  D8A48184 99A49793           5416          DC    CL96'Quadruple Compare and Swap (QCS: 34): 2nd NE'
000371FD  8540C396 94978199
00037205  85408195 8440E2A6
0003720D  8197404D D8C3E27A
00037215  40F3F45D 7A40F295
0003721D  8440D5C5 40404040
00037225  40404040 40404040
0003722D  40404040 40404040
00037235  40404040 40404040
0003723D  40404040 40404040
00037245  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003724D  40404040 40404040                        5417
                                                   5418          TSTEND
00037255  D7D9C940 4040C5D5                        5419+TEND30   DC    CL32'PRI   END of test: 30 <<<'
0003725D  C4409686 40A385A2
00037265  A37A40F3 F0404C4C
0003726D  4C404040 40404040                        5420
                                                   5421 * ------------------------------------------------------------------
                                                   5422 * Test Quadruple Compare and Swap (QCS: 34) instruction : 3rd NE
                                                   5423 * ------------------------------------------------------------------
                                                   5424          TSTHDR 2,3,34             3rd Operand NOT Equal (cc=2, UR=3)
00037275                    00037275  00037400     5425+         ORG   PLOETST+(128*K)+31*(3*K)
00037400  D7D9C940 4040E2E3                        5426+         DC    CL32'PRI   START of test: 31'
00037408  C1D9E340 968640A3
00037410  85A2A37A 40F3F140
00037418  40404040 40404040
00037420                    00037420              5427+         USING *,R5                base for test data and test routine
00037420                    00000000              5428+         USING PLOETEST,R5
00037420  00037450                                 5429+T31      DC    A(CODE31)           address of test code
00037424  001F                                     5430+         DC    H'31'               test number
00037426  00                                       5431+         DC    X'00'
00037427  22                                       5432+         DC    HL1'34'             Function
00037428  03                                       5433+         DC    HL1'3'              expected unequal results (UR)
00037429  00                                       5434+         DC    HL1'00'             found UR
0003742A  02                                       5435+         DC    HL1'2'              cc
0003742B  0D                                       5436+         DC    HL1'13'             not expected CC mask
0003742C  00                                       5437+         DS    X                   extracted cc
00037430  00000000 00000000                        5438+         DS    2F                  extract PSW after test (has CC)
00037438  00037DE0                                 5439+         DC    A(FMSG31)           address of test fail message
0003743C  00037B40                                 5440+         DC    A(OPSRC31)          address of operand source
00037440  00037A20                                 5441+         DC    A(OPDST31)          address of operand destination
00037444  00037C60                                 5442+         DC    A(PL31)             address of parameter list
00037448  00004197                                 5443+         DC    A(RANDX+31)         address of random source data
0003744C  00037E55                                 5444+         DC    A(TEND31)           address of test end eye catche
                                                   5445
                                                   5446          TSTCODE
00037450                                           5447+CODE31   DS    0F
                                                   5448 *                                  load random source
00037450  E310 501C 0014                0000001C   5449          LGF   R1,OPSRCA           R1: OPSRC - Primary
00037456  B904 0061                                5450          LGR   R6,R1               R6: OPSRC - Secondary
0003745A  B904 0071                                5451          LGR   R7,R1               R7: OPSRC - Home
                                                   5452
0003745E  E320 5028 0014                00000028   5453          LGF   R2,RANDXA           R2: address random data
00037464  E330 5024 0014                00000024   5454          LGF   R3,PLA              R3: address PL
                                                   5455
0003746A  B219 0200                     00000200   5456          SAC   ARMODE
0003746E  D203 1000 2000      00000000  00000000   5457          MVC   0(4,R1),0(R2)       OP1 - Primary
00037474  D203 6004 2004      00000004  00000004   5458          MVC   4(4,R6),4(R2)       OP2 - Secondary
0003747A  D203 7008 2008      00000008  00000008   5459          MVC   8(4,R7),8(R2)       OP3 - Home
00037480  D203 600C 200C      0000000C  0000000C   5460          MVC   12(4,R6),12(R2)     OP4 - Secondary
                                                   5461 *
                                                   5462 * Quadruple Compare and Swap (QCS: 34)
                                                   5463 * Build Parambter List
                                                   5464 *
00037486  D203 3040 2000      00000040  00000000   5465          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003748C  D203 3048 2004      00000048  00000004  5466          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00037492  D203 3050 2008      00000050  00000008  5467          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
00037498  D203 3058 200C      00000058  0000000C  5468          MVC   88(4,R3),12(R2)     Load PL OP4-Compare
                                                  5469
0003749E  D203 3060 2080      00000060  00000080  5470          MVC   96(4,R3),128+0(R2)   Load PL OP1-Replacement
000374A4  D203 3068 2084      00000068  00000084  5471          MVC   104(4,R3),128+4(R2)  Load PL OP2-Replacement
000374AA  D203 3070 2088      00000070  00000088  5472          MVC   112(4,R3),128+8(R2)  Load PL OP3-Replacement
000374B0  D203 3078 208C      00000078  0000008C  5473          MVC   120(4,R3),128+12(R2) Load PL OP4-Replacement
                                                  5474 ****
                                                  5475 * force a failure by changing the 3rd operand
                                                  5476 ****
000374B6  D201 7008 2100      00000008  00000100  5477          MVC   8(2,R7),256(R2)     some random data
                                                  5478 ****
000374BC  E313 0000 0024                00000000  5479          STG   R1,0(R3)            load OP1 address in PL
000374C2  E54C 3020 0000                00000020  5480          MVHI  32(R3),PRIALET
                                                  5481
000374C8  4111 0004                     00000004  5482          LA    R1,4(R1)
000374CC  E313 0008 0024                00000008  5483          STG   R1,8(R3)            load OP2 address in PL
000374D2  E54C 3024 0001                00000024  5484          MVHI  36(R3),SECALET
                                                  5485
000374D8  4111 0004                     00000004  5486          LA    R1,4(R1)
000374DC  E313 0010 0024                00000010  5487          STG   R1,16(R3)           load OP3 address in PL
000374E2  E54C 3028 0002                00000028  5488          MVHI  40(R3),HOMEALET
                                                  5489
000374E8  4111 0004                     00000004  5490          LA    R1,4(R1)
000374EC  E313 0018 0024                00000018  5491          STG   R1,24(R3)           load OP4 address in PL
000374F2  E54C 302C 0001                0000002C  5492          MVHI  44(R3),SECALET
                                                  5493
000374F8  B982 0000                               5494          XGR   R0,R0
000374FC  4100 0022                     00000022  5495          LA    R0,34
                                                  5496
00037500  B219 0200                     00000200  5497          SAC   ARMODE
00037504  EE0F 0000 3000      00000000  00000000  5498          PLO   0,0,R15,0(R3)       Quadruple C/S (QCS: 34)
0003750A  B219 0000                     00000000  5499          SAC   PRIMODE
                                                  5500
0003750E  B98D 0020                               5501          EPSW  R2,R0               extract psw
00037512  5020 5010                     00000010  5502          ST    R2,CCPSW              to save CC
00037516  E310 500B 0076                0000000B  5503          LB    R1,CCMASK           (failure CC mask)
0003751C  8910 0004                     00000004  5504          SLL   R1,4                (shift to BC instr CC position)
00037520  4410 5160                     00037580  5505          EX    R1,F343TSTCC            fail if...
                                                  5506
                                00037524  00000001  5507 F343CKUR EQU   *
00037524  D200 5009 3030      00000009  00000030  5508          MVC   URFOUND,48(R3)      Extract UR
0003752A  D500 5008 3030      00000008  00000030  5509          CLC   UR,48(R3)
00037530  4780 5118                     00037538  5510          BE    F343CKRES
00037534  45F0 8388                     00002388  5511          BAL   R15,URMSG
                                                  5512 *
                                                  5513 * compare results
                                                  5514 *
                                00037538  00000001  5515 F343CKRES EQU   *
                                                  5516 *                              reset addresses
00037538  E310 501C 0014                0000001C  5517          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003753E  B904 0061                               5518          LGR   R6,R1               R6:  OPSRC - Secondary
00037542  B904 0071                               5519          LGR   R7,R1               R7:  OPSRC - Home
                                                  5520
00037546  B219 0200                     00000200  5521          SAC   ARMODE
```

```
  LOC         OBJECT CODE        ADDR1     ADDR2     STMT
0003754A  D503 1000 3040       00000000  00000040  5522           CLC   0(4,R1),64(R3)
00037550  4770 5156                        00037576  5523           BNE   F343XX                OP1C Different, error
00037554  D503 6004 3048       00000004  00000048  5524           CLC   4(4,R6),72(R3)
0003755A  4770 5156                        00037576  5525           BNE   F343XX                OP2C Different, error
0003755E  D503 7008 3050       00000008  00000050  5526           CLC   8(4,R7),80(R3)
00037564  4770 5156                        00037576  5527           BNE   F343XX                OP3C Different, error
00037568  D503 600C 3058       0000000C  00000058  5528           CLC   12(4,R6),88(R3)
0003756E  B219 0000                        00000000  5529           SAC   PRIMODE
00037572  478B 0000                        00000000  5530           BE    0(R11)                OP4Csame, return
                                                     5531
                                 00037576  00000001  5532 F343XX    EQU   *
00037576  B219 0000                        00000000  5533           SAC   PRIMODE
0003757A  45F0 8450                        00002450  5534           BAL   R15,FAILMSG
0003757E  07FB                                       5535           BR    R11
                                                     5536
00037580  4700 5164                        00037584  5537 F343TSTCC BC     0,F343CCMSG           fail if unexpected condition code
                                 00037584  00000001  5538 F343CCMSG EQU   *
00037584  45F0 82B2                        000022B2  5539           BAL   R15,CCMSG
00037588  47F0 5104                        00037524  5540           B     F343CKUR
                                                     5541
                                                     5542           TSTDATA
0003758C                         0003758C  00037A00  5543+          ORG   PLOETST+(128*K)+31*(3*K)+(K+512)
                                                     5544           TSTLBL OPDST                Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00037A00  D7D9C940 40406E6E                          5545+          DC    CL32'PRI   >>>OPDST31<<<' eye catcher
00037A08  6ED6D7C4 E2E3F3F1
00037A10  4C4C4C40 40404040
00037A18  40404040 40404040
00037A20                                             5546+OPDST31   DS    0C
00037A20  00000000 00000000                          5547           DC    XL256'0'
00037A28  00000000 00000000
00037A30  00000000 00000000
00037A38  00000000 00000000
00037A40  00000000 00000000
00037A48  00000000 00000000
00037A50  00000000 00000000
00037A58  00000000 00000000
00037A60  00000000 00000000
00037A68  00000000 00000000
00037A70  00000000 00000000
00037A78  00000000 00000000
00037A80  00000000 00000000
00037A88  00000000 00000000
00037A90  00000000 00000000
00037A98  00000000 00000000
00037AA0  00000000 00000000
00037AA8  00000000 00000000
00037AB0  00000000 00000000
00037AB8  00000000 00000000
00037AC0  00000000 00000000
00037AC8  00000000 00000000
00037AD0  00000000 00000000
00037AD8  00000000 00000000
00037AE0  00000000 00000000
00037AE8  00000000 00000000
00037AF0  00000000 00000000
00037AF8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00037B00  00000000 00000000
00037B08  00000000 00000000
00037B10  00000000 00000000
00037B18  00000000 00000000
                                          5548
                                          5549           TSTLBL OPSRC              Source: OP1-OP4
00037B20  D7D9C940 40406E6E               5550+          DC    CL32'PRI   >>>OPSRC31<<<' eye catcher
00037B28  6ED6D7E2 D9C3F3F1
00037B30  4C4C4C40 40404040
00037B38  40404040 40404040
00037B40                                  5551+OPSRC31   DS    0C
00037B40  00000000 00000000               5552           DC    XL256'0'
00037B48  00000000 00000000
00037B50  00000000 00000000
00037B58  00000000 00000000
00037B60  00000000 00000000
00037B68  00000000 00000000
00037B70  00000000 00000000
00037B78  00000000 00000000
00037B80  00000000 00000000
00037B88  00000000 00000000
00037B90  00000000 00000000
00037B98  00000000 00000000
00037BA0  00000000 00000000
00037BA8  00000000 00000000
00037BB0  00000000 00000000
00037BB8  00000000 00000000
00037BC0  00000000 00000000
00037BC8  00000000 00000000
00037BD0  00000000 00000000
00037BD8  00000000 00000000
00037BE0  00000000 00000000
00037BE8  00000000 00000000
00037BF0  00000000 00000000
00037BF8  00000000 00000000
00037C00  00000000 00000000
00037C08  00000000 00000000
00037C10  00000000 00000000
00037C18  00000000 00000000
00037C20  00000000 00000000
00037C28  00000000 00000000
00037C30  00000000 00000000
00037C38  00000000 00000000
                                        5553
                                        5554           TSTLBL PL                 Parameter List
00037C40  D7D9C940 40406E6E             5555+          DC    CL32'PRI   >>>PL31<<<'    eye catcher
00037C48  6ED7D3F3 F14C4C4C
00037C50  40404040 40404040
00037C58  40404040 40404040
00037C60                                5556+PL31      DS    0C
00037C60  00000000 00000000             5557           DC    XL352'00'
00037C68  00000000 00000000
00037C70  00000000 00000000
00037C78  00000000 00000000
00037C80  00000000 00000000
00037C88  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00037C90  00000000 00000000
00037C98  00000000 00000000
00037CA0  00000000 00000000
00037CA8  00000000 00000000
00037CB0  00000000 00000000
00037CB8  00000000 00000000
00037CC0  00000000 00000000
00037CC8  00000000 00000000
00037CD0  00000000 00000000
00037CD8  00000000 00000000
00037CE0  00000000 00000000
00037CE8  00000000 00000000
00037CF0  00000000 00000000
00037CF8  00000000 00000000
00037D00  00000000 00000000
00037D08  00000000 00000000
00037D10  00000000 00000000
00037D18  00000000 00000000
00037D20  00000000 00000000
00037D28  00000000 00000000
00037D30  00000000 00000000
00037D38  00000000 00000000
00037D40  00000000 00000000
00037D48  00000000 00000000
00037D50  00000000 00000000
00037D58  00000000 00000000
00037D60  00000000 00000000
00037D68  00000000 00000000
00037D70  00000000 00000000
00037D78  00000000 00000000
00037D80  00000000 00000000
00037D88  00000000 00000000
00037D90  00000000 00000000
00037D98  00000000 00000000
00037DA0  00000000 00000000
00037DA8  00000000 00000000
00037DB0  00000000 00000000
00037DB8  00000000 00000000
                                      5558
                                      5559          TSTLBL FMSG               fail message
00037DC0  D7D9C940 40406E6E           5560+         DC    CL32'PRI   >>>FMSG31<<<'  eye catcher
00037DC8  6EC6D4E2 C7F3F14C
00037DD0  4C4C4040 40404040
00037DD8  40404040 40404040
00037DE0                              5561+FMSG31   DS    0C
00037DE0  86818993 858440D7           5562          DC    C'failed PLO-Extension '
00037DE8  D3D660C5 A7A38595
00037DF0  A2899695 40
00037DF5  D8A48184 99A49793           5563          DC    CL96'Quadruple Compare and Swap (QCS: 34): 3rd NE'
00037DFD  8540C396 94978199
00037E05  85408195 8440E2A6
00037E0D  8197404D D8C3E27A
00037E15  40F3F45D 7A40F399
00037E1D  8440D5C5 40404040
00037E25  40404040 40404040
00037E2D  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00037E35  40404040 40404040
00037E3D  40404040 40404040
00037E45  40404040 40404040
00037E4D  40404040 40404040
                                          5564
                                          5565          TSTEND
00037E55  D7D9C940 4040C5D5               5566+TEND31   DC    CL32'PRI   END of test: 31 <<<'
00037E5D  C4409686 40A385A2
00037E65  A37A40F3 F1404C4C
00037E6D  4C404040 40404040
                                          5567
                                          5568 * ------------------------------------------------------------------
                                          5569 * Test Quadruple Compare and Swap (QCS: 34) instruction : 4th NE
                                          5570 * ------------------------------------------------------------------
                                          5571          TSTHDR 2,4,34             4th Operand NOT Equal (cc=2, UR=4)
00037E75                  00037E75  00038000  5572+         ORG   PLOETST+(128*K)+32*(3*K)
00038000  D7D9C940 4040E2E3               5573+         DC    CL32'PRI   START of test: 32'
00038008  C1D9E340 968640A3
00038010  85A2A37A 40F3F240
00038018  40404040 40404040
00038020                  00038020            5574+         USING *,R5                base for test data and test routine
00038020                  00000000            5575+         USING PLOETEST,R5
00038020  00038050                            5576+T32      DC    A(CODE32)           address of test code
00038024  0020                                5577+         DC    H'32'               test number
00038026  00                                  5578+         DC    X'00'
00038027  22                                  5579+         DC    HL1'34'             Function
00038028  04                                  5580+         DC    HL1'4'              expected unequal results (UR)
00038029  00                                  5581+         DC    HL1'00'             found UR
0003802A  02                                  5582+         DC    HL1'2'              cc
0003802B  0D                                  5583+         DC    HL1'13'             not expected CC mask
0003802C  00                                  5584+         DS    X                   extracted cc
00038030  00000000 00000000                   5585+         DS    2F                  extract PSW after test (has CC)
00038038  000389E0                            5586+         DC    A(FMSG32)           address of test fail message
0003803C  00038740                            5587+         DC    A(OPSRC32)          address of operand source
00038040  00038620                            5588+         DC    A(OPDST32)          address of operand destination
00038044  00038860                            5589+         DC    A(PL32)             address of parameter list
00038048  00004198                            5590+         DC    A(RANDX+32)         address of random source data
0003804C  00038A55                            5591+         DC    A(TEND32)           address of test end eye catche
                                              5592
                                              5593          TSTCODE
00038050                                      5594+CODE32   DS    0F
                                              5595 *                                  load random source
00038050  E310 501C 0014                0000001C  5596          LGF   R1,OPSRCA           R1: OPSRC - Primary
00038056  B904 0061                             5597          LGR   R6,R1               R6: OPSRC - Secondary
0003805A  B904 0071                             5598          LGR   R7,R1               R7: OPSRC - Home
                                                5599
0003805E  E320 5028 0014                00000028  5600          LGF   R2,RANDXA           R2: address random data
00038064  E330 5024 0014                00000024  5601          LGF   R3,PLA              R3: address PL
                                                5602
0003806A  B219 0200                     00000200  5603          SAC   ARMODE
0003806E  D203 1000 2000      00000000  00000000  5604          MVC   0(4,R1),0(R2)       OP1 - Primary
00038074  D203 6004 2004      00000004  00000004  5605          MVC   4(4,R6),4(R2)       OP2 - Secondary
0003807A  D203 7008 2008      00000008  00000008  5606          MVC   8(4,R7),8(R2)       OP3 - Home
00038080  D203 600C 200C      0000000C  0000000C  5607          MVC   12(4,R6),12(R2)     OP4 - Secondary
                                                5608 *
                                                5609 * Quadruple Compare and Swap (QCS: 34)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          5610 * Build Parambter List
                                          5611 *
00038086  D203 3040 2000      00000040  00000000  5612          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
0003808C  D203 3048 2004      00000048  00000004  5613          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00038092  D203 3050 2008      00000050  00000008  5614          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
00038098  D203 3058 200C      00000058  0000000C  5615          MVC   88(4,R3),12(R2)     Load PL OP4-Compare
                                          5616
0003809E  D203 3060 2080      00000060  00000080  5617          MVC   96(4,R3),128+0(R2)   Load PL OP1-Replacement
000380A4  D203 3068 2084      00000068  00000084  5618          MVC   104(4,R3),128+4(R2)  Load PL OP2-Replacement
000380AA  D203 3070 2088      00000070  00000088  5619          MVC   112(4,R3),128+8(R2)  Load PL OP3-Replacement
000380B0  D203 3078 208C      00000078  0000008C  5620          MVC   120(4,R3),128+12(R2) Load PL OP4-Replacement
                                          5621 ****
                                          5622 * force a failure by changing the 4th operand
                                          5623 ****
000380B6  D201 600C 2100      0000000C  00000100  5624          MVC   12(2,R6),256(R2)     some random data
                                          5625 ****
000380BC  E313 0000 0024                00000000  5626          STG   R1,0(R3)            load OP1 address in PL
000380C2  E54C 3020 0000                00000020  5627          MVHI  32(R3),PRIALET
                                          5628
000380C8  4111 0004                     00000004  5629          LA    R1,4(R1)
000380CC  E313 0008 0024                00000008  5630          STG   R1,8(R3)            load OP2 address in PL
000380D2  E54C 3024 0001                00000024  5631          MVHI  36(R3),SECALET
                                          5632
000380D8  4111 0004                     00000004  5633          LA    R1,4(R1)
000380DC  E313 0010 0024                00000010  5634          STG   R1,16(R3)           load OP3 address in PL
000380E2  E54C 3028 0002                00000028  5635          MVHI  40(R3),HOMEALET
                                          5636
000380E8  4111 0004                     00000004  5637          LA    R1,4(R1)
000380EC  E313 0018 0024                00000018  5638          STG   R1,24(R3)           load OP4 address in PL
000380F2  E54C 302C 0001                0000002C  5639          MVHI  44(R3),SECALET
                                          5640
000380F8  B982 0000                               5641          XGR   R0,R0
000380FC  4100 0022                     00000022  5642          LA    R0,34
                                          5643
00038100  B219 0200                     00000200  5644          SAC   ARMODE
00038104  EE0F 0000 3000      00000000  00000000  5645          PLO   0,0,R15,0(R3)       Quadruple C/S (QCS: 34)
0003810A  B219 0000                     00000000  5646          SAC   PRIMODE
                                          5647
0003810E  B98D 0020                               5648          EPSW  R2,R0               extract psw
00038112  5020 5010                     00000010  5649          ST    R2,CCPSW              to save CC
00038116  E310 500B 0076                0000000B  5650          LB    R1,CCMASK           (failure CC mask)
0003811C  8910 0004                     00000004  5651          SLL   R1,4                (shift to BC instr CC position)
00038120  4410 5160                     00038180  5652          EX    R1,F344TSTCC            fail if...
                                          5653
                          00038124      00000001  5654 F344CKUR EQU   *
00038124  D200 5009 3030  00000009      00000030  5655          MVC   URFOUND,48(R3)      Extract UR
0003812A  D500 5008 3030  00000008      00000030  5656          CLC   UR,48(R3)
00038130  4780 5118                     00038138  5657          BE    F344CKRES
00038134  45F0 8388                     00002388  5658          BAL   R15,URMSG
                                          5659 *
                                          5660 * compare results
                                          5661 *
                          00038138      00000001  5662 F344CKRES EQU   *
                                          5663 *                                    reset addresses
00038138  E310 501C 0014                0000001C  5664          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003813E  B904 0061                               5665          LGR   R6,R1               R6:  OPSRC - Secondary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038142  B904 0071                       5666           LGR   R7,R1               R7:  OPSRC - Home
                                           5667
00038146  B219 0200                 00000200  5668           SAC   ARMODE
0003814A  D503 1000 3040  00000000  00000040  5669           CLC   0(4,R1),64(R3)
00038150  4770 5156                 00038176  5670           BNE   F344XX              OP1C Different, error
00038154  D503 6004 3048  00000004  00000048  5671           CLC   4(4,R6),72(R3)
0003815A  4770 5156                 00038176  5672           BNE   F344XX              OP2C Different, error
0003815E  D503 7008 3050  00000008  00000050  5673           CLC   8(4,R7),80(R3)
00038164  4770 5156                 00038176  5674           BNE   F344XX              OP3C Different, error
00038168  D503 600C 3058  0000000C  00000058  5675           CLC   12(4,R6),88(R3)
0003816E  B219 0000                 00000000  5676           SAC   PRIMODE
00038172  478B 0000                 00000000  5677           BE    0(R11)              OP4Csame, return
                                           5678
                          00038176  00000001  5679 F344XX    EQU   *
00038176  B219 0000                 00000000  5680           SAC   PRIMODE
0003817A  45F0 8450                 00002450  5681           BAL   R15,FAILMSG
0003817E  07FB                              5682           BR    R11
                                           5683
00038180  4700 5164                 00038184  5684 F344TSTCC BC     0,F344CCMSG        fail if unexpected condition code
                          00038184  00000001  5685 F344CCMSG EQU    *
00038184  45F0 82B2                 000022B2  5686           BAL   R15,CCMSG
00038188  47F0 5104                 00038124  5687           B     F344CKUR
                                           5688
                                           5689           TSTDATA
0003818C                  0003818C  00038600  5690+          ORG   PLOETST+(128*K)+32*(3*K)+(K+512)
                                           5691           TSTLBL OPDST              Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00038600  D7D9C940 40406E6E           5692+          DC    CL32'PRI   >>>OPDST32<<<' eye catcher
00038608  6ED6D7C4 E2E3F3F2
00038610  4C4C4C40 40404040
00038618  40404040 40404040
00038620                            5693+OPDST32  DS    0C
00038620  00000000 00000000         5694          DC    XL256'0'
00038628  00000000 00000000
00038630  00000000 00000000
00038638  00000000 00000000
00038640  00000000 00000000
00038648  00000000 00000000
00038650  00000000 00000000
00038658  00000000 00000000
00038660  00000000 00000000
00038668  00000000 00000000
00038670  00000000 00000000
00038678  00000000 00000000
00038680  00000000 00000000
00038688  00000000 00000000
00038690  00000000 00000000
00038698  00000000 00000000
000386A0  00000000 00000000
000386A8  00000000 00000000
000386B0  00000000 00000000
000386B8  00000000 00000000
000386C0  00000000 00000000
000386C8  00000000 00000000
000386D0  00000000 00000000
000386D8  00000000 00000000
000386E0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000386E8  00000000 00000000
000386F0  00000000 00000000
000386F8  00000000 00000000
00038700  00000000 00000000
00038708  00000000 00000000
00038710  00000000 00000000
00038718  00000000 00000000
                                          5695
                                          5696           TSTLBL OPSRC              Source: OP1-OP4
00038720  D7D9C940 40406E6E               5697+          DC    CL32'PRI   >>>OPSRC32<<<' eye catcher
00038728  6ED6D7E2 D9C3F3F2
00038730  4C4C4C40 40404040
00038738  40404040 40404040
00038740                                  5698+OPSRC32   DS    0C
00038740  00000000 00000000               5699           DC    XL256'0'
00038748  00000000 00000000
00038750  00000000 00000000
00038758  00000000 00000000
00038760  00000000 00000000
00038768  00000000 00000000
00038770  00000000 00000000
00038778  00000000 00000000
00038780  00000000 00000000
00038788  00000000 00000000
00038790  00000000 00000000
00038798  00000000 00000000
000387A0  00000000 00000000
000387A8  00000000 00000000
000387B0  00000000 00000000
000387B8  00000000 00000000
000387C0  00000000 00000000
000387C8  00000000 00000000
000387D0  00000000 00000000
000387D8  00000000 00000000
000387E0  00000000 00000000
000387E8  00000000 00000000
000387F0  00000000 00000000
000387F8  00000000 00000000
00038800  00000000 00000000
00038808  00000000 00000000
00038810  00000000 00000000
00038818  00000000 00000000
00038820  00000000 00000000
00038828  00000000 00000000
00038830  00000000 00000000
00038838  00000000 00000000
                                          5700
                                          5701           TSTLBL PL                 Parameter List
00038840  D7D9C940 40406E6E               5702+          DC    CL32'PRI   >>>PL32<<<'    eye catcher
00038848  6ED7D3F3 F24C4C4C
00038850  40404040 40404040
00038858  40404040 40404040
00038860                                  5703+PL32      DS    0C
00038860  00000000 00000000               5704           DC    XL352'00'
00038868  00000000 00000000
00038870  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00038878  00000000 00000000
00038880  00000000 00000000
00038888  00000000 00000000
00038890  00000000 00000000
00038898  00000000 00000000
000388A0  00000000 00000000
000388A8  00000000 00000000
000388B0  00000000 00000000
000388B8  00000000 00000000
000388C0  00000000 00000000
000388C8  00000000 00000000
000388D0  00000000 00000000
000388D8  00000000 00000000
000388E0  00000000 00000000
000388E8  00000000 00000000
000388F0  00000000 00000000
000388F8  00000000 00000000
00038900  00000000 00000000
00038908  00000000 00000000
00038910  00000000 00000000
00038918  00000000 00000000
00038920  00000000 00000000
00038928  00000000 00000000
00038930  00000000 00000000
00038938  00000000 00000000
00038940  00000000 00000000
00038948  00000000 00000000
00038950  00000000 00000000
00038958  00000000 00000000
00038960  00000000 00000000
00038968  00000000 00000000
00038970  00000000 00000000
00038978  00000000 00000000
00038980  00000000 00000000
00038988  00000000 00000000
00038990  00000000 00000000
00038998  00000000 00000000
000389A0  00000000 00000000
000389A8  00000000 00000000
000389B0  00000000 00000000
000389B8  00000000 00000000
                             5705
                             5706          TSTLBL FMSG               fail message
000389C0  D7D9C940 40406E6E  5707+         DC    CL32'PRI   >>>FMSG32<<<'  eye catcher
000389C8  6EC6D4E2 C7F3F24C
000389D0  4C4C4040 40404040
000389D8  40404040 40404040
000389E0                     5708+FMSG32   DS    0C
000389E0  86818993 858440D7  5709          DC    C'failed PLO-Extension '
000389E8  D3D660C5 A7A38595
000389F0  A2899695 40
000389F5  D8A48184 99A49793  5710          DC    CL96'Quadruple Compare and Swap (QCS: 34): 4th NE'
000389FD  8540C396 94978199
00038A05  85408195 8440E2A6
00038A0D  8197404D D8C3E27A
00038A15  40F3F45D 7A40F4A3
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038A1D  8840D5C5 40404040
00038A25  40404040 40404040
00038A2D  40404040 40404040
00038A35  40404040 40404040
00038A3D  40404040 40404040
00038A45  40404040 40404040
00038A4D  40404040 40404040
                                        5711
                                        5712          TSTEND
00038A55  D7D9C940 4040C5D5             5713+TEND32   DC    CL32'PRI   END of test: 32 <<<'
00038A5D  C4409686 40A385A2
00038A65  A37A40F3 F2404C4C
00038A6D  4C404040 40404040
                                        5714
                                        5715 * ------------------------------------------------------------------
                                        5716 * Test Quadruple Compare and Swap (QCS: 34) instruction : All EQ
                                        5717 * ------------------------------------------------------------------
                                        5718          TSTHDR 0,99,34             All EQ (cc=0, UR=99)
00038A75                00038A75  00038C00  5719+         ORG   PLOETST+(128*K)+33*(3*K)
00038C00  D7D9C940 4040E2E3             5720+         DC    CL32'PRI   START of test: 33'
00038C08  C1D9E340 968640A3
00038C10  85A2A37A 40F3F340
00038C18  40404040 40404040
00038C20                00038C20        5721+         USING *,R5                 base for test data and test routine
00038C20                00000000        5722+         USING PLOETEST,R5
00038C20  00038C50                      5723+T33      DC    A(CODE33)            address of test code
00038C24  0021                          5724+         DC    H'33'                test number
00038C26  00                            5725+         DC    X'00'
00038C27  22                            5726+         DC    HL1'34'              Function
00038C28  63                            5727+         DC    HL1'99'              expected unequal results (UR)
00038C29  00                            5728+         DC    HL1'00'              found UR
00038C2A  00                            5729+         DC    HL1'0'               cc
00038C2B  07                            5730+         DC    HL1'7'               not expected CC mask
00038C2C  00                            5731+         DS    X                    extracted cc
00038C30  00000000 00000000             5732+         DS    2F                   extract PSW after test (has CC)
00038C38  000395E0                      5733+         DC    A(FMSG33)            address of test fail message
00038C3C  00039340                      5734+         DC    A(OPSRC33)           address of operand source
00038C40  00039220                      5735+         DC    A(OPDST33)           address of operand destination
00038C44  00039460                      5736+         DC    A(PL33)              address of parameter list
00038C48  00004199                      5737+         DC    A(RANDX+33)          address of random source data
00038C4C  00039655                      5738+         DC    A(TEND33)            address of test end eye catche
                                        5739
                                        5740          TSTCODE
00038C50                                5741+CODE33   DS    0F
                                        5742 *                                   load random source
00038C50  E310 501C 0014                0000001C  5743          LGF   R1,OPSRCA    R1: OPSRC - Primary
00038C56  B904 0061                       5744          LGR   R6,R1              R6: OPSRC - Secondary
00038C5A  B904 0071                       5745          LGR   R7,R1              R7: OPSRC - Home
                                          5746
00038C5E  E320 5028 0014                00000028  5747          LGF   R2,RANDXA    R2: address random data
00038C64  E330 5024 0014                00000024  5748          LGF   R3,PLA       R3: address PL
                                          5749
00038C6A  B219 0200                     00000200  5750          SAC   ARMODE
00038C6E  D203 1000 2000      00000000  00000000  5751          MVC   0(4,R1),0(R2)      OP1 - Primary
00038C74  D203 6004 2004      00000004  00000004  5752          MVC   4(4,R6),4(R2)      OP2 - Secondary
00038C7A  D203 7008 2008      00000008  00000008  5753          MVC   8(4,R7),8(R2)      OP3 - Home
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00038C80  D203 600C 200C      0000000C  0000000C  5754          MVC   12(4,R6),12(R2)     OP4 - Secondary
                                                  5755 *
                                                  5756 * Quadruple Compare and Swap (QCS: 34)
                                                  5757 * Build Parambter List
                                                  5758 *
00038C86  D203 3040 2000      00000040  00000000  5759          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
00038C8C  D203 3048 2004      00000048  00000004  5760          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00038C92  D203 3050 2008      00000050  00000008  5761          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
00038C98  D203 3058 200C      00000058  0000000C  5762          MVC   88(4,R3),12(R2)     Load PL OP4-Compare
                                                  5763
00038C9E  D203 3060 2080      00000060  00000080  5764          MVC   96(4,R3),128+0(R2)   Load PL OP1-Replacement
00038CA4  D203 3068 2084      00000068  00000084  5765          MVC   104(4,R3),128+4(R2)  Load PL OP2-Replacement
00038CAA  D203 3070 2088      00000070  00000088  5766          MVC   112(4,R3),128+8(R2)  Load PL OP3-Replacement
00038CB0  D203 3078 208C      00000078  0000008C  5767          MVC   120(4,R3),128+12(R2) Load PL OP4-Replacement
                                                  5768 ****
                                                  5769 * ALL EQ: do not force a failure by changing an operand
                                                  5770 ****
                                                  5771 *         MVC   12(2,R6),256(R2)     some random data
                                                  5772 ****
00038CB6  E313 0000 0024                  00000000  5773          STG   R1,0(R3)            load OP1 address in PL
00038CBC  E54C 3020 0000                  00000020  5774          MVHI  32(R3),PRIALET
                                                  5775
00038CC2  4111 0004                       00000004  5776          LA    R1,4(R1)
00038CC6  E313 0008 0024                  00000008  5777          STG   R1,8(R3)            load OP2 address in PL
00038CCC  E54C 3024 0001                  00000024  5778          MVHI  36(R3),SECALET
                                                  5779
00038CD2  4111 0004                       00000004  5780          LA    R1,4(R1)
00038CD6  E313 0010 0024                  00000010  5781          STG   R1,16(R3)           load OP3 address in PL
00038CDC  E54C 3028 0002                  00000028  5782          MVHI  40(R3),HOMEALET
                                                  5783
00038CE2  4111 0004                       00000004  5784          LA    R1,4(R1)
00038CE6  E313 0018 0024                  00000018  5785          STG   R1,24(R3)           load OP4 address in PL
00038CEC  E54C 302C 0001                  0000002C  5786          MVHI  44(R3),SECALET
                                                  5787
00038CF2  B982 0000                               5788          XGR   R0,R0
00038CF6  4100 0022                       00000022  5789          LA    R0,34
                                                  5790
00038CFA  B219 0200                       00000200  5791          SAC   ARMODE
00038CFE  EE0F 0000 3000      00000000    00000000  5792          PLO   0,0,R15,0(R3)       Quadruple C/S (QCS: 34)
00038D04  B219 0000                       00000000  5793          SAC   PRIMODE
                                                  5794
00038D08  B98D 0020                               5795          EPSW  R2,R0               extract psw
00038D0C  5020 5010                       00000010  5796          ST    R2,CCPSW              to save CC
00038D10  E310 500B 0076                  0000000B  5797          LB    R1,CCMASK           (failure CC mask)
00038D16  8910 0004                       00000004  5798          SLL   R1,4                (shift to BC instr CC position)
00038D1A  4410 515A                       00038D7A  5799          EX    R1,F345TSTCC            fail if...
                                                  5800
                          00038D1E        00000001  5801 F345CKUR EQU   *
00038D1E  D200 5009 3030  00000009        00000030  5802          MVC   URFOUND,48(R3)      Extract UR
00038D24  D500 5008 3030  00000008        00000030  5803          CLC   UR,48(R3)
00038D2A  4780 5112                       00038D32  5804          BE    F345CKRES
00038D2E  45F0 8388                       00002388  5805          BAL   R15,URMSG
                                                  5806 *
                                                  5807 * compare results
                                                  5808 *
                          00038D32        00000001  5809 F345CKRES EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5810 *                                 reset addresses
00038D32  E310 501C 0014                0000001C  5811          LGF   R1,OPSRCA          R1:  OPSRC - Primary
00038D38  B904 0061                             0  5812          LGR   R6,R1              R6:  OPSRC - Secondary
00038D3C  B904 0071                             1  5813          LGR   R7,R1              R7:  OPSRC - Home
                                                    5814
                                                    5815 * All equal; replacement values copied to operand source
                                                    5816
00038D40  B219 0200                     00000200  5817          SAC   ARMODE
00038D44  D503 1000 3060  00000000      00000060  5818          CLC   0(4,R1),96(R3)
00038D4A  4770 5150                     00038D70  5819          BNE   F345XX             OP1R Different, error
00038D4E  D503 6004 3068  00000004      00000068  5820          CLC   4(4,R6),104(R3)
00038D54  4770 5150                     00038D70  5821          BNE   F345XX             OP2R Different, error
00038D58  D503 7008 3070  00000008      00000070  5822          CLC   8(4,R7),112(R3)
00038D5E  4770 5150                     00038D70  5823          BNE   F345XX             OP3R Different, error
00038D62  D503 600C 3078  0000000C      00000078  5824          CLC   12(4,R6),120(R3)
00038D68  B219 0000                     00000000  5825          SAC   PRIMODE
00038D6C  478B 0000                     00000000  5826          BE    0(R11)             OP4R same, return
                                                    5827
                          00038D70      00000001  5828 F345XX   EQU   *
00038D70  B219 0000                     00000000  5829          SAC   PRIMODE
00038D74  45F0 8450                     00002450  5830          BAL   R15,FAILMSG
00038D78  07FB                                    5831          BR    R11
                                                    5832
00038D7A  4700 515E                     00038D7E  5833 F345TSTCC BC    0,F345CCMSG        fail if unexpected condition code
                          00038D7E      00000001  5834 F345CCMSG EQU   *
00038D7E  45F0 82B2                     000022B2  5835          BAL   R15,CCMSG
00038D82  47F0 50FE                     00038D1E  5836          B     F345CKUR
                                                    5837
                                                    5838          TSTDATA
00038D86                  00038D86      00039200  5839+         ORG   PLOETST+(128*K)+33*(3*K)+(K+512)
                                                    5840          TSTLBL OPDST             Dest/Data Op1-R, Op2-R, Op3-R, Op4-R
00039200  D7D9C940 40406E6E                        5841+         DC    CL32'PRI   >>>OPDST33<<<' eye catcher
00039208  6ED6D7C4 E2E3F3F3
00039210  4C4C4C40 40404040
00039218  40404040 40404040
00039220                                           5842+OPDST33  DS    0C
00039220  00000000 00000000                        5843          DC    XL256'0'
00039228  00000000 00000000
00039230  00000000 00000000
00039238  00000000 00000000
00039240  00000000 00000000
00039248  00000000 00000000
00039250  00000000 00000000
00039258  00000000 00000000
00039260  00000000 00000000
00039268  00000000 00000000
00039270  00000000 00000000
00039278  00000000 00000000
00039280  00000000 00000000
00039288  00000000 00000000
00039290  00000000 00000000
00039298  00000000 00000000
000392A0  00000000 00000000
000392A8  00000000 00000000
000392B0  00000000 00000000
000392B8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000392C0  00000000 00000000
000392C8  00000000 00000000
000392D0  00000000 00000000
000392D8  00000000 00000000
000392E0  00000000 00000000
000392E8  00000000 00000000
000392F0  00000000 00000000
000392F8  00000000 00000000
00039300  00000000 00000000
00039308  00000000 00000000
00039310  00000000 00000000
00039318  00000000 00000000
                                          5844
                                          5845           TSTLBL OPSRC              Source: OP1-OP4
00039320  D7D9C940 40406E6E               5846+          DC    CL32'PRI   >>>OPSRC33<<<' eye catcher
00039328  6ED6D7E2 D9C3F3F3
00039330  4C4C4C40 40404040
00039338  40404040 40404040
00039340                                  5847+OPSRC33   DS    0C
00039340  00000000 00000000               5848           DC    XL256'0'
00039348  00000000 00000000
00039350  00000000 00000000
00039358  00000000 00000000
00039360  00000000 00000000
00039368  00000000 00000000
00039370  00000000 00000000
00039378  00000000 00000000
00039380  00000000 00000000
00039388  00000000 00000000
00039390  00000000 00000000
00039398  00000000 00000000
000393A0  00000000 00000000
000393A8  00000000 00000000
000393B0  00000000 00000000
000393B8  00000000 00000000
000393C0  00000000 00000000
000393C8  00000000 00000000
000393D0  00000000 00000000
000393D8  00000000 00000000
000393E0  00000000 00000000
000393E8  00000000 00000000
000393F0  00000000 00000000
000393F8  00000000 00000000
00039400  00000000 00000000
00039408  00000000 00000000
00039410  00000000 00000000
00039418  00000000 00000000
00039420  00000000 00000000
00039428  00000000 00000000
00039430  00000000 00000000
00039438  00000000 00000000
                                       5849
                                       5850           TSTLBL PL                 Parameter List
00039440  D7D9C940 40406E6E            5851+          DC    CL32'PRI   >>>PL33<<<'    eye catcher
00039448  6ED7D3F3 F34C4C4C
00039450  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00039458  40404040 40404040
00039460                                           5852+PL33     DS    0C
00039460  00000000 00000000                        5853          DC    XL352'00'
00039468  00000000 00000000
00039470  00000000 00000000
00039478  00000000 00000000
00039480  00000000 00000000
00039488  00000000 00000000
00039490  00000000 00000000
00039498  00000000 00000000
000394A0  00000000 00000000
000394A8  00000000 00000000
000394B0  00000000 00000000
000394B8  00000000 00000000
000394C0  00000000 00000000
000394C8  00000000 00000000
000394D0  00000000 00000000
000394D8  00000000 00000000
000394E0  00000000 00000000
000394E8  00000000 00000000
000394F0  00000000 00000000
000394F8  00000000 00000000
00039500  00000000 00000000
00039508  00000000 00000000
00039510  00000000 00000000
00039518  00000000 00000000
00039520  00000000 00000000
00039528  00000000 00000000
00039530  00000000 00000000
00039538  00000000 00000000
00039540  00000000 00000000
00039548  00000000 00000000
00039550  00000000 00000000
00039558  00000000 00000000
00039560  00000000 00000000
00039568  00000000 00000000
00039570  00000000 00000000
00039578  00000000 00000000
00039580  00000000 00000000
00039588  00000000 00000000
00039590  00000000 00000000
00039598  00000000 00000000
000395A0  00000000 00000000
000395A8  00000000 00000000
000395B0  00000000 00000000
000395B8  00000000 00000000
                                                   5854
                                                   5855          TSTLBL FMSG               fail message
000395C0  D7D9C940 40406E6E                        5856+         DC    CL32'PRI   >>>FMSG33<<<'  eye catcher
000395C8  6EC6D4E2 C7F3F34C
000395D0  4C4C4040 40404040
000395D8  40404040 40404040
000395E0                                           5857+FMSG33   DS    0C
000395E0  86818993 858440D7                        5858          DC    C'failed PLO-Extension '
000395E8  D3D660C5 A7A38595
000395F0  A2899695 40
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000395F5  D8A48184 99A49793                        5859          DC    CL96'Quadruple Compare and Swap (QCS: 34): All EQ'
000395FD  8540C396 94978199
00039605  85408195 8440E2A6
0003960D  8197404D D8C3E27A
00039615  40F3F45D 7A40C193
0003961D  9340C5D8 40404040
00039625  40404040 40404040
0003962D  40404040 40404040
00039635  40404040 40404040
0003963D  40404040 40404040
00039645  40404040 40404040
0003964D  40404040 40404040
                                                   5860
                                                   5861          TSTEND
00039655  D7D9C940 4040C5D5                        5862+TEND33   DC    CL32'PRI   END of test: 33 <<<'
0003965D  C4409686 40A385A2
00039665  A37A40F3 F3404C4C
0003966D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   5864 *=============================================================================
                                                   5865
                                                   5866 * -----------------------------------------------------------------------
                                                   5867 * Test Triple Compare and Swap (TCSO: 33) instruction : 1st NE
                                                   5868 * -----------------------------------------------------------------------
                                                   5869          TSTHDR 2,1,33             1st Operand NOT Equal (cc=2, UR=1)
00039675                          00039675  00039800 5870+         ORG   PLOETST+(128*K)+34*(3*K)
00039800  D7D9C940 4040E2E3                        5871+         DC    CL32'PRI   START of test: 34'
00039808  C1D9E340 968640A3
00039810  85A2A37A 40F3F440
00039818  40404040 40404040
00039820                          00039820         5872+         USING *,R5                base for test data and test routine
00039820                          00000000         5873+         USING PLOETEST,R5
00039820  00039850                                 5874+T34      DC    A(CODE34)           address of test code
00039824  0022                                     5875+         DC    H'34'               test number
00039826  00                                       5876+         DC    X'00'
00039827  21                                       5877+         DC    HL1'33'             Function
00039828  01                                       5878+         DC    HL1'1'              expected unequal results (UR)
00039829  00                                       5879+         DC    HL1'00'             found UR
0003982A  02                                       5880+         DC    HL1'2'              cc
0003982B  0D                                       5881+         DC    HL1'13'             not expected CC mask
0003982C  00                                       5882+         DS    X                   extracted cc
00039830  00000000 00000000                        5883+         DS    2F                  extract PSW after test (has CC)
00039838  0003A1E0                                 5884+         DC    A(FMSG34)           address of test fail message
0003983C  00039F40                                 5885+         DC    A(OPSRC34)          address of operand source
00039840  00039E20                                 5886+         DC    A(OPDST34)          address of operand destination
00039844  0003A060                                 5887+         DC    A(PL34)             address of parameter list
00039848  0000419A                                 5888+         DC    A(RANDX+34)         address of random source data
0003984C  0003A255                                 5889+         DC    A(TEND34)           address of test end eye catche
                                                   5890
                                                   5891          TSTCODE
00039850                                           5892+CODE34   DS    0F
                                                   5893 *                                  load random source
00039850  E310 501C 0014                0000001C   5894          LGF   R1,OPSRCA           R1: OPSRC - Primary
00039856  B904 0061                               5895          LGR   R6,R1               R6: OPSRC - Secondary
0003985A  B904 0071                               5896          LGR   R7,R1               R7: OPSRC - Home
                                                   5897
0003985E  E320 5028 0014                00000028   5898          LGF   R2,RANDXA           R2: address random data
00039864  E330 5024 0014                00000024   5899          LGF   R3,PLA              R3: address PL
                                                   5900
0003986A  B219 0200                     00000200   5901          SAC   ARMODE
0003986E  D21F 1000 2000        00000000 00000000  5902          MVC   0(32,R1),0(R2)      OP1 - Primary
00039874  D21F 6020 2020        00000020 00000020  5903          MVC   32(32,R6),32(R2)    OP2 - Secondary
0003987A  D21F 7040 2040        00000040 00000040  5904          MVC   64(32,R7),64(R2)    OP3 - Home
                                                   5905 *
                                                   5906 * Triple Compare and Swap (TCSO: 33)
                                                   5907 * Build Parameter List
                                                   5908 *
00039880  D25F 3040 2000        00000040 00000000  5909          MVC   64(96,R3),0(R2)     Load PL OP1-Op3 Compare
00039886  D25F 30A0 2080        000000A0 00000080  5910          MVC   160(96,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   5911 ****
                                                   5912 * force a failure by changing the first operand
                                                   5913 ****
0003988C  D207 1000 2100        00000000 00000100  5914          MVC   0(8,R1),256(R2)     some random data
                                                   5915 ****
00039892  E313 0000 0024                00000000   5916          STG   R1,0(R3)            load OP1 address in PL (pri space)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00039898  E54C 3020 0000                 00000020  5917           MVHI  32(R3),PRIALET
                                                   5918
0003989E  4111 0020                      00000020  5919           LA    R1,32(R1)
000398A2  E313 0008 0024                 00000008  5920           STG   R1,8(R3)           load OP2 address in PL (sec space)
000398A8  E54C 3024 0001                 00000024  5921           MVHI  36(R3),SECALET
                                                   5922
000398AE  4111 0020                      00000020  5923           LA    R1,32(R1)
000398B2  E313 0010 0024                 00000010  5924           STG   R1,16(R3)          load OP3 address in PL (home space)
000398B8  E54C 3028 0002                 00000028  5925           MVHI  40(R3),HOMEALET
                                                   5926
000398BE  B982 0000                               5927           XGR   R0,R0
000398C2  4100 0021                      00000021  5928           LA    R0,33
                                                   5929
000398C6  B219 0200                      00000200  5930           SAC   ARMODE
000398CA  EE0F 0000 3000      00000000   00000000  5931           PLO   0,0,R15,0(R3)      Triple C/S (TCSO: 33)
000398D0  B219 0000                      00000000  5932           SAC   PRIMODE
                                                   5933
000398D4  B98D 0020                               5934           EPSW  R2,R0              extract psw
000398D8  5020 5010                      00000010  5935           ST    R2,CCPSW             to save CC
000398DC  E310 500B 0076                 0000000B  5936           LB    R1,CCMASK          (failure CC mask)
000398E2  8910 0004                      00000004  5937           SLL   R1,4               (shift to BC instr CC position)
000398E6  4410 511C                      0003993C  5938           EX    R1,F331TSTCC           fail if...
                                                   5939
                                000398EA  00000001  5940 F331CKUR EQU   *
000398EA  D200 5009 3030      00000009   00000030  5941           MVC   URFOUND,48(R3)     Extract UR
000398F0  D500 5008 3030      00000008   00000030  5942           CLC   UR,48(R3)
000398F6  4780 50DE                      000398FE  5943           BE    F331CKRES
000398FA  45F0 8388                      00002388  5944           BAL   R15,URMSG
                                                   5945 *
                                                   5946 * compare results
                                                   5947 *
                                000398FE  00000001  5948 F331CKRES EQU   *
                                                   5949 *                             reset addresses
000398FE  E310 501C 0014                 0000001C  5950           LGF   R1,OPSRCA          R1:  OPSRC - Primary
00039904  B904 0061                               5951           LGR   R6,R1              R6:  OPSRC - Secondary
00039908  B904 0071                               5952           LGR   R7,R1              R7:  OPSRC - Home
                                                   5953
0003990C  B219 0200                      00000200  5954           SAC   ARMODE
00039910  D51F 1000 3040      00000000   00000040  5955           CLC   0(32,R1),64(R3)
00039916  4770 5112                      00039932  5956           BNE   F331XX             OP1C Different, error
0003991A  D51F 6020 3060      00000020   00000060  5957           CLC   32(32,R6),96(R3)
00039920  4770 5112                      00039932  5958           BNE   F331XX             OP2C Different, error
00039924  D51F 7040 3080      00000040   00000080  5959           CLC   64(32,R7),128(R3)
0003992A  B219 0000                      00000000  5960           SAC   PRIMODE
0003992E  478B 0000                      00000000  5961           BE    0(R11)             OP3C same, return
                                                   5962
                                00039932  00000001  5963 F331XX   EQU   *
00039932  B219 0000                      00000000  5964           SAC   PRIMODE
00039936  45F0 8450                      00002450  5965           BAL   R15,FAILMSG
0003993A  07FB                                    5966           BR    R11
                                                   5967
0003993C  4700 5120                      00039940  5968 F331TSTCC BC    0,F331CCMSG        fail if unexpected condition code
                                00039940  00000001  5969 F331CCMSG EQU   *
00039940  45F0 82B2                      000022B2  5970           BAL   R15,CCMSG
00039944  47F0 50CA                      000398EA  5971           B     F331CKUR
                                                   5972
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    5973           TSTDATA
00039948                      00039948  00039E00  5974+          ORG   PLOETST+(128*K)+34*(3*K)+(K+512)
                                                    5975           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00039E00  D7D9C940 40406E6E                         5976+          DC    CL32'PRI   >>>OPDST34<<<' eye catcher
00039E08  6ED6D7C4 E2E3F3F4
00039E10  4C4C4C40 40404040
00039E18  40404040 40404040
00039E20                                            5977+OPDST34   DS    0C
00039E20  00000000 00000000                         5978           DC    XL256'0'
00039E28  00000000 00000000
00039E30  00000000 00000000
00039E38  00000000 00000000
00039E40  00000000 00000000
00039E48  00000000 00000000
00039E50  00000000 00000000
00039E58  00000000 00000000
00039E60  00000000 00000000
00039E68  00000000 00000000
00039E70  00000000 00000000
00039E78  00000000 00000000
00039E80  00000000 00000000
00039E88  00000000 00000000
00039E90  00000000 00000000
00039E98  00000000 00000000
00039EA0  00000000 00000000
00039EA8  00000000 00000000
00039EB0  00000000 00000000
00039EB8  00000000 00000000
00039EC0  00000000 00000000
00039EC8  00000000 00000000
00039ED0  00000000 00000000
00039ED8  00000000 00000000
00039EE0  00000000 00000000
00039EE8  00000000 00000000
00039EF0  00000000 00000000
00039EF8  00000000 00000000
00039F00  00000000 00000000
00039F08  00000000 00000000
00039F10  00000000 00000000
00039F18  00000000 00000000
                                                    5979
                                                    5980           TSTLBL OPSRC              Source: OP1-OP3
00039F20  D7D9C940 40406E6E                         5981+          DC    CL32'PRI   >>>OPSRC34<<<' eye catcher
00039F28  6ED6D7E2 D9C3F3F4
00039F30  4C4C4C40 40404040
00039F38  40404040 40404040
00039F40                                            5982+OPSRC34   DS    0C
00039F40  00000000 00000000                         5983           DC    XL256'0'
00039F48  00000000 00000000
00039F50  00000000 00000000
00039F58  00000000 00000000
00039F60  00000000 00000000
00039F68  00000000 00000000
00039F70  00000000 00000000
00039F78  00000000 00000000
00039F80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00039F88  00000000 00000000
00039F90  00000000 00000000
00039F98  00000000 00000000
00039FA0  00000000 00000000
00039FA8  00000000 00000000
00039FB0  00000000 00000000
00039FB8  00000000 00000000
00039FC0  00000000 00000000
00039FC8  00000000 00000000
00039FD0  00000000 00000000
00039FD8  00000000 00000000
00039FE0  00000000 00000000
00039FE8  00000000 00000000
00039FF0  00000000 00000000
00039FF8  00000000 00000000
0003A000  00000000 00000000
0003A008  00000000 00000000
0003A010  00000000 00000000
0003A018  00000000 00000000
0003A020  00000000 00000000
0003A028  00000000 00000000
0003A030  00000000 00000000
0003A038  00000000 00000000
                                          5984
                                          5985          TSTLBL PL                 Parameter List
0003A040  D7D9C940 40406E6E               5986+         DC    CL32'PRI   >>>PL34<<<'    eye catcher
0003A048  6ED7D3F3 F44C4C4C
0003A050  40404040 40404040
0003A058  40404040 40404040
0003A060                                  5987+PL34     DS    0C
0003A060  00000000 00000000               5988          DC    XL352'00'
0003A068  00000000 00000000
0003A070  00000000 00000000
0003A078  00000000 00000000
0003A080  00000000 00000000
0003A088  00000000 00000000
0003A090  00000000 00000000
0003A098  00000000 00000000
0003A0A0  00000000 00000000
0003A0A8  00000000 00000000
0003A0B0  00000000 00000000
0003A0B8  00000000 00000000
0003A0C0  00000000 00000000
0003A0C8  00000000 00000000
0003A0D0  00000000 00000000
0003A0D8  00000000 00000000
0003A0E0  00000000 00000000
0003A0E8  00000000 00000000
0003A0F0  00000000 00000000
0003A0F8  00000000 00000000
0003A100  00000000 00000000
0003A108  00000000 00000000
0003A110  00000000 00000000
0003A118  00000000 00000000
0003A120  00000000 00000000
0003A128  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003A130  00000000 00000000
0003A138  00000000 00000000
0003A140  00000000 00000000
0003A148  00000000 00000000
0003A150  00000000 00000000
0003A158  00000000 00000000
0003A160  00000000 00000000
0003A168  00000000 00000000
0003A170  00000000 00000000
0003A178  00000000 00000000
0003A180  00000000 00000000
0003A188  00000000 00000000
0003A190  00000000 00000000
0003A198  00000000 00000000
0003A1A0  00000000 00000000
0003A1A8  00000000 00000000
0003A1B0  00000000 00000000
0003A1B8  00000000 00000000
                                        5989
                                        5990          TSTLBL FMSG               fail message
0003A1C0  D7D9C940 40406E6E             5991+         DC    CL32'PRI   >>>FMSG34<<<'  eye catcher
0003A1C8  6EC6D4E2 C7F3F44C
0003A1D0  4C4C4040 40404040
0003A1D8  40404040 40404040
0003A1E0                                5992+FMSG34   DS    0C
0003A1E0  86818993 858440D7             5993          DC    C'failed PLO-Extension '
0003A1E8  D3D660C5 A7A38595
0003A1F0  A2899695 40
0003A1F5  E3998997 938540C3                           5994          DC    CL96'Triple Compare and Swap (TCSO: 33): 1st NE'
0003A1FD  96949781 99854081
0003A205  958440E2 A6819740
0003A20D  4DE3C3E2 D67A40F3
0003A215  F35D7A40 F1A2A340
0003A21D  D5C54040 40404040
0003A225  40404040 40404040
0003A22D  40404040 40404040
0003A235  40404040 40404040
0003A23D  40404040 40404040
0003A245  40404040 40404040
0003A24D  40404040 40404040
                                        5995
                                        5996          TSTEND
0003A255  D7D9C940 4040C5D5             5997+TEND34   DC    CL32'PRI   END of test: 34 <<<'
0003A25D  C4409686 40A385A2
0003A265  A37A40F3 F4404C4C
0003A26D  4C404040 40404040
                                        5998
                                        5999 * ------------------------------------------------------------------
                                        6000 * Test Triple Compare and Swap (TCSO: 33) instruction : 2nd NE
                                        6001 * ------------------------------------------------------------------
                                        6002          TSTHDR 2,2,33             2nd Operand NOT Equal (cc=2, UR=2)
0003A275                    0003A275  0003A400  6003+         ORG   PLOETST+(128*K)+35*(3*K)
0003A400  D7D9C940 4040E2E3             6004+         DC    CL32'PRI   START of test: 35'
0003A408  C1D9E340 968640A3
0003A410  85A2A37A 40F3F540
0003A418  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003A420                     0003A420              6005+          USING *,R5                base for test data and test routine
0003A420                     00000000              6006+          USING PLOETEST,R5
0003A420  0003A450                                 6007+T35       DC    A(CODE35)           address of test code
0003A424  0023                                     6008+          DC    H'35'               test number
0003A426  00                                       6009+          DC    X'00'
0003A427  21                                       6010+          DC    HL1'33'             Function
0003A428  02                                       6011+          DC    HL1'2'              expected unequal results (UR)
0003A429  00                                       6012+          DC    HL1'00'             found UR
0003A42A  02                                       6013+          DC    HL1'2'              cc
0003A42B  0D                                       6014+          DC    HL1'13'             not expected CC mask
0003A42C  00                                       6015+          DS    X                   extracted cc
0003A430  00000000 00000000                        6016+          DS    2F                  extract PSW after test (has CC)
0003A438  0003ADE0                                 6017+          DC    A(FMSG35)           address of test fail message
0003A43C  0003AB40                                 6018+          DC    A(OPSRC35)          address of operand source
0003A440  0003AA20                                 6019+          DC    A(OPDST35)          address of operand destination
0003A444  0003AC60                                 6020+          DC    A(PL35)             address of parameter list
0003A448  0000419B                                 6021+          DC    A(RANDX+35)         address of random source data
0003A44C  0003AE55                                 6022+          DC    A(TEND35)           address of test end eye catche
                                                   6023
                                                   6024           TSTCODE
0003A450                                           6025+CODE35    DS    0F
                                                   6026 *                                   load random source
0003A450  E310 501C 0014                 0000001C  6027           LGF   R1,OPSRCA           R1: OPSRC - Primary
0003A456  B904 0061                                6028           LGR   R6,R1               R6: OPSRC - Secondary
0003A45A  B904 0071                                6029           LGR   R7,R1               R7: OPSRC - Home
                                                   6030
0003A45E  E320 5028 0014                 00000028  6031           LGF   R2,RANDXA           R2: address random data
0003A464  E330 5024 0014                 00000024  6032           LGF   R3,PLA              R3: address PL
                                                   6033
0003A46A  B219 0200                      00000200  6034           SAC   ARMODE
0003A46E  D21F 1000 2000       00000000  00000000  6035           MVC   0(32,R1),0(R2)      OP1 - Primary
0003A474  D21F 6020 2020       00000020  00000020  6036           MVC   32(32,R6),32(R2)    OP2 - Secondary
0003A47A  D21F 7040 2040       00000040  00000040  6037           MVC   64(32,R7),64(R2)    OP3 - Home
                                                   6038 *
                                                   6039 * Triple Compare and Swap (TCSO: 33)
                                                   6040 * Build Parambter List
                                                   6041 *
0003A480  D25F 3040 2000       00000040  00000000  6042           MVC   64(96,R3),0(R2)     Load PL OP1-Op3 Compare
0003A486  D25F 30A0 2080       000000A0  00000080  6043           MVC   160(96,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6044 ****
                                                   6045 * force a failure by changing the 2nd operand
                                                   6046 ****
0003A48C  D207 6020 2100       00000020  00000100  6047           MVC   32(8,R6),256(R2)    some random data
                                                   6048 ****
0003A492  E313 0000 0024                 00000000  6049           STG   R1,0(R3)            load OP1 address in PL (pri space)
0003A498  E54C 3020 0000                 00000020  6050           MVHI  32(R3),PRIALET
                                                   6051
0003A49E  4111 0020                      00000020  6052           LA    R1,32(R1)
0003A4A2  E313 0008 0024                 00000008  6053           STG   R1,8(R3)            load OP2 address in PL (sec space)
0003A4A8  E54C 3024 0001                 00000024  6054           MVHI  36(R3),SECALET
                                                   6055
0003A4AE  4111 0020                      00000020  6056           LA    R1,32(R1)
0003A4B2  E313 0010 0024                 00000010  6057           STG   R1,16(R3)           load OP3 address in PL (home space)
0003A4B8  E54C 3028 0002                 00000028  6058           MVHI  40(R3),HOMEALET
                                                   6059
0003A4BE  B982 0000                                6060           XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003A4C2  4100 0021                    00000021  6061          LA    R0,33
                                                 6062
0003A4C6  B219 0200                    00000200  6063          SAC   ARMODE
0003A4CA  EE0F 0000 3000      00000000 00000000  6064          PLO   0,0,R15,0(R3)       Triple C/S (TCSO: 33)
0003A4D0  B219 0000                    00000000  6065          SAC   PRIMODE
                                                 6066
0003A4D4  B98D 0020                              6067          EPSW  R2,R0              extract psw
0003A4D8  5020 5010                    00000010  6068          ST    R2,CCPSW             to save CC
0003A4DC  E310 500B 0076               0000000B  6069          LB    R1,CCMASK          (failure CC mask)
0003A4E2  8910 0004                    00000004  6070          SLL   R1,4               (shift to BC instr CC position)
0003A4E6  4410 511C                    0003A53C  6071          EX    R1,F332TSTCC         fail if...
                                                 6072
                              0003A4EA 00000001  6073 F332CKUR EQU   *
0003A4EA  D200 5009 3030      00000009 00000030  6074          MVC   URFOUND,48(R3)     Extract UR
0003A4F0  D500 5008 3030      00000008 00000030  6075          CLC   UR,48(R3)
0003A4F6  4780 50DE                    0003A4FE  6076          BE    F332CKRES
0003A4FA  45F0 8388                    00002388  6077          BAL   R15,URMSG
                                                 6078 *
                                                 6079 * compare results
                                                 6080 *
                              0003A4FE 00000001  6081 F332CKRES EQU   *
                                                 6082 *                               reset addresses
0003A4FE  E310 501C 0014               0000001C  6083          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003A504  B904 0061                              6084          LGR   R6,R1              R6:  OPSRC - Secondary
0003A508  B904 0071                              6085          LGR   R7,R1              R7:  OPSRC - Home
                                                 6086
0003A50C  B219 0200                    00000200  6087          SAC   ARMODE
0003A510  D51F 1000 3040      00000000 00000040  6088          CLC   0(32,R1),64(R3)
0003A516  4770 5112                    0003A532  6089          BNE   F332XX             OP1C Different, error
0003A51A  D51F 6020 3060      00000020 00000060  6090          CLC   32(32,R6),96(R3)
0003A520  4770 5112                    0003A532  6091          BNE   F332XX             OP2C Different, error
0003A524  D51F 7040 3080      00000040 00000080  6092          CLC   64(32,R7),128(R3)
0003A52A  B219 0000                    00000000  6093          SAC   PRIMODE
0003A52E  478B 0000                    00000000  6094          BE    0(R11)             OP3C same, return
                                                 6095
                              0003A532 00000001  6096 F332XX   EQU   *
0003A532  B219 0000                    00000000  6097          SAC   PRIMODE
0003A536  45F0 8450                    00002450  6098          BAL   R15,FAILMSG
0003A53A  07FB                                   6099          BR    R11
                                                 6100
0003A53C  4700 5120                    0003A540  6101 F332TSTCC BC    0,F332CCMSG        fail if unexpected condition code
                              0003A540 00000001  6102 F332CCMSG EQU   *
0003A540  45F0 82B2                    000022B2  6103          BAL   R15,CCMSG
0003A544  47F0 50CA                    0003A4EA  6104          B     F332CKUR
                                                 6105
                                                 6106          TSTDATA
0003A548                      0003A548 0003AA00  6107+         ORG   PLOETST+(128*K)+35*(3*K)+(K+512)
                                                 6108          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
0003AA00  D7D9C940 40406E6E                      6109+         DC    CL32'PRI   >>>OPDST35<<<' eye catcher
0003AA08  6ED6D7C4 E2E3F3F5
0003AA10  4C4C4C40 40404040
0003AA18  40404040 40404040
0003AA20                                         6110+OPDST35  DS    0C
0003AA20  00000000 00000000                      6111          DC    XL256'0'
0003AA28  00000000 00000000
0003AA30  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003AA38  00000000 00000000
0003AA40  00000000 00000000
0003AA48  00000000 00000000
0003AA50  00000000 00000000
0003AA58  00000000 00000000
0003AA60  00000000 00000000
0003AA68  00000000 00000000
0003AA70  00000000 00000000
0003AA78  00000000 00000000
0003AA80  00000000 00000000
0003AA88  00000000 00000000
0003AA90  00000000 00000000
0003AA98  00000000 00000000
0003AAA0  00000000 00000000
0003AAA8  00000000 00000000
0003AAB0  00000000 00000000
0003AAB8  00000000 00000000
0003AAC0  00000000 00000000
0003AAC8  00000000 00000000
0003AAD0  00000000 00000000
0003AAD8  00000000 00000000
0003AAE0  00000000 00000000
0003AAE8  00000000 00000000
0003AAF0  00000000 00000000
0003AAF8  00000000 00000000
0003AB00  00000000 00000000
0003AB08  00000000 00000000
0003AB10  00000000 00000000
0003AB18  00000000 00000000
                                      6112
                                      6113          TSTLBL OPSRC              Source: OP1-OP3
0003AB20  D7D9C940 40406E6E           6114+         DC    CL32'PRI   >>>OPSRC35<<<' eye catcher
0003AB28  6ED6D7E2 D9C3F3F5
0003AB30  4C4C4C40 40404040
0003AB38  40404040 40404040
0003AB40                              6115+OPSRC35  DS    0C
0003AB40  00000000 00000000           6116          DC    XL256'0'
0003AB48  00000000 00000000
0003AB50  00000000 00000000
0003AB58  00000000 00000000
0003AB60  00000000 00000000
0003AB68  00000000 00000000
0003AB70  00000000 00000000
0003AB78  00000000 00000000
0003AB80  00000000 00000000
0003AB88  00000000 00000000
0003AB90  00000000 00000000
0003AB98  00000000 00000000
0003ABA0  00000000 00000000
0003ABA8  00000000 00000000
0003ABB0  00000000 00000000
0003ABB8  00000000 00000000
0003ABC0  00000000 00000000
0003ABC8  00000000 00000000
0003ABD0  00000000 00000000
0003ABD8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003ABE0  00000000 00000000
0003ABE8  00000000 00000000
0003ABF0  00000000 00000000
0003ABF8  00000000 00000000
0003AC00  00000000 00000000
0003AC08  00000000 00000000
0003AC10  00000000 00000000
0003AC18  00000000 00000000
0003AC20  00000000 00000000
0003AC28  00000000 00000000
0003AC30  00000000 00000000
0003AC38  00000000 00000000
                                        6117
                                        6118           TSTLBL PL                  Parameter List
0003AC40  D7D9C940 40406E6E             6119+          DC     CL32'PRI   >>>PL35<<<'    eye catcher
0003AC48  6ED7D3F3 F54C4C4C
0003AC50  40404040 40404040
0003AC58  40404040 40404040
0003AC60                                6120+PL35      DS     0C
0003AC60  00000000 00000000             6121           DC     XL352'00'
0003AC68  00000000 00000000
0003AC70  00000000 00000000
0003AC78  00000000 00000000
0003AC80  00000000 00000000
0003AC88  00000000 00000000
0003AC90  00000000 00000000
0003AC98  00000000 00000000
0003ACA0  00000000 00000000
0003ACA8  00000000 00000000
0003ACB0  00000000 00000000
0003ACB8  00000000 00000000
0003ACC0  00000000 00000000
0003ACC8  00000000 00000000
0003ACD0  00000000 00000000
0003ACD8  00000000 00000000
0003ACE0  00000000 00000000
0003ACE8  00000000 00000000
0003ACF0  00000000 00000000
0003ACF8  00000000 00000000
0003AD00  00000000 00000000
0003AD08  00000000 00000000
0003AD10  00000000 00000000
0003AD18  00000000 00000000
0003AD20  00000000 00000000
0003AD28  00000000 00000000
0003AD30  00000000 00000000
0003AD38  00000000 00000000
0003AD40  00000000 00000000
0003AD48  00000000 00000000
0003AD50  00000000 00000000
0003AD58  00000000 00000000
0003AD60  00000000 00000000
0003AD68  00000000 00000000
0003AD70  00000000 00000000
0003AD78  00000000 00000000
0003AD80  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003AD88  00000000 00000000
0003AD90  00000000 00000000
0003AD98  00000000 00000000
0003ADA0  00000000 00000000
0003ADA8  00000000 00000000
0003ADB0  00000000 00000000
0003ADB8  00000000 00000000
                                        6122
                                        6123          TSTLBL FMSG                fail message
0003ADC0  D7D9C940 40406E6E             6124+         DC    CL32'PRI   >>>FMSG35<<<'  eye catcher
0003ADC8  6EC6D4E2 C7F3F54C
0003ADD0  4C4C4040 40404040
0003ADD8  40404040 40404040
0003ADE0                                6125+FMSG35   DS    0C
0003ADE0  86818993 858440D7             6126          DC    C'failed PLO-Extension '
0003ADE8  D3D660C5 A7A38595
0003ADF0  A2899695 40
0003ADF5  E3998997 938540C3             6127          DC    CL96'Triple Compare and Swap (TCSO: 33): 2nd NE'
0003ADFD  96949781 99854081
0003AE05  958440E2 A6819740
0003AE0D  4DE3C3E2 D67A40F3
0003AE15  F35D7A40 F2958440
0003AE1D  D5C54040 40404040
0003AE25  40404040 40404040
0003AE2D  40404040 40404040
0003AE35  40404040 40404040
0003AE3D  40404040 40404040
0003AE45  40404040 40404040
0003AE4D  40404040 40404040
                                        6128
                                        6129          TSTEND
0003AE55  D7D9C940 4040C5D5             6130+TEND35   DC    CL32'PRI   END of test: 35 <<<'
0003AE5D  C4409686 40A385A2
0003AE65  A37A40F3 F5404C4C
0003AE6D  4C404040 40404040
                                        6131
                                        6132 * ------------------------------------------------------------------
                                        6133 * Test Triple Compare and Swap (TCSO: 33) instruction : 3rd NE
                                        6134 * ------------------------------------------------------------------
                                        6135          TSTHDR 2,3,33              3rd Operand NOT Equal (cc=2, UR=3)
0003AE75                  0003AE75  0003B000  6136+         ORG   PLOETST+(128*K)+36*(3*K)
0003B000  D7D9C940 4040E2E3             6137+         DC    CL32'PRI   START of test: 36'
0003B008  C1D9E340 968640A3
0003B010  85A2A37A 40F3F640
0003B018  40404040 40404040
0003B020                  0003B020              6138+         USING *,R5                 base for test data and test routine
0003B020                  00000000              6139+         USING PLOETEST,R5
0003B020  0003B050                              6140+T36      DC    A(CODE36)            address of test code
0003B024  0024                                  6141+         DC    H'36'                test number
0003B026  00                                    6142+         DC    X'00'
0003B027  21                                    6143+         DC    HL1'33'              Function
0003B028  03                                    6144+         DC    HL1'3'               expected unequal results (UR)
0003B029  00                                    6145+         DC    HL1'00'              found UR
0003B02A  02                                    6146+         DC    HL1'2'               cc
0003B02B  0D                                    6147+         DC    HL1'13'              not expected CC mask
0003B02C  00                                    6148+         DS    X                    extracted cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B030  00000000 00000000                        6149+          DS    2F                  extract PSW after test (has CC)
0003B038  0003B9E0                                 6150+          DC    A(FMSG36)           address of test fail message
0003B03C  0003B740                                 6151+          DC    A(OPSRC36)          address of operand source
0003B040  0003B620                                 6152+          DC    A(OPDST36)          address of operand destination
0003B044  0003B860                                 6153+          DC    A(PL36)             address of parameter list
0003B048  0000419C                                 6154+          DC    A(RANDX+36)         address of random source data
0003B04C  0003BA55                                 6155+          DC    A(TEND36)           address of test end eye catche
                                                   6156
                                                   6157          TSTCODE
0003B050                                           6158+CODE36    DS    0F
                                                   6159 *                                   load random source
0003B050  E310 501C 0014                0000001C   6160           LGF   R1,OPSRCA           R1: OPSRC - Primary
0003B056  B904 0061                                6161           LGR   R6,R1               R6: OPSRC - Secondary
0003B05A  B904 0071                                6162           LGR   R7,R1               R7: OPSRC - Home
                                                   6163
0003B05E  E320 5028 0014                00000028   6164           LGF   R2,RANDXA           R2: address random data
0003B064  E330 5024 0014                00000024   6165           LGF   R3,PLA              R3: address PL
                                                   6166
0003B06A  B219 0200                     00000200   6167           SAC   ARMODE
0003B06E  D21F 1000 2000      00000000  00000000   6168           MVC   0(32,R1),0(R2)      OP1 - Primary
0003B074  D21F 6020 2020      00000020  00000020   6169           MVC   32(32,R6),32(R2)    OP2 - Secondary
0003B07A  D21F 7040 2040      00000040  00000040   6170           MVC   64(32,R7),64(R2)    OP3 - Home
                                                   6171 *
                                                   6172 * Triple Compare and Swap (TCSO: 33)
                                                   6173 * Build Parambter List
                                                   6174 *
0003B080  D25F 3040 2000      00000040  00000000   6175           MVC   64(96,R3),0(R2)     Load PL OP1-Op3 Compare
0003B086  D25F 30A0 2080      000000A0  00000080   6176           MVC   160(96,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6177 ****
                                                   6178 * force a failure by changing the 3rd operand
                                                   6179 ****
0003B08C  D207 7040 2100      00000040  00000100   6180           MVC   64(8,R7),256(R2)    some random data
                                                   6181 ****
0003B092  E313 0000 0024                00000000   6182           STG   R1,0(R3)            load OP1 address in PL (pri space)
0003B098  E54C 3020 0000                00000020   6183           MVHI  32(R3),PRIALET
                                                   6184
0003B09E  4111 0020                     00000020   6185           LA    R1,32(R1)
0003B0A2  E313 0008 0024                00000008   6186           STG   R1,8(R3)            load OP2 address in PL (sec space)
0003B0A8  E54C 3024 0001                00000024   6187           MVHI  36(R3),SECALET
                                                   6188
0003B0AE  4111 0020                     00000020   6189           LA    R1,32(R1)
0003B0B2  E313 0010 0024                00000010   6190           STG   R1,16(R3)           load OP3 address in PL (home space)
0003B0B8  E54C 3028 0002                00000028   6191           MVHI  40(R3),HOMEALET
                                                   6192
0003B0BE  B982 0000                                6193           XGR   R0,R0
0003B0C2  4100 0021                     00000021   6194           LA    R0,33
                                                   6195
0003B0C6  B219 0200                     00000200   6196           SAC   ARMODE
0003B0CA  EE0F 0000 3000      00000000  00000000   6197           PLO   0,0,R15,0(R3)       Triple C/S (TCSO: 33)
0003B0D0  B219 0000                     00000000   6198           SAC   PRIMODE
                                                   6199
0003B0D4  B98D 0020                                6200           EPSW  R2,R0               extract psw
0003B0D8  5020 5010                     00000010   6201           ST    R2,CCPSW              to save CC
0003B0DC  E310 500B 0076                0000000B   6202           LB    R1,CCMASK           (failure CC mask)
0003B0E2  8910 0004                     00000004   6203           SLL   R1,4                (shift to BC instr CC position)
0003B0E6  4410 511C                     0003B13C   6204           EX    R1,F333TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6205
                               0003B0EA  00000001  6206 F333CKUR EQU   *
0003B0EA  D200 5009 3030       00000009  00000030  6207          MVC   URFOUND,48(R3)      Extract UR
0003B0F0  D500 5008 3030       00000008  00000030  6208          CLC   UR,48(R3)
0003B0F6  4780 50DE                      0003B0FE  6209          BE    F333CKRES
0003B0FA  45F0 8388                      00002388  6210          BAL   R15,URMSG
                                                    6211 *
                                                    6212 * compare results
                                                    6213 *
                               0003B0FE  00000001  6214 F333CKRES EQU   *
                                                    6215 *                           reset addresses
0003B0FE  E310 501C 0014                 0000001C  6216          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003B104  B904 0061                                6217          LGR   R6,R1              R6:  OPSRC - Secondary
0003B108  B904 0071                                6218          LGR   R7,R1              R7:  OPSRC - Home
                                                    6219
0003B10C  B219 0200                      00000200  6220          SAC   ARMODE
0003B110  D51F 1000 3040       00000000  00000040  6221          CLC   0(32,R1),64(R3)
0003B116  4770 5112                      0003B132  6222          BNE   F333XX             OP1C Different, error
0003B11A  D51F 6020 3060       00000020  00000060  6223          CLC   32(32,R6),96(R3)
0003B120  4770 5112                      0003B132  6224          BNE   F333XX             OP2C Different, error
0003B124  D51F 7040 3080       00000040  00000080  6225          CLC   64(32,R7),128(R3)
0003B12A  B219 0000                      00000000  6226          SAC   PRIMODE
0003B12E  478B 0000                      00000000  6227          BE    0(R11)             OP3C same, return
                                                    6228
                               0003B132  00000001  6229 F333XX   EQU   *
0003B132  B219 0000                      00000000  6230          SAC   PRIMODE
0003B136  45F0 8450                      00002450  6231          BAL   R15,FAILMSG
0003B13A  07FB                                     6232          BR    R11
                                                    6233
0003B13C  4700 5120                      0003B140  6234 F333TSTCC BC    0,F333CCMSG        fail if unexpected condition code
                               0003B140  00000001  6235 F333CCMSG EQU   *
0003B140  45F0 82B2                      000022B2  6236          BAL   R15,CCMSG
0003B144  47F0 50CA                      0003B0EA  6237          B     F333CKUR
                                                    6238
                                                    6239          TSTDATA
0003B148                       0003B148  0003B600  6240+         ORG   PLOETST+(128*K)+36*(3*K)+(K+512)
                                                    6241          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
0003B600  D7D9C940 40406E6E                        6242+         DC    CL32'PRI   >>>OPDST36<<<' eye catcher
0003B608  6ED6D7C4 E2E3F3F6
0003B610  4C4C4C40 40404040
0003B618  40404040 40404040
0003B620                                           6243+OPDST36  DS    0C
0003B620  00000000 00000000                        6244          DC    XL256'0'
0003B628  00000000 00000000
0003B630  00000000 00000000
0003B638  00000000 00000000
0003B640  00000000 00000000
0003B648  00000000 00000000
0003B650  00000000 00000000
0003B658  00000000 00000000
0003B660  00000000 00000000
0003B668  00000000 00000000
0003B670  00000000 00000000
0003B678  00000000 00000000
0003B680  00000000 00000000
0003B688  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003B690  00000000 00000000
0003B698  00000000 00000000
0003B6A0  00000000 00000000
0003B6A8  00000000 00000000
0003B6B0  00000000 00000000
0003B6B8  00000000 00000000
0003B6C0  00000000 00000000
0003B6C8  00000000 00000000
0003B6D0  00000000 00000000
0003B6D8  00000000 00000000
0003B6E0  00000000 00000000
0003B6E8  00000000 00000000
0003B6F0  00000000 00000000
0003B6F8  00000000 00000000
0003B700  00000000 00000000
0003B708  00000000 00000000
0003B710  00000000 00000000
0003B718  00000000 00000000
                                          6245
                                          6246          TSTLBL OPSRC              Source: OP1-OP3
0003B720  D7D9C940 40406E6E               6247+         DC    CL32'PRI   >>>OPSRC36<<<' eye catcher
0003B728  6ED6D7E2 D9C3F3F6
0003B730  4C4C4C40 40404040
0003B738  40404040 40404040
0003B740                                  6248+OPSRC36  DS    0C
0003B740  00000000 00000000               6249          DC    XL256'0'
0003B748  00000000 00000000
0003B750  00000000 00000000
0003B758  00000000 00000000
0003B760  00000000 00000000
0003B768  00000000 00000000
0003B770  00000000 00000000
0003B778  00000000 00000000
0003B780  00000000 00000000
0003B788  00000000 00000000
0003B790  00000000 00000000
0003B798  00000000 00000000
0003B7A0  00000000 00000000
0003B7A8  00000000 00000000
0003B7B0  00000000 00000000
0003B7B8  00000000 00000000
0003B7C0  00000000 00000000
0003B7C8  00000000 00000000
0003B7D0  00000000 00000000
0003B7D8  00000000 00000000
0003B7E0  00000000 00000000
0003B7E8  00000000 00000000
0003B7F0  00000000 00000000
0003B7F8  00000000 00000000
0003B800  00000000 00000000
0003B808  00000000 00000000
0003B810  00000000 00000000
0003B818  00000000 00000000
0003B820  00000000 00000000
0003B828  00000000 00000000
0003B830  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003B838  00000000 00000000                          6250
                                                     6251          TSTLBL PL                 Parameter List
0003B840  D7D9C940 40406E6E                          6252+         DC    CL32'PRI   >>>PL36<<<'    eye catcher
0003B848  6ED7D3F3 F64C4C4C
0003B850  40404040 40404040
0003B858  40404040 40404040
0003B860                                             6253+PL36     DS    0C
0003B860  00000000 00000000                          6254          DC    XL352'00'
0003B868  00000000 00000000
0003B870  00000000 00000000
0003B878  00000000 00000000
0003B880  00000000 00000000
0003B888  00000000 00000000
0003B890  00000000 00000000
0003B898  00000000 00000000
0003B8A0  00000000 00000000
0003B8A8  00000000 00000000
0003B8B0  00000000 00000000
0003B8B8  00000000 00000000
0003B8C0  00000000 00000000
0003B8C8  00000000 00000000
0003B8D0  00000000 00000000
0003B8D8  00000000 00000000
0003B8E0  00000000 00000000
0003B8E8  00000000 00000000
0003B8F0  00000000 00000000
0003B8F8  00000000 00000000
0003B900  00000000 00000000
0003B908  00000000 00000000
0003B910  00000000 00000000
0003B918  00000000 00000000
0003B920  00000000 00000000
0003B928  00000000 00000000
0003B930  00000000 00000000
0003B938  00000000 00000000
0003B940  00000000 00000000
0003B948  00000000 00000000
0003B950  00000000 00000000
0003B958  00000000 00000000
0003B960  00000000 00000000
0003B968  00000000 00000000
0003B970  00000000 00000000
0003B978  00000000 00000000
0003B980  00000000 00000000
0003B988  00000000 00000000
0003B990  00000000 00000000
0003B998  00000000 00000000
0003B9A0  00000000 00000000
0003B9A8  00000000 00000000
0003B9B0  00000000 00000000
0003B9B8  00000000 00000000
                                                     6255
                                                     6256          TSTLBL FMSG               fail message
0003B9C0  D7D9C940 40406E6E                          6257+         DC    CL32'PRI   >>>FMSG36<<<'  eye catcher
0003B9C8  6EC6D4E2 C7F3F64C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003B9D0  4C4C4040 40404040
0003B9D8  40404040 40404040
0003B9E0                                          6258+FMSG36   DS    0C
0003B9E0  86818993 858440D7                       6259          DC    C'failed PLO-Extension '
0003B9E8  D3D660C5 A7A38595
0003B9F0  A2899695 40
0003B9F5  E3998997 938540C3                       6260          DC    CL96'Triple Compare and Swap (TCSO: 33): 3rd NE'
0003B9FD  96949781 99854081
0003BA05  958440E2 A6819740
0003BA0D  4DE3C3E2 D67A40F3
0003BA15  F35D7A40 F3998440
0003BA1D  D5C54040 40404040
0003BA25  40404040 40404040
0003BA2D  40404040 40404040
0003BA35  40404040 40404040
0003BA3D  40404040 40404040
0003BA45  40404040 40404040
0003BA4D  40404040 40404040
                                                  6261
                                                  6262          TSTEND
0003BA55  D7D9C940 4040C5D5                       6263+TEND36   DC    CL32'PRI   END of test: 36 <<<'
0003BA5D  C4409686 40A385A2
0003BA65  A37A40F3 F6404C4C
0003BA6D  4C404040 40404040
                                                  6264
                                                  6265 * ------------------------------------------------------------------
                                                  6266 * Test Triple Compare and Swap (TCSO: 33) instruction : ALL EQ
                                                  6267 * ------------------------------------------------------------------
                                                  6268          TSTHDR 0,99,33              ALL EQ (cc=0, UR=99)
0003BA75                          0003BA75  0003BC00  6269+         ORG   PLOETST+(128*K)+37*(3*K)
0003BC00  D7D9C940 4040E2E3                       6270+         DC    CL32'PRI   START of test: 37'
0003BC08  C1D9E340 968640A3
0003BC10  85A2A37A 40F3F740
0003BC18  40404040 40404040
0003BC20                          0003BC20        6271+         USING *,R5                base for test data and test routine
0003BC20                          00000000        6272+         USING PLOETEST,R5
0003BC20  0003BC50                                6273+T37      DC    A(CODE37)           address of test code
0003BC24  0025                                    6274+         DC    H'37'               test number
0003BC26  00                                      6275+         DC    X'00'
0003BC27  21                                      6276+         DC    HL1'33'             Function
0003BC28  63                                      6277+         DC    HL1'99'             expected unequal results (UR)
0003BC29  00                                      6278+         DC    HL1'00'             found UR
0003BC2A  00                                      6279+         DC    HL1'0'              cc
0003BC2B  07                                      6280+         DC    HL1'7'              not expected CC mask
0003BC2C  00                                      6281+         DS    X                   extracted cc
0003BC30  00000000 00000000                       6282+         DS    2F                  extract PSW after test (has CC)
0003BC38  0003C5E0                                6283+         DC    A(FMSG37)           address of test fail message
0003BC3C  0003C340                                6284+         DC    A(OPSRC37)          address of operand source
0003BC40  0003C220                                6285+         DC    A(OPDST37)          address of operand destination
0003BC44  0003C460                                6286+         DC    A(PL37)             address of parameter list
0003BC48  0000419D                                6287+         DC    A(RANDX+37)         address of random source data
0003BC4C  0003C655                                6288+         DC    A(TEND37)           address of test end eye catche
                                                  6289
                                                  6290          TSTCODE
0003BC50                                          6291+CODE37   DS    0F
                                                  6292 *                                    load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003BC50  E310 501C 0014                 0000001C  6293          LGF   R1,OPSRCA           R1: OPSRC - Primary
0003BC56  B904 0061                      0000001C  6294          LGR   R6,R1               R6: OPSRC - Secondary
0003BC5A  B904 0071                      0000001C  6295          LGR   R7,R1               R7: OPSRC - Home
                                                   6296
0003BC5E  E320 5028 0014                 00000028  6297          LGF   R2,RANDXA           R2: address random data
0003BC64  E330 5024 0014                 00000024  6298          LGF   R3,PLA              R3: address PL
                                                   6299
0003BC6A  B219 0200                      00000200  6300          SAC   ARMODE
0003BC6E  D21F 1000 2000       00000000  00000000  6301          MVC   0(32,R1),0(R2)      OP1 - Primary
0003BC74  D21F 6020 2020       00000020  00000020  6302          MVC   32(32,R6),32(R2)    OP2 - Secondary
0003BC7A  D21F 7040 2040       00000040  00000040  6303          MVC   64(32,R7),64(R2)    OP3 - Home
                                                   6304 *
                                                   6305 * Triple Compare and Swap (TCSO: 33)
                                                   6306 * Build Parambter List
                                                   6307 *
0003BC80  D25F 3040 2000       00000040  00000000  6308          MVC   64(96,R3),0(R2)     Load PL OP1-Op3 Compare
0003BC86  D25F 30A0 2080       000000A0  00000080  6309          MVC   160(96,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6310 ****
                                                   6311 * Do NOTforce a failure by changing an operand
                                                   6312 ****
                                                   6313 *        MVC   0(8,R1),256(R2)     some random data
                                                   6314 ****
0003BC8C  E313 0000 0024                 00000000  6315          STG   R1,0(R3)            load OP1 address in PL (pri space)
0003BC92  E54C 3020 0000                 00000020  6316          MVHI  32(R3),PRIALET
                                                   6317
0003BC98  4111 0020                      00000020  6318          LA    R1,32(R1)
0003BC9C  E313 0008 0024                 00000008  6319          STG   R1,8(R3)            load OP2 address in PL (sec space)
0003BCA2  E54C 3024 0001                 00000024  6320          MVHI  36(R3),SECALET
                                                   6321
0003BCA8  4111 0020                      00000020  6322          LA    R1,32(R1)
0003BCAC  E313 0010 0024                 00000010  6323          STG   R1,16(R3)           load OP3 address in PL (home space)
0003BCB2  E54C 3028 0002                 00000028  6324          MVHI  40(R3),HOMEALET
                                                   6325
0003BCB8  B982 0000                                6326          XGR   R0,R0
0003BCBC  4100 0021                      00000021  6327          LA    R0,33
                                                   6328
0003BCC0  B219 0200                      00000200  6329          SAC   ARMODE
0003BCC4  EE0F 0000 3000       00000000  00000000  6330          PLO   0,0,R15,0(R3)       Triple C/S (TCSO: 33)
0003BCCA  B219 0000                      00000000  6331          SAC   PRIMODE
                                                   6332
0003BCCE  B98D 0020                                6333          EPSW  R2,R0               extract psw
0003BCD2  5020 5010                      00000010  6334          ST    R2,CCPSW              to save CC
0003BCD6  E310 500B 0076                 0000000B  6335          LB    R1,CCMASK           (failure CC mask)
0003BCDC  8910 0004                      00000004  6336          SLL   R1,4                (shift to BC instr CC position)
0003BCE0  4410 5116                      0003BD36  6337          EX    R1,F334TSTCC            fail if...
                                                   6338
                                0003BCE4  00000001  6339 F334CKUR EQU   *
0003BCE4  D200 5009 3030       00000009  00000030  6340          MVC   URFOUND,48(R3)      Extract UR
0003BCEA  D500 5008 3030       00000008  00000030  6341          CLC   UR,48(R3)
0003BCF0  4780 50D8                      0003BCF8  6342          BE    F334CKRES
0003BCF4  45F0 8388                      00002388  6343          BAL   R15,URMSG
                                                   6344 *
                                                   6345 * compare results
                                                   6346 *
                                0003BCF8  00000001  6347 F334CKRES EQU   *
                                                   6348 *                                reset addresses
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003BCF8  E310 501C 0014                  0000001C  6349           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003BCFE  B904 0061                               6350           LGR   R6,R1               R6:  OPSRC - Secondary
0003BD02  B904 0071                               6351           LGR   R7,R1               R7:  OPSRC - Home
                                                  6352
                                                  6353 * All equal; replacement values copied to operand source
                                                  6354
0003BD06  B219 0200                       00000200  6355           SAC   ARMODE
0003BD0A  D51F 1000 30A0      00000000    000000A0  6356           CLC   0(32,R1),160(R3)
0003BD10  4770 510C                       0003BD2C  6357           BNE   F334XX              OP1R Different, error
0003BD14  D51F 6020 30C0      00000020    000000C0  6358           CLC   32(32,R6),192(R3)
0003BD1A  4770 510C                       0003BD2C  6359           BNE   F334XX              OP2R Different, error
0003BD1E  D51F 7040 30E0      00000040    000000E0  6360           CLC   64(32,R7),224(R3)
0003BD24  B219 0000                       00000000  6361           SAC   PRIMODE
0003BD28  478B 0000                       00000000  6362           BE    0(R11)
                                                  6363
                              0003BD2C    00000001  6364 F334XX    EQU   *
0003BD2C  B219 0000                       00000000  6365           SAC   PRIMODE
0003BD30  45F0 8450                       00002450  6366           BAL   R15,FAILMSG
0003BD34  07FB                                    6367           BR    R11
                                                  6368
0003BD36  4700 511A                       0003BD3A  6369 F334TSTCC BC    0,F334CCMSG         fail if unexpected condition code
                              0003BD3A    00000001  6370 F334CCMSG EQU   *
0003BD3A  45F0 82B2                       000022B2  6371           BAL   R15,CCMSG
0003BD3E  47F0 50C4                       0003BCE4  6372           B     F334CKUR
                                                  6373
                                                  6374           TSTDATA
0003BD42                      0003BD42    0003C200  6375+          ORG   PLOETST+(128*K)+37*(3*K)+(K+512)
                                                  6376           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003C200  D7D9C940 40406E6E                       6377+          DC    CL32'PRI   >>>OPDST37<<<' eye catcher
0003C208  6ED6D7C4 E2E3F3F7
0003C210  4C4C4C40 40404040
0003C218  40404040 40404040
0003C220                                          6378+OPDST37   DS    0C
0003C220  00000000 00000000                       6379           DC    XL256'0'
0003C228  00000000 00000000
0003C230  00000000 00000000
0003C238  00000000 00000000
0003C240  00000000 00000000
0003C248  00000000 00000000
0003C250  00000000 00000000
0003C258  00000000 00000000
0003C260  00000000 00000000
0003C268  00000000 00000000
0003C270  00000000 00000000
0003C278  00000000 00000000
0003C280  00000000 00000000
0003C288  00000000 00000000
0003C290  00000000 00000000
0003C298  00000000 00000000
0003C2A0  00000000 00000000
0003C2A8  00000000 00000000
0003C2B0  00000000 00000000
0003C2B8  00000000 00000000
0003C2C0  00000000 00000000
0003C2C8  00000000 00000000
0003C2D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C2D8  00000000 00000000
0003C2E0  00000000 00000000
0003C2E8  00000000 00000000
0003C2F0  00000000 00000000
0003C2F8  00000000 00000000
0003C300  00000000 00000000
0003C308  00000000 00000000
0003C310  00000000 00000000
0003C318  00000000 00000000
                                          6380
                                          6381          TSTLBL OPSRC              Source: OP1-OP3
0003C320  D7D9C940 40406E6E               6382+         DC    CL32'PRI   >>>OPSRC37<<<' eye catcher
0003C328  6ED6D7E2 D9C3F3F7
0003C330  4C4C4C40 40404040
0003C338  40404040 40404040
0003C340                                  6383+OPSRC37  DS    0C
0003C340  00000000 00000000               6384          DC    XL256'0'
0003C348  00000000 00000000
0003C350  00000000 00000000
0003C358  00000000 00000000
0003C360  00000000 00000000
0003C368  00000000 00000000
0003C370  00000000 00000000
0003C378  00000000 00000000
0003C380  00000000 00000000
0003C388  00000000 00000000
0003C390  00000000 00000000
0003C398  00000000 00000000
0003C3A0  00000000 00000000
0003C3A8  00000000 00000000
0003C3B0  00000000 00000000
0003C3B8  00000000 00000000
0003C3C0  00000000 00000000
0003C3C8  00000000 00000000
0003C3D0  00000000 00000000
0003C3D8  00000000 00000000
0003C3E0  00000000 00000000
0003C3E8  00000000 00000000
0003C3F0  00000000 00000000
0003C3F8  00000000 00000000
0003C400  00000000 00000000
0003C408  00000000 00000000
0003C410  00000000 00000000
0003C418  00000000 00000000
0003C420  00000000 00000000
0003C428  00000000 00000000
0003C430  00000000 00000000
0003C438  00000000 00000000
                                       6385
                                       6386          TSTLBL PL                 Parameter List
0003C440  D7D9C940 40406E6E            6387+         DC    CL32'PRI   >>>PL37<<<'    eye catcher
0003C448  6ED7D3F3 F74C4C4C
0003C450  40404040 40404040
0003C458  40404040 40404040
0003C460                               6388+PL37     DS    0C
0003C460  00000000 00000000            6389          DC    XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C468  00000000 00000000
0003C470  00000000 00000000
0003C478  00000000 00000000
0003C480  00000000 00000000
0003C488  00000000 00000000
0003C490  00000000 00000000
0003C498  00000000 00000000
0003C4A0  00000000 00000000
0003C4A8  00000000 00000000
0003C4B0  00000000 00000000
0003C4B8  00000000 00000000
0003C4C0  00000000 00000000
0003C4C8  00000000 00000000
0003C4D0  00000000 00000000
0003C4D8  00000000 00000000
0003C4E0  00000000 00000000
0003C4E8  00000000 00000000
0003C4F0  00000000 00000000
0003C4F8  00000000 00000000
0003C500  00000000 00000000
0003C508  00000000 00000000
0003C510  00000000 00000000
0003C518  00000000 00000000
0003C520  00000000 00000000
0003C528  00000000 00000000
0003C530  00000000 00000000
0003C538  00000000 00000000
0003C540  00000000 00000000
0003C548  00000000 00000000
0003C550  00000000 00000000
0003C558  00000000 00000000
0003C560  00000000 00000000
0003C568  00000000 00000000
0003C570  00000000 00000000
0003C578  00000000 00000000
0003C580  00000000 00000000
0003C588  00000000 00000000
0003C590  00000000 00000000
0003C598  00000000 00000000
0003C5A0  00000000 00000000
0003C5A8  00000000 00000000
0003C5B0  00000000 00000000
0003C5B8  00000000 00000000
                                        6390
                                        6391          TSTLBL FMSG                fail message
0003C5C0  D7D9C940 40406E6E             6392+         DC    CL32'PRI   >>>FMSG37<<<'  eye catcher
0003C5C8  6EC6D4E2 C7F3F74C
0003C5D0  4C4C4040 40404040
0003C5D8  40404040 40404040
0003C5E0                                6393+FMSG37   DS    0C
0003C5E0  86818993 858440D7             6394          DC    C'failed PLO-Extension '
0003C5E8  D3D660C5 A7A38595
0003C5F0  A2899695 40
0003C5F5  E3998997 938540C3             6395          DC    CL96'Triple Compare and Swap (TCSO: 33): ALL EQ'
0003C5FD  96949781 99854081
0003C605  958440E2 A6819740
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003C60D  4DE3C3E2 D67A40F3
0003C615  F35D7A40 C1D3D340
0003C61D  C5D84040 40404040
0003C625  40404040 40404040
0003C62D  40404040 40404040
0003C635  40404040 40404040
0003C63D  40404040 40404040
0003C645  40404040 40404040
0003C64D  40404040 40404040
                                          6396
                                          6397           TSTEND
0003C655  D7D9C940 4040C5D5               6398+TEND37    DC    CL32'PRI   END of test: 37 <<<'
0003C65D  C4409686 40A385A2
0003C665  A37A40F3 F7404C4C
0003C66D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6400 *================================================================================
                                                    6401
                                                    6402 * ------------------------------------------------------------------------
                                                    6403 * Test Triple Compare and Swap (TCSX: 32) instruction : 1st NE
                                                    6404 * ------------------------------------------------------------------------
                                                    6405          TSTHDR 2,1,32             1st Operand NOT Equal (cc=2, UR=1)
0003C675                          0003C675  0003C800  6406+         ORG   PLOETST+(128*K)+38*(3*K)
0003C800  D7D9C940 4040E2E3                         6407+         DC    CL32'PRI   START of test: 38'
0003C808  C1D9E340 968640A3
0003C810  85A2A37A 40F3F840
0003C818  40404040 40404040
0003C820                          0003C820          6408+         USING *,R5                base for test data and test routine
0003C820                          00000000          6409+         USING PLOETEST,R5
0003C820  0003C850                                  6410+T38      DC    A(CODE38)           address of test code
0003C824  0026                                      6411+         DC    H'38'               test number
0003C826  00                                        6412+         DC    X'00'
0003C827  20                                        6413+         DC    HL1'32'             Function
0003C828  01                                        6414+         DC    HL1'1'              expected unequal results (UR)
0003C829  00                                        6415+         DC    HL1'00'             found UR
0003C82A  02                                        6416+         DC    HL1'2'              cc
0003C82B  0D                                        6417+         DC    HL1'13'             not expected CC mask
0003C82C  00                                        6418+         DS    X                   extracted cc
0003C830  00000000 00000000                         6419+         DS    2F                  extract PSW after test (has CC)
0003C838  0003D1E0                                  6420+         DC    A(FMSG38)           address of test fail message
0003C83C  0003CF40                                  6421+         DC    A(OPSRC38)          address of operand source
0003C840  0003CE20                                  6422+         DC    A(OPDST38)          address of operand destination
0003C844  0003D060                                  6423+         DC    A(PL38)             address of parameter list
0003C848  0000419E                                  6424+         DC    A(RANDX+38)         address of random source data
0003C84C  0003D255                                  6425+         DC    A(TEND38)           address of test end eye catche
                                                    6426
                                                    6427          TSTCODE
0003C850                                            6428+CODE38   DS    0F
                                                    6429 *                                  load random source
0003C850  E310 501C 0014                 0000001C   6430          LGF   R1,OPSRCA           R1: OPSRC - Primary
0003C856  B904 0061                                 6431          LGR   R6,R1               R6: OPSRC - Secondary
0003C85A  B904 0071                                 6432          LGR   R7,R1               R7: OPSRC - Home
                                                    6433
0003C85E  E320 5028 0014                 00000028   6434          LGF   R2,RANDXA           R2: address random data
0003C864  E330 5024 0014                 00000024   6435          LGF   R3,PLA              R3: address PL
                                                    6436
0003C86A  B219 0200                      00000200   6437          SAC   ARMODE
0003C86E  D20F 1000 2000       00000000  00000000   6438          MVC   0(16,R1),0(R2)      OP1 - Primary
0003C874  D20F 6010 2010       00000010  00000010   6439          MVC   16(16,R6),16(R2)    OP2 - Secondary
0003C87A  D20F 7020 2020       00000020  00000020   6440          MVC   32(16,R7),32(R2)    OP3 - Home
                                                    6441 *
                                                    6442 * Triple Compare and Swap (TCSX: 32)
                                                    6443 * Build Parambter List
                                                    6444 *
0003C880  D22F 3040 2000       00000040  00000000   6445          MVC   64(48,R3),0(R2)     Load PL OP1-Op3 Compare
0003C886  D22F 3070 2080       00000070  00000080   6446          MVC   112(48,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                    6447 ****
                                                    6448 * force a failure by changing the 1st operand
                                                    6449 ****
0003C88C  D207 1000 2100       00000000  00000100   6450          MVC   0(8,R1),256(R2)     some random data
                                                    6451 ****
0003C892  E313 0000 0024                 00000000   6452          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003C898  E54C 3020 0000                 00000020  6453           MVHI  32(R3),PRIALET
                                                   6454
0003C89E  4111 0010                      00000010  6455           LA    R1,16(R1)
0003C8A2  E313 0008 0024                 00000008  6456           STG   R1,8(R3)            load OP2 address in PL
0003C8A8  E54C 3024 0001                 00000024  6457           MVHI  36(R3),SECALET
                                                   6458
0003C8AE  4111 0010                      00000010  6459           LA    R1,16(R1)
0003C8B2  E313 0010 0024                 00000010  6460           STG   R1,16(R3)           load OP3 address in PL
0003C8B8  E54C 3028 0002                 00000028  6461           MVHI  40(R3),HOMEALET
                                                   6462
0003C8BE  B982 0000                                6463           XGR   R0,R0
0003C8C2  4100 0020                      00000020  6464           LA    R0,32
                                                   6465
0003C8C6  B219 0200                      00000200  6466           SAC   ARMODE
0003C8CA  EE0F 0000 3000 00000000 00000000 6467           PLO   0,0,R15,0(R3)       Triple C/S (TCSX: 32)
0003C8D0  B219 0000                      00000000  6468           SAC   PRIMODE
                                                   6469
0003C8D4  B98D 0020                                6470           EPSW  R2,R0               extract psw
0003C8D8  5020 5010                      00000010  6471           ST    R2,CCPSW              to save CC
0003C8DC  E310 500B 0076                 0000000B  6472           LB    R1,CCMASK           (failure CC mask)
0003C8E2  8910 0004                      00000004  6473           SLL   R1,4                (shift to BC instr CC position)
0003C8E6  4410 511C                      0003C93C  6474           EX    R1,F321TSTCC            fail if...
                                                   6475
                                0003C8EA  00000001  6476 F321CKUR EQU   *
0003C8EA  D200 5009 3030 00000009 00000030 6477           MVC   URFOUND,48(R3)      Extract UR
0003C8F0  D500 5008 3030 00000008 00000030 6478           CLC   UR,48(R3)
0003C8F6  4780 50DE                      0003C8FE  6479           BE    F321CKRES
0003C8FA  45F0 8388                      00002388  6480           BAL   R15,URMSG
                                                   6481 *
                                                   6482 * compare results
                                                   6483 *
                                0003C8FE  00000001  6484 F321CKRES EQU   *
                                                   6485 *                                reset addresses
0003C8FE  E310 501C 0014                 0000001C  6486           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003C904  B904 0061                                6487           LGR   R6,R1               R6:  OPSRC - Secondary
0003C908  B904 0071                                6488           LGR   R7,R1               R7:  OPSRC - Home
                                                   6489
0003C90C  B219 0200                      00000200  6490           SAC   ARMODE
0003C910  D50F 1000 3040 00000000 00000040 6491           CLC   0(16,R1),64(R3)
0003C916  4770 5112                      0003C932  6492           BNE   F321XX              OP1C Different, error
0003C91A  D50F 6010 3050 00000010 00000050 6493           CLC   16(16,R6),80(R3)
0003C920  4770 5112                      0003C932  6494           BNE   F321XX              OP2C Different, error
0003C924  D50F 7020 3060 00000020 00000060 6495           CLC   32(16,R7),96(R3)
0003C92A  B219 0000                      00000000  6496           SAC   PRIMODE
0003C92E  478B 0000                      00000000  6497           BE    0(R11)              OP3C same, return
                                                   6498
                                0003C932  00000001  6499 F321XX   EQU   *
0003C932  B219 0000                      00000000  6500           SAC   PRIMODE
0003C936  45F0 8450                      00002450  6501           BAL   R15,FAILMSG
0003C93A  07FB                                     6502           BR    R11
                                                   6503
0003C93C  4700 5120                      0003C940  6504 F321TSTCC BC    0,F321CCMSG         fail if unexpected condition code
                                0003C940  00000001  6505 F321CCMSG EQU  *
0003C940  45F0 82B2                      000022B2  6506           BAL   R15,CCMSG
0003C944  47F0 50CA                      0003C8EA  6507           B     F321CKUR
                                                   6508
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6509           TSTDATA
0003C948                      0003C948  0003CE00  6510+          ORG   PLOETST+(128*K)+38*(3*K)+(K+512)
                                                    6511           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003CE00  D7D9C940 40406E6E                         6512+          DC    CL32'PRI   >>>OPDST38<<<' eye catcher
0003CE08  6ED6D7C4 E2E3F3F8
0003CE10  4C4C4C40 40404040
0003CE18  40404040 40404040
0003CE20                                            6513+OPDST38   DS    0C
0003CE20  00000000 00000000                         6514           DC    XL256'0'
0003CE28  00000000 00000000
0003CE30  00000000 00000000
0003CE38  00000000 00000000
0003CE40  00000000 00000000
0003CE48  00000000 00000000
0003CE50  00000000 00000000
0003CE58  00000000 00000000
0003CE60  00000000 00000000
0003CE68  00000000 00000000
0003CE70  00000000 00000000
0003CE78  00000000 00000000
0003CE80  00000000 00000000
0003CE88  00000000 00000000
0003CE90  00000000 00000000
0003CE98  00000000 00000000
0003CEA0  00000000 00000000
0003CEA8  00000000 00000000
0003CEB0  00000000 00000000
0003CEB8  00000000 00000000
0003CEC0  00000000 00000000
0003CEC8  00000000 00000000
0003CED0  00000000 00000000
0003CED8  00000000 00000000
0003CEE0  00000000 00000000
0003CEE8  00000000 00000000
0003CEF0  00000000 00000000
0003CEF8  00000000 00000000
0003CF00  00000000 00000000
0003CF08  00000000 00000000
0003CF10  00000000 00000000
0003CF18  00000000 00000000
                                                    6515
                                                    6516           TSTLBL OPSRC              Source: OP1-OP3
0003CF20  D7D9C940 40406E6E                         6517+          DC    CL32'PRI   >>>OPSRC38<<<' eye catcher
0003CF28  6ED6D7E2 D9C3F3F8
0003CF30  4C4C4C40 40404040
0003CF38  40404040 40404040
0003CF40                                            6518+OPSRC38   DS    0C
0003CF40  00000000 00000000                         6519           DC    XL256'0'
0003CF48  00000000 00000000
0003CF50  00000000 00000000
0003CF58  00000000 00000000
0003CF60  00000000 00000000
0003CF68  00000000 00000000
0003CF70  00000000 00000000
0003CF78  00000000 00000000
0003CF80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003CF88  00000000 00000000
0003CF90  00000000 00000000
0003CF98  00000000 00000000
0003CFA0  00000000 00000000
0003CFA8  00000000 00000000
0003CFB0  00000000 00000000
0003CFB8  00000000 00000000
0003CFC0  00000000 00000000
0003CFC8  00000000 00000000
0003CFD0  00000000 00000000
0003CFD8  00000000 00000000
0003CFE0  00000000 00000000
0003CFE8  00000000 00000000
0003CFF0  00000000 00000000
0003CFF8  00000000 00000000
0003D000  00000000 00000000
0003D008  00000000 00000000
0003D010  00000000 00000000
0003D018  00000000 00000000
0003D020  00000000 00000000
0003D028  00000000 00000000
0003D030  00000000 00000000
0003D038  00000000 00000000
                                        6520
                                        6521          TSTLBL PL                 Parameter List
0003D040  D7D9C940 40406E6E             6522+         DC    CL32'PRI   >>>PL38<<<'    eye catcher
0003D048  6ED7D3F3 F84C4C4C
0003D050  40404040 40404040
0003D058  40404040 40404040
0003D060                                6523+PL38     DS    0C
0003D060  00000000 00000000             6524          DC    XL352'00'
0003D068  00000000 00000000
0003D070  00000000 00000000
0003D078  00000000 00000000
0003D080  00000000 00000000
0003D088  00000000 00000000
0003D090  00000000 00000000
0003D098  00000000 00000000
0003D0A0  00000000 00000000
0003D0A8  00000000 00000000
0003D0B0  00000000 00000000
0003D0B8  00000000 00000000
0003D0C0  00000000 00000000
0003D0C8  00000000 00000000
0003D0D0  00000000 00000000
0003D0D8  00000000 00000000
0003D0E0  00000000 00000000
0003D0E8  00000000 00000000
0003D0F0  00000000 00000000
0003D0F8  00000000 00000000
0003D100  00000000 00000000
0003D108  00000000 00000000
0003D110  00000000 00000000
0003D118  00000000 00000000
0003D120  00000000 00000000
0003D128  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003D130  00000000 00000000
0003D138  00000000 00000000
0003D140  00000000 00000000
0003D148  00000000 00000000
0003D150  00000000 00000000
0003D158  00000000 00000000
0003D160  00000000 00000000
0003D168  00000000 00000000
0003D170  00000000 00000000
0003D178  00000000 00000000
0003D180  00000000 00000000
0003D188  00000000 00000000
0003D190  00000000 00000000
0003D198  00000000 00000000
0003D1A0  00000000 00000000
0003D1A8  00000000 00000000
0003D1B0  00000000 00000000
0003D1B8  00000000 00000000
                                        6525
                                        6526          TSTLBL FMSG               fail message
0003D1C0  D7D9C940 40406E6E             6527+         DC    CL32'PRI   >>>FMSG38<<<'  eye catcher
0003D1C8  6EC6D4E2 C7F3F84C
0003D1D0  4C4C4040 40404040
0003D1D8  40404040 40404040
0003D1E0                                6528+FMSG38   DS    0C
0003D1E0  86818993 858440D7             6529          DC    C'failed PLO-Extension '
0003D1E8  D3D660C5 A7A38595
0003D1F0  A2899695 40
0003D1F5  E3998997 938540C3                           6530          DC    CL96'Triple Compare and Swap (TCSX: 32): 1st NE'
0003D1FD  96949781 99854081
0003D205  958440E2 A6819740
0003D20D  4DE3C3E2 E77A40F3
0003D215  F25D7A40 F1A2A340
0003D21D  D5C54040 40404040
0003D225  40404040 40404040
0003D22D  40404040 40404040
0003D235  40404040 40404040
0003D23D  40404040 40404040
0003D245  40404040 40404040
0003D24D  40404040 40404040
                                        6531
                                        6532          TSTEND
0003D255  D7D9C940 4040C5D5             6533+TEND38   DC    CL32'PRI   END of test: 38 <<<'
0003D25D  C4409686 40A385A2
0003D265  A37A40F3 F8404C4C
0003D26D  4C404040 40404040
                                        6534
                                        6535 * ------------------------------------------------------------------
                                        6536 * Test Triple Compare and Swap (TCSX: 32) instruction : 2nd NE
                                        6537 * ------------------------------------------------------------------
                                        6538          TSTHDR 2,2,32             2nd Operand NOT Equal (cc=2, UR=2)
0003D275                  0003D275  0003D400  6539+         ORG   PLOETST+(128*K)+39*(3*K)
0003D400  D7D9C940 4040E2E3             6540+         DC    CL32'PRI   START of test: 39'
0003D408  C1D9E340 968640A3
0003D410  85A2A37A 40F3F940
0003D418  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D420                     0003D420              6541+          USING *,R5                base for test data and test routine
0003D420                     00000000              6542+          USING PLOETEST,R5
0003D420  0003D450                                 6543+T39       DC    A(CODE39)           address of test code
0003D424  0027                                     6544+          DC    H'39'               test number
0003D426  00                                       6545+          DC    X'00'
0003D427  20                                       6546+          DC    HL1'32'             Function
0003D428  02                                       6547+          DC    HL1'2'              expected unequal results (UR)
0003D429  00                                       6548+          DC    HL1'00'             found UR
0003D42A  02                                       6549+          DC    HL1'2'              cc
0003D42B  0D                                       6550+          DC    HL1'13'             not expected CC mask
0003D42C  00                                       6551+          DS    X                   extracted cc
0003D430  00000000 00000000                        6552+          DS    2F                  extract PSW after test (has CC)
0003D438  0003DDE0                                 6553+          DC    A(FMSG39)           address of test fail message
0003D43C  0003DB40                                 6554+          DC    A(OPSRC39)          address of operand source
0003D440  0003DA20                                 6555+          DC    A(OPDST39)          address of operand destination
0003D444  0003DC60                                 6556+          DC    A(PL39)             address of parameter list
0003D448  0000419F                                 6557+          DC    A(RANDX+39)         address of random source data
0003D44C  0003DE55                                 6558+          DC    A(TEND39)           address of test end eye catche
                                                   6559
                                                   6560           TSTCODE
0003D450                                           6561+CODE39    DS    0F
                                                   6562 *                                   load random source
0003D450  E310 501C 0014                 0000001C  6563           LGF   R1,OPSRCA           R1: OPSRC - Primary
0003D456  B904 0061                                6564           LGR   R6,R1               R6: OPSRC - Secondary
0003D45A  B904 0071                                6565           LGR   R7,R1               R7: OPSRC - Home
                                                   6566
0003D45E  E320 5028 0014                 00000028  6567           LGF   R2,RANDXA           R2: address random data
0003D464  E330 5024 0014                 00000024  6568           LGF   R3,PLA              R3: address PL
                                                   6569
0003D46A  B219 0200                      00000200  6570           SAC   ARMODE
0003D46E  D20F 1000 2000       00000000  00000000  6571           MVC   0(16,R1),0(R2)      OP1 - Primary
0003D474  D20F 6010 2010       00000010  00000010  6572           MVC   16(16,R6),16(R2)    OP2 - Secondary
0003D47A  D20F 7020 2020       00000020  00000020  6573           MVC   32(16,R7),32(R2)    OP3 - Home
                                                   6574 *
                                                   6575 * Triple Compare and Swap (TCSX: 32)
                                                   6576 * Build Parambter List
                                                   6577 *
0003D480  D22F 3040 2000       00000040  00000000  6578           MVC   64(48,R3),0(R2)     Load PL OP1-Op3 Compare
0003D486  D22F 3070 2080       00000070  00000080  6579           MVC   112(48,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6580 ****
                                                   6581 * force a failure by changing the 2nd operand
                                                   6582 ****
0003D48C  D207 6010 2100       00000010  00000100  6583           MVC   16(8,R6),256(R2)    some random data
                                                   6584 ****
0003D492  E313 0000 0024                 00000000  6585           STG   R1,0(R3)            load OP1 address in PL
0003D498  E54C 3020 0000                 00000020  6586           MVHI  32(R3),PRIALET
                                                   6587
0003D49E  4111 0010                      00000010  6588           LA    R1,16(R1)
0003D4A2  E313 0008 0024                 00000008  6589           STG   R1,8(R3)            load OP2 address in PL
0003D4A8  E54C 3024 0001                 00000024  6590           MVHI  36(R3),SECALET
                                                   6591
0003D4AE  4111 0010                      00000010  6592           LA    R1,16(R1)
0003D4B2  E313 0010 0024                 00000010  6593           STG   R1,16(R3)           load OP3 address in PL
0003D4B8  E54C 3028 0002                 00000028  6594           MVHI  40(R3),HOMEALET
                                                   6595
0003D4BE  B982 0000                                6596           XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003D4C2  4100 0020                    00000020  6597           LA    R0,32
                                                 6598
0003D4C6  B219 0200                    00000200  6599           SAC   ARMODE
0003D4CA  EE0F 0000 3000      00000000 00000000  6600           PLO   0,0,R15,0(R3)       Triple C/S (TCSX: 32)
0003D4D0  B219 0000                    00000000  6601           SAC   PRIMODE
                                                 6602
0003D4D4  B98D 0020                              6603           EPSW  R2,R0              extract psw
0003D4D8  5020 5010                    00000010  6604           ST    R2,CCPSW             to save CC
0003D4DC  E310 500B 0076               0000000B  6605           LB    R1,CCMASK          (failure CC mask)
0003D4E2  8910 0004                    00000004  6606           SLL   R1,4               (shift to BC instr CC position)
0003D4E6  4410 511C                    0003D53C  6607           EX    R1,F322TSTCC         fail if...
                                                 6608
                                0003D4EA 00000001  6609 F322CKUR EQU   *
0003D4EA  D200 5009 3030      00000009 00000030  6610           MVC   URFOUND,48(R3)     Extract UR
0003D4F0  D500 5008 3030      00000008 00000030  6611           CLC   UR,48(R3)
0003D4F6  4780 50DE                    0003D4FE  6612           BE    F322CKRES
0003D4FA  45F0 8388                    00002388  6613           BAL   R15,URMSG
                                                 6614 *
                                                 6615 * compare results
                                                 6616 *
                                0003D4FE 00000001  6617 F322CKRES EQU   *
                                                 6618 *                           reset addresses
0003D4FE  E310 501C 0014               0000001C  6619           LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003D504  B904 0061                              6620           LGR   R6,R1              R6:  OPSRC - Secondary
0003D508  B904 0071                              6621           LGR   R7,R1              R7:  OPSRC - Home
                                                 6622
0003D50C  B219 0200                    00000200  6623           SAC   ARMODE
0003D510  D50F 1000 3040      00000000 00000040  6624           CLC   0(16,R1),64(R3)
0003D516  4770 5112                    0003D532  6625           BNE   F322XX             OP1C Different, error
0003D51A  D50F 6010 3050      00000010 00000050  6626           CLC   16(16,R6),80(R3)
0003D520  4770 5112                    0003D532  6627           BNE   F322XX             OP2C Different, error
0003D524  D50F 7020 3060      00000020 00000060  6628           CLC   32(16,R7),96(R3)
0003D52A  B219 0000                    00000000  6629           SAC   PRIMODE
0003D52E  478B 0000                    00000000  6630           BE    0(R11)             OP3C same, return
                                                 6631
                                0003D532 00000001  6632 F322XX   EQU   *
0003D532  B219 0000                    00000000  6633           SAC   PRIMODE
0003D536  45F0 8450                    00002450  6634           BAL   R15,FAILMSG
0003D53A  07FB                                   6635           BR    R11
                                                 6636
0003D53C  4700 5120                    0003D540  6637 F322TSTCC BC    0,F322CCMSG        fail if unexpected condition code
                                0003D540 00000001  6638 F322CCMSG EQU  *
0003D540  45F0 82B2                    000022B2  6639           BAL   R15,CCMSG
0003D544  47F0 50CA                    0003D4EA  6640           B     F322CKUR
                                                 6641
                                                 6642           TSTDATA
0003D548                       0003D548 0003DA00  6643+          ORG   PLOETST+(128*K)+39*(3*K)+(K+512)
                                                 6644           TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
0003DA00  D7D9C940 40406E6E                      6645+          DC    CL32'PRI   >>>OPDST39<<<' eye catcher
0003DA08  6ED6D7C4 E2E3F3F9
0003DA10  4C4C4C40 40404040
0003DA18  40404040 40404040
0003DA20                                         6646+OPDST39   DS    0C
0003DA20  00000000 00000000                      6647           DC    XL256'0'
0003DA28  00000000 00000000
0003DA30  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003DA38  00000000 00000000
0003DA40  00000000 00000000
0003DA48  00000000 00000000
0003DA50  00000000 00000000
0003DA58  00000000 00000000
0003DA60  00000000 00000000
0003DA68  00000000 00000000
0003DA70  00000000 00000000
0003DA78  00000000 00000000
0003DA80  00000000 00000000
0003DA88  00000000 00000000
0003DA90  00000000 00000000
0003DA98  00000000 00000000
0003DAA0  00000000 00000000
0003DAA8  00000000 00000000
0003DAB0  00000000 00000000
0003DAB8  00000000 00000000
0003DAC0  00000000 00000000
0003DAC8  00000000 00000000
0003DAD0  00000000 00000000
0003DAD8  00000000 00000000
0003DAE0  00000000 00000000
0003DAE8  00000000 00000000
0003DAF0  00000000 00000000
0003DAF8  00000000 00000000
0003DB00  00000000 00000000
0003DB08  00000000 00000000
0003DB10  00000000 00000000
0003DB18  00000000 00000000
                                        6648
                                        6649           TSTLBL OPSRC              Source: OP1-OP3
0003DB20  D7D9C940 40406E6E             6650+          DC     CL32'PRI   >>>OPSRC39<<<' eye catcher
0003DB28  6ED6D7E2 D9C3F3F9
0003DB30  4C4C4C40 40404040
0003DB38  40404040 40404040
0003DB40                                6651+OPSRC39   DS     0C
0003DB40  00000000 00000000             6652           DC     XL256'0'
0003DB48  00000000 00000000
0003DB50  00000000 00000000
0003DB58  00000000 00000000
0003DB60  00000000 00000000
0003DB68  00000000 00000000
0003DB70  00000000 00000000
0003DB78  00000000 00000000
0003DB80  00000000 00000000
0003DB88  00000000 00000000
0003DB90  00000000 00000000
0003DB98  00000000 00000000
0003DBA0  00000000 00000000
0003DBA8  00000000 00000000
0003DBB0  00000000 00000000
0003DBB8  00000000 00000000
0003DBC0  00000000 00000000
0003DBC8  00000000 00000000
0003DBD0  00000000 00000000
0003DBD8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003DBE0  00000000 00000000
0003DBE8  00000000 00000000
0003DBF0  00000000 00000000
0003DBF8  00000000 00000000
0003DC00  00000000 00000000
0003DC08  00000000 00000000
0003DC10  00000000 00000000
0003DC18  00000000 00000000
0003DC20  00000000 00000000
0003DC28  00000000 00000000
0003DC30  00000000 00000000
0003DC38  00000000 00000000
                                          6653
                                          6654           TSTLBL PL                  Parameter List
0003DC40  D7D9C940 40406E6E               6655+          DC     CL32'PRI   >>>PL39<<<'    eye catcher
0003DC48  6ED7D3F3 F94C4C4C
0003DC50  40404040 40404040
0003DC58  40404040 40404040
0003DC60                                  6656+PL39      DS     0C
0003DC60  00000000 00000000               6657           DC     XL352'00'
0003DC68  00000000 00000000
0003DC70  00000000 00000000
0003DC78  00000000 00000000
0003DC80  00000000 00000000
0003DC88  00000000 00000000
0003DC90  00000000 00000000
0003DC98  00000000 00000000
0003DCA0  00000000 00000000
0003DCA8  00000000 00000000
0003DCB0  00000000 00000000
0003DCB8  00000000 00000000
0003DCC0  00000000 00000000
0003DCC8  00000000 00000000
0003DCD0  00000000 00000000
0003DCD8  00000000 00000000
0003DCE0  00000000 00000000
0003DCE8  00000000 00000000
0003DCF0  00000000 00000000
0003DCF8  00000000 00000000
0003DD00  00000000 00000000
0003DD08  00000000 00000000
0003DD10  00000000 00000000
0003DD18  00000000 00000000
0003DD20  00000000 00000000
0003DD28  00000000 00000000
0003DD30  00000000 00000000
0003DD38  00000000 00000000
0003DD40  00000000 00000000
0003DD48  00000000 00000000
0003DD50  00000000 00000000
0003DD58  00000000 00000000
0003DD60  00000000 00000000
0003DD68  00000000 00000000
0003DD70  00000000 00000000
0003DD78  00000000 00000000
0003DD80  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003DD88  00000000 00000000
0003DD90  00000000 00000000
0003DD98  00000000 00000000
0003DDA0  00000000 00000000
0003DDA8  00000000 00000000
0003DDB0  00000000 00000000
0003DDB8  00000000 00000000
                                        6658
                                        6659          TSTLBL FMSG                fail message
0003DDC0  D7D9C940 40406E6E             6660+         DC    CL32'PRI   >>>FMSG39<<<'  eye catcher
0003DDC8  6EC6D4E2 C7F3F94C
0003DDD0  4C4C4040 40404040
0003DDD8  40404040 40404040
0003DDE0                                6661+FMSG39   DS    0C
0003DDE0  86818993 858440D7             6662          DC    C'failed PLO-Extension '
0003DDE8  D3D660C5 A7A38595
0003DDF0  A2899695 40
0003DDF5  E3998997 938540C3             6663          DC    CL96'Triple Compare and Swap (TCSX: 32): 2nd NE'
0003DDFD  96949781 99854081
0003DE05  958440E2 A6819740
0003DE0D  4DE3C3E2 E77A40F3
0003DE15  F25D7A40 F2958440
0003DE1D  D5C54040 40404040
0003DE25  40404040 40404040
0003DE2D  40404040 40404040
0003DE35  40404040 40404040
0003DE3D  40404040 40404040
0003DE45  40404040 40404040
0003DE4D  40404040 40404040
                                        6664
                                        6665          TSTEND
0003DE55  D7D9C940 4040C5D5             6666+TEND39   DC    CL32'PRI   END of test: 39 <<<'
0003DE5D  C4409686 40A385A2
0003DE65  A37A40F3 F9404C4C
0003DE6D  4C404040 40404040
                                        6667
                                        6668 * ------------------------------------------------------------------
                                        6669 * Test Triple Compare and Swap (TCSX: 32) instruction : 3nd NE
                                        6670 * ------------------------------------------------------------------
                                        6671          TSTHDR 2,3,32              3nd Operand NOT Equal (cc=2, UR=3)
0003DE75                  0003DE75  0003E000  6672+         ORG   PLOETST+(128*K)+40*(3*K)
0003E000  D7D9C940 4040E2E3             6673+         DC    CL32'PRI   START of test: 40'
0003E008  C1D9E340 968640A3
0003E010  85A2A37A 40F4F040
0003E018  40404040 40404040
0003E020                  0003E020      6674+         USING *,R5                 base for test data and test routine
0003E020                  00000000      6675+         USING PLOETEST,R5
0003E020  0003E050                      6676+T40      DC    A(CODE40)            address of test code
0003E024  0028                          6677+         DC    H'40'                test number
0003E026  00                            6678+         DC    X'00'
0003E027  20                            6679+         DC    HL1'32'              Function
0003E028  03                            6680+         DC    HL1'3'               expected unequal results (UR)
0003E029  00                            6681+         DC    HL1'00'              found UR
0003E02A  02                            6682+         DC    HL1'2'               cc
0003E02B  0D                            6683+         DC    HL1'13'              not expected CC mask
0003E02C  00                            6684+         DS    X                    extracted cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0003E030  00000000 00000000                        6685+          DS    2F                  extract PSW after test (has CC)
0003E038  0003E9E0                                 6686+          DC    A(FMSG40)           address of test fail message
0003E03C  0003E740                                 6687+          DC    A(OPSRC40)          address of operand source
0003E040  0003E620                                 6688+          DC    A(OPDST40)          address of operand destination
0003E044  0003E860                                 6689+          DC    A(PL40)             address of parameter list
0003E048  000041A0                                 6690+          DC    A(RANDX+40)         address of random source data
0003E04C  0003EA55                                 6691+          DC    A(TEND40)           address of test end eye catche
                                                   6692
                                                   6693           TSTCODE
0003E050                                           6694+CODE40    DS    0F
                                                   6695 *                                   load random source
0003E050  E310 501C 0014                0000001C   6696           LGF   R1,OPSRCA           R1: OPSRC - Primary
0003E056  B904 0061                                6697           LGR   R6,R1               R6: OPSRC - Secondary
0003E05A  B904 0071                                6698           LGR   R7,R1               R7: OPSRC - Home
                                                   6699
0003E05E  E320 5028 0014                00000028   6700           LGF   R2,RANDXA           R2: address random data
0003E064  E330 5024 0014                00000024   6701           LGF   R3,PLA              R3: address PL
                                                   6702
0003E06A  B219 0200                     00000200   6703           SAC   ARMODE
0003E06E  D20F 1000 2000      00000000  00000000   6704           MVC   0(16,R1),0(R2)      OP1 - Primary
0003E074  D20F 6010 2010      00000010  00000010   6705           MVC   16(16,R6),16(R2)    OP2 - Secondary
0003E07A  D20F 7020 2020      00000020  00000020   6706           MVC   32(16,R7),32(R2)    OP3 - Home
                                                   6707 *
                                                   6708 * Triple Compare and Swap (TCSX: 32)
                                                   6709 * Build Parambter List
                                                   6710 *
0003E080  D22F 3040 2000      00000040  00000000   6711           MVC   64(48,R3),0(R2)     Load PL OP1-Op3 Compare
0003E086  D22F 3070 2080      00000070  00000080   6712           MVC   112(48,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6713 ****
                                                   6714 * force a failure by changing the 3nd operand
                                                   6715 ****
0003E08C  D207 7020 2100      00000020  00000100   6716           MVC   32(8,R7),256(R2)    some random data
                                                   6717 ****
0003E092  E313 0000 0024                00000000   6718           STG   R1,0(R3)            load OP1 address in PL
0003E098  E54C 3020 0000                00000020   6719           MVHI  32(R3),PRIALET
                                                   6720
0003E09E  4111 0010                     00000010   6721           LA    R1,16(R1)
0003E0A2  E313 0008 0024                00000008   6722           STG   R1,8(R3)            load OP2 address in PL
0003E0A8  E54C 3024 0001                00000024   6723           MVHI  36(R3),SECALET
                                                   6724
0003E0AE  4111 0010                     00000010   6725           LA    R1,16(R1)
0003E0B2  E313 0010 0024                00000010   6726           STG   R1,16(R3)           load OP3 address in PL
0003E0B8  E54C 3028 0002                00000028   6727           MVHI  40(R3),HOMEALET
                                                   6728
0003E0BE  B982 0000                                6729           XGR   R0,R0
0003E0C2  4100 0020                     00000020   6730           LA    R0,32
                                                   6731
0003E0C6  B219 0200                     00000200   6732           SAC   ARMODE
0003E0CA  EE0F 0000 3000      00000000  00000000   6733           PLO   0,0,R15,0(R3)       Triple C/S (TCSX: 32)
0003E0D0  B219 0000                     00000000   6734           SAC   PRIMODE
                                                   6735
0003E0D4  B98D 0020                                6736           EPSW  R2,R0               extract psw
0003E0D8  5020 5010                     00000010   6737           ST    R2,CCPSW              to save CC
0003E0DC  E310 500B 0076                0000000B   6738           LB    R1,CCMASK           (failure CC mask)
0003E0E2  8910 0004                     00000004   6739           SLL   R1,4                (shift to BC instr CC position)
0003E0E6  4410 511C                     0003E13C   6740           EX    R1,F323TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6741
                               0003E0EA  00000001  6742 F323CKUR EQU   *
0003E0EA  D200 5009 3030       00000009  00000030  6743          MVC   URFOUND,48(R3)      Extract UR
0003E0F0  D500 5008 3030       00000008  00000030  6744          CLC   UR,48(R3)
0003E0F6  4780 50DE                      0003E0FE  6745          BE    F323CKRES
0003E0FA  45F0 8388                      00002388  6746          BAL   R15,URMSG
                                                    6747 *
                                                    6748 * compare results
                                                    6749 *
                               0003E0FE  00000001  6750 F323CKRES EQU   *
                                                    6751 *                           reset addresses
0003E0FE  E310 501C 0014                 0000001C  6752          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003E104  B904 0061                                6753          LGR   R6,R1              R6:  OPSRC - Secondary
0003E108  B904 0071                                6754          LGR   R7,R1              R7:  OPSRC - Home
                                                    6755
0003E10C  B219 0200                      00000200  6756          SAC   ARMODE
0003E110  D50F 1000 3040       00000000  00000040  6757          CLC   0(16,R1),64(R3)
0003E116  4770 5112                      0003E132  6758          BNE   F323XX             OP1C Different, error
0003E11A  D50F 6010 3050       00000010  00000050  6759          CLC   16(16,R6),80(R3)
0003E120  4770 5112                      0003E132  6760          BNE   F323XX             OP2C Different, error
0003E124  D50F 7020 3060       00000020  00000060  6761          CLC   32(16,R7),96(R3)
0003E12A  B219 0000                      00000000  6762          SAC   PRIMODE
0003E12E  478B 0000                      00000000  6763          BE    0(R11)             OP3C same, return
                                                    6764
                               0003E132  00000001  6765 F323XX   EQU   *
0003E132  B219 0000                      00000000  6766          SAC   PRIMODE
0003E136  45F0 8450                      00002450  6767          BAL   R15,FAILMSG
0003E13A  07FB                                     6768          BR    R11
                                                    6769
0003E13C  4700 5120                      0003E140  6770 F323TSTCC BC    0,F323CCMSG        fail if unexpected condition code
                               0003E140  00000001  6771 F323CCMSG EQU   *
0003E140  45F0 82B2                      000022B2  6772          BAL   R15,CCMSG
0003E144  47F0 50CA                      0003E0EA  6773          B     F323CKUR
                                                    6774
                                                    6775          TSTDATA
0003E148                       0003E148  0003E600  6776+         ORG   PLOETST+(128*K)+40*(3*K)+(K+512)
                                                    6777          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
0003E600  D7D9C940 40406E6E                         6778+         DC    CL32'PRI   >>>OPDST40<<<' eye catcher
0003E608  6ED6D7C4 E2E3F4F0
0003E610  4C4C4C40 40404040
0003E618  40404040 40404040
0003E620                                            6779+OPDST40  DS    0C
0003E620  00000000 00000000                         6780          DC    XL256'0'
0003E628  00000000 00000000
0003E630  00000000 00000000
0003E638  00000000 00000000
0003E640  00000000 00000000
0003E648  00000000 00000000
0003E650  00000000 00000000
0003E658  00000000 00000000
0003E660  00000000 00000000
0003E668  00000000 00000000
0003E670  00000000 00000000
0003E678  00000000 00000000
0003E680  00000000 00000000
0003E688  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003E690  00000000 00000000
0003E698  00000000 00000000
0003E6A0  00000000 00000000
0003E6A8  00000000 00000000
0003E6B0  00000000 00000000
0003E6B8  00000000 00000000
0003E6C0  00000000 00000000
0003E6C8  00000000 00000000
0003E6D0  00000000 00000000
0003E6D8  00000000 00000000
0003E6E0  00000000 00000000
0003E6E8  00000000 00000000
0003E6F0  00000000 00000000
0003E6F8  00000000 00000000
0003E700  00000000 00000000
0003E708  00000000 00000000
0003E710  00000000 00000000
0003E718  00000000 00000000
                                        6781
                                        6782          TSTLBL OPSRC              Source: OP1-OP3
0003E720  D7D9C940 40406E6E             6783+         DC    CL32'PRI   >>>OPSRC40<<<' eye catcher
0003E728  6ED6D7E2 D9C3F4F0
0003E730  4C4C4C40 40404040
0003E738  40404040 40404040
0003E740                                6784+OPSRC40  DS    0C
0003E740  00000000 00000000             6785          DC    XL256'0'
0003E748  00000000 00000000
0003E750  00000000 00000000
0003E758  00000000 00000000
0003E760  00000000 00000000
0003E768  00000000 00000000
0003E770  00000000 00000000
0003E778  00000000 00000000
0003E780  00000000 00000000
0003E788  00000000 00000000
0003E790  00000000 00000000
0003E798  00000000 00000000
0003E7A0  00000000 00000000
0003E7A8  00000000 00000000
0003E7B0  00000000 00000000
0003E7B8  00000000 00000000
0003E7C0  00000000 00000000
0003E7C8  00000000 00000000
0003E7D0  00000000 00000000
0003E7D8  00000000 00000000
0003E7E0  00000000 00000000
0003E7E8  00000000 00000000
0003E7F0  00000000 00000000
0003E7F8  00000000 00000000
0003E800  00000000 00000000
0003E808  00000000 00000000
0003E810  00000000 00000000
0003E818  00000000 00000000
0003E820  00000000 00000000
0003E828  00000000 00000000
0003E830  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0003E838  00000000 00000000                          6786
                                                     6787          TSTLBL PL                  Parameter List
0003E840  D7D9C940 40406E6E                          6788+         DC    CL32'PRI   >>>PL40<<<'    eye catcher
0003E848  6ED7D3F4 F04C4C4C
0003E850  40404040 40404040
0003E858  40404040 40404040
0003E860                                             6789+PL40     DS    0C
0003E860  00000000 00000000                          6790          DC    XL352'00'
0003E868  00000000 00000000
0003E870  00000000 00000000
0003E878  00000000 00000000
0003E880  00000000 00000000
0003E888  00000000 00000000
0003E890  00000000 00000000
0003E898  00000000 00000000
0003E8A0  00000000 00000000
0003E8A8  00000000 00000000
0003E8B0  00000000 00000000
0003E8B8  00000000 00000000
0003E8C0  00000000 00000000
0003E8C8  00000000 00000000
0003E8D0  00000000 00000000
0003E8D8  00000000 00000000
0003E8E0  00000000 00000000
0003E8E8  00000000 00000000
0003E8F0  00000000 00000000
0003E8F8  00000000 00000000
0003E900  00000000 00000000
0003E908  00000000 00000000
0003E910  00000000 00000000
0003E918  00000000 00000000
0003E920  00000000 00000000
0003E928  00000000 00000000
0003E930  00000000 00000000
0003E938  00000000 00000000
0003E940  00000000 00000000
0003E948  00000000 00000000
0003E950  00000000 00000000
0003E958  00000000 00000000
0003E960  00000000 00000000
0003E968  00000000 00000000
0003E970  00000000 00000000
0003E978  00000000 00000000
0003E980  00000000 00000000
0003E988  00000000 00000000
0003E990  00000000 00000000
0003E998  00000000 00000000
0003E9A0  00000000 00000000
0003E9A8  00000000 00000000
0003E9B0  00000000 00000000
0003E9B8  00000000 00000000
                                                     6791
                                                     6792          TSTLBL FMSG                fail message
0003E9C0  D7D9C940 40406E6E                          6793+         DC    CL32'PRI   >>>FMSG40<<<'  eye catcher
0003E9C8  6EC6D4E2 C7F4F04C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003E9D0  4C4C4040 40404040
0003E9D8  40404040 40404040
0003E9E0                                          6794+FMSG40   DS    0C
0003E9E0  86818993 858440D7                       6795          DC    C'failed PLO-Extension '
0003E9E8  D3D660C5 A7A38595
0003E9F0  A2899695 40
0003E9F5  E3998997 938540C3                       6796          DC    CL96'Triple Compare and Swap (TCSX: 32): 3nd NE'
0003E9FD  96949781 99854081
0003EA05  958440E2 A6819740
0003EA0D  4DE3C3E2 E77A40F3
0003EA15  F25D7A40 F3958440
0003EA1D  D5C54040 40404040
0003EA25  40404040 40404040
0003EA2D  40404040 40404040
0003EA35  40404040 40404040
0003EA3D  40404040 40404040
0003EA45  40404040 40404040
0003EA4D  40404040 40404040
                                                  6797
                                                  6798          TSTEND
0003EA55  D7D9C940 4040C5D5                       6799+TEND40   DC    CL32'PRI   END of test: 40 <<<'
0003EA5D  C4409686 40A385A2
0003EA65  A37A40F4 F0404C4C
0003EA6D  4C404040 40404040
                                                  6800
                                                  6801 * ------------------------------------------------------------------
                                                  6802 * Test Triple Compare and Swap (TCSX: 32) instruction : All EQ
                                                  6803 * ------------------------------------------------------------------
                                                  6804          TSTHDR 0,99,32             All Equal (cc=0, UR=99)
0003EA75                          0003EA75  0003EC00  6805+         ORG   PLOETST+(128*K)+41*(3*K)
0003EC00  D7D9C940 4040E2E3                       6806+         DC    CL32'PRI   START of test: 41'
0003EC08  C1D9E340 968640A3
0003EC10  85A2A37A 40F4F140
0003EC18  40404040 40404040
0003EC20                          0003EC20        6807+         USING *,R5                base for test data and test routine
0003EC20                          00000000        6808+         USING PLOETEST,R5
0003EC20  0003EC50                                6809+T41      DC    A(CODE41)           address of test code
0003EC24  0029                                    6810+         DC    H'41'               test number
0003EC26  00                                      6811+         DC    X'00'
0003EC27  20                                      6812+         DC    HL1'32'             Function
0003EC28  63                                      6813+         DC    HL1'99'             expected unequal results (UR)
0003EC29  00                                      6814+         DC    HL1'00'             found UR
0003EC2A  00                                      6815+         DC    HL1'0'              cc
0003EC2B  07                                      6816+         DC    HL1'7'              not expected CC mask
0003EC2C  00                                      6817+         DS    X                   extracted cc
0003EC30  00000000 00000000                       6818+         DS    2F                  extract PSW after test (has CC)
0003EC38  0003F5E0                                6819+         DC    A(FMSG41)           address of test fail message
0003EC3C  0003F340                                6820+         DC    A(OPSRC41)          address of operand source
0003EC40  0003F220                                6821+         DC    A(OPDST41)          address of operand destination
0003EC44  0003F460                                6822+         DC    A(PL41)             address of parameter list
0003EC48  000041A1                                6823+         DC    A(RANDX+41)         address of random source data
0003EC4C  0003F655                                6824+         DC    A(TEND41)           address of test end eye catche
                                                  6825
                                                  6826          TSTCODE
0003EC50                                          6827+CODE41   DS    0F
                                                  6828 *                                    load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003EC50  E310 501C 0014                 0000001C  6829          LGF   R1,OPSRCA           R1: OPSRC - Primary
0003EC56  B904 0061                      0000001C  6830          LGR   R6,R1               R6: OPSRC - Secondary
0003EC5A  B904 0071                      0000001C  6831          LGR   R7,R1               R7: OPSRC - Home
                                                   6832
0003EC5E  E320 5028 0014                 00000028  6833          LGF   R2,RANDXA           R2: address random data
0003EC64  E330 5024 0014                 00000024  6834          LGF   R3,PLA              R3: address PL
                                                   6835
0003EC6A  B219 0200                      00000200  6836          SAC   ARMODE
0003EC6E  D20F 1000 2000       00000000  00000000  6837          MVC   0(16,R1),0(R2)      OP1 - Primary
0003EC74  D20F 6010 2010       00000010  00000010  6838          MVC   16(16,R6),16(R2)    OP2 - Secondary
0003EC7A  D20F 7020 2020       00000020  00000020  6839          MVC   32(16,R7),32(R2)    OP3 - Home
                                                   6840 *
                                                   6841 * Triple Compare and Swap (TCSX: 32)
                                                   6842 * Build Parambter List
                                                   6843 *
0003EC80  D22F 3040 2000       00000040  00000000  6844          MVC   64(48,R3),0(R2)     Load PL OP1-Op3 Compare
0003EC86  D22F 3070 2080       00000070  00000080  6845          MVC   112(48,R3),128(R2)  Load PL Op1-Op3 Replacement
                                                   6846 ****
                                                   6847 * Do Not force a failure by changing the 3nd operand
                                                   6848 ****
                                                   6849 *        MVC   0(8,R1),256(R2)     some random data
                                                   6850 ****
0003EC8C  E313 0000 0024                 00000000  6851          STG   R1,0(R3)            load OP1 address in PL
0003EC92  E54C 3020 0000                 00000020  6852          MVHI  32(R3),PRIALET
                                                   6853
0003EC98  4111 0010                      00000010  6854          LA    R1,16(R1)
0003EC9C  E313 0008 0024                 00000008  6855          STG   R1,8(R3)            load OP2 address in PL
0003ECA2  E54C 3024 0001                 00000024  6856          MVHI  36(R3),SECALET
                                                   6857
0003ECA8  4111 0010                      00000010  6858          LA    R1,16(R1)
0003ECAC  E313 0010 0024                 00000010  6859          STG   R1,16(R3)           load OP3 address in PL
0003ECB2  E54C 3028 0002                 00000028  6860          MVHI  40(R3),HOMEALET
                                                   6861
0003ECB8  B982 0000                                6862          XGR   R0,R0
0003ECBC  4100 0020                      00000020  6863          LA    R0,32
                                                   6864
0003ECC0  B219 0200                      00000200  6865          SAC   ARMODE
0003ECC4  EE0F 0000 3000       00000000  00000000  6866          PLO   0,0,R15,0(R3)       Triple C/S (TCSX: 32)
0003ECCA  B219 0000                      00000000  6867          SAC   PRIMODE
                                                   6868
0003ECCE  B98D 0020                                6869          EPSW  R2,R0               extract psw
0003ECD2  5020 5010                      00000010  6870          ST    R2,CCPSW              to save CC
0003ECD6  E310 500B 0076                 0000000B  6871          LB    R1,CCMASK           (failure CC mask)
0003ECDC  8910 0004                      00000004  6872          SLL   R1,4                (shift to BC instr CC position)
0003ECE0  4410 5116                      0003ED36  6873          EX    R1,F324TSTCC            fail if...
                                                   6874
                               0003ECE4  00000001  6875 F324CKUR EQU   *
0003ECE4  D200 5009 3030       00000009  00000030  6876          MVC   URFOUND,48(R3)      Extract UR
0003ECEA  D500 5008 3030       00000008  00000030  6877          CLC   UR,48(R3)
0003ECF0  4780 50D8                      0003ECF8  6878          BE    F324CKRES
0003ECF4  45F0 8388                      00002388  6879          BAL   R15,URMSG
                                                   6880 *
                                                   6881 * compare results
                                                   6882 *
                               0003ECF8  00000001  6883 F324CKRES EQU   *
                                                   6884 *                              reset addresses
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003ECF8  E310 501C 0014                  0000001C  6885           LGF   R1,OPSRCA           R1:  OPSRC - Primary
0003ECFE  B904 0061                               6886           LGR   R6,R1               R6:  OPSRC - Secondary
0003ED02  B904 0071                               6887           LGR   R7,R1               R7:  OPSRC - Home
                                                  6888
                                                  6889 * All equal; replacement values copied to operand source
                                                  6890
0003ED06  B219 0200                       00000200  6891           SAC   ARMODE
0003ED0A  D50F 1000 3070      00000000    00000070  6892           CLC   0(16,R1),112(R3)
0003ED10  4770 510C                       0003ED2C  6893           BNE   F324XX               OP1R Different, error
0003ED14  D50F 6010 3080      00000010    00000080  6894           CLC   16(16,R6),128(R3)
0003ED1A  4770 510C                       0003ED2C  6895           BNE   F324XX               OP2R Different, error
0003ED1E  D50F 7020 3090      00000020    00000090  6896           CLC   32(16,R7),144(R3)
0003ED24  B219 0000                       00000000  6897           SAC   PRIMODE
0003ED28  478B 0000                       00000000  6898           BE    0(R11)
                                                  6899
                              0003ED2C    00000001  6900 F324XX    EQU   *
0003ED2C  B219 0000                       00000000  6901           SAC   PRIMODE
0003ED30  45F0 8450                       00002450  6902           BAL   R15,FAILMSG
0003ED34  07FB                                    6903           BR    R11
                                                  6904
0003ED36  4700 511A                       0003ED3A  6905 F324TSTCC BC    0,F324CCMSG         fail if unexpected condition code
                              0003ED3A    00000001  6906 F324CCMSG EQU   *
0003ED3A  45F0 82B2                       000022B2  6907           BAL   R15,CCMSG
0003ED3E  47F0 50C4                       0003ECE4  6908           B     F324CKUR
                                                  6909
                                                  6910           TSTDATA
0003ED42                      0003ED42    0003F200  6911+          ORG   PLOETST+(128*K)+41*(3*K)+(K+512)
                                                  6912           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003F200  D7D9C940 40406E6E                       6913+          DC    CL32'PRI   >>>OPDST41<<<' eye catcher
0003F208  6ED6D7C4 E2E3F4F1
0003F210  4C4C4C40 40404040
0003F218  40404040 40404040
0003F220                                          6914+OPDST41   DS    0C
0003F220  00000000 00000000                       6915           DC    XL256'0'
0003F228  00000000 00000000
0003F230  00000000 00000000
0003F238  00000000 00000000
0003F240  00000000 00000000
0003F248  00000000 00000000
0003F250  00000000 00000000
0003F258  00000000 00000000
0003F260  00000000 00000000
0003F268  00000000 00000000
0003F270  00000000 00000000
0003F278  00000000 00000000
0003F280  00000000 00000000
0003F288  00000000 00000000
0003F290  00000000 00000000
0003F298  00000000 00000000
0003F2A0  00000000 00000000
0003F2A8  00000000 00000000
0003F2B0  00000000 00000000
0003F2B8  00000000 00000000
0003F2C0  00000000 00000000
0003F2C8  00000000 00000000
0003F2D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003F2D8  00000000 00000000
0003F2E0  00000000 00000000
0003F2E8  00000000 00000000
0003F2F0  00000000 00000000
0003F2F8  00000000 00000000
0003F300  00000000 00000000
0003F308  00000000 00000000
0003F310  00000000 00000000
0003F318  00000000 00000000
                                          6916
                                          6917           TSTLBL OPSRC              Source: OP1-OP3
0003F320  D7D9C940 40406E6E               6918+          DC     CL32'PRI   >>>OPSRC41<<<' eye catcher
0003F328  6ED6D7E2 D9C3F4F1
0003F330  4C4C4C40 40404040
0003F338  40404040 40404040
0003F340                                  6919+OPSRC41   DS     0C
0003F340  00000000 00000000               6920           DC     XL256'0'
0003F348  00000000 00000000
0003F350  00000000 00000000
0003F358  00000000 00000000
0003F360  00000000 00000000
0003F368  00000000 00000000
0003F370  00000000 00000000
0003F378  00000000 00000000
0003F380  00000000 00000000
0003F388  00000000 00000000
0003F390  00000000 00000000
0003F398  00000000 00000000
0003F3A0  00000000 00000000
0003F3A8  00000000 00000000
0003F3B0  00000000 00000000
0003F3B8  00000000 00000000
0003F3C0  00000000 00000000
0003F3C8  00000000 00000000
0003F3D0  00000000 00000000
0003F3D8  00000000 00000000
0003F3E0  00000000 00000000
0003F3E8  00000000 00000000
0003F3F0  00000000 00000000
0003F3F8  00000000 00000000
0003F400  00000000 00000000
0003F408  00000000 00000000
0003F410  00000000 00000000
0003F418  00000000 00000000
0003F420  00000000 00000000
0003F428  00000000 00000000
0003F430  00000000 00000000
0003F438  00000000 00000000
                                       6921
                                       6922           TSTLBL PL                 Parameter List
0003F440  D7D9C940 40406E6E            6923+          DC     CL32'PRI   >>>PL41<<<'    eye catcher
0003F448  6ED7D3F4 F14C4C4C
0003F450  40404040 40404040
0003F458  40404040 40404040
0003F460                               6924+PL41      DS     0C
0003F460  00000000 00000000            6925           DC     XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003F468  00000000 00000000
0003F470  00000000 00000000
0003F478  00000000 00000000
0003F480  00000000 00000000
0003F488  00000000 00000000
0003F490  00000000 00000000
0003F498  00000000 00000000
0003F4A0  00000000 00000000
0003F4A8  00000000 00000000
0003F4B0  00000000 00000000
0003F4B8  00000000 00000000
0003F4C0  00000000 00000000
0003F4C8  00000000 00000000
0003F4D0  00000000 00000000
0003F4D8  00000000 00000000
0003F4E0  00000000 00000000
0003F4E8  00000000 00000000
0003F4F0  00000000 00000000
0003F4F8  00000000 00000000
0003F500  00000000 00000000
0003F508  00000000 00000000
0003F510  00000000 00000000
0003F518  00000000 00000000
0003F520  00000000 00000000
0003F528  00000000 00000000
0003F530  00000000 00000000
0003F538  00000000 00000000
0003F540  00000000 00000000
0003F548  00000000 00000000
0003F550  00000000 00000000
0003F558  00000000 00000000
0003F560  00000000 00000000
0003F568  00000000 00000000
0003F570  00000000 00000000
0003F578  00000000 00000000
0003F580  00000000 00000000
0003F588  00000000 00000000
0003F590  00000000 00000000
0003F598  00000000 00000000
0003F5A0  00000000 00000000
0003F5A8  00000000 00000000
0003F5B0  00000000 00000000
0003F5B8  00000000 00000000
                                        6926
                                        6927          TSTLBL FMSG                fail message
0003F5C0  D7D9C940 40406E6E             6928+         DC    CL32'PRI   >>>FMSG41<<<'  eye catcher
0003F5C8  6EC6D4E2 C7F4F14C
0003F5D0  4C4C4040 40404040
0003F5D8  40404040 40404040
0003F5E0                                6929+FMSG41   DS    0C
0003F5E0  86818993 858440D7             6930          DC    C'failed PLO-Extension '
0003F5E8  D3D660C5 A7A38595
0003F5F0  A2899695 40
0003F5F5  E3998997 938540C3             6931          DC    CL96'Triple Compare and Swap (TCSX: 32): All EQ'
0003F5FD  96949781 99854081
0003F605  958440E2 A6819740
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003F60D  4DE3C3E2 E77A40F3
0003F615  F25D7A40 C1939340
0003F61D  C5D84040 40404040
0003F625  40404040 40404040
0003F62D  40404040 40404040
0003F635  40404040 40404040
0003F63D  40404040 40404040
0003F645  40404040 40404040
0003F64D  40404040 40404040
                                          6932
                                          6933           TSTEND
0003F655  D7D9C940 4040C5D5               6934+TEND41    DC    CL32'PRI   END of test: 41 <<<'
0003F65D  C4409686 40A385A2
0003F665  A37A40F4 F1404C4C
0003F66D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    6936 *=============================================================================
                                                    6937
                                                    6938 * -----------------------------------------------------------------------
                                                    6939 * Test Triple Compare and Swap (TCSG: 31) instruction : 1st NE
                                                    6940 * -----------------------------------------------------------------------
                                                    6941          TSTHDR 2,1,31             1st Operand NOT Equal (cc=2, UR=1)
0003F675                          0003F675  0003F800  6942+         ORG   PLOETST+(128*K)+42*(3*K)
0003F800  D7D9C940 4040E2E3                         6943+         DC    CL32'PRI   START of test: 42'
0003F808  C1D9E340 968640A3
0003F810  85A2A37A 40F4F240
0003F818  40404040 40404040
0003F820                          0003F820          6944+         USING *,R5                base for test data and test routine
0003F820                          00000000          6945+         USING PLOETEST,R5
0003F820  0003F850                                  6946+T42      DC    A(CODE42)           address of test code
0003F824  002A                                      6947+         DC    H'42'               test number
0003F826  00                                        6948+         DC    X'00'
0003F827  1F                                        6949+         DC    HL1'31'             Function
0003F828  01                                        6950+         DC    HL1'1'              expected unequal results (UR)
0003F829  00                                        6951+         DC    HL1'00'             found UR
0003F82A  02                                        6952+         DC    HL1'2'              cc
0003F82B  0D                                        6953+         DC    HL1'13'             not expected CC mask
0003F82C  00                                        6954+         DS    X                   extracted cc
0003F830  00000000 00000000                         6955+         DS    2F                  extract PSW after test (has CC)
0003F838  000401E0                                  6956+         DC    A(FMSG42)           address of test fail message
0003F83C  0003FF40                                  6957+         DC    A(OPSRC42)          address of operand source
0003F840  0003FE20                                  6958+         DC    A(OPDST42)          address of operand destination
0003F844  00040060                                  6959+         DC    A(PL42)             address of parameter list
0003F848  000041A2                                  6960+         DC    A(RANDX+42)         address of random source data
0003F84C  00040255                                  6961+         DC    A(TEND42)           address of test end eye catche
                                                    6962
                                                    6963          TSTCODE
0003F850                                            6964+CODE42   DS    0F
                                                    6965 *                                  load random source
0003F850  E310 501C 0014                 0000001C   6966          LGF   R1,OPSRCA           R1: OPSRC - Primary
0003F856  B904 0061                                 6967          LGR   R6,R1               R6: OPSRC - Secondary
0003F85A  B904 0071                                 6968          LGR   R7,R1               R7: OPSRC - Home
                                                    6969
0003F85E  E320 5028 0014                 00000028   6970          LGF   R2,RANDXA           R2: address random data
0003F864  E330 5024 0014                 00000024   6971          LGF   R3,PLA              R3: address PL
                                                    6972
0003F86A  B219 0200                      00000200   6973          SAC   ARMODE
0003F86E  D207 1000 2000       00000000  00000000   6974          MVC   0(8,R1),0(R2)       OP1 - Primary
0003F874  D207 6008 2008       00000008  00000008   6975          MVC   8(8,R6),8(R2)       OP2 - Secondary
0003F87A  D207 7010 2010       00000010  00000010   6976          MVC   16(8,R7),16(R2)     OP3 - Home
                                                    6977 *
                                                    6978 * Triple Compare and Swap (TCSG: 31)
                                                    6979 * Build Parambter List
                                                    6980 *
0003F880  D217 3040 2000       00000040  00000000   6981          MVC   64(24,R3),0(R2)     Load PL OP1-Op3 Compare
0003F886  D217 3058 2080       00000058  00000080   6982          MVC   88(24,R3),128(R2)   Load PL Op1-Op3 Replacement
                                                    6983 ****
                                                    6984 * force a failure by changing the 1st operand
                                                    6985 ****
0003F88C  D203 1000 2100       00000000  00000100   6986          MVC   0(4,R1),256(R2)     some random data
                                                    6987 ****
0003F892  E313 0000 0024                 00000000   6988          STG   R1,0(R3)            load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0003F898  E54C 3020 0000                 00000020  6989           MVHI  32(R3),PRIALET
                                                   6990
0003F89E  4111 0008                      00000008  6991           LA    R1,8(R1)
0003F8A2  E313 0008 0024                 00000008  6992           STG   R1,8(R3)           load OP2 address in PL
0003F8A8  E54C 3024 0001                 00000024  6993           MVHI  36(R3),SECALET
                                                   6994
0003F8AE  4111 0008                      00000008  6995           LA    R1,8(R1)
0003F8B2  E313 0010 0024                 00000010  6996           STG   R1,16(R3)          load OP3 address in PL
0003F8B8  E54C 3028 0002                 00000028  6997           MVHI  40(R3),HOMEALET
                                                   6998
0003F8BE  B982 0000                                6999           XGR   R0,R0
0003F8C2  4100 001F                      0000001F  7000           LA    R0,31
                                                   7001
0003F8C6  B219 0200                      00000200  7002           SAC   ARMODE
0003F8CA  EE0F 0000 3000       00000000  00000000  7003           PLO   0,0,R15,0(R3)      Triple C/S (TCSG: 31)
0003F8D0  B219 0000                      00000000  7004           SAC   PRIMODE
                                                   7005
0003F8D4  B98D 0020                                7006           EPSW  R2,R0              extract psw
0003F8D8  5020 5010                      00000010  7007           ST    R2,CCPSW             to save CC
0003F8DC  E310 500B 0076                 0000000B  7008           LB    R1,CCMASK          (failure CC mask)
0003F8E2  8910 0004                      00000004  7009           SLL   R1,4               (shift to BC instr CC position)
0003F8E6  4410 511C                      0003F93C  7010           EX    R1,F311TSTCC         fail if...
                                                   7011
                               0003F8EA  00000001  7012 F311CKUR EQU   *
0003F8EA  D200 5009 3030       00000009  00000030  7013           MVC   URFOUND,48(R3)     Extract UR
0003F8F0  D500 5008 3030       00000008  00000030  7014           CLC   UR,48(R3)
0003F8F6  4780 50DE                      0003F8FE  7015           BE    F311CKRES
0003F8FA  45F0 8388                      00002388  7016           BAL   R15,URMSG
                                                   7017 *
                                                   7018 * compare results
                                                   7019 *
                               0003F8FE  00000001  7020 F311CKRES EQU   *
                                                   7021 *                               reset addresses
0003F8FE  E310 501C 0014                 0000001C  7022           LGF   R1,OPSRCA          R1:  OPSRC - Primary
0003F904  B904 0061                                7023           LGR   R6,R1              R6:  OPSRC - Secondary
0003F908  B904 0071                                7024           LGR   R7,R1              R7:  OPSRC - Home
                                                   7025
0003F90C  B219 0200                      00000200  7026           SAC   ARMODE
0003F910  D507 1000 3040       00000000  00000040  7027           CLC   0(8,R1),64(R3)
0003F916  4770 5112                      0003F932  7028           BNE   F311XX             OP1C Different, error
0003F91A  D507 6008 3048       00000008  00000048  7029           CLC   8(8,R6),72(R3)
0003F920  4770 5112                      0003F932  7030           BNE   F311XX             OP2C Different, error
0003F924  D507 7010 3050       00000010  00000050  7031           CLC   16(8,R7),80(R3)
0003F92A  B219 0000                      00000000  7032           SAC   PRIMODE
0003F92E  478B 0000                      00000000  7033           BE    0(R11)             OP3C same, return
                                                   7034
                               0003F932  00000001  7035 F311XX   EQU   *
0003F932  B219 0000                      00000000  7036           SAC   PRIMODE
0003F936  45F0 8450                      00002450  7037           BAL   R15,FAILMSG
0003F93A  07FB                                     7038           BR    R11
                                                   7039
0003F93C  4700 5120                      0003F940  7040 F311TSTCC BC    0,F311CCMSG        fail if unexpected condition code
                               0003F940  00000001  7041 F311CCMSG EQU   *
0003F940  45F0 82B2                      000022B2  7042           BAL   R15,CCMSG
0003F944  47F0 50CA                      0003F8EA  7043           B     F311CKUR
                                                   7044
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7045           TSTDATA
0003F948                      0003F948  0003FE00  7046+          ORG   PLOETST+(128*K)+42*(3*K)+(K+512)
                                                    7047           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
0003FE00  D7D9C940 40406E6E                         7048+          DC    CL32'PRI   >>>OPDST42<<<' eye catcher
0003FE08  6ED6D7C4 E2E3F4F2
0003FE10  4C4C4C40 40404040
0003FE18  40404040 40404040
0003FE20                                            7049+OPDST42   DS    0C
0003FE20  00000000 00000000                         7050           DC    XL256'0'
0003FE28  00000000 00000000
0003FE30  00000000 00000000
0003FE38  00000000 00000000
0003FE40  00000000 00000000
0003FE48  00000000 00000000
0003FE50  00000000 00000000
0003FE58  00000000 00000000
0003FE60  00000000 00000000
0003FE68  00000000 00000000
0003FE70  00000000 00000000
0003FE78  00000000 00000000
0003FE80  00000000 00000000
0003FE88  00000000 00000000
0003FE90  00000000 00000000
0003FE98  00000000 00000000
0003FEA0  00000000 00000000
0003FEA8  00000000 00000000
0003FEB0  00000000 00000000
0003FEB8  00000000 00000000
0003FEC0  00000000 00000000
0003FEC8  00000000 00000000
0003FED0  00000000 00000000
0003FED8  00000000 00000000
0003FEE0  00000000 00000000
0003FEE8  00000000 00000000
0003FEF0  00000000 00000000
0003FEF8  00000000 00000000
0003FF00  00000000 00000000
0003FF08  00000000 00000000
0003FF10  00000000 00000000
0003FF18  00000000 00000000
                                                    7051
                                                    7052           TSTLBL OPSRC              Source: OP1-OP3
0003FF20  D7D9C940 40406E6E                         7053+          DC    CL32'PRI   >>>OPSRC42<<<' eye catcher
0003FF28  6ED6D7E2 D9C3F4F2
0003FF30  4C4C4C40 40404040
0003FF38  40404040 40404040
0003FF40                                            7054+OPSRC42   DS    0C
0003FF40  00000000 00000000                         7055           DC    XL256'0'
0003FF48  00000000 00000000
0003FF50  00000000 00000000
0003FF58  00000000 00000000
0003FF60  00000000 00000000
0003FF68  00000000 00000000
0003FF70  00000000 00000000
0003FF78  00000000 00000000
0003FF80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0003FF88  00000000 00000000
0003FF90  00000000 00000000
0003FF98  00000000 00000000
0003FFA0  00000000 00000000
0003FFA8  00000000 00000000
0003FFB0  00000000 00000000
0003FFB8  00000000 00000000
0003FFC0  00000000 00000000
0003FFC8  00000000 00000000
0003FFD0  00000000 00000000
0003FFD8  00000000 00000000
0003FFE0  00000000 00000000
0003FFE8  00000000 00000000
0003FFF0  00000000 00000000
0003FFF8  00000000 00000000
00040000  00000000 00000000
00040008  00000000 00000000
00040010  00000000 00000000
00040018  00000000 00000000
00040020  00000000 00000000
00040028  00000000 00000000
00040030  00000000 00000000
00040038  00000000 00000000
                                        7056
                                        7057          TSTLBL PL                 Parameter List
00040040  D7D9C940 40406E6E             7058+         DC    CL32'PRI   >>>PL42<<<'    eye catcher
00040048  6ED7D3F4 F24C4C4C
00040050  40404040 40404040
00040058  40404040 40404040
00040060                                7059+PL42     DS    0C
00040060  00000000 00000000             7060          DC    XL352'00'
00040068  00000000 00000000
00040070  00000000 00000000
00040078  00000000 00000000
00040080  00000000 00000000
00040088  00000000 00000000
00040090  00000000 00000000
00040098  00000000 00000000
000400A0  00000000 00000000
000400A8  00000000 00000000
000400B0  00000000 00000000
000400B8  00000000 00000000
000400C0  00000000 00000000
000400C8  00000000 00000000
000400D0  00000000 00000000
000400D8  00000000 00000000
000400E0  00000000 00000000
000400E8  00000000 00000000
000400F0  00000000 00000000
000400F8  00000000 00000000
00040100  00000000 00000000
00040108  00000000 00000000
00040110  00000000 00000000
00040118  00000000 00000000
00040120  00000000 00000000
00040128  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00040130  00000000 00000000
00040138  00000000 00000000
00040140  00000000 00000000
00040148  00000000 00000000
00040150  00000000 00000000
00040158  00000000 00000000
00040160  00000000 00000000
00040168  00000000 00000000
00040170  00000000 00000000
00040178  00000000 00000000
00040180  00000000 00000000
00040188  00000000 00000000
00040190  00000000 00000000
00040198  00000000 00000000
000401A0  00000000 00000000
000401A8  00000000 00000000
000401B0  00000000 00000000
000401B8  00000000 00000000
                                        7061
                                        7062          TSTLBL FMSG               fail message
000401C0  D7D9C940 40406E6E             7063+         DC    CL32'PRI   >>>FMSG42<<<'  eye catcher
000401C8  6EC6D4E2 C7F4F24C
000401D0  4C4C4040 40404040
000401D8  40404040 40404040
000401E0                                7064+FMSG42   DS    0C
000401E0  86818993 858440D7             7065          DC    C'failed PLO-Extension '
000401E8  D3D660C5 A7A38595
000401F0  A2899695 40
000401F5  E3998997 938540C3                           7066          DC    CL96'Triple Compare and Swap (TCSG: 31): 1st NE'
000401FD  96949781 99854081
00040205  958440E2 A6819740
0004020D  4DE3C3E2 C77A40F3
00040215  F15D7A40 F1A2A340
0004021D  D5C54040 40404040
00040225  40404040 40404040
0004022D  40404040 40404040
00040235  40404040 40404040
0004023D  40404040 40404040
00040245  40404040 40404040
0004024D  40404040 40404040
                                        7067
                                        7068          TSTEND
00040255  D7D9C940 4040C5D5             7069+TEND42   DC    CL32'PRI   END of test: 42 <<<'
0004025D  C4409686 40A385A2
00040265  A37A40F4 F2404C4C
0004026D  4C404040 40404040
                                        7070
                                        7071 * ------------------------------------------------------------------
                                        7072 * Test Triple Compare and Swap (TCSG: 31) instruction : 2nd NE
                                        7073 * ------------------------------------------------------------------
                                        7074          TSTHDR 2,2,31             2nd Operand NOT Equal (cc=2, UR=2)
00040275                      00040275  00040400      7075+         ORG   PLOETST+(128*K)+43*(3*K)
00040400  D7D9C940 4040E2E3             7076+         DC    CL32'PRI   START of test: 43'
00040408  C1D9E340 968640A3
00040410  85A2A37A 40F4F340
00040418  40404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00040420                     00040420            7077+          USING *,R5                base for test data and test routine
00040420                     00000000            7078+          USING PLOETEST,R5
00040420  00040450                               7079+T43       DC    A(CODE43)           address of test code
00040424  002B                                   7080+          DC    H'43'               test number
00040426  00                                     7081+          DC    X'00'
00040427  1F                                     7082+          DC    HL1'31'             Function
00040428  02                                     7083+          DC    HL1'2'              expected unequal results (UR)
00040429  00                                     7084+          DC    HL1'00'             found UR
0004042A  02                                     7085+          DC    HL1'2'              cc
0004042B  0D                                     7086+          DC    HL1'13'             not expected CC mask
0004042C  00                                     7087+          DS    X                   extracted cc
00040430  00000000 00000000                      7088+          DS    2F                  extract PSW after test (has CC)
00040438  00040DE0                               7089+          DC    A(FMSG43)           address of test fail message
0004043C  00040B40                               7090+          DC    A(OPSRC43)          address of operand source
00040440  00040A20                               7091+          DC    A(OPDST43)          address of operand destination
00040444  00040C60                               7092+          DC    A(PL43)             address of parameter list
00040448  000041A3                               7093+          DC    A(RANDX+43)         address of random source data
0004044C  00040E55                               7094+          DC    A(TEND43)           address of test end eye catche
                                                 7095
                                                 7096           TSTCODE
00040450                                         7097+CODE43    DS    0F
                                                 7098 *                                   load random source
00040450  E310 501C 0014             0000001C    7099           LGF   R1,OPSRCA           R1: OPSRC - Primary
00040456  B904 0061                              7100           LGR   R6,R1               R6: OPSRC - Secondary
0004045A  B904 0071                              7101           LGR   R7,R1               R7: OPSRC - Home
                                                 7102
0004045E  E320 5028 0014             00000028    7103           LGF   R2,RANDXA           R2: address random data
00040464  E330 5024 0014             00000024    7104           LGF   R3,PLA              R3: address PL
                                                 7105
0004046A  B219 0200                  00000200    7106           SAC   ARMODE
0004046E  D207 1000 2000  00000000   00000000    7107           MVC   0(8,R1),0(R2)       OP1 - Primary
00040474  D207 6008 2008  00000008   00000008    7108           MVC   8(8,R6),8(R2)       OP2 - Secondary
0004047A  D207 7010 2010  00000010   00000010    7109           MVC   16(8,R7),16(R2)     OP3 - Home
                                                 7110 *
                                                 7111 * Triple Compare and Swap (TCSG: 31)
                                                 7112 * Build Parambter List
                                                 7113 *
00040480  D217 3040 2000  00000040   00000000    7114           MVC   64(24,R3),0(R2)     Load PL OP1-Op3 Compare
00040486  D217 3058 2080  00000058   00000080    7115           MVC   88(24,R3),128(R2)   Load PL Op1-Op3 Replacement
                                                 7116 ****
                                                 7117 * force a failure by changing the 2nd operand
                                                 7118 ****
0004048C  D203 6008 2100  00000008   00000100    7119           MVC   8(4,R6),256(R2)     some random data
                                                 7120 ****
00040492  E313 0000 0024             00000000    7121           STG   R1,0(R3)            load OP1 address in PL
00040498  E54C 3020 0000             00000020    7122           MVHI  32(R3),PRIALET
                                                 7123
0004049E  4111 0008                  00000008    7124           LA    R1,8(R1)
000404A2  E313 0008 0024             00000008    7125           STG   R1,8(R3)            load OP2 address in PL
000404A8  E54C 3024 0001             00000024    7126           MVHI  36(R3),SECALET
                                                 7127
000404AE  4111 0008                  00000008    7128           LA    R1,8(R1)
000404B2  E313 0010 0024             00000010    7129           STG   R1,16(R3)           load OP3 address in PL
000404B8  E54C 3028 0002             00000028    7130           MVHI  40(R3),HOMEALET
                                                 7131
000404BE  B982 0000                              7132           XGR   R0,R0
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000404C2  4100 001F                      0000001F  7133           LA    R0,31
                                                   7134
000404C6  B219 0200                      00000200  7135           SAC   ARMODE
000404CA  EE0F 0000 3000      00000000   00000000  7136           PLO   0,0,R15,0(R3)       Triple C/S (TCSG: 31)
000404D0  B219 0000                      00000000  7137           SAC   PRIMODE
                                                   7138
000404D4  B98D 0020                                7139           EPSW  R2,R0              extract psw
000404D8  5020 5010                      00000010  7140           ST    R2,CCPSW             to save CC
000404DC  E310 500B 0076                 0000000B  7141           LB    R1,CCMASK          (failure CC mask)
000404E2  8910 0004                      00000004  7142           SLL   R1,4               (shift to BC instr CC position)
000404E6  4410 511C                      0004053C  7143           EX    R1,F312TSTCC         fail if...
                                                   7144
                                000404EA  00000001  7145 F312CKUR EQU   *
000404EA  D200 5009 3030      00000009   00000030  7146           MVC   URFOUND,48(R3)     Extract UR
000404F0  D500 5008 3030      00000008   00000030  7147           CLC   UR,48(R3)
000404F6  4780 50DE                      000404FE  7148           BE    F312CKRES
000404FA  45F0 8388                      00002388  7149           BAL   R15,URMSG
                                                   7150 *
                                                   7151 * compare results
                                                   7152 *
                                000404FE  00000001  7153 F312CKRES EQU   *
                                                   7154 *                               reset addresses
000404FE  E310 501C 0014                 0000001C  7155           LGF   R1,OPSRCA          R1:  OPSRC - Primary
00040504  B904 0061                                7156           LGR   R6,R1              R6:  OPSRC - Secondary
00040508  B904 0071                                7157           LGR   R7,R1              R7:  OPSRC - Home
                                                   7158
0004050C  B219 0200                      00000200  7159           SAC   ARMODE
00040510  D507 1000 3040      00000000   00000040  7160           CLC   0(8,R1),64(R3)
00040516  4770 5112                      00040532  7161           BNE   F312XX             OP1C Different, error
0004051A  D507 6008 3048      00000008   00000048  7162           CLC   8(8,R6),72(R3)
00040520  4770 5112                      00040532  7163           BNE   F312XX             OP2C Different, error
00040524  D507 7010 3050      00000010   00000050  7164           CLC   16(8,R7),80(R3)
0004052A  B219 0000                      00000000  7165           SAC   PRIMODE
0004052E  478B 0000                      00000000  7166           BE    0(R11)             OP3C same, return
                                                   7167
                                00040532  00000001  7168 F312XX   EQU   *
00040532  B219 0000                      00000000  7169           SAC   PRIMODE
00040536  45F0 8450                      00002450  7170           BAL   R15,FAILMSG
0004053A  07FB                                     7171           BR    R11
                                                   7172
0004053C  4700 5120                      00040540  7173 F312TSTCC BC    0,F312CCMSG        fail if unexpected condition code
                                00040540  00000001  7174 F312CCMSG EQU  *
00040540  45F0 82B2                      000022B2  7175           BAL   R15,CCMSG
00040544  47F0 50CA                      000404EA  7176           B     F312CKUR
                                                   7177
                                                   7178           TSTDATA
00040548                        00040548  00040A00  7179+          ORG   PLOETST+(128*K)+43*(3*K)+(K+512)
                                                   7180           TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
00040A00  D7D9C940 40406E6E                        7181+          DC    CL32'PRI   >>>OPDST43<<<' eye catcher
00040A08  6ED6D7C4 E2E3F4F3
00040A10  4C4C4C40 40404040
00040A18  40404040 40404040
00040A20                                           7182+OPDST43   DS    0C
00040A20  00000000 00000000                        7183           DC    XL256'0'
00040A28  00000000 00000000
00040A30  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00040A38  00000000 00000000
00040A40  00000000 00000000
00040A48  00000000 00000000
00040A50  00000000 00000000
00040A58  00000000 00000000
00040A60  00000000 00000000
00040A68  00000000 00000000
00040A70  00000000 00000000
00040A78  00000000 00000000
00040A80  00000000 00000000
00040A88  00000000 00000000
00040A90  00000000 00000000
00040A98  00000000 00000000
00040AA0  00000000 00000000
00040AA8  00000000 00000000
00040AB0  00000000 00000000
00040AB8  00000000 00000000
00040AC0  00000000 00000000
00040AC8  00000000 00000000
00040AD0  00000000 00000000
00040AD8  00000000 00000000
00040AE0  00000000 00000000
00040AE8  00000000 00000000
00040AF0  00000000 00000000
00040AF8  00000000 00000000
00040B00  00000000 00000000
00040B08  00000000 00000000
00040B10  00000000 00000000
00040B18  00000000 00000000
                                      7184
                                      7185           TSTLBL OPSRC              Source: OP1-OP3
00040B20  D7D9C940 40406E6E           7186+          DC     CL32'PRI   >>>OPSRC43<<<' eye catcher
00040B28  6ED6D7E2 D9C3F4F3
00040B30  4C4C4C40 40404040
00040B38  40404040 40404040
00040B40                              7187+OPSRC43   DS     0C
00040B40  00000000 00000000           7188           DC     XL256'0'
00040B48  00000000 00000000
00040B50  00000000 00000000
00040B58  00000000 00000000
00040B60  00000000 00000000
00040B68  00000000 00000000
00040B70  00000000 00000000
00040B78  00000000 00000000
00040B80  00000000 00000000
00040B88  00000000 00000000
00040B90  00000000 00000000
00040B98  00000000 00000000
00040BA0  00000000 00000000
00040BA8  00000000 00000000
00040BB0  00000000 00000000
00040BB8  00000000 00000000
00040BC0  00000000 00000000
00040BC8  00000000 00000000
00040BD0  00000000 00000000
00040BD8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00040BE0  00000000 00000000
00040BE8  00000000 00000000
00040BF0  00000000 00000000
00040BF8  00000000 00000000
00040C00  00000000 00000000
00040C08  00000000 00000000
00040C10  00000000 00000000
00040C18  00000000 00000000
00040C20  00000000 00000000
00040C28  00000000 00000000
00040C30  00000000 00000000
00040C38  00000000 00000000
                                        7189
                                        7190          TSTLBL PL                 Parameter List
00040C40  D7D9C940 40406E6E             7191+         DC    CL32'PRI   >>>PL43<<<'    eye catcher
00040C48  6ED7D3F4 F34C4C4C
00040C50  40404040 40404040
00040C58  40404040 40404040
00040C60                                7192+PL43     DS    0C
00040C60  00000000 00000000             7193          DC    XL352'00'
00040C68  00000000 00000000
00040C70  00000000 00000000
00040C78  00000000 00000000
00040C80  00000000 00000000
00040C88  00000000 00000000
00040C90  00000000 00000000
00040C98  00000000 00000000
00040CA0  00000000 00000000
00040CA8  00000000 00000000
00040CB0  00000000 00000000
00040CB8  00000000 00000000
00040CC0  00000000 00000000
00040CC8  00000000 00000000
00040CD0  00000000 00000000
00040CD8  00000000 00000000
00040CE0  00000000 00000000
00040CE8  00000000 00000000
00040CF0  00000000 00000000
00040CF8  00000000 00000000
00040D00  00000000 00000000
00040D08  00000000 00000000
00040D10  00000000 00000000
00040D18  00000000 00000000
00040D20  00000000 00000000
00040D28  00000000 00000000
00040D30  00000000 00000000
00040D38  00000000 00000000
00040D40  00000000 00000000
00040D48  00000000 00000000
00040D50  00000000 00000000
00040D58  00000000 00000000
00040D60  00000000 00000000
00040D68  00000000 00000000
00040D70  00000000 00000000
00040D78  00000000 00000000
00040D80  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00040D88  00000000 00000000
00040D90  00000000 00000000
00040D98  00000000 00000000
00040DA0  00000000 00000000
00040DA8  00000000 00000000
00040DB0  00000000 00000000
00040DB8  00000000 00000000
                                        7194
                                        7195          TSTLBL FMSG                fail message
00040DC0  D7D9C940 40406E6E             7196+         DC    CL32'PRI   >>>FMSG43<<<'  eye catcher
00040DC8  6EC6D4E2 C7F4F34C
00040DD0  4C4C4040 40404040
00040DD8  40404040 40404040
00040DE0                                7197+FMSG43   DS    0C
00040DE0  86818993 858440D7             7198          DC    C'failed PLO-Extension '
00040DE8  D3D660C5 A7A38595
00040DF0  A2899695 40
00040DF5  E3998997 938540C3             7199          DC    CL96'Triple Compare and Swap (TCSG: 31): 2nd NE'
00040DFD  96949781 99854081
00040E05  958440E2 A6819740
00040E0D  4DE3C3E2 C77A40F3
00040E15  F15D7A40 F2958440
00040E1D  D5C54040 40404040
00040E25  40404040 40404040
00040E2D  40404040 40404040
00040E35  40404040 40404040
00040E3D  40404040 40404040
00040E45  40404040 40404040
00040E4D  40404040 40404040
                                        7200
                                        7201          TSTEND
00040E55  D7D9C940 4040C5D5             7202+TEND43   DC    CL32'PRI   END of test: 43 <<<'
00040E5D  C4409686 40A385A2
00040E65  A37A40F4 F3404C4C
00040E6D  4C404040 40404040
                                        7203
                                        7204 * ------------------------------------------------------------------
                                        7205 * Test Triple Compare and Swap (TCSG: 31) instruction : 3rd NE
                                        7206 * ------------------------------------------------------------------
                                        7207          TSTHDR 2,3,31              3rd Operand NOT Equal (cc=2, UR=3)
00040E75                  00040E75  00041000  7208+         ORG   PLOETST+(128*K)+44*(3*K)
00041000  D7D9C940 4040E2E3             7209+         DC    CL32'PRI   START of test: 44'
00041008  C1D9E340 968640A3
00041010  85A2A37A 40F4F440
00041018  40404040 40404040
00041020                  00041020              7210+         USING *,R5                 base for test data and test routine
00041020                  00000000              7211+         USING PLOETEST,R5
00041020  00041050                              7212+T44      DC    A(CODE44)            address of test code
00041024  002C                                  7213+         DC    H'44'                test number
00041026  00                                    7214+         DC    X'00'
00041027  1F                                    7215+         DC    HL1'31'              Function
00041028  03                                    7216+         DC    HL1'3'               expected unequal results (UR)
00041029  00                                    7217+         DC    HL1'00'              found UR
0004102A  02                                    7218+         DC    HL1'2'               cc
0004102B  0D                                    7219+         DC    HL1'13'              not expected CC mask
0004102C  00                                    7220+         DS    X                    extracted cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00041030  00000000 00000000                        7221+          DS    2F                  extract PSW after test (has CC)
00041038  000419E0                                 7222+          DC    A(FMSG44)           address of test fail message
0004103C  00041740                                 7223+          DC    A(OPSRC44)          address of operand source
00041040  00041620                                 7224+          DC    A(OPDST44)          address of operand destination
00041044  00041860                                 7225+          DC    A(PL44)             address of parameter list
00041048  000041A4                                 7226+          DC    A(RANDX+44)         address of random source data
0004104C  00041A55                                 7227+          DC    A(TEND44)           address of test end eye catche
                                                   7228
                                                   7229           TSTCODE
00041050                                           7230+CODE44    DS    0F
                                                   7231 *                                   load random source
00041050  E310 501C 0014                0000001C   7232           LGF   R1,OPSRCA           R1: OPSRC - Primary
00041056  B904 0061                                7233           LGR   R6,R1               R6: OPSRC - Secondary
0004105A  B904 0071                                7234           LGR   R7,R1               R7: OPSRC - Home
                                                   7235
0004105E  E320 5028 0014                00000028   7236           LGF   R2,RANDXA           R2: address random data
00041064  E330 5024 0014                00000024   7237           LGF   R3,PLA              R3: address PL
                                                   7238
0004106A  B219 0200                     00000200   7239           SAC   ARMODE
0004106E  D207 1000 2000      00000000  00000000   7240           MVC   0(8,R1),0(R2)       OP1 - Primary
00041074  D207 6008 2008      00000008  00000008   7241           MVC   8(8,R6),8(R2)       OP2 - Secondary
0004107A  D207 7010 2010      00000010  00000010   7242           MVC   16(8,R7),16(R2)     OP3 - Home
                                                   7243 *
                                                   7244 * Triple Compare and Swap (TCSG: 31)
                                                   7245 * Build Parambter List
                                                   7246 *
00041080  D217 3040 2000      00000040  00000000   7247           MVC   64(24,R3),0(R2)     Load PL OP1-Op3 Compare
00041086  D217 3058 2080      00000058  00000080   7248           MVC   88(24,R3),128(R2)   Load PL Op1-Op3 Replacement
                                                   7249 ****
                                                   7250 * force a failure by changing the 3rd operand
                                                   7251 ****
0004108C  D203 7010 2100      00000010  00000100   7252           MVC   16(4,R7),256(R2)    some random data
                                                   7253 ****
00041092  E313 0000 0024                00000000   7254           STG   R1,0(R3)            load OP1 address in PL
00041098  E54C 3020 0000                00000020   7255           MVHI  32(R3),PRIALET
                                                   7256
0004109E  4111 0008                     00000008   7257           LA    R1,8(R1)
000410A2  E313 0008 0024                00000008   7258           STG   R1,8(R3)            load OP2 address in PL
000410A8  E54C 3024 0001                00000024   7259           MVHI  36(R3),SECALET
                                                   7260
000410AE  4111 0008                     00000008   7261           LA    R1,8(R1)
000410B2  E313 0010 0024                00000010   7262           STG   R1,16(R3)           load OP3 address in PL
000410B8  E54C 3028 0002                00000028   7263           MVHI  40(R3),HOMEALET
                                                   7264
000410BE  B982 0000                                7265           XGR   R0,R0
000410C2  4100 001F                     0000001F   7266           LA    R0,31
                                                   7267
000410C6  B219 0200                     00000200   7268           SAC   ARMODE
000410CA  EE0F 0000 3000      00000000  00000000   7269           PLO   0,0,R15,0(R3)       Triple C/S (TCSG: 31)
000410D0  B219 0000                     00000000   7270           SAC   PRIMODE
                                                   7271
000410D4  B98D 0020                                7272           EPSW  R2,R0               extract psw
000410D8  5020 5010                     00000010   7273           ST    R2,CCPSW              to save CC
000410DC  E310 500B 0076                0000000B   7274           LB    R1,CCMASK           (failure CC mask)
000410E2  8910 0004                     00000004   7275           SLL   R1,4                (shift to BC instr CC position)
000410E6  4410 511C                     0004113C   7276           EX    R1,F313TSTCC            fail if...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7277
                               000410EA  00000001  7278 F313CKUR EQU   *
000410EA  D200 5009 3030       00000009  00000030  7279          MVC   URFOUND,48(R3)      Extract UR
000410F0  D500 5008 3030       00000008  00000030  7280          CLC   UR,48(R3)
000410F6  4780 50DE                      000410FE  7281          BE    F313CKRES
000410FA  45F0 8388                      00002388  7282          BAL   R15,URMSG
                                                    7283 *
                                                    7284 * compare results
                                                    7285 *
                               000410FE  00000001  7286 F313CKRES EQU   *
                                                    7287 *                             reset addresses
000410FE  E310 501C 0014                 0000001C  7288          LGF   R1,OPSRCA          R1:  OPSRC - Primary
00041104  B904 0061                                 7289          LGR   R6,R1              R6:  OPSRC - Secondary
00041108  B904 0071                                 7290          LGR   R7,R1              R7:  OPSRC - Home
                                                    7291
0004110C  B219 0200                      00000200  7292          SAC   ARMODE
00041110  D507 1000 3040       00000000  00000040  7293          CLC   0(8,R1),64(R3)
00041116  4770 5112                      00041132  7294          BNE   F313XX             OP1C Different, error
0004111A  D507 6008 3048       00000008  00000048  7295          CLC   8(8,R6),72(R3)
00041120  4770 5112                      00041132  7296          BNE   F313XX             OP2C Different, error
00041124  D507 7010 3050       00000010  00000050  7297          CLC   16(8,R7),80(R3)
0004112A  B219 0000                      00000000  7298          SAC   PRIMODE
0004112E  478B 0000                      00000000  7299          BE    0(R11)             OP3C same, return
                                                    7300
                               00041132  00000001  7301 F313XX   EQU   *
00041132  B219 0000                      00000000  7302          SAC   PRIMODE
00041136  45F0 8450                      00002450  7303          BAL   R15,FAILMSG
0004113A  07FB                                      7304          BR    R11
                                                    7305
0004113C  4700 5120                      00041140  7306 F313TSTCC BC    0,F313CCMSG         fail if unexpected condition code
                               00041140  00000001  7307 F313CCMSG EQU   *
00041140  45F0 82B2                      000022B2  7308          BAL   R15,CCMSG
00041144  47F0 50CA                      000410EA  7309          B     F313CKUR
                                                    7310
                                                    7311          TSTDATA
00041148                       00041148  00041600  7312+         ORG   PLOETST+(128*K)+44*(3*K)+(K+512)
                                                    7313          TSTLBL OPDST             Dest/Data: Op1-R, Op2-R, Op3-R
00041600  D7D9C940 40406E6E                         7314+         DC    CL32'PRI   >>>OPDST44<<<' eye catcher
00041608  6ED6D7C4 E2E3F4F4
00041610  4C4C4C40 40404040
00041618  40404040 40404040
00041620                                            7315+OPDST44  DS    0C
00041620  00000000 00000000                         7316          DC    XL256'0'
00041628  00000000 00000000
00041630  00000000 00000000
00041638  00000000 00000000
00041640  00000000 00000000
00041648  00000000 00000000
00041650  00000000 00000000
00041658  00000000 00000000
00041660  00000000 00000000
00041668  00000000 00000000
00041670  00000000 00000000
00041678  00000000 00000000
00041680  00000000 00000000
00041688  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00041690  00000000 00000000
00041698  00000000 00000000
000416A0  00000000 00000000
000416A8  00000000 00000000
000416B0  00000000 00000000
000416B8  00000000 00000000
000416C0  00000000 00000000
000416C8  00000000 00000000
000416D0  00000000 00000000
000416D8  00000000 00000000
000416E0  00000000 00000000
000416E8  00000000 00000000
000416F0  00000000 00000000
000416F8  00000000 00000000
00041700  00000000 00000000
00041708  00000000 00000000
00041710  00000000 00000000
00041718  00000000 00000000
                                        7317
                                        7318          TSTLBL OPSRC              Source: OP1-OP3
00041720  D7D9C940 40406E6E             7319+         DC    CL32'PRI   >>>OPSRC44<<<' eye catcher
00041728  6ED6D7E2 D9C3F4F4
00041730  4C4C4C40 40404040
00041738  40404040 40404040
00041740                                7320+OPSRC44  DS    0C
00041740  00000000 00000000             7321          DC    XL256'0'
00041748  00000000 00000000
00041750  00000000 00000000
00041758  00000000 00000000
00041760  00000000 00000000
00041768  00000000 00000000
00041770  00000000 00000000
00041778  00000000 00000000
00041780  00000000 00000000
00041788  00000000 00000000
00041790  00000000 00000000
00041798  00000000 00000000
000417A0  00000000 00000000
000417A8  00000000 00000000
000417B0  00000000 00000000
000417B8  00000000 00000000
000417C0  00000000 00000000
000417C8  00000000 00000000
000417D0  00000000 00000000
000417D8  00000000 00000000
000417E0  00000000 00000000
000417E8  00000000 00000000
000417F0  00000000 00000000
000417F8  00000000 00000000
00041800  00000000 00000000
00041808  00000000 00000000
00041810  00000000 00000000
00041818  00000000 00000000
00041820  00000000 00000000
00041828  00000000 00000000
00041830  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00041838  00000000 00000000                          7322
                                                     7323          TSTLBL PL                 Parameter List
00041840  D7D9C940 40406E6E                          7324+         DC    CL32'PRI   >>>PL44<<<'    eye catcher
00041848  6ED7D3F4 F44C4C4C
00041850  40404040 40404040
00041858  40404040 40404040
00041860                                             7325+PL44     DS    0C
00041860  00000000 00000000                          7326          DC    XL352'00'
00041868  00000000 00000000
00041870  00000000 00000000
00041878  00000000 00000000
00041880  00000000 00000000
00041888  00000000 00000000
00041890  00000000 00000000
00041898  00000000 00000000
000418A0  00000000 00000000
000418A8  00000000 00000000
000418B0  00000000 00000000
000418B8  00000000 00000000
000418C0  00000000 00000000
000418C8  00000000 00000000
000418D0  00000000 00000000
000418D8  00000000 00000000
000418E0  00000000 00000000
000418E8  00000000 00000000
000418F0  00000000 00000000
000418F8  00000000 00000000
00041900  00000000 00000000
00041908  00000000 00000000
00041910  00000000 00000000
00041918  00000000 00000000
00041920  00000000 00000000
00041928  00000000 00000000
00041930  00000000 00000000
00041938  00000000 00000000
00041940  00000000 00000000
00041948  00000000 00000000
00041950  00000000 00000000
00041958  00000000 00000000
00041960  00000000 00000000
00041968  00000000 00000000
00041970  00000000 00000000
00041978  00000000 00000000
00041980  00000000 00000000
00041988  00000000 00000000
00041990  00000000 00000000
00041998  00000000 00000000
000419A0  00000000 00000000
000419A8  00000000 00000000
000419B0  00000000 00000000
000419B8  00000000 00000000
                                                     7327
                                                     7328          TSTLBL FMSG               fail message
000419C0  D7D9C940 40406E6E                          7329+         DC    CL32'PRI   >>>FMSG44<<<'  eye catcher
000419C8  6EC6D4E2 C7F4F44C
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000419D0  4C4C4040 40404040
000419D8  40404040 40404040
000419E0                                         7330+FMSG44   DS    0C
000419E0  86818993 858440D7                      7331          DC    C'failed PLO-Extension '
000419E8  D3D660C5 A7A38595
000419F0  A2899695 40
000419F5  E3998997 938540C3                      7332          DC    CL96'Triple Compare and Swap (TCSG: 31): 3rd NE'
000419FD  96949781 99854081
00041A05  958440E2 A6819740
00041A0D  4DE3C3E2 C77A40F3
00041A15  F15D7A40 F3998440
00041A1D  D5C54040 40404040
00041A25  40404040 40404040
00041A2D  40404040 40404040
00041A35  40404040 40404040
00041A3D  40404040 40404040
00041A45  40404040 40404040
00041A4D  40404040 40404040
                                                 7333
                                                 7334          TSTEND
00041A55  D7D9C940 4040C5D5                      7335+TEND44   DC    CL32'PRI   END of test: 44 <<<'
00041A5D  C4409686 40A385A2
00041A65  A37A40F4 F4404C4C
00041A6D  4C404040 40404040
                                                 7336
                                                 7337 * ------------------------------------------------------------------
                                                 7338 * Test Triple Compare and Swap (TCSG: 31) instruction : All EQ
                                                 7339 * ------------------------------------------------------------------
                                                 7340          TSTHDR 0,99,31              All Operands Equal (cc=0, UR=99)
00041A75                          00041A75  00041C00 7341+         ORG   PLOETST+(128*K)+45*(3*K)
00041C00  D7D9C940 4040E2E3                      7342+         DC    CL32'PRI   START of test: 45'
00041C08  C1D9E340 968640A3
00041C10  85A2A37A 40F4F540
00041C18  40404040 40404040
00041C20                          00041C20       7343+         USING *,R5                  base for test data and test routine
00041C20                          00000000       7344+         USING PLOETEST,R5
00041C20  00041C50                               7345+T45      DC    A(CODE45)             address of test code
00041C24  002D                                   7346+         DC    H'45'                 test number
00041C26  00                                     7347+         DC    X'00'
00041C27  1F                                     7348+         DC    HL1'31'               Function
00041C28  63                                     7349+         DC    HL1'99'               expected unequal results (UR)
00041C29  00                                     7350+         DC    HL1'00'               found UR
00041C2A  00                                     7351+         DC    HL1'0'                cc
00041C2B  07                                     7352+         DC    HL1'7'                not expected CC mask
00041C2C  00                                     7353+         DS    X                     extracted cc
00041C30  00000000 00000000                      7354+         DS    2F                    extract PSW after test (has CC)
00041C38  000425E0                               7355+         DC    A(FMSG45)             address of test fail message
00041C3C  00042340                               7356+         DC    A(OPSRC45)            address of operand source
00041C40  00042220                               7357+         DC    A(OPDST45)            address of operand destination
00041C44  00042460                               7358+         DC    A(PL45)               address of parameter list
00041C48  000041A5                               7359+         DC    A(RANDX+45)           address of random source data
00041C4C  00042655                               7360+         DC    A(TEND45)             address of test end eye catche
                                                 7361
                                                 7362          TSTCODE
00041C50                                         7363+CODE45   DS    0F
                                                 7364 *                                    load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00041C50  E310 501C 0014                 0000001C  7365          LGF   R1,OPSRCA           R1: OPSRC - Primary
00041C56  B904 0061                      00000000  7366          LGR   R6,R1               R6: OPSRC - Secondary
00041C5A  B904 0071                      00000000  7367          LGR   R7,R1               R7: OPSRC - Home
                                                   7368
00041C5E  E320 5028 0014                 00000028  7369          LGF   R2,RANDXA           R2: address random data
00041C64  E330 5024 0014                 00000024  7370          LGF   R3,PLA              R3: address PL
                                                   7371
00041C6A  B219 0200                      00000200  7372          SAC   ARMODE
00041C6E  D207 1000 2000       00000000  00000000  7373          MVC   0(8,R1),0(R2)       OP1 - Primary
00041C74  D207 6008 2008       00000008  00000008  7374          MVC   8(8,R6),8(R2)       OP2 - Secondary
00041C7A  D207 7010 2010       00000010  00000010  7375          MVC   16(8,R7),16(R2)     OP3 - Home
                                                   7376 *
                                                   7377 * Triple Compare and Swap (TCSG: 31)
                                                   7378 * Build Parambter List
                                                   7379 *
00041C80  D217 3040 2000       00000040  00000000  7380          MVC   64(24,R3),0(R2)     Load PL OP1-Op3 Compare
00041C86  D217 3058 2080       00000058  00000080  7381          MVC   88(24,R3),128(R2)   Load PL Op1-Op3 Replacement
                                                   7382 ****
                                                   7383 * do not force a failure by changing the 3rd operand
                                                   7384 ****
                                                   7385 *        MVC   0(4,R1),256(R2)     some random data
                                                   7386 ****
00041C8C  E313 0000 0024                 00000000  7387          STG   R1,0(R3)            load OP1 address in PL
00041C92  E54C 3020 0000                 00000020  7388          MVHI  32(R3),PRIALET
                                                   7389
00041C98  4111 0008                      00000008  7390          LA    R1,8(R1)
00041C9C  E313 0008 0024                 00000008  7391          STG   R1,8(R3)            load OP2 address in PL
00041CA2  E54C 3024 0001                 00000024  7392          MVHI  36(R3),SECALET
                                                   7393
00041CA8  4111 0008                      00000008  7394          LA    R1,8(R1)
00041CAC  E313 0010 0024                 00000010  7395          STG   R1,16(R3)           load OP3 address in PL
00041CB2  E54C 3028 0002                 00000028  7396          MVHI  40(R3),HOMEALET
                                                   7397
00041CB8  B982 0000                      00000000  7398          XGR   R0,R0
00041CBC  4100 001F                      0000001F  7399          LA    R0,31
                                                   7400
00041CC0  B219 0200                      00000200  7401          SAC   ARMODE
00041CC4  EE0F 0000 3000       00000000  00000000  7402          PLO   0,0,R15,0(R3)       Triple C/S (TCSG: 31)
00041CCA  B219 0000                      00000000  7403          SAC   PRIMODE
                                                   7404
00041CCE  B98D 0020                      00000000  7405          EPSW  R2,R0               extract psw
00041CD2  5020 5010                      00000010  7406          ST    R2,CCPSW              to save CC
00041CD6  E310 500B 0076                 0000000B  7407          LB    R1,CCMASK           (failure CC mask)
00041CDC  8910 0004                      00000004  7408          SLL   R1,4                (shift to BC instr CC position)
00041CE0  4410 5116                      00041D36  7409          EX    R1,F314TSTCC            fail if...
                                                   7410
                               00041CE4  00000001  7411 F314CKUR EQU   *
00041CE4  D200 5009 3030       00000009  00000030  7412          MVC   URFOUND,48(R3)      Extract UR
00041CEA  D500 5008 3030       00000008  00000030  7413          CLC   UR,48(R3)
00041CF0  4780 50D8                      00041CF8  7414          BE    F314CKRES
00041CF4  45F0 8388                      00002388  7415          BAL   R15,URMSG
                                                   7416 *
                                                   7417 * compare results
                                                   7418 *
                               00041CF8  00000001  7419 F314CKRES EQU   *
                                                   7420 *                                reset addresses
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00041CF8  E310 501C 0014                 0000001C  7421          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00041CFE  B904 0061                              7422          LGR   R6,R1               R6:  OPSRC - Secondary
00041D02  B904 0071                              7423          LGR   R7,R1               R7:  OPSRC - Home
                                                 7424
                                                 7425 * All equal; replacement values copied to operand source
                                                 7426
00041D06  B219 0200                      00000200  7427          SAC   ARMODE
00041D0A  D507 1000 3058       00000000  00000058  7428          CLC   0(8,R1),88(R3)
00041D10  4770 510C                      00041D2C  7429          BNE   F314XX              OP1R Different, error
00041D14  D507 6008 3060       00000008  00000060  7430          CLC   8(8,R6),96(R3)
00041D1A  4770 510C                      00041D2C  7431          BNE   F314XX              OP2R Different, error
00041D1E  D507 7010 3068       00000010  00000068  7432          CLC   16(8,R7),104(R3)
00041D24  B219 0000                      00000000  7433          SAC   PRIMODE
00041D28  478B 0000                      00000000  7434          BE    0(R11)
                                                 7435
                               00041D2C  00000001  7436 F314XX   EQU   *
00041D2C  B219 0000                      00000000  7437          SAC   PRIMODE
00041D30  45F0 8450                      00002450  7438          BAL   R15,FAILMSG
00041D34  07FB                                   7439          BR    R11
                                                 7440
00041D36  4700 511A                      00041D3A  7441 F314TSTCC BC    0,F314CCMSG         fail if unexpected condition code
                               00041D3A  00000001  7442 F314CCMSG EQU   *
00041D3A  45F0 82B2                      000022B2  7443          BAL   R15,CCMSG
00041D3E  47F0 50C4                      00041CE4  7444          B     F314CKUR
                                                 7445
                                                 7446          TSTDATA
00041D42                       00041D42  00042200  7447+         ORG   PLOETST+(128*K)+45*(3*K)+(K+512)
                                                 7448          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00042200  D7D9C940 40406E6E                      7449+         DC    CL32'PRI   >>>OPDST45<<<' eye catcher
00042208  6ED6D7C4 E2E3F4F5
00042210  4C4C4C40 40404040
00042218  40404040 40404040
00042220                                         7450+OPDST45  DS    0C
00042220  00000000 00000000                      7451          DC    XL256'0'
00042228  00000000 00000000
00042230  00000000 00000000
00042238  00000000 00000000
00042240  00000000 00000000
00042248  00000000 00000000
00042250  00000000 00000000
00042258  00000000 00000000
00042260  00000000 00000000
00042268  00000000 00000000
00042270  00000000 00000000
00042278  00000000 00000000
00042280  00000000 00000000
00042288  00000000 00000000
00042290  00000000 00000000
00042298  00000000 00000000
000422A0  00000000 00000000
000422A8  00000000 00000000
000422B0  00000000 00000000
000422B8  00000000 00000000
000422C0  00000000 00000000
000422C8  00000000 00000000
000422D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
000422D8  00000000 00000000
000422E0  00000000 00000000
000422E8  00000000 00000000
000422F0  00000000 00000000
000422F8  00000000 00000000
00042300  00000000 00000000
00042308  00000000 00000000
00042310  00000000 00000000
00042318  00000000 00000000
                                        7452
                                        7453          TSTLBL OPSRC              Source: OP1-OP3
00042320  D7D9C940 40406E6E             7454+         DC    CL32'PRI   >>>OPSRC45<<<' eye catcher
00042328  6ED6D7E2 D9C3F4F5
00042330  4C4C4C40 40404040
00042338  40404040 40404040
00042340                                7455+OPSRC45  DS    0C
00042340  00000000 00000000             7456          DC    XL256'0'
00042348  00000000 00000000
00042350  00000000 00000000
00042358  00000000 00000000
00042360  00000000 00000000
00042368  00000000 00000000
00042370  00000000 00000000
00042378  00000000 00000000
00042380  00000000 00000000
00042388  00000000 00000000
00042390  00000000 00000000
00042398  00000000 00000000
000423A0  00000000 00000000
000423A8  00000000 00000000
000423B0  00000000 00000000
000423B8  00000000 00000000
000423C0  00000000 00000000
000423C8  00000000 00000000
000423D0  00000000 00000000
000423D8  00000000 00000000
000423E0  00000000 00000000
000423E8  00000000 00000000
000423F0  00000000 00000000
000423F8  00000000 00000000
00042400  00000000 00000000
00042408  00000000 00000000
00042410  00000000 00000000
00042418  00000000 00000000
00042420  00000000 00000000
00042428  00000000 00000000
00042430  00000000 00000000
00042438  00000000 00000000
                                     7457
                                     7458          TSTLBL PL                 Parameter List
00042440  D7D9C940 40406E6E          7459+         DC    CL32'PRI   >>>PL45<<<'    eye catcher
00042448  6ED7D3F4 F54C4C4C
00042450  40404040 40404040
00042458  40404040 40404040
00042460                             7460+PL45     DS    0C
00042460  00000000 00000000          7461          DC    XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00042468  00000000 00000000
00042470  00000000 00000000
00042478  00000000 00000000
00042480  00000000 00000000
00042488  00000000 00000000
00042490  00000000 00000000
00042498  00000000 00000000
000424A0  00000000 00000000
000424A8  00000000 00000000
000424B0  00000000 00000000
000424B8  00000000 00000000
000424C0  00000000 00000000
000424C8  00000000 00000000
000424D0  00000000 00000000
000424D8  00000000 00000000
000424E0  00000000 00000000
000424E8  00000000 00000000
000424F0  00000000 00000000
000424F8  00000000 00000000
00042500  00000000 00000000
00042508  00000000 00000000
00042510  00000000 00000000
00042518  00000000 00000000
00042520  00000000 00000000
00042528  00000000 00000000
00042530  00000000 00000000
00042538  00000000 00000000
00042540  00000000 00000000
00042548  00000000 00000000
00042550  00000000 00000000
00042558  00000000 00000000
00042560  00000000 00000000
00042568  00000000 00000000
00042570  00000000 00000000
00042578  00000000 00000000
00042580  00000000 00000000
00042588  00000000 00000000
00042590  00000000 00000000
00042598  00000000 00000000
000425A0  00000000 00000000
000425A8  00000000 00000000
000425B0  00000000 00000000
000425B8  00000000 00000000
                                      7462
                                      7463          TSTLBL FMSG                fail message
000425C0  D7D9C940 40406E6E           7464+         DC    CL32'PRI   >>>FMSG45<<<'  eye catcher
000425C8  6EC6D4E2 C7F4F54C
000425D0  4C4C4040 40404040
000425D8  40404040 40404040
000425E0                              7465+FMSG45   DS    0C
000425E0  86818993 858440D7           7466          DC    C'failed PLO-Extension '
000425E8  D3D660C5 A7A38595
000425F0  A2899695 40
000425F5  E3998997 938540C3           7467          DC    CL96'Triple Compare and Swap (TCSG: 31): All EQ'
000425FD  96949781 99854081
00042605  958440E2 A6819740
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004260D  4DE3C3E2 C77A40F3
00042615  F15D7A40 C1939340
0004261D  C5D84040 40404040
00042625  40404040 40404040
0004262D  40404040 40404040
00042635  40404040 40404040
0004263D  40404040 40404040
00042645  40404040 40404040
0004264D  40404040 40404040
                                          7468
                                          7469           TSTEND
00042655  D7D9C940 4040C5D5               7470+TEND45    DC    CL32'PRI   END of test: 45 <<<'
0004265D  C4409686 40A385A2
00042665  A37A40F4 F5404C4C
0004266D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   7472 *==============================================================================
                                                   7473
                                                   7474 * ------------------------------------------------------------------------
                                                   7475 * Test Triple Compare and Swap (TCS: 30) instruction : 1st NE
                                                   7476 * ------------------------------------------------------------------------
                                                   7477          TSTHDR 2,1,30             1st Operand NOT Equal (cc=2, UR=1)
00042675                          00042675  00042800 7478+         ORG   PLOETST+(128*K)+46*(3*K)
00042800  D7D9C940 4040E2E3                        7479+         DC    CL32'PRI   START of test: 46'
00042808  C1D9E340 968640A3
00042810  85A2A37A 40F4F640
00042818  40404040 40404040
00042820                          00042820         7480+         USING *,R5                base for test data and test routine
00042820                          00000000         7481+         USING PLOETEST,R5
00042820  00042850                                 7482+T46      DC    A(CODE46)           address of test code
00042824  002E                                     7483+         DC    H'46'               test number
00042826  00                                       7484+         DC    X'00'
00042827  1E                                       7485+         DC    HL1'30'             Function
00042828  01                                       7486+         DC    HL1'1'              expected unequal results (UR)
00042829  00                                       7487+         DC    HL1'00'             found UR
0004282A  02                                       7488+         DC    HL1'2'              cc
0004282B  0D                                       7489+         DC    HL1'13'             not expected CC mask
0004282C  00                                       7490+         DS    X                   extracted cc
00042830  00000000 00000000                        7491+         DS    2F                  extract PSW after test (has CC)
00042838  000431E0                                 7492+         DC    A(FMSG46)           address of test fail message
0004283C  00042F40                                 7493+         DC    A(OPSRC46)          address of operand source
00042840  00042E20                                 7494+         DC    A(OPDST46)          address of operand destination
00042844  00043060                                 7495+         DC    A(PL46)             address of parameter list
00042848  000041A6                                 7496+         DC    A(RANDX+46)         address of random source data
0004284C  00043255                                 7497+         DC    A(TEND46)           address of test end eye catche
                                                   7498
                                                   7499          TSTCODE
00042850                                           7500+CODE46   DS    0F
                                                   7501 *                                  load random source
00042850  E310 501C 0014                0000001C   7502          LGF   R1,OPSRCA           R1: OPSRC - Primary
00042856  B904 0061                                7503          LGR   R6,R1               R6: OPSRC - Secondary
0004285A  B904 0071                                7504          LGR   R7,R1               R7: OPSRC - Home
                                                   7505
0004285E  E320 5028 0014                00000028   7506          LGF   R2,RANDXA           R2: address random data
00042864  E330 5024 0014                00000024   7507          LGF   R3,PLA              R3: address PL
                                                   7508
0004286A  B219 0200                     00000200   7509          SAC   ARMODE
0004286E  D203 1000 2000        00000000 00000000  7510          MVC   0(4,R1),0(R2)       OP1 - Primary
00042874  D203 6004 2004        00000004 00000004  7511          MVC   4(4,R6),4(R2)       OP2 - Secondary
0004287A  D203 7008 2008        00000008 00000008  7512          MVC   8(4,R7),8(R2)       OP3 - Home
                                                   7513 *
                                                   7514 * Triple Compare and Swap (TCS: 30)
                                                   7515 * Build Parambter List
                                                   7516 *
00042880  D203 3040 2000        00000040 00000000  7517          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
00042886  D203 3048 2004        00000048 00000004  7518          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
0004288C  D203 3050 2008        00000050 00000008  7519          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
                                                   7520
00042892  D203 3058 2080        00000058 00000080  7521          MVC   88(4,R3),128+0(R2)  Load PL OP1-Replacement
00042898  D203 3060 2084        00000060 00000084  7522          MVC   96(4,R3),128+4(R2)  Load PL OP2-Replacement
0004289E  D203 3068 2088        00000068 00000088  7523          MVC   104(4,R3),128+8(R2) Load PL OP3-Replacement
                                                   7524
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          7525 ****
                                          7526 * force a failure by changing the 1st operand
                                          7527 ****
000428A4  D201 1000 2100      00000000  00000100  7528          MVC   0(2,R1),256(R2)     some random data
                                          7529 ****
000428AA  E313 0000 0024                00000000  7530          STG   R1,0(R3)           load OP1 address in PL
000428B0  E54C 3020 0000                00000020  7531          MVHI  32(R3),PRIALET
                                          7532
000428B6  4111 0004                     00000004  7533          LA    R1,4(R1)
000428BA  E313 0008 0024                00000008  7534          STG   R1,8(R3)           load OP2 address in PL
000428C0  E54C 3024 0001                00000024  7535          MVHI  36(R3),SECALET
                                          7536
000428C6  4111 0004                     00000004  7537          LA    R1,4(R1)
000428CA  E313 0010 0024                00000010  7538          STG   R1,16(R3)          load OP3 address in PL
000428D0  E54C 3028 0002                00000028  7539          MVHI  40(R3),HOMEALET
                                          7540
000428D6  B982 0000                               7541          XGR   R0,R0
000428DA  4100 001E                     0000001E  7542          LA    R0,30
                                          7543
000428DE  B219 0200                     00000200  7544          SAC   ARMODE
000428E2  EE0F 0000 3000      00000000  00000000  7545          PLO   0,0,R15,0(R3)      Triple C/S (TCS: 30)
000428E8  B219 0000                     00000000  7546          SAC   PRIMODE
                                          7547
000428EC  B98D 0020                               7548          EPSW  R2,R0              extract psw
000428F0  5020 5010                     00000010  7549          ST    R2,CCPSW             to save CC
000428F4  E310 500B 0076                0000000B  7550          LB    R1,CCMASK          (failure CC mask)
000428FA  8910 0004                     00000004  7551          SLL   R1,4               (shift to BC instr CC position)
000428FE  4410 5134                     00042954  7552          EX    R1,F301TSTCC           fail if...
                                          7553
                                00042902  00000001  7554 F301CKUR EQU   *
00042902  D200 5009 3030      00000009  00000030  7555          MVC   URFOUND,48(R3)     Extract UR
00042908  D500 5008 3030      00000008  00000030  7556          CLC   UR,48(R3)
0004290E  4780 50F6                     00042916  7557          BE    F301CKRES
00042912  45F0 8388                     00002388  7558          BAL   R15,URMSG
                                          7559 *
                                          7560 * compare results
                                          7561 *
                                00042916  00000001  7562 F301CKRES EQU   *
                                          7563 *                           reset addresses
00042916  E310 501C 0014                0000001C  7564          LGF   R1,OPSRCA          R1:  OPSRC - Primary
0004291C  B904 0061                               7565          LGR   R6,R1              R6:  OPSRC - Secondary
00042920  B904 0071                               7566          LGR   R7,R1              R7:  OPSRC - Home
                                          7567
00042924  B219 0200                     00000200  7568          SAC   ARMODE
00042928  D503 1000 3040      00000000  00000040  7569          CLC   0(4,R1),64(R3)
0004292E  4770 512A                     0004294A  7570          BNE   F301XX             OP1C Different, error
00042932  D503 6004 3048      00000004  00000048  7571          CLC   4(4,R6),72(R3)
00042938  4770 512A                     0004294A  7572          BNE   F301XX             OP2C Different, error
0004293C  D503 7008 3050      00000008  00000050  7573          CLC   8(4,R7),80(R3)
00042942  B219 0000                     00000000  7574          SAC   PRIMODE
00042946  478B 0000                     00000000  7575          BE    0(R11)             OP3C same, return
                                          7576
                                0004294A  00000001  7577 F301XX   EQU   *
0004294A  B219 0000                     00000000  7578          SAC   PRIMODE
0004294E  45F0 8450                     00002450  7579          BAL   R15,FAILMSG
00042952  07FB                                    7580          BR    R11
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
                                                     7581
00042954  4700 5138                         00042958  7582 F301TSTCC BC   0,F301CCMSG         fail if unexpected condition code
                                00042958  00000001  7583 F301CCMSG EQU  *
00042958  45F0 82B2                         000022B2  7584           BAL  R15,CCMSG
0004295C  47F0 50E2                         00042902  7585           B    F301CKUR
                                                     7586
                                                     7587           TSTDATA
00042960                        00042960  00042E00  7588+          ORG  PLOETST+(128*K)+46*(3*K)+(K+512)
                                                     7589           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00042E00  D7D9C940 40406E6E                          7590+          DC   CL32'PRI   >>>OPDST46<<<' eye catcher
00042E08  6ED6D7C4 E2E3F4F6
00042E10  4C4C4C40 40404040
00042E18  40404040 40404040
00042E20                                             7591+OPDST46   DS   0C
00042E20  00000000 00000000                          7592           DC   XL256'0'
00042E28  00000000 00000000
00042E30  00000000 00000000
00042E38  00000000 00000000
00042E40  00000000 00000000
00042E48  00000000 00000000
00042E50  00000000 00000000
00042E58  00000000 00000000
00042E60  00000000 00000000
00042E68  00000000 00000000
00042E70  00000000 00000000
00042E78  00000000 00000000
00042E80  00000000 00000000
00042E88  00000000 00000000
00042E90  00000000 00000000
00042E98  00000000 00000000
00042EA0  00000000 00000000
00042EA8  00000000 00000000
00042EB0  00000000 00000000
00042EB8  00000000 00000000
00042EC0  00000000 00000000
00042EC8  00000000 00000000
00042ED0  00000000 00000000
00042ED8  00000000 00000000
00042EE0  00000000 00000000
00042EE8  00000000 00000000
00042EF0  00000000 00000000
00042EF8  00000000 00000000
00042F00  00000000 00000000
00042F08  00000000 00000000
00042F10  00000000 00000000
00042F18  00000000 00000000
                                                     7593
                                                     7594           TSTLBL OPSRC              Source: OP1-OP3
00042F20  D7D9C940 40406E6E                          7595+          DC   CL32'PRI   >>>OPSRC46<<<' eye catcher
00042F28  6ED6D7E2 D9C3F4F6
00042F30  4C4C4C40 40404040
00042F38  40404040 40404040
00042F40                                             7596+OPSRC46   DS   0C
00042F40  00000000 00000000                          7597           DC   XL256'0'
00042F48  00000000 00000000
00042F50  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00042F58  00000000 00000000
00042F60  00000000 00000000
00042F68  00000000 00000000
00042F70  00000000 00000000
00042F78  00000000 00000000
00042F80  00000000 00000000
00042F88  00000000 00000000
00042F90  00000000 00000000
00042F98  00000000 00000000
00042FA0  00000000 00000000
00042FA8  00000000 00000000
00042FB0  00000000 00000000
00042FB8  00000000 00000000
00042FC0  00000000 00000000
00042FC8  00000000 00000000
00042FD0  00000000 00000000
00042FD8  00000000 00000000
00042FE0  00000000 00000000
00042FE8  00000000 00000000
00042FF0  00000000 00000000
00042FF8  00000000 00000000
00043000  00000000 00000000
00043008  00000000 00000000
00043010  00000000 00000000
00043018  00000000 00000000
00043020  00000000 00000000
00043028  00000000 00000000
00043030  00000000 00000000
00043038  00000000 00000000
                                      7598
                                      7599          TSTLBL PL                 Parameter List
00043040  D7D9C940 40406E6E           7600+         DC    CL32'PRI   >>>PL46<<<'    eye catcher
00043048  6ED7D3F4 F64C4C4C
00043050  40404040 40404040
00043058  40404040 40404040
00043060                              7601+PL46     DS    0C
00043060  00000000 00000000           7602          DC    XL352'00'
00043068  00000000 00000000
00043070  00000000 00000000
00043078  00000000 00000000
00043080  00000000 00000000
00043088  00000000 00000000
00043090  00000000 00000000
00043098  00000000 00000000
000430A0  00000000 00000000
000430A8  00000000 00000000
000430B0  00000000 00000000
000430B8  00000000 00000000
000430C0  00000000 00000000
000430C8  00000000 00000000
000430D0  00000000 00000000
000430D8  00000000 00000000
000430E0  00000000 00000000
000430E8  00000000 00000000
000430F0  00000000 00000000
000430F8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00043100  00000000 00000000
00043108  00000000 00000000
00043110  00000000 00000000
00043118  00000000 00000000
00043120  00000000 00000000
00043128  00000000 00000000
00043130  00000000 00000000
00043138  00000000 00000000
00043140  00000000 00000000
00043148  00000000 00000000
00043150  00000000 00000000
00043158  00000000 00000000
00043160  00000000 00000000
00043168  00000000 00000000
00043170  00000000 00000000
00043178  00000000 00000000
00043180  00000000 00000000
00043188  00000000 00000000
00043190  00000000 00000000
00043198  00000000 00000000
000431A0  00000000 00000000
000431A8  00000000 00000000
000431B0  00000000 00000000
000431B8  00000000 00000000
                                        7603
                                        7604          TSTLBL FMSG               fail message
000431C0  D7D9C940 40406E6E             7605+         DC    CL32'PRI   >>>FMSG46<<<'  eye catcher
000431C8  6EC6D4E2 C7F4F64C
000431D0  4C4C4040 40404040
000431D8  40404040 40404040
000431E0                                7606+FMSG46   DS    0C
000431E0  86818993 858440D7             7607          DC    C'failed PLO-Extension '
000431E8  D3D660C5 A7A38595
000431F0  A2899695 40
000431F5  E3998997 938540C3             7608          DC    CL96'Triple Compare and Swap (TCS: 30): 1st NE'
000431FD  96949781 99854081
00043205  958440E2 A6819740
0004320D  4DE3C3E2 7A40F3F0
00043215  5D7A40F1 A2A340D5
0004321D  C5404040 40404040
00043225  40404040 40404040
0004322D  40404040 40404040
00043235  40404040 40404040
0004323D  40404040 40404040
00043245  40404040 40404040
0004324D  40404040 40404040
                                        7609          TSTEND
00043255  D7D9C940 4040C5D5             7610+TEND46   DC    CL32'PRI   END of test: 46 <<<'
0004325D  C4409686 40A385A2
00043265  A37A40F4 F6404C4C
0004326D  4C404040 40404040
                                        7611
                                        7612 * ------------------------------------------------------------------
                                        7613 * Test Triple Compare and Swap (TCS: 30) instruction : 2nd NE
                                        7614 * ------------------------------------------------------------------
                                        7615          TSTHDR 2,2,30             2nd Operand NOT Equal (cc=2, UR=2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00043275                     00043275  00043400  7616+          ORG   PLOETST+(128*K)+47*(3*K)
00043400  D7D9C940 4040E2E3                       7617+          DC    CL32'PRI   START of test: 47'
00043408  C1D9E340 968640A3
00043410  85A2A37A 40F4F740
00043418  40404040 40404040
00043420                     00043420             7618+          USING *,R5                 base for test data and test routine
00043420                     00000000             7619+          USING PLOETEST,R5
00043420  00043450                                7620+T47       DC    A(CODE47)            address of test code
00043424  002F                                    7621+          DC    H'47'                test number
00043426  00                                      7622+          DC    X'00'
00043427  1E                                      7623+          DC    HL1'30'              Function
00043428  02                                      7624+          DC    HL1'2'               expected unequal results (UR)
00043429  00                                      7625+          DC    HL1'00'              found UR
0004342A  02                                      7626+          DC    HL1'2'               cc
0004342B  0D                                      7627+          DC    HL1'13'              not expected CC mask
0004342C  00                                      7628+          DS    X                    extracted cc
00043430  00000000 00000000                       7629+          DS    2F                   extract PSW after test (has CC)
00043438  00043DE0                                7630+          DC    A(FMSG47)            address of test fail message
0004343C  00043B40                                7631+          DC    A(OPSRC47)           address of operand source
00043440  00043A20                                7632+          DC    A(OPDST47)           address of operand destination
00043444  00043C60                                7633+          DC    A(PL47)              address of parameter list
00043448  000041A7                                7634+          DC    A(RANDX+47)          address of random source data
0004344C  00043E55                                7635+          DC    A(TEND47)            address of test end eye catche
                                                  7636
                                                  7637           TSTCODE
00043450                                          7638+CODE47    DS    0F                   load random source
00043450  E310 501C 0014                0000001C  7640           LGF   R1,OPSRCA            R1: OPSRC - Primary
00043456  B904 0061                               7641           LGR   R6,R1                R6: OPSRC - Secondary
0004345A  B904 0071                               7642           LGR   R7,R1                R7: OPSRC - Home
                                                  7643
0004345E  E320 5028 0014                00000028  7644           LGF   R2,RANDXA            R2: address random data
00043464  E330 5024 0014                00000024  7645           LGF   R3,PLA               R3: address PL
                                                  7646
0004346A  B219 0200                     00000200  7647           SAC   ARMODE
0004346E  D203 1000 2000      00000000  00000000  7648           MVC   0(4,R1),0(R2)        OP1 - Primary
00043474  D203 6004 2004      00000004  00000004  7649           MVC   4(4,R6),4(R2)        OP2 - Secondary
0004347A  D203 7008 2008      00000008  00000008  7650           MVC   8(4,R7),8(R2)        OP3 - Home
                                                  7651 *
                                                  7652 * Triple Compare and Swap (TCS: 30)
                                                  7653 * Build Parambter List
                                                  7654 *
00043480  D203 3040 2000      00000040  00000000  7655           MVC   64(4,R3),0(R2)       Load PL OP1-Compare
00043486  D203 3048 2004      00000048  00000004  7656           MVC   72(4,R3),4(R2)       Load PL OP2-Compare
0004348C  D203 3050 2008      00000050  00000008  7657           MVC   80(4,R3),8(R2)       Load PL OP3-Compare
                                                  7658
00043492  D203 3058 2080      00000058  00000080  7659           MVC   88(4,R3),128+0(R2)   Load PL OP1-Replacement
00043498  D203 3060 2084      00000060  00000084  7660           MVC   96(4,R3),128+4(R2)   Load PL OP2-Replacement
0004349E  D203 3068 2088      00000068  00000088  7661           MVC   104(4,R3),128+8(R2)  Load PL OP3-Replacement
                                                  7662
                                                  7663 ****
                                                  7664 * force a failure by changing the 2nd operand
                                                  7665 ****
000434A4  D201 6004 2100      00000004  00000100  7666           MVC   4(2,R6),256(R2)      some random data
                                                  7667 ****
000434AA  E313 0000 0024                00000000  7668           STG   R1,0(R3)             load OP1 address in PL
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000434B0  E54C 3020 0000                 00000020  7669           MVHI  32(R3),PRIALET
                                                   7670
000434B6  4111 0004                      00000004  7671           LA    R1,4(R1)
000434BA  E313 0008 0024                 00000008  7672           STG   R1,8(R3)           load OP2 address in PL
000434C0  E54C 3024 0001                 00000024  7673           MVHI  36(R3),SECALET
                                                   7674
000434C6  4111 0004                      00000004  7675           LA    R1,4(R1)
000434CA  E313 0010 0024                 00000010  7676           STG   R1,16(R3)          load OP3 address in PL
000434D0  E54C 3028 0002                 00000028  7677           MVHI  40(R3),HOMEALET
                                                   7678
000434D6  B982 0000                               7679           XGR   R0,R0
000434DA  4100 001E                      0000001E  7680           LA    R0,30
                                                   7681
000434DE  B219 0200                      00000200  7682           SAC   ARMODE
000434E2  EE0F 0000 3000       00000000  00000000  7683           PLO   0,0,R15,0(R3)      Triple C/S (TCS: 30)
000434E8  B219 0000                      00000000  7684           SAC   PRIMODE
                                                   7685
000434EC  B98D 0020                               7686           EPSW  R2,R0              extract psw
000434F0  5020 5010                      00000010  7687           ST    R2,CCPSW             to save CC
000434F4  E310 500B 0076                 0000000B  7688           LB    R1,CCMASK          (failure CC mask)
000434FA  8910 0004                      00000004  7689           SLL   R1,4               (shift to BC instr CC position)
000434FE  4410 5134                      00043554  7690           EX    R1,F302TSTCC         fail if...
                                                   7691
                                00043502  00000001  7692 F302CKUR  EQU   *
00043502  D200 5009 3030       00000009  00000030  7693           MVC   URFOUND,48(R3)     Extract UR
00043508  D500 5008 3030       00000008  00000030  7694           CLC   UR,48(R3)
0004350E  4780 50F6                      00043516  7695           BE    F302CKRES
00043512  45F0 8388                      00002388  7696           BAL   R15,URMSG
                                                   7697 *
                                                   7698 * compare results
                                                   7699 *
                                00043516  00000001  7700 F302CKRES EQU   *
                                                   7701 *                           reset addresses
00043516  E310 501C 0014                 0000001C  7702           LGF   R1,OPSRCA          R1:  OPSRC - Primary
0004351C  B904 0061                               7703           LGR   R6,R1              R6:  OPSRC - Secondary
00043520  B904 0071                               7704           LGR   R7,R1              R7:  OPSRC - Home
                                                   7705
00043524  B219 0200                      00000200  7706           SAC   ARMODE
00043528  D503 1000 3040       00000000  00000040  7707           CLC   0(4,R1),64(R3)
0004352E  4770 512A                      0004354A  7708           BNE   F302XX             OP1C Different, error
00043532  D503 6004 3048       00000004  00000048  7709           CLC   4(4,R6),72(R3)
00043538  4770 512A                      0004354A  7710           BNE   F302XX             OP2C Different, error
0004353C  D503 7008 3050       00000008  00000050  7711           CLC   8(4,R7),80(R3)
00043542  B219 0000                      00000000  7712           SAC   PRIMODE
00043546  478B 0000                      00000000  7713           BE    0(R11)             OP3C same, return
                                                   7714
                                0004354A  00000001  7715 F302XX   EQU   *
0004354A  B219 0000                      00000000  7716           SAC   PRIMODE
0004354E  45F0 8450                      00002450  7717           BAL   R15,FAILMSG
00043552  07FB                                     7718           BR    R11
                                                   7719
00043554  4700 5138                      00043558  7720 F302TSTCC BC    0,F302CCMSG        fail if unexpected condition code
                                00043558  00000001  7721 F302CCMSG EQU  *
00043558  45F0 82B2                      000022B2  7722           BAL   R15,CCMSG
0004355C  47F0 50E2                      00043502  7723           B     F302CKUR
                                                   7724
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7725           TSTDATA
00043560                      00043560  00043A00  7726+          ORG   PLOETST+(128*K)+47*(3*K)+(K+512)
                                                    7727           TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00043A00  D7D9C940 40406E6E                         7728+          DC    CL32'PRI   >>>OPDST47<<<' eye catcher
00043A08  6ED6D7C4 E2E3F4F7
00043A10  4C4C4C40 40404040
00043A18  40404040 40404040
00043A20                                            7729+OPDST47   DS    0C
00043A20  00000000 00000000                         7730           DC    XL256'0'
00043A28  00000000 00000000
00043A30  00000000 00000000
00043A38  00000000 00000000
00043A40  00000000 00000000
00043A48  00000000 00000000
00043A50  00000000 00000000
00043A58  00000000 00000000
00043A60  00000000 00000000
00043A68  00000000 00000000
00043A70  00000000 00000000
00043A78  00000000 00000000
00043A80  00000000 00000000
00043A88  00000000 00000000
00043A90  00000000 00000000
00043A98  00000000 00000000
00043AA0  00000000 00000000
00043AA8  00000000 00000000
00043AB0  00000000 00000000
00043AB8  00000000 00000000
00043AC0  00000000 00000000
00043AC8  00000000 00000000
00043AD0  00000000 00000000
00043AD8  00000000 00000000
00043AE0  00000000 00000000
00043AE8  00000000 00000000
00043AF0  00000000 00000000
00043AF8  00000000 00000000
00043B00  00000000 00000000
00043B08  00000000 00000000
00043B10  00000000 00000000
00043B18  00000000 00000000
                                                    7731
                                                    7732           TSTLBL OPSRC              OSource: OP1-OP3
00043B20  D7D9C940 40406E6E                         7733+          DC    CL32'PRI   >>>OPSRC47<<<' eye catcher
00043B28  6ED6D7E2 D9C3F4F7
00043B30  4C4C4C40 40404040
00043B38  40404040 40404040
00043B40                                            7734+OPSRC47   DS    0C
00043B40  00000000 00000000                         7735           DC    XL256'0'
00043B48  00000000 00000000
00043B50  00000000 00000000
00043B58  00000000 00000000
00043B60  00000000 00000000
00043B68  00000000 00000000
00043B70  00000000 00000000
00043B78  00000000 00000000
00043B80  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00043B88  00000000 00000000
00043B90  00000000 00000000
00043B98  00000000 00000000
00043BA0  00000000 00000000
00043BA8  00000000 00000000
00043BB0  00000000 00000000
00043BB8  00000000 00000000
00043BC0  00000000 00000000
00043BC8  00000000 00000000
00043BD0  00000000 00000000
00043BD8  00000000 00000000
00043BE0  00000000 00000000
00043BE8  00000000 00000000
00043BF0  00000000 00000000
00043BF8  00000000 00000000
00043C00  00000000 00000000
00043C08  00000000 00000000
00043C10  00000000 00000000
00043C18  00000000 00000000
00043C20  00000000 00000000
00043C28  00000000 00000000
00043C30  00000000 00000000
00043C38  00000000 00000000
                                        7736
                                        7737          TSTLBL PL                 Parameter List
00043C40  D7D9C940 40406E6E             7738+         DC    CL32'PRI   >>>PL47<<<'    eye catcher
00043C48  6ED7D3F4 F74C4C4C
00043C50  40404040 40404040
00043C58  40404040 40404040
00043C60                                7739+PL47     DS    0C
00043C60  00000000 00000000             7740          DC    XL352'00'
00043C68  00000000 00000000
00043C70  00000000 00000000
00043C78  00000000 00000000
00043C80  00000000 00000000
00043C88  00000000 00000000
00043C90  00000000 00000000
00043C98  00000000 00000000
00043CA0  00000000 00000000
00043CA8  00000000 00000000
00043CB0  00000000 00000000
00043CB8  00000000 00000000
00043CC0  00000000 00000000
00043CC8  00000000 00000000
00043CD0  00000000 00000000
00043CD8  00000000 00000000
00043CE0  00000000 00000000
00043CE8  00000000 00000000
00043CF0  00000000 00000000
00043CF8  00000000 00000000
00043D00  00000000 00000000
00043D08  00000000 00000000
00043D10  00000000 00000000
00043D18  00000000 00000000
00043D20  00000000 00000000
00043D28  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00043D30  00000000 00000000
00043D38  00000000 00000000
00043D40  00000000 00000000
00043D48  00000000 00000000
00043D50  00000000 00000000
00043D58  00000000 00000000
00043D60  00000000 00000000
00043D68  00000000 00000000
00043D70  00000000 00000000
00043D78  00000000 00000000
00043D80  00000000 00000000
00043D88  00000000 00000000
00043D90  00000000 00000000
00043D98  00000000 00000000
00043DA0  00000000 00000000
00043DA8  00000000 00000000
00043DB0  00000000 00000000
00043DB8  00000000 00000000
                                        7741
                                        7742          TSTLBL FMSG               fail message
00043DC0  D7D9C940 40406E6E             7743+         DC    CL32'PRI   >>>FMSG47<<<'  eye catcher
00043DC8  6EC6D4E2 C7F4F74C
00043DD0  4C4C4040 40404040
00043DD8  40404040 40404040
00043DE0                                7744+FMSG47   DS    0C
00043DE0  86818993 858440D7             7745          DC    C'failed PLO-Extension '
00043DE8  D3D660C5 A7A38595
00043DF0  A2899695 40
00043DF5  E3998997 938540C3                           7746          DC    CL96'Triple Compare and Swap (TCS: 30): 2nd NE'
00043DFD  96949781 99854081
00043E05  958440E2 A6819740
00043E0D  4DE3C3E2 7A40F3F0
00043E15  5D7A40F2 958440D5
00043E1D  C5404040 40404040
00043E25  40404040 40404040
00043E2D  40404040 40404040
00043E35  40404040 40404040
00043E3D  40404040 40404040
00043E45  40404040 40404040
00043E4D  40404040 40404040
                                        7747          TSTEND
00043E55  D7D9C940 4040C5D5             7748+TEND47   DC    CL32'PRI   END of test: 47 <<<'
00043E5D  C4409686 40A385A2
00043E65  A37A40F4 F7404C4C
00043E6D  4C404040 40404040
                                        7749
                                        7750 * ------------------------------------------------------------------
                                        7751 * Test Triple Compare and Swap (TCS: 30) instruction : 3rd NE
                                        7752 * ------------------------------------------------------------------
                                        7753          TSTHDR 2,3,30             3rd Operand NOT Equal (cc=2, UR=3)
00043E75                        00043E75  00044000    7754+         ORG   PLOETST+(128*K)+48*(3*K)
00044000  D7D9C940 4040E2E3             7755+         DC    CL32'PRI   START of test: 48'
00044008  C1D9E340 968640A3
00044010  85A2A37A 40F4F840
00044018  40404040 40404040
00044020                        00044020              7756+         USING *,R5                base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00044020                     00000000            7757+         USING PLOETEST,R5
00044020  00044050                               7758+T48      DC    A(CODE48)           address of test code
00044024  0030                                   7759+         DC    H'48'               test number
00044026  00                                     7760+         DC    X'00'
00044027  1E                                     7761+         DC    HL1'30'             Function
00044028  03                                     7762+         DC    HL1'3'              expected unequal results (UR)
00044029  00                                     7763+         DC    HL1'00'             found UR
0004402A  02                                     7764+         DC    HL1'2'              cc
0004402B  0D                                     7765+         DC    HL1'13'             not expected CC mask
0004402C  00                                     7766+         DS    X                   extracted cc
00044030  00000000 00000000                      7767+         DS    2F                  extract PSW after test (has CC)
00044038  000449E0                               7768+         DC    A(FMSG48)           address of test fail message
0004403C  00044740                               7769+         DC    A(OPSRC48)          address of operand source
00044040  00044620                               7770+         DC    A(OPDST48)          address of operand destination
00044044  00044860                               7771+         DC    A(PL48)             address of parameter list
00044048  000041A8                               7772+         DC    A(RANDX+48)         address of random source data
0004404C  00044A55                               7773+         DC    A(TEND48)           address of test end eye catche
                                                 7774
                                                 7775          TSTCODE
00044050                                         7776+CODE48   DS    0F
                                                 7777 *                                  load random source
00044050  E310 501C 0014           0000001C      7778          LGF   R1,OPSRCA           R1: OPSRC - Primary
00044056  B904 0061                              7779          LGR   R6,R1               R6: OPSRC - Secondary
0004405A  B904 0071                              7780          LGR   R7,R1               R7: OPSRC - Home
                                                 7781
0004405E  E320 5028 0014           00000028      7782          LGF   R2,RANDXA           R2: address random data
00044064  E330 5024 0014           00000024      7783          LGF   R3,PLA              R3: address PL
                                                 7784
0004406A  B219 0200                00000200      7785          SAC   ARMODE
0004406E  D203 1000 2000  00000000 00000000      7786          MVC   0(4,R1),0(R2)       OP1 - Primary
00044074  D203 6004 2004  00000004 00000004      7787          MVC   4(4,R6),4(R2)       OP2 - Secondary
0004407A  D203 7008 2008  00000008 00000008      7788          MVC   8(4,R7),8(R2)       OP3 - Home
                                                 7789 *
                                                 7790 * Triple Compare and Swap (TCS: 30)
                                                 7791 * Build Parambter List
                                                 7792 *
00044080  D203 3040 2000  00000040 00000000      7793          MVC   64(4,R3),0(R2)      Load PL OP1-Compare
00044086  D203 3048 2004  00000048 00000004      7794          MVC   72(4,R3),4(R2)      Load PL OP2-Compare
0004408C  D203 3050 2008  00000050 00000008      7795          MVC   80(4,R3),8(R2)      Load PL OP3-Compare
                                                 7796
00044092  D203 3058 2080  00000058 00000080      7797          MVC   88(4,R3),128+0(R2)  Load PL OP1-Replacement
00044098  D203 3060 2084  00000060 00000084      7798          MVC   96(4,R3),128+4(R2)  Load PL OP2-Replacement
0004409E  D203 3068 2088  00000068 00000088      7799          MVC   104(4,R3),128+8(R2) Load PL OP3-Replacement
                                                 7800
                                                 7801 ****
                                                 7802 * force a failure by changing the 3rd operand
                                                 7803 ****
000440A4  D201 7008 2100  00000008 00000100      7804          MVC   8(2,R7),256(R2)     some random data
                                                 7805 ****
000440AA  E313 0000 0024                         7806          STG   R1,0(R3)            load OP1 address in PL
000440B0  E54C 3020 0000                         7807          MVHI  32(R3),PRIALET
                                                 7808
000440B6  4111 0004                              7809          LA    R1,4(R1)
000440BA  E313 0008 0024                         7810          STG   R1,8(R3)            load OP2 address in PL
000440C0  E54C 3024 0001                         7811          MVHI  36(R3),SECALET
                                                 7812
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000440C6  4111 0004                    00000004  7813          LA    R1,4(R1)
000440CA  E313 0010 0024               00000010  7814          STG   R1,16(R3)          load OP3 address in PL
000440D0  E54C 3028 0002               00000028  7815          MVHI  40(R3),HOMEALET
                                                 7816
000440D6  B982 0000                              7817          XGR   R0,R0
000440DA  4100 001E                    0000001E  7818          LA    R0,30
                                                 7819
000440DE  B219 0200                    00000200  7820          SAC   ARMODE
000440E2  EE0F 0000 3000  00000000     00000000  7821          PLO   0,0,R15,0(R3)       Triple C/S (TCS: 30)
000440E8  B219 0000                    00000000  7822          SAC   PRIMODE
                                                 7823
000440EC  B98D 0020                              7824          EPSW  R2,R0               extract psw
000440F0  5020 5010                    00000010  7825          ST    R2,CCPSW              to save CC
000440F4  E310 500B 0076               0000000B  7826          LB    R1,CCMASK           (failure CC mask)
000440FA  8910 0004                    00000004  7827          SLL   R1,4                (shift to BC instr CC position)
000440FE  4410 5134                    00044154  7828          EX    R1,F303TSTCC            fail if...
                                                 7829
                          00044102     00000001  7830 F303CKUR EQU   *
00044102  D200 5009 3030  00000009     00000030  7831          MVC   URFOUND,48(R3)      Extract UR
00044108  D500 5008 3030  00000008     00000030  7832          CLC   UR,48(R3)
0004410E  4780 50F6                    00044116  7833          BE    F303CKRES
00044112  45F0 8388                    00002388  7834          BAL   R15,URMSG
                                                 7835 *
                                                 7836 * compare results
                                                 7837 *
                          00044116     00000001  7838 F303CKRES EQU   *
                                                 7839 *                              reset addresses
00044116  E310 501C 0014               0000001C  7840          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0004411C  B904 0061                              7841          LGR   R6,R1               R6:  OPSRC - Secondary
00044120  B904 0071                              7842          LGR   R7,R1               R7:  OPSRC - Home
                                                 7843
00044124  B219 0200                    00000200  7844          SAC   ARMODE
00044128  D503 1000 3040  00000000     00000040  7845          CLC   0(4,R1),64(R3)
0004412E  4770 512A                    0004414A  7846          BNE   F303XX              OP1C Different, error
00044132  D503 6004 3048  00000004     00000048  7847          CLC   4(4,R6),72(R3)
00044138  4770 512A                    0004414A  7848          BNE   F303XX              OP2C Different, error
0004413C  D503 7008 3050  00000008     00000050  7849          CLC   8(4,R7),80(R3)
00044142  478B 0000                    00000000  7850          BE    0(R11)              OP3C same, return
00044146  B219 0000                    00000000  7851          SAC   PRIMODE
                                                 7852
                          0004414A     00000001  7853 F303XX   EQU   *
0004414A  B219 0000                    00000000  7854          SAC   PRIMODE
0004414E  45F0 8450                    00002450  7855          BAL   R15,FAILMSG
00044152  07FB                                   7856          BR    R11
                                                 7857
00044154  4700 5138                    00044158  7858 F303TSTCC BC    0,F303CCMSG         fail if unexpected condition code
                          00044158     00000001  7859 F303CCMSG EQU   *
00044158  45F0 82B2                    000022B2  7860          BAL   R15,CCMSG
0004415C  47F0 50E2                    00044102  7861          B     F303CKUR
                                                 7862
                                                 7863          TSTDATA
00044160                  00044160     00044600  7864+         ORG   PLOETST+(128*K)+48*(3*K)+(K+512)
                                                 7865          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00044600  D7D9C940 40406E6E                      7866+         DC    CL32'PRI   >>>OPDST48<<<' eye catcher
00044608  6ED6D7C4 E2E3F4F8
00044610  4C4C4C40 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00044618  40404040 40404040
00044620                                             7867+OPDST48  DS    0C
00044620  00000000 00000000                          7868          DC    XL256'0'
00044628  00000000 00000000
00044630  00000000 00000000
00044638  00000000 00000000
00044640  00000000 00000000
00044648  00000000 00000000
00044650  00000000 00000000
00044658  00000000 00000000
00044660  00000000 00000000
00044668  00000000 00000000
00044670  00000000 00000000
00044678  00000000 00000000
00044680  00000000 00000000
00044688  00000000 00000000
00044690  00000000 00000000
00044698  00000000 00000000
000446A0  00000000 00000000
000446A8  00000000 00000000
000446B0  00000000 00000000
000446B8  00000000 00000000
000446C0  00000000 00000000
000446C8  00000000 00000000
000446D0  00000000 00000000
000446D8  00000000 00000000
000446E0  00000000 00000000
000446E8  00000000 00000000
000446F0  00000000 00000000
000446F8  00000000 00000000
00044700  00000000 00000000
00044708  00000000 00000000
00044710  00000000 00000000
00044718  00000000 00000000
                                                     7869
                                                     7870          TSTLBL OPSRC              Source: OP1-OP3
00044720  D7D9C940 40406E6E                          7871+         DC    CL32'PRI   >>>OPSRC48<<<' eye catcher
00044728  6ED6D7E2 D9C3F4F8
00044730  4C4C4C40 40404040
00044738  40404040 40404040
00044740                                             7872+OPSRC48  DS    0C
00044740  00000000 00000000                          7873          DC    XL256'0'
00044748  00000000 00000000
00044750  00000000 00000000
00044758  00000000 00000000
00044760  00000000 00000000
00044768  00000000 00000000
00044770  00000000 00000000
00044778  00000000 00000000
00044780  00000000 00000000
00044788  00000000 00000000
00044790  00000000 00000000
00044798  00000000 00000000
000447A0  00000000 00000000
000447A8  00000000 00000000
000447B0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000447B8  00000000 00000000
000447C0  00000000 00000000
000447C8  00000000 00000000
000447D0  00000000 00000000
000447D8  00000000 00000000
000447E0  00000000 00000000
000447E8  00000000 00000000
000447F0  00000000 00000000
000447F8  00000000 00000000
00044800  00000000 00000000
00044808  00000000 00000000
00044810  00000000 00000000
00044818  00000000 00000000
00044820  00000000 00000000
00044828  00000000 00000000
00044830  00000000 00000000
00044838  00000000 00000000
                                        7874
                                        7875          TSTLBL PL                 Parameter List
00044840  D7D9C940 40406E6E             7876+         DC    CL32'PRI   >>>PL48<<<'    eye catcher
00044848  6ED7D3F4 F84C4C4C
00044850  40404040 40404040
00044858  40404040 40404040
00044860                                7877+PL48     DS    0C
00044860  00000000 00000000             7878          DC    XL352'00'
00044868  00000000 00000000
00044870  00000000 00000000
00044878  00000000 00000000
00044880  00000000 00000000
00044888  00000000 00000000
00044890  00000000 00000000
00044898  00000000 00000000
000448A0  00000000 00000000
000448A8  00000000 00000000
000448B0  00000000 00000000
000448B8  00000000 00000000
000448C0  00000000 00000000
000448C8  00000000 00000000
000448D0  00000000 00000000
000448D8  00000000 00000000
000448E0  00000000 00000000
000448E8  00000000 00000000
000448F0  00000000 00000000
000448F8  00000000 00000000
00044900  00000000 00000000
00044908  00000000 00000000
00044910  00000000 00000000
00044918  00000000 00000000
00044920  00000000 00000000
00044928  00000000 00000000
00044930  00000000 00000000
00044938  00000000 00000000
00044940  00000000 00000000
00044948  00000000 00000000
00044950  00000000 00000000
00044958  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00044960  00000000 00000000
00044968  00000000 00000000
00044970  00000000 00000000
00044978  00000000 00000000
00044980  00000000 00000000
00044988  00000000 00000000
00044990  00000000 00000000
00044998  00000000 00000000
000449A0  00000000 00000000
000449A8  00000000 00000000
000449B0  00000000 00000000
000449B8  00000000 00000000
                                        7879
                                        7880          TSTLBL FMSG               fail message
000449C0  D7D9C940 40406E6E             7881+         DC    CL32'PRI   >>>FMSG48<<<'  eye catcher
000449C8  6EC6D4E2 C7F4F84C
000449D0  4C4C4040 40404040
000449D8  40404040 40404040
000449E0                                7882+FMSG48   DS    0C
000449E0  86818993 858440D7             7883          DC    C'failed PLO-Extension '
000449E8  D3D660C5 A7A38595
000449F0  A2899695 40
000449F5  E3998997 938540C3             7884          DC    CL96'Triple Compare and Swap (TCS: 30): 3rd NE'
000449FD  96949781 99854081
00044A05  958440E2 A6819740
00044A0D  4DE3C3E2 7A40F3F0
00044A15  5D7A40F3 998440D5
00044A1D  C5404040 40404040
00044A25  40404040 40404040
00044A2D  40404040 40404040
00044A35  40404040 40404040
00044A3D  40404040 40404040
00044A45  40404040 40404040
00044A4D  40404040 40404040
                                        7885          TSTEND
00044A55  D7D9C940 4040C5D5             7886+TEND48   DC    CL32'PRI   END of test: 48 <<<'
00044A5D  C4409686 40A385A2
00044A65  A37A40F4 F8404C4C
00044A6D  4C404040 40404040
                                        7887
                                        7888 * ------------------------------------------------------------------
                                        7889 * Test Triple Compare and Swap (TCS: 30) instruction : All EQ
                                        7890 * ------------------------------------------------------------------
                                        7891          TSTHDR 0,99,30            All Operand Equal (cc=0, UR=99)
00044A75                  00044A75  00044C00  7892+         ORG   PLOETST+(128*K)+49*(3*K)
00044C00  D7D9C940 4040E2E3             7893+         DC    CL32'PRI   START of test: 49'
00044C08  C1D9E340 968640A3
00044C10  85A2A37A 40F4F940
00044C18  40404040 40404040
00044C20                  00044C20              7894+         USING *,R5                base for test data and test routine
00044C20                  00000000              7895+         USING PLOETEST,R5
00044C20  00044C50                              7896+T49      DC    A(CODE49)           address of test code
00044C24  0031                                  7897+         DC    H'49'               test number
00044C26  00                                    7898+         DC    X'00'
00044C27  1E                                    7899+         DC    HL1'30'             Function
00044C28  63                                    7900+         DC    HL1'99'             expected unequal results (UR)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00044C29  00                                       7901+          DC    HL1'00'             found UR
00044C2A  00                                       7902+          DC    HL1'0'              cc
00044C2B  07                                       7903+          DC    HL1'7'              not expected CC mask
00044C2C  00                                       7904+          DS    X                   extracted cc
00044C30  00000000 00000000                        7905+          DS    2F                  extract PSW after test (has CC)
00044C38  000455E0                                 7906+          DC    A(FMSG49)           address of test fail message
00044C3C  00045340                                 7907+          DC    A(OPSRC49)          address of operand source
00044C40  00045220                                 7908+          DC    A(OPDST49)          address of operand destination
00044C44  00045460                                 7909+          DC    A(PL49)             address of parameter list
00044C48  000041A9                                 7910+          DC    A(RANDX+49)         address of random source data
00044C4C  00045655                                 7911+          DC    A(TEND49)           address of test end eye catche
                                                   7912
                                                   7913           TSTCODE
00044C50                                           7914+CODE49    DS    0F
                                                   7915 *                                   load random source
00044C50  E310 501C 0014                0000001C   7916           LGF   R1,OPSRCA           R1: OPSRC - Primary
00044C56  B904 0061                                7917           LGR   R6,R1               R6: OPSRC - Secondary
00044C5A  B904 0071                                7918           LGR   R7,R1               R7: OPSRC - Home
                                                   7919
00044C5E  E320 5028 0014                00000028   7920           LGF   R2,RANDXA           R2: address random data
00044C64  E330 5024 0014                00000024   7921           LGF   R3,PLA              R3: address PL
                                                   7922
00044C6A  B219 0200                     00000200   7923           SAC   ARMODE
00044C6E  D203 1000 2000      00000000  00000000   7924           MVC   0(4,R1),0(R2)       OP1 - Primary
00044C74  D203 6004 2004      00000004  00000004   7925           MVC   4(4,R6),4(R2)       OP2 - Secondary
00044C7A  D203 7008 2008      00000008  00000008   7926           MVC   8(4,R7),8(R2)       OP3 - Home
                                                   7927 *
                                                   7928 * Triple Compare and Swap (TCS: 30)
                                                   7929 * Build Parambter List
                                                   7930 *
00044C80  D203 3040 2000      00000040  00000000   7931           MVC   64(4,R3),0(R2)      Load PL OP1-Compare
00044C86  D203 3048 2004      00000048  00000004   7932           MVC   72(4,R3),4(R2)      Load PL OP2-Compare
00044C8C  D203 3050 2008      00000050  00000008   7933           MVC   80(4,R3),8(R2)      Load PL OP3-Compare
                                                   7934
00044C92  D203 3058 2080      00000058  00000080   7935           MVC   88(4,R3),128+0(R2)  Load PL OP1-Replacement
00044C98  D203 3060 2084      00000060  00000084   7936           MVC   96(4,R3),128+4(R2)  Load PL OP2-Replacement
00044C9E  D203 3068 2088      00000068  00000088   7937           MVC   104(4,R3),128+8(R2) Load PL OP3-Replacement
                                                   7938
                                                   7939 ****
                                                   7940 * Do NOT force a failure by changing the 3rd operand
                                                   7941 ****
                                                   7942 *         MVC   8(2,R7),256(R2)     some random data
                                                   7943 ****
00044CA4  E313 0000 0024                00000000   7944           STG   R1,0(R3)            load OP1 address in PL
00044CAA  E54C 3020 0000                00000020   7945           MVHI  32(R3),PRIALET
                                                   7946
00044CB0  4111 0004                     00000004   7947           LA    R1,4(R1)
00044CB4  E313 0008 0024                00000008   7948           STG   R1,8(R3)            load OP2 address in PL
00044CBA  E54C 3024 0001                00000024   7949           MVHI  36(R3),SECALET
                                                   7950
00044CC0  4111 0004                     00000004   7951           LA    R1,4(R1)
00044CC4  E313 0010 0024                00000010   7952           STG   R1,16(R3)           load OP3 address in PL
00044CCA  E54C 3028 0002                00000028   7953           MVHI  40(R3),HOMEALET
                                                   7954
00044CD0  B982 0000                                7955           XGR   R0,R0
00044CD4  4100 001E                     0000001E   7956           LA    R0,30
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    7957
00044CD8  B219 0200                     00000200  7958          SAC   ARMODE
00044CDC  EE0F 0000 3000      00000000  00000000  7959          PLO   0,0,R15,0(R3)       Triple C/S (TCS: 30)
00044CE2  B219 0000                     00000000  7960          SAC   PRIMODE
                                                    7961
                                                    7962
00044CE6  B98D 0020                               7963          EPSW  R2,R0               extract psw
00044CEA  5020 5010                     00000010  7964          ST    R2,CCPSW              to save CC
00044CEE  E310 500B 0076                0000000B  7965          LB    R1,CCMASK           (failure CC mask)
00044CF4  8910 0004                     00000004  7966          SLL   R1,4                (shift to BC instr CC position)
00044CF8  4410 5132                     00044D52  7967          EX    R1,F304TSTCC          fail if...
                                                    7968
                              00044CFC  00000001  7969 F304CKUR EQU   *
00044CFC  D200 5009 3030      00000009  00000030  7970          MVC   URFOUND,48(R3)      Extract UR
00044D02  D500 5008 3030      00000008  00000030  7971          CLC   UR,48(R3)
00044D08  4780 50F0                     00044D10  7972          BE    F304CKRES
00044D0C  45F0 8388                     00002388  7973          BAL   R15,URMSG
                                                    7974 *
                                                    7975 * compare results
                                                    7976 *
                              00044D10  00000001  7977 F304CKRES EQU   *
                                                    7978 *                            reset addresses
00044D10  E310 501C 0014                0000001C  7979          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00044D16  B904 0061                               7980          LGR   R6,R1               R6:  OPSRC - Secondary
00044D1A  B904 0071                               7981          LGR   R7,R1               R7:  OPSRC - Home
00044D1E  5840 5020                     00000020  7982          L     R4,OPREPA
                                                    7983
                                                    7984 * All equal; replacement values copied to operand source
                                                    7985
00044D22  B219 0200                     00000200  7986          SAC   ARMODE
00044D26  D503 1000 3058      00000000  00000058  7987          CLC   0(4,R1),88(R3)
00044D2C  4770 5128                     00044D48  7988          BNE   F304XX              OP1R Different, error
00044D30  D503 6004 3060      00000004  00000060  7989          CLC   4(4,R6),96(R3)
00044D36  4770 5128                     00044D48  7990          BNE   F304XX              OP2R Different, error
00044D3A  D503 7008 3068      00000008  00000068  7991          CLC   8(4,R7),104(R3)
00044D40  B219 0000                     00000000  7992          SAC   PRIMODE
00044D44  478B 0000                     00000000  7993          BE    0(R11)              OP4R same, return
                                                    7994
                              00044D48  00000001  7995 F304XX   EQU   *
00044D48  B219 0000                     00000000  7996          SAC   PRIMODE
00044D4C  45F0 8450                     00002450  7997          BAL   R15,FAILMSG
00044D50  07FB                                    7998          BR    R11
                                                    7999
00044D52  4700 5136                     00044D56  8000 F304TSTCC BC    0,F304CCMSG          fail if unexpected condition code
                              00044D56  00000001  8001 F304CCMSG EQU   *
00044D56  45F0 82B2                     000022B2  8002          BAL   R15,CCMSG
00044D5A  47F0 50DC                     00044CFC  8003          B     F304CKUR
                                                    8004
                                                    8005          TSTDATA
00044D5E                      00044D5E  00045200  8006+         ORG   PLOETST+(128*K)+49*(3*K)+(K+512)
                                                    8007          TSTLBL OPDST              Dest/Data: Op1-R, Op2-R, Op3-R
00045200  D7D9C940 40406E6E                         8008+         DC    CL32'PRI   >>>OPDST49<<<' eye catcher
00045208  6ED6D7C4 E2E3F4F9
00045210  4C4C4C40 40404040
00045218  40404040 40404040
00045220                                            8009+OPDST49  DS    0C
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045220  00000000 00000000                        8010          DC    XL256'0'
00045228  00000000 00000000
00045230  00000000 00000000
00045238  00000000 00000000
00045240  00000000 00000000
00045248  00000000 00000000
00045250  00000000 00000000
00045258  00000000 00000000
00045260  00000000 00000000
00045268  00000000 00000000
00045270  00000000 00000000
00045278  00000000 00000000
00045280  00000000 00000000
00045288  00000000 00000000
00045290  00000000 00000000
00045298  00000000 00000000
000452A0  00000000 00000000
000452A8  00000000 00000000
000452B0  00000000 00000000
000452B8  00000000 00000000
000452C0  00000000 00000000
000452C8  00000000 00000000
000452D0  00000000 00000000
000452D8  00000000 00000000
000452E0  00000000 00000000
000452E8  00000000 00000000
000452F0  00000000 00000000
000452F8  00000000 00000000
00045300  00000000 00000000
00045308  00000000 00000000
00045310  00000000 00000000
00045318  00000000 00000000
                                                   8011
                                                   8012          TSTLBL OPSRC              Source: OP1-OP3
00045320  D7D9C940 40406E6E                        8013+         DC    CL32'PRI   >>>OPSRC49<<<' eye catcher
00045328  6ED6D7E2 D9C3F4F9
00045330  4C4C4C40 40404040
00045338  40404040 40404040
00045340                                           8014+OPSRC49  DS    0C
00045340  00000000 00000000                        8015          DC    XL256'0'
00045348  00000000 00000000
00045350  00000000 00000000
00045358  00000000 00000000
00045360  00000000 00000000
00045368  00000000 00000000
00045370  00000000 00000000
00045378  00000000 00000000
00045380  00000000 00000000
00045388  00000000 00000000
00045390  00000000 00000000
00045398  00000000 00000000
000453A0  00000000 00000000
000453A8  00000000 00000000
000453B0  00000000 00000000
000453B8  00000000 00000000
000453C0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000453C8  00000000 00000000
000453D0  00000000 00000000
000453D8  00000000 00000000
000453E0  00000000 00000000
000453E8  00000000 00000000
000453F0  00000000 00000000
000453F8  00000000 00000000
00045400  00000000 00000000
00045408  00000000 00000000
00045410  00000000 00000000
00045418  00000000 00000000
00045420  00000000 00000000
00045428  00000000 00000000
00045430  00000000 00000000
00045438  00000000 00000000
                                        8016
                                        8017          TSTLBL PL                 Parameter List
00045440  D7D9C940 40406E6E             8018+         DC    CL32'PRI   >>>PL49<<<'    eye catcher
00045448  6ED7D3F4 F94C4C4C
00045450  40404040 40404040
00045458  40404040 40404040
00045460                                8019+PL49     DS    0C
00045460  00000000 00000000             8020          DC    XL352'00'
00045468  00000000 00000000
00045470  00000000 00000000
00045478  00000000 00000000
00045480  00000000 00000000
00045488  00000000 00000000
00045490  00000000 00000000
00045498  00000000 00000000
000454A0  00000000 00000000
000454A8  00000000 00000000
000454B0  00000000 00000000
000454B8  00000000 00000000
000454C0  00000000 00000000
000454C8  00000000 00000000
000454D0  00000000 00000000
000454D8  00000000 00000000
000454E0  00000000 00000000
000454E8  00000000 00000000
000454F0  00000000 00000000
000454F8  00000000 00000000
00045500  00000000 00000000
00045508  00000000 00000000
00045510  00000000 00000000
00045518  00000000 00000000
00045520  00000000 00000000
00045528  00000000 00000000
00045530  00000000 00000000
00045538  00000000 00000000
00045540  00000000 00000000
00045548  00000000 00000000
00045550  00000000 00000000
00045558  00000000 00000000
00045560  00000000 00000000
00045568  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045570  00000000 00000000
00045578  00000000 00000000
00045580  00000000 00000000
00045588  00000000 00000000
00045590  00000000 00000000
00045598  00000000 00000000
000455A0  00000000 00000000
000455A8  00000000 00000000
000455B0  00000000 00000000
000455B8  00000000 00000000
                                          8021
                                          8022           TSTLBL FMSG                fail message
000455C0  D7D9C940 40406E6E               8023+          DC    CL32'PRI   >>>FMSG49<<<'  eye catcher
000455C8  6EC6D4E2 C7F4F94C
000455D0  4C4C4040 40404040
000455D8  40404040 40404040
000455E0                                  8024+FMSG49    DS    0C
000455E0  86818993 858440D7               8025           DC    C'failed PLO-Extension '
000455E8  D3D660C5 A7A38595
000455F0  A2899695 40
000455F5  E3998997 938540C3               8026           DC    CL96'Triple Compare and Swap (TCS: 30): All EQ'
000455FD  96949781 99854081
00045605  958440E2 A6819740
0004560D  4DE3C3E2 7A40F3F0
00045615  5D7A40C1 939340C5
0004561D  D8404040 40404040
00045625  40404040 40404040
0004562D  40404040 40404040
00045635  40404040 40404040
0004563D  40404040 40404040
00045645  40404040 40404040
0004564D  40404040 40404040
                                          8027           TSTEND
00045655  D7D9C940 4040C5D5               8028+TEND49    DC    CL32'PRI   END of test: 49 <<<'
0004565D  C4409686 40A385A2
00045665  A37A40F4 F9404C4C
0004566D  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8030 *=============================================================================
                                                   8031
                                                   8032 * -----------------------------------------------------------------------
                                                   8033 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : NE
                                                   8034 * -----------------------------------------------------------------------
                                                   8035          TSTHDR 1,99,29           1st Operand NOT Equal (cc=1, UR=99)
00045675                          00045675  00045800 8036+         ORG   PLOETST+(128*K)+50*(3*K)
00045800  D7D9C940 4040E2E3                        8037+         DC    CL32'PRI   START of test: 50'
00045808  C1D9E340 968640A3
00045810  85A2A37A 40F5F040
00045818  40404040 40404040
00045820                          00045820         8038+         USING *,R5               base for test data and test routine
00045820                          00000000         8039+         USING PLOETEST,R5
00045820  00045850                                 8040+T50      DC    A(CODE50)          address of test code
00045824  0032                                     8041+         DC    H'50'              test number
00045826  00                                       8042+         DC    X'00'
00045827  1D                                       8043+         DC    HL1'29'            Function
00045828  63                                       8044+         DC    HL1'99'            expected unequal results (UR)
00045829  00                                       8045+         DC    HL1'00'            found UR
0004582A  01                                       8046+         DC    HL1'1'             cc
0004582B  0B                                       8047+         DC    HL1'11'            not expected CC mask
0004582C  00                                       8048+         DS    X                  extracted cc
00045830  00000000 00000000                        8049+         DS    2F                 extract PSW after test (has CC)
00045838  000461E0                                 8050+         DC    A(FMSG50)          address of test fail message
0004583C  00045F40                                 8051+         DC    A(OPSRC50)         address of operand source
00045840  00045E20                                 8052+         DC    A(OPDST50)         address of operand destination
00045844  00046060                                 8053+         DC    A(PL50)            address of parameter list
00045848  000041AA                                 8054+         DC    A(RANDX+50)        address of random source data
0004584C  0004625A                                 8055+         DC    A(TEND50)          address of test end eye catche
                                                   8056
                                                   8057          TSTCODE
00045850                                           8058+CODE50   DS    0F
                                                   8059 *                                 load random source
00045850  E310 501C 0014                0000001C   8060          LGF   R1,OPSRCA          R1: OPSRC - Primary
00045856  B904 0061                                8061          LGR   R6,R1              R6: OPSRC - Secondary
0004585A  B904 0071                                8062          LGR   R7,R1              R7: OPSRC - Home
                                                   8063
0004585E  E320 5028 0014                00000028   8064          LGF   R2,RANDXA          R2: address random data
00045864  E330 5024 0014                00000024   8065          LGF   R3,PLA             R3: address PL
0004586A  E340 5020 0014                00000020   8066          LGF   R4,OPREPA          R4: OPREPA - PrimaryL
                                                   8067
00045870  B219 0200                     00000200   8068          SAC   ARMODE
00045874  D21F 4000 2000      00000000  00000000   8069          MVC   0(32,R4),0(R2)     load OP2  (same as OP1C below)
                                                   8070
0004587A  D21F 1000 20C0      00000000  000000C0   8071          MVC   0(32,R1),192+0(R2)       OP4 - Primary
00045880  D21F 6020 20E0      00000020  000000E0   8072          MVC   32(32,R6),192+32(R2)     OP6 - Secondary
00045886  D21F 7040 2100      00000040  00000100   8073          MVC   64(32,R7),192+64(R2)     OP8 - Home
                                                   8074 *
                                                   8075 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                   8076 * Build Parambter List
                                                   8077 *
0004588C  D21F 3000 2000      00000000  00000000   8078          MVC   0(32,R3),0(R2)     Load PL OP1-C
00045892  D21F 3020 2020      00000020  00000020   8079          MVC   32(32,R3),32(R2)   load PL OP1-R
00045898  D21F 3090 2040      00000090  00000040   8080          MVC   144(32,R3),64(R2)  load PL OP3
0004589E  D21F 30B0 2060      000000B0  00000060   8081          MVC   176(32,R3),96(R2)  load PL OP5
000458A4  D21F 30D0 2080      000000D0  00000080   8082          MVC   208(32,R3),128(R2) load PL OP7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          8083 ****
                                          8084 * force a failure by changing the Op1-c
                                          8085 ****
000458AA  D203 3000 2021      00000000  00000021  8086          MVC   0(4,R3),33(R2)
                                          8087 ****
000458B0  E313 0048 0024                00000048  8088          STG   R1,72(R3)           load OP4 address in PL (pri space)
000458B6  E54C 3044 0000                00000044  8089          MVHI  68(R3),PRIALET
                                          8090
000458BC  4111 0020                     00000020  8091          LA    R1,32(R1)
000458C0  E313 0068 0024                00000068  8092          STG   R1,104(R3)          load OP6 address in PL (sec space)
000458C6  E54C 3064 0001                00000064  8093          MVHI  100(R3),SECALET
                                          8094
000458CC  4111 0020                     00000020  8095          LA    R1,32(R1)
000458D0  E313 0088 0024                00000088  8096          STG   R1,136(R3)          load OP8 address in PL (home space)
000458D6  E54C 3084 0002                00000084  8097          MVHI  132(R3),HOMEALET
                                          8098
000458DC  B982 0000                             8099          XGR   R0,R0
000458E0  4100 001D                     0000001D  8100          LA    R0,29
                                          8101
000458E4  B219 0200                     00000200  8102          SAC   ARMODE
000458E8  EE0B 4000 3000      00000000  00000000  8103          PLO   0,0(R4),R11,0(R3)   C/S Triple Store (CSTSTO: 29)
000458EE  B219 0000                     00000000  8104          SAC   PRIMODE
                                          8105
000458F2  B98D 0020                             8106          EPSW  R2,R0               extract psw
000458F6  5020 5010                     00000010  8107          ST    R2,CCPSW              to save CC
000458FA  E310 500B 0076                0000000B  8108          LB    R1,CCMASK           (failure CC mask)
00045900  8910 0004                     00000004  8109          SLL   R1,4                (shift to BC instr CC position)
00045904  4410 5132                     00045952  8110          EX    R1,F291TSTCC            fail if...
                                          8111
                                00045908  00000001  8112 F291CKUR EQU   *
00045908  D200 5009 3030      00000009  00000030  8113          MVC   URFOUND,48(R3)      Extract UR
0004590E  D500 5008 3030      00000008  00000030  8114          CLC   UR,48(R3)
00045914  4780 50FC                     0004591C  8115          BE    F291CKRES
00045918  45F0 8388                     00002388  8116          BAL   R15,URMSG
                                          8117 *
                                          8118 * compare results
                                          8119 *
                                0004591C  00000001  8120 F291CKRES EQU   *
                                          8121 *                            reset addresses
0004591C  E310 501C 0014                0000001C  8122          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00045922  B904 0061                             8123          LGR   R6,R1               R6:  OPSRC - Secondary
00045926  B904 0071                             8124          LGR   R7,R1               R7:  OPSRC - Home
0004592A  E330 5024 0014                00000024  8125          LGF   R3,PLA              R3:  address PL
00045930  E340 5020 0014                00000020  8126          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                          8127
00045936  B219 0200                     00000200  8128          SAC   ARMODE
0004593A  D51F 4000 3000      00000000  00000000  8129          CLC   0(32,R4),0(R3)      Op2 == OP1C?
00045940  B219 0000                     00000000  8130          SAC   PRIMODE
00045944  478B 0000                     00000000  8131          BE    0(R11)              same, return
                                          8132
                                00045948  00000001  8133 F291XX   EQU   *
00045948  B219 0000                     00000000  8134          SAC   PRIMODE
0004594C  45F0 8450                     00002450  8135          BAL   R15,FAILMSG
00045950  07FB                                    8136          BR    R11
                                          8137
00045952  4700 5136                     00045956  8138 F291TSTCC BC    0,F291CCMSG          fail if unexpected condition code
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                               00045956  00000001  8139 F291CCMSG EQU  *
00045956  45F0 82B2                      000022B2  8140           BAL   R15,CCMSG
0004595A  47F0 50E8                      00045908  8141           B     F291CKUR
                                                   8142
                                                   8143           TSTDATA
0004595E                     0004595E  00045E00  8144+          ORG   PLOETST+(128*K)+50*(3*K)+(K+512)
                                                   8145           TSTLBL OPDST              OP2, Op4, Op6, Op8
00045E00  D7D9C940 40406E6E                        8146+          DC    CL32'PRI   >>>OPDST50<<<' eye catcher
00045E08  6ED6D7C4 E2E3F5F0
00045E10  4C4C4C40 40404040
00045E18  40404040 40404040
00045E20                                           8147+OPDST50   DS    0C
00045E20  00000000 00000000                        8148           DC    XL256'0'
00045E28  00000000 00000000
00045E30  00000000 00000000
00045E38  00000000 00000000
00045E40  00000000 00000000
00045E48  00000000 00000000
00045E50  00000000 00000000
00045E58  00000000 00000000
00045E60  00000000 00000000
00045E68  00000000 00000000
00045E70  00000000 00000000
00045E78  00000000 00000000
00045E80  00000000 00000000
00045E88  00000000 00000000
00045E90  00000000 00000000
00045E98  00000000 00000000
00045EA0  00000000 00000000
00045EA8  00000000 00000000
00045EB0  00000000 00000000
00045EB8  00000000 00000000
00045EC0  00000000 00000000
00045EC8  00000000 00000000
00045ED0  00000000 00000000
00045ED8  00000000 00000000
00045EE0  00000000 00000000
00045EE8  00000000 00000000
00045EF0  00000000 00000000
00045EF8  00000000 00000000
00045F00  00000000 00000000
00045F08  00000000 00000000
00045F10  00000000 00000000
00045F18  00000000 00000000
                                                   8149
                                                   8150           TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00045F20  D7D9C940 40406E6E                        8151+          DC    CL32'PRI   >>>OPSRC50<<<' eye catcher
00045F28  6ED6D7E2 D9C3F5F0
00045F30  4C4C4C40 40404040
00045F38  40404040 40404040
00045F40                                           8152+OPSRC50   DS    0C
00045F40  00000000 00000000                        8153           DC    XL256'0'
00045F48  00000000 00000000
00045F50  00000000 00000000
00045F58  00000000 00000000
00045F60  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00045F68  00000000 00000000
00045F70  00000000 00000000
00045F78  00000000 00000000
00045F80  00000000 00000000
00045F88  00000000 00000000
00045F90  00000000 00000000
00045F98  00000000 00000000
00045FA0  00000000 00000000
00045FA8  00000000 00000000
00045FB0  00000000 00000000
00045FB8  00000000 00000000
00045FC0  00000000 00000000
00045FC8  00000000 00000000
00045FD0  00000000 00000000
00045FD8  00000000 00000000
00045FE0  00000000 00000000
00045FE8  00000000 00000000
00045FF0  00000000 00000000
00045FF8  00000000 00000000
00046000  00000000 00000000
00046008  00000000 00000000
00046010  00000000 00000000
00046018  00000000 00000000
00046020  00000000 00000000
00046028  00000000 00000000
00046030  00000000 00000000
00046038  00000000 00000000
                                        8154
                                        8155           TSTLBL PL                  Parameter List
00046040  D7D9C940 40406E6E             8156+          DC    CL32'PRI   >>>PL50<<<'    eye catcher
00046048  6ED7D3F5 F04C4C4C
00046050  40404040 40404040
00046058  40404040 40404040
00046060                                8157+PL50      DS    0C
00046060  00000000 00000000             8158           DC    XL352'00'
00046068  00000000 00000000
00046070  00000000 00000000
00046078  00000000 00000000
00046080  00000000 00000000
00046088  00000000 00000000
00046090  00000000 00000000
00046098  00000000 00000000
000460A0  00000000 00000000
000460A8  00000000 00000000
000460B0  00000000 00000000
000460B8  00000000 00000000
000460C0  00000000 00000000
000460C8  00000000 00000000
000460D0  00000000 00000000
000460D8  00000000 00000000
000460E0  00000000 00000000
000460E8  00000000 00000000
000460F0  00000000 00000000
000460F8  00000000 00000000
00046100  00000000 00000000
00046108  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00046110  00000000 00000000
00046118  00000000 00000000
00046120  00000000 00000000
00046128  00000000 00000000
00046130  00000000 00000000
00046138  00000000 00000000
00046140  00000000 00000000
00046148  00000000 00000000
00046150  00000000 00000000
00046158  00000000 00000000
00046160  00000000 00000000
00046168  00000000 00000000
00046170  00000000 00000000
00046178  00000000 00000000
00046180  00000000 00000000
00046188  00000000 00000000
00046190  00000000 00000000
00046198  00000000 00000000
000461A0  00000000 00000000
000461A8  00000000 00000000
000461B0  00000000 00000000
000461B8  00000000 00000000
                                          8159
                                          8160          TSTLBL FMSG                fail message
000461C0  D7D9C940 40406E6E               8161+         DC    CL32'PRI   >>>FMSG50<<<'  eye catcher
000461C8  6EC6D4E2 C7F5F04C
000461D0  4C4C4040 40404040
000461D8  40404040 40404040
000461E0                                  8162+FMSG50   DS    0C
000461E0  86818993 858440D7               8163          DC    C'failed PLO-Extension '
000461E8  D3D660C5 A7A38595
000461F0  A2899695 40
000461F5  C3969497 81998540               8164          DC    C'Compare and Swap and Triple Store (CSTSTO: 29) : NE'
000461FD  81958440 E2A68197
00046205  40819584 40E39989
0004620D  97938540 E2A39699
00046215  85404DC3 E2E3E2E3
0004621D  D67A40F2 F95D407A
00046225  40D5C5
00046228  40404040 40404040               8165          DC    CL50' '
00046230  40404040 40404040
00046238  40404040 40404040
00046240  40404040 40404040
00046248  40404040 40404040
00046250  40404040 40404040
00046258  4040
                                          8166          TSTEND
0004625A  D7D9C940 4040C5D5               8167+TEND50   DC    CL32'PRI   END of test: 50 <<<'
00046262  C4409686 40A385A2
0004626A  A37A40F5 F0404C4C
00046272  4C404040 40404040
                                          8168
                                          8169 * ------------------------------------------------------------------
                                          8170 * Test Compare and Swap and Triple Store (CSTSTO: 29) instruction : EQ
                                          8171 * ------------------------------------------------------------------
                                          8172          TSTHDR 0,99,29             1st Operand NOT Equal (cc=1, UR=99)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004627A                       0004627A  00046400  8173+          ORG   PLOETST+(128*K)+51*(3*K)
00046400  D7D9C940 4040E2E3                        8174+          DC    CL32'PRI   START of test: 51'
00046408  C1D9E340 968640A3
00046410  85A2A37A 40F5F140
00046418  40404040 40404040
00046420                       00046420            8175+          USING *,R5                 base for test data and test routine
00046420                       00000000            8176+          USING PLOETEST,R5
00046420  00046450                                 8177+T51       DC    A(CODE51)            address of test code
00046424  0033                                     8178+          DC    H'51'                test number
00046426  00                                       8179+          DC    X'00'
00046427  1D                                       8180+          DC    HL1'29'              Function
00046428  63                                       8181+          DC    HL1'99'              expected unequal results (UR)
00046429  00                                       8182+          DC    HL1'00'              found UR
0004642A  00                                       8183+          DC    HL1'0'               cc
0004642B  07                                       8184+          DC    HL1'7'               not expected CC mask
0004642C  00                                       8185+          DS    X                    extracted cc
00046430  00000000 00000000                        8186+          DS    2F                   extract PSW after test (has CC)
00046438  00046DE0                                 8187+          DC    A(FMSG51)            address of test fail message
0004643C  00046B40                                 8188+          DC    A(OPSRC51)           address of operand source
00046440  00046A20                                 8189+          DC    A(OPDST51)           address of operand destination
00046444  00046C60                                 8190+          DC    A(PL51)              address of parameter list
00046448  000041AB                                 8191+          DC    A(RANDX+51)          address of random source data
0004644C  00046E5A                                 8192+          DC    A(TEND51)            address of test end eye catche
                                                   8193
                                                   8194           TSTCODE
00046450                                           8195+CODE51    DS    0F
                                                   8196 *                                    load random source
00046450  E310 501C 0014                 0000001C  8197           LGF   R1,OPSRCA            R1: OPSRC - Primary
00046456  B904 0061                                8198           LGR   R6,R1                R6: OPSRC - Secondary
0004645A  B904 0071                                8199           LGR   R7,R1                R7: OPSRC - Home
                                                   8200
0004645E  E320 5028 0014                 00000028  8201           LGF   R2,RANDXA            R2: address random data
00046464  E330 5024 0014                 00000024  8202           LGF   R3,PLA               R3: address PL
0004646A  E340 5020 0014                 00000020  8203           LGF   R4,OPREPA            R4: OPREPA - PrimaryL
                                                   8204
00046470  B219 0200                      00000200  8205           SAC   ARMODE
00046474  D21F 4000 2000       00000000  00000000  8206           MVC   0(32,R4),0(R2)       load OP2  (same as OP1C below)
                                                   8207
0004647A  D21F 1000 20C0       00000000  000000C0  8208           MVC   0(32,R1),192+0(R2)       OP4 - Primary
00046480  D21F 6020 20E0       00000020  000000E0  8209           MVC   32(32,R6),192+32(R2)     OP6 - Secondary
00046486  D21F 7040 2100       00000040  00000100  8210           MVC   64(32,R7),192+64(R2)     OP8 - Home
                                                   8211 *
                                                   8212 * Compare and Swap and Triple Store (CSTSTO: 29)
                                                   8213 * Build Parambter List
                                                   8214 *
0004648C  D21F 3000 2000       00000000  00000000  8215           MVC   0(32,R3),0(R2)       Load PL OP1-C
00046492  D21F 3020 2020       00000020  00000020  8216           MVC   32(32,R3),32(R2)     load PL OP1-R
00046498  D21F 3090 2040       00000090  00000040  8217           MVC   144(32,R3),64(R2)    load PL OP3
0004649E  D21F 30B0 2060       000000B0  00000060  8218           MVC   176(32,R3),96(R2)    load PL OP5
000464A4  D21F 30D0 2080       000000D0  00000080  8219           MVC   208(32,R3),128(R2)   load PL OP7
                                                   8220
                                                   8221 ****
                                                   8222 * Force a EQ by changing the Op1-C == Op2
                                                   8223 ****
000464AA  D21F 3000 4000       00000000  00000000  8224           MVC   0(32,R3),0(R4)
                                                   8225 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000464B0  E313 0048 0024                 00000048  8226           STG   R1,72(R3)           load OP4 address in PL (pri space)
000464B6  E54C 3044 0000                 00000044  8227           MVHI  68(R3),PRIALET
                                                   8228
000464BC  4111 0020                      00000020  8229           LA    R1,32(R1)
000464C0  E313 0068 0024                 00000068  8230           STG   R1,104(R3)          load OP6 address in PL (sec space)
000464C6  E54C 3064 0001                 00000064  8231           MVHI  100(R3),SECALET
                                                   8232
000464CC  4111 0020                      00000020  8233           LA    R1,32(R1)
000464D0  E313 0088 0024                 00000088  8234           STG   R1,136(R3)          load OP8 address in PL (home space)
000464D6  E54C 3084 0002                 00000084  8235           MVHI  132(R3),HOMEALET
                                                   8236
000464DC  B982 0000                              8237           XGR   R0,R0
000464E0  4100 001D                      0000001D  8238           LA    R0,29
                                                   8239
000464E4  B219 0200                      00000200  8240           SAC   ARMODE
000464E8  EE0B 4000 3000       00000000  00000000  8241           PLO   0,0(R4),R11,0(R3)   C/S Triple Store (CSTSTO: 29)
000464EE  B219 0000                      00000000  8242           SAC   PRIMODE
                                                   8243
000464F2  B98D 0020                              8244           EPSW  R2,R0               extract psw
000464F6  5020 5010                      00000010  8245           ST    R2,CCPSW              to save CC
000464FA  E310 500B 0076                 0000000B  8246           LB    R1,CCMASK           (failure CC mask)
00046500  8910 0004                      00000004  8247           SLL   R1,4                (shift to BC instr CC position)
00046504  4410 5150                      00046570  8248           EX    R1,F292TSTCC            fail if...
                                                   8249
                                00046508  00000001  8250 F292CKUR EQU   *
00046508  D200 5009 3030       00000009  00000030  8251           MVC   URFOUND,48(R3)      Extract UR
0004650E  D500 5008 3030       00000008  00000030  8252           CLC   UR,48(R3)
00046514  4780 50FC                      0004651C  8253           BE    F292CKRES
00046518  45F0 8388                      00002388  8254           BAL   R15,URMSG
                                                   8255 *
                                                   8256 * compare results
                                                   8257 *
                                0004651C  00000001  8258 F292CKRES EQU   *
                                                   8259 *                              reset addresses
0004651C  E310 501C 0014                 0000001C  8260           LGF   R1,OPSRCA           R1:  OPSRC - Primary
00046522  B904 0061                              8261           LGR   R6,R1               R6:  OPSRC - Secondary
00046526  B904 0071                              8262           LGR   R7,R1               R7:  OPSRC - Home
0004652A  E330 5024 0014                 00000024  8263           LGF   R3,PLA              R3:  address PL
00046530  E340 5020 0014                 00000020  8264           LGF   R4,OPREPA           R4:  OPREPA - Primary
                                                   8265
00046536  B219 0200                      00000200  8266           SAC   ARMODE
0004653A  D51F 1000 3090       00000000  00000090  8267           CLC   0(32,R1),144(R3)    OP4 ==  PL Op3
00046540  4770 5146                      00046566  8268           BNE   F292XX                   different, error
00046544  D51F 6020 30B0       00000020  000000B0  8269           CLC   32(32,R6),176(R3)   OP6 ==  PL Op5
0004654A  4770 5146                      00046566  8270           BNE   F292XX                   different, error
0004654E  D51F 7040 30D0       00000040  000000D0  8271           CLC   64(32,R7),208(R3)   OP8 ==  PL Op7
00046554  4770 5146                      00046566  8272           BNE   F292XX                   different, error
00046558  D51F 4000 3020       00000000  00000020  8273           CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
0004655E  B219 0000                      00000000  8274           SAC   PRIMODE
00046562  478B 0000                      00000000  8275           BE    0(R11)                   same, return
                                                   8276
                                00046566  00000001  8277 F292XX   EQU   *
00046566  B219 0000                      00000000  8278           SAC   PRIMODE
0004656A  45F0 8450                      00002450  8279           BAL   R15,FAILMSG
0004656E  07FB                                     8280           BR    R11
                                                   8281
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT
00046570  4700 5154                      00046574  8282 F292TSTCC BC   0,F292CCMSG          fail if unexpected condition code
                               00046574  00000001  8283 F292CCMSG EQU  *
00046574  45F0 82B2                      000022B2  8284           BAL   R15,CCMSG
00046578  47F0 50E8                      00046508  8285           B     F292CKUR
                                                   8286
                                                   8287           TSTDATA
0004657C                       0004657C  00046A00  8288+          ORG   PLOETST+(128*K)+51*(3*K)+(K+512)
                                                   8289           TSTLBL OPDST              OP2, Op4, Op6, Op8
00046A00  D7D9C940 40406E6E                        8290+          DC    CL32'PRI   >>>OPDST51<<<' eye catcher
00046A08  6ED6D7C4 E2E3F5F1
00046A10  4C4C4C40 40404040
00046A18  40404040 40404040
00046A20                                           8291+OPDST51   DS    0C
00046A20  00000000 00000000                        8292           DC    XL256'0'
00046A28  00000000 00000000
00046A30  00000000 00000000
00046A38  00000000 00000000
00046A40  00000000 00000000
00046A48  00000000 00000000
00046A50  00000000 00000000
00046A58  00000000 00000000
00046A60  00000000 00000000
00046A68  00000000 00000000
00046A70  00000000 00000000
00046A78  00000000 00000000
00046A80  00000000 00000000
00046A88  00000000 00000000
00046A90  00000000 00000000
00046A98  00000000 00000000
00046AA0  00000000 00000000
00046AA8  00000000 00000000
00046AB0  00000000 00000000
00046AB8  00000000 00000000
00046AC0  00000000 00000000
00046AC8  00000000 00000000
00046AD0  00000000 00000000
00046AD8  00000000 00000000
00046AE0  00000000 00000000
00046AE8  00000000 00000000
00046AF0  00000000 00000000
00046AF8  00000000 00000000
00046B00  00000000 00000000
00046B08  00000000 00000000
00046B10  00000000 00000000
00046B18  00000000 00000000
                                                   8293
                                                   8294           TSTLBL OPSRC              Op1-C, Op3, Op5, Op7, Op1-R
00046B20  D7D9C940 40406E6E                        8295+          DC    CL32'PRI   >>>OPSRC51<<<' eye catcher
00046B28  6ED6D7E2 D9C3F5F1
00046B30  4C4C4C40 40404040
00046B38  40404040 40404040
00046B40                                           8296+OPSRC51   DS    0C
00046B40  00000000 00000000                        8297           DC    XL256'0'
00046B48  00000000 00000000
00046B50  00000000 00000000
00046B58  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00046B60  00000000 00000000
00046B68  00000000 00000000
00046B70  00000000 00000000
00046B78  00000000 00000000
00046B80  00000000 00000000
00046B88  00000000 00000000
00046B90  00000000 00000000
00046B98  00000000 00000000
00046BA0  00000000 00000000
00046BA8  00000000 00000000
00046BB0  00000000 00000000
00046BB8  00000000 00000000
00046BC0  00000000 00000000
00046BC8  00000000 00000000
00046BD0  00000000 00000000
00046BD8  00000000 00000000
00046BE0  00000000 00000000
00046BE8  00000000 00000000
00046BF0  00000000 00000000
00046BF8  00000000 00000000
00046C00  00000000 00000000
00046C08  00000000 00000000
00046C10  00000000 00000000
00046C18  00000000 00000000
00046C20  00000000 00000000
00046C28  00000000 00000000
00046C30  00000000 00000000
00046C38  00000000 00000000
                                                  8298
                                                  8299           TSTLBL PL                 Parameter List
00046C40  D7D9C940 40406E6E                       8300+          DC    CL32'PRI   >>>PL51<<<'    eye catcher
00046C48  6ED7D3F5 F14C4C4C
00046C50  40404040 40404040
00046C58  40404040 40404040
00046C60                                          8301+PL51      DS    0C
00046C60  00000000 00000000                       8302           DC    XL352'00'
00046C68  00000000 00000000
00046C70  00000000 00000000
00046C78  00000000 00000000
00046C80  00000000 00000000
00046C88  00000000 00000000
00046C90  00000000 00000000
00046C98  00000000 00000000
00046CA0  00000000 00000000
00046CA8  00000000 00000000
00046CB0  00000000 00000000
00046CB8  00000000 00000000
00046CC0  00000000 00000000
00046CC8  00000000 00000000
00046CD0  00000000 00000000
00046CD8  00000000 00000000
00046CE0  00000000 00000000
00046CE8  00000000 00000000
00046CF0  00000000 00000000
00046CF8  00000000 00000000
00046D00  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00046D08  00000000 00000000
00046D10  00000000 00000000
00046D18  00000000 00000000
00046D20  00000000 00000000
00046D28  00000000 00000000
00046D30  00000000 00000000
00046D38  00000000 00000000
00046D40  00000000 00000000
00046D48  00000000 00000000
00046D50  00000000 00000000
00046D58  00000000 00000000
00046D60  00000000 00000000
00046D68  00000000 00000000
00046D70  00000000 00000000
00046D78  00000000 00000000
00046D80  00000000 00000000
00046D88  00000000 00000000
00046D90  00000000 00000000
00046D98  00000000 00000000
00046DA0  00000000 00000000
00046DA8  00000000 00000000
00046DB0  00000000 00000000
00046DB8  00000000 00000000
                                        8303
                                        8304          TSTLBL FMSG                fail message
00046DC0  D7D9C940 40406E6E             8305+         DC    CL32'PRI   >>>FMSG51<<<'  eye catcher
00046DC8  6EC6D4E2 C7F5F14C
00046DD0  4C4C4040 40404040
00046DD8  40404040 40404040
00046DE0                                8306+FMSG51   DS    0C
00046DE0  86818993 858440D7             8307          DC    C'failed PLO-Extension '
00046DE8  D3D660C5 A7A38595
00046DF0  A2899695 40
00046DF5  C3969497 81998540             8308          DC    C'Compare and Swap and Triple Store (CSTSTO: 29) : EQ'
00046DFD  81958440 E2A68197
00046E05  40819584 40E39989
00046E0D  97938540 E2A39699
00046E15  85404DC3 E2E3E2E3
00046E1D  D67A40F2 F95D407A
00046E25  40C5D8
00046E28  40404040 40404040             8309          DC    CL50' '
00046E30  40404040 40404040
00046E38  40404040 40404040
00046E40  40404040 40404040
00046E48  40404040 40404040
00046E50  40404040 40404040
00046E58  4040
                                        8310          TSTEND
00046E5A  D7D9C940 4040C5D5             8311+TEND51   DC    CL32'PRI   END of test: 51 <<<'
00046E62  C4409686 40A385A2
00046E6A  A37A40F5 F1404C4C
00046E72  4C404040 40404040
                                        8312
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                  8314 *================================================================================
                                                  8315
                                                  8316 * ------------------------------------------------------------------------
                                                  8317 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : NE
                                                  8318 * ------------------------------------------------------------------------
                                                  8319          TSTHDR 1,99,28           1Op1-C != OP2 (cc=1, UR=99)
00046E7A                          00046E7A  00047000  8320+         ORG   PLOETST+(128*K)+52*(3*K)
00047000  D7D9C940 4040E2E3                       8321+         DC    CL32'PRI   START of test: 52'
00047008  C1D9E340 968640A3
00047010  85A2A37A 40F5F240
00047018  40404040 40404040
00047020                          00047020        8322+         USING *,R5               base for test data and test routine
00047020                          00000000        8323+         USING PLOETEST,R5
00047020  00047050                                8324+T52      DC    A(CODE52)          address of test code
00047024  0034                                    8325+         DC    H'52'              test number
00047026  00                                      8326+         DC    X'00'
00047027  1C                                      8327+         DC    HL1'28'            Function
00047028  63                                      8328+         DC    HL1'99'            expected unequal results (UR)
00047029  00                                      8329+         DC    HL1'00'            found UR
0004702A  01                                      8330+         DC    HL1'1'             cc
0004702B  0B                                      8331+         DC    HL1'11'            not expected CC mask
0004702C  00                                      8332+         DS    X                  extracted cc
00047030  00000000 00000000                       8333+         DS    2F                 extract PSW after test (has CC)
00047038  000479E0                                8334+         DC    A(FMSG52)          address of test fail message
0004703C  00047740                                8335+         DC    A(OPSRC52)         address of operand source
00047040  00047620                                8336+         DC    A(OPDST52)         address of operand destination
00047044  00047860                                8337+         DC    A(PL52)            address of parameter list
00047048  000041AC                                8338+         DC    A(RANDX+52)        address of random source data
0004704C  00047A5A                                8339+         DC    A(TEND52)          address of test end eye catche
                                                  8340
                                                  8341          TSTCODE
00047050                                          8342+CODE52   DS    0F
                                                  8343 *                                 load random source
00047050  E310 501C 0014                0000001C  8344          LGF   R1,OPSRCA          R1: OPSRC - Primary
00047056  B904 0061                               8345          LGR   R6,R1              R6: OPSRC - Secondary
0004705A  B904 0071                               8346          LGR   R7,R1              R7: OPSRC - Home
                                                  8347
0004705E  E320 5028 0014                00000028  8348          LGF   R2,RANDXA          R2: address random data
00047064  E330 5024 0014                00000024  8349          LGF   R3,PLA             R3: address PL
0004706A  E340 5020 0014                00000020  8350          LGF   R4,OPREPA          R4: OPREPA - PrimaryL
                                                  8351
00047070  B219 0200                     00000200  8352          SAC   ARMODE
00047074  D21F 4000 2000        00000000  00000000  8353          MVC   0(32,R4),0(R2)     load OP2  (same as OP1C below)
                                                  8354
0004707A  D21F 1000 20C0        00000000  000000C0  8355          MVC   0(32,R1),192+0(R2)       OP4 - Primary
00047080  D21F 6020 20E0        00000020  000000E0  8356          MVC   32(32,R6),192+32(R2)     OP6 - Secondary
                                                  8357 *
                                                  8358 * Compare and Swap and Double Store (CSDSTO: 28)
                                                  8359 * Build Parambter List
                                                  8360 *
00047086  D21F 3000 2000        00000000  00000000  8361          MVC   0(32,R3),0(R2)     Load PL OP1-C
0004708C  D21F 3020 2020        00000020  00000020  8362          MVC   32(32,R3),32(R2)   load PL OP1-R
00047092  D21F 3070 2040        00000070  00000040  8363          MVC   112(32,R3),64(R2)  load PL OP3
00047098  D21F 3090 2060        00000090  00000060  8364          MVC   144(32,R3),96(R2)  load PL OP5
                                                  8365 ****
                                                  8366 * force a failure by changing the Op1-c
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          8367 ****
0004709E  D203 3000 2021      00000000  00000021  8368          MVC   0(4,R3),33(R2)
                                          8369 ****
000470A4  E313 0048 0024                00000048  8370          STG   R1,72(R3)           load OP4 address in PL (pri space)
000470AA  E54C 3044 0000                00000044  8371          MVHI  68(R3),PRIALET
                                          8372
000470B0  4111 0020                     00000020  8373          LA    R1,32(R1)
000470B4  E313 0068 0024                00000068  8374          STG   R1,104(R3)          load OP6 address in PL (sec space)
000470BA  E54C 3064 0001                00000064  8375          MVHI  100(R3),SECALET
                                          8376
000470C0  B982 0000                       8377          XGR   R0,R0
000470C4  4100 001C                     0000001C  8378          LA    R0,28
                                          8379
000470C8  B219 0200                     00000200  8380          SAC   ARMODE
000470CC  EE0B 4000 3000      00000000  00000000  8381          PLO   0,0(R4),R11,0(R3)   C/S Double Store (CSDSTO: 28)
000470D2  B219 0000                     00000000  8382          SAC   PRIMODE
                                          8383
000470D6  B98D 0020                       8384          EPSW  R2,R0               extract psw
000470DA  5020 5010                     00000010  8385          ST    R2,CCPSW              to save CC
000470DE  E310 500B 0076                0000000B  8386          LB    R1,CCMASK           (failure CC mask)
000470E4  8910 0004                     00000004  8387          SLL   R1,4                (shift to BC instr CC position)
000470E8  4410 5116                     00047136  8388          EX    R1,F281TSTCC            fail if...
                                          8389
                          000470EC  00000001  8390 F281CKUR EQU   *
000470EC  D200 5009 3030      00000009  00000030  8391          MVC   URFOUND,48(R3)      Extract UR
000470F2  D500 5008 3030      00000008  00000030  8392          CLC   UR,48(R3)
000470F8  4780 50E0                     00047100  8393          BE    F281CKRES
000470FC  45F0 8388                     00002388  8394          BAL   R15,URMSG
                                          8395 *
                                          8396 * compare results
                                          8397 *
                          00047100  00000001  8398 F281CKRES EQU   *
                                          8399 *                                    reset addresses
00047100  E310 501C 0014                0000001C  8400          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00047106  B904 0061                       8401          LGR   R6,R1               R6:  OPSRC - Secondary
0004710A  B904 0071                       8402          LGR   R7,R1               R7:  OPSRC - Home
0004710E  E330 5024 0014                00000024  8403          LGF   R3,PLA              R3:  address PL
00047114  E340 5020 0014                00000020  8404          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                          8405
0004711A  B219 0200                     00000200  8406          SAC   ARMODE
0004711E  D51F 4000 3000      00000000  00000000  8407          CLC   0(32,R4),0(R3)      Op2 == OP1C?
00047124  B219 0000                     00000000  8408          SAC   PRIMODE
00047128  478B 0000                     00000000  8409          BE    0(R11)              same, return
                                          8410
                          0004712C  00000001  8411 F281XX   EQU   *
0004712C  B219 0000                     00000000  8412          SAC   PRIMODE
00047130  45F0 8450                     00002450  8413          BAL   R15,FAILMSG
00047134  07FB                            8414          BR    R11
                                          8415
00047136  4700 511A                     0004713A  8416 F281TSTCC BC    0,F281CCMSG          fail if unexpected condition code
                          0004713A  00000001  8417 F281CCMSG EQU  *
0004713A  45F0 82B2                     000022B2  8418          BAL   R15,CCMSG
0004713E  47F0 50CC                     000470EC  8419          B     F281CKUR
                                          8420
                                          8421          TSTDATA
00047142                  00047142  00047600  8422+         ORG   PLOETST+(128*K)+52*(3*K)+(K+512)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8423           TSTLBL OPDST              OP2, Op4, Op6
00047600  D7D9C940 40406E6E                       8424+          DC     CL32'PRI   >>>OPDST52<<<' eye catcher
00047608  6ED6D7C4 E2E3F5F2
00047610  4C4C4C40 40404040
00047618  40404040 40404040
00047620                                         8425+OPDST52   DS     0C
00047620  00000000 00000000                      8426           DC     XL256'0'
00047628  00000000 00000000
00047630  00000000 00000000
00047638  00000000 00000000
00047640  00000000 00000000
00047648  00000000 00000000
00047650  00000000 00000000
00047658  00000000 00000000
00047660  00000000 00000000
00047668  00000000 00000000
00047670  00000000 00000000
00047678  00000000 00000000
00047680  00000000 00000000
00047688  00000000 00000000
00047690  00000000 00000000
00047698  00000000 00000000
000476A0  00000000 00000000
000476A8  00000000 00000000
000476B0  00000000 00000000
000476B8  00000000 00000000
000476C0  00000000 00000000
000476C8  00000000 00000000
000476D0  00000000 00000000
000476D8  00000000 00000000
000476E0  00000000 00000000
000476E8  00000000 00000000
000476F0  00000000 00000000
000476F8  00000000 00000000
00047700  00000000 00000000
00047708  00000000 00000000
00047710  00000000 00000000
00047718  00000000 00000000
                                                 8427
                                                 8428           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00047720  D7D9C940 40406E6E                      8429+          DC     CL32'PRI   >>>OPSRC52<<<' eye catcher
00047728  6ED6D7E2 D9C3F5F2
00047730  4C4C4C40 40404040
00047738  40404040 40404040
00047740                                         8430+OPSRC52   DS     0C
00047740  00000000 00000000                      8431           DC     XL256'0'
00047748  00000000 00000000
00047750  00000000 00000000
00047758  00000000 00000000
00047760  00000000 00000000
00047768  00000000 00000000
00047770  00000000 00000000
00047778  00000000 00000000
00047780  00000000 00000000
00047788  00000000 00000000
00047790  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00047798  00000000 00000000
000477A0  00000000 00000000
000477A8  00000000 00000000
000477B0  00000000 00000000
000477B8  00000000 00000000
000477C0  00000000 00000000
000477C8  00000000 00000000
000477D0  00000000 00000000
000477D8  00000000 00000000
000477E0  00000000 00000000
000477E8  00000000 00000000
000477F0  00000000 00000000
000477F8  00000000 00000000
00047800  00000000 00000000
00047808  00000000 00000000
00047810  00000000 00000000
00047818  00000000 00000000
00047820  00000000 00000000
00047828  00000000 00000000
00047830  00000000 00000000
00047838  00000000 00000000
                                        8432
                                        8433           TSTLBL PL                  Parameter List
00047840  D7D9C940 40406E6E             8434+          DC    CL32'PRI   >>>PL52<<<'    eye catcher
00047848  6ED7D3F5 F24C4C4C
00047850  40404040 40404040
00047858  40404040 40404040
00047860                                8435+PL52      DS    0C
00047860  00000000 00000000             8436           DC    XL352'00'
00047868  00000000 00000000
00047870  00000000 00000000
00047878  00000000 00000000
00047880  00000000 00000000
00047888  00000000 00000000
00047890  00000000 00000000
00047898  00000000 00000000
000478A0  00000000 00000000
000478A8  00000000 00000000
000478B0  00000000 00000000
000478B8  00000000 00000000
000478C0  00000000 00000000
000478C8  00000000 00000000
000478D0  00000000 00000000
000478D8  00000000 00000000
000478E0  00000000 00000000
000478E8  00000000 00000000
000478F0  00000000 00000000
000478F8  00000000 00000000
00047900  00000000 00000000
00047908  00000000 00000000
00047910  00000000 00000000
00047918  00000000 00000000
00047920  00000000 00000000
00047928  00000000 00000000
00047930  00000000 00000000
00047938  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00047940  00000000 00000000
00047948  00000000 00000000
00047950  00000000 00000000
00047958  00000000 00000000
00047960  00000000 00000000
00047968  00000000 00000000
00047970  00000000 00000000
00047978  00000000 00000000
00047980  00000000 00000000
00047988  00000000 00000000
00047990  00000000 00000000
00047998  00000000 00000000
000479A0  00000000 00000000
000479A8  00000000 00000000
000479B0  00000000 00000000
000479B8  00000000 00000000
                                        8437
                                        8438          TSTLBL FMSG               fail message
000479C0  D7D9C940 40406E6E             8439+         DC    CL32'PRI   >>>FMSG52<<<'  eye catcher
000479C8  6EC6D4E2 C7F5F24C
000479D0  4C4C4040 40404040
000479D8  40404040 40404040
000479E0                                8440+FMSG52   DS    0C
000479E0  86818993 858440D7             8441          DC    C'failed PLO-Extension '
000479E8  D3D660C5 A7A38595
000479F0  A2899695 40
000479F5  C3969497 81998540             8442          DC    C'Compare and Swap and Double Store (CSDSTO: 28) : NE'
000479FD  81958440 E2A68197
00047A05  40819584 40C496A4
00047A0D  82938540 E2A39699
00047A15  85404DC3 E2C4E2E3
00047A1D  D67A40F2 F85D407A
00047A25  40D5C5
00047A28  40404040 40404040             8443          DC    CL50' '
00047A30  40404040 40404040
00047A38  40404040 40404040
00047A40  40404040 40404040
00047A48  40404040 40404040
00047A50  40404040 40404040
00047A58  4040
                                        8444          TSTEND
00047A5A  D7D9C940 4040C5D5             8445+TEND52   DC    CL32'PRI   END of test: 52 <<<'
00047A62  C4409686 40A385A2
00047A6A  A37A40F5 F2404C4C
00047A72  4C404040 40404040
                                        8446
                                        8447 * ------------------------------------------------------------------
                                        8448 * Test Compare and Swap and Double Store (CSDSTO: 28) instruction : EQ
                                        8449 * ------------------------------------------------------------------
                                        8450          TSTHDR 0,99,28            Op1-C == OP2 (cc=0, UR=99)
00047A7A                  00047A7A  00047C00  8451+         ORG   PLOETST+(128*K)+53*(3*K)
00047C00  D7D9C940 4040E2E3             8452+         DC    CL32'PRI   START of test: 53'
00047C08  C1D9E340 968640A3
00047C10  85A2A37A 40F5F340
00047C18  40404040 40404040
00047C20                  00047C20      8453+         USING *,R5                base for test data and test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00047C20                       00000000            8454+         USING PLOETEST,R5
00047C20  00047C50                                 8455+T53      DC    A(CODE53)           address of test code
00047C24  0035                                     8456+         DC    H'53'               test number
00047C26  00                                       8457+         DC    X'00'
00047C27  1C                                       8458+         DC    HL1'28'             Function
00047C28  63                                       8459+         DC    HL1'99'             expected unequal results (UR)
00047C29  00                                       8460+         DC    HL1'00'             found UR
00047C2A  00                                       8461+         DC    HL1'0'              cc
00047C2B  07                                       8462+         DC    HL1'7'              not expected CC mask
00047C2C  00                                       8463+         DS    X                   extracted cc
00047C30  00000000 00000000                        8464+         DS    2F                  extract PSW after test (has CC)
00047C38  000485E0                                 8465+         DC    A(FMSG53)           address of test fail message
00047C3C  00048340                                 8466+         DC    A(OPSRC53)          address of operand source
00047C40  00048220                                 8467+         DC    A(OPDST53)          address of operand destination
00047C44  00048460                                 8468+         DC    A(PL53)             address of parameter list
00047C48  000041AD                                 8469+         DC    A(RANDX+53)         address of random source data
00047C4C  0004865A                                 8470+         DC    A(TEND53)           address of test end eye catche
                                                   8471
                                                   8472          TSTCODE
00047C50                                           8473+CODE53   DS    0F
                                                   8474 *                                  load random source
00047C50  E310 501C 0014                 0000001C  8475          LGF   R1,OPSRCA           R1: OPSRC - Primary
00047C56  B904 0061                                8476          LGR   R6,R1               R6: OPSRC - Secondary
00047C5A  B904 0071                                8477          LGR   R7,R1               R7: OPSRC - Home
                                                   8478
00047C5E  E320 5028 0014                 00000028  8479          LGF   R2,RANDXA           R2: address random data
00047C64  E330 5024 0014                 00000024  8480          LGF   R3,PLA              R3: address PL
00047C6A  E340 5020 0014                 00000020  8481          LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   8482
00047C70  B219 0200                      00000200  8483          SAC   ARMODE
00047C74  D21F 4000 2000       00000000  00000000  8484          MVC   0(32,R4),0(R2)      load OP2  (same as OP1C below)
                                                   8485
00047C7A  D21F 1000 20C0       00000000  000000C0  8486          MVC   0(32,R1),192+0(R2)        OP4 - Primary
00047C80  D21F 6020 20E0       00000020  000000E0  8487          MVC   32(32,R6),192+32(R2)      OP6 - Secondary
                                                   8488 *
                                                   8489 * Compare and Swap and Double Store (CSDSTO: 28)
                                                   8490 * Build Parambter List
                                                   8491 *
00047C86  D21F 3000 2000       00000000  00000000  8492          MVC   0(32,R3),0(R2)      Load PL OP1-C
00047C8C  D21F 3020 2020       00000020  00000020  8493          MVC   32(32,R3),32(R2)    load PL OP1-R
00047C92  D21F 3070 2040       00000070  00000040  8494          MVC   112(32,R3),64(R2)   load PL OP3
00047C98  D21F 3090 2060       00000090  00000060  8495          MVC   144(32,R3),96(R2)   load PL OP5
                                                   8496 ****
                                                   8497 * Force a EQ by changing the Op1-C == Op2
                                                   8498 ****
00047C9E  D21F 3000 4000       00000000  00000000  8499          MVC   0(32,R3),0(R4)
                                                   8500 ****
00047CA4  E313 0048 0024                 00000048  8501          STG   R1,72(R3)           load OP4 address in PL (pri space)
00047CAA  E54C 3044 0000                 00000044  8502          MVHI  68(R3),PRIALET
                                                   8503
00047CB0  4111 0020                      00000020  8504          LA    R1,32(R1)
00047CB4  E313 0068 0024                 00000068  8505          STG   R1,104(R3)          load OP6 address in PL (sec space)
00047CBA  E54C 3064 0001                 00000064  8506          MVHI  100(R3),SECALET
                                                   8507
00047CC0  B982 0000                                8508          XGR   R0,R0
00047CC4  4100 001C                      0000001C  8509          LA    R0,28
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    8510
00047CC8  B219 0200                     00000200  8511          SAC   ARMODE
00047CCC  EE0B 4000 3000      00000000  00000000  8512          PLO   0,0(R4),R11,0(R3)   C/S Double Store (CSDSTO: 28)
00047CD2  B219 0000                     00000000  8513          SAC   PRIMODE
                                                    8514
00047CD6  B98D 0020                               8515          EPSW  R2,R0               extract psw
00047CDA  5020 5010                     00000010  8516          ST    R2,CCPSW              to save CC
00047CDE  E310 500B 0076                0000000B  8517          LB    R1,CCMASK           (failure CC mask)
00047CE4  8910 0004                     00000004  8518          SLL   R1,4                (shift to BC instr CC position)
00047CE8  4410 512A                     00047D4A  8519          EX    R1,F282TSTCC            fail if...
                                                    8520
                          00047CEC  00000001  8521 F282CKUR EQU   *
00047CEC  D200 5009 3030  00000009  00000030  8522          MVC   URFOUND,48(R3)      Extract UR
00047CF2  D500 5008 3030  00000008  00000030  8523          CLC   UR,48(R3)
00047CF8  4780 50E0                     00047D00  8524          BE    F282CKRES
00047CFC  45F0 8388                     00002388  8525          BAL   R15,URMSG
                                                    8526 *
                                                    8527 * compare results
                                                    8528 *
                          00047D00  00000001  8529 F282CKRES EQU   *
                                                    8530 *                              reset addresses
00047D00  E310 501C 0014                0000001C  8531          LGF   R1,OPSRCA           R1:  OPSRC - Primary
00047D06  B904 0061                               8532          LGR   R6,R1               R6:  OPSRC - Secondary
00047D0A  B904 0071                               8533          LGR   R7,R1               R7:  OPSRC - Home
00047D0E  E330 5024 0014                00000024  8534          LGF   R3,PLA              R3:  address PL
00047D14  E340 5020 0014                00000020  8535          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                                    8536
00047D1A  B219 0200                     00000200  8537          SAC   ARMODE
00047D1E  D51F 1000 3070      00000000  00000070  8538          CLC   0(32,R1),112(R3)    OP4 ==  PL Op3
00047D24  4770 5120                     00047D40  8539          BNE   F282XX                   different, error
00047D28  D51F 6020 3090      00000020  00000090  8540          CLC   32(32,R6),144(R3)   OP6 ==  PL Op5
00047D2E  4770 5120                     00047D40  8541          BNE   F282XX                   different, error
00047D32  D51F 4000 3020      00000000  00000020  8542          CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
00047D38  B219 0000                     00000000  8543          SAC   PRIMODE
00047D3C  478B 0000                     00000000  8544          BE    0(R11)                   same, return
                                                    8545
                          00047D40  00000001  8546 F282XX   EQU   *
00047D40  B219 0000                     00000000  8547          SAC   PRIMODE
00047D44  45F0 8450                     00002450  8548          BAL   R15,FAILMSG
00047D48  07FB                                    8549          BR    R11
                                                    8550
00047D4A  4700 512E                     00047D4E  8551 F282TSTCC BC    0,F282CCMSG          fail if unexpected condition code
                          00047D4E  00000001  8552 F282CCMSG EQU   *
00047D4E  45F0 82B2                     000022B2  8553          BAL   R15,CCMSG
00047D52  47F0 50CC                     00047CEC  8554          B     F282CKUR
                                                    8555
                                                    8556          TSTDATA
00047D56                      00047D56  00048200  8557+         ORG   PLOETST+(128*K)+53*(3*K)+(K+512)
                                                    8558          TSTLBL OPDST              OP2, Op4, Op6
00048200  D7D9C940 40406E6E                         8559+         DC    CL32'PRI   >>>OPDST53<<<' eye catcher
00048208  6ED6D7C4 E2E3F5F3
00048210  4C4C4C40 40404040
00048218  40404040 40404040
00048220                                            8560+OPDST53  DS    0C
00048220  00000000 00000000                         8561          DC    XL256'0'
00048228  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048230  00000000 00000000
00048238  00000000 00000000
00048240  00000000 00000000
00048248  00000000 00000000
00048250  00000000 00000000
00048258  00000000 00000000
00048260  00000000 00000000
00048268  00000000 00000000
00048270  00000000 00000000
00048278  00000000 00000000
00048280  00000000 00000000
00048288  00000000 00000000
00048290  00000000 00000000
00048298  00000000 00000000
000482A0  00000000 00000000
000482A8  00000000 00000000
000482B0  00000000 00000000
000482B8  00000000 00000000
000482C0  00000000 00000000
000482C8  00000000 00000000
000482D0  00000000 00000000
000482D8  00000000 00000000
000482E0  00000000 00000000
000482E8  00000000 00000000
000482F0  00000000 00000000
000482F8  00000000 00000000
00048300  00000000 00000000
00048308  00000000 00000000
00048310  00000000 00000000
00048318  00000000 00000000
                                      8562
                                      8563           TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
00048320  D7D9C940 40406E6E           8564+          DC     CL32'PRI   >>>OPSRC53<<<' eye catcher
00048328  6ED6D7E2 D9C3F5F3
00048330  4C4C4C40 40404040
00048338  40404040 40404040
00048340                              8565+OPSRC53   DS     0C
00048340  00000000 00000000           8566           DC     XL256'0'
00048348  00000000 00000000
00048350  00000000 00000000
00048358  00000000 00000000
00048360  00000000 00000000
00048368  00000000 00000000
00048370  00000000 00000000
00048378  00000000 00000000
00048380  00000000 00000000
00048388  00000000 00000000
00048390  00000000 00000000
00048398  00000000 00000000
000483A0  00000000 00000000
000483A8  00000000 00000000
000483B0  00000000 00000000
000483B8  00000000 00000000
000483C0  00000000 00000000
000483C8  00000000 00000000
000483D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

000483D8  00000000 00000000
000483E0  00000000 00000000
000483E8  00000000 00000000
000483F0  00000000 00000000
000483F8  00000000 00000000
00048400  00000000 00000000
00048408  00000000 00000000
00048410  00000000 00000000
00048418  00000000 00000000
00048420  00000000 00000000
00048428  00000000 00000000
00048430  00000000 00000000
00048438  00000000 00000000
                                          8567
                                          8568           TSTLBL PL                  Parameter List
00048440  D7D9C940 40406E6E               8569+          DC     CL32'PRI   >>>PL53<<<'    eye catcher
00048448  6ED7D3F5 F34C4C4C
00048450  40404040 40404040
00048458  40404040 40404040
00048460                                  8570+PL53      DS     0C
00048460  00000000 00000000               8571           DC     XL352'00'
00048468  00000000 00000000
00048470  00000000 00000000
00048478  00000000 00000000
00048480  00000000 00000000
00048488  00000000 00000000
00048490  00000000 00000000
00048498  00000000 00000000
000484A0  00000000 00000000
000484A8  00000000 00000000
000484B0  00000000 00000000
000484B8  00000000 00000000
000484C0  00000000 00000000
000484C8  00000000 00000000
000484D0  00000000 00000000
000484D8  00000000 00000000
000484E0  00000000 00000000
000484E8  00000000 00000000
000484F0  00000000 00000000
000484F8  00000000 00000000
00048500  00000000 00000000
00048508  00000000 00000000
00048510  00000000 00000000
00048518  00000000 00000000
00048520  00000000 00000000
00048528  00000000 00000000
00048530  00000000 00000000
00048538  00000000 00000000
00048540  00000000 00000000
00048548  00000000 00000000
00048550  00000000 00000000
00048558  00000000 00000000
00048560  00000000 00000000
00048568  00000000 00000000
00048570  00000000 00000000
00048578  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00048580  00000000 00000000
00048588  00000000 00000000
00048590  00000000 00000000
00048598  00000000 00000000
000485A0  00000000 00000000
000485A8  00000000 00000000
000485B0  00000000 00000000
000485B8  00000000 00000000
                                        8572
                                        8573          TSTLBL FMSG               fail message
000485C0  D7D9C940 40406E6E             8574+         DC    CL32'PRI   >>>FMSG53<<<'  eye catcher
000485C8  6EC6D4E2 C7F5F34C
000485D0  4C4C4040 40404040
000485D8  40404040 40404040
000485E0                                8575+FMSG53   DS    0C
000485E0  86818993 858440D7             8576          DC    C'failed PLO-Extension '
000485E8  D3D660C5 A7A38595
000485F0  A2899695 40
000485F5  C3969497 81998540             8577          DC    C'Compare and Swap and Double Store (CSDSTO: 28) : EQ'
000485FD  81958440 E2A68197
00048605  40819584 40C496A4
0004860D  82938540 E2A39699
00048615  85404DC3 E2C4E2E3
0004861D  D67A40F2 F85D407A
00048625  40C5D8
00048628  40404040 40404040             8578          DC    CL50' '
00048630  40404040 40404040
00048638  40404040 40404040
00048640  40404040 40404040
00048648  40404040 40404040
00048650  40404040 40404040
00048658  4040
                                        8579          TSTEND
0004865A  D7D9C940 4040C5D5             8580+TEND53   DC    CL32'PRI   END of test: 53 <<<'
00048662  C4409686 40A385A2
0004866A  A37A40F5 F3404C4C
00048672  4C404040 40404040
                                        8581
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8583 *================================================================================
                                                   8584
                                                   8585 * ------------------------------------------------------------------------
                                                   8586 * Test Compare and Swap and Store (CSSTO: 27) instruction : NE
                                                   8587 * ------------------------------------------------------------------------
                                                   8588          TSTHDR 1,99,26            Op1-C != OP2 (cc=1, UR=99)
0004867A                          0004867A  00048800 8589+         ORG   PLOETST+(128*K)+54*(3*K)
00048800  D7D9C940 4040E2E3                        8590+         DC    CL32'PRI   START of test: 54'
00048808  C1D9E340 968640A3
00048810  85A2A37A 40F5F440
00048818  40404040 40404040
00048820                          00048820         8591+         USING *,R5                base for test data and test routine
00048820                          00000000         8592+         USING PLOETEST,R5
00048820  00048850                                 8593+T54      DC    A(CODE54)           address of test code
00048824  0036                                     8594+         DC    H'54'               test number
00048826  00                                       8595+         DC    X'00'
00048827  1A                                       8596+         DC    HL1'26'             Function
00048828  63                                       8597+         DC    HL1'99'             expected unequal results (UR)
00048829  00                                       8598+         DC    HL1'00'             found UR
0004882A  01                                       8599+         DC    HL1'1'              cc
0004882B  0B                                       8600+         DC    HL1'11'             not expected CC mask
0004882C  00                                       8601+         DS    X                   extracted cc
00048830  00000000 00000000                        8602+         DS    2F                  extract PSW after test (has CC)
00048838  000491E0                                 8603+         DC    A(FMSG54)           address of test fail message
0004883C  00048F40                                 8604+         DC    A(OPSRC54)          address of operand source
00048840  00048E20                                 8605+         DC    A(OPDST54)          address of operand destination
00048844  00049060                                 8606+         DC    A(PL54)             address of parameter list
00048848  000041AE                                 8607+         DC    A(RANDX+54)         address of random source data
0004884C  00049252                                 8608+         DC    A(TEND54)           address of test end eye catche
                                                   8609
                                                   8610          TSTCODE
00048850                                           8611+CODE54   DS    0F
                                                   8612 *                                  load random source
00048850  E310 501C 0014                 0000001C  8613          LGF   R1,OPSRCA           R1: OPSRC - Primary
00048856  B904 0061                                8614          LGR   R6,R1               R6: OPSRC - Secondary
0004885A  B904 0071                                8615          LGR   R7,R1               R7: OPSRC - Home
                                                   8616
0004885E  E320 5028 0014                 00000028  8617          LGF   R2,RANDXA           R2: address random data
00048864  E330 5024 0014                 00000024  8618          LGF   R3,PLA              R3: address PL
0004886A  E340 5020 0014                 00000020  8619          LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   8620
00048870  B219 0200                      00000200  8621          SAC   ARMODE
00048874  D21F 4000 2000       00000000  00000000  8622          MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004887A  D21F 6000 20C0       00000000  000000C0  8623          MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   8624 *
                                                   8625 * Compare and Swap and Store (CSSTO: 27)
                                                   8626 * Build Parambter List
                                                   8627 *
00048880  D21F 3000 2000       00000000  00000000  8628          MVC   0(32,R3),0(R2)      Load PL OP1-C
00048886  D21F 3020 2020       00000020  00000020  8629          MVC   32(32,R3),32(R2)    load PL OP1-R
0004888C  D21F 3050 2040       00000050  00000040  8630          MVC   80(32,R3),64(R2)    load PL OP3
                                                   8631 ****
                                                   8632 * force a failure by changing Op1-c
                                                   8633 ****
00048892  D203 3000 2021       00000000  00000021  8634          MVC   0(4,R3),33(R2)
                                                   8635 ****
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00048898  E313 0048 0024                 00000048  8636          STG   R1,72(R3)           load OP4 address in PL (sec space)
0004889E  E54C 3044 0001                 00000044  8637          MVHI  68(R3),SECALET
                                                   8638
000488A4  B982 0000                              8639          XGR   R0,R0
000488A8  4100 001B                      0000001B  8640          LA    R0,27
                                                   8641
000488AC  B219 0200                      00000200  8642          SAC   ARMODE
000488B0  EE0B 4000 3000       00000000  00000000  8643          PLO   0,0(R4),R11,0(R3)   C/S Store (CSSTO: 27)
000488B6  B219 0000                      00000000  8644          SAC   PRIMODE
                                                   8645
000488BA  B98D 0020                              8646          EPSW  R2,R0               extract psw
000488BE  5020 5010                      00000010  8647          ST    R2,CCPSW              to save CC
000488C2  E310 500B 0076                 0000000B  8648          LB    R1,CCMASK           (failure CC mask)
000488C8  8910 0004                      00000004  8649          SLL   R1,4                (shift to BC instr CC position)
000488CC  4410 50FA                      0004891A  8650          EX    R1,F271TSTCC            fail if...
                                                   8651
                                000488D0  00000001  8652 F271CKUR EQU   *
000488D0  D200 5009 3030       00000009  00000030  8653          MVC   URFOUND,48(R3)      Extract UR
000488D6  D500 5008 3030       00000008  00000030  8654          CLC   UR,48(R3)
000488DC  4780 50C4                      000488E4  8655          BE    F271CKRES
000488E0  45F0 8388                      00002388  8656          BAL   R15,URMSG
                                                   8657 *
                                                   8658 * compare results
                                                   8659 *
                                000488E4  00000001  8660 F271CKRES EQU   *
                                                   8661 *                              reset addresses
000488E4  E310 501C 0014                 0000001C  8662          LGF   R1,OPSRCA           R1:  OPSRC - Primary
000488EA  B904 0061                              8663          LGR   R6,R1               R6:  OPSRC - Secondary
000488EE  B904 0071                              8664          LGR   R7,R1               R7:  OPSRC - Home
000488F2  E330 5024 0014                 00000024  8665          LGF   R3,PLA              R3:  address PL
000488F8  E340 5020 0014                 00000020  8666          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                                   8667
000488FE  B219 0200                      00000200  8668          SAC   ARMODE
00048902  D51F 4000 3000       00000000  00000000  8669          CLC   0(32,R4),0(R3)      Op2 == OP1C?
00048908  B219 0000                      00000000  8670          SAC   PRIMODE
0004890C  478B 0000                      00000000  8671          BE    0(R11)              same, return
                                                   8672
                                00048910  00000001  8673 F271XX   EQU   *
00048910  B219 0000                      00000000  8674          SAC   PRIMODE
00048914  45F0 8450                      00002450  8675          BAL   R15,FAILMSG
00048918  07FB                                     8676          BR    R11
                                                   8677
0004891A  4700 50FE                      0004891E  8678 F271TSTCC BC    0,F271CCMSG          fail if unexpected condition code
                                0004891E  00000001  8679 F271CCMSG EQU  *
0004891E  45F0 82B2                      000022B2  8680          BAL   R15,CCMSG
00048922  47F0 50B0                      000488D0  8681          B     F271CKUR
                                                   8682
                                                   8683          TSTDATA
00048926                        00048926  00048E00  8684+         ORG   PLOETST+(128*K)+54*(3*K)+(K+512)
                                                   8685          TSTLBL OPDST              OP2, Op4
00048E00  D7D9C940 40406E6E                        8686+         DC    CL32'PRI   >>>OPDST54<<<' eye catcher
00048E08  6ED6D7C4 E2E3F5F4
00048E10  4C4C4C40 40404040
00048E18  40404040 40404040
00048E20                                           8687+OPDST54  DS    0C
00048E20  00000000 00000000                        8688          DC    XL256'0'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048E28  00000000 00000000
00048E30  00000000 00000000
00048E38  00000000 00000000
00048E40  00000000 00000000
00048E48  00000000 00000000
00048E50  00000000 00000000
00048E58  00000000 00000000
00048E60  00000000 00000000
00048E68  00000000 00000000
00048E70  00000000 00000000
00048E78  00000000 00000000
00048E80  00000000 00000000
00048E88  00000000 00000000
00048E90  00000000 00000000
00048E98  00000000 00000000
00048EA0  00000000 00000000
00048EA8  00000000 00000000
00048EB0  00000000 00000000
00048EB8  00000000 00000000
00048EC0  00000000 00000000
00048EC8  00000000 00000000
00048ED0  00000000 00000000
00048ED8  00000000 00000000
00048EE0  00000000 00000000
00048EE8  00000000 00000000
00048EF0  00000000 00000000
00048EF8  00000000 00000000
00048F00  00000000 00000000
00048F08  00000000 00000000
00048F10  00000000 00000000
00048F18  00000000 00000000
                                          8689
                                          8690           TSTLBL OPSRC              Op1-C, Op3, Op1-R
00048F20  D7D9C940 40406E6E               8691+          DC    CL32'PRI   >>>OPSRC54<<<' eye catcher
00048F28  6ED6D7E2 D9C3F5F4
00048F30  4C4C4C40 40404040
00048F38  40404040 40404040
00048F40                                  8692+OPSRC54   DS    0C
00048F40  00000000 00000000               8693           DC    XL256'0'
00048F48  00000000 00000000
00048F50  00000000 00000000
00048F58  00000000 00000000
00048F60  00000000 00000000
00048F68  00000000 00000000
00048F70  00000000 00000000
00048F78  00000000 00000000
00048F80  00000000 00000000
00048F88  00000000 00000000
00048F90  00000000 00000000
00048F98  00000000 00000000
00048FA0  00000000 00000000
00048FA8  00000000 00000000
00048FB0  00000000 00000000
00048FB8  00000000 00000000
00048FC0  00000000 00000000
00048FC8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00048FD0  00000000 00000000
00048FD8  00000000 00000000
00048FE0  00000000 00000000
00048FE8  00000000 00000000
00048FF0  00000000 00000000
00048FF8  00000000 00000000
00049000  00000000 00000000
00049008  00000000 00000000
00049010  00000000 00000000
00049018  00000000 00000000
00049020  00000000 00000000
00049028  00000000 00000000
00049030  00000000 00000000
00049038  00000000 00000000
                                          8694
                                          8695          TSTLBL PL                 Parameter List
00049040  D7D9C940 40406E6E               8696+         DC    CL32'PRI   >>>PL54<<<'    eye catcher
00049048  6ED7D3F5 F44C4C4C
00049050  40404040 40404040
00049058  40404040 40404040
00049060                                  8697+PL54     DS    0C
00049060  00000000 00000000               8698          DC    XL352'00'
00049068  00000000 00000000
00049070  00000000 00000000
00049078  00000000 00000000
00049080  00000000 00000000
00049088  00000000 00000000
00049090  00000000 00000000
00049098  00000000 00000000
000490A0  00000000 00000000
000490A8  00000000 00000000
000490B0  00000000 00000000
000490B8  00000000 00000000
000490C0  00000000 00000000
000490C8  00000000 00000000
000490D0  00000000 00000000
000490D8  00000000 00000000
000490E0  00000000 00000000
000490E8  00000000 00000000
000490F0  00000000 00000000
000490F8  00000000 00000000
00049100  00000000 00000000
00049108  00000000 00000000
00049110  00000000 00000000
00049118  00000000 00000000
00049120  00000000 00000000
00049128  00000000 00000000
00049130  00000000 00000000
00049138  00000000 00000000
00049140  00000000 00000000
00049148  00000000 00000000
00049150  00000000 00000000
00049158  00000000 00000000
00049160  00000000 00000000
00049168  00000000 00000000
00049170  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
00049178  00000000 00000000
00049180  00000000 00000000
00049188  00000000 00000000
00049190  00000000 00000000
00049198  00000000 00000000
000491A0  00000000 00000000
000491A8  00000000 00000000
000491B0  00000000 00000000
000491B8  00000000 00000000
                                        8699
                                        8700          TSTLBL FMSG               fail message
000491C0  D7D9C940 40406E6E             8701+         DC    CL32'PRI   >>>FMSG54<<<'  eye catcher
000491C8  6EC6D4E2 C7F5F44C
000491D0  4C4C4040 40404040
000491D8  40404040 40404040
000491E0                                8702+FMSG54   DS    0C
000491E0  86818993 858440D7             8703          DC    C'failed PLO-Extension '
000491E8  D3D660C5 A7A38595
000491F0  A2899695 40
000491F5  C3969497 81998540             8704          DC    C'Compare and Swap and Store (CSSTO: 27) : NE'
000491FD  81958440 E2A68197
00049205  40819584 40E2A396
0004920D  9985404D C3E2E2E3
00049215  D67A40F2 F75D407A
0004921D  40D5C5
00049220  40404040 40404040             8705          DC    CL50' '
00049228  40404040 40404040
00049230  40404040 40404040
00049238  40404040 40404040
00049240  40404040 40404040
00049248  40404040 40404040
00049250  4040
                                        8706          TSTEND
00049252  D7D9C940 4040C5D5             8707+TEND54   DC    CL32'PRI   END of test: 54 <<<'
0004925A  C4409686 40A385A2
00049262  A37A40F5 F4404C4C
0004926A  4C404040 40404040
                                        8708
                                        8709 * ------------------------------------------------------------------
                                        8710 * Test Compare and Swap and Store (CSDSTO: 27) instruction : EQ
                                        8711 * ------------------------------------------------------------------
                                        8712          TSTHDR 0,99,27            Op1-C == OP2 (cc=0, UR=99)
00049272                  00049272  00049400  8713+         ORG   PLOETST+(128*K)+55*(3*K)
00049400  D7D9C940 4040E2E3             8714+         DC    CL32'PRI   START of test: 55'
00049408  C1D9E340 968640A3
00049410  85A2A37A 40F5F540
00049418  40404040 40404040
00049420                  00049420      8715+         USING *,R5                base for test data and test routine
00049420                  00000000      8716+         USING PLOETEST,R5
00049420  00049450                      8717+T55      DC    A(CODE55)           address of test code
00049424  0037                          8718+         DC    H'55'               test number
00049426  00                            8719+         DC    X'00'
00049427  1B                            8720+         DC    HL1'27'             Function
00049428  63                            8721+         DC    HL1'99'             expected unequal results (UR)
00049429  00                            8722+         DC    HL1'00'             found UR
0004942A  00                            8723+         DC    HL1'0'              cc
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004942B  07                                       8724+          DC    HL1'7'              not expected CC mask
0004942C  00                                       8725+          DS    X                   extracted cc
00049430  00000000 00000000                        8726+          DS    2F                  extract PSW after test (has CC)
00049438  00049DE0                                 8727+          DC    A(FMSG55)           address of test fail message
0004943C  00049B40                                 8728+          DC    A(OPSRC55)          address of operand source
00049440  00049A20                                 8729+          DC    A(OPDST55)          address of operand destination
00049444  00049C60                                 8730+          DC    A(PL55)             address of parameter list
00049448  000041AF                                 8731+          DC    A(RANDX+55)         address of random source data
0004944C  00049E52                                 8732+          DC    A(TEND55)           address of test end eye catche
                                                   8733
                                                   8734           TSTCODE
00049450                                           8735+CODE55    DS    0F
                                                   8736 *                                   load random source
00049450  E310 501C 0014                0000001C   8737           LGF   R1,OPSRCA           R1: OPSRC - Primary
00049456  B904 0061                                8738           LGR   R6,R1               R6: OPSRC - Secondary
0004945A  B904 0071                                8739           LGR   R7,R1               R7: OPSRC - Home
                                                   8740
0004945E  E320 5028 0014                00000028   8741           LGF   R2,RANDXA           R2: address random data
00049464  E330 5024 0014                00000024   8742           LGF   R3,PLA              R3: address PL
0004946A  E340 5020 0014                00000020   8743           LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   8744
00049470  B219 0200                     00000200   8745           SAC   ARMODE
00049474  D21F 4000 2000      00000000  00000000   8746           MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004947A  D21F 6000 20C0      00000000  000000C0   8747           MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   8748 *
                                                   8749 * Compare and Swap and Double Store (CSDSTO: 28)
                                                   8750 * Build Parambter List
                                                   8751 *
00049480  D21F 3000 2000      00000000  00000000   8752           MVC   0(32,R3),0(R2)      Load PL OP1-C
00049486  D21F 3020 2020      00000020  00000020   8753           MVC   32(32,R3),32(R2)    load PL OP1-R
0004948C  D21F 3050 2040      00000050  00000040   8754           MVC   80(32,R3),64(R2)    load PL OP3
                                                   8755
00049492  D21F 6000 20C0      00000000  000000C0   8756           MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   8757
                                                   8758 ****
                                                   8759 * Force a EQ by changing Op1-C == Op2
                                                   8760 ****
00049498  D21F 3000 4000      00000000  00000000   8761           MVC   0(32,R3),0(R4)
                                                   8762 ****
0004949E  E313 0048 0024                00000048   8763           STG   R1,72(R3)           load OP4 address in PL (sec space)
000494A4  E54C 3044 0001                00000044   8764           MVHI  68(R3),SECALET
                                                   8765
000494AA  B982 0000                                8766           XGR   R0,R0
000494AE  4100 001B                     0000001B   8767           LA    R0,27
                                                   8768
000494B2  B219 0200                     00000200   8769           SAC   ARMODE
000494B6  EE0B 4000 3000      00000000  00000000   8770           PLO   0,0(R4),R11,0(R3)   C/S Store (CSSTO: 27)
000494BC  B219 0000                     00000000   8771           SAC   PRIMODE
                                                   8772
000494C0  B98D 0020                                8773           EPSW  R2,R0               extract psw
000494C4  5020 5010                     00000010   8774           ST    R2,CCPSW              to save CC
000494C8  E310 500B 0076                0000000B   8775           LB    R1,CCMASK           (failure CC mask)
000494CE  8910 0004                     00000004   8776           SLL   R1,4                (shift to BC instr CC position)
000494D2  4410 510A                     0004952A   8777           EX    R1,F272TSTCC            fail if...
                                                   8778
                                        000494D6   00000001   8779 F272CKUR EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
000494D6  D200 5009 3030      00000009  00000030  8780          MVC   URFOUND,48(R3)      Extract UR
000494DC  D500 5008 3030      00000008  00000030  8781          CLC   UR,48(R3)
000494E2  4780 50CA                     000494EA  8782          BE    F272CKRES
000494E6  45F0 8388                     00002388  8783          BAL   R15,URMSG
                                                  8784 *
                                                  8785 * compare results
                                                  8786 *
                          000494EA      00000001  8787 F272CKRES EQU   *
                                                  8788 *                                  reset addresses
000494EA  E310 501C 0014                0000001C  8789          LGF   R1,OPSRCA           R1:  OPSRC - Primary
000494F0  B904 0061                               8790          LGR   R6,R1               R6:  OPSRC - Secondary
000494F4  B904 0071                               8791          LGR   R7,R1               R7:  OPSRC - Home
000494F8  E330 5024 0014                00000024  8792          LGF   R3,PLA              R3:  address PL
000494FE  E340 5020 0014                00000020  8793          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                                  8794
00049504  B219 0200                     00000200  8795          SAC   ARMODE
00049508  D51F 6000 3050      00000000  00000050  8796          CLC   0(32,R6),80(R3)     OP4 ==  PL Op3
0004950E  4770 5100                     00049520  8797          BNE   F272XX                   different, error
00049512  D51F 4000 3020      00000000  00000020  8798          CLC   0(32,R4),32(R3)     OP2 ==  PL Op-R
00049518  B219 0000                     00000000  8799          SAC   PRIMODE
0004951C  478B 0000                     00000000  8800          BE    0(R11)                   same, return
                                                  8801
                          00049520      00000001  8802 F272XX   EQU   *
00049520  B219 0000                     00000000  8803          SAC   PRIMODE
00049524  45F0 8450                     00002450  8804          BAL   R15,FAILMSG
00049528  07FB                                    8805          BR    R11
                                                  8806
0004952A  4700 510E                     0004952E  8807 F272TSTCC BC    0,F272CCMSG         fail if unexpected condition code
                          0004952E      00000001  8808 F272CCMSG EQU   *
0004952E  45F0 82B2                     000022B2  8809          BAL   R15,CCMSG
00049532  47F0 50B6                     000494D6  8810          B     F272CKUR
                                                  8811
                                                  8812          TSTDATA
00049536                  00049536      00049A00  8813+         ORG   PLOETST+(128*K)+55*(3*K)+(K+512)
                                                  8814          TSTLBL OPDST                OP2, Op4
00049A00  D7D9C940 40406E6E                       8815+         DC    CL32'PRI   >>>OPDST55<<<' eye catcher
00049A08  6ED6D7C4 E2E3F5F5
00049A10  4C4C4C40 40404040
00049A18  40404040 40404040
00049A20                                  8816+OPDST55  DS    0C
00049A20  00000000 00000000               8817          DC    XL256'0'
00049A28  00000000 00000000
00049A30  00000000 00000000
00049A38  00000000 00000000
00049A40  00000000 00000000
00049A48  00000000 00000000
00049A50  00000000 00000000
00049A58  00000000 00000000
00049A60  00000000 00000000
00049A68  00000000 00000000
00049A70  00000000 00000000
00049A78  00000000 00000000
00049A80  00000000 00000000
00049A88  00000000 00000000
00049A90  00000000 00000000
00049A98  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

00049AA0  00000000 00000000
00049AA8  00000000 00000000
00049AB0  00000000 00000000
00049AB8  00000000 00000000
00049AC0  00000000 00000000
00049AC8  00000000 00000000
00049AD0  00000000 00000000
00049AD8  00000000 00000000
00049AE0  00000000 00000000
00049AE8  00000000 00000000
00049AF0  00000000 00000000
00049AF8  00000000 00000000
00049B00  00000000 00000000
00049B08  00000000 00000000
00049B10  00000000 00000000
00049B18  00000000 00000000
                                          8818
                                          8819          TSTLBL OPSRC              Op1-C, Op3, Op1-R
00049B20  D7D9C940 40406E6E               8820+         DC    CL32'PRI   >>>OPSRC55<<<' eye catcher
00049B28  6ED6D7E2 D9C3F5F5
00049B30  4C4C4C40 40404040
00049B38  40404040 40404040
00049B40                                  8821+OPSRC55  DS    0C
00049B40  00000000 00000000               8822          DC    XL256'0'
00049B48  00000000 00000000
00049B50  00000000 00000000
00049B58  00000000 00000000
00049B60  00000000 00000000
00049B68  00000000 00000000
00049B70  00000000 00000000
00049B78  00000000 00000000
00049B80  00000000 00000000
00049B88  00000000 00000000
00049B90  00000000 00000000
00049B98  00000000 00000000
00049BA0  00000000 00000000
00049BA8  00000000 00000000
00049BB0  00000000 00000000
00049BB8  00000000 00000000
00049BC0  00000000 00000000
00049BC8  00000000 00000000
00049BD0  00000000 00000000
00049BD8  00000000 00000000
00049BE0  00000000 00000000
00049BE8  00000000 00000000
00049BF0  00000000 00000000
00049BF8  00000000 00000000
00049C00  00000000 00000000
00049C08  00000000 00000000
00049C10  00000000 00000000
00049C18  00000000 00000000
00049C20  00000000 00000000
00049C28  00000000 00000000
00049C30  00000000 00000000
00049C38  00000000 00000000
                                          8823
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8824           TSTLBL PL                  Parameter List
00049C40  D7D9C940 40406E6E                        8825+          DC     CL32'PRI   >>>PL55<<<'    eye catcher
00049C48  6ED7D3F5 F54C4C4C
00049C50  40404040 40404040
00049C58  40404040 40404040
00049C60                                           8826+PL55      DS     0C
00049C60  00000000 00000000                        8827           DC     XL352'00'
00049C68  00000000 00000000
00049C70  00000000 00000000
00049C78  00000000 00000000
00049C80  00000000 00000000
00049C88  00000000 00000000
00049C90  00000000 00000000
00049C98  00000000 00000000
00049CA0  00000000 00000000
00049CA8  00000000 00000000
00049CB0  00000000 00000000
00049CB8  00000000 00000000
00049CC0  00000000 00000000
00049CC8  00000000 00000000
00049CD0  00000000 00000000
00049CD8  00000000 00000000
00049CE0  00000000 00000000
00049CE8  00000000 00000000
00049CF0  00000000 00000000
00049CF8  00000000 00000000
00049D00  00000000 00000000
00049D08  00000000 00000000
00049D10  00000000 00000000
00049D18  00000000 00000000
00049D20  00000000 00000000
00049D28  00000000 00000000
00049D30  00000000 00000000
00049D38  00000000 00000000
00049D40  00000000 00000000
00049D48  00000000 00000000
00049D50  00000000 00000000
00049D58  00000000 00000000
00049D60  00000000 00000000
00049D68  00000000 00000000
00049D70  00000000 00000000
00049D78  00000000 00000000
00049D80  00000000 00000000
00049D88  00000000 00000000
00049D90  00000000 00000000
00049D98  00000000 00000000
00049DA0  00000000 00000000
00049DA8  00000000 00000000
00049DB0  00000000 00000000
00049DB8  00000000 00000000
                                                   8828
                                                   8829           TSTLBL FMSG                fail message
00049DC0  D7D9C940 40406E6E                        8830+          DC     CL32'PRI   >>>FMSG55<<<'  eye catcher
00049DC8  6EC6D4E2 C7F5F54C
00049DD0  4C4C4040 40404040
00049DD8  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
00049DE0                                            8831+FMSG55   DS    0C
00049DE0  86818993 858440D7                         8832          DC    C'failed PLO-Extension '
00049DE8  D3D660C5 A7A38595
00049DF0  A2899695 40
00049DF5  C3969497 81998540                         8833          DC    C'Compare and Swap and Store (CSSTO: 27) : EQ'
00049DFD  81958440 E2A68197
00049E05  40819584 40E2A396
00049E0D  9985404D C3E2E2E3
00049E15  D67A40F2 F75D407A
00049E1D  40C5D8
00049E20  40404040 40404040                         8834          DC    CL50' '
00049E28  40404040 40404040
00049E30  40404040 40404040
00049E38  40404040 40404040
00049E40  40404040 40404040
00049E48  40404040 40404040
00049E50  4040
                                                    8835          TSTEND
00049E52  D7D9C940 4040C5D5                         8836+TEND55   DC    CL32'PRI   END of test: 55 <<<'
00049E5A  C4409686 40A385A2
00049E62  A37A40F5 F5404C4C
00049E6A  4C404040 40404040
                                                    8837
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   8839 *================================================================================
                                                   8840
                                                   8841 * ------------------------------------------------------------------------
                                                   8842 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op1 != Op2
                                                   8843 * ------------------------------------------------------------------------
                                                   8844          TSTHDR 1,99,26            Op1-C != OP2 (cc=1, UR=99)
00049E72                          00049E72  0004A000 8845+         ORG   PLOETST+(128*K)+56*(3*K)
0004A000  D7D9C940 4040E2E3                        8846+         DC    CL32'PRI   START of test: 56'
0004A008  C1D9E340 968640A3
0004A010  85A2A37A 40F5F640
0004A018  40404040 40404040
0004A020                          0004A020         8847+         USING *,R5                base for test data and test routine
0004A020                          00000000         8848+         USING PLOETEST,R5
0004A020  0004A050                                 8849+T56      DC    A(CODE56)           address of test code
0004A024  0038                                     8850+         DC    H'56'               test number
0004A026  00                                       8851+         DC    X'00'
0004A027  1A                                       8852+         DC    HL1'26'             Function
0004A028  63                                       8853+         DC    HL1'99'             expected unequal results (UR)
0004A029  00                                       8854+         DC    HL1'00'             found UR
0004A02A  01                                       8855+         DC    HL1'1'              cc
0004A02B  0B                                       8856+         DC    HL1'11'             not expected CC mask
0004A02C  00                                       8857+         DS    X                   extracted cc
0004A030  00000000 00000000                        8858+         DS    2F                  extract PSW after test (has CC)
0004A038  0004A9E0                                 8859+         DC    A(FMSG56)           address of test fail message
0004A03C  0004A740                                 8860+         DC    A(OPSRC56)          address of operand source
0004A040  0004A620                                 8861+         DC    A(OPDST56)          address of operand destination
0004A044  0004A860                                 8862+         DC    A(PL56)             address of parameter list
0004A048  000041B0                                 8863+         DC    A(RANDX+56)         address of random source data
0004A04C  0004AA59                                 8864+         DC    A(TEND56)           address of test end eye catche
                                                   8865
                                                   8866          TSTCODE
0004A050                                           8867+CODE56   DS    0F
                                                   8868 *                                  load random source
0004A050  E310 501C 0014                0000001C   8869          LGF   R1,OPSRCA           R1: OPSRC - Primary
0004A056  B904 0061                                8870          LGR   R6,R1               R6: OPSRC - Secondary
0004A05A  B904 0071                                8871          LGR   R7,R1               R7: OPSRC - Home
                                                   8872
0004A05E  E320 5028 0014                00000028   8873          LGF   R2,RANDXA           R2: address random data
0004A064  E330 5024 0014                00000024   8874          LGF   R3,PLA              R3: address PL
0004A06A  E340 5020 0014                00000020   8875          LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   8876
0004A070  B219 0200                     00000200   8877          SAC   ARMODE
0004A074  D21F 4000 2000        00000000 00000000  8878          MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004A07A  D21F 6000 20C0        00000000 000000C0  8879          MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   8880 *
                                                   8881 * Double Compare and Swap (DCSO: 26)
                                                   8882 * Build Parambter List
                                                   8883 *
0004A080  D21F 3000 2000        00000000 00000000  8884          MVC   0(32,R3),0(R2)      Load PL OP1-C
0004A086  D21F 3020 2020        00000020 00000020  8885          MVC   32(32,R3),32(R2)    load PL OP1-R
0004A08C  D21F 3050 2040        00000050 00000040  8886          MVC   80(32,R3),64(R2)    load PL OP3
0004A092  D21F 3070 2060        00000070 00000060  8887          MVC   112(32,R3),96(R2)   load PL OP3-R
                                                   8888 ****
                                                   8889 * force a failure by changing the Op1-c
                                                   8890 ****
0004A098  D203 3000 2021        00000000 00000021  8891          MVC   0(4,R3),33(R2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          8892 ****
0004A09E  E363 0048 0024                  00000048  8893          STG   R6,72(R3)           load OP4 address in PL (sec space)
0004A0A4  E54C 3044 0001                  00000044  8894          MVHI  68(R3),SECALET
                                          8895
0004A0AA  B982 0000                       00000200  8896          XGR   R0,R0
0004A0AE  4100 001A                       0000001A  8897          LA    R0,26
                                          8898
0004A0B2  B219 0200                       00000200  8899          SAC   ARMODE
0004A0B6  EE0B 4000 3000       00000000   00000000  8900          PLO   0,0(R4),R11,0(R3)   Double C/S (DCSO: 26)
0004A0BC  B219 0000                       00000000  8901          SAC   PRIMODE
                                          8902
0004A0C0  B98D 0020                       00000200  8903          EPSW  R2,R0               extract psw
0004A0C4  5020 5010                       00000010  8904          ST    R2,CCPSW              to save CC
0004A0C8  E310 500B 0076                  0000000B  8905          LB    R1,CCMASK           (failure CC mask)
0004A0CE  8910 0004                       00000004  8906          SLL   R1,4                (shift to BC instr CC position)
0004A0D2  4410 5106                       0004A126  8907          EX    R1,F261TSTCC            fail if...
                                          8908
                                0004A0D6  00000001  8909 F261CKUR EQU   *
0004A0D6  D200 5009 3030       00000009   00000030  8910          MVC   URFOUND,48(R3)      Extract UR
0004A0DC  D500 5008 3030       00000008   00000030  8911          CLC   UR,48(R3)
0004A0E2  4780 50CA                       0004A0EA  8912          BE    F261CKRES
0004A0E6  45F0 8388                       00002388  8913          BAL   R15,URMSG
                                          8914 *
                                          8915 * compare results
                                          8916 *
                                0004A0EA  00000001  8917 F261CKRES EQU   *
                                          8918 *                        reset addresses
0004A0EA  E310 501C 0014                  0000001C  8919          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0004A0F0  B904 0061                       00000200  8920          LGR   R6,R1               R6:  OPSRC - Secondary
0004A0F4  B904 0071                       00000200  8921          LGR   R7,R1               R7:  OPSRC - Home
0004A0F8  E330 5024 0014                  00000024  8922          LGF   R3,PLA              R3:  address PL
0004A0FE  E340 5020 0014                  00000020  8923          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                          8924
0004A104  B219 0200                       00000200  8925          SAC   ARMODE
0004A108  D51F 4000 3000       00000000   00000000  8926          CLC   0(32,R4),0(R3)      Op2 == OP1C?
0004A10E  D51F 4000 3000       00000000   00000000  8927          CLC   0(32,R4),0(R3)      Op2 == OP1C?
0004A114  B219 0000                       00000000  8928          SAC   PRIMODE
0004A118  478B 0000                       00000000  8929          BE    0(R11)              same, return
                                          8930
                                0004A11C  00000001  8931 F261XX   EQU   *
0004A11C  B219 0000                       00000000  8932          SAC   PRIMODE
0004A120  45F0 8450                       00002450  8933          BAL   R15,FAILMSG
0004A124  07FB                            00000200  8934          BR    R11
                                          8935
0004A126  4700 510A                       0004A12A  8936 F261TSTCC BC    0,F261CCMSG          fail if unexpected condition code
                                0004A12A  00000001  8937 F261CCMSG EQU   *
0004A12A  45F0 82B2                       000022B2  8938          BAL   R15,CCMSG
0004A12E  47F0 50B6                       0004A0D6  8939          B     F261CKUR
                                          8940
                                          8941          TSTDATA
0004A132                        0004A132  0004A600  8942+         ORG   PLOETST+(128*K)+56*(3*K)+(K+512)
                                          8943          TSTLBL OPDST              OP2, Op4
0004A600  D7D9C940 40406E6E               8944+         DC    CL32'PRI   >>>OPDST56<<<' eye catcher
0004A608  6ED6D7C4 E2E3F5F6
0004A610  4C4C4C40 40404040
0004A618  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004A620                                             8945+OPDST56  DS    0C
0004A620  00000000 00000000                          8946          DC    XL256'0'
0004A628  00000000 00000000
0004A630  00000000 00000000
0004A638  00000000 00000000
0004A640  00000000 00000000
0004A648  00000000 00000000
0004A650  00000000 00000000
0004A658  00000000 00000000
0004A660  00000000 00000000
0004A668  00000000 00000000
0004A670  00000000 00000000
0004A678  00000000 00000000
0004A680  00000000 00000000
0004A688  00000000 00000000
0004A690  00000000 00000000
0004A698  00000000 00000000
0004A6A0  00000000 00000000
0004A6A8  00000000 00000000
0004A6B0  00000000 00000000
0004A6B8  00000000 00000000
0004A6C0  00000000 00000000
0004A6C8  00000000 00000000
0004A6D0  00000000 00000000
0004A6D8  00000000 00000000
0004A6E0  00000000 00000000
0004A6E8  00000000 00000000
0004A6F0  00000000 00000000
0004A6F8  00000000 00000000
0004A700  00000000 00000000
0004A708  00000000 00000000
0004A710  00000000 00000000
0004A718  00000000 00000000
                                                     8947
                                                     8948          TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0004A720  D7D9C940 40406E6E                          8949+         DC    CL32'PRI   >>>OPSRC56<<<' eye catcher
0004A728  6ED6D7E2 D9C3F5F6
0004A730  4C4C4C40 40404040
0004A738  40404040 40404040
0004A740                                             8950+OPSRC56  DS    0C
0004A740  00000000 00000000                          8951          DC    XL256'0'
0004A748  00000000 00000000
0004A750  00000000 00000000
0004A758  00000000 00000000
0004A760  00000000 00000000
0004A768  00000000 00000000
0004A770  00000000 00000000
0004A778  00000000 00000000
0004A780  00000000 00000000
0004A788  00000000 00000000
0004A790  00000000 00000000
0004A798  00000000 00000000
0004A7A0  00000000 00000000
0004A7A8  00000000 00000000
0004A7B0  00000000 00000000
0004A7B8  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004A7C0  00000000 00000000
0004A7C8  00000000 00000000
0004A7D0  00000000 00000000
0004A7D8  00000000 00000000
0004A7E0  00000000 00000000
0004A7E8  00000000 00000000
0004A7F0  00000000 00000000
0004A7F8  00000000 00000000
0004A800  00000000 00000000
0004A808  00000000 00000000
0004A810  00000000 00000000
0004A818  00000000 00000000
0004A820  00000000 00000000
0004A828  00000000 00000000
0004A830  00000000 00000000
0004A838  00000000 00000000
                                          8952
                                          8953          TSTLBL PL                  Parameter List
0004A840  D7D9C940 40406E6E               8954+         DC    CL32'PRI   >>>PL56<<<'    eye catcher
0004A848  6ED7D3F5 F64C4C4C
0004A850  40404040 40404040
0004A858  40404040 40404040
0004A860                                  8955+PL56     DS    0C
0004A860  00000000 00000000               8956          DC    XL352'00'
0004A868  00000000 00000000
0004A870  00000000 00000000
0004A878  00000000 00000000
0004A880  00000000 00000000
0004A888  00000000 00000000
0004A890  00000000 00000000
0004A898  00000000 00000000
0004A8A0  00000000 00000000
0004A8A8  00000000 00000000
0004A8B0  00000000 00000000
0004A8B8  00000000 00000000
0004A8C0  00000000 00000000
0004A8C8  00000000 00000000
0004A8D0  00000000 00000000
0004A8D8  00000000 00000000
0004A8E0  00000000 00000000
0004A8E8  00000000 00000000
0004A8F0  00000000 00000000
0004A8F8  00000000 00000000
0004A900  00000000 00000000
0004A908  00000000 00000000
0004A910  00000000 00000000
0004A918  00000000 00000000
0004A920  00000000 00000000
0004A928  00000000 00000000
0004A930  00000000 00000000
0004A938  00000000 00000000
0004A940  00000000 00000000
0004A948  00000000 00000000
0004A950  00000000 00000000
0004A958  00000000 00000000
0004A960  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004A968  00000000 00000000
0004A970  00000000 00000000
0004A978  00000000 00000000
0004A980  00000000 00000000
0004A988  00000000 00000000
0004A990  00000000 00000000
0004A998  00000000 00000000
0004A9A0  00000000 00000000
0004A9A8  00000000 00000000
0004A9B0  00000000 00000000
0004A9B8  00000000 00000000
                                      8957
                                      8958          TSTLBL FMSG               fail message
0004A9C0  D7D9C940 40406E6E           8959+         DC    CL32'PRI   >>>FMSG56<<<'  eye catcher
0004A9C8  6EC6D4E2 C7F5F64C
0004A9D0  4C4C4040 40404040
0004A9D8  40404040 40404040
0004A9E0                              8960+FMSG56   DS    0C
0004A9E0  86818993 858440D7           8961          DC    C'failed PLO-Extension '
0004A9E8  D3D660C5 A7A38595
0004A9F0  A2899695 40
0004A9F5  C496A482 938540C3           8962          DC    C'Double Compare and Swap (DCSO: 26) : NE Op1 != Op2'
0004A9FD  96949781 99854081
0004AA05  958440E2 A6819740
0004AA0D  4DC4C3E2 D67A40F2
0004AA15  F65D407A 40D5C540
0004AA1D  D697F140 5A7E40D6
0004AA25  97F2
0004AA27  40404040 40404040           8963          DC    CL50' '
0004AA2F  40404040 40404040
0004AA37  40404040 40404040
0004AA3F  40404040 40404040
0004AA47  40404040 40404040
0004AA4F  40404040 40404040
0004AA57  4040
                                      8964          TSTEND
0004AA59  D7D9C940 4040C5D5           8965+TEND56   DC    CL32'PRI   END of test: 56 <<<'
0004AA61  C4409686 40A385A2
0004AA69  A37A40F5 F6404C4C
0004AA71  4C404040 40404040
                                      8966
                                      8967 * ------------------------------------------------------------------
                                      8968 * Test Double Compare and Swap (DCSO: 26) instruction : NE Op3 != Op4
                                      8969 * ------------------------------------------------------------------
                                      8970          TSTHDR 2,99,26            Op3 != OP4 (cc=2, UR=99)
0004AA79                 0004AA79  0004AC00  8971+         ORG   PLOETST+(128*K)+57*(3*K)
0004AC00  D7D9C940 4040E2E3           8972+         DC    CL32'PRI   START of test: 57'
0004AC08  C1D9E340 968640A3
0004AC10  85A2A37A 40F5F740
0004AC18  40404040 40404040
0004AC20                 0004AC20     8973+         USING *,R5                base for test data and test routine
0004AC20                 00000000     8974+         USING PLOETEST,R5
0004AC20  0004AC50                    8975+T57      DC    A(CODE57)           address of test code
0004AC24  0039                        8976+         DC    H'57'               test number
0004AC26  00                          8977+         DC    X'00'
0004AC27  1A                          8978+         DC    HL1'26'             Function
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004AC28  63                                       8979+          DC    HL1'99'             expected unequal results (UR)
0004AC29  00                                       8980+          DC    HL1'00'             found UR
0004AC2A  02                                       8981+          DC    HL1'2'              cc
0004AC2B  0D                                       8982+          DC    HL1'13'             not expected CC mask
0004AC2C  00                                       8983+          DS    X                   extracted cc
0004AC30  00000000 00000000                        8984+          DS    2F                  extract PSW after test (has CC)
0004AC38  0004B5E0                                 8985+          DC    A(FMSG57)           address of test fail message
0004AC3C  0004B340                                 8986+          DC    A(OPSRC57)          address of operand source
0004AC40  0004B220                                 8987+          DC    A(OPDST57)          address of operand destination
0004AC44  0004B460                                 8988+          DC    A(PL57)             address of parameter list
0004AC48  000041B1                                 8989+          DC    A(RANDX+57)         address of random source data
0004AC4C  0004B658                                 8990+          DC    A(TEND57)           address of test end eye catche
                                                   8991
                                                   8992           TSTCODE
0004AC50                                           8993+CODE57    DS    0F
                                                   8994 *                                   load random source
0004AC50  E310 501C 0014                0000001C   8995           LGF   R1,OPSRCA           R1: OPSRC - Primary
0004AC56  B904 0061                                8996           LGR   R6,R1               R6: OPSRC - Secondary
0004AC5A  B904 0071                                8997           LGR   R7,R1               R7: OPSRC - Home
                                                   8998
0004AC5E  E320 5028 0014                00000028   8999           LGF   R2,RANDXA           R2: address random data
0004AC64  E330 5024 0014                00000024   9000           LGF   R3,PLA              R3: address PL
0004AC6A  E340 5020 0014                00000020   9001           LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   9002
0004AC70  B219 0200                     00000200   9003           SAC   ARMODE
0004AC74  D21F 4000 2000      00000000  00000000   9004           MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004AC7A  D21F 6000 20C0      00000000  000000C0   9005           MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   9006 *
                                                   9007 * Double Compare and Swap (DCSO: 26)
                                                   9008 * Build Parambter List
                                                   9009 *
0004AC80  D21F 3000 2000      00000000  00000000   9010           MVC   0(32,R3),0(R2)      Load PL OP1-C
0004AC86  D21F 3020 2020      00000020  00000020   9011           MVC   32(32,R3),32(R2)    load PL OP1-R
0004AC8C  D21F 3050 2040      00000050  00000040   9012           MVC   80(32,R3),64(R2)    load PL OP3
0004AC92  D21F 3070 2060      00000070  00000060   9013           MVC   112(32,R3),96(R2)   load PL OP3-R
                                                   9014 ****
                                                   9015 * force a failure by changing the Op3-C
                                                   9016 ****
0004AC98  D203 3050 2021      00000050  00000021   9017           MVC   80(4,R3),33(R2)
                                                   9018 ****
0004AC9E  E363 0048 0024                00000048   9019           STG   R6,72(R3)           load OP4 address in PL (sec space)
0004ACA4  E54C 3044 0001                00000044   9020           MVHI  68(R3),SECALET
                                                   9021
0004ACAA  B982 0000                                9022           XGR   R0,R0
0004ACAE  4100 001A                     0000001A   9023           LA    R0,26
                                                   9024
0004ACB2  B219 0200                     00000200   9025           SAC   ARMODE
0004ACB6  EE0B 4000 3000      00000000  00000000   9026           PLO   0,0(R4),R11,0(R3)   Double C/S (DCSO: 26)
0004ACBC  B219 0000                     00000000   9027           SAC   PRIMODE
                                                   9028
0004ACC0  B98D 0020                                9029           EPSW  R2,R0               extract psw
0004ACC4  5020 5010                     00000010   9030           ST    R2,CCPSW              to save CC
0004ACC8  E310 500B 0076                0000000B   9031           LB    R1,CCMASK           (failure CC mask)
0004ACCE  8910 0004                     00000004   9032           SLL   R1,4                (shift to BC instr CC position)
0004ACD2  4410 510A                     0004AD2A   9033           EX    R1,F262TSTCC            fail if...
                                                   9034
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    9035 F262CKUR EQU   *
                               0004ACD6  00000001
0004ACD6  D200 5009 3030       00000009  00000030  9036          MVC   URFOUND,48(R3)      Extract UR
0004ACDC  D500 5008 3030       00000008  00000030  9037          CLC   UR,48(R3)
0004ACE2  4780 50CA                      0004ACEA  9038          BE    F262CKRES
0004ACE6  45F0 8388                      00002388  9039          BAL   R15,URMSG
                                                    9040 *
                                                    9041 * compare results
                                                    9042 *
                               0004ACEA  00000001  9043 F262CKRES EQU   *
                                                    9044 *                              reset addresses
0004ACEA  E310 501C 0014                 0000001C  9045          LGF   R1,OPSRCA         R1:  OPSRC - Primary
0004ACF0  B904 0061                                 9046          LGR   R6,R1             R6:  OPSRC - Secondary
0004ACF4  B904 0071                                 9047          LGR   R7,R1             R7:  OPSRC - Home
0004ACF8  E330 5024 0014                 00000024  9048          LGF   R3,PLA            R3:  address PL
0004ACFE  E340 5020 0014                 00000020  9049          LGF   R4,OPREPA         R4:  OPREPA - Primary
                                                    9050
0004AD04  B219 0200                      00000200  9051          SAC   ARMODE
0004AD08  D51F 4000 3000       00000000  00000000  9052          CLC   0(32,R4),0(R3)    Op2 == OP1-C?
0004AD0E  4770 5100                      0004AD20  9053          BNE   F262XX
0004AD12  D51F 6000 3050       00000000  00000050  9054          CLC   0(32,R6),80(R3)   OP4 == Op3-C (replaced?)
0004AD18  B219 0000                      00000000  9055          SAC   PRIMODE
0004AD1C  478B 0000                      00000000  9056          BE    0(R11)            EQ, matches test of Op3 != OP4
                                                    9057
                               0004AD20  00000001  9058 F262XX   EQU   *
0004AD20  B219 0000                      00000000  9059          SAC   PRIMODE
0004AD24  45F0 8450                      00002450  9060          BAL   R15,FAILMSG
0004AD28  07FB                                      9061          BR    R11
                                                    9062
0004AD2A  4700 510E                      0004AD2E  9063 F262TSTCC BC    0,F262CCMSG       fail if unexpected condition code
                               0004AD2E  00000001  9064 F262CCMSG EQU   *
0004AD2E  45F0 82B2                      000022B2  9065          BAL   R15,CCMSG
0004AD32  47F0 50B6                      0004ACD6  9066          B     F262CKUR
                                                    9067
                                                    9068          TSTDATA
0004AD36                       0004AD36  0004B200  9069+         ORG   PLOETST+(128*K)+57*(3*K)+(K+512)
                                                    9070          TSTLBL OPDST              OP2, OP4
0004B200  D7D9C940 40406E6E                         9071+         DC    CL32'PRI   >>>OPDST57<<<' eye catcher
0004B208  6ED6D7C4 E2E3F5F7
0004B210  4C4C4C40 40404040
0004B218  40404040 40404040
0004B220                                            9072+OPDST57  DS    0C
0004B220  00000000 00000000                         9073          DC    XL256'0'
0004B228  00000000 00000000
0004B230  00000000 00000000
0004B238  00000000 00000000
0004B240  00000000 00000000
0004B248  00000000 00000000
0004B250  00000000 00000000
0004B258  00000000 00000000
0004B260  00000000 00000000
0004B268  00000000 00000000
0004B270  00000000 00000000
0004B278  00000000 00000000
0004B280  00000000 00000000
0004B288  00000000 00000000
0004B290  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004B298  00000000 00000000
0004B2A0  00000000 00000000
0004B2A8  00000000 00000000
0004B2B0  00000000 00000000
0004B2B8  00000000 00000000
0004B2C0  00000000 00000000
0004B2C8  00000000 00000000
0004B2D0  00000000 00000000
0004B2D8  00000000 00000000
0004B2E0  00000000 00000000
0004B2E8  00000000 00000000
0004B2F0  00000000 00000000
0004B2F8  00000000 00000000
0004B300  00000000 00000000
0004B308  00000000 00000000
0004B310  00000000 00000000
0004B318  00000000 00000000
                                        9074
                                        9075          TSTLBL OPSRC              Op1-C, Op3, Op5, Op1-R
0004B320  D7D9C940 40406E6E             9076+         DC    CL32'PRI   >>>OPSRC57<<<' eye catcher
0004B328  6ED6D7E2 D9C3F5F7
0004B330  4C4C4C40 40404040
0004B338  40404040 40404040
0004B340                                9077+OPSRC57  DS    0C
0004B340  00000000 00000000             9078          DC    XL256'0'
0004B348  00000000 00000000
0004B350  00000000 00000000
0004B358  00000000 00000000
0004B360  00000000 00000000
0004B368  00000000 00000000
0004B370  00000000 00000000
0004B378  00000000 00000000
0004B380  00000000 00000000
0004B388  00000000 00000000
0004B390  00000000 00000000
0004B398  00000000 00000000
0004B3A0  00000000 00000000
0004B3A8  00000000 00000000
0004B3B0  00000000 00000000
0004B3B8  00000000 00000000
0004B3C0  00000000 00000000
0004B3C8  00000000 00000000
0004B3D0  00000000 00000000
0004B3D8  00000000 00000000
0004B3E0  00000000 00000000
0004B3E8  00000000 00000000
0004B3F0  00000000 00000000
0004B3F8  00000000 00000000
0004B400  00000000 00000000
0004B408  00000000 00000000
0004B410  00000000 00000000
0004B418  00000000 00000000
0004B420  00000000 00000000
0004B428  00000000 00000000
0004B430  00000000 00000000
0004B438  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    9079
                                                    9080          TSTLBL PL                 Parameter List
0004B440  D7D9C940 40406E6E                         9081+         DC    CL32'PRI   >>>PL57<<<'    eye catcher
0004B448  6ED7D3F5 F74C4C4C
0004B450  40404040 40404040
0004B458  40404040 40404040
0004B460                                            9082+PL57     DS    0C
0004B460  00000000 00000000                         9083          DC    XL352'00'
0004B468  00000000 00000000
0004B470  00000000 00000000
0004B478  00000000 00000000
0004B480  00000000 00000000
0004B488  00000000 00000000
0004B490  00000000 00000000
0004B498  00000000 00000000
0004B4A0  00000000 00000000
0004B4A8  00000000 00000000
0004B4B0  00000000 00000000
0004B4B8  00000000 00000000
0004B4C0  00000000 00000000
0004B4C8  00000000 00000000
0004B4D0  00000000 00000000
0004B4D8  00000000 00000000
0004B4E0  00000000 00000000
0004B4E8  00000000 00000000
0004B4F0  00000000 00000000
0004B4F8  00000000 00000000
0004B500  00000000 00000000
0004B508  00000000 00000000
0004B510  00000000 00000000
0004B518  00000000 00000000
0004B520  00000000 00000000
0004B528  00000000 00000000
0004B530  00000000 00000000
0004B538  00000000 00000000
0004B540  00000000 00000000
0004B548  00000000 00000000
0004B550  00000000 00000000
0004B558  00000000 00000000
0004B560  00000000 00000000
0004B568  00000000 00000000
0004B570  00000000 00000000
0004B578  00000000 00000000
0004B580  00000000 00000000
0004B588  00000000 00000000
0004B590  00000000 00000000
0004B598  00000000 00000000
0004B5A0  00000000 00000000
0004B5A8  00000000 00000000
0004B5B0  00000000 00000000
0004B5B8  00000000 00000000
                                                    9084
                                                    9085          TSTLBL FMSG               fail message
0004B5C0  D7D9C940 40406E6E                         9086+         DC    CL32'PRI   >>>FMSG57<<<'  eye catcher
0004B5C8  6EC6D4E2 C7F5F74C
0004B5D0  4C4C4040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B5D8  40404040 40404040
0004B5E0                                          9087+FMSG57   DS    0C
0004B5E0  86818993 858440D7                       9088          DC    C'failed PLO-Extension '
0004B5E8  D3D660C5 A7A38595
0004B5F0  A2899695 40
0004B5F5  C496A482 938540C3                       9089          DC    C'Double Compare and Swap (DCSO: 26) : Op3-C != OP4'
0004B5FD  96949781 99854081
0004B605  958440E2 A6819740
0004B60D  4DC4C3E2 D67A40F2
0004B615  F65D407A 40D697F3
0004B61D  60C3405A 7E40D6D7
0004B625  F4
0004B626  40404040 40404040                       9090          DC    CL50' '
0004B62E  40404040 40404040
0004B636  40404040 40404040
0004B63E  40404040 40404040
0004B646  40404040 40404040
0004B64E  40404040 40404040
0004B656  4040
                                                  9091          TSTEND
0004B658  D7D9C940 4040C5D5                       9092+TEND57   DC    CL32'PRI   END of test: 57 <<<'
0004B660  C4409686 40A385A2
0004B668  A37A40F5 F7404C4C
0004B670  4C404040 40404040
                                                  9093
                                                  9094 * ------------------------------------------------------------------
                                                  9095 * Test Double Compare and Swap (DCSO: 26) instruction : EQ
                                                  9096 * ------------------------------------------------------------------
                                                  9097          TSTHDR 0,99,28              Op1-C == OP2 && Op3 == OP4
0004B678                          0004B678  0004B800  9098+         ORG   PLOETST+(128*K)+58*(3*K)
0004B800  D7D9C940 4040E2E3                       9099+         DC    CL32'PRI   START of test: 58'
0004B808  C1D9E340 968640A3
0004B810  85A2A37A 40F5F840
0004B818  40404040 40404040
0004B820                          0004B820        9100+         USING *,R5                  base for test data and test routine
0004B820                          00000000        9101+         USING PLOETEST,R5
0004B820  0004B850                                9102+T58      DC    A(CODE58)             address of test code
0004B824  003A                                    9103+         DC    H'58'                 test number
0004B826  00                                      9104+         DC    X'00'
0004B827  1C                                      9105+         DC    HL1'28'               Function
0004B828  63                                      9106+         DC    HL1'99'               expected unequal results (UR)
0004B829  00                                      9107+         DC    HL1'00'               found UR
0004B82A  00                                      9108+         DC    HL1'0'                cc
0004B82B  07                                      9109+         DC    HL1'7'                not expected CC mask
0004B82C  00                                      9110+         DS    X                     extracted cc
0004B830  00000000 00000000                       9111+         DS    2F                    extract PSW after test (has CC)
0004B838  0004C1E0                                9112+         DC    A(FMSG58)             address of test fail message
0004B83C  0004BF40                                9113+         DC    A(OPSRC58)            address of operand source
0004B840  0004BE20                                9114+         DC    A(OPDST58)            address of operand destination
0004B844  0004C060                                9115+         DC    A(PL58)               address of parameter list
0004B848  000041B2                                9116+         DC    A(RANDX+58)           address of random source data
0004B84C  0004C24E                                9117+         DC    A(TEND58)             address of test end eye catche
                                                  9118 *                                          (cc=0, UR=99)
                                                  9119          TSTCODE
0004B850                                          9120+CODE58   DS    0F
                                                  9121 *                                      load random source
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004B850  E310 501C 0014                 0000001C  9122          LGF   R1,OPSRCA           R1: OPSRC - Primary
0004B856  B904 0061                              9123          LGR   R6,R1               R6: OPSRC - Secondary
0004B85A  B904 0071                              9124          LGR   R7,R1               R7: OPSRC - Home
                                                 9125
0004B85E  E320 5028 0014                 00000028  9126          LGF   R2,RANDXA           R2: address random data
0004B864  E330 5024 0014                 00000024  9127          LGF   R3,PLA              R3: address PL
0004B86A  E340 5020 0014                 00000020  9128          LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                 9129
0004B870  B219 0200                      00000200  9130          SAC   ARMODE
0004B874  D21F 4000 2000       00000000  00000000  9131          MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004B87A  D21F 6000 20C0       00000000  000000C0  9132          MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                 9133 *
                                                 9134 * Compare and Swap and Double Store (CSDSTO: 28)
                                                 9135 * Build Parambter List
                                                 9136 *
0004B880  D21F 3000 2000       00000000  00000000  9137          MVC   0(32,R3),0(R2)      Load PL OP1-C
0004B886  D21F 3020 2020       00000020  00000020  9138          MVC   32(32,R3),32(R2)    load PL OP1-R
0004B88C  D21F 3050 2040       00000050  00000040  9139          MVC   80(32,R3),64(R2)    load PL OP3
0004B892  D21F 3070 2060       00000070  00000060  9140          MVC   112(32,R3),96(R2)   load PL OP3-R
                                                 9141 ****
                                                 9142 * Force a EQ by changing Op1-C == OP2 && Op3 == OP4
                                                 9143 ****
0004B898  D21F 3000 4000       00000000  00000000  9144          MVC   0(32,R3),0(R4)      Op1-C <-- Op2
0004B89E  D21F 3050 6000       00000050  00000000  9145          MVC   80(32,R3),0(R6)     Op3-C <-- Op4
                                                 9146 ****
0004B8A4  E363 0048 0024                 00000048  9147          STG   R6,72(R3)           load OP4 address in PL (sec space)
0004B8AA  E54C 3044 0001                 00000044  9148          MVHI  68(R3),SECALET
                                                 9149
0004B8B0  B982 0000                              9150          XGR   R0,R0
0004B8B4  4100 001A                      0000001A  9151          LA    R0,26
                                                 9152
0004B8B8  B219 0200                      00000200  9153          SAC   ARMODE
0004B8BC  EE0B 4000 3000       00000000  00000000  9154          PLO   0,0(R4),R11,0(R3)   Double C/S (DCSO: 26)
0004B8C2  B219 0000                      00000000  9155          SAC   PRIMODE
                                                 9156
0004B8C6  B98D 0020                              9157          EPSW  R2,R0               extract psw
0004B8CA  5020 5010                      00000010  9158          ST    R2,CCPSW              to save CC
0004B8CE  E310 500B 0076                 0000000B  9159          LB    R1,CCMASK           (failure CC mask)
0004B8D4  8910 0004                      00000004  9160          SLL   R1,4                (shift to BC instr CC position)
0004B8D8  4410 5110                      0004B930  9161          EX    R1,F263TSTCC            fail if...
                                                 9162
                                0004B8DC  00000001  9163 F263CKUR EQU   *
0004B8DC  D200 5009 3030       00000009  00000030  9164          MVC   URFOUND,48(R3)      Extract UR
0004B8E2  D500 5008 3030       00000008  00000030  9165          CLC   UR,48(R3)
0004B8E8  4780 50D0                      0004B8F0  9166          BE    F263CKRES
0004B8EC  45F0 8388                      00002388  9167          BAL   R15,URMSG
                                                 9168 *
                                                 9169 * compare results
                                                 9170 *
                                0004B8F0  00000001  9171 F263CKRES EQU   *
                                                 9172 *                              reset addresses
0004B8F0  E310 501C 0014                 0000001C  9173          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0004B8F6  B904 0061                              9174          LGR   R6,R1               R6:  OPSRC - Secondary
0004B8FA  B904 0071                              9175          LGR   R7,R1               R7:  OPSRC - Home
0004B8FE  E330 5024 0014                 00000024  9176          LGF   R3,PLA              R3:  address PL
0004B904  E340 5020 0014                 00000020  9177          LGF   R4,OPREPA           R4:  OPREPA - Primary
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                    9178
0004B90A  B219 0200                       00000200  9179           SAC   ARMODE
0004B90E  D51F 6000 3070      00000000    00000070  9180           CLC   0(32,R6),112(R3)    OP4 ==  PL Op3-R
0004B914  4770 5106                       0004B926  9181           BNE   F263XX                  different, error
0004B918  D51F 4000 3020      00000000    00000020  9182           CLC   0(32,R4),32(R3)     OP2 ==  PL Op1-R
0004B91E  B219 0000                       00000000  9183           SAC   PRIMODE
0004B922  478B 0000                       00000000  9184           BE    0(R11)                   same, return
                                                    9185
                              0004B926    00000001  9186 F263XX    EQU   *
0004B926  B219 0000                       00000000  9187           SAC   PRIMODE
0004B92A  45F0 8450                       00002450  9188           BAL   R15,FAILMSG
0004B92E  07FB                                      9189           BR    R11
                                                    9190
0004B930  4700 5114                       0004B934  9191 F263TSTCC BC    0,F263CCMSG         fail if unexpected condition code
                              0004B934    00000001  9192 F263CCMSG EQU   *
0004B934  45F0 82B2                       000022B2  9193           BAL   R15,CCMSG
0004B938  47F0 50BC                       0004B8DC  9194           B     F263CKUR
                                                    9195
                                                    9196           TSTDATA
0004B93C                      0004B93C    0004BE00  9197+          ORG   PLOETST+(128*K)+58*(3*K)+(K+512)
                                                    9198           TSTLBL OPDST              OP2, Op4
0004BE00  D7D9C940 40406E6E                         9199+          DC    CL32'PRI   >>>OPDST58<<<' eye catcher
0004BE08  6ED6D7C4 E2E3F5F8
0004BE10  4C4C4C40 40404040
0004BE18  40404040 40404040
0004BE20                                            9200+OPDST58   DS    0C
0004BE20  00000000 00000000                         9201           DC    XL256'0'
0004BE28  00000000 00000000
0004BE30  00000000 00000000
0004BE38  00000000 00000000
0004BE40  00000000 00000000
0004BE48  00000000 00000000
0004BE50  00000000 00000000
0004BE58  00000000 00000000
0004BE60  00000000 00000000
0004BE68  00000000 00000000
0004BE70  00000000 00000000
0004BE78  00000000 00000000
0004BE80  00000000 00000000
0004BE88  00000000 00000000
0004BE90  00000000 00000000
0004BE98  00000000 00000000
0004BEA0  00000000 00000000
0004BEA8  00000000 00000000
0004BEB0  00000000 00000000
0004BEB8  00000000 00000000
0004BEC0  00000000 00000000
0004BEC8  00000000 00000000
0004BED0  00000000 00000000
0004BED8  00000000 00000000
0004BEE0  00000000 00000000
0004BEE8  00000000 00000000
0004BEF0  00000000 00000000
0004BEF8  00000000 00000000
0004BF00  00000000 00000000
0004BF08  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004BF10  00000000 00000000
0004BF18  00000000 00000000
                                          9202
                                          9203           TSTLBL OPSRC              Op1-C, Op3, Op1-R, Op3-R
0004BF20  D7D9C940 40406E6E               9204+          DC    CL32'PRI   >>>OPSRC58<<<' eye catcher
0004BF28  6ED6D7E2 D9C3F5F8
0004BF30  4C4C4C40 40404040
0004BF38  40404040 40404040
0004BF40                                  9205+OPSRC58   DS    0C
0004BF40  00000000 00000000               9206           DC    XL256'0'
0004BF48  00000000 00000000
0004BF50  00000000 00000000
0004BF58  00000000 00000000
0004BF60  00000000 00000000
0004BF68  00000000 00000000
0004BF70  00000000 00000000
0004BF78  00000000 00000000
0004BF80  00000000 00000000
0004BF88  00000000 00000000
0004BF90  00000000 00000000
0004BF98  00000000 00000000
0004BFA0  00000000 00000000
0004BFA8  00000000 00000000
0004BFB0  00000000 00000000
0004BFB8  00000000 00000000
0004BFC0  00000000 00000000
0004BFC8  00000000 00000000
0004BFD0  00000000 00000000
0004BFD8  00000000 00000000
0004BFE0  00000000 00000000
0004BFE8  00000000 00000000
0004BFF0  00000000 00000000
0004BFF8  00000000 00000000
0004C000  00000000 00000000
0004C008  00000000 00000000
0004C010  00000000 00000000
0004C018  00000000 00000000
0004C020  00000000 00000000
0004C028  00000000 00000000
0004C030  00000000 00000000
0004C038  00000000 00000000
                                          9207
                                          9208           TSTLBL PL                 Parameter List
0004C040  D7D9C940 40406E6E               9209+          DC    CL32'PRI   >>>PL58<<<'    eye catcher
0004C048  6ED7D3F5 F84C4C4C
0004C050  40404040 40404040
0004C058  40404040 40404040
0004C060                                  9210+PL58      DS    0C
0004C060  00000000 00000000               9211           DC    XL352'00'
0004C068  00000000 00000000
0004C070  00000000 00000000
0004C078  00000000 00000000
0004C080  00000000 00000000
0004C088  00000000 00000000
0004C090  00000000 00000000
0004C098  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004C0A0  00000000 00000000
0004C0A8  00000000 00000000
0004C0B0  00000000 00000000
0004C0B8  00000000 00000000
0004C0C0  00000000 00000000
0004C0C8  00000000 00000000
0004C0D0  00000000 00000000
0004C0D8  00000000 00000000
0004C0E0  00000000 00000000
0004C0E8  00000000 00000000
0004C0F0  00000000 00000000
0004C0F8  00000000 00000000
0004C100  00000000 00000000
0004C108  00000000 00000000
0004C110  00000000 00000000
0004C118  00000000 00000000
0004C120  00000000 00000000
0004C128  00000000 00000000
0004C130  00000000 00000000
0004C138  00000000 00000000
0004C140  00000000 00000000
0004C148  00000000 00000000
0004C150  00000000 00000000
0004C158  00000000 00000000
0004C160  00000000 00000000
0004C168  00000000 00000000
0004C170  00000000 00000000
0004C178  00000000 00000000
0004C180  00000000 00000000
0004C188  00000000 00000000
0004C190  00000000 00000000
0004C198  00000000 00000000
0004C1A0  00000000 00000000
0004C1A8  00000000 00000000
0004C1B0  00000000 00000000
0004C1B8  00000000 00000000
                             9212
                             9213          TSTLBL FMSG               fail message
0004C1C0  D7D9C940 40406E6E  9214+         DC    CL32'PRI   >>>FMSG58<<<'  eye catcher
0004C1C8  6EC6D4E2 C7F5F84C
0004C1D0  4C4C4040 40404040
0004C1D8  40404040 40404040
0004C1E0                     9215+FMSG58   DS    0C
0004C1E0  86818993 858440D7  9216          DC    C'failed PLO-Extension '
0004C1E8  D3D660C5 A7A38595
0004C1F0  A2899695 40
0004C1F5  C496A482 938540C3  9217          DC    C'Double Compare and Swap (DCSO: 26) : EQ'
0004C1FD  96949781 99854081
0004C205  958440E2 A6819740
0004C20D  4DC4C3E2 D67A40F2
0004C215  F65D407A 40C5D8
0004C21C  40404040 40404040  9218          DC    CL50' '
0004C224  40404040 40404040
0004C22C  40404040 40404040
0004C234  40404040 40404040
0004C23C  40404040 40404040
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004C244  40404040 40404040
0004C24C  4040
                                                     9219          TSTEND
0004C24E  D7D9C940 4040C5D5                          9220+TEND58   DC    CL32'PRI   END of test: 58 <<<'
0004C256  C4409686 40A385A2
0004C25E  A37A40F5 F8404C4C
0004C266  4C404040 40404040
                                                     9221
                                                     9222
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                                                    9224 * ------------------------------------------------------------------------
                                                    9225 * Test Compare and Swap (CSO: 25) instruction
                                                    9226 * ------------------------------------------------------------------------
                                                    9227 *
                                                    9228 *       Compare and Swap (CSO: 25) instruction does not have any
                                                    9229 *       ARMODE  (alet) operands, therefore no test.
                                                    9230 *
                                                    9231 * ------------------------------------------------------------------------
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   9233 *================================================================================
                                                   9234
                                                   9235 * ------------------------------------------------------------------------
                                                   9236 * Test Compare and Load (CLO: 24) instruction : NE Op1-C != Op2
                                                   9237 * ------------------------------------------------------------------------
                                                   9238          TSTHDR 1,99,24            Op1-C != OP2 (cc=1, UR=99)
0004C26E                          0004C26E  0004C400 9239+         ORG   PLOETST+(128*K)+59*(3*K)
0004C400  D7D9C940 4040E2E3                        9240+         DC    CL32'PRI   START of test: 59'
0004C408  C1D9E340 968640A3
0004C410  85A2A37A 40F5F940
0004C418  40404040 40404040
0004C420                          0004C420         9241+         USING *,R5                base for test data and test routine
0004C420                          00000000         9242+         USING PLOETEST,R5
0004C420  0004C450                                 9243+T59      DC    A(CODE59)           address of test code
0004C424  003B                                     9244+         DC    H'59'               test number
0004C426  00                                       9245+         DC    X'00'
0004C427  18                                       9246+         DC    HL1'24'             Function
0004C428  63                                       9247+         DC    HL1'99'             expected unequal results (UR)
0004C429  00                                       9248+         DC    HL1'00'             found UR
0004C42A  01                                       9249+         DC    HL1'1'              cc
0004C42B  0B                                       9250+         DC    HL1'11'             not expected CC mask
0004C42C  00                                       9251+         DS    X                   extracted cc
0004C430  00000000 00000000                        9252+         DS    2F                  extract PSW after test (has CC)
0004C438  0004CDE0                                 9253+         DC    A(FMSG59)           address of test fail message
0004C43C  0004CB40                                 9254+         DC    A(OPSRC59)          address of operand source
0004C440  0004CA20                                 9255+         DC    A(OPDST59)          address of operand destination
0004C444  0004CC60                                 9256+         DC    A(PL59)             address of parameter list
0004C448  000041B3                                 9257+         DC    A(RANDX+59)         address of random source data
0004C44C  0004CE53                                 9258+         DC    A(TEND59)           address of test end eye catche
                                                   9259
                                                   9260          TSTCODE
0004C450                                           9261+CODE59   DS    0F
                                                   9262 *                                  load random source
0004C450  E310 501C 0014                0000001C   9263          LGF   R1,OPSRCA           R1: OPSRC - Primary
0004C456  B904 0061                                9264          LGR   R6,R1               R6: OPSRC - Secondary
0004C45A  B904 0071                                9265          LGR   R7,R1               R7: OPSRC - Home
                                                   9266
0004C45E  E320 5028 0014                00000028   9267          LGF   R2,RANDXA           R2: address random data
0004C464  E330 5024 0014                00000024   9268          LGF   R3,PLA              R3: address PL
0004C46A  E340 5020 0014                00000020   9269          LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   9270
0004C470  B219 0200                     00000200   9271          SAC   ARMODE
0004C474  D21F 4000 2000        00000000 00000000  9272          MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004C47A  D21F 6000 20C0        00000000 000000C0  9273          MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   9274 *
                                                   9275 * Compare and Load (CLO: 24)
                                                   9276 * Build Parambter List
                                                   9277 *
0004C480  D21F 3000 2000        00000000 00000000  9278          MVC   0(32,R3),0(R2)      Load PL OP1-C
0004C486  D21F 3020 2020        00000020 00000020  9279          MVC   32(32,R3),32(R2)    load PL OP3
                                                   9280 ****
                                                   9281 * force a NE by changing the Op1-c
                                                   9282 ****
0004C48C  D203 3000 2021        00000000 00000021  9283          MVC   0(4,R3),33(R2)
                                                   9284 ****
0004C492  E363 0048 0024                00000048   9285          STG   R6,72(R3)           load OP4 address in PL (sec space)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004C498  E54C 3044 0001                 00000044  9286          MVHI  68(R3),SECALET
                                                   9287
0004C49E  B982 0000                              9288          XGR   R0,R0
0004C4A2  4100 0018                      00000018  9289          LA    R0,24
                                                   9290
0004C4A6  B219 0200                      00000200  9291          SAC   ARMODE
0004C4AA  EE0B 4000 3000       00000000  00000000  9292          PLO   0,0(R4),R11,0(R3)   C/L (CLO: 24)
0004C4B0  B219 0000                      00000000  9293          SAC   PRIMODE
                                                   9294
0004C4B4  B98D 0020                              9295          EPSW  R2,R0               extract psw
0004C4B8  5020 5010                      00000010  9296          ST    R2,CCPSW              to save CC
0004C4BC  E310 500B 0076                 0000000B  9297          LB    R1,CCMASK           (failure CC mask)
0004C4C2  8910 0004                      00000004  9298          SLL   R1,4                (shift to BC instr CC position)
0004C4C6  4410 50F4                      0004C514  9299          EX    R1,F241TSTCC            fail if...
                                                   9300
                               0004C4CA  00000001  9301 F241CKUR EQU   *
0004C4CA  D200 5009 3030       00000009  00000030  9302          MVC   URFOUND,48(R3)      Extract UR
0004C4D0  D500 5008 3030       00000008  00000030  9303          CLC   UR,48(R3)
0004C4D6  4780 50BE                      0004C4DE  9304          BE    F241CKRES
0004C4DA  45F0 8388                      00002388  9305          BAL   R15,URMSG
                                                   9306 *
                                                   9307 * compare results
                                                   9308 *
                               0004C4DE  00000001  9309 F241CKRES EQU   *
                                                   9310 *                            reset addresses
0004C4DE  E310 501C 0014                 0000001C  9311          LGF   R1,OPSRCA           R1:  OPSRC - Primary
0004C4E4  B904 0061                              9312          LGR   R6,R1               R6:  OPSRC - Secondary
0004C4E8  B904 0071                              9313          LGR   R7,R1               R7:  OPSRC - Home
0004C4EC  E330 5024 0014                 00000024  9314          LGF   R3,PLA              R3:  address PL
0004C4F2  E340 5020 0014                 00000020  9315          LGF   R4,OPREPA           R4:  OPREPA - Primary
                                                   9316
0004C4F8  B219 0200                      00000200  9317          SAC   ARMODE
0004C4FC  D51F 4000 3000       00000000  00000000  9318          CLC   0(32,R4),0(R3)      Op2 == OP1-C (replaced)?
0004C502  B219 0000                      00000000  9319          SAC   PRIMODE
0004C506  478B 0000                      00000000  9320          BE    0(R11)              same, return
                                                   9321
                               0004C50A  00000001  9322 F241XX   EQU   *
0004C50A  B219 0000                      00000000  9323          SAC   PRIMODE
0004C50E  45F0 8450                      00002450  9324          BAL   R15,FAILMSG
0004C512  07FB                                   9325          BR    R11
                                                   9326
0004C514  4700 50F8                      0004C518  9327 F241TSTCC BC    0,F241CCMSG         fail if unexpected condition code
                               0004C518  00000001  9328 F241CCMSG EQU   *
0004C518  45F0 82B2                      000022B2  9329          BAL   R15,CCMSG
0004C51C  47F0 50AA                      0004C4CA  9330          B     F241CKUR
                                                   9331
                                                   9332          TSTDATA
0004C520                       0004C520  0004CA00  9333+         ORG   PLOETST+(128*K)+59*(3*K)+(K+512)
                                                   9334          TSTLBL OPDST              OP2, Op4
0004CA00  D7D9C940 40406E6E                        9335+         DC    CL32'PRI   >>>OPDST59<<<' eye catcher
0004CA08  6ED6D7C4 E2E3F5F9
0004CA10  4C4C4C40 40404040
0004CA18  40404040 40404040
0004CA20                                           9336+OPDST59  DS    0C
0004CA20  00000000 00000000                        9337          DC    XL256'0'
0004CA28  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004CA30  00000000 00000000
0004CA38  00000000 00000000
0004CA40  00000000 00000000
0004CA48  00000000 00000000
0004CA50  00000000 00000000
0004CA58  00000000 00000000
0004CA60  00000000 00000000
0004CA68  00000000 00000000
0004CA70  00000000 00000000
0004CA78  00000000 00000000
0004CA80  00000000 00000000
0004CA88  00000000 00000000
0004CA90  00000000 00000000
0004CA98  00000000 00000000
0004CAA0  00000000 00000000
0004CAA8  00000000 00000000
0004CAB0  00000000 00000000
0004CAB8  00000000 00000000
0004CAC0  00000000 00000000
0004CAC8  00000000 00000000
0004CAD0  00000000 00000000
0004CAD8  00000000 00000000
0004CAE0  00000000 00000000
0004CAE8  00000000 00000000
0004CAF0  00000000 00000000
0004CAF8  00000000 00000000
0004CB00  00000000 00000000
0004CB08  00000000 00000000
0004CB10  00000000 00000000
0004CB18  00000000 00000000
                                      9338
                                      9339           TSTLBL OPSRC              Op1-C, Op3
0004CB20  D7D9C940 40406E6E           9340+          DC     CL32'PRI   >>>OPSRC59<<<' eye catcher
0004CB28  6ED6D7E2 D9C3F5F9
0004CB30  4C4C4C40 40404040
0004CB38  40404040 40404040
0004CB40                              9341+OPSRC59   DS     0C
0004CB40  00000000 00000000           9342           DC     XL256'0'
0004CB48  00000000 00000000
0004CB50  00000000 00000000
0004CB58  00000000 00000000
0004CB60  00000000 00000000
0004CB68  00000000 00000000
0004CB70  00000000 00000000
0004CB78  00000000 00000000
0004CB80  00000000 00000000
0004CB88  00000000 00000000
0004CB90  00000000 00000000
0004CB98  00000000 00000000
0004CBA0  00000000 00000000
0004CBA8  00000000 00000000
0004CBB0  00000000 00000000
0004CBB8  00000000 00000000
0004CBC0  00000000 00000000
0004CBC8  00000000 00000000
0004CBD0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004CBD8  00000000 00000000
0004CBE0  00000000 00000000
0004CBE8  00000000 00000000
0004CBF0  00000000 00000000
0004CBF8  00000000 00000000
0004CC00  00000000 00000000
0004CC08  00000000 00000000
0004CC10  00000000 00000000
0004CC18  00000000 00000000
0004CC20  00000000 00000000
0004CC28  00000000 00000000
0004CC30  00000000 00000000
0004CC38  00000000 00000000
                                          9343
                                          9344           TSTLBL PL                  Parameter List
0004CC40  D7D9C940 40406E6E               9345+          DC     CL32'PRI   >>>PL59<<<'    eye catcher
0004CC48  6ED7D3F5 F94C4C4C
0004CC50  40404040 40404040
0004CC58  40404040 40404040
0004CC60                                  9346+PL59      DS     0C
0004CC60  00000000 00000000               9347           DC     XL352'00'
0004CC68  00000000 00000000
0004CC70  00000000 00000000
0004CC78  00000000 00000000
0004CC80  00000000 00000000
0004CC88  00000000 00000000
0004CC90  00000000 00000000
0004CC98  00000000 00000000
0004CCA0  00000000 00000000
0004CCA8  00000000 00000000
0004CCB0  00000000 00000000
0004CCB8  00000000 00000000
0004CCC0  00000000 00000000
0004CCC8  00000000 00000000
0004CCD0  00000000 00000000
0004CCD8  00000000 00000000
0004CCE0  00000000 00000000
0004CCE8  00000000 00000000
0004CCF0  00000000 00000000
0004CCF8  00000000 00000000
0004CD00  00000000 00000000
0004CD08  00000000 00000000
0004CD10  00000000 00000000
0004CD18  00000000 00000000
0004CD20  00000000 00000000
0004CD28  00000000 00000000
0004CD30  00000000 00000000
0004CD38  00000000 00000000
0004CD40  00000000 00000000
0004CD48  00000000 00000000
0004CD50  00000000 00000000
0004CD58  00000000 00000000
0004CD60  00000000 00000000
0004CD68  00000000 00000000
0004CD70  00000000 00000000
0004CD78  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0004CD80  00000000 00000000
0004CD88  00000000 00000000
0004CD90  00000000 00000000
0004CD98  00000000 00000000
0004CDA0  00000000 00000000
0004CDA8  00000000 00000000
0004CDB0  00000000 00000000
0004CDB8  00000000 00000000
                                        9348
                                        9349          TSTLBL FMSG               fail message
0004CDC0  D7D9C940 40406E6E             9350+         DC    CL32'PRI   >>>FMSG59<<<'  eye catcher
0004CDC8  6EC6D4E2 C7F5F94C
0004CDD0  4C4C4040 40404040
0004CDD8  40404040 40404040
0004CDE0                                9351+FMSG59   DS    0C
0004CDE0  86818993 858440D7             9352          DC    C'failed PLO-Extension '
0004CDE8  D3D660C5 A7A38595
0004CDF0  A2899695 40
0004CDF5  C3969497 81998540             9353          DC    C'Compare and Load (CLO: 24) : NE Op1-C != Op2'
0004CDFD  81958440 D3968184
0004CE05  404DC3D3 D67A40F2
0004CE0D  F45D407A 40D5C540
0004CE15  D697F160 C3405A7E
0004CE1D  40D697F2
0004CE21  40404040 40404040             9354          DC    CL50' '
0004CE29  40404040 40404040
0004CE31  40404040 40404040
0004CE39  40404040 40404040
0004CE41  40404040 40404040
0004CE49  40404040 40404040
0004CE51  4040
                                        9355          TSTEND
0004CE53  D7D9C940 4040C5D5             9356+TEND59   DC    CL32'PRI   END of test: 59 <<<'
0004CE5B  C4409686 40A385A2
0004CE63  A37A40F5 F9404C4C
0004CE6B  4C404040 40404040
                                        9357
                                        9358 * ------------------------------------------------------------------
                                        9359 * Test Compare and Load (CLO: 24) instruction : EQ Op1-C == OP2
                                        9360 * ------------------------------------------------------------------
                                        9361          TSTHDR 0,99,25            EQ Op1-C == OP2 (cc=0, UR=99)
0004CE73                  0004CE73  0004D000  9362+         ORG   PLOETST+(128*K)+60*(3*K)
0004D000  D7D9C940 4040E2E3             9363+         DC    CL32'PRI   START of test: 60'
0004D008  C1D9E340 968640A3
0004D010  85A2A37A 40F6F040
0004D018  40404040 40404040
0004D020                  0004D020      9364+         USING *,R5                base for test data and test routine
0004D020                  00000000      9365+         USING PLOETEST,R5
0004D020  0004D050                      9366+T60      DC    A(CODE60)           address of test code
0004D024  003C                          9367+         DC    H'60'               test number
0004D026  00                            9368+         DC    X'00'
0004D027  19                            9369+         DC    HL1'25'             Function
0004D028  63                            9370+         DC    HL1'99'             expected unequal results (UR)
0004D029  00                            9371+         DC    HL1'00'             found UR
0004D02A  00                            9372+         DC    HL1'0'              cc
0004D02B  07                            9373+         DC    HL1'7'              not expected CC mask
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0004D02C  00                                       9374+          DS    X                   extracted cc
0004D030  00000000 00000000                        9375+          DS    2F                  extract PSW after test (has CC)
0004D038  0004D9E0                                 9376+          DC    A(FMSG60)           address of test fail message
0004D03C  0004D740                                 9377+          DC    A(OPSRC60)          address of operand source
0004D040  0004D620                                 9378+          DC    A(OPDST60)          address of operand destination
0004D044  0004D860                                 9379+          DC    A(PL60)             address of parameter list
0004D048  000041B4                                 9380+          DC    A(RANDX+60)         address of random source data
0004D04C  0004DA54                                 9381+          DC    A(TEND60)           address of test end eye catche
                                                   9382
                                                   9383           TSTCODE
0004D050                                           9384+CODE60    DS    0F
                                                   9385 *                                   load random source
0004D050  E310 501C 0014                0000001C   9386           LGF   R1,OPSRCA           R1: OPSRC - Primary
0004D056  B904 0061                                9387           LGR   R6,R1               R6: OPSRC - Secondary
0004D05A  B904 0071                                9388           LGR   R7,R1               R7: OPSRC - Home
                                                   9389
0004D05E  E320 5028 0014                00000028   9390           LGF   R2,RANDXA           R2: address random data
0004D064  E330 5024 0014                00000024   9391           LGF   R3,PLA              R3: address PL
0004D06A  E340 5020 0014                00000020   9392           LGF   R4,OPREPA           R4: OPREPA - PrimaryL
                                                   9393
0004D070  B219 0200                     00000200   9394           SAC   ARMODE
0004D074  D21F 4000 2000      00000000  00000000   9395           MVC   0(32,R4),0(R2)      load OP2  (same as OP1C)
0004D07A  D21F 6000 20C0      00000000  000000C0   9396           MVC   0(32,R6),192+0(R2)  OP4 - Secondary
                                                   9397 *
                                                   9398 * Compare and Swap (DCSO: 25)
                                                   9399 * Build Parambter List
                                                   9400 *
0004D080  D21F 3000 2000      00000000  00000000   9401           MVC   0(32,R3),0(R2)      Load PL OP1-C
0004D086  D21F 3020 2020      00000020  00000020   9402           MVC   32(32,R3),32(R2)    load PL OP3
                                                   9403 ****
                                                   9404 * force an EQ by changing the Op1-C
                                                   9405 ****
0004D08C  D21F 3000 4000      00000000  00000000   9406           MVC   0(32,R3),0(R4)      Op1-C <-- Op2
                                                   9407 ****
0004D092  E363 0048 0024                00000048   9408           STG   R6,72(R3)           load OP4 address in PL (sec space)
0004D098  E54C 3044 0001                00000044   9409           MVHI  68(R3),SECALET
                                                   9410
0004D09E  B982 0000                                9411           XGR   R0,R0
0004D0A2  4100 0018                     00000018   9412           LA    R0,24
                                                   9413
0004D0A6  B219 0200                     00000200   9414           SAC   ARMODE
0004D0AA  EE0B 4000 3000      00000000  00000000   9415           PLO   0,0(R4),R11,0(R3)   C/L (CLO: 24)
0004D0B0  B219 0000                     00000000   9416           SAC   PRIMODE
                                                   9417
0004D0B4  B98D 0020                                9418           EPSW  R2,R0               extract psw
0004D0B8  5020 5010                     00000010   9419           ST    R2,CCPSW              to save CC
0004D0BC  E310 500B 0076                0000000B   9420           LB    R1,CCMASK           (failure CC mask)
0004D0C2  8910 0004                     00000004   9421           SLL   R1,4                (shift to BC instr CC position)
0004D0C6  4410 50F4                     0004D114   9422           EX    R1,F242TSTCC            fail if...
                                                   9423
                                        0004D0CA   00000001   9424 F242CKUR EQU   *
0004D0CA  D200 5009 3030      00000009  00000030   9425           MVC   URFOUND,48(R3)      Extract UR
0004D0D0  D500 5008 3030      00000008  00000030   9426           CLC   UR,48(R3)
0004D0D6  4780 50BE                     0004D0DE   9427           BE    F242CKRES
0004D0DA  45F0 8388                     00002388   9428           BAL   R15,URMSG
                                                   9429 *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                                   9430 * compare results
                                                   9431 *
                               0004D0DE  00000001  9432 F242CKRES EQU   *
                                                   9433 *                                reset addresses
0004D0DE  E310 501C 0014                 0000001C  9434          LGF   R1,OPSRCA         R1:  OPSRC - Primary
0004D0E4  B904 0061                              0  9435          LGR   R6,R1             R6:  OPSRC - Secondary
0004D0E8  B904 0071                              0  9436          LGR   R7,R1             R7:  OPSRC - Home
0004D0EC  E330 5024 0014                 00000024  9437          LGF   R3,PLA            R3:  address PL
0004D0F2  E340 5020 0014                 00000020  9438          LGF   R4,OPREPA         R4:  OPREPA - Primary
                                                   9439
0004D0F8  B219 0200                      00000200  9440          SAC   ARMODE
0004D0FC  D51F 6000 3020  00000000       00000020  9441          CLC   0(32,R6),32(R3)   OP4 == OP3
0004D102  B219 0000                      00000000  9442          SAC   PRIMODE
0004D106  478B 0000                      00000000  9443          BE    0(R11)            same, return
                                                   9444
                               0004D10A  00000001  9445 F242XX   EQU   *
0004D10A  B219 0000                      00000000  9446          SAC   PRIMODE
0004D10E  45F0 8450                      00002450  9447          BAL   R15,FAILMSG
0004D112  07FB                                   0  9448          BR    R11
                                                   9449
0004D114  4700 50F8                      0004D118  9450 F242TSTCC BC    0,F242CCMSG       fail if unexpected condition code
                               0004D118  00000001  9451 F242CCMSG EQU   *
0004D118  45F0 82B2                      000022B2  9452          BAL   R15,CCMSG
0004D11C  47F0 50AA                      0004D0CA  9453          B     F242CKUR
                                                   9454
                                                   9455          TSTDATA
0004D120                       0004D120  0004D600  9456+         ORG   PLOETST+(128*K)+60*(3*K)+(K+512)
                                                   9457          TSTLBL OPDST            OP2, Op4
0004D600  D7D9C940 40406E6E                        9458+         DC    CL32'PRI   >>>OPDST60<<<' eye catcher
0004D608  6ED6D7C4 E2E3F6F0
0004D610  4C4C4C40 40404040
0004D618  40404040 40404040
0004D620                                           9459+OPDST60  DS    0C
0004D620  00000000 00000000                        9460          DC    XL256'0'
0004D628  00000000 00000000
0004D630  00000000 00000000
0004D638  00000000 00000000
0004D640  00000000 00000000
0004D648  00000000 00000000
0004D650  00000000 00000000
0004D658  00000000 00000000
0004D660  00000000 00000000
0004D668  00000000 00000000
0004D670  00000000 00000000
0004D678  00000000 00000000
0004D680  00000000 00000000
0004D688  00000000 00000000
0004D690  00000000 00000000
0004D698  00000000 00000000
0004D6A0  00000000 00000000
0004D6A8  00000000 00000000
0004D6B0  00000000 00000000
0004D6B8  00000000 00000000
0004D6C0  00000000 00000000
0004D6C8  00000000 00000000
0004D6D0  00000000 00000000
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT
0004D6D8  00000000 00000000
0004D6E0  00000000 00000000
0004D6E8  00000000 00000000
0004D6F0  00000000 00000000
0004D6F8  00000000 00000000
0004D700  00000000 00000000
0004D708  00000000 00000000
0004D710  00000000 00000000
0004D718  00000000 00000000
                                          9461
                                          9462           TSTLBL OPSRC              Op1-C, Op3
0004D720  D7D9C940 40406E6E               9463+          DC    CL32'PRI   >>>OPSRC60<<<' eye catcher
0004D728  6ED6D7E2 D9C3F6F0
0004D730  4C4C4C40 40404040
0004D738  40404040 40404040
0004D740                                  9464+OPSRC60   DS    0C
0004D740  00000000 00000000               9465           DC    XL256'0'
0004D748  00000000 00000000
0004D750  00000000 00000000
0004D758  00000000 00000000
0004D760  00000000 00000000
0004D768  00000000 00000000
0004D770  00000000 00000000
0004D778  00000000 00000000
0004D780  00000000 00000000
0004D788  00000000 00000000
0004D790  00000000 00000000
0004D798  00000000 00000000
0004D7A0  00000000 00000000
0004D7A8  00000000 00000000
0004D7B0  00000000 00000000
0004D7B8  00000000 00000000
0004D7C0  00000000 00000000
0004D7C8  00000000 00000000
0004D7D0  00000000 00000000
0004D7D8  00000000 00000000
0004D7E0  00000000 00000000
0004D7E8  00000000 00000000
0004D7F0  00000000 00000000
0004D7F8  00000000 00000000
0004D800  00000000 00000000
0004D808  00000000 00000000
0004D810  00000000 00000000
0004D818  00000000 00000000
0004D820  00000000 00000000
0004D828  00000000 00000000
0004D830  00000000 00000000
0004D838  00000000 00000000
                                          9466
                                          9467           TSTLBL PL                 Parameter List
0004D840  D7D9C940 40406E6E               9468+          DC    CL32'PRI   >>>PL60<<<'    eye catcher
0004D848  6ED7D3F6 F04C4C4C
0004D850  40404040 40404040
0004D858  40404040 40404040
0004D860                                  9469+PL60      DS    0C
0004D860  00000000 00000000               9470           DC    XL352'00'
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004D868  00000000 00000000
0004D870  00000000 00000000
0004D878  00000000 00000000
0004D880  00000000 00000000
0004D888  00000000 00000000
0004D890  00000000 00000000
0004D898  00000000 00000000
0004D8A0  00000000 00000000
0004D8A8  00000000 00000000
0004D8B0  00000000 00000000
0004D8B8  00000000 00000000
0004D8C0  00000000 00000000
0004D8C8  00000000 00000000
0004D8D0  00000000 00000000
0004D8D8  00000000 00000000
0004D8E0  00000000 00000000
0004D8E8  00000000 00000000
0004D8F0  00000000 00000000
0004D8F8  00000000 00000000
0004D900  00000000 00000000
0004D908  00000000 00000000
0004D910  00000000 00000000
0004D918  00000000 00000000
0004D920  00000000 00000000
0004D928  00000000 00000000
0004D930  00000000 00000000
0004D938  00000000 00000000
0004D940  00000000 00000000
0004D948  00000000 00000000
0004D950  00000000 00000000
0004D958  00000000 00000000
0004D960  00000000 00000000
0004D968  00000000 00000000
0004D970  00000000 00000000
0004D978  00000000 00000000
0004D980  00000000 00000000
0004D988  00000000 00000000
0004D990  00000000 00000000
0004D998  00000000 00000000
0004D9A0  00000000 00000000
0004D9A8  00000000 00000000
0004D9B0  00000000 00000000
0004D9B8  00000000 00000000
                                        9471
                                        9472          TSTLBL FMSG               fail message
0004D9C0  D7D9C940 40406E6E             9473+         DC    CL32'PRI   >>>FMSG60<<<'  eye catcher
0004D9C8  6EC6D4E2 C7F6F04C
0004D9D0  4C4C4040 40404040
0004D9D8  40404040 40404040
0004D9E0                                9474+FMSG60   DS    0C
0004D9E0  86818993 858440D7             9475          DC    C'failed PLO-Extension '
0004D9E8  D3D660C5 A7A38595
0004D9F0  A2899695 40
0004D9F5  C3969497 81998540             9476          DC    C'Compare and Load (CLO: 24)  : EQ Op1-C == OP2'
0004D9FD  81958440 D3968184
0004DA05  404DC3D3 D67A40F2
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

0004DA0D  F45D4040 7A40C5D8
0004DA15  40D697F1 60C3407E
0004DA1D  7E40D6D7 F2
0004DA22  40404040 40404040                          9477          DC    CL50' '
0004DA2A  40404040 40404040
0004DA32  40404040 40404040
0004DA3A  40404040 40404040
0004DA42  40404040 40404040
0004DA4A  40404040 40404040
0004DA52  4040
                                                     9478          TSTEND
0004DA54  D7D9C940 4040C5D5                          9479+TEND60   DC    CL32'PRI   END of test: 60 <<<'
0004DA5C  C4409686 40A385A2
0004DA64  A37A40F6 F0404C4C
0004DA6C  4C404040 40404040
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   9481 ********************************************************************************
                                                   9482 * table of pointers to individual tests
                                                   9483 ********************************************************************************
                                                   9484
0004DA74                      0004DA74  00004578   9485          ORG   PLOTESTS            (1k reserved for test pointers)
                                                   9486          PTTABLE
00004578                                           9487+TTABLE   DS    0F
00004578  00020C20                                 9488+         DC    A(T1)
0000457C  00021820                                 9489+         DC    A(T2)
00004580  00022420                                 9490+         DC    A(T3)
00004584  00023020                                 9491+         DC    A(T4)
00004588  00023C20                                 9492+         DC    A(T5)
0000458C  00024820                                 9493+         DC    A(T6)
00004590  00025420                                 9494+         DC    A(T7)
00004594  00026020                                 9495+         DC    A(T8)
00004598  00026C20                                 9496+         DC    A(T9)
0000459C  00027820                                 9497+         DC    A(T10)
000045A0  00028420                                 9498+         DC    A(T11)
000045A4  00029020                                 9499+         DC    A(T12)
000045A8  00029C20                                 9500+         DC    A(T13)
000045AC  0002A820                                 9501+         DC    A(T14)
000045B0  0002B420                                 9502+         DC    A(T15)
000045B4  0002C020                                 9503+         DC    A(T16)
000045B8  0002CC20                                 9504+         DC    A(T17)
000045BC  0002D820                                 9505+         DC    A(T18)
000045C0  0002E420                                 9506+         DC    A(T19)
000045C4  0002F020                                 9507+         DC    A(T20)
000045C8  0002FC20                                 9508+         DC    A(T21)
000045CC  00030820                                 9509+         DC    A(T22)
000045D0  00031420                                 9510+         DC    A(T23)
000045D4  00032020                                 9511+         DC    A(T24)
000045D8  00032C20                                 9512+         DC    A(T25)
000045DC  00033820                                 9513+         DC    A(T26)
000045E0  00034420                                 9514+         DC    A(T27)
000045E4  00035020                                 9515+         DC    A(T28)
000045E8  00035C20                                 9516+         DC    A(T29)
000045EC  00036820                                 9517+         DC    A(T30)
000045F0  00037420                                 9518+         DC    A(T31)
000045F4  00038020                                 9519+         DC    A(T32)
000045F8  00038C20                                 9520+         DC    A(T33)
000045FC  00039820                                 9521+         DC    A(T34)
00004600  0003A420                                 9522+         DC    A(T35)
00004604  0003B020                                 9523+         DC    A(T36)
00004608  0003BC20                                 9524+         DC    A(T37)
0000460C  0003C820                                 9525+         DC    A(T38)
00004610  0003D420                                 9526+         DC    A(T39)
00004614  0003E020                                 9527+         DC    A(T40)
00004618  0003EC20                                 9528+         DC    A(T41)
0000461C  0003F820                                 9529+         DC    A(T42)
00004620  00040420                                 9530+         DC    A(T43)
00004624  00041020                                 9531+         DC    A(T44)
00004628  00041C20                                 9532+         DC    A(T45)
0000462C  00042820                                 9533+         DC    A(T46)
00004630  00043420                                 9534+         DC    A(T47)
00004634  00044020                                 9535+         DC    A(T48)
00004638  00044C20                                 9536+         DC    A(T49)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
0000463C  00045820                                9537+          DC    A(T50)
00004640  00046420                                9538+          DC    A(T51)
00004644  00047020                                9539+          DC    A(T52)
00004648  00047C20                                9540+          DC    A(T53)
0000464C  00048820                                9541+          DC    A(T54)
00004650  00049420                                9542+          DC    A(T55)
00004654  0004A020                                9543+          DC    A(T56)
00004658  0004AC20                                9544+          DC    A(T57)
0000465C  0004B820                                9545+          DC    A(T58)
00004660  0004C420                                9546+          DC    A(T59)
00004664  0004D020                                9547+          DC    A(T60)
                                                  9548+*
00004668  00000000                                9549+          DC    A(0)                END OF TABLE
0000466C  00000000                                9550+          DC    A(0)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          9552 ********************************************************************************
                                          9553 *        Registers
                                          9554 ********************************************************************************


                             00000000  00000001  9556 R0       EQU   0
                             00000001  00000001  9557 R1       EQU   1
                             00000002  00000001  9558 R2       EQU   2
                             00000003  00000001  9559 R3       EQU   3
                             00000004  00000001  9560 R4       EQU   4
                             00000005  00000001  9561 R5       EQU   5
                             00000006  00000001  9562 R6       EQU   6
                             00000007  00000001  9563 R7       EQU   7
                             00000008  00000001  9564 R8       EQU   8
                             00000009  00000001  9565 R9       EQU   9
                             0000000A  00000001  9566 R10      EQU   10
                             0000000B  00000001  9567 R11      EQU   11
                             0000000C  00000001  9568 R12      EQU   12
                             0000000D  00000001  9569 R13      EQU   13
                             0000000E  00000001  9570 R14      EQU   14
                             0000000F  00000001  9571 R15      EQU   15


                                          9573 ********************************************************************************
                                          9574 *        Vector Registers
                                          9575 ********************************************************************************


                             00000000  00000001  9577 V0       EQU   0
                             00000001  00000001  9578 V1       EQU   1
                             00000002  00000001  9579 V2       EQU   2
                             00000003  00000001  9580 V3       EQU   3
                             00000004  00000001  9581 V4       EQU   4
                             00000005  00000001  9582 V5       EQU   5
                             00000006  00000001  9583 V6       EQU   6
                             00000007  00000001  9584 V7       EQU   7
                             00000008  00000001  9585 V8       EQU   8
                             00000009  00000001  9586 V9       EQU   9
                             0000000A  00000001  9587 V10      EQU   10
                             0000000B  00000001  9588 V11      EQU   11
                             0000000C  00000001  9589 V12      EQU   12
                             0000000D  00000001  9590 V13      EQU   13
                             0000000E  00000001  9591 V14      EQU   14
                             0000000F  00000001  9592 V15      EQU   15
                             00000010  00000001  9593 V16      EQU   16
                             00000011  00000001  9594 V17      EQU   17
                             00000012  00000001  9595 V18      EQU   18
                             00000013  00000001  9596 V19      EQU   19
                             00000014  00000001  9597 V20      EQU   20
                             00000015  00000001  9598 V21      EQU   21
                             00000016  00000001  9599 V22      EQU   22
                             00000017  00000001  9600 V23      EQU   23
                             00000018  00000001  9601 V24      EQU   24
                             00000019  00000001  9602 V25      EQU   25
                             0000001A  00000001  9603 V26      EQU   26
                             0000001B  00000001  9604 V27      EQU   27
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2     STMT

                              0000001C  00000001  9605 V28      EQU   28
                              0000001D  00000001  9606 V29      EQU   29
                              0000001E  00000001  9607 V30      EQU   30
                              0000001F  00000001  9608 V31      EQU   31
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT
                                          9610 ********************************************************************************
                                          9611 *        Control Registers
                                          9612 ********************************************************************************

                            00000000  00000001  9614 CR0       EQU   0
                            00000001  00000001  9615 CR1       EQU   1
                            00000002  00000001  9616 CR2       EQU   2
                            00000003  00000001  9617 CR3       EQU   3
                            00000004  00000001  9618 CR4       EQU   4
                            00000005  00000001  9619 CR5       EQU   5
                            00000006  00000001  9620 CR6       EQU   6
                            00000007  00000001  9621 CR7       EQU   7
                            00000008  00000001  9622 CR8       EQU   8
                            00000009  00000001  9623 CR9       EQU   9
                            0000000A  00000001  9624 CR10      EQU   10
                            0000000B  00000001  9625 CR11      EQU   11
                            0000000C  00000001  9626 CR12      EQU   12
                            0000000D  00000001  9627 CR13      EQU   13
                            0000000E  00000001  9628 CR14      EQU   14
                            0000000F  00000001  9629 CR15      EQU   15

                                          9631 ********************************************************************************
                                          9632 *        Access Registers
                                          9633 ********************************************************************************

                            00000000  00000001  9635 AR0       EQU   0
                            00000001  00000001  9636 AR1       EQU   1
                            00000002  00000001  9637 AR2       EQU   2
                            00000003  00000001  9638 AR3       EQU   3
                            00000004  00000001  9639 AR4       EQU   4
                            00000005  00000001  9640 AR5       EQU   5
                            00000006  00000001  9641 AR6       EQU   6
                            00000007  00000001  9642 AR7       EQU   7
                            00000008  00000001  9643 AR8       EQU   8
                            00000009  00000001  9644 AR9       EQU   9
                            0000000A  00000001  9645 AR10      EQU   10
                            0000000B  00000001  9646 AR11      EQU   11
                            0000000C  00000001  9647 AR12      EQU   12
                            0000000D  00000001  9648 AR13      EQU   13
                            0000000E  00000001  9649 AR14      EQU   14
                            0000000F  00000001  9650 AR15      EQU   15
                                          9651
                                          9652          END
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
A000LABL	U	000000	1	172	935 938 941 947 954 961 971 996 1021 1043
ALE0	X	019200	1	1023	
ALE1	X	019210	1	1024	
ALE2	X	019220	16	1026	
ALE3	X	019230	1	1032	
ALE4	X	019240	1	1033	
ALE5	X	019250	1	1034	
ALE6	X	019260	1	1035	
ALE7	X	019270	1	1036	
AR0	U	000000	1	9635	343 380
AR1	U	000001	1	9636	
AR10	U	00000A	1	9645	
AR11	U	00000B	1	9646	
AR12	U	00000C	1	9647	584
AR13	U	00000D	1	9648	586
AR14	U	00000E	1	9649	
AR15	U	00000F	1	9650	343 380
AR2	U	000002	1	9637	
AR3	U	000003	1	9638	
AR4	U	000004	1	9639	
AR5	U	000005	1	9640	
AR6	U	000006	1	9641	580
AR7	U	000007	1	9642	582
AR8	U	000008	1	9643	
AR9	U	000009	1	9644	
AREGS	F	004A00	4	929	343 380
ARMODE	U	000200	1	1068	587 1323 1350 1377 1391 1468 1495 1522 1536 1613 1640 1667 1681 1759 1786 1813 1827 1906 1928 1950 1964 2039 2060 2083 2098 2174 2196 2219 2233 2308 2330 2353 2368 2445 2462 2481 2495 2569 2586 2605 2619 2692 2720 2735 2808 2836 2851 2928 2940 2955 2969 3041 3076 3100 3182 3217 3241 3323 3359 3383 3465 3500 3524 3607 3642 3667 3751 3786 3810 3892 3926 3950 4032 4066 4090 4172 4206 4230 4313 4347 4373 4457 4491 4515 4597 4631 4655 4737 4771 4795 4877 4912 4936 5018 5052 5078 5162 5203 5227 5309 5350 5374 5456 5497 5521 5603 5644 5668 5750 5791 5817 5901 5930 5954 6034 6063 6087 6167 6196 6220 6300 6329 6355 6437 6466 6490 6570 6599 6623 6703 6732 6756 6836 6865 6891 6973 7002 7026 7106 7135 7159 7239 7268 7292 7372 7401 7427 7509 7544 7568 7647 7682 7706 7785 7820 7844 7923 7958 7986 8068 8102 8128 8205 8240 8266 8352 8380 8406 8483 8511 8537 8621 8642 8668 8745 8769 8795 8877 8899 8925 9003 9025 9051 9130 9153 9179 9271 9291 9317 9394 9414 9440
BEGIN	I	002000	2	215	179 211 212 213
CC	U	00000A	1	1090	409 438
CCFOUND	X	00000C	1	1093	420 445
CCMASK	U	00000B	1	1091	3082 3223 3365 3506 3648 3792 3932 4072 4212 4353 4497 4637 4777 4918 5058 5209 5356 5503 5650 5797 5936 6069 6202 6335 6472 6605 6738 6871 7008 7141 7274 7407 7550 7688 7826 7965 8108 8246 8386 8517 8648 8775 8905 9031 9159 9297 9420
CCMSG	U	0022B2	1	402	3118 3259 3401 3542 3685 3828 3968 4108 4248 4391 4533 4673 4813 4954 5096 5245 5392 5539 5686 5835 5970 6103 6236 6371 6506 6639 6772 6907 7042 7175 7308 7443 7584 7722 7860 8002 8140 8284 8418 8553 8680 8809 8938 9065 9193 9329 9452
CCPRTEXP	C	0040C4	1	787	442
CCPRTFUN	C	0040B3	2	784	435
CCPRTGOT	C	0040D4	1	790	449
CCPRTLINE	C	004077	16	780	792 452
CCPRTLNG	U	00005F	1	792	451

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CCPRTNUM	C	004087	3	781	428
CCPSW	F	000010	4	1094	417 3081 3222 3364 3505 3647 3791 3931 4071 4211 4352 4496 4636 4776 4917 5057 5208 5355 5502 5649 5796 5935 6068 6201 6334 6471 6604 6737 6870 7007 7140 7273 7406 7549 7687 7825 7964 8107 8245 8385 8516 8647 8774 8904 9030 9158 9296 9419
CODE1	F	020C50	4	1314	1296
CODE10	F	027850	4	2559	2541
CODE11	F	028450	4	2683	2665
CODE12	F	029050	4	2799	2781
CODE13	F	029C50	4	2918	2900
CODE14	F	02A850	4	3032	3014
CODE15	F	02B450	4	3173	3155
CODE16	F	02C050	4	3314	3296
CODE17	F	02CC50	4	3456	3438
CODE18	F	02D850	4	3598	3580
CODE19	F	02E450	4	3742	3724
CODE2	F	021850	4	1459	1441
CODE20	F	02F050	4	3883	3865
CODE21	F	02FC50	4	4023	4005
CODE22	F	030850	4	4163	4145
CODE23	F	031450	4	4304	4286
CODE24	F	032050	4	4448	4430
CODE25	F	032C50	4	4588	4570
CODE26	F	033850	4	4728	4710
CODE27	F	034450	4	4868	4850
CODE28	F	035050	4	5009	4991
CODE29	F	035C50	4	5153	5135
CODE3	F	022450	4	1604	1586
CODE30	F	036850	4	5300	5282
CODE31	F	037450	4	5447	5429
CODE32	F	038050	4	5594	5576
CODE33	F	038C50	4	5741	5723
CODE34	F	039850	4	5892	5874
CODE35	F	03A450	4	6025	6007
CODE36	F	03B050	4	6158	6140
CODE37	F	03BC50	4	6291	6273
CODE38	F	03C850	4	6428	6410
CODE39	F	03D450	4	6561	6543
CODE4	F	023050	4	1750	1732
CODE40	F	03E050	4	6694	6676
CODE41	F	03EC50	4	6827	6809
CODE42	F	03F850	4	6964	6946
CODE43	F	040450	4	7097	7079
CODE44	F	041050	4	7230	7212
CODE45	F	041C50	4	7363	7345
CODE46	F	042850	4	7500	7482
CODE47	F	043450	4	7638	7620
CODE48	F	044050	4	7776	7758
CODE49	F	044C50	4	7914	7896
CODE5	F	023C50	4	1897	1879
CODE50	F	045850	4	8058	8040
CODE51	F	046450	4	8195	8177
CODE52	F	047050	4	8342	8324
CODE53	F	047C50	4	8473	8455
CODE54	F	048850	4	8611	8593
CODE55	F	049450	4	8735	8717

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
CODE56	F	04A050	4	8867	8849
CODE57	F	04AC50	4	8993	8975
CODE58	F	04B850	4	9120	9102
CODE59	F	04C450	4	9261	9243
CODE6	F	024850	4	2030	2012
CODE60	F	04D050	4	9384	9366
CODE7	F	025450	4	2165	2147
CODE8	F	026050	4	2299	2281
CODE9	F	026C50	4	2436	2418
CR0	U	000000	1	9614	344
CR1	U	000001	1	9615	
CR10	U	00000A	1	9624	
CR11	U	00000B	1	9625	
CR12	U	00000C	1	9626	
CR13	U	00000D	1	9627	
CR14	U	00000E	1	9628	
CR15	U	00000F	1	9629	344
CR2	U	000002	1	9616	
CR3	U	000003	1	9617	
CR4	U	000004	1	9618	
CR5	U	000005	1	9619	
CR6	U	000006	1	9620	
CR7	U	000007	1	9621	
CR8	U	000008	1	9622	
CR9	U	000009	1	9623	
CREG0	X	004980	4	912	
CREG1	X	004988	4	913	
CREG10	X	0049D0	4	922	
CREG11	X	0049D8	4	923	
CREG12	X	0049E0	4	924	
CREG13	X	0049E8	4	925	
CREG14	X	0049F0	4	926	
CREG15	X	0049F8	4	927	
CREG2	X	004990	4	914	
CREG3	X	004998	4	915	
CREG4	X	0049A0	4	916	
CREG5	X	0049A8	4	917	
CREG6	X	0049B0	4	918	
CREG7	X	0049B8	4	919	
CREG8	X	0049C0	4	920	
CREG9	X	0049C8	4	921	
CREGS	D	004980	8	911	344
CTLR0	F	00279C	4	738	347 348 349 350
DECNUM	C	004161	16	823	425 427 432 434 439 441 446 448 488 490 495 497 502 504 509 511 538 540
DUAL	U	019200	1	1022	1002
DUALD	A	019110	4	1002	
DUCT	X	019100	64	997	914
EDIT	X	004135	18	818	426 433 440 447 489 496 503 510 539
ENDTEST	U	0022A0	1	390	361
EOJ	I	0026C8	4	714	255 393 395
EOJPSW	D	0026B8	8	712	714
F241CCMSG	U	04C518	1	9328	9327
F241CKRES	U	04C4DE	1	9309	9304
F241CKUR	U	04C4CA	1	9301	9330
F241TSTCC	I	04C514	4	9327	9299

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
F241XX	U	04C50A	1	9322	
F242CCMSG	U	04D118	1	9451	9450
F242CKRES	U	04D0DE	1	9432	9427
F242CKUR	U	04D0CA	1	9424	9453
F242TSTCC	I	04D114	4	9450	9422
F242XX	U	04D10A	1	9445	
F261CCMSG	U	04A12A	1	8937	8936
F261CKRES	U	04A0EA	1	8917	8912
F261CKUR	U	04A0D6	1	8909	8939
F261TSTCC	I	04A126	4	8936	8907
F261XX	U	04A11C	1	8931	
F262CCMSG	U	04AD2E	1	9064	9063
F262CKRES	U	04ACEA	1	9043	9038
F262CKUR	U	04ACD6	1	9035	9066
F262TSTCC	I	04AD2A	4	9063	9033
F262XX	U	04AD20	1	9058	9053
F263CCMSG	U	04B934	1	9192	9191
F263CKRES	U	04B8F0	1	9171	9166
F263CKUR	U	04B8DC	1	9163	9194
F263TSTCC	I	04B930	4	9191	9161
F263XX	U	04B926	1	9186	9181
F271CCMSG	U	04891E	1	8679	8678
F271CKRES	U	0488E4	1	8660	8655
F271CKUR	U	0488D0	1	8652	8681
F271TSTCC	I	04891A	4	8678	8650
F271XX	U	048910	1	8673	
F272CCMSG	U	04952E	1	8808	8807
F272CKRES	U	0494EA	1	8787	8782
F272CKUR	U	0494D6	1	8779	8810
F272TSTCC	I	04952A	4	8807	8777
F272XX	U	049520	1	8802	8797
F281CCMSG	U	04713A	1	8417	8416
F281CKRES	U	047100	1	8398	8393
F281CKUR	U	0470EC	1	8390	8419
F281TSTCC	I	047136	4	8416	8388
F281XX	U	04712C	1	8411	
F282CCMSG	U	047D4E	1	8552	8551
F282CKRES	U	047D00	1	8529	8524
F282CKUR	U	047CEC	1	8521	8554
F282TSTCC	I	047D4A	4	8551	8519
F282XX	U	047D40	1	8546	8539 8541
F291CCMSG	U	045956	1	8139	8138
F291CKRES	U	04591C	1	8120	8115
F291CKUR	U	045908	1	8112	8141
F291TSTCC	I	045952	4	8138	8110
F291XX	U	045948	1	8133	
F292CCMSG	U	046574	1	8283	8282
F292CKRES	U	04651C	1	8258	8253
F292CKUR	U	046508	1	8250	8285
F292TSTCC	I	046570	4	8282	8248
F292XX	U	046566	1	8277	8268 8270 8272
F301CCMSG	U	042958	1	7583	7582
F301CKRES	U	042916	1	7562	7557
F301CKUR	U	042902	1	7554	7585
F301TSTCC	I	042954	4	7582	7552
F301XX	U	04294A	1	7577	7570 7572

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
F302CCMSG	U	043558	1	7721	7720
F302CKRES	U	043516	1	7700	7695
F302CKUR	U	043502	1	7692	7723
F302TSTCC	I	043554	4	7720	7690
F302XX	U	04354A	1	7715	7708 7710
F303CCMSG	U	044158	1	7859	7858
F303CKRES	U	044116	1	7838	7833
F303CKUR	U	044102	1	7830	7861
F303TSTCC	I	044154	4	7858	7828
F303XX	U	04414A	1	7853	7846 7848
F304CCMSG	U	044D56	1	8001	8000
F304CKRES	U	044D10	1	7977	7972
F304CKUR	U	044CFC	1	7969	8003
F304TSTCC	I	044D52	4	8000	7967
F304XX	U	044D48	1	7995	7988 7990
F311CCMSG	U	03F940	1	7041	7040
F311CKRES	U	03F8FE	1	7020	7015
F311CKUR	U	03F8EA	1	7012	7043
F311TSTCC	I	03F93C	4	7040	7010
F311XX	U	03F932	1	7035	7028 7030
F312CCMSG	U	040540	1	7174	7173
F312CKRES	U	0404FE	1	7153	7148
F312CKUR	U	0404EA	1	7145	7176
F312TSTCC	I	04053C	4	7173	7143
F312XX	U	040532	1	7168	7161 7163
F313CCMSG	U	041140	1	7307	7306
F313CKRES	U	0410FE	1	7286	7281
F313CKUR	U	0410EA	1	7278	7309
F313TSTCC	I	04113C	4	7306	7276
F313XX	U	041132	1	7301	7294 7296
F314CCMSG	U	041D3A	1	7442	7441
F314CKRES	U	041CF8	1	7419	7414
F314CKUR	U	041CE4	1	7411	7444
F314TSTCC	I	041D36	4	7441	7409
F314XX	U	041D2C	1	7436	7429 7431
F321CCMSG	U	03C940	1	6505	6504
F321CKRES	U	03C8FE	1	6484	6479
F321CKUR	U	03C8EA	1	6476	6507
F321TSTCC	I	03C93C	4	6504	6474
F321XX	U	03C932	1	6499	6492 6494
F322CCMSG	U	03D540	1	6638	6637
F322CKRES	U	03D4FE	1	6617	6612
F322CKUR	U	03D4EA	1	6609	6640
F322TSTCC	I	03D53C	4	6637	6607
F322XX	U	03D532	1	6632	6625 6627
F323CCMSG	U	03E140	1	6771	6770
F323CKRES	U	03E0FE	1	6750	6745
F323CKUR	U	03E0EA	1	6742	6773
F323TSTCC	I	03E13C	4	6770	6740
F323XX	U	03E132	1	6765	6758 6760
F324CCMSG	U	03ED3A	1	6906	6905
F324CKRES	U	03ECF8	1	6883	6878
F324CKUR	U	03ECE4	1	6875	6908
F324TSTCC	I	03ED36	4	6905	6873
F324XX	U	03ED2C	1	6900	6893 6895
F331CCMSG	U	039940	1	5969	5968

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F331CKRES	U	0398FE	1	5948	5943		
F331CKUR	U	0398EA	1	5940	5971		
F331TSTCC	I	03993C	4	5968	5938		
F331XX	U	039932	1	5963	5956	5958	
F332CCMSG	U	03A540	1	6102	6101		
F332CKRES	U	03A4FE	1	6081	6076		
F332CKUR	U	03A4EA	1	6073	6104		
F332TSTCC	I	03A53C	4	6101	6071		
F332XX	U	03A532	1	6096	6089	6091	
F333CCMSG	U	03B140	1	6235	6234		
F333CKRES	U	03B0FE	1	6214	6209		
F333CKUR	U	03B0EA	1	6206	6237		
F333TSTCC	I	03B13C	4	6234	6204		
F333XX	U	03B132	1	6229	6222	6224	
F334CCMSG	U	03BD3A	1	6370	6369		
F334CKRES	U	03BCF8	1	6347	6342		
F334CKUR	U	03BCE4	1	6339	6372		
F334TSTCC	I	03BD36	4	6369	6337		
F334XX	U	03BD2C	1	6364	6357	6359	
F341CCMSG	U	035D84	1	5244	5243		
F341CKRES	U	035D38	1	5221	5216		
F341CKUR	U	035D24	1	5213	5246		
F341TSTCC	I	035D80	4	5243	5211		
F341XX	U	035D76	1	5238	5229	5231	5233
F342CCMSG	U	036984	1	5391	5390		
F342CKRES	U	036938	1	5368	5363		
F342CKUR	U	036924	1	5360	5393		
F342TSTCC	I	036980	4	5390	5358		
F342XX	U	036976	1	5385	5376	5378	5380
F343CCMSG	U	037584	1	5538	5537		
F343CKRES	U	037538	1	5515	5510		
F343CKUR	U	037524	1	5507	5540		
F343TSTCC	I	037580	4	5537	5505		
F343XX	U	037576	1	5532	5523	5525	5527
F344CCMSG	U	038184	1	5685	5684		
F344CKRES	U	038138	1	5662	5657		
F344CKUR	U	038124	1	5654	5687		
F344TSTCC	I	038180	4	5684	5652		
F344XX	U	038176	1	5679	5670	5672	5674
F345CCMSG	U	038D7E	1	5834	5833		
F345CKRES	U	038D32	1	5809	5804		
F345CKUR	U	038D1E	1	5801	5836		
F345TSTCC	I	038D7A	4	5833	5799		
F345XX	U	038D70	1	5828	5819	5821	5823
F351CCMSG	U	032160	1	4532	4531		
F351CKRES	U	032114	1	4509	4504		
F351CKUR	U	032100	1	4501	4534		
F351TSTCC	I	03215C	4	4531	4499		
F351XX	U	032152	1	4526	4517	4519	4521
F352CCMSG	U	032D60	1	4672	4671		
F352CKRES	U	032D14	1	4649	4644		
F352CKUR	U	032D00	1	4641	4674		
F352TSTCC	I	032D5C	4	4671	4639		
F352XX	U	032D52	1	4666	4657	4659	4661
F353CCMSG	U	033960	1	4812	4811		
F353CKRES	U	033914	1	4789	4784		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES		
F353CKUR	U	033900	1	4781	4814		
F353TSTCC	I	03395C	4	4811	4779		
F353XX	U	033952	1	4806	4797	4799	4801
F354CCMSG	U	034560	1	4953	4952		
F354CKRES	U	034514	1	4930	4925		
F354CKUR	U	034500	1	4922	4955		
F354TSTCC	I	03455C	4	4952	4920		
F354XX	U	034552	1	4947	4938	4940	4942
F355CCMSG	U	03515A	1	5095	5094		
F355CKRES	U	03510E	1	5070	5065		
F355CKUR	U	0350FA	1	5062	5097		
F355TSTCC	I	035156	4	5094	5060		
F355XX	U	03514C	1	5089	5080	5082	5084
F361CCMSG	U	02E560	1	3827	3826		
F361CKRES	U	02E514	1	3804	3799		
F361CKUR	U	02E500	1	3796	3829		
F361TSTCC	I	02E55C	4	3826	3794		
F361XX	U	02E552	1	3821	3812	3814	3816
F362CCMSG	U	02F160	1	3967	3966		
F362CKRES	U	02F114	1	3944	3939		
F362CKUR	U	02F100	1	3936	3969		
F362TSTCC	I	02F15C	4	3966	3934		
F362XX	U	02F152	1	3961	3952	3954	3956
F363CCMSG	U	02FD60	1	4107	4106		
F363CKRES	U	02FD14	1	4084	4079		
F363CKUR	U	02FD00	1	4076	4109		
F363TSTCC	I	02FD5C	4	4106	4074		
F363XX	U	02FD52	1	4101	4092	4094	4096
F364CCMSG	U	030960	1	4247	4246		
F364CKRES	U	030914	1	4224	4219		
F364CKUR	U	030900	1	4216	4249		
F364TSTCC	I	03095C	4	4246	4214		
F365CCMSG	U	03155A	1	4390	4389		
F365CKRES	U	03150E	1	4365	4360		
F365CKUR	U	0314FA	1	4357	4392		
F365TSTCC	I	031556	4	4389	4355		
F371CCMSG	U	02A960	1	3117	3116		
F371CKRES	U	02A914	1	3094	3089		
F371CKUR	U	02A900	1	3086	3119		
F371TSTCC	I	02A95C	4	3116	3084		
F372CCMSG	U	02B560	1	3258	3257		
F372CKRES	U	02B514	1	3235	3230		
F372CKUR	U	02B500	1	3227	3260		
F372TSTCC	I	02B55C	4	3257	3225		
F373CCMSG	U	02C160	1	3400	3399		
F373CKRES	U	02C114	1	3377	3372		
F373CKUR	U	02C100	1	3369	3402		
F373TSTCC	I	02C15C	4	3399	3367		
F373XX	U	02C152	1	3394	3385	3387	3389
F374CCMSG	U	02CD60	1	3541	3540		
F374CKRES	U	02CD14	1	3518	3513		
F374CKUR	U	02CD00	1	3510	3543		
F374TSTCC	I	02CD5C	4	3540	3508		
F374XX	U	02CD52	1	3535	3526	3528	3530
F375CCMSG	U	02D95A	1	3684	3683		
F375CKRES	U	02D90E	1	3660	3655		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
F375CKUR	U	02D8FA	1	3652	3686
F375TSTCC	I	02D956	4	3683	3650
F375XX	U	02D94C	1	3678	3669 3671 3673
F4053	U	026D26	1	2502	2497
F4154	U	026156	1	2377	2370 2372
F4255	U	025556	1	2242	2235 2237
F4352	U	027926	1	2626	2621
F4356	U	024956	1	2108	2100 2102
F4457	U	023D56	1	1973	1966 1968
F4558	U	023186	1	1838	1829 1831 1833
F4659	U	022586	1	1693	1683 1685 1687
F4760	U	021986	1	1547	1538 1540 1542
F4861	U	020D86	1	1402	1393 1395 1397
FAILED	F	004000	4	756	391 413 483 535
FAILHLNG	U	000015	1	768	
FAILLINE	C	004008	16	765	768 771 556
FAILLNG	U	00006F	1	771	555
FAILMSG	U	002450	1	533	1404 1549 1695 1840 1975 2110 2244 2379 2504 2628 2744 2860 2974 3113 3254 3396 3537 3680 3823 3963 4103 4243 4386 4528 4668 4808 4949 5091 5240 5387 5534 5681 5830 5965 6098 6231 6366 6501 6634 6767 6902 7037 7170 7303 7438 7579 7717 7855 7997 8135 8279 8413 8548 8675 8804 8933 9060 9188 9324 9447
FAILPSW	D	0026D0	8	716	718
FAILTEST	I	0026E0	4	718	394
FAILTNUM	C	004018	3	766	541
FAILTXT	C	00401D	90	770	543 546
FAILTXTA	A	000018	4	1096	545
FB0001	F	002070	8	239	243 244 246
FCCSAV	D	002378	8	463	405 455
FCCSVR15	F	00236C	4	460	403 457
FCCSVR5	F	002370	4	461	404 456
FLDWSAV	D	0024B8	8	567	554 559
FLSVR15	F	0024AC	4	564	
FLSVR5	F	0024B0	4	565	
FMSG1	C	021480	1	1426	1306
FMSG10	C	028080	1	2650	2551
FMSG11	C	028C80	1	2766	2675
FMSG12	C	029880	1	2882	2791
FMSG13	C	02A480	1	2996	2910
FMSG14	C	02B1E0	1	3140	3024
FMSG15	C	02BDE0	1	3281	3165
FMSG16	C	02C9E0	1	3423	3306
FMSG17	C	02D5E0	1	3564	3448
FMSG18	C	02E1E0	1	3707	3590
FMSG19	C	02EDE0	1	3850	3734
FMSG2	C	022080	1	1571	1451
FMSG20	C	02F9E0	1	3990	3875
FMSG21	C	0305E0	1	4130	4015
FMSG22	C	0311E0	1	4270	4155
FMSG23	C	031DE0	1	4413	4296
FMSG24	C	0329E0	1	4555	4440
FMSG25	C	0335E0	1	4695	4580
FMSG26	C	0341E0	1	4835	4720
FMSG27	C	034DE0	1	4976	4860
FMSG28	C	0359E0	1	5118	5001
FMSG29	C	0365E0	1	5267	5145

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
FMSG3	C	022C80	1	1717	1596
FMSG30	C	0371E0	1	5414	5292
FMSG31	C	037DE0	1	5561	5439
FMSG32	C	0389E0	1	5708	5586
FMSG33	C	0395E0	1	5857	5733
FMSG34	C	03A1E0	1	5992	5884
FMSG35	C	03ADE0	1	6125	6017
FMSG36	C	03B9E0	1	6258	6150
FMSG37	C	03C5E0	1	6393	6283
FMSG38	C	03D1E0	1	6528	6420
FMSG39	C	03DDE0	1	6661	6553
FMSG4	C	023880	1	1862	1742
FMSG40	C	03E9E0	1	6794	6686
FMSG41	C	03F5E0	1	6929	6819
FMSG42	C	0401E0	1	7064	6956
FMSG43	C	040DE0	1	7197	7089
FMSG44	C	0419E0	1	7330	7222
FMSG45	C	0425E0	1	7465	7355
FMSG46	C	0431E0	1	7606	7492
FMSG47	C	043DE0	1	7744	7630
FMSG48	C	0449E0	1	7882	7768
FMSG49	C	0455E0	1	8024	7906
FMSG5	C	024480	1	1997	1889
FMSG50	C	0461E0	1	8162	8050
FMSG51	C	046DE0	1	8306	8187
FMSG52	C	0479E0	1	8440	8334
FMSG53	C	0485E0	1	8575	8465
FMSG54	C	0491E0	1	8702	8603
FMSG55	C	049DE0	1	8831	8727
FMSG56	C	04A9E0	1	8960	8859
FMSG57	C	04B5E0	1	9087	8985
FMSG58	C	04C1E0	1	9215	9112
FMSG59	C	04CDE0	1	9351	9253
FMSG6	C	025080	1	2132	2022
FMSG60	C	04D9E0	1	9474	9376
FMSG7	C	025C80	1	2266	2157
FMSG8	C	026880	1	2401	2291
FMSG9	C	027480	1	2526	2428
FUN	U	000007	1	1086	431 494
FXX137	U	02A952	1	3111	3102 3104 3106
FXX237	U	02B552	1	3252	3243 3245 3247
FXX437	U	030952	1	4241	4232 4234 4236
FXX50	U	029104	1	2858	2853
FXX51	U	02850C	1	2742	2737
FXX537	U	03154C	1	4384	4375 4377 4379
HASTEO	X	01A000	64	1044	1029
HOMEALET	U	000002	1	1076	581 585 1341 1360 1364 1486 1505 1509 1631 1650 1654 1777 1796 1800 1923 1938 1942 2056 2070 2074 2191 2206 2210 2325 2340 2344 2472 2476 2596 2600 2711 2715 2827 2831 2950 3067 3208 3350 3491 3633 3777 3917 4057 4197 4338 4482 4622 4626 4762 4903 5043 5194 5341 5488 5635 5782 5925 6058 6191 6324 6461 6594 6727 6860 6997 7130 7263 7396 7539 7677 7815 7953 8097 8235
HOMEMODE	U	000300	1	1067	
IMAGE	1	000000	318068	0	
K	U	000400	1	743	744 745 746 1292 1408 1437 1553 1582 1699 1728 1844 1875 1979 2008 2114 2143 2248 2277 2383 2414 2508 2537 2632 2661 2748 2777 2864 2896

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
					2978 3010 3122 3151 3263 3292 3405 3434 3546 3576 3689 3720 3832 3861
					3972 4001 4112 4141 4252 4282 4395 4426 4537 4566 4677 4706 4817 4846
					4958 4987 5100 5131 5249 5278 5396 5425 5543 5572 5690 5719 5839 5870
					5974 6003 6107 6136 6240 6269 6375 6406 6510 6539 6643 6672 6776 6805
					6911 6942 7046 7075 7179 7208 7312 7341 7447 7478 7588 7616 7726 7754
					7864 7892 8006 8036 8144 8173 8288 8320 8422 8451 8557 8589 8684 8713
					8813 8845 8942 8971 9069 9098 9197 9239 9333 9362 9456 277 304 331
K64	U	010000	1	745	
MB	U	100000	1	746	
MSG	I	002600	4	673	254 557 657
MSGCMD	C	00264E	9	703	686 687
MSGMSG	C	002657	95	704	680 701 678
MSGMVC	I	002648	6	701	684
MSGOK	I	002616	2	682	679
MSGRET	I	002636	4	697	690 693
MSGSAVE	F	00263C	4	700	676 697
NEXTTEST	U	0021DE	1	358	383
OPDST1	C	021220	1	1411	1308
OPDST10	C	027E20	1	2635	2553
OPDST11	C	028A20	1	2751	2677
OPDST12	C	029620	1	2867	2793
OPDST13	C	02A220	1	2981	2912
OPDST14	C	02AE20	1	3125	3026
OPDST15	C	02BA20	1	3266	3167
OPDST16	C	02C620	1	3408	3308
OPDST17	C	02D220	1	3549	3450
OPDST18	C	02DE20	1	3692	3592
OPDST19	C	02EA20	1	3835	3736
OPDST2	C	021E20	1	1556	1453
OPDST20	C	02F620	1	3975	3877
OPDST21	C	030220	1	4115	4017
OPDST22	C	030E20	1	4255	4157
OPDST23	C	031A20	1	4398	4298
OPDST24	C	032620	1	4540	4442
OPDST25	C	033220	1	4680	4582
OPDST26	C	033E20	1	4820	4722
OPDST27	C	034A20	1	4961	4862
OPDST28	C	035620	1	5103	5003
OPDST29	C	036220	1	5252	5147
OPDST3	C	022A20	1	1702	1598
OPDST30	C	036E20	1	5399	5294
OPDST31	C	037A20	1	5546	5441
OPDST32	C	038620	1	5693	5588
OPDST33	C	039220	1	5842	5735
OPDST34	C	039E20	1	5977	5886
OPDST35	C	03AA20	1	6110	6019
OPDST36	C	03B620	1	6243	6152
OPDST37	C	03C220	1	6378	6285
OPDST38	C	03CE20	1	6513	6422
OPDST39	C	03DA20	1	6646	6555
OPDST4	C	023620	1	1847	1744
OPDST40	C	03E620	1	6779	6688
OPDST41	C	03F220	1	6914	6821
OPDST42	C	03FE20	1	7049	6958
OPDST43	C	040A20	1	7182	7091
OPDST44	C	041620	1	7315	7224

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
OPDST45	C	042220	1	7450	7357
OPDST46	C	042E20	1	7591	7494
OPDST47	C	043A20	1	7729	7632
OPDST48	C	044620	1	7867	7770
OPDST49	C	045220	1	8009	7908
OPDST5	C	024220	1	1982	1891
OPDST50	C	045E20	1	8147	8052
OPDST51	C	046A20	1	8291	8189
OPDST52	C	047620	1	8425	8336
OPDST53	C	048220	1	8560	8467
OPDST54	C	048E20	1	8687	8605
OPDST55	C	049A20	1	8816	8729
OPDST56	C	04A620	1	8945	8861
OPDST57	C	04B220	1	9072	8987
OPDST58	C	04BE20	1	9200	9114
OPDST59	C	04CA20	1	9336	9255
OPDST6	C	024E20	1	2117	2024
OPDST60	C	04D620	1	9459	9378
OPDST7	C	025A20	1	2251	2159
OPDST8	C	026620	1	2386	2293
OPDST9	C	027220	1	2511	2430
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OPSRC51	C	046B40	1	8296	8188
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OPSRC6	C	024EC0	1	2122	2023
OPSRC60	C	04D740	1	9464	9377
OPSRC7	C	025AC0	1	2256	2158
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PL3	C	022B60	1	1712	1599
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PL31	C	037C60	1	5556	5442
PL32	C	038860	1	5703	5589
PL33	C	039460	1	5852	5736
PL34	C	03A060	1	5987	5887
PL35	C	03AC60	1	6120	6020
PL36	C	03B860	1	6253	6153
PL37	C	03C460	1	6388	6286
PL38	C	03D060	1	6523	6423
PL39	C	03DC60	1	6656	6556
PL4	C	023760	1	1857	1745
PL40	C	03E860	1	6789	6689
PL41	C	03F460	1	6924	6822
PL42	C	040060	1	7059	6959
PL43	C	040C60	1	7192	7092
PL44	C	041860	1	7325	7225
PL45	C	042460	1	7460	7358
PL46	C	043060	1	7601	7495
PL47	C	043C60	1	7739	7633
PL48	C	044860	1	7877	7771
PL49	C	045460	1	8019	7909
PL5	C	024360	1	1992	1892
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PL52	C	047860	1	8435	8337
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PL54	C	049060	1	8697	8606
PL55	C	049C60	1	8826	8730
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PL58	C	04C060	1	9210	9115
PL59	C	04CC60	1	9346	9256
PL6	C	024F60	1	2127	2025
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PL7	C	025B60	1	2261	2160
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SYMBOL TYPE VALUE LENGTH DEFN REFERENCES

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R9	U	000009	1	9565	212 219 220 222
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RPTERROR	I	0025C6	4	651	453 516
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SEGSEC	X	00D000	4	939	919
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T13	A	029C20	4	2900	9500
T14	A	02A820	4	3014	9501
T15	A	02B420	4	3155	9502
T16	A	02C020	4	3296	9503
T17	A	02CC20	4	3438	9504
T18	A	02D820	4	3580	9505
T19	A	02E420	4	3724	9506
T2	A	021820	4	1441	9489
T20	A	02F020	4	3865	9507
T21	A	02FC20	4	4005	9508
T22	A	030820	4	4145	9509
T23	A	031420	4	4286	9510
T24	A	032020	4	4430	9511
T25	A	032C20	4	4570	9512
T26	A	033820	4	4710	9513

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
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T3	A	022420	4	1586	9490
T30	A	036820	4	5282	9517
T31	A	037420	4	5429	9518
T32	A	038020	4	5576	9519
T33	A	038C20	4	5723	9520
T34	A	039820	4	5874	9521
T35	A	03A420	4	6007	9522
T36	A	03B020	4	6140	9523
T37	A	03BC20	4	6273	9524
T38	A	03C820	4	6410	9525
T39	A	03D420	4	6543	9526
T4	A	023020	4	1732	9491
T40	A	03E020	4	6676	9527
T41	A	03EC20	4	6809	9528
T42	A	03F820	4	6946	9529
T43	A	040420	4	7079	9530
T44	A	041020	4	7212	9531
T45	A	041C20	4	7345	9532
T46	A	042820	4	7482	9533
T47	A	043420	4	7620	9534
T48	A	044020	4	7758	9535
T49	A	044C20	4	7896	9536
T5	A	023C20	4	1879	9492
T50	A	045820	4	8040	9537
T51	A	046420	4	8177	9538
T52	A	047020	4	8324	9539
T53	A	047C20	4	8455	9540
T54	A	048820	4	8593	9541
T55	A	049420	4	8717	9542
T56	A	04A020	4	8849	9543
T57	A	04AC20	4	8975	9544
T58	A	04B820	4	9102	9545
T59	A	04C420	4	9243	9546
T6	A	024820	4	2012	9493
T60	A	04D020	4	9366	9547
T7	A	025420	4	2147	9494
T8	A	026020	4	2281	9495
T9	A	026C20	4	2418	9496
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TEND10	C	0280F5	32	2655	2556
TEND11	C	028CF5	32	2771	2680
TEND12	C	0298F5	32	2887	2796
TEND13	C	02A4F5	32	3001	2915
TEND14	C	02B255	32	3145	3029
TEND15	C	02BE55	32	3286	3170
TEND16	C	02CA55	32	3428	3311
TEND17	C	02D655	32	3569	3453
TEND18	C	02E255	32	3712	3595
TEND19	C	02EE55	32	3855	3739
TEND2	C	0220F5	32	1576	1456
TEND20	C	02FA55	32	3995	3880
TEND21	C	030655	32	4135	4020
TEND22	C	031255	32	4275	4160

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
TEND23	C	031E55	32	4418	4301
TEND24	C	032A55	32	4560	4445
TEND25	C	033655	32	4700	4585
TEND26	C	034255	32	4840	4725
TEND27	C	034E55	32	4981	4865
TEND28	C	035A55	32	5123	5006
TEND29	C	036655	32	5272	5150
TEND3	C	022CF5	32	1722	1601
TEND30	C	037255	32	5419	5297
TEND31	C	037E55	32	5566	5444
TEND32	C	038A55	32	5713	5591
TEND33	C	039655	32	5862	5738
TEND34	C	03A255	32	5997	5889
TEND35	C	03AE55	32	6130	6022
TEND36	C	03BA55	32	6263	6155
TEND37	C	03C655	32	6398	6288
TEND38	C	03D255	32	6533	6425
TEND39	C	03DE55	32	6666	6558
TEND4	C	0238F5	32	1867	1747
TEND40	C	03EA55	32	6799	6691
TEND41	C	03F655	32	6934	6824
TEND42	C	040255	32	7069	6961
TEND43	C	040E55	32	7202	7094
TEND44	C	041A55	32	7335	7227
TEND45	C	042655	32	7470	7360
TEND46	C	043255	32	7610	7497
TEND47	C	043E55	32	7748	7635
TEND48	C	044A55	32	7886	7773
TEND49	C	045655	32	8028	7911
TEND5	C	0244F5	32	2002	1894
TEND50	C	04625A	32	8167	8055
TEND51	C	046E5A	32	8311	8192
TEND52	C	047A5A	32	8445	8339
TEND53	C	04865A	32	8580	8470
TEND54	C	049252	32	8707	8608
TEND55	C	049E52	32	8836	8732
TEND56	C	04AA59	32	8965	8864
TEND57	C	04B658	32	9092	8990
TEND58	C	04C24E	32	9220	9117
TEND59	C	04CE53	32	9356	9258
TEND6	C	0250F5	32	2137	2027
TEND60	C	04DA54	32	9479	9381
TEND7	C	025CF5	32	2271	2162
TEND8	C	0268F5	32	2406	2296
TEND9	C	0274F5	32	2531	2433
TENDA	A	00002C	4	1102	635
TESTING	F	004004	4	757	370
TNUM	H	000004	2	1084	369 424 487 537
TSUB	A	000000	4	1083	374
TTABLE	F	004578	4	9487	
UR	U	000008	1	1087	479 501 3088 3229 3371 3512 3654 3798 3938 4078 4218 4359 4503 4643 4783 4924 5064 5215 5362 5509 5656 5803 5942 6075 6208 6341 6478 6611 6744 6877 7014 7147 7280 7413 7556 7694 7832 7971 8114 8252 8392 8523 8654 8781 8911 9037 9165 9303 9426
URFOUND	U	000009	1	1088	508 3087 3228 3370 3511 3653 3797 3937 4077 4217 4358 4502 4642 4782 4923 5063 5214 5361 5508 5655 5802 5941 6074 6207 6340 6477 6610 6743

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
					6876 7013 7146 7279 7412 7555 7693 7831 7970 8113 8251 8391 8522 8653 8780 8910 9036 9164 9302 9425
URMSG	U	002388	1	472	3090 3231 3373 3514 3656 3800 3940 4080 4220 4361 4505 4645 4785 4926 5066 5217 5364 5511 5658 5805 5944 6077 6210 6343 6480 6613 6746 6879 7016 7149 7282 7415 7558 7696 7834 7973 8116 8254 8394 8525 8656 8783 8913 9039 9167 9305 9428
URPRTEXP	C	004123	1	807	505
URPRTFUN	C	004112	2	804	498
URPRTGOT	C	004133	1	810	512
URPRTLINE	C	0040D6	16	800	812 515
URPRTLNG	U	00005F	1	812	514
URPRTNUM	C	0040E6	3	801	491
URSAV	D	002440	8	526	475 518
URSVR15	F	002434	4	523	473 520
URSVR5	F	002438	4	524	474 519
V0	U	000000	1	9577	
V1	U	000001	1	9578	
V10	U	00000A	1	9587	
V11	U	00000B	1	9588	
V12	U	00000C	1	9589	
V13	U	00000D	1	9590	
V14	U	00000E	1	9591	
V15	U	00000F	1	9592	
V16	U	000010	1	9593	
V17	U	000011	1	9594	
V18	U	000012	1	9595	
V19	U	000013	1	9596	
V2	U	000002	1	9579	
V20	U	000014	1	9597	
V21	U	000015	1	9598	
V22	U	000016	1	9599	
V23	U	000017	1	9600	
V24	U	000018	1	9601	
V25	U	000019	1	9602	
V26	U	00001A	1	9603	
V27	U	00001B	1	9604	
V28	U	00001C	1	9605	
V29	U	00001D	1	9606	
V3	U	000003	1	9580	
V30	U	00001E	1	9607	
V31	U	00001F	1	9608	
V4	U	000004	1	9581	
V5	U	000005	1	9582	
V6	U	000006	1	9583	
V7	U	000007	1	9584	
V8	U	000008	1	9585	
V9	U	000009	1	9586	
X0001	U	002098	1	242	230 243
XC0001	U	0020C0	1	256	248
=A(PLOTESTS)	A	002704	4	729	356
=AD(PAGHOM)	A	0026F8	8	727	319
=AD(PAGPRI)	A	0026E8	8	725	265
=AD(PAGSEC)	A	0026F0	8	726	292
=AL2(L'MSGMSG)	R	002716	2	734	678
=CL128' '	C	002719	128	736	543
=CL4'HOME'	C	002710	4	732	598 609 620 631 641

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
=CL4'SEC '	C	00270C	4	731	596 607 618 629 639
=F'1'	F	002700	4	728	247 412 482 534
=H'0'	H	002714	2	733	673
=X'04'	X	002718	1	735	345
=XL4'3'	X	002708	4	730	419

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 MACRO    DEFN  REFERENCES

FCHECK    122    229
PTTABLE  1217   9486
TSTCODE  1136   1313   1458   1603   1749   1896   2029   2164   2298   2435   2558   2682   2798   2917   3031   3172   3313   3455
                3597   3741   3882   4022   4162   4303   4447   4587   4727   4867   5008   5152   5299   5446   5593   5740   5891
                6024   6157   6290   6427   6560   6693   6826   6963   7096   7229   7362   7499   7637   7775   7913   8057   8194
                8341   8472   8610   8734   8866   8992   9119   9260   9383
TSTDATA  1148   1407   1552   1698   1843   1978   2113   2247   2382   2507   2631   2747   2863   2977   3121   3262   3404   3545
                3688   3831   3971   4111   4251   4394   4536   4676   4816   4957   5099   5248   5395   5542   5689   5838   5973
                6106   6239   6374   6509   6642   6775   6910   7045   7178   7311   7446   7587   7725   7863   8005   8143   8287
                8421   8556   8683   8812   8941   9068   9196   9332   9455
TSTEND   1127   1430   1575   1721   1866   2001   2136   2270   2405   2530   2654   2770   2886   3000   3144   3285   3427   3568
                3711   3854   3994   4134   4274   4417   4559   4699   4839   4980   5122   5271   5418   5565   5712   5861   5996
                6129   6262   6397   6532   6665   6798   6933   7068   7201   7334   7469   7609   7747   7885   8027   8166   8310
                8444   8579   8706   8835   8964   9091   9219   9355   9478
TSTHDR   1161   1291   1436   1581   1727   1874   2007   2142   2276   2413   2536   2660   2776   2895   3009   3150   3291   3433
                3575   3719   3860   4000   4140   4281   4425   4565   4705   4845   4986   5130   5277   5424   5571   5718   5869
                6002   6135   6268   6405   6538   6671   6804   6941   7074   7207   7340   7477   7615   7753   7891   8035   8172
                8319   8450   8588   8712   8844   8970   9097   9238   9361
TSTLBL   1115   1409   1414   1419   1424   1554   1559   1564   1569   1700   1705   1710   1715   1845   1850   1855   1860   1980
                1985   1990   1995   2115   2120   2125   2130   2249   2254   2259   2264   2384   2389   2394   2399   2509   2514
                2519   2524   2633   2638   2643   2648   2749   2754   2759   2764   2865   2870   2875   2880   2979   2984   2989
                2994   3123   3128   3133   3138   3264   3269   3274   3279   3406   3411   3416   3421   3547   3552   3557   3562
                3690   3695   3700   3705   3833   3838   3843   3848   3973   3978   3983   3988   4113   4118   4123   4128   4253
                4258   4263   4268   4396   4401   4406   4411   4538   4543   4548   4553   4678   4683   4688   4693   4818   4823
                4828   4833   4959   4964   4969   4974   5101   5106   5111   5116   5250   5255   5260   5265   5397   5402   5407
                5412   5544   5549   5554   5559   5691   5696   5701   5706   5840   5845   5850   5855   5975   5980   5985   5990
                6108   6113   6118   6123   6241   6246   6251   6256   6376   6381   6386   6391   6511   6516   6521   6526   6644
                6649   6654   6659   6777   6782   6787   6792   6912   6917   6922   6927   7047   7052   7057   7062   7180   7185
                7190   7195   7313   7318   7323   7328   7448   7453   7458   7463   7589   7594   7599   7604   7727   7732   7737
                7742   7865   7870   7875   7880   8007   8012   8017   8022   8145   8150   8155   8160   8289   8294   8299   8304
                8423   8428   8433   8438   8558   8563   8568   8573   8685   8690   8695   8700   8814   8819   8824   8829   8943
                8948   8953   8958   9070   9075   9080   9085   9198   9203   9208   9213   9334   9339   9344   9349   9457   9462
                9467   9472
```

```
   DESC      SYMBOL    SIZE        POS           ADDR

Entry: 0

Image       IMAGE     318068   00000-4DA73   00000-4DA73
  Region              318068   00000-4DA73   00000-4DA73
    CSECT   PLOETST   318068   00000-4DA73   00000-4DA73
```

```
   STMT                         FILE NAME
1     /devstor/sharedvfp/tests/plo-02-extension-armode.asm

** NO ERRORS FOUND **
```